# Contents

**Polyfold Theory, verified: the machine-checked edition** **1**

# Polyfold Theory, verified: the machine-checked edition

*A companion to the expanded exposition of Part I, and of Chapters 7–10 of Part II, of H. Hofer, K. Wysocki and E. Zehnder's* Polyfold and Fredholm Theory *(Springer 2021). Statement numbers and pages are those of the Springer edition, followed in parentheses by those of the text the formalization was first checked against: for Part I the 2014 preprint, arXiv 1407.3185 ("preprint"), for Part II, which has no preprint, the book's arXiv version, 1707.08941 ("2017"); a page that could not be located in the Springer text is given in the old numbering only, marked as such. The V- and G-numbers are the project's own, and the Lean names carry the preprint's (Part I) and the 2017 (Part II) numbers;* `Exposition/BookConcordance.md` *gives the correspondence). That exposition is written to be read; this edition is written to be checked. It records, chapter by chapter, exactly which statements of the text have been verified by the Lean 4 proof assistant against the Mathlib library, what each verified statement says when read back in ordinary language, which hypotheses the text left silent, where the formal statement or proof departs from the text's, and what remains unverified and why. Nothing here needs to be read in Lean; every checked statement is restated in prose, and the Lean names are given only so that the claims can be audited.*

**Where to start.** The project has three parts:

- the expanded exposition (`Exposition/Chapter1.md`, …), the book for reading;
- *Polyfold Theory, Checked*, the Lean formalization (`Polyfold/`) and the reports on it;
- the sign-off page, the list of statements a reader is asked to confirm.

This edition is the way into the second part. Its reports each answer one question:

- **What is proved?** This edition: every checked result in plain language, with all its hypotheses, and the list of statements to sign off.
- **How is it proved?** *Polyfold Theory, proof by proof* (`Exposition/ProofByProof.md`): the formal proofs retold as mathematics, with a trace tag on each step naming the Lean declaration that carries it.
- **How does it differ from the text?** *Polyfold Theory beside Lean*, the comparison (`Exposition/polyfold-beside`) each result of the book beside the checked statement, with a verdict (the same, restated, hypotheses made explicit, stronger, narrower) and the differences.

A result has the same number, V1.5 say, in this edition, the comparison and the sign-off page.

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## 0. Summary

| Chapter | HWZ | Content | Status | Lean lines |
|---|---|---|---|---|
| Chapter 1 | Ch. 1, §§1.1–1.4 | sc-Banach spaces, sc-operators and sc-Fredholm operators, sc-differentiability and its relation to classical differentiability, the chain rule, the degeneracy index | **verified** | 7,010 |
| Chapter 2 | Ch. 2 | sc-smooth retractions and retracts, M-polyfolds, the degeneracy index, tame M-polyfolds, strong bundles | **verified except Proposition 2.1.2 (preprint 2.2) and Lemma 2.4.9 (preprint 2.29) as stated; Remark 2.4.8's (preprint 2.28) negative claim is false** | 23,045 |
| Chapter 3 | Ch. 3 | sc-Fredholm sections and the implicit function theorem | **partly verified** | 56,006 |
| Chapter 4 | Ch. 4, §§4.1–4.4 | manifolds with boundary with corners: $sc^+$-retracts and the characterization, smooth finite-dimensional submanifolds, families and Sard's theorem (vendored) | **verified except Lemma 4.1.11 (preprint 4.10) and §4.4** | 14,051 |

| Chapter | HWZ | Content | Status | Lean lines |
|---|---|---|---|---|
| Chapter 5 | Ch. 5, §§5.1–5.3, 5.5 (preprint §§5.1–5.4) | the Fredholm package for M-polyfolds: auxiliary norms, compactness, $sc^+$-sections and bump functions, transversality, general and good position, homotopies, partitions of unity | **verified except Theorem 5.3.15 (preprint 5.22) and Theorem 5.27 (preprint; not in the 2021 edition) and the cited Theorems 5.5.5, 5.5.13, 5.5.16 (preprint 5.32, 5.40, 5.43)** | 12,687 |
| Chapter 6 | Ch. 6, §§6.2–6.6, 6.8 (preprint §§6.1–6.5, 6.7) | linearizations and orientations: determinant lines and exact sequences, the determinant of a Fredholm operator, good left-projections, stabilization, the determinant line bundle, the propagation of orientations along paths, orientable and oriented sc-Fredholm sections, the orientation of the solution set | **verified except Proposition 6.6.3 (1) (preprint 6.28), the global parts of Theorem 6.6.18 (preprint 6.41) and Remark 6.6.19 (preprint 6.42), and Theorem 6.7.1 (preprint 6.43)** | 22,675 |

| Chapter | HWZ | Content | Status | Lean lines |
|---|---|---|---|---|
| Chapter 7 | Part II, Ch. 7, §§7.1–7.3, 7.5–7.6 (2017 §§7.1–7.5) | ep-groupoids: groupoids of M-polyfolds with local sc-diffeomorphisms as source and target and a local properness, the orbit space, the natural representation of the isotropy groups, local uniformizers, ep-subgroupoids, effective and reduced ep-groupoids, the topology of the orbit space, index lifts, invariant sc-smooth partitions of unity, proper subsets | **verified except the tangent $T\varphi$ of Definition 7.1.4 (2017 Definition 7.4), the remark after Definition 7.1.7 (2017 Definition 7.7) in its 2017 form, the Hausdorff property of $\mathbf{X}^R$ in Proposition 7.2.10 (2017 Proposition 7.9) and Proposition 7.10 (2017; rewritten in 2021 as Theorem 7.2.14, which assumes it), and the index lifts $X^i$, $i \geq 2$, of Appendix 7.6.4** | 17,447 |

| Chapter | HWZ | Content | Status | Lean lines |
|---|---|---|---|---|
| Chapter 8 | Part II, Ch. 8, §§8.1–8.3, 8.5–8.6 (2017 §§8.1–8.5) | bundles and covering functors: the tangent ep-groupoid and the tangent functor, natural transformations, the orbit space of the tangent, strong bundles over ep-groupoids and their functorial picture, strong bundle maps and section functors, proper covering functors, their local structure and geometric lifts, proper strong bundle covering functors, the counting of the fibres | **verified except the sc-differential forms of Definitions 8.2.1 and 8.2.2 (2017 Definitions 8.2 and 8.3), the linearity of $T\varphi$ in Remark 8.1.10 (2017 Remark 8.1), and Lemma 8.1.2 (2017 Lemma 8.1) in the form $m_{T\mathbf{X}} = Tm$, and, of the 2021 additions, the step $T(g,h) = \mathrm{id} \Rightarrow Tg = \mathrm{id}$ of Proposition 8.1.16, Proposition 8.4.1 (1) □ (2), (3), Proposition 8.4.3 and the strong bundle structure of $F^*V$ in Lemma 8.5.10** | 10,995 |

| Chapter | HWZ | Content | Status | Lean lines |
|---|---|---|---|---|
| Chapter 9 | Part II, Ch. 9, §§9.1–9.5 (2017 Chapter 9) | branched ep$^+$-subgroupoids: local branching structures, the tangent and the boundary, good points, orientations, the geometry of local branching structures, sets of measure zero and the canonical σ-algebras (integration and Stokes not formalized) | **verified for §§9.1–9.4 and §9.6.1 with the readings and hypotheses recorded below, unconditionally for ep-groupoids whose object M-polyfold is $\mathbb{R}^m$, except the global part of Proposition 9.1.11, Proposition 9.1.9, Lemma 9.4.3 and the boundary-orientation geometry of Definitions 9.3.8 and 9.3.10; for §9.5 the sets of measure zero, the σ-algebras and the isotropy factors are verified, branched integration, the boundary measures and Stokes' theorem are not** | 16,925 |

| Chapter | HWZ | Content | Status | Lean lines |
|---|---|---|---|---|
| Chapter 10 | Part II, Ch. 10, §§10.1–10.6 (2017 Chapter 10) | equivalences and localization: equivalences of ep-groupoids, natural equivalences, the weak fibred product, localization at the equivalences and generalized maps, strong bundle equivalences, localization in the strong bundle case | **verified for equivalences, natural equivalences, the weak fibred product, the localization $\mathcal{EP}(E^{-1})$ and strong bundle equivalences, with the readings recorded below, except Remark 10.1.8, Propositions 10.1.9 and 10.1.10, the local regularity part of Proposition 10.1.19, the universal property of the localization, Proposition 10.3.16, Theorem 10.3.19, the (LR) and (SLR) parts of Theorem 10.3.21, the Fred part of Theorem 10.4.5 and Propositions 10.4.6 and 10.4.7; the strong bundle localization (§10.5) is not yet covered by this section** | 7,828 |

Total: 188,669 lines of Lean in 308 files under `Polyfold/` (`wc -l`, excluding unfinished `WIP_` files). Toolchain: Lean v4.33.1, Mathlib v4.33.1. To reproduce: `lake build`, then `lake env lean -DmaxSynthPendingDepth=3 Polyfold/Test.lean`, which prints the axioms of every main theorem. Edition of 8 November 2026.

## 1. How to read a verified statement

**The statements are the contract.** A machine-checked proof guarantees that the *stated* theorem follows from Lean's axioms. It guarantees nothing about whether the statement is the one the authors meant. The reader's job, and the only mathematical judgement this document asks for, is to read each prose restatement and confirm that it says what the text means. Where the restatement is weaker or stronger than the text, or uses a different but equivalent definition, this is flagged as a **Deviation**.

**The setting.** An sc-Banach space is formalized as an indexed family of real Banach spaces $E_m$, $m \geq 0$, with injective compact continuous linear inclusions $E_{m+1} \to E_m$ and the smooth points dense in every level; the text's subspace picture $E_0 \supset E_1 \supset \cdots$ is recovered by identifying $E_m$ with its image in $E_0$, and "$x \in E_m$" for $x \in E_0$ means that $x$ is in that image. A map $f : U \to F$ on a subset $U \subseteq E_0$ is a total function $E_0 \to F_0$ whose values off $U$ play no role; its level maps $f_m : U_m \to F_m$ are determined by the injectivity of the inclusions. "Relatively open in a partial quadrant" enters most theorems only through its consequences on each level, that $U_m$ is the intersection of an open and a convex set and has unique differentiability at each of its points (an *sc-domain*); whenever the text's hypothesis is replaced by this weaker one the item says so.

**Classical differentiability.** $C^k$ on a relatively open subset of a partial quadrant means Mathlib's $C^k$ *within the set*, with derivatives within the set; on these sets derivatives are unique, so the notion agrees with the exposition's Definition 3.C.

**Axioms.** Every verified theorem depends only on the three standard axioms of Lean's Mathlib (`propext`, `Classical.choice`, `Quot.sound`), as `Polyfold/Test.lean` prints; there is no `sorry`.

## 2. Chapter 1 (HWZ §§1.1–1.4): verified

Chapter 1 of Part I (pp. 5–20 (preprint pp. 6–17)) sets up the sc-calculus: sc-Banach spaces, sc-operators, sc-subspaces and quotients (Definitions 1.1.1–1.1.6 (preprint 1.1–1.5), Propositions 1.1.7–1.1.8 (preprint 1.6–1.7)), sc-Fredholm operators and their stability under $\mathrm{sc}^+$-perturbations (Definition 1.1.9 (preprint 1.8), Propositions 1.1.10–1.1.11 (preprint 1.9–1.10)), partial quadrants, $\mathrm{sc}^0$, $\mathrm{sc}^1$, $\mathrm{sc}^k$ and sc-smooth maps (Definitions 1.1.12–1.1.15 and the unnumbered definition after them (preprint Definitions 1.11–1.15)), the relation between sc-differentiability and classical differentiability on the levels (Propositions 1.2.1–1.2.4 (preprint 1.16–1.19), Corollary 1.2.5 (preprint 1.20)), the chain rule (Theorem 1.3.1 (preprint 1.21)), the degeneracy index of a partial quadrant (Definition 1.3.3 (preprint 1.23), Lemma 1.3.4 (preprint 1.24)) and boundary recognition (Theorem 1.3.5 (preprint 1.25)). Every numbered statement is verified, Proposition 1.2.4 (preprint 1.19) and Corollary 1.2.5 (preprint 1.20) under the text's own hypothesis ($C^{l+1}$ from $U_{m+l}$ to $F_m$ for $l \leq k$), by the 2010 paper's induction $(\mathrm{S}_k)$ on elementary maps (V1.37–V1.39). One is verified in a stronger form: Theorem 1.3.5 (preprint 1.25), which the text quotes from another paper, is proved here for $\mathrm{sc}^1$ germs. Example 1.1.3 (preprint 1.2), the weighted Sobolev spaces, is not formalized (Mathlib has no Sobolev spaces). The remarks of the text that the exposition turns into lemmas are verified as well (V1.28–V1.36): the invariance of the index under $\mathrm{sc}^+$-perturbations and on every level, the uniqueness of the sc-structure in finite dimensions, sc-splittings as sc-projections, the smooth points of a quotient, the compact-open continuity of $Df$, $\mathrm{sc}^1 = C^1$ in finite dimensions on quadrants

with boundary, and the level shift and tangent of a tuple. The setting throughout: an sc-Banach space is an indexed family of real Banach spaces $E_m$ with injective compact continuous linear inclusions $E_{m+1} \to E_m$ and dense smooth points; $\iota_m : E_m \to E_0$ is the composite inclusion, and "$x \in E_m$" for $x \in E_0$ means $x \in \iota_m(E_m)$. A map $f : U \to F$, $U \subseteq E_0$, is a total function $E_0 \to F_0$ whose values off $U$ do not matter, with level maps $f_m : U_m \to F_m$ determined by the injectivity of $\iota_m$. The text's hypothesis "$U$ relatively open in a partial quadrant $C$ of $E$" is, in most theorems, weakened to the *sc-domain* condition it implies: each $U_m$ is the intersection of an open and a convex subset of $E_m$ and has unique differentiability at each of its points. Where a theorem is stated for sc-domains the text's hypothesis is a special case. The infrastructure that Mathlib lacked and that is built here, the Riesz–Schauder theorem and the stability of Fredholm operators under compact perturbations (used without citation in §1.4.1), the compact-open lemma behind §1.2, and sc-domains, has its own items (V1.24–V1.27); so has the machinery of the proof of Proposition 1.2.4 (preprint 1.19) under the text's hypothesis, the derivative tower, the elementary maps with the induction $(\mathrm{S}_k)$, and the sum rule for $\mathrm{sc}^k$ maps (V1.37–V1.39).

**V1.1 (sc-Banach spaces, sums and shifts; `Polyfold.ScScale`, `Polyfold.ScScale.toBase`, `Polyfold.ScScale.smooth`, `Polyfold.ScScale.smoothLvl`, `Polyfold.ScBanach`, `Polyfold.ScBanach.denseRan` `Polyfold.ScBanach.const`, `Polyfold.ScBanach.prod`, `Polyfold.ScBanach.prod_smoothLvl`, `Polyfold.ScBanach.shift`, `Polyfold.ScBanach.shift_smoothLvl`, `Polyfold.ScBanach.shiftK`, `Polyfold.ScBanach.lvlSet`).** A *scale* consists of real Banach spaces $E_m$, $m \in \mathbb{N}$, and injective compact continuous linear maps $E_{m+1} \to E_m$; $\iota_m : E_m \to E_0$ is the composite, $E_\infty \subseteq E_0$ the subspace of points that lie in $\iota_m(E_m)$ for every $m$, and $E_\infty \cap E_m := \iota_m^{-1}(E_\infty)$. An *sc-Banach space* is a scale in which $E_\infty \cap E_m$ is dense in $E_m$ for every $m$. Consequences: the image of $E_{m+k}$ in $E_m$ is dense. Constructions: for a finite-dimensional normed space $V$ the constant family $E_m = V$, with identity inclusions, is an sc-Banach space all of whose points are smooth; for sc-Banach spaces $E$, $F$, the family $E_m \times F_m$ with the product inclusions is an sc-Banach space $E \oplus F$ whose smooth points at level $m$ are the pairs of smooth points; the family $E_{m+1}$ with the inclusions of $E$ is an sc-Banach space $E^1$ whose smooth points at level $m$ are those of $E$ at level $m+1$; $E^k$ is the $k$-fold iterate. For $U \subseteq E_0$, the level set $U_m := \iota_m^{-1}(U) \subseteq E_m$. *Book:* Definition 1.1.1 (preprint 1.1), p. 5 (preprint p. 6): a decreasing sequence of Banach spaces $E = E_0 \supset E_1 \supset \cdots$ with compact inclusions and $E_\infty = \bigcap E_m$ dense in every $E_m$; the constant structure on a finite-dimensional space; $(E \oplus F)_m = E_m \oplus F_m$; the filtration $A_m = A \cap E_m$ of a subset and the shift $A^k$. *Hidden hypotheses:* the inclusions are injective (implicit in "$\subset$") and continuous (part of compactness). *Deviation (restated):* the $E_m$ are separate Banach spaces, not subsets of one space; the two pictures are equivalent and nothing in Chapter 1 uses more than the family. The text's remarks that a finite-dimensional space has *only* the constant structure and that the constant family on an infinite-dimensional space is not an sc-structure are V1.31.

**V1.2 (sc-operators and sc-isomorphisms; `Polyfold.ScOperator`, `Polyfold.ScOperator.ext_base`, `Polyfold.IsScOperator`, `Polyfold.ScOperator.isScOperator`, `Polyfold.IsScOperator.toScOperator`, `Polyfold.ScOperator.comp`, `Polyfold.ScOperator.mapsTo_smoothLvl`, `Polyfold.ScIso`, `Polyfold.ScIso.symm`, `Polyfold.ScIso.trans`, `Polyfold.isScOperator_of_mapsTo`, `Polyfold.ScIso.ofBijec`** An *sc-operator* $T : E \to F$ between sc-Banach spaces is a family of continuous linear maps $T_m : E_m \to F_m$ with $\iota_m^F T_{m+1} = T_m \iota_m^E$; the family is determined by $T_0$, and a continuous linear $T_0 : E_0 \to F_0$ *is an sc-operator* exactly when for each $m$ some continuous linear $T_m : E_m \to F_m$ satisfies $\iota_m T_m = T_0 \iota_m$ (the two formulations correspond). Sc-operators compose, add, and map smooth points to smooth points. An *sc-isomorphism* is a family of continuous linear equivalences $E_m \simeq F_m$ commuting with the inclusions; inverses and composites are sc-isomorphisms. A continuous linear $T_0 : E_0 \to F_0$ with $T_0(E_m) \subseteq F_m$ for every $m$ is an sc-operator (the continuity

of $T_0 : E_m \to F_m$ is the closed graph theorem), and an sc-operator bijective on every level is an sc-isomorphism (the open mapping theorem). *Book:* Definition 1.1.4 (preprint 1.3), p. 6 (preprint p. 7). *Hidden hypotheses:* none. *Deviation (restated):* the level maps are data, with the text's predicate on the base map recovered as `Polyfold.IsScOperator`; the continuity of the inverse of a bijective sc-operator on each level, which the text's definition of an sc-isomorphism demands, is built into the formal notion, and derived from the open mapping theorem separately (`Polyfold.ScIso.ofBijective`); the continuity on the levels asked by Definition 1.1.4 (preprint 1.3) is automatic for a bounded $T_0$ that maps $E_m$ into $F_m$ (Findings, Chapter 3, item 233).

**V1.3 (sc⁺-operators; `Polyfold.ScPlus`, `Polyfold.ScBanach.shiftIncl`, `Polyfold.ScPlus.toScOperator`, `Polyfold.ScPlus.isCompactOperator`, `Polyfold.ScPlus.toScOperator_mem_range_toBase_succ`).** An sc$^+$*-operator* $E \to F$ is an sc-operator $E \to F^1$, i.e. continuous linear $S_m : E_m \to F_{m+1}$ commuting with the inclusions. Composed with the inclusion $F^1 \to F$ (an sc-operator) it is an sc-operator $E \to F$ whose level-$m$ map $E_m \to F_m$ is compact for every $m$, and which raises the level: $x \in \iota_m(E_m)$ implies $S_0 x \in \iota_{m+1}(F_{m+1})$. *Book:* Definition 1.1.5 (preprint 1.4), p. 7 (preprint p. 7), and the remark that follows it: an sc-operator $S$ with $S(E_m) \subset F_{m+1}$ and $S : E \to F^1$ an sc-operator is sc-compact. *Hidden hypotheses:* none; the condition $S(E_m) \subset F_{m+1}$ is contained in "$S : E \to F^1$ is an sc-operator" and is dropped.

**V1.4 (sc-subspaces and sc-splittings; `Polyfold.ScBanach.levelSub`, `Polyfold.ScBanach.isClosed_levelSub`, `Polyfold.ScBanach.IsScSubspace`, `Polyfold.ScBanach.ofSubspaceScale`, `Polyfold.ScBanach.ofSubspace`, `Polyfold.ScBanach.IsScSplitting`, `Polyfold.ScBanach.IsScSplitting.closedComplemented`, `Polyfold.ScBanach.IsScSplitting.symm`).** For a subspace $A \subseteq E_0$ let $A_m := \iota_m^{-1}(A) \subseteq E_m$, closed if $A$ is. $A$ is an *sc-subspace* if $A$ is closed and $A_m$ is contained in the closure of $A_m \cap E_\infty$ for every $m$. A closed $A$ with the levels $A_m$ and the restricted inclusions is a scale (the restricted inclusions are compact), and an sc-subspace is in this way an sc-Banach space. $E = F \oplus G$ is an *sc-splitting* if $F$ and $G$ are sc-subspaces and $F_m$, $G_m$ are complementary subspaces of $E_m$ for every $m$; then $F_m$ is closed complemented in $E_m$, and the splitting is symmetric in $F$ and $G$. *Book:* Definition 1.1.6 (preprint 1.5), p. 7 (preprint p. 7): $F$ closed with $(F \cap E_m)$ an sc-structure; an sc-complement $G$ with $(G \cap E_m)$ an sc-structure and $E_m = F_m \oplus G_m$ a topological direct sum on every level. *Hidden hypotheses:* none. *Deviation (restated):* the closedness of $F_m$ in $E_m$ and the compactness of $F_{m+1} \to F_m$ are automatic and are not conditions; "topological direct sum" is read as complementary closed subspaces, the continuity of the projections following from the closed graph theorem (`Polyfold.ScBanach.IsScSplitting.closedComplemented`).

**V1.5 (Proposition 1.1.7 (preprint 1.6); `Polyfold.ScBanach.proposition_1_6_iff`, `Polyfold.ScBanach.proposi`** Let $E$ be an sc-Banach space and $F \subseteq E_0$ a finite-dimensional subspace. $F$ is an sc-subspace if and only if $F \leq E_\infty$; and if $F \leq E_\infty$ there is a subspace $G \subseteq E_0$ such that $E = F \oplus G$ is an sc-splitting. *Book:* preprint p. 7, the same. *Hidden hypotheses:* none. *Proof:* as in the text, with a continuous projection $E_0 \to F$ from Mathlib (Hahn–Banach), lifted to each level through $F_m \cong F$.

**V1.6 (Proposition 1.1.8 (preprint 1.7); `Polyfold.ScBanach.quotientScale`, `Polyfold.ScBanach.quotient`, `Polyfold.ScBanach.quotientMap`, `Polyfold.ScBanach.proposition_1_7`).** Let $E$ be an sc-Banach space and $A \subseteq E_0$ a closed subspace. The quotient Banach spaces $E_m/A_m$, with the inclusions $E_{m+1}/A_{m+1} \to E_m/A_m$ induced by $\iota$ (injective and compact), and with the smooth points dense on every level, form an sc-Banach space $E/A$, and the quotient maps $E_m \to E_m/A_m$ form an sc-operator $E \to E/A$. As a statement: there exist an sc-Banach space $Q$ and an sc-operator $q : E \to Q$ with every $q_m$ surjective and $\ker q_m = A_m$. *Book:* preprint p. 8, for an sc-subspace $A$: $E/A$ with the filtration $E_m/A_m$ is an sc-Banach space, where $E_m/A_m$

is identified with $\{(x + A) \cap E_m : x \in E\}$. *Hidden hypotheses:* none. *Deviation (stronger):* only closedness of $A$ is used, not that $A$ is an sc-subspace. The text's identification of $E_m/A_m$ with a subset of $E/A$ is the injectivity of the induced inclusion, which holds because $A_m$ is the full preimage of $A$. The text's description of $(E/A)_\infty$ as the classes of points of $E_\infty$ is V1.33; only the trivial inclusion is used here.

**V1.7 (sc-Fredholm operators and the index; `Polyfold.ScFredholmSplitting`, `Polyfold.IsScFredholm`, `Polyfold.ScFredholmSplitting.index`, `Polyfold.scIndex`, `Polyfold.ScFredholmSplitting.index_eq`, `Polyfold.ScFredholmSplitting.K_le_smooth`, `Polyfold.ScFredholmSplitting.ker_lvl`, `Polyfold.ScFredholmSplitting.range_lvl`, `Polyfold.ScFredholmSplitting.isFredholm`).** An *sc-Fredholm splitting* of an sc-operator $T : E \to F$ is given by subspaces $K, X \subseteq E_0$ and $C, Y \subseteq F_0$ such that $E = K \oplus X$ and $F = C \oplus Y$ are sc-splittings, $K = \ker T_0$, $Y = \operatorname{range} T_0$, $K$ and $C$ are finite-dimensional, and for every $m$ the map $T_m$ is a bijection of $X_m$ onto $Y_m$. $T$ is *sc-Fredholm* if a splitting exists. The index $\dim K - \dim C$ of a splitting equals $\dim \ker T_0 - \dim(F_0/\operatorname{range} T_0)$, hence does not depend on the splitting. Moreover $K \le E_\infty$; $\ker T_m = K_m$ and $\operatorname{range} T_m = Y_m$; and every $T_m : E_m \to F_m$ is a Fredholm operator in Mathlib's sense (finite-dimensional kernel, closed range of finite codimension, strict, with closed complemented kernel). *Book:* Definition 1.1.9 (preprint 1.8), p. 8 (preprint p. 9), with the remarks that $K \subset E_\infty$, $T(X_m) = Y_m$, $E_m = K \oplus X_m$ and "$F = C \oplus T(E_m)$" (read $F_m$), and $\operatorname{ind}(T) = \dim K - \dim C$. *Hidden hypotheses:* none. *Deviation (restated):* condition (3), "$T : X \to Y$ is an sc-isomorphism", is taken as the bijectivity of $T_m : X_m \to Y_m$ for every $m$; the continuity of the inverses is automatic by the open mapping theorem on the closed subspaces $X_m$, $Y_m$. The independence of the index from the splitting and the level-wise Fredholm property are not stated in the text.

**V1.8 (Proposition 1.1.10 (preprint 1.9); `Polyfold.proposition_1_9`, `Polyfold.ScOperator.IsRegularizing`, `Polyfold.IsScFredholm.isRegularizing`, `Polyfold.isScFredholm_iff`).** Let $T : E \to F$ be an sc-operator with an sc-Fredholm splitting, $e \in E_0$ and $m \in \mathbb{N}$. If $T_0 e \in \iota_m(F_m)$ then $e \in \iota_m(E_m)$. As a property of sc-operators (`Polyfold.ScOperator.IsRegularizing`): every sc-Fredholm operator is regularizing. Conversely, an sc-operator $T$ with $T_0$ Fredholm that is regularizing is sc-Fredholm (V1.9); so $T$ is sc-Fredholm if and only if $T_0 : E_0 \to F_0$ is Fredholm and $T$ is regularizing. *Book:* preprint p. 9, the same ("an sc-Fredholm operator is regularizing"). *Hidden hypotheses:* none; the step "$c = 0$" uses $C \cap Y = 0$ on level 0, as in the text. The converse is not stated in the text; it is the tool for the text's unargued "is sc-Fredholm" for operators obtained by conjugation and restriction in Chapter 3 (V3.8, V3.46, V3.96; Findings, Chapter 3, item 233).

**V1.9 (Proposition 1.1.11 (preprint 1.10); `Polyfold.proposition_1_10`, `Polyfold.ScPlus.regularizing_add`, `Polyfold.isScFredholm_of_isRegularizing`, `Polyfold.isScFredholm_of_isRegularizing_index`).** Let $E$, $F$ be sc-Banach spaces, $T : E \to F$ an sc-Fredholm operator and $S : E \to F$ an $\mathrm{sc}^+$-operator. Then $T + S$ (with $S$ read as an sc-operator $E \to F$) is sc-Fredholm. Along the way, $T + S$ is regularizing in the sense of V1.8. *Book:* preprint p. 10, the same; the proof is in §1.4.1, pp. 14–16 (preprint pp. 14–15). *Hidden hypotheses:* none. *Proof:* the text's, with three additions. The opening claim that $T + S : E_m \to F_m$ is Fredholm is the classical compact-perturbation theorem, proved here since Mathlib lacks it (V1.24–V1.25). The step "$y = (T + S)e$, so by Proposition 1.9 $e \in E_m$" needs a bootstrap, since $Se$ is only one level up: $Te = y - Se \in F_1$ gives $e \in E_1$, then $Se \in F_2$, and so on $m$ times; the same bootstrap gives $K_m = K_{m+1}$. The complement $C$ is the range of a linear right inverse of $F_0 \to F_0/Y$ with values in $F_\infty$ rather than the span of a lifted basis, and condition (3) of Definition 1.1.9 (preprint 1.8) for $T + S$, which the text does not verify, is proved as the bijectivity of $(T + S)_m : X_m \to Y_m$. The text does not assert that the index

is preserved; it is, and this is V1.30. The construction uses only two properties of $T + S$, that it is Fredholm on level 0 and regularizing; with these as hypotheses it is proved as a criterion: an sc-operator that is Fredholm on level 0 and regularizing is sc-Fredholm, and every level map is then Fredholm of the level-0 index (not stated in the text).

**V1.10 (partial quadrants and tuples; `Polyfold.quadrant`, `Polyfold.quadrant_convex`, `Polyfold.quadrant_isClosed`, `Polyfold.lvlSet_quadrant`, `Polyfold.IsPartialQuadrant`, `Polyfold.IsPartialQuadrant.convex`, `Polyfold.IsPartialQuadrant.isClosed`, `Polyfold.IsPartialQuadran` `Polyfold.isPartialQuadrant_univ`, `Polyfold.IsRelOpenInQuadrant`).** In the sc-Banach space $\mathbb{R}^n \oplus W$ (with $\mathbb{R}^n$ constant) the *quadrant* on level $m$ is $\{(a, w) : a_i \geq 0 \text{ for all } i\}$; it is closed and convex with non-empty interior, and the level sets of the level-0 quadrant are the level quadrants. $C \subseteq E_0$ is a *partial quadrant* if there are $n$, an sc-Banach space $W$ and an sc-isomorphism $T : E \to \mathbb{R}^n \oplus W$ with $C = T_0^{-1}(\text{quadrant})$. Then every $C_m$ is closed, convex, with non-empty interior in $E_m$, and $C_m$ has unique differentiability (in Mathlib's sense) at each of its points; $E_0$ itself is a partial quadrant ($n = 0$). $(U, C, E)$ is a *tuple*, i.e. $U$ is a *relatively open subset of the partial quadrant* $C$, if $C$ is a partial quadrant and $U = O \cap C$ with $O \subseteq E_0$ open. *Book:* Definition 1.1.12 (preprint 1.11), p. 9 (preprint p. 10), and the sentence after it. *Hidden hypotheses:* none; "closed convex" is derived from the model. *Deviation:* none in the definition. The text's tacit use of $C_k$ as a partial quadrant in $E^k$ is V1.36 (for $k = 1$, and by iteration).

**V1.11 (sc$^0$ maps; `Polyfold.levelMap`, `Polyfold.IsSc0`, `Polyfold.IsSc0.levelMap_mapsTo`, `Polyfold.IsSc0.continuousOn_base`, `Polyfold.IsSc0.congr`, `Polyfold.IsSc0.mono`, `Polyfold.IsSc0.comp`, `Polyfold.isSc0_id`, `Polyfold.ScOperator.isSc0`).** For $f : E_0 \to F_0$ the *level map* $f_m : E_m \to F_m$ sends $x$ to the unique $y \in F_m$ with $\iota_m y = f(\iota_m x)$, when such $y$ exists (and to a junk value otherwise). $f$ is sc$^0$ *on* $U \subseteq E_0$ if for every $m$ and $x \in U_m$ the point $f(\iota_m x)$ lies in $\iota_m(F_m)$, and $f_m$ is continuous on $U_m$. Then $f$ is continuous on $U$; if $f(U) \subseteq V$ then $f_m(U_m) \subseteq V_m$; the property depends only on the values of $f$ on $U$ and passes to subsets of $U$; the composition of an sc$^0$ map on $U$ with values in $V$ and an sc$^0$ map on $V$ is sc$^0$ on $U$; the identity and every sc-operator are sc$^0$, the latter with level maps $T_m$. *Book:* Definition 1.1.13 (preprint 1.12), p. 9 (preprint p. 10): $f(U_m) \subset V_m$ (meaning $U'_m$) and $f : U_m \to U'_m$ continuous for all $m$. *Hidden hypotheses:* none. *Deviation (restated):* maps are total functions on $E_0$, and the target tuple enters only through $f(U) \subseteq U'$.

**V1.12 (the tangent; `Polyfold.ScBanach.tangent`, `Polyfold.ScBanach.tangentSet`, `Polyfold.ScBanach.lvlSet` `Polyfold.ScBanach.tangentSet_mono`, `Polyfold.IsScDomain.tangentSet`, `Polyfold.ScBanach.shift_tangen`** The *tangent* of an sc-Banach space $E$ is $TE := E^1 \oplus E$, with levels $E_{m+1} \times E_m$. For $U \subseteq E_0$, $TU \subseteq (TE)_0 = E_1 \times E_0$ is $\{(x, h) : x \in U_1\}$, with level-$m$ part $U_{m+1} \times E_m$; $U \subseteq V$ implies $TU \subseteq TV$. If $U$ is an sc-domain (V1.27), $TU$ is an sc-domain in $TE$; and the level-1 part of $TU$ is $T(U_1)$ in $T(E^1) = (TE)^1$. *Book:* Definition 1.1.14 (preprint 1.13), p. 9 (preprint p. 10): $T(U, C, E) = (U^1 \oplus E, C^1 \oplus E, E^1 \oplus E)$, "again a tuple". *Hidden hypotheses:* none. *Deviation:* none; that $TC$ is a partial quadrant in $TE$, so that $T(U, C, E)$ is again a tuple, is V1.36, while the proofs of §§1.2–1.3 propagate the sc-domain property (V1.27) instead.

**V1.13 (sc$^1$ maps and the tangent map; `Polyfold.tangentMapWith`, `Polyfold.IsSc1With`, `Polyfold.IsSc1`, `Polyfold.scDeriv`, `Polyfold.tangentMap`, `Polyfold.IsSc1With.eq_of_uniqueDiffWithinAt` `Polyfold.IsSc1.scDeriv_eqOn`, `Polyfold.IsSc1.tangentMap_eqOn`, `Polyfold.IsSc1.congr`, `Polyfold.IsSc1With.continuousOn_deriv`, `Polyfold.IsSc1With.exists_delta_deriv`, `Polyfold.exists_isSc1_not_continuousOn_deriv`, `Polyfold.ScOneExample.isSc1With`, `Polyfold.ScOneExample.not_continuousAt_D`).** Let $f : E_0 \to F_0$, $U \subseteq E_0$ and $D : E_1 \to \mathcal{L}(E_0, F_0)$ a family of bounded operators. The tangent map $T_D f : (TE)_0 \to (TF)_0$ is

$(x,h) \mapsto (f_1(x), D(x)h)$. $f$ is sc$^1$ *on $U$ with derivatives $D$* if (1) $f$ is sc$^0$ on $U$; (2) for every $x \in U_1$, the map $f \circ \iota_1 : E_1 \to F_0$ has derivative $D(x) \circ \iota_1$ at $x$ within $U_1$; (3) $T_D f$ is sc$^0$ on $TU$. $f$ is sc$^1$ *on $U$* if some $D$ makes it so; `Polyfold.scDeriv` is a chosen such $D$ (junk if none exists) and `Polyfold.tangentMap` the tangent map built from it. If $D$ and $D'$ both satisfy (2) at a point $x \in U_1$ where $U_1$ has unique differentiability, then $D(x) = D'(x)$, because $E_1$ has dense image in $E_0$; hence on an sc-domain the chosen derivative agrees on $U_1$ with any admissible one, and the tangent map on $TU$. The property depends only on the values of $f$ on $U$. Level 0 of condition (3) is the continuity of $(x,h) \mapsto D(x)h$ on $U_1 \times E_0$, and it implies: for $x \in U_1$, a compact $K \subseteq E_0$ and $c > 0$ there is $\delta > 0$ with $\|D(x')w - D(x)w\| \le c$ for all $x' \in U_1$ with $\|x' - x\|_1 < \delta$ and all $w \in K$ (continuity of $x \mapsto D(x)$ in the compact-open topology). It need not be norm continuous: on $E = \mathbb{R} \oplus X$, with $X$ the sc-Banach space whose level $m$ consists of the real sequences with $(k+1)^m x_k \to 0$, normed by $\sup_k (k+1)^m |x_k|$ (every level a copy of $c_0$, the inclusions $\mathrm{diag}(1/(k+1))$), the map $f(t,x)_k = \cos(kt)x_k$ is sc$^1$ with $Df(t,x)(\tau,\xi)_k = -k\sin(kt)x_k\tau + \cos(kt)\xi_k$, and for every derivative family $D$ of $f$ the map $x \mapsto D(x)$ is not continuous on $E_1$ into $\mathcal{L}(E_0, X_0)$ with the operator norm: $(\pi/(n+1), 0) \to 0$ in $E_1$, but $\|Df(\pi/(n+1),0) - Df(0,0)\| \ge 2$. *Book:* Definition 1.1.15 (preprint 1.14), pp. 10 ff. (preprint pp. 10–11), with conditions (1)–(3) as stated, and the remarks that $x \mapsto Df(x)$ is continuous for the compact-open but not in general for the norm topology, and that sc¹ maps between finite-dimensional spaces are the $C^1$ maps. *Hidden hypotheses:* the uniqueness of $Df(x)$, without which "the tangent map" is undefined; it holds because $E_1$ is dense in $E_0$ and $U_1$ has unique differentiability (true for relatively open subsets of partial quadrants, V1.10). *Deviation (hypotheses explicit):* the derivative family is an explicit parameter of `Polyfold.IsSc1With`, with `Polyfold.IsSc1` the text's notion. The compact-open remark is V1.34, its negative half the example above, which the text does not give (Findings, Chapter 1, follow-up item 1), and the finite-dimensional remark is V1.35 (on relatively open subsets of quadrants, with boundary; the open case is also in `Polyfold/Test.lean`, `isSc1_iff_contDiffOn_of_finiteDimensional`).

**V1.14 (sc^k and sc-smooth maps; `Polyfold.IsSck`, `Polyfold.IsScSmooth`, `Polyfold.IsSck.of_succ`, `Polyfold.IsSck.of_le`, `Polyfold.isScSmooth_iff`, `Polyfold.IsSck.congr`).** By recursion: $f$ is sc$^0$ on $U$ in the sense of V1.11; $f$ is sc$^{k+1}$ on $U$ if it is sc$^1$ on $U$ and its tangent map (built from the chosen derivative) is sc$^k$ on $TU \subseteq (TE)_0$ as a map into $(TF)_0$. $f$ is *sc-smooth* on $U$ if it is sc$^k$ for every $k$. Then sc$^{k+1}$ implies sc$^k$, and sc$^l$ for $l \le k$; $f$ is sc-smooth iff it is sc$^1$ with sc-smooth tangent map; on an sc-domain, sc$^k$ depends only on the values on $U$. *Book:* preprint p. 11, the inductive definition after Definition 1.1.15 (preprint 1.14) (numbered 1.15 here, as in the Lean files; Remark 1.1.17 (preprint 1.15) on partial cones is not formalized). *Hidden hypotheses:* none beyond V1.13.

**V1.15 (Proposition 1.2.1 (preprint 1.16); `Polyfold.IsSc1Levels`, `Polyfold.IsSc1Levels.continuousOn_clm`, `Polyfold.IsSc1Levels.contDiffOn`, `Polyfold.IsSc1Levels.toBase_apply`, `Polyfold.IsSc1Levels.isSc1Wit` `Polyfold.IsSc1With.isSc1Levels`, `Polyfold.proposition_1_16`, `Polyfold.IsSc1With.contDiffOn_base`, `Polyfold.IsSc1With.scOperator_deriv`).** Let $U$ be an sc-domain in $E$ and $f$ sc$^0$ on $U$. Then $f$ is sc$^1$ on $U$ if and only if there is a family $D_m(x) \in \mathcal{L}(E_m, F_m)$, $m \in \mathbb{N}$, $x \in U_{m+1}$, such that for every $m$: (a) $f_m \circ \iota : U_{m+1} \to F_m$ has derivative $D_m(x) \circ \iota$ at every $x \in U_{m+1}$ within $U_{m+1}$; (b) $(x,h) \mapsto D_m(x)h$ is continuous on $U_{m+1} \times E_m$. For any such family, $x \mapsto D_m(x) \circ \iota$ is continuous $U_{m+1} \to \mathcal{L}(E_{m+1}, F_m)$ in the operator norm, so $f : U_{m+1} \to F_m$ is $C^1$ within $U_{m+1}$; and $\iota_m D_m(x) = D_0(\iota x)\iota_m$, so all $D_m(x)$ are restrictions of the single operator $Df(x) := D_0(x)$. In the direction "sc¹ □ levels" the family is the level operators of V1.26; in particular $f \circ \iota_1 : U_1 \to F_0$ is $C^1$. If $f$ is sc$^1$ with derivatives $Df$ and $x \in U_1$ is a smooth point, $Df(x) : E_0 \to F_0$ is an

sc-operator. *Book:* preprint p. 12: an $\mathrm{sc}^0$ map on a relatively open subset of a partial quadrant is $\mathrm{sc}^1$ iff (1) for $m \geq 1$, $f : U_m \to F_{m-1}$ is $C^1$ with $df : U_m \to \mathcal{L}(E_m, F_{m-1})$ continuous and (2) $df(x)$ extends to a bounded $Df(x) : E_{m-1} \to F_{m-1}$ with $(x, h) \mapsto Df(x)h$ continuous on $U_m \oplus E_{m-1}$; in particular $Df(x)$ is an sc-operator at smooth points. *Hidden hypotheses:* the compatibility of the extensions $Df(x)$ across levels, which the text's notation assumes and the direction ($\square$) needs (Lemma 3.H of the exposition); it is proved. *Deviation (restated):* the extensions form an explicit family; the $C^1$ property and the norm continuity of $df$ are consequences of (a)–(b) rather than part of the condition; the hypothesis on $U$ is the weaker sc-domain condition.

**V1.16 (Proposition 1.2.2 (preprint 1.17); `Polyfold.IsSc0.shift`, `Polyfold.IsSc1With.shift`, `Polyfold.IsSck.shift`, `Polyfold.IsScSmooth.shift`, `Polyfold.IsScDomain.shift`).** Let $U$ be an sc-domain in $E$ and $f$ $\mathrm{sc}^k$ on $U$. Then the level-1 map $f_1 : E_1 \to F_1$ is $\mathrm{sc}^k$ on $U_1$ as a map between the shifted sc-Banach spaces $E^1$ and $F^1$ (whose level sets of $U_1$ are the $U_{m+1}$). For $k = 0$ no hypothesis on $U$ is needed; for an $\mathrm{sc}^1$ map with derivatives $Df$, the derivatives of $f_1$ are the level-1 operators $Df(x) : E_1 \to F_1$, $x \in U_2$; sc-smooth maps lift to sc-smooth maps; and $U_1$ is an sc-domain in $E^1$. *Book:* preprint p. 12: if $f : U \to V$ is $\mathrm{sc}^k$ between relatively open subsets of partial quadrants then $f : U^1 \to V^1$ is $\mathrm{sc}^k$. *Hidden hypotheses:* none. *Deviation:* none; the target $V$ plays no role and the hypothesis on $U$ is weaker.

**V1.17 (Proposition 1.2.3 (preprint 1.18); `Polyfold.IsSck.contDiffOn_levels'`, `Polyfold.IsSck.contDiffOn_l`
`Polyfold.IsSck.contDiffOn_levels_of_le`, `Polyfold.IsSck.contDiffOn_base`, `Polyfold.IsScSmooth.contD`**
Let $U$ be an sc-domain in $E$ and $f$ $\mathrm{sc}^k$ on $U$. Then for every $m$, the map $f_m \circ \iota : U_{m+k} \to F_m$ is $C^k$ within $U_{m+k}$; for $l \leq k$, $f_m \circ \iota : U_{m+l} \to F_m$ is $C^l$; in particular $f \circ \iota_k : U_k \to F_0$ is $C^k$. If $f$ is sc-smooth, $f : U_{m+k} \to F_m$ is $C^k$ for all $m$ and $k$. *Book:* preprint p. 13, the same for $f : U \to V$ between relatively open subsets of partial quadrants. *Hidden hypotheses:* the meaning of "$C^k$" on a relatively open subset of a quadrant, never defined in the text: here Mathlib's $C^k$ within the set, with unique derivatives on these sets. *Deviation:* none in the statement. *Proof:* a direct induction on $k$ over all sc-domains, with a compatible level-lowering operator $\rho : E_{m+k} \to E_m$ as parameter, and the compact-open lemma (V1.26) applied to the second component of the level tangent map; the 2010 paper's multilinear maps $\Gamma(x)$ and its differentiation under the integral sign are not used.

**V1.18 (Proposition 1.2.4 (preprint 1.19); `Polyfold.proposition_1_19_text`, `Polyfold.IsScSmooth.of_contDif`
`Polyfold.proposition_1_19_zero`, `Polyfold.proposition_1_19`, `Polyfold.IsScSmooth.of_contDiffOn_leve`**
Let $U$ be an sc-domain in $E$ and $f$ $\mathrm{sc}^0$ on $U$. If for every $m \geq 0$ and $0 \leq l \leq k$ the map $f_m \circ \iota : U_{m+l} \to F_m$ (the level-$m$ map restricted along the inclusion $E_{m+l} \to E_m$) is $C^{l+1}$ within $U_{m+l}$, then $f$ is $\mathrm{sc}^{k+1}$ on $U$; if this holds for every $l$, then $f$ is sc-smooth. Also proved, under the stronger hypothesis that every level map $f_m : U_m \to F_m$ is $C^{k+1}$ within $U_m$: $f$ is $\mathrm{sc}^{k+1}$; for $k = 0$, where the two hypotheses agree, $f$ is $\mathrm{sc}^1$ with $Df(x)$ the derivative of $f : U_0 \to F_0$ within $U$ at $\iota_1 x$; and if $f$ maps $U_m$ into $F_m$ for every $m$ and every $f_m$ is $C^\infty$, then $f$ is sc-smooth. *Book:* preprint p. 13: if $f : U_{m+l} \to F_m$ is $C^{l+1}$ for all $m \geq 0$ and $0 \leq l \leq k$, then $f$ is $\mathrm{sc}^{k+1}$. *Hidden hypotheses:* none ("$f$ induces a map $U_{m+l} \to F_m$" is the $\mathrm{sc}^0$ hypothesis). *Deviation:* none; the sc-smooth statement is a direct consequence. *Proof:* the 2010 paper's (Proposition 2.4): $f$ is the elementary map with no vectors, and every elementary map with $j$ vectors and $j + s \leq k + 1$ is $\mathrm{sc}^s$ (V1.38), with the derivatives $d^j f(x)$ supplied by the derivative tower of $f$ (V1.37). The induction cannot be run on the hypothesis itself: if $f$ satisfies it with $k$, then $Tf$ in general does not satisfy it with $k - 1$, since its second component $(x, h) \mapsto Df(x)h : U_{m+l+1} \times E_{m+l} \to F_m$ is only $C^l$. Under the stronger hypothesis the level maps of $Tf$ are $C^k$ and the induction is immediate.

**V1.19 (Corollary 1.2.5 (preprint 1.20); `Polyfold.corollary_1_20_text`, `Polyfold.IsScSmooth.of_contDiffOn_`
`Polyfold.IsScSmooth.of_contDiffOn_finiteDimensional`, `Polyfold.corollary_1_20_zero`,
`Polyfold.corollary_1_20`, `Polyfold.corollary_1_20_base`).** Let $U$ be an sc-domain in $E$, $V$ a finite-dimensional normed space with its constant sc-structure and $f : E_0 \to V$. If for some $k$ and all $0 \le l \le k$ the map $f \circ \iota_l : U_l \to V$ is $C^{l+1}$ within $U_l$, then $f$ is $\mathrm{sc}^{k+1}$ on $U$. If $f \circ \iota_l$ is $C^{l+1}$ within $U_l$ for every $l$, then $f$ is sc-smooth; if $f$ is $C^\infty$ within $U$, then $f$ is sc-smooth. Also proved, under the stronger hypothesis: if $f \circ \iota_m$ is $C^{k+1}$ within $U_m$ for every $m$, in particular if $f$ is $C^{k+1}$ within $U$, then $f$ is $\mathrm{sc}^{k+1}$; for $k = 0$ this is the text's hypothesis. *Book:* preprint p. 13: if $f : U_l \to \mathbb{R}^N$ is $C^{l+1}$ for $0 \le l \le k$ then $f$ is $\mathrm{sc}^{k+1}$. *Hidden hypotheses:* none; no $\mathrm{sc}^0$ hypothesis is needed, since $f \circ \iota_m$ is the continuous $f \circ \iota_0$ composed with an inclusion. *Deviation:* none ($\mathbb{R}^N$ is any finite-dimensional space); the two sc-smooth statements are the exposition's additions. *Proof:* Proposition 1.2.4 (preprint 1.19) (V1.18) for the constant structure, the hypothesis on $U_{m+l} \to V$ being the restriction of the one on $U_l$.

**V1.20 (Theorem 1.3.1 (preprint 1.21), the chain rule; `Polyfold.IsSc1With.hasFDerivWithinAt_comp`,
`Polyfold.IsSc1With.comp`, `Polyfold.theorem_1_21'`, `Polyfold.theorem_1_21`, `Polyfold.IsSck.comp`,
`Polyfold.IsSck.comp_of_isRelOpenInQuadrant`, `Polyfold.IsScSmooth.comp`).** Let $U \subseteq C \subseteq E$ and $V \subseteq D \subseteq F$ be relatively open subsets of partial quadrants, $G$ an sc-Banach space, $f : E_0 \to F_0$ $\mathrm{sc}^1$ on $U$ with $f(U) \subseteq V$, and $g : F_0 \to G_0$ $\mathrm{sc}^1$ on $V$. Then $g \circ f$ is $\mathrm{sc}^1$ on $U$, and $T(g \circ f) = Tg \circ Tf$ on $TU$. More precisely: if $g$ is $\mathrm{sc}^1$ on $V$ with derivatives $Dg$, $f$ is $\mathrm{sc}^1$ on $U$ with derivatives $Df$, $f(U) \subseteq V$, and $V_1$ is the intersection of an open and a convex set, then $g \circ f$ is $\mathrm{sc}^1$ on $U$ with derivatives $x \mapsto Dg(f_1(x)) \circ Df(x)$, with no hypothesis on $U$; and $g \circ f \circ \iota_1$ has this derivative at every $x \in U_1$ within $U_1$. Iterated: if $U$, $V$ are sc-domains (in particular relatively open subsets of partial quadrants), $f$ $\mathrm{sc}^k$ on $U$ with $f(U) \subseteq V$ and $g$ $\mathrm{sc}^k$ on $V$, then $g \circ f$ is $\mathrm{sc}^k$ on $U$; sc-smooth maps compose to sc-smooth maps. *Book:* preprint p. 13, with a third tuple $W \subset Q \subset G$ for the target of $g$; the proof is §1.4.2, pp. 15–16 (preprint pp. 15–16). *Hidden hypotheses:* the segment from $f(x)$ to $f(x+h)$ lies in $V_1$ for $|h|_1$ small, which uses the convexity of $D_1$, the openness of $V_1$ in $D_1$ and the continuity of $f : U_1 \to F_1$; the uniqueness of $D(g \circ f)(x)$ behind "$T(g \circ f) = Tg \circ Tf$". *Deviation:* none in the statement; $W \subset Q \subset G$ plays no role. *Proof:* the text's two integrals are replaced by the mean value inequality for $t \mapsto g(y_t) - t\,Dg(f(x))(f(x+h) - f(x))$ on $[0,1]$; the first integral is not needed since $Dg(f(x))$ is bounded on $F_0$; the uniform convergence on the compact closure of $\{Df(x)h/|h|_1\}$ is Mathlib's `IsCompact.eventually_forall_of_forall_eventually` rather than a subsequence argument. The text's appeal to Proposition 1.2.3 (preprint 1.18) is not used.

**V1.21 (Definition 1.3.3 (preprint 1.23); `Polyfold.quadrantDeg`, `Polyfold.degIndexWith`,
`Polyfold.degIndex`, `Polyfold.degIndex_eq`, `Polyfold.degIndex_quadrant`).** For $p = (a, w) \in \mathbb{R}^n \oplus W$ let $\#(p)$ be the number of $i$ with $a_i = 0$; for an sc-isomorphism $T : E \to \mathbb{R}^n \oplus W$ presenting a partial quadrant $C$ let $d_T(x) := \#(T_0 x)$. The *degeneracy index* $d_C : E_0 \to \mathbb{N}$ is $d_T$ for a chosen presentation $T$ of $C$ (and 0 if $C$ is not a partial quadrant); for every presentation $T$ and every $x \in C$, $d_C(x) = d_T(x)$. On the quadrant itself $d(p) = \#(p)$. *Book:* preprint p. 14: $d_C(x) = \#\{i : a_i = 0\}$ where $T(x) = (a_1, \dots, a_k, w)$; interior points, boundary points and corners. *Hidden hypotheses:* none. *Deviation:* none; $d_C$ is total on $E_0$ with junk values off $C$, and the classification into interior, boundary and corner points is terminology.

**V1.22 (Lemma 1.3.4 (preprint 1.24); `Polyfold.quadrantDeg_eq_of_equiv`, `Polyfold.lemma_1_24`,
`Polyfold.lemma_1_24_dim`).** Let $T : E \to \mathbb{R}^n \oplus W$ and $T' : E \to \mathbb{R}^{n'} \oplus W'$ be sc-isomorphisms with $C = T_0^{-1}(Q_n) = T_0'^{-1}(Q_{n'})$. Then $d_T(x) = d_{T'}(x)$ for every $x \in C$, and $n = n'$. Core: if $S : \mathbb{R}^n \times W_0 \to \mathbb{R}^{n'} \times W_0'$ is a continuous linear equivalence with $q \in Q_n \iff Sq \in Q_{n'}$,

then $\#(p) = \#(Sp)$ for every $p \in Q_n$. *Book:* p. 14 (preprint p. 14), "$d_C$ does not depend on the choice of $T$", proof on preprint p. 17. *Hidden hypotheses:* none; the proof uses only linear algebra on level 0. *Proof:* the text's, except that $k = k'$ is not proved first but derived from the lemma at $x = 0$; the codimension of $E_x$ is $\dim(V/E_x) = \#I(x)$ with $E_x = \ker((b, v) \mapsto (b_i)_{i \in I(x)})$, and "$S(E_x) \subset E'_y$ hence equality" is completed by the same inclusion for $S^{-1}$.

**V1.23 (Theorem 1.3.5 (preprint 1.25), boundary recognition; `Polyfold.theorem_1_25`, `Polyfold.theorem_1_25_lvl1`, `Polyfold.degIndexWith_le_of_germ`, `Polyfold.eventually_degIndexWith_le`, `Polyfold.exists_tendsto_lvl1_degIndexWith`, `Polyfold.IsSc1With.mapsTo_minimalSubspace`).** Let $U \subseteq C \subseteq E$ and $U' \subseteq C' \subseteq E'$ be relatively open subsets of partial quadrants, $f : E_0 \to E'_0$ $\mathrm{sc}^1$ on $U$ with $f(U) \subseteq U'$, $g : E'_0 \to E_0$ $\mathrm{sc}^1$ on $U'$ with $g(U') \subseteq U$, $g(f(x)) = x$ for $x \in U$ and $f(g(y)) = y$ for $y \in U'$. Then $d_C(x) = d_{C'}(f(x))$ for every $x \in U$. Ingredients: $d_{C'}$ is upper semicontinuous ($d_{C'}(y') \le d_{C'}(y)$ for $y'$ near $y$); every $x \in C$ is a limit of points of $C$ on level 1 with the same index; and for $x \in U_1$ the derivative $Df(x)$ maps $E_x := \{v : (T_0 v)_i = 0 \text{ whenever } (T_0 x)_i = 0\}$ into $E'_{f(x)}$, from which $d_C(x) \le d_{C'}(f(x))$ follows by comparing codimensions through the chain rule. *Book:* preprint p. 14: for tuples $(U, C, E)$, $(U', C', E')$ and a germ $f : (U, x) \to (U', x')$ of a local sc-diffeomorphism, $d_C(x) = d_{C'}(f(x))$; no proof, a reference to "[20], Theorem 1.19". *Hidden hypotheses:* the notion "germ of a local sc-diffeomorphism", undefined in Chapter 1, read as an $\mathrm{sc}^1$ map $f : U \to U'$ with $\mathrm{sc}^1$ inverse $g : U' \to U$ on relatively open neighbourhoods $U$, $U'$ of $x$, $f(x)$ (which may be shrunk at will). *Deviation (stronger):* only $\mathrm{sc}^1$ is required of $f$ and $g$, where the text has sc-smooth. *Proof:* the argument of §2.4 (preprint §2.3) (Lemma 2.4.9 (preprint 2.29), Theorem 2.4.12 (preprint 2.31), Corollary 2.4.13 (preprint 2.32)) for the identity retraction, with level-1 points in place of smooth points; the Lean file is `Polyfold/Chapter2/DegeneracyIndex.lean`.

**V1.24 (Fredholm operators between Banach spaces, Riesz–Schauder; `Polyfold.isStrictMap_of_isClosed_ran`, `Polyfold.isFredholm_of_banach`, `Polyfold.isCompactOperator_of_finiteDimensional_range`, `Polyfold.finiteDimensional_ker_one_add`, `Polyfold.isClosed_range_one_add`, `Polyfold.coFG_range_one_`, `Polyfold.isFredholm_one_add`).** Let $E$, $F$ be real Banach spaces. A continuous linear $u : E \to F$ with closed range is a strict map; if moreover $\ker u$ is finite-dimensional and range $u$ has finite codimension, $u$ is a Fredholm operator in Mathlib's sense. A continuous linear map of finite rank is a compact operator. For a compact operator $c : E \to E$: $\ker(1 + c)$ is finite-dimensional, $\mathrm{range}(1 + c)$ is closed and of finite codimension, and $1 + c$ is Fredholm. *Book:* not stated; used in §1.4.1, preprint p. 14 ("$T + S : E_m \to F_m$ is Fredholm", "its image $Y$ is closed"). *Deviation (not in the book):* infrastructure Mathlib v4.33.1 lacks.

**V1.25 (compact perturbations of Fredholm operators; `Polyfold.isFredholm_add_isCompactOperator`).** Let $E$, $F$ be real Banach spaces, $u : E \to F$ Fredholm and $k : E \to F$ compact. Then $u + k$ is Fredholm. *Book:* not stated; the first line of §1.4.1. *Deviation (not in the book):* infrastructure; the invariance of the index, which the exposition records (Kato IV.5.26) and the text does not, is V1.28.

**V1.26 (the compact-open lemma and the level operators; `Polyfold.exists_delta_of_continuousOn`, `Polyfold.continuousOn_clm_comp_of_isCompactOperator`, `Polyfold.contDiffOn_clm_comp_of_isCompactOp`, `Polyfold.levelDeriv`, `Polyfold.IsSc1With.exists_levelDeriv`, `Polyfold.IsSc1With.toBase_levelDeriv`, `Polyfold.IsSc1With.levelMap_tangentMapWith`, `Polyfold.IsSc1With.continuousOn_levelDeriv`, `Polyfold.IsSc1With.continuousOn_levelDeriv_clm`, `Polyfold.IsSc1With.hasFDerivWithinAt_level`, `Polyfold.IsSc1With.contDiffOn_level`, `Polyfold.IsSc1With.levelDeriv_eq`).** *Compact-open lemma.* Let $X$, $Y$, $Y'$, $Z$ be real normed spaces, $\iota : Y' \to Y$ a compact operator and $A : X \to \mathcal{L}(Y, Z)$. If $(x, y) \mapsto A(x)y$ is continuous on $s \times Y$, then $x \mapsto A(x) \circ \iota$ is continuous

on $s$ into $\mathcal{L}(Y', Z)$ with the operator norm; if $s$ is the intersection of an open and a convex set with unique differentiability at every point and $(x, y) \mapsto A(x)y$ is $C^k$ within $s \times Y$, then $x \mapsto A(x) \circ \iota$ is $C^k$ within $s$. *Level operators.* Let $f$ be sc$^1$ on $U$ with derivatives $Df$, $m \in \mathbb{N}$ and $x \in U_{m+1}$. There is a unique continuous linear $Df(x) : E_m \to F_m$ (written `levelDeriv`) with $\iota_m \circ Df(x) = Df(\iota x) \circ \iota_m$, namely the second component of the level-$m$ tangent map, which is $(x, h) \mapsto (f_{m+1}(x), Df(x)h)$ on $U_{m+1} \times E_m$; $(x, h) \mapsto Df(x)h$ is continuous on $U_{m+1} \times E_m$; $x \mapsto Df(x) \circ \iota$ is continuous $U_{m+1} \to \mathcal{L}(E_{m+1}, F_m)$. If $U$ is an sc-domain, $f_m \circ \iota : U_{m+1} \to F_m$ has derivative $Df(x) \circ \iota$ at $x$ within $U_{m+1}$, hence is $C^1$ within $U_{m+1}$; and if $f_m : U_m \to F_m$ has derivative $L$ at $\iota x$ within $U_m$, then $L = Df(x)$. *Book:* not stated as such; the compactness argument of the 2010 paper's Propositions 2.1–2.3 (quoted in §1.2) and its integral identity $f(x + h) - f(x) = \int_0^1 Df(x + \tau h)h \, d\tau$. *Deviation (not in the book):* the $C^k$ case of the lemma, proved by induction with the mean value inequality, replaces the multilinear maps $\Gamma(x)$ and the differentiation under the integral sign of the 2010 paper; the integral identity is formed as a Bochner integral in $F_m$ and identified with $f(x + h) - f(x)$ by the fundamental theorem of calculus for the $C^1$ map $f : U_1 \to F_0$.

**V1.27 (sc-domains; `Polyfold.IsScDomain`, `Polyfold.IsRelOpenInQuadrant.isScDomain`, `Polyfold.IsScDomain.tangentSet`, `Polyfold.IsScDomain.shift`).** $U \subseteq E_0$ is an *sc-domain* if for every $m$ the level set $U_m$ is the intersection of an open and a convex subset of $E_m$, and $U_m$ has unique differentiability at each of its points. A relatively open subset of a partial quadrant is an sc-domain; if $U$ is an sc-domain, so are $TU$ in $TE$ and $U_1$ in $E^1$. *Book:* not stated; the text works with tuples $(U, C, E)$ and asserts that $T(U, C, E)$ is a tuple. *Deviation (not in the book):* the level-wise regularity that the proofs of §§1.2–1.3 use, stable under the tangent construction and the level shift, stated in place of the tuple.

**V1.28 (the Fredholm index and its invariance; `Polyfold.linIndex`, `Polyfold.HasFiniteIndex`, `Polyfold.linIndex_comp`, `Polyfold.linIndex_eq_zero_of_finiteRange_sub_id`, `Polyfold.HasFiniteIndex.`
`Polyfold.HasFiniteIndex.of_comp_of_comp`, `Polyfold.fredholmIndex`, `Polyfold.IsFredholm.hasFiniteInd`
`Polyfold.fredholmIndex_add_of_norm_lt`, `Polyfold.fredholmIndex_add_isCompactOperator`).**
For a linear map $u : E \to F$ over a field let $\operatorname{ind}(u) := \dim \ker u - \dim(F/\operatorname{range} u) \in \mathbb{Z}$, meaningful when both dimensions are finite ("finite index"). If $u$ and $v : F \to G$ have finite index, so has $v \circ u$, and $\operatorname{ind}(v \circ u) = \operatorname{ind}(u) + \operatorname{ind}(v)$. If $a$ is bijective and $r$ has finite rank, $a + r$ has finite index 0 (in particular $1 + r$). If $v \circ u$ and $u \circ v$ have finite index, so has $u$. For real Banach spaces $E$, $F$: a Fredholm operator $u : E \to F$ has finite index, and $\operatorname{ind}(u)$ is its *Fredholm index*. If $v : F \to E$ is a quasi-inverse of $u$ ($vu - 1$ and $uv - 1$ have finite rank) and $\|v\| \, \|s\| < 1$, then $u + s$ has finite index and $\operatorname{ind}(u + s) = \operatorname{ind}(u)$. If $u$ is Fredholm and $k : E \to F$ is compact, then $\operatorname{ind}(u + k) = \operatorname{ind}(u)$. *Book:* not stated; the classical fact behind the exposition's (F2) (Kato IV.5.26), which the text's §1.4.1 would need to conclude that the index is preserved. *Deviation (not in the book):* infrastructure Mathlib v4.33.1 lacks (it has Fredholm operators but no index). *Proof:* additivity by rank–nullity along the exact sequence $0 \to \ker u \to \ker vu \to \operatorname{range} u \cap \ker v \to F/\operatorname{range} u \to G/\operatorname{range} vu \to G/\operatorname{range} v \to 0$; small perturbations through $v(u+s) = (1+vs) + (vu-1)$ and $(u+s)v = (1+sv) + (uv-1)$, where $1 + vs$, $1 + sv$ are invertible by the Neumann series; compact perturbations by the local constancy of $t \mapsto \operatorname{ind}(u + tk)$ on $\mathbb{R}$ (each $u + t_0 k$ is Fredholm by V1.25), not by Kato's argument.

**V1.29 (Lemma 2.D, the level operators of an sc-Fredholm operator; `Polyfold.ScFredholmSplitting.C_le_smoo`
`Polyfold.ScBanach.finrank_levelSub_of_le_smooth`, `Polyfold.ScFredholmSplitting.fredholmIndex_lvl`,
`Polyfold.IsScFredholm.fredholmIndex_lvl`, `Polyfold.scIndex_eq_fredholmIndex`).** Let $T : E \to F$ be an sc-operator with an sc-Fredholm splitting $(K, X, C, Y)$ (V1.7). Then $C \leq F_\infty$

(and $K \leq E_\infty$, V1.7); a subspace $A \leq E_\infty$ has $\dim A_m = \dim A$ for every $m$; and for every $m$ the level map $T_m : E_m \to F_m$ is a Fredholm operator with $\operatorname{ind}(T_m) = \dim K - \dim C$. Hence for an sc-Fredholm $T$, $\operatorname{ind}(T_m) = \operatorname{ind}(T) := \dim\ker T_0 - \dim(F_0/\operatorname{range} T_0)$ for every $m$; the latter is the Fredholm index of $T_0$. *Book:* preprint p. 9, after Definition 1.1.9 (preprint 1.8): $K \subset E_\infty$, $T(X_m) = Y_m$, $E_m = K \oplus X_m$, "$F = C \oplus T(E_m)$"; the level-wise index is not stated. *Hidden hypotheses:* none. *Deviation (stronger):* the Fredholm property and the index on every level are added (the exposition's Lemma 2.D). *Proof:* $\ker T_m = K_m \cong K$ and $F_m/\operatorname{range} T_m = F_m/Y_m \cong C_m \cong C$, the isomorphisms $A_m \cong A$ coming from $A \leq E_\infty$.

**V1.30 (Proposition 1.1.11 (preprint 1.10), the index; `Polyfold.proposition_1_10_index`, `Polyfold.proposition_1_10_index_lvl`).** Let $T : E \to F$ be sc-Fredholm and $S : E \to F$ an sc$^+$-operator. Then $\operatorname{ind}(T+S) = \operatorname{ind}(T)$, the index being $\dim\ker(\cdot)_0 - \dim\operatorname{coker}(\cdot)_0$; and for every $m$ the level map $(T+S)_m : E_m \to F_m$ is Fredholm of index $\operatorname{ind}(T)$. *Book:* p. 9 (preprint p. 10), Proposition 1.1.11 (preprint 1.10) states only that $T+S$ is sc-Fredholm (V1.9). *Hidden hypotheses:* none. *Deviation (stronger):* the index statement is the exposition's addition. *Proof:* on level 0, $T_0$ is Fredholm (V1.7) and $S_0$ compact (V1.3), so V1.28 applies; on level $m$, V1.29 for the sc-Fredholm operator $T+S$ (V1.9).

**V1.31 (finite-dimensional spaces; `Polyfold.ScBanach.smooth_eq_top_of_finiteDimensional`, `Polyfold.ScBanach.toBase_bijective_of_finiteDimensional`, `Polyfold.ScBanach.isoConstOfFiniteDimen` `Polyfold.ScScale.finiteDimensional_of_surjective_incl`).** Let $E$ be an sc-Banach space with $E_0$ finite-dimensional. Then every point of $E_0$ is smooth, every $\iota_m : E_m \to E_0$ is bijective, and the maps $\iota_m$ form an sc-isomorphism of $E$ onto $E_0$ with its constant structure. For a scale $E$ (no density assumed) and any $m$: if the inclusion $E_{m+1} \to E_m$ is surjective, then $E_m$ is finite-dimensional. *Book:* preprint p. 6, after Definition 1.1.1 (preprint 1.1): a finite-dimensional Banach space has precisely one sc-structure, the constant one; on an infinite-dimensional space the constant structure is not an sc-structure, because the inclusions are not compact. *Hidden hypotheses:* none. *Deviation (restated):* "precisely one" is read, for the indexed families of V1.1, as "sc-isomorphic to the constant structure by the inclusions" (so the level norms are all equivalent), and "the constant structure fails (1)" as "an onto inclusion forces finite dimension". *Proof:* $E_\infty$ is a dense, hence (finite dimension) closed, hence full subspace of $E_0$; an onto compact inclusion is an isomorphism (open mapping theorem), so the identity of $E_m$ is compact.

**V1.32 (sc-splittings are sc-projections; `Polyfold.ScBanach.IsScSplitting.isTopCompl`, `Polyfold.ScBanach.IsScSplitting.scProj`, `Polyfold.ScBanach.IsScSplitting.scProj_idem`, `Polyfold.ScBanach.IsScSplitting.range_scProj`, `Polyfold.ScBanach.IsScSplitting.ker_scProj`, `Polyfold.ScOperator.idem_lvl`, `Polyfold.ScOperator.levelSub_range_eq`, `Polyfold.ScBanach.isScSplitt` `Polyfold.ScBanach.isScSplitting_iff_exists_scProj`).** Let $A, B \subseteq E_0$. If $E = A \oplus B$ is an sc-splitting, then on every level $A_m$, $B_m$ are topological complements, and the continuous projections of $E_m$ onto $A_m$ along $B_m$ form an sc-operator $P : E \to E$ with $P_m^2 = P_m$, $P_m(E_m) = A_m$ and $\ker P_m = B_m$. Conversely, if $P : E \to E$ is an sc-operator with $P_0^2 = P_0$, then $P_m^2 = P_m$ and $(P_0(E_0))_m = P_m(E_m)$ for every $m$, and $E = P_0(E_0) \oplus \ker P_0$ is an sc-splitting. Hence $E = A \oplus B$ is an sc-splitting iff some sc-operator $P$ with $P_0^2 = P_0$ has range $A$ and kernel $B$. *Book:* pp. 6–8 (preprint pp. 7–8): the definition of an sc-splitting, and in the proof of Proposition 1.1.7 (preprint 1.6) the step "since $P \circ P = P$, it is a sc-projection" with $G := (1-P)(E)$. *Hidden hypotheses:* none. *Deviation (hypotheses explicit):* the equivalence the text uses is stated and proved (the exposition's Lemma 2.B); $\ker P_0$ is the text's $(1-P)(E)$. The formal Proposition 1.1.7 (preprint 1.6) (V1.5) does not depend on it.

**V1.33 (the smooth points of a quotient; `Polyfold.ScBanach.exists_smooth_mem_coset`,**

`Polyfold.ScBanach.mk_mem_quotient_smooth_iff`**).** Let $A$ be an sc-subspace of $E$ and $x \in E_0$. If for every $j$ there is $y_j \in E_j$ with $\iota_j y_j - x \in A$, then there is $y \in E_\infty$ with $y - x \in A$. Equivalently: the class of $x$ is a smooth point of the quotient $E/A$ (V1.6) if and only if $x + A$ contains a smooth point of $E$. *Book:* preprint p. 9, in the proof of Proposition 1.1.8 (preprint 1.7): "$(E/A)_\infty$ consists of all elements of the form $x + A_\infty$ with $x \in E_\infty$". *Hidden hypotheses:* that $A$ is an sc-subspace (the density of $A_\infty$ in every $A_j$) is used; the text has it. *Deviation:* none. *Proof:* a Mittag-Leffler construction (the exposition's Lemma 2.C), with $\prod_{i<j} \max(1, \|E_{i+1} \to E_i\|)$ bounding all the inclusions $E_j \to E_i$, $i \le j$.

**V1.34 (continuity of $Df$ for the compact-open topology;** `Polyfold.continuousOn_apply_iff_uniformOnCompac` `Polyfold.IsSc1With.deriv_uniformOnCompact`, `Polyfold.exists_isSc1_not_continuousOn_deriv`**).** Let $X$ be a metric space, $Y$, $Z$ real normed spaces, $s \subseteq X$ and $A : X \to \mathcal{L}(Y, Z)$. Then $(x, h) \mapsto A(x)h$ is continuous on $s \times Y$ if and only if for every $x \in s$, every compact $K \subseteq Y$ and every $\varepsilon > 0$ there is a neighbourhood of $x$ in $s$ on which $\|A(x')w - A(x)w\| < \varepsilon$ for all $w \in K$. Applied to an sc$^1$ map with derivatives $Df$: $x \mapsto Df(x)$ is continuous on $U_1$ for the topology of uniform convergence on compact subsets of $E_0$. There is an sc$^1$ map on a relatively open subset of a partial quadrant for which no derivative family is continuous on $U_1$ into $\mathcal{L}(E_0, F_0)$ with the operator norm (the example of V1.13). *Book:* preprint p. 11: $x \mapsto Df(x)$ is in general not continuous for the operator norm, but it is continuous for the compact-open topology. *Hidden hypotheses:* none. *Deviation (restated):* the compact-open topology on $\mathcal{L}(Y, Z)$ is written out as uniform convergence on compact sets, and the remark becomes an equivalence with level 0 of condition (3) of Definition 1.1.15 (preprint 1.14) (the exposition's Lemma 3.F). The negative half of the remark is proved by a sequence-space example (V1.13), not by the shift map of the exposition's §6, which needs weighted Sobolev spaces.

**V1.35 (finite dimensions: sc¹ is $C^1$, also at the boundary;** `Polyfold.ScBanach.lvlSet_const`, `Polyfold.isSc1_iff_contDiffOn_const`, `Polyfold.isSc1_iff_fderivWithin_const`, `Polyfold.isScSmooth_i` `Polyfold.isSc1_iff_contDiffOn_of_isRelOpenInQuadrant`, `Polyfold.isScSmooth_iff_contDiffOn_of_isRe`**).** Let $V$, $W$ be finite-dimensional real normed spaces with their constant sc-structures, $f : V \to W$ and $U \subseteq V$ an sc-domain (V1.27), for instance a relatively open subset of a partial quadrant of $V$. Then $f$ is sc$^1$ on $U$ if and only if $f$ is $C^1$ within $U$; equivalently, $f$ is differentiable within $U$ at every point of $U$ and the derivative within $U$, $x \mapsto$ fderivWithin $f\, U\, x$, is continuous on $U$. And $f$ is sc-smooth on $U$ if and only if it is $C^\infty$ within $U$. *Book:* preprint p. 11: "the sc$^1$-maps between finite dimensional Banach spaces are the familiar $C^1$-maps". *Hidden hypotheses:* the meaning of $C^1$ at boundary points of a quadrant, never defined in the text; here the derivative within $U$, which at a boundary point is the one-sided derivative along $U$ and is unique because the quadrant has non-empty interior. *Deviation (hypotheses explicit):* none beyond this; by V1.31 every sc-structure on a finite-dimensional space is the constant one up to sc-isomorphism. *Proof:* Propositions 1.2.1 and 1.2.4 (preprint 1.16 and 1.19) at $k = 0$ (V1.15, V1.18) for the levels of the constant structure, which all equal $U$.

**V1.36 (level shifts and tangents of tuples;** `Polyfold.ScIso.shift`, `Polyfold.IsPartialQuadrant.shift`, `Polyfold.IsRelOpenInQuadrant.shift`, `Polyfold.tangentQuadrantIso`, `Polyfold.IsPartialQuadrant.tange` `Polyfold.IsRelOpenInQuadrant.tangentSet`, `Polyfold.ScBanach.shift_tangentSet`**).** Let $C$ be a partial quadrant in $E$, presented by an sc-isomorphism $T : E \to \mathbb{R}^n \oplus W$. Then $C_1$ is a partial quadrant in $E^1$, presented by the level shift $T^1 : E^1 \to \mathbb{R}^n \oplus W^1$; and $TC = \{(x, h) \in E_1 \times E_0 : x \in C_1\}$ is a partial quadrant in $TE$, presented by $(x, h) \mapsto (a, (w, T_0 h))$ where $T_1 x = (a, w)$, an sc-isomorphism $TE \to \mathbb{R}^n \oplus (W^1 \oplus (\mathbb{R}^n \oplus W))$. If $U$ is relatively open in $C$, then $U_1$ is relatively open in $C_1$ and $TU$ in $TC$; and the level-1 part

of $TU$ is $T(U_1)$. *Book:* preprint p. 10, after Definition 1.1.14 (preprint 1.13): "$T(U, C, E)$ is a tuple consisting again of a relatively open subset $TU$ in the partial quadrant $TC$"; the corresponding fact for $C^1 \subset E^1$ is used without being stated. *Hidden hypotheses:* none. *Deviation:* none; $C^k$ for $k \geq 2$ is the iterate. These are the exposition's Lemmas 3.A(2) and 3.B, proved in the Chapter 2 files (`Polyfold/Chapter2/LevelPolyfold.lean`, `Polyfold/Chapter2/TangentBundle.lean`).

**V1.37 (the derivative tower; `Polyfold.IsDerivTower`, `Polyfold.levelDerivTower`, `Polyfold.baseDerivTower`, `Polyfold.contDiffOn_levels_min`, `Polyfold.levelDerivTower_toBase`, `Polyfold.baseDerivTower_toBase`, `Polyfold.isDerivTower_of_contDiffOn_levels`, `Polyfold.IsDerivTower.eq_of_toBase`).** A derivative tower of order $k+1$ for $f : E_0 \to F_0$ on $U$ consists of maps $\Phi_i : E_0 \times E_0^i \to F_0$ with $\Phi_0(x) = f(x)$ and, for $x \in U_{m+l}$, continuous $i$-linear maps $\Psi_{i,m,l}(x) : E_{m+l}^i \to F_m$, for $i \leq l+1$ and $i \leq k+1$, such that (a) $\iota_m \Psi_{i,m,l}(x)(h_1, \dots, h_i) = \Phi_i(\iota_{m+l}x)(\iota_{m+l}h_1, \dots, \iota_{m+l}h_i)$; (b) $\Psi_{i,m,l}$ is continuous on $U_{m+l}$ into the space of continuous $i$-linear maps; (c) for $i \leq l$ and $i \leq k$, $\Psi_{i,m,l}$ is differentiable within $U_{m+l}$ at every point, with derivative $\xi \mapsto \Psi_{i+1,m,l}(x)(\xi, \cdot)$. If $U$ is an sc-domain, $f$ is $\mathrm{sc}^0$ on $U$ and $f_m \circ \iota : U_{m+l} \to F_m$ is $C^{l+1}$ within $U_{m+l}$ for all $m$ and $l \leq k$, then the iterated derivatives within the level sets, $\Psi_{i,m,l} := d^i(f_m \circ \iota)$ on $U_{m+l}$ and $\Phi_i$ the $i$-th derivative of $f : U_{i-1} \to F_0$ at lifts to $E_{i-1}$, form a derivative tower of order $k+1$. In a tower, two values $\Psi_{i,m,l}(x)(h)$, $\Psi_{i,m,l'}(x')(h')$ with the same images in $E_0$ are equal. *Book:* not stated; the 2010 paper's proof of Proposition 2.4 (pp. 51–53) writes one symbol $d^j f(x)$ for the derivatives on all level pairs. *Hidden hypotheses:* the compatibility (a) is the identification the text makes tacitly. *Deviation (not in the book):* a tool of the formalization. *Proof:* (a) from the chain rule for iterated derivatives within a set composed on the left and on the right with continuous linear maps (the inclusions), and the uniqueness of derivatives on the level sets; for $l > k$ the hypothesis is used with order $k+1$, restricted from $U_{m+k}$.

**V1.38 (elementary maps and the induction $(\mathrm{S}_k)$; `Polyfold.ElemData`, `Polyfold.ElemData.Good`, `Polyfold.ElemData.dom`, `Polyfold.ElemData.map`, `Polyfold.ElemData.map_toBase`, `Polyfold.ElemData.leve`
`Polyfold.ElemData.isSc0`, `Polyfold.ElemData.deriv`, `Polyfold.ElemData.hasFDerivWithinAt_level`, `Polyfold.ElemData.continuousOn_deriv`, `Polyfold.ElemData.isSc1Levels`, `Polyfold.ElemData.isSc1With`, `Polyfold.ElemData.shiftData`, `Polyfold.ElemData.derivData`, `Polyfold.ElemData.slotData`, `Polyfold.ElemData.tangentMap_eqOn`, `Polyfold.ElemData.isSck`, `Polyfold.ElemData.base`, `Polyfold.ScBanach.shiftAdd`, `Polyfold.ScBanach.shiftAddZeroIso`, `Polyfold.ScBanach.shiftAddShiftIsc`** Fix a derivative tower of order $k+1$ for $f$ on $U$ (V1.37), and write $E^p$ for the sc-Banach space with levels $(E^p)_l = E_{p+l}$ ($E \cong E^0$ and $(E^p)^1 \cong E^{p+1}$ as sc-Banach spaces). Elementary data on an sc-Banach space $X$ are an sc-operator $\pi_0 : X \to E^p$, sc-operators $\pi_a : X \to E^{n_a}$ ($1 \leq a \leq i$) and a number $m$; the elementary map is $g(z) = \Phi_i(\pi_0 z)(\pi_1 z, \dots, \pi_i z)$, a map $X \to F^m$, on $W := \pi_0^{-1}(U_p)$. Assume $m + i \leq p$, $m + i \leq n_a + 1$ for all $a$, and $i \leq k$. Then $g$ is $\mathrm{sc}^0$ on $W$, its level-$l$ map being $\Psi_i$ on level $(m+l, i-1)$ at the slots. If $W$ is an sc-domain, $g$ is $\mathrm{sc}^1$ with

$$Dg(z)\zeta = d^{i+1}f(\pi_0 z)(\pi_0\zeta, \pi_1 z, \dots, \pi_i z) + \sum_{a=1}^{i} d^i f(\pi_0 z)(\pi_1 z, \dots, \pi_a \zeta, \dots, \pi_i z),$$

and, if moreover $i+1 \leq k$, its tangent map on $TW$ is $(g', g_0 + \sum_a g_a)$, through $(F^m)^1 \cong F^{m+1}$ in the first component, with $g'$ the elementary map of the data $(\pi_0, \pi_a)$ taken on the first factor of $TX = X^1 \oplus X$ (levels $p+1$, $n_a+1$, $m+1$), $g_0$ the one with $i+1$ vectors ($\pi_0$ on the second factor, $\pi_a$ on the first) and target shift $m$, and $g_a$ the one with $\pi_a$ moved to the second factor; all satisfy the three conditions. Finally, if $W$ is an sc-domain and $i + s \leq k+1$, then $g$ is $\mathrm{sc}^s$ on $W$. *Book:* 2010 paper, Proposition 2.4, pp. 51–53, the inductive statement $(\mathrm{S}_k)$: the components of $T^{k+1}f$ are linear combinations of $d^j f(x)(h_1, \dots, h_j)$ on $U^{k+1} \oplus E^{n_1} \oplus \cdots \oplus E^{n_j} \to F^l$

with $j + l - 1 \le n_i \le k$, each $\mathrm{sc}^1$ with the derivative (11), and those of $T^{k+2}f$ are again of this form. *Hidden hypotheses:* the base-point condition $j + l \le k + 1$, which the text derives from its bounds. *Deviation (not in the book):* the induction runs on $s$ over all elementary maps on all $X$, not on the description of $T^{k+1}f$; the upper bound $n_i \le k$ is not needed, and its consequences $j+l \le p$ and $j \le k$ are conditions of their own. *Proof:* $\mathrm{sc}^1$ by the level criterion of Proposition 1.2.1 (preprint 1.16) (V1.15): $Dg$ exists one level down because $d^{i+1}f$ and $d^i f$ do, by the derivative of the multilinear evaluation (V1.39); $\mathrm{sc}^{s+1}$ from the tangent map, the pairing, the sum rule (V1.39) and the induction hypothesis.

**V1.39 (sums of $\mathrm{sc}^k$ maps and the derivative of multilinear evaluation; `Polyfold.IsSck.zero`, `Polyfold.IsSck.add`, `Polyfold.IsSck.sum`, `Polyfold.hasFDerivAt_cmm_eval`).** Let $W$ be an sc-domain in $X$. The constant map $0$ is $\mathrm{sc}^k$ on $W$; if $g_1$, $g_2$ are $\mathrm{sc}^k$ on $W$, so is $g_1 + g_2$; a finite sum of $\mathrm{sc}^k$ maps is $\mathrm{sc}^k$. For continuous multilinear maps $A : \prod_a M_a \to Z$ between normed spaces, the evaluation $(A, h) \mapsto A(h)$ is differentiable at every $(A, h)$ with derivative $(B, \eta) \mapsto B(h) + \sum_a A(h_1, \dots, \eta_a, \dots)$. *Book:* not stated; the 2010 paper's proof of Proposition 2.4 uses both (the terms of $\pi \circ T^{k+1}f$ treated one at a time, and the derivative (11)). *Hidden hypotheses:* none. *Deviation (not in the book):* tools; only sums, not scalar multiples, are needed. *Proof:* $g_1 + g_2$ is the sc-smooth addition $Y \oplus Y \to Y$ composed with the pairing $(g_1, g_2)$ (`Polyfold.IsSck.scPair`), by the chain rule (V1.20); the sum by induction.

**V1.40 (Remark 1.1.2, Banachable spaces and equivalent norms; `Polyfold.remark_1_1_2_2021`, `Polyfold.remark_1_1_2_2021_openMapping`).** Let $E_1$, $E_2$ be real normed spaces and $e : E_1 \to E_2$ a linear equivalence (one vector space carrying two norms). (1) If $e$ and $e^{-1}$ are continuous (the two norms induce the same topology), there is $c > 0$ with $\|ex\| \le c\|x\|$ and $\|x\| \le c\|ex\|$ for all $x$. (2) If both spaces are complete and $e$ is continuous, then $e^{-1}$ is continuous, and the two norms are equivalent as in (1). *Book:* Remark 1.1.2, p. 5, new in the 2021 edition: "Banach space" means a Banachable space, a vector space with a topology induced by a complete norm, and all norms inducing that topology are equivalent "in view of the open mapping theorem". *Hidden hypotheses:* none. *Deviation (restated):* the remark is a convention, not a statement; the Lean records its mathematical content in two readings. Part (1) needs neither completeness nor the open mapping theorem (a continuous linear map between normed spaces is bounded); the open mapping theorem is what gives part (2), two complete norms one of which dominates the other. *Proof:* (1) by the operator norms of $e$ and $e^{-1}$; (2) by Mathlib's open mapping theorem (`ContinuousLinearEquiv.ofBijective`), then (1).

**V1.41 (Proposition 1.1.11 (preprint 1.10), the 2021 statement with the index; `Polyfold.proposition_1_1_11_202`** Let $T : E \to F$ be sc-Fredholm and $S : E \to F$ an $\mathrm{sc}^+$-operator. Then $T + S$ is sc-Fredholm, $\mathrm{ind}(T + S) = \mathrm{ind}(T)$, and for every $m$ the level map $(T + S)_m : E_m \to F_m$ is Fredholm of index $\mathrm{ind}(T)$. *Book:* Proposition 1.1.11, p. 9 (preprint 1.10, p. 10); the 2021 statement adds that $T + S$ "has the same index". The proof in Appendix 1.4.1, pp. 15–16, is the one of the earlier texts and does not argue the index. *Hidden hypotheses:* none. *Deviation (stronger):* the level statement is the formalization's addition; the index statement, the exposition's addition in V1.30, is now the book's own. *Proof:* the conjunction of V1.9 (`Polyfold.proposition_1_10`) and V1.30 (`Polyfold.proposition_1_10_index`, `Polyfold.proposition_1_10_index_lvl`); the index rests on the invariance of the Fredholm index under compact perturbations (V1.28).

**What is not verified in this chapter**

- pp. 6 ff. (preprint pp. 6–7), Example 1.1.3 (preprint 1.2): the weighted Sobolev spaces $H^{m,\delta_m}(\mathbb{R}\times S^1)$ as an sc-structure on $L^2$. Mathlib has no weighted Sobolev spaces and no Rellich–Kondrachov theorem; the exposition proves it (Proposition 1.C). Planned, with the shift map of the 2010 paper (exposition §6), as the project's step H2.
- p. 10 (preprint p. 11), Remark 1.1.17 (preprint 1.15): partial cones. Not formalized, as in the exposition.
- preprint p. 11, after Definition 1.1.15 (preprint 1.14): the text's own instance of a derivative that is not norm continuous, the shift map of the exposition's §6, needs the weighted Sobolev spaces of Example 1.1.3 (preprint 1.2). The remark itself is verified, the negative half by a sequence-space example (V1.13, V1.34), the positive half by V1.34.

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## 3. Chapter 2 (HWZ §§2.1, 2.3–2.7 (preprint §§2.1–2.6)): verified except Proposition 2.1.2 (preprint 2.2) and Lemma 2.4.9 (preprint 2.29) as stated; Remark 2.4.8's (preprint 2.28) negative claim is false

Chapter 2 of Part I (pp. 21–88 (preprint pp. 18–60)) builds differential geometry on the images of sc-smooth retractions. §2.1 defines retractions and retracts, proves that the tangent $TO = Tr(TU)$ does not depend on the retraction (Proposition 2.1.11 (preprint 2.4)), defines sc-smooth maps between retracts and proves the chain rule (Definitions 2.1.1–2.1.14 (preprint 2.1–2.7), Proposition 2.1.13 (preprint 2.6), Theorem 2.1.15 (preprint 2.8)). §2.3 (preprint §2.2) glues retracts into M-polyfolds, gives them a filtration $X_m$ and a tangent bundle $TX$, proves metrizability (Theorem 2.3.10 (preprint 2.16), Proposition 2.3.15 (preprint 2.19)) and treats sub-M-polyfolds (Definitions 2.3.2–2.3.18 (preprint 2.11–2.20), Proposition 2.3.19 (preprint 2.21)). §2.4 (preprint §2.3) transports the degeneracy index to M-polyfolds and compares it with the tangent space (Definitions 2.4.1–2.4.7 (preprint 2.22–2.27), Lemmas 2.4.3, 2.4.6, 2.4.9 (preprint 2.23, 2.26, 2.29), Propositions 2.4.4, 2.4.10 (preprint 2.24, 2.30), Theorem 2.4.12 (preprint 2.31), Corollary 2.4.13 (preprint 2.32)). §2.5 (preprint §2.4) introduces tame retracts and the faces of tame M-polyfolds (Definitions 2.5.1–2.5.10 (preprint 2.33–2.39), Propositions 2.5.3–2.5.20 (preprint 2.35–2.46), with the products of M-polyfolds that Proposition 2.5.20 (preprint 2.46) uses), §2.6 (preprint §2.5) strong bundles (Definitions 2.6.1–2.6.7 (preprint 2.47–2.53), Proposition 2.6.8 (preprint 2.54)), and the appendix §2.7 (preprint §2.6) proves Proposition 2.1.2 (preprint 2.2), Theorem 2.3.10 (preprint 2.16) and Proposition 2.3.15 (preprint 2.19) with Smirnov's theorem (Theorem 2.7.1 (preprint 2.55)), Lemma 2.7.3 (preprint 2.56) and Proposition 2.7.4 (preprint 2.57). Verified: every definition except 2.9 and 2.38 (which are given only in the forms the theory uses; 2.9, the 2021 Definition 2.1.16, is formalized since, apart from its categories, V2.92) and every numbered result of §§2.1, 2.3, 2.4, 2.6 (preprint §§2.1–2.3 and §2.5), with these exceptions: Proposition 2.1.2 (preprint 2.2) (the classical submanifold statement, motivation only), Lemma 2.4.9 (preprint 2.29) in its path formulation (its content, the sign arguments, is verified and Proposition 2.4.10 (preprint 2.30) is proved from it), the openness of $p : TX \to X_1$ in Proposition 2.3.15(2) (preprint 2.19). The second statement of Theorem 2.4.12 (preprint 2.31) (equality makes $C_xO$ a partial quadrant) is verified without its hypothesis, and **Remark 2.4.8's (preprint 2.28) claim that $C_xO$ is in general not a partial quadrant in $T_xO$ is false**: it is one at every smooth point of every sc-smooth retract (V2.33), by Flor's theorem on nonnegative idempotent matrices, proved in the needed case. Every numbered result of §2.5 (preprint §2.4) is verified: Propositions 2.5.3, 2.5.6 (preprint 2.35, 2.36) (with "$C_xO$ is a partial quadrant", which holds for every retract, V2.33), Lemma 2.5.12 (preprint 2.40), Corollary 2.5.13 (preprint 2.41),

Propositions 2.5.14, 2.5.15 (preprint 2.42, 2.43; the retraction onto a face, and the face as a tame sub-M-polyfold with $d_F = d_O - 1$), 2.5.16 (preprint 2.44; both sentences, the intersections of local faces included), Theorem 2.5.17 (preprint 2.45) and Proposition 2.5.20 (preprint 2.46) (fibred products, with the products of M-polyfolds it uses). In §2.6 (preprint §2.5) the M-polyfold $Y[1]$, the sc-smoothness of $P[1]$ and the chart-independence of $Y_{m,k}$ are verified, and the chart-wise definition of sc$^+$-sections is shown to be the text's; equivalence of atlases, of M-polyfolds and of strong bundles, is an equivalence relation (V2.11, V2.57), strong bundle charts restrict to open subsets of the base (V2.57), and the charts of a strong bundle depend only on its class of atlases (V2.58). **Theorem 2.3.10 (preprint 2.16) holds with the text's hypotheses**: the formal proof adds none, avoids Lemma 2.7.3 (preprint 2.56) and Proposition 2.7.4 (preprint 2.57) (which need a regularity of $X_m$ that the text does not check) and proves Smirnov's theorem, which Mathlib lacks, in a pull-back form. Several statements are proved in a stronger form: Propositions 2.4.10, 2.5.3 (preprint 2.30, 2.35) and Theorem 2.4.12 (preprint 2.31) at points on level 1 and for sc$^1$ germs, Proposition 2.5.3 (preprint 2.35) from condition (2) of tameness alone, Corollary 2.4.13 (preprint 2.32) for every relatively open subset of a partial quadrant, and Proposition 2.5.6's (preprint 2.36) $d_O = d_C$ in the form "a tame chart realizes $d_X$" for every M-polyfold.

The setting. Sc-Banach spaces and maps are those of Chapter 1. A *retraction* is a total map $r : E_0 \to E_0$ with an explicit domain $U$; an *M-polyfold* is a Hausdorff paracompact space with one sc-smooth atlas representing its class, and every notion is defined through the maximal atlas of all charts compatible with it. Sc-smoothness between retracts quantifies over all retractions onto the source (equivalent to the text's "one"). Where the text computes in the model $C = [0,\infty)^k \oplus W$, the formal statements use a presentation $T : E \cong \mathbb{R}^n \oplus W$ of $C$ and the coordinate functionals $\lambda_i = (T\,\cdot)_i$; the subspaces $E_x$, the cones $C_x$ and tameness are defined through $T$. The tangent bundle $TX$ is a quotient of tuples $(x, \varphi, h)$ with its topology and charts; its fibres $T_xX$ are sets, identified with $T_{\varphi(x)}O$ by the charts, with no linear structure of their own. Theorem 1.3.5 (preprint 1.25) of Chapter 1 (V1.23) is proved in `Polyfold/Chapter2/DegeneracyIndex.lean` by the argument of §2.4 (preprint §2.3).

**V2.1 (sc-smooth retractions; `Polyfold.ScRetraction`, `Polyfold.ScRetraction.retract`, `Polyfold.ScRetraction.mem_retract_iff`, `Polyfold.ScRetraction.apply_eq_self_of_mem_retract`, `Polyfold.ScRetraction.lvlSet_retract_subset`, `Polyfold.ScRetraction.scDeriv_comp_self`).** Let $C$ be a partial quadrant in $E$. An *sc-smooth retraction* on $C$ consists of a set $U \subseteq E_0$ relatively open in $C$ and a map $r : E_0 \to E_0$ with $r(U) \subseteq U$, $r$ sc-smooth on $U$ and $r(r(x)) = r(x)$ for $x \in U$. Its *retract* is $O = r(U)$; $O \subseteq U$, $x \in O$ iff $x \in U$ and $r(x) = x$, and $O_m \subseteq U_m$. For $x \in O_1$, $Dr(x) \circ Dr(x) = Dr(x)$. *Book:* Definition 2.1.1 (preprint 2.1), p. 21 (preprint p. 18), and the remark after Definition 2.1.3 (preprint 2.3), p. 26 (preprint p. 19). *Hidden hypotheses:* none. *Deviation:* none; the values of $r$ off $U$ play no role.

**V2.2 (sc-smooth retracts; `Polyfold.IsScRetract`, `Polyfold.ScRetraction.isScRetract`, `Polyfold.ScRetraction.id`, `Polyfold.IsRelOpenInQuadrant.isScRetract`, `Polyfold.LocalModel`).** $(O, C, E)$ is an *sc-smooth retract* if $O = r(U)$ for an sc-smooth retraction on $C$. A relatively open $U \subseteq C$ is a retract ($r = \mathrm{id}$). A *local model* is a triple $(E, C, O)$ with $C$ a partial quadrant and $O$ a retract in $C$. *Book:* Definition 2.1.3 (preprint 2.3), p. 22 (preprint p. 18). *Hidden hypotheses:* none.

**V2.3 (Proposition 2.1.11 (preprint 2.4); `Polyfold.proposition_2_4`, `Polyfold.ScRetraction.comp_eq_self_of`, `Polyfold.ScRetraction.tangent_subset_of_retract_subset`, `Polyfold.ScRetraction.tangent`, `Polyfold.ScRetraction.mem_tangent_iff`).** For sc-smooth retractions $r$ on $U$ and $r'$ on $U'$, both relatively open in $C$, with $r(U) = r'(U')$: $Tr(TU) = Tr'(TU')$. If only $r(U) \subseteq r'(U')$, then $r' \circ r = r$ on $U$ and $Tr(TU) \subseteq Tr'(TU')$. A point $p$ lies in $Tr(TU)$ iff $p \in TU$ and $Tr(p) = p$.

*Book:* Proposition 2.1.11 (preprint 2.4), p. 26 (preprint p. 19). *Hidden hypotheses:* the composition $r' \circ r$ is defined because $O \subseteq U'$. *Proof:* as in the text, by Theorem 1.3.1 (preprint 1.21) applied to $r' \circ r = r$.

**V2.4 (Definition 2.1.12 (preprint 2.5), the tangent of a retract;** `Polyfold.retractTangent`, `Polyfold.retractTangent_eq`, `Polyfold.ScRetraction.mem_tangent_iff_exists`, `Polyfold.ScRetraction.m` `Polyfold.tangentSpace`, `Polyfold.tangentSpace_eq_range`, `Polyfold.tangentSpace_eq_coe_submodule`**).** For a retract $O$, $TO \subseteq E_1 \times E_0$ is $Tr(TU)$ for any retraction $r$ onto $O$ (V2.3); $TO = \{(r(x), Dr(x)h) : x \in U_1, h \in E_0\}$, and for $x \in O_1$, $(x,h) \in TO$ iff $h$ is in the range of $Dr(x)$. The tangent space $T_xO = \{h : (x,h) \in TO\}$ is this range, a closed subspace of $E_0$. *Book:* Definition 2.1.12 (preprint 2.5), p. 26 (preprint p. 19), and Definition 2.3.12 (preprint 2.17), p. 38 (preprint p. 25) (the fibre $T_xO$). *Hidden hypotheses:* none. *Deviation (restated):* $TO$ is defined for every subset $O$ by a choice of retraction, and is empty if $O$ is not a retract.

**V2.5 (Proposition 2.1.13 (preprint 2.6);** `Polyfold.proposition_2_6_1`, `Polyfold.proposition_2_6_2`, `Polyfold.proposition_2_6_3`, `Polyfold.ScRetraction.comp_comp_eqOn`**).** Let $r$ on $U$ and $s$ on $V$ be retractions onto the same $O$ and $f : E_0 \to E_0'$. (1) If $f \circ r$ is sc-smooth on $U$, then $f \circ s$ is sc-smooth on $V$. (2) Then $T(f \circ r) = T(f \circ s)$ on $Tr(TU)$. (3) If $f \circ r$ is sc-smooth and $f$ maps $O$ into the retract $O' = r'(U')$, then $T(f \circ r)$ maps $Tr(TU)$ into $Tr'(TU')$. *Book:* Proposition 2.1.13 (preprint 2.6), p. 27 (preprint p. 20). *Hidden hypotheses:* in (3), $f(O) \subseteq O'$ (contained in "$f : O \to O'$"). *Proof:* as in the text, with $f \circ r \circ s = f \circ s$ and $\rho \circ f \circ r = f \circ r$ and Theorem 1.3.1 (preprint 1.21).

**V2.6 (Definition 2.1.14 (preprint 2.7), sc-smooth maps between retracts;** `Polyfold.IsScSmoothRetractMap`, `Polyfold.isScSmoothRetractMap_iff_exists`, `Polyfold.retractTangentMap`, `Polyfold.retractTangentMap_` `Polyfold.IsScSmoothRetractMap.tangentMap_mapsTo`, `Polyfold.retractTangentMap_apply`, `Polyfold.IsScSmoothRetractMap.continuousOn`, `Polyfold.IsScSmoothRetractMap.mem_range_toBase`**).** $f$ is *sc-smooth from* $(O,C,E)$ *to* $(O',C',E')$ if $f(O) \subseteq O'$ and $f \circ r$ is sc-smooth on $U$ for every sc-smooth retraction $r$ on $C$ with $r(U) = O$; for a retract $O$ this holds iff $f(O) \subseteq O'$ and $f \circ r$ is sc-smooth for one such $r$. The tangent map $Tf$ is $T(f \circ r)$ for a chosen $r$; on $TO$ it equals $T(f \circ r)$ for every $r$ onto $O$, maps $TO$ into $TO'$ when $O, O'$ are retracts, and $Tf(x,h) = (f(x), D(f \circ r)(x)h)$. Such an $f$ is continuous on $O$ and maps points of $O$ on level $m$ to points on level $m$. *Book:* Definition 2.1.14 (preprint 2.7), pp. 27–28 (preprint pp. 20–21). *Hidden hypotheses:* none. *Deviation (restated):* the definition quantifies over all retractions and contains $f(O) \subseteq O'$; the text's form is `Polyfold.isScSmoothRetractMap_iff_exists`.

**V2.7 (Theorem 2.1.15 (preprint 2.8), the chain rule for retracts;** `Polyfold.theorem_2_8`, `Polyfold.IsScSmoothRetractMap.comp`, `Polyfold.retractTangentMap_comp`, `Polyfold.isScSmoothRetractMa` `Polyfold.ScRetraction.isScSmoothRetractMap`, `Polyfold.IsScSmoothRetractMap.congr`**).** Let $(O,C,E)$ and $(O',C',E')$ be retracts, $f$ sc-smooth from $O$ to $O'$ and $g$ sc-smooth from $O'$ to $(O'',C'',E'')$. Then $g \circ f$ is sc-smooth from $O$ to $O''$ and $T(g \circ f) = Tg \circ Tf$ on $TO$. The identity is sc-smooth, a retraction is sc-smooth as a map $(U,C,E) \to (O,C,E)$, and sc-smoothness depends only on the values on $O$. *Book:* Theorem 2.1.15 (preprint 2.8), p. 28 (preprint p. 21), which writes $(O'',C',E'')$ for $(O'',C'',E'')$. *Hidden hypotheses:* $r' \circ f \circ r = f \circ r$, which needs $f(O) \subseteq O'$. *Proof:* the text gives none ("follows from Theorem 1.21"); $(g \circ f) \circ r = (g \circ r') \circ (f \circ r)$ on $U$ and Theorem 1.3.1 (preprint 1.21).

**V2.8 (the tangent functor on retracts;** `Polyfold.IsPartialQuadrant.tangentSet`, `Polyfold.IsRelOpenInQuadra` `Polyfold.ScRetraction.tangentRetraction`, `Polyfold.ScRetraction.tangentRetraction_retract`, `Polyfold.IsScRetract.tangentSet`, `Polyfold.LocalModel.tangent`, `Polyfold.IsScSmoothRetractMap.tange`

`Polyfold.retractTangentMap_id`, `Polyfold.retractTangent_inter_isOpen`, `Polyfold.retractTangentMap_i`
$TC$ is a partial quadrant in $TE$ and $TU$ is relatively open in $TC$; $Tr$ is an sc-smooth retraction on $TU$ with retract $TO$, so $(TO, TC, TE)$ is a retract. If $f$ is sc-smooth between the retracts $O$ and $O'$, then $Tf$ is sc-smooth from $(TO, TC, TE)$ to $(TO', TC', TE')$; $T$ of a map equal to the identity on $O$ is the identity on $TO$. For $W \subseteq E_0$ open: $T(O \cap W) = TO \cap (W_1 \times E_0)$, and $Tf$ computed on $(O \cap W, C, E)$ equals $Tf$ computed on $(O, C, E)$ there. *Book:* preprint p. 21, after Theorem 2.1.15 (preprint 2.8) ($T$ a functor $\mathcal{R} \to \mathcal{R}$), and preprint p. 22, after Definition 2.1.16 (preprint 2.9) ($Tr$ a bundle retraction). *Hidden hypotheses:* the locality of $Tf$ under restriction to open subsets, used in Definitions 2.3.12–2.3.14 (preprint 2.17–2.18). *Deviation (restated):* the bundle structure is V2.92 ($Tr$ a bundle retraction covering $r^1$, $Tf$ a bundle map; gap G.5, formalized since); the category $\mathcal{BR}$ is not formalized.

**V2.9 (Proposition 2.3.1 (preprint 2.10);** `Polyfold.proposition_2_10_1`, `Polyfold.proposition_2_10_2`, `Polyfold.ScRetraction.restrict`, `Polyfold.ScRetraction.restrict_retract`, `Polyfold.IsScSmoothRetrac` `Polyfold.IsScSmoothRetractMap.of_locally`, `Polyfold.LocalModel.restrict`**).** (1) If $(O, C, E)$ is a retract and $W \subseteq E_0$ is open, $(O \cap W, C, E)$ is a retract (with $r$ restricted to $U \cap r^{-1}(W)$). (2) If $(V, C, E)$ is a retract and $s$ is sc-smooth from $(V, C, E)$ to itself with $s \circ s = s$ on $V$, then $(s(V), C, E)$ is a retract, with the retraction $s \circ r$. Sc-smooth maps between retracts restrict to $O \cap W$, and are sc-smooth as soon as every point of $O$ has an open neighbourhood $W$ with $f$ sc-smooth on $O \cap W$. *Book:* Proposition 2.3.1 (preprint 2.10), p. 33 (preprint p. 23). *Hidden hypotheses:* the restriction of an sc-smooth map to a smaller relatively open set is sc-smooth (V2.63). *Deviation (restated):* an open subset of $O$ is written $O \cap W$.

**V2.10 (Definition 2.3.2 (preprint 2.11), charts;** `Polyfold.MChart`, `Polyfold.MChart.overlap`, `Polyfold.MChart.isScRetract_overlap`, `Polyfold.MChart.transition`, `Polyfold.MChart.Compatible`, `Polyfold.MChart.compatible_self`, `Polyfold.MChart.Compatible.restrict`, `Polyfold.MChart.compatible_`
A *chart* on a space $X$ is a local model $(O, C, E)$ and a partial equivalence $\varphi$ between $X$ and $E_0$ with open source $V$ and target $O$, continuous on $V$, with inverse continuous on $O$. The overlap $\varphi(V \cap V')$ is a retract. Two charts are *compatible* if $\psi \circ \varphi^{-1}$ is sc-smooth from $(\varphi(V \cap V'), C, E)$ to $(\psi(V \cap V'), C', E')$ and $\varphi \circ \psi^{-1}$ back. A chart is compatible with itself, compatibility passes to restrictions, and two charts each compatible with every chart of a family covering $V \cap V'$ are compatible. *Book:* Definition 2.3.2 (preprint 2.11), pp. 34 ff. (preprint pp. 23–24). *Hidden hypotheses:* none. *Deviation (restated):* the chart is a partial equivalence; the last statement, the locality of compatibility, is not in the text and is what makes the maximal atlas (V2.12) work.

**V2.11 (Definition 2.3.3 (preprint 2.12), atlases;** `Polyfold.MAtlas`, `Polyfold.MAtlas.Equiv`, `Polyfold.MAtlas.equiv_iff`, `Polyfold.MAtlas.Equiv.refl`, `Polyfold.MAtlas.Equiv.symm`, `Polyfold.MAtlas.Equiv.trans`, `Polyfold.MAtlas.equivalence`**).** An *atlas* is a set of pairwise compatible charts whose sources cover $X$; two atlases are *equivalent* if their union is an atlas, equivalently if every chart of one is compatible with every chart of the other; the relation is an equivalence relation. *Book:* Definition 2.3.3 (preprint 2.12), p. 35 (preprint p. 24). *Hidden hypotheses:* none; the transitivity of the equivalence, which "an equivalence class of atlases" (Definition 2.3.4 (preprint 2.13)) presupposes, is not discussed in the text and is proved from the locality of compatibility (Findings, Chapter 2, item 41).

**V2.12 (Definition 2.3.4 (preprint 2.13), M-polyfolds;** `Polyfold.MPolyfold`, `Polyfold.MPolyfold.IsChart`, `Polyfold.MPolyfold.IsChart.compatible`, `Polyfold.MPolyfold.IsChart.restrict`, `Polyfold.MPolyfold.is` `Polyfold.MPolyfold.exists_isChart_mem_source`, `Polyfold.LocalModel.toMPolyfold`**).** An *M-polyfold* structure on a Hausdorff paracompact space $X$ is an atlas. The *charts of the structure* are all charts compatible with every chart of this atlas; any two are compatible, restrictions of

charts are charts, every point lies in the source of one, and equivalent atlases give the same charts. A nonempty retract $O$ is an M-polyfold with the chart id. *Book:* Definition 2.3.4 (preprint 2.13), p. 35 (preprint p. 24). *Hidden hypotheses:* charts compatible with an atlas are compatible with each other (the text does not discuss it; it needs V2.10). *Deviation (restated):* one atlas represents its equivalence class.

**V2.13 (Definition 2.3.8 (preprint 2.14), sc-smooth maps between M-polyfolds;** `Polyfold.MPolyfold.IsScSmooth` `Polyfold.MPolyfold.IsScSmoothMap`, `Polyfold.MPolyfold.IsScSmoothOn.of_exists_charts`, `Polyfold.MPolyfold.isScSmoothMap_id`, `Polyfold.MPolyfold.IsScSmoothOn.comp`, `Polyfold.MPolyfold.IsS` $f : X \to Y$ is *sc-smooth on the open set* $U$ if it is continuous on $U$ and, for all charts $\varphi$ of $X$, $\psi$ of $Y$ and every open $V \subseteq U \cap \operatorname{dom}\varphi$ with $f(V) \subseteq \operatorname{dom}\psi$, the map $\psi \circ f \circ \varphi^{-1}$ is sc-smooth from $(\varphi(V), C, E)$ to $(O', C', E')$; $f$ is sc-smooth if it is sc-smooth on $X$. It suffices that every $x \in U$ has one such pair of charts and one such $V \ni x$. The identity is sc-smooth; if $f$ is sc-smooth on $U$ with $f(U) \subseteq U'$ and $g$ on $U'$, then $g \circ f$ is sc-smooth on $U$. *Book:* Definition 2.3.8 (preprint 2.14), p. 37 (preprint p. 24), and "the definition does not depend on the choice of charts". *Hidden hypotheses:* $f$ continuous (for charts with $f(V) \subseteq V'$ to exist around every point). *Deviation (hypotheses explicit):* continuity is part of the definition; maps on open subsets are included.

**V2.14 (the filtration $X_m$;** `Polyfold.MPolyfold.levelSet`, `Polyfold.MPolyfold.mem_levelSet_iff`, `Polyfold.MPolyfold.levelSet_zero`, `Polyfold.MPolyfold.levelSet_succ_subset`, `Polyfold.MPolyfold.smo` `Polyfold.MPolyfold.levelTopology`, `Polyfold.MPolyfold.Level`, `Polyfold.MPolyfold.continuous_levelVa` `Polyfold.MPolyfold.isEmbedding_levelLift`**).** $X_m$ is the set of points $x$ with $\varphi(x) \in \iota_m(E_m)$ for some chart $\varphi$ of the structure around $x$; then this holds for every chart around $x$. $X_0 = X$, $X_{m+1} \subseteq X_m$, $X_\infty = \bigcap_m X_m$. $X_m$ carries the topology generated by the sets $\{x \in V : \varphi(x) \in W\}$, $\varphi$ a chart, $W \subseteq E_m$ open (in terms of the lift of $\varphi(x)$ to $E_m$). The inclusion $X_m \to X$ is continuous, $X_m$ is Hausdorff (`Polyfold.MPolyfold.instT2SpaceLevel`), and on $V_m = V \cap X_m$ the lift of $\varphi$ to $E_m$ is a topological embedding. *Book:* preprint p. 25, the paragraph after Definition 2.3.8 (preprint 2.14). *Hidden hypotheses:* that the level is independent of the chart needs that transition maps are $\mathrm{sc}^0$ on the overlap retracts (the text: "of course"). *Deviation (restated):* the topology is the one generated by the text's family; that the family is a basis is not proved.

**V2.15 (Lemma 2.3.9 (preprint 2.15);** `Polyfold.lemma_2_15`, `Polyfold.MPolyfold.levelIncl`**).** The inclusion $X_{m+1} \to X_m$ is continuous for every $m$. *Book:* Lemma 2.3.9 (preprint 2.15), p. 37 (preprint p. 25). *Hidden hypotheses:* none.

**V2.16 (the M-polyfold $X^1$;** `Polyfold.ScRetraction.shift`, `Polyfold.ScRetraction.shift_retract`, `Polyfold.IsScRetract.shift`, `Polyfold.LocalModel.shift`, `Polyfold.IsScSmoothRetractMap.shift`, `Polyfold.MPolyfold.levelChart`, `Polyfold.MPolyfold.levelChart_compatible`, `Polyfold.MPolyfold.level` `Polyfold.MPolyfold.shift`, `Polyfold.MPolyfold.shift_isChart`, `Polyfold.MPolyfold.mem_shift_levelSet` A retraction $r$ on $U$ induces an sc-smooth retraction on $U_1 \subseteq C_1 \subseteq E^1$ with retract $O_1$, so $(O_1, C_1, E^1)$ is a retract; an sc-smooth map between retracts induces an sc-smooth map between their level-1 parts (Proposition 1.2.2 (preprint 1.17) for retracts). For every chart $\varphi$ of $X$, $(V_1, \varphi, (O_1, C_1, E^1))$ is a chart of $X_1$; any two are compatible, they form an atlas, and $X_1$ with it is an M-polyfold $X^1$. *Book:* preprint p. 25 ("$(V_m, \varphi, (O_m, C_m, E_m))$ is a chart on $X_m$ … the collection of such charts is an atlas on $X_m$"). *Hidden hypotheses:* the paracompactness of $X_1$ (Theorem 2.3.10 (preprint 2.16)), an instance argument supplied by V2.17. The levels of $X^1$ are those of $X$ shifted by one, as sets: $(X^1)_m = X_{m+1}$. *Deviation (narrower):* $m = 1$ here; for $m \geq 2$ the iterate $(X^1)^1 \cdots$ has the points of $X_m$, and its topology and charts are those of $X_m$ by V2.93

(gap G.6, closed; the set identity was added in a fourth addendum, Findings, Chapter 5, item 142).

**V2.17 (Theorem 2.3.10 (preprint 2.16);** `Polyfold.theorem_2_16`, `Polyfold.MPolyfold.metrizableSpace`, `Polyfold.MPolyfold.metrizableSpace_level`, `Polyfold.MPolyfold.paracompactSpace_level`, `Polyfold.MPolyfold.SmoothLevel`, `Polyfold.MPolyfold.smoothLevel_topology_eq`, `Polyfold.MPolyfold.me` `Polyfold.MPolyfold.paracompactSpace_subtype`**).** Let $X$ be an M-polyfold. Then $X$ is metrizable; for every $m$, the level $X_m$ with the topology of V2.14 is metrizable and paracompact; and $X_\infty$, with the initial topology of the inclusions $X_\infty \to X_m$ ($m \geq 0$), is metrizable. Every subset of $X$ is paracompact. *Book:* Theorem 2.3.10 (preprint 2.16), p. 37 (preprint p. 25), proof pp. 74–76 (preprint pp. 57–58). *Hidden hypotheses:* the topology of $X_\infty$, which the text does not define. *Deviation (hypotheses explicit):* **no hypothesis is added.** *Proof:* not the text's. The text needs $X_m$ regular for Proposition 2.7.4 (preprint 2.57) and does not check it. Here a partition of unity of the paracompact Hausdorff $X$ subordinate to the chart domains is pulled back to $X_m$ along the continuous inclusion; with the metrizable chart domains $V_m \cong O_m$ it gives an explicit metric (V2.62). Paracompactness of $X_m$ follows by Stone's theorem; $X_\infty$ embeds in $\prod_m X_m$.

**V2.18 (Definition 2.3.12 (preprint 2.17), tangent vectors in charts;** `Polyfold.MPolyfold.TangentPre`, `Polyfold.MPolyfold.TangentPre.transport`, `Polyfold.MPolyfold.TangentPre.transport_self`, `Polyfold.MPolyfold.TangentPre.transport_mem`, `Polyfold.MPolyfold.TangentPre.transport_transport`, `Polyfold.MPolyfold.TangentPre.Rel`, `Polyfold.MPolyfold.TangentPre.Rel.symm`, `Polyfold.MPolyfold.Tan` `Polyfold.MPolyfold.TangentPre.setoid`**).** A tangent tuple is $(x, \varphi, h)$ with $x \in X_1$, $\varphi$ a chart of the structure around $x$ and $(\varphi(x), h) \in TO$. Its transport to a chart $\psi$ around $x$ is $T(\psi \circ \varphi^{-1})(\varphi(x), h)$, computed on the overlap retract; it lies in $TO'$ over $\psi(x)$, transport to $\varphi$ is the identity, and transport to $\psi$ followed by transport to $\chi$ is transport to $\chi$. Two tuples are equivalent if their points agree and the transport of the first to the chart of the second is the second; this is an equivalence relation. *Book:* Definition 2.3.12 (preprint 2.17), pp. 38–39 (preprint pp. 25–26). *Hidden hypotheses:* that the relation is an equivalence relation (the text does not check it); it needs Theorem 2.1.15 (preprint 2.8) on the triple overlap and the locality of $Tf$ (V2.8). *Deviation (restated):* none in substance.

**V2.19 (Definition 2.3.14 (preprint 2.18), the tangent space $TX$;** `Polyfold.MPolyfold.TangentSpace`, `Polyfold.MPolyfold.tangentProj`, `Polyfold.MPolyfold.tangentFiber`, `Polyfold.MPolyfold.tangentChartS` `Polyfold.MPolyfold.tangentChartMap`, `Polyfold.MPolyfold.tangentChartMap_mk_self`, `Polyfold.MPolyfold.tangentChartMap_injOn`, `Polyfold.MPolyfold.tangentChartMap_mk_ofVec`, `Polyfold.MPolyfold.tangentChartMap_mk_ofVec_of_mem`, `Polyfold.MPolyfold.tangentBasis`**).** $TX$ is the set of equivalence classes of tangent tuples, with $p : TX \to X_1$, $[(x, \varphi, h)] \mapsto x$, and $T_xX = p^{-1}(x)$. For a chart $\varphi$, $TV = p^{-1}(V_1)$ and $T\varphi : TV \to TO$ sends a class to its transport to $\varphi$; $T\varphi[(x, \varphi, h)] = (\varphi(x), h)$, $T\varphi$ is injective on $TV$ with inverse $a \mapsto [(\varphi^{-1}(a_1), \varphi, a_2)]$ on $TO$, and $T\psi \circ (T\varphi)^{-1} = T(\psi \circ \varphi^{-1})$ on $T(\varphi(V \cap V'))$. $TX$ carries the topology generated by the sets $\{\alpha \in TV : T\varphi(\alpha) \in W\}$, $W$ open. *Book:* Definition 2.3.14 (preprint 2.18) and the paragraphs after it, pp. 38–40 (preprint pp. 26–27). *Hidden hypotheses:* none. *Deviation (narrower):* the vector space, Banach and sc-Banach structures of $T_xX$ are not formalized.

**V2.20 (Proposition 2.3.15 (preprint 2.19);** `Polyfold.MPolyfold.continuous_tangentProj`, `Polyfold.MPolyfold.continuous_levelVal_tangentProj`, `Polyfold.MPolyfold.isOpen_tangentChartSet`, `Polyfold.MPolyfold.metrizableSpace_tangentSpace`, `Polyfold.MPolyfold.paracompactSpace_tangentSpac` `Polyfold.proposition_2_19_3`**).** (1) $TX$ is Hausdorff (the instance `Polyfold.MPolyfold.instT2SpaceTangentS` (2) $p : TX \to X_1$ is continuous, and so is $TX \to X$; each $TV$ is open. (3) $TX$ is metrizable and paracompact. *Book:* Proposition 2.3.15 (preprint 2.19), p. 39 (preprint p. 27), proof pp. 75–78 (preprint pp. 58–60). *Hidden hypotheses:* none for (3). *Deviation (narrower):* the openness of $p$ in (2)

is not proved; the basis property in (1) is not needed. *Proof:* (3) as for Theorem 2.3.10 (preprint 2.16), pulling a partition of unity of $X$ back along $TX \to X_1 \to X$; the charts $TV \cong TO$ are metrizable.

**V2.21 ($TX$ as an M-polyfold; `Polyfold.MPolyfold.tangentChart`, `Polyfold.MPolyfold.tangentChart_overla` `Polyfold.MPolyfold.tangentChart_transition`, `Polyfold.MPolyfold.tangentChart_compatible`, `Polyfold.MPolyfold.tangentAtlas`, `Polyfold.MPolyfold.tangentPolyfold`, `Polyfold.MPolyfold.isScSmoot` `Polyfold.MPolyfold.tangentProj_isScSmooth`).** For a chart $\varphi$ of $X$, $(TV, T\varphi, (TO, TC, TE))$ is a chart of $TX$, with $T\varphi(TV \cap TV') = T(\varphi(V \cap V'))$ and transition map $T(\varphi' \circ \varphi^{-1})$ (formula (5)); these charts are pairwise compatible, form an atlas, and make $TX$ an M-polyfold. The projection $TO \to O_1$ is sc-smooth from $(TO, TC, TE)$ to $(O_1, C_1, E^1)$, and $p : TX \to X^1$ is sc-smooth. *Book:* preprint p. 27, after Proposition 2.3.15 (preprint 2.19). *Hidden hypotheses:* the paracompactness of $TX$ and $X_1$ (instance arguments, supplied by V2.20 and V2.17). *Deviation (narrower):* "sc-smooth M-polyfold bundle" is read as "$p$ is sc-smooth", and through the bundle models and the transition maps in V2.94; the fibrewise linear structure is not formalized.

**V2.22 (tangent maps of sc-smooth maps between M-polyfolds; `Polyfold.MPolyfold.tangentMap`, `Polyfold.MPolyfold.tangentMap_mk`, `Polyfold.MPolyfold.levelVal_tangentProj_tangentMap`, `Polyfold.MPolyfold.tangentMap_mapsTo_fiber`, `Polyfold.MPolyfold.tangentChartMap_tangentMap`, `Polyfold.MPolyfold.tangentMap_id`, `Polyfold.MPolyfold.tangentMap_comp`, `Polyfold.MPolyfold.continuc` `Polyfold.MPolyfold.tangentMap_isScSmooth`, `Polyfold.MPolyfold.IsScSmoothMap.mapsTo_levelSet`).** An sc-smooth $f : X \to Y$ maps $X_m$ into $Y_m$. Its tangent map $Tf : TX \to TY$, $[(x, \varphi, h)] \mapsto [(f(x), \psi, T(\psi \circ f \circ \varphi^{-1})(\varphi(x), h)_2)]$, is independent of the chart $\psi$ around $f(x)$ and of the representative; $p \circ Tf = f \circ p$, $Tf(T_xX) \subseteq T_{f(x)}Y$, $T\psi \circ Tf = T(\psi \circ f \circ \varphi^{-1}) \circ T\varphi$ in charts, $T(\mathrm{id}) = \mathrm{id}$, $T(g \circ f) = Tg \circ Tf$, and $Tf$ is continuous and sc-smooth $TX \to TY$. *Book:* not stated; the text uses $Tf$ for M-polyfolds from §2.4 (preprint §2.3) on (Proposition 2.3.19(4) (preprint 2.21), Definition 2.5.8 (preprint 2.38)). *Deviation (not in the book):* the construction the text leaves implicit.

**V2.23 (Definition 2.3.18 (preprint 2.20), sub-M-polyfolds; `Polyfold.MPolyfold.IsSubMPolyfold`, `Polyfold.MPolyfold.isSubMPolyfold_univ`).** $A \subseteq X$ is a *sub-M-polyfold* if every $a \in A$ has an open $V \ni a$ and a map $r$ with $r(V) \subseteq V$, $r$ sc-smooth on $V$, $r \circ r = r$ on $V$ and $r(V) = A \cap V$. *Book:* Definition 2.3.18 (preprint 2.20), p. 42 (preprint p. 28). *Hidden hypotheses:* none.

**V2.24 (Proposition 2.3.19 (1), (2) (preprint 2.21); `Polyfold.MPolyfold.SubData`, `Polyfold.MPolyfold.SubData.s` `Polyfold.MPolyfold.SubData.image_shrink`, `Polyfold.MPolyfold.SubData.rho_isScSmooth`, `Polyfold.MPolyfold.SubData.isScRetract_image`, `Polyfold.MPolyfold.subChart`, `Polyfold.MPolyfold.sub` `Polyfold.MPolyfold.subAtlas`, `Polyfold.MPolyfold.subPolyfold`, `Polyfold.MPolyfold.subPolyfold_inclu` `Polyfold.MPolyfold.subPolyfold_retraction_isScSmooth`, `Polyfold.IsScSmoothRetractMap.mono_source`, `Polyfold.MPolyfold.paracompactSpace_of_isClosed`).** Let $A$ be a sub-M-polyfold with the subspace topology. For data $(V, r)$ at a point and a chart $\varphi$, the set $W = (V \cap \mathrm{dom}\,\varphi) \cap r^{-1}(V \cap \mathrm{dom}\,\varphi)$ is open with $r(W) \subseteq W$ and $r(W) = W \cap A$; $\rho = \varphi \circ r \circ \varphi^{-1}$ is sc-smooth and idempotent on $\varphi(W)$, so $\varphi(W \cap A)$ is a retract and $\varphi$ restricted to $W \cap A$ is a chart of $A$ with model $(\varphi(W \cap A), C, E)$. Any two such charts are compatible; they form an atlas, and $A$ is an M-polyfold. (1) The inclusion $A \to X$ is sc-smooth. (2) For every data $(V, r)$, the map $i^{-1} \circ r$ is sc-smooth on $V$. A closed subset of $X$ is paracompact. *Book:* Proposition 2.3.19 (1), (2) (preprint 2.21), pp. 42–44 (preprint pp. 28–30). *Hidden hypotheses:* the topology of $A$ (the subspace topology; then (1)'s homeomorphism is by construction) and the paracompactness of $A$, an instance argument of the structure, supplied for every $A$ by `Polyfold.MPolyfold.paracompactSpace_subtype`

(V2.17). *Deviation (hypotheses explicit):* none otherwise.

**V2.25 (Proposition 2.3.19 (3), (4) (preprint 2.21), in a chart; `Polyfold.ScRetraction.compRetraction`, `Polyfold.ScRetraction.compRetraction_retract`, `Polyfold.proposition_2_21_3`, `Polyfold.proposition_2`**
Let $r$ be a retraction onto $O$, $s$ sc-smooth from $(O, C, E)$ to itself with $s \circ s = s$, $t = s \circ r$ (a retraction onto $s(O)$) and $a \in s(O)_1$. If $a$ is smooth: $T_aO = Dr(a)E_0$, $T_as(O) = Dt(a)E_0 \subseteq T_aO$, the subspaces $Dt(a)E_0$ and $(Dr(a) - Dt(a))E_0$ are sc-subspaces of $E$, and on every level $m$ their level-$m$ parts are disjoint with sum the level-$m$ part of $T_aO$. For every $a \in s(O)_1$: $s$ is sc-smooth from $O$ to $s(O)$ and $Ts$ maps $T_aO$ onto $T_as(O)$. *Book:* Proposition 2.3.19 (3), (4) (preprint 2.21), preprint p. 30, proved in coordinates with "$T_aO = Ds(a)Dr(a)E \oplus (I - Ds(a))Dr(a)E$". *Hidden hypotheses:* $Ds(a)$ means $Dt(a)$; the splitting is an sc-splitting because $Dr(a)$, $Dt(a)$ are commuting sc-projections with $Dr(a)Dt(a) = Dt(a) = Dt(a)Dr(a)$. *Deviation (narrower):* in a chart only, as the text proves it; the transport to $T_aX$ is not formalized (no sc-structure on the fibres of $TX$).

**V2.26 (Definition 2.4.1 (preprint 2.22), the degeneracy index; `Polyfold.MPolyfold.chartDegIndex`, `Polyfold.MPolyfold.degIndex`, `Polyfold.MPolyfold.degIndex_le`, `Polyfold.MPolyfold.exists_isChart_de`, `Polyfold.MPolyfold.eventually_chartDegIndex_le`).** For a chart $\varphi$ around $x$, $d(x, \varphi) = d_C(\varphi(x))$; $d_X(x)$ is the least $d(x, \varphi)$ over the charts of the structure around $x$. The minimum is attained, $d_X(x) \le d(x, \varphi)$ for every such chart, and $d(\cdot, \varphi)$ is upper semicontinuous on the source of $\varphi$. *Book:* Definition 2.4.1 (preprint 2.22), p. 44 (preprint p. 31) ("smooth charts"). *Hidden hypotheses:* none. *Deviation (restated):* "smooth charts" are the charts of the maximal atlas.

**V2.27 (Lemma 2.4.3 (preprint 2.23); `Polyfold.lemma_2_23`, `Polyfold.eventually_degIndexWith_le`, `Polyfold.MPolyfold.isOpen_setOf_degIndex_lt`, `Polyfold.MPolyfold.isClosed_setOf_le_degIndex`).** For every $x \in X$, $d_X(y) \le d_X(x)$ for all $y$ near $x$; hence $\{d_X < k\}$ is open and $\{d_X \ge k\}$ closed. The degeneracy index $d_C$ of a partial quadrant is upper semicontinuous. *Book:* Lemma 2.4.3 (preprint 2.23), p. 45 (preprint p. 31) ("evident"). *Hidden hypotheses:* none.

**V2.28 (Proposition 2.4.4 (preprint 2.24); `Polyfold.MPolyfold.IsScDiffeoOn`, `Polyfold.proposition_2_24`, `Polyfold.MPolyfold.transportChart`, `Polyfold.MPolyfold.transportChart_isChart`, `Polyfold.MPolyfold.degIndex_le_of_isScDiffeoOn`).** Let $U \subseteq X$, $V \subseteq Y$ be open, $f$ sc-smooth on $U$ with $f(U) \subseteq V$ and $g$ sc-smooth on $V$ with $g(V) \subseteq U$, inverse to each other. Then $d_X(x) = d_Y(f(x))$ for every $x \in U$. For a chart $\psi$ of $Y$, the map $\psi \circ f$ on $U \cap f^{-1}(\operatorname{dom}\psi)$ is a chart of the structure of $X$. *Book:* Proposition 2.4.4 (preprint 2.24), p. 45 (preprint p. 31) ("from the definitions one deduces immediately"). *Hidden hypotheses:* none. *Deviation (restated):* a germ is represented by an sc-diffeomorphism of open sets.

**V2.29 (Definition 2.4.5 (preprint 2.25), the boundary; `Polyfold.MPolyfold.boundary`, `Polyfold.MPolyfold.IsWithoutBoundary`, `Polyfold.MPolyfold.isWithoutBoundary_iff`, `Polyfold.MPolyfold.isClosed_boundary`, `Polyfold.MPolyfold.isOpen_interior`, `Polyfold.MPolyfold.corn`, `Polyfold.MPolyfold.isClosed_corners`, `Polyfold.MPolyfold.IsScDiffeoOn.mem_boundary_iff`).** $\partial X = \{d_X \ge 1\}$; $X$ is without boundary iff $d_X \equiv 0$ iff $\partial X = \emptyset$. $\partial X$ and $\{d_X \ge 2\}$ are closed, $\{d_X = 0\}$ is open, and sc-diffeomorphisms of open sets preserve $\partial X$. *Book:* Definition 2.4.5 (preprint 2.25), p. 45 (preprint p. 31). *Hidden hypotheses:* none.

**V2.30 (Lemma 2.4.6 (preprint 2.26); `Polyfold.lemma_2_26`).** For a sub-M-polyfold $A$ with the structure of V2.24, $d_A(a) \le d_X(a)$ for every $a \in A$. *Book:* Lemma 2.4.6 (preprint 2.26), pp. 45–46 (preprint pp. 31–32). *Hidden hypotheses:* the paracompactness of $A$, as in V2.24.

**V2.31 (the subspaces $E_x$ and the cones $C_x$; `Polyfold.minimalSubspaceWith`, `Polyfold.mem_minimalSubspaceWi`**

`Polyfold.isClosed_minimalSubspaceWith`, `Polyfold.finrank_quotient_minimalSubspaceWith`, `Polyfold.partialQuadrantAtWith`, `Polyfold.minimalSubspaceWith_subset_partialQuadrantAtWith`, `Polyfold.isClosed_partialQuadrantAtWith`, `Polyfold.exists_tendsto_lvl1_degIndexWith`, `Polyfold.exists_tendsto_smooth_degIndexWith`, `Polyfold.ScBanach.subset_closure_ker_inter_range_to`, `Polyfold.partialQuadrantAtWith_subset_closure`). Let $T : E \cong \mathbb{R}^n \oplus W$ present $C$. $E_x = \{v : (Tv)_i = 0 \text{ whenever } (Tx)_i = 0\}$ is a closed subspace of $E_0$ with $\dim(E_0/E_x) = d_C(x)$; $C_x = \{v : (Tv)_i \geq 0 \text{ whenever } (Tx)_i = 0\}$ is closed and convex and contains $E_x$. Every $x \in C$ is a limit of points of level 1 (and of smooth points) with the same index and in $C$. The level-1 points of the kernel of any continuous linear map into a finite-dimensional space are dense in it, and the level-1 points of $C_x$ are dense in $C_x$. *Book:* preprint p. 32 ($E_i$, $E_I$, $I(x)$, $E_x$, $H_i$, $C_x$ in the model $\mathbb{R}^k \oplus W$). *Hidden hypotheses:* the densities, which the text uses without proof ("the closure of $(T_x^R O)_\infty$ is $T_x^R O$"). *Deviation (restated):* defined through a presentation of $C$.

**V2.32 (Definition 2.4.7 (preprint 2.27);** `Polyfold.reducedTangentSpaceWith`, `Polyfold.partialConeWith`, `Polyfold.quotDim`, `Polyfold.quotDim_inf_minimalSubspace_le`, `Polyfold.quotDim_reducedTangentSpaceWi` For a retraction $r$, a presentation $T$ and $x \in O_1$: $T_x^R O = Dr(x)E_0 \cap E_x$ and $C_x O = Dr(x)E_0 \cap C_x$. For subspaces $A, B$, $\dim(A/(A \cap B))$ is the natural number $\mathrm{quotDim}(A, B)$; $\dim(T_x O / T_x^R O) \leq d_C(x)$. *Book:* Definition 2.4.7 (preprint 2.27), p. 47 (preprint p. 33). *Hidden hypotheses:* none. *Deviation (stronger):* defined at every $x \in O_1$, not only at smooth points.

**V2.33 (Remark 2.4.8 (preprint 2.28);** `Polyfold.reducedTangentSpaceWith_subset_partialConeWith`, `Polyfold.remark_2_28_isPartialQuadrant`, `Polyfold.remark_2_28_partialConeWith`, `Polyfold.flor_nonnegRange_simplicial`, `Polyfold.remark_2_28_quotient_simplicial`, `Polyfold.ScRetraction.isScSubspace_reducedTangentSpaceWith`, `Polyfold.partialConeWith_eq_image_sc` $T_x^R O \subseteq C_x O \subseteq T_x O$. Moreover, for every sc-smooth retraction $r$ in $C = T^{-1}([0,\infty)^n \oplus W)$ and every smooth $x \in O$ (no tameness), $T_x O$ and $T_x^R O$ are sc-subspaces of $E$ and $C_x O$ is a partial quadrant in the sc-Banach space $T_x O$ (with the sc-structure induced from $E$), of dimension $r = \operatorname{rank} M = \dim(T_x O / T_x^R O) \leq d_C(x)$, where $M$ is the matrix of $Dr(x)$ on $E/E_x \cong \mathbb{R}^{I(x)}$. The core is Flor's theorem in the needed case: for a linear $M$ on $\mathbb{R}^\iota$ with $M^2 = M$ mapping the orthant into itself, the cone $\operatorname{range} M \cap [0,\infty)^\iota$ is the set of nonnegative combinations of linearly independent columns of $M$. *Book:* Remark 2.4.8 (preprint 2.28), p. 47 (preprint p. 33): the inclusions, and "naively one might expect that $C_x O$ is a partial quadrant in $T_x O$. However, this is in general not the case", without an example. *Hidden hypotheses:* none. *Deviation (stronger):* **the remark's negative claim does not hold in our formal reading** (checked in Lean); the remark may mean the narrower point below. *Proof:* $Dr(x)$ maps $C_x$ and $E_x$ into themselves (the sign arguments of Lemma 2.4.9 (preprint 2.29)), so $C_x O = Dr(x)(C_x)$; modulo $T_x^R O$ it is the column cone of the nonnegative idempotent matrix $M$, simplicial by Flor's theorem (minimal column supports, generation by descent on the support, independence through a private index of each minimal support); the sc-isomorphism $T_x O \cong \mathbb{R}^r \oplus T_x^R O$ of the proof of Theorem 2.4.12 (preprint 2.31) (V2.42) carries $C_x O$ onto $[0,\infty)^r \oplus T_x^R O$. What can fail for a non-tame retract is only $r = d_C(x)$ (Theorem 2.4.12 (preprint 2.31)), which the text proves for tame retracts (Proposition 2.5.6 (preprint 2.36), V2.42) (Findings, Chapter 2, item 50; round-trip item V2.33).

**V2.34 (the sign arguments;** `Polyfold.IsSc1With.deriv_mem_minimalSubspace_of_lvl1`, `Polyfold.IsSc1With.mapsTo_minimalSubspace`, `Polyfold.IsSc1With.deriv_mem_partialQuadrant_of_lvl1`, `Polyfold.IsSc1With.mapsTo_partialQuadrant`, `Polyfold.eventually_add_smul_mem_lvlSet`, `Polyfold.eventually_add_smul_mem_lvlSet_of_cone`, `Polyfold.ScRetraction.scDeriv_mapsTo_minimalSub` Let $f$ be sc$^1$ on $U$, relatively open in $C$ (presented by $T$), with values in $C'$ (presented by $T'$), and

$x \in U_1$. For $v \in E_1$ with $\iota_1 v \in E_x$: $x + tv \in U_1$ for $|t|$ small and $Df(x)v \in E'_{f(x)}$; hence $Df(x)$ maps $E_x$ into $E'_{f(x)}$. For $v \in E_1$ with $\iota_1 v \in C_x$: $x + tv \in U_1$ for small $t \geq 0$ and $Df(x)v \in C'_{f(x)}$; hence $Df(x)$ maps $C_x$ into $C'_{f(x)}$. For a retraction $r$ and $x \in O_1$, $Dr(x)$ maps $E_x$ into $E_x$. *Book:* the proofs of Lemma 2.4.9 (preprint 2.29), pp. 48–49 (preprint pp. 33–35), and Proposition 2.5.3 (preprint 2.35), p. 54 (preprint p. 40). *Deviation (not in the book):* the derivative-level content of Lemma 2.4.9 (preprint 2.29), isolated; Lemma 2.4.9 (preprint 2.29) itself, about sc-smooth paths, is not formalized. *Proof:* the function $t \mapsto (T'f(x + tv))_j$ is $\geq 0$ near 0 and vanishes at 0; its derivative at 0 vanishes (two-sided) or is $\geq 0$ (one-sided).

**V2.35 (Proposition 2.4.10 (preprint 2.30); `Polyfold.ScRetraction.IsDiffeoGerm`, `Polyfold.IsScSmoothRetract` `Polyfold.ScRetraction.IsDiffeoGerm.deriv_inv_deriv`, `Polyfold.ScRetraction.IsDiffeoGerm.deriv_der` `Polyfold.proposition_2_30_reduced`, `Polyfold.proposition_2_30_cone`).** A *germ of sc-diffeomorphisms* between the retracts $O = r(U)$ and $O' = r'(U')$ is a pair $F, G$ with $F \circ r$, $G \circ r'$ sc¹, $F(O) \subseteq O'$, $G(O') \subseteq O$, $G \circ F =$ id on $O$ and $F \circ G =$ id on $O'$; an sc-smooth map between retracts with an sc-smooth inverse is one. For $x \in O_1$ and $y = (F \circ r)_1(x)$: $D(G \circ r')(y) \circ D(F \circ r)(x) = Dr(x)$, $D(F \circ r)(x) \circ D(G \circ r')(y) = Dr'(y)$, and, with $Tf(x) = D(F \circ r)(x)$, $Tf(x)(T^R_x O) = T^R_y O'$ and $Tf(x)(C_x O) = C_y O'$. *Book:* Proposition 2.4.10 (preprint 2.30), p. 49 (preprint p. 35) (smooth $x$, sc-smooth germs; deduced from Lemma 2.4.9 (preprint 2.29)). *Hidden hypotheses:* none. *Deviation (stronger):* level-1 points and sc¹ germs. *Proof:* from the sign arguments (V2.34) and the two derivative identities, not from Lemma 2.4.9 (preprint 2.29).

**V2.36 (a retract as an M-polyfold; `Polyfold.LocalModel.selfChart`, `Polyfold.LocalModel.toMPolyfold`, `Polyfold.IsRelOpenInQuadrant.toMPolyfold`, `Polyfold.IsRelOpenInQuadrant.selfChart_isChart`, `Polyfold.MChart.Compatible.isDiffeoGerm`).** A nonempty retract $O$, and a nonempty relatively open $U \subseteq C$, is an M-polyfold on the subtype with the single chart id. Two compatible charts give a germ of sc-diffeomorphisms (in the sense of V2.35) between the overlap retracts, with the restricted retractions. *Book:* preprint p. 35, before Theorem 2.4.12 (preprint 2.31). *Deviation (restated):* the text's $d_O(x) = \min d_{C'}(\varphi(x))$ over germs into retracts is Definition 2.4.1 (preprint 2.22) for this structure.

**V2.37 (Theorem 2.4.12 (preprint 2.31), first statement; `Polyfold.theorem_2_31`, `Polyfold.theorem_2_31_retrac` `Polyfold.ScRetraction.IsDiffeoGerm.quotDim_le`, `Polyfold.MPolyfold.quotDim_le_chartDegIndex`, `Polyfold.ScRetraction.scDeriv_restrict`, `Polyfold.degIndexWith_le_of_germ`, `Polyfold.remark_2_28_is`** Let $X$ be an M-polyfold, $\varphi$ a chart of the structure around $x$ with $\varphi(x) = \iota_1 z$ on level 1, $r$ a retraction onto the model $O$ and $T$ a presentation of $C$. Then $\dim(T_z O / T^R_z O) \leq d_X(x)$. For a retract $O$ itself as an M-polyfold and $x \in O_1$: $\dim(T_x O / T^R_x O) \leq d_O(x)$. Along a germ of sc-diffeomorphisms $F$ between retracts, $\dim(T_x O / T^R_x O) \leq d_{C'}(F(x))$. *Book:* Theorem 2.4.12 (preprint 2.31), p. 50 (preprint p. 35) (a retract, a smooth $x$). *Hidden hypotheses:* none. *Deviation (stronger):* every M-polyfold, every chart, points on level 1. The second statement of the theorem (equality makes $C_x O$ a partial quadrant) holds without its hypothesis: $C_x O$ is a partial quadrant at every smooth point of every retract (V2.33). *Proof:* for every chart $\psi$ around $x$, $D(\psi \circ \varphi^{-1})$ induces $T_z O / T^R_z O \cong T_y O' / T^R_y O'$, which embeds in $E'/E'_y$ of dimension $d_{C'}(\psi(x))$; the text instead reduces to a chart realizing $d_O(x)$.

**V2.38 (Corollary 2.4.13 (preprint 2.32); `Polyfold.corollary_2_32`, `Polyfold.IsRelOpenInQuadrant.degIndex_` `Polyfold.IsRelOpenInQuadrant.degIndex_le_degIndex_of_lvl1`, `Polyfold.degIndexWith_le_of_germ`).** For a nonempty relatively open subset $U$ of a partial quadrant $C$, viewed as the M-polyfold $X_U$, $d_{X_U}(x) = d_C(x)$ for every $x \in U$. *Book:* Corollary 2.4.13 (preprint 2.32), p. 51 (preprint p. 37)

($U = C$). *Hidden hypotheses:* the semicontinuity of $d_{C'}$ and Proposition 2.4.4 (preprint 2.24) at the approximating points. *Deviation (stronger):* every relatively open $U$; level-1 approximants suffice. Theorem 1.3.5 (preprint 1.25) (V1.23, `Polyfold.theorem_1_25`) is proved in the same file by the same argument.

**V2.39 (Definition 2.5.1 (preprint 2.33); `Polyfold.partialQuadrantAtWith`, `Polyfold.minimalSubspaceWith`, `Polyfold.minimalSubspaceWith_subset_partialQuadrantAtWith`, `Polyfold.convex_partialQuadrantAtWith`, `Polyfold.finrank_quotient_minimalSubspaceWith`, `Polyfold.minimalSubspaceWith_le`, `Polyfold.minimalSubspaceWith_eq`).** For a presentation $T$ of $C$ and $e \in C$: the partial quadrant $C_e$ and the minimal subspace $E_e$ of V2.31, with $E_e \subseteq C_e$, $C_e$ convex and $\dim(E_0/E_e) = d_C(e)$. For two presentations $T$, $T'$ of the same $C$, the two subspaces $E_e$ agree. *Book:* Definition 2.5.1 (preprint 2.33), p. 53 (preprint p. 39), through the extended faces. *Hidden hypotheses:* none. *Deviation (restated):* through a presentation, of which $E_e$ is independent (the argument of Lemma 1.3.4 (preprint 1.24)); the intrinsic description by extended faces (preprint p. 39) is not formalized.

**V2.40 (Definition 2.5.2 (preprint 2.34), tame retractions; `Polyfold.ScRetraction.IsTameWith`, `Polyfold.ScRetraction.IsTame`, `Polyfold.IsTameRetract`, `Polyfold.IsTameRetract.isScRetract`, `Polyfold.ScRetraction.id_isTame`, `Polyfold.ScRetraction.IsTameWith.isTame`, `Polyfold.ScRetraction.I`**
A retraction $r$ is *tame for* $T$ if (1) $d_C(r(x)) = d_C(x)$ for $x \in U$ and (2) for every smooth $x \in O_1$ there is a subspace $A \leq E_x$ such that $E = Dr(x)E_0 \oplus A$ is an sc-splitting. $r$ is *tame* if it is tame for every presentation of $C$; tameness for one presentation implies it. A *tame retract* is the retract of a tame retraction. Identities of relatively open subsets are tame, and the restriction of a tame retraction to $U \cap r^{-1}(V)$, $V$ open, is tame. *Book:* Definition 2.5.2 (preprint 2.34), pp. 53–54 (preprint pp. 39–40). *Hidden hypotheses:* that the definition does not depend on the presentation, which the text uses tacitly. *Deviation (restated):* none in substance.

**V2.41 (Proposition 2.5.3 (preprint 2.35); `Polyfold.proposition_2_35`, `Polyfold.ScRetraction.scDeriv_mapsT`, `Polyfold.ScRetraction.isScOperator_scDeriv`, `Polyfold.ScRetraction.isScSplitting_range_scDeriv`, `Polyfold.coord_scDeriv_eq`).** Let $r$ be tame for $T$ and $x \in O_1$ smooth. Then $(1 - Dr(x))E_0 \leq E_{x'}$ and $E = Dr(x)E_0 \oplus (1 - Dr(x))E_0$ is an sc-splitting; equivalently $\lambda_j(Dr(x)v) = \lambda_j(v)$ for $j \in I(x)$ and all $v$. *Book:* Proposition 2.5.3 (preprint 2.35), p. 54 (preprint p. 41), with the discussion on p. 54 (preprint p. 40). *Hidden hypotheses:* none. *Deviation (stronger):* only condition (2) of tameness is used; $Dr(x)E_x \subseteq E_x$ holds for every retraction (V2.34), where the text uses condition (1).

**V2.42 (Proposition 2.5.6 (preprint 2.36); `Polyfold.proposition_2_36`, `Polyfold.proposition_2_36_degIndex`, `Polyfold.quotDim_reducedTangentSpaceWith_eq`, `Polyfold.MPolyfold.degIndex_eq_chartDegIndex_of_isT`, `Polyfold.MPolyfold.degIndex_eq_chartDegIndex_of_isTameRetract`, `Polyfold.LocalModel.degIndex_toMP`, `Polyfold.quotDim_eq_finrank_quotient_of_isCompl`, `Polyfold.proposition_2_36_partialCone`, `Polyfold.splitIso`, `Polyfold.isPartialQuadrant_ofSubspace`).** Let $r$ be tame for $T$. (13) At a smooth $x \in O_1$, $\dim(T_xO/T^R_xO) = d_O(x) = d_C(x)$, where $d_O$ is the index of the M-polyfold $O$; moreover $T_xO$ and $T^R_xO$ are sc-subspaces of $E$, and $C_xO = T_xO \cap C_x$ is a partial quadrant in the sc-Banach space $T_xO$ (with the sc-structure induced from $E$). (14) $d_O(x) = d_C(x)$ for every $x \in O$. More generally, a chart $\varphi$ of any M-polyfold whose model is a tame retract satisfies $d_X(y) = d_C(\varphi(y))$ for every $y$ in its source. For sc-subspaces $B \leq A$ with $B = A \cap \bigcap_j \ker \ell_j$ and smooth $e_j \in A$ with $\ell_i(e_j) = \delta_{ij}$, the map $v \mapsto ((\ell_j v)_j, v - \sum_j \ell_j(v)e_j)$ is an sc-isomorphism $A \cong \mathbb{R}^d \oplus B$, and $\{v \in A : \ell_j(v) \geq 0 \ \forall j\}$ is a partial quadrant in $A$. *Book:* Proposition 2.5.6 (preprint 2.36), p. 56 (preprint p. 41). *Hidden hypotheses:* none. *Deviation (stronger):* the chart form, which Proposition 2.5.14 (preprint 2.42) needs; $T^R_xO$ is an sc-subspace, where the text says a

Banach space. *Proof:* (13), (14) as in the text: $E_x = T_x^R O \oplus A$ gives $\dim(T_xO/T_x^RO) = d_C(x)$, Theorem 2.4.12 (preprint 2.31) gives $\leq d_X$; a general point is approximated by smooth points $r(T^{-1}(a, w_j))$ of the same index, with Lemma 2.4.3 (preprint 2.23). The partial quadrant not from the equality case of Theorem 2.4.12 (preprint 2.31), as in the text, but directly (it holds without tameness, V2.33; what tameness adds is the dimension $d_O(x)$ of the quadrant): the smooth vectors $\hat{e}_j = Dr(x)T^{-1}(e_{i_j}, 0)$ satisfy $\lambda_{i_k}(\hat{e}_j) = \delta_{jk}$ by Proposition 2.5.3 (preprint 2.35), and the splitting applies to $A = T_xO$, $B = T_x^RO$; $T_x^RO$ is an sc-subspace because the smooth points $Dr(x)T^{-1}(a, w')$, $w' \in W_\infty$, lie in it ($Dr(x)E_x \subseteq E_x$) and approximate $u$, $Tu = (a, w)$, on every level.

**V2.43 (Definition 2.5.7 (preprint 2.37); `Polyfold.MPolyfold.IsTame`, `Polyfold.MPolyfold.isChart_of_mem_eq`, `Polyfold.MPolyfold.IsTame.exists_chart`).** $X$ is *tame* if some atlas equivalent to its atlas consists of charts with tame models; then every point lies in a chart of the structure whose model is a tame retract. *Book:* Definition 2.5.7 (preprint 2.37), p. 57 (preprint p. 42). *Hidden hypotheses:* none.

**V2.44 (Definition 2.5.10 (preprint 2.39); `Polyfold.MPolyfold.IsFace`, `Polyfold.MPolyfold.IsFaceStructured`, `Polyfold.MPolyfold.IsFace.isClosed`, `Polyfold.MPolyfold.IsFace.subset_boundary`).** A *face* is the closure of the connected component in $\{d_X = 1\}$ of a point with $d_X = 1$; $X$ is *face-structured* if every $x$ lies in exactly $d_X(x)$ faces. Faces are closed and contained in $\partial X$. *Book:* Definition 2.5.10 (preprint 2.39), p. 57 (preprint p. 43). *Hidden hypotheses:* none.

**V2.45 (the local geometry of faces; `Polyfold.subset_coordHyper_of_isPreconnected`, `Polyfold.IsSc1With.apply_deriv_eq_zero_of_ray`, `Polyfold.ray_zeroSet`, `Polyfold.local_face_structur`, `Polyfold.eventually_zeroSet_subset`, `Polyfold.convex_ball_inter_zeroSet_eq`, `Polyfold.ScRetraction.`, `Polyfold.coordHyper`).** With $\lambda_i = (T\,\cdot)_i$, $I(z) = \{i : \lambda_i(z) = 0\}$ and $f_i = \{\lambda_i = 0\}$: (a) a preconnected subset of $\{z \in C : d_C(z) = 1\}$ containing a point of $f_i$ lies in $f_i$; (b) if $r$ is tame for $T$, $x \in O_1$ smooth, $i \in I(x)$, $\lambda_i(y) = 0$ and $\lambda_l(y) > 0$ for the other $l \in I(x)$, then for all small $\tau > 0$, $x + \tau y \in U$, $I(x + \tau y) = \{i\}$ and $I(r(x + \tau y)) = \{i\}$; (c) for $e \in O$ with $\lambda_i(e) = 0$ there are $\varepsilon > 0$ and $p \in S = \{z \in O : d_C(z) = 1\}$ with $\lambda_i(p) = 0$ such that $B = B_\varepsilon(e) \cap C \subseteq U$, $I(z) \subseteq I(e)$ and $I(r(z)) \subseteq I(e)$ on $B$, $r$ maps the points of $B$ with $I = \{i\}$ into the component of $S$ through $p$, and maps $f_i \cap B$ into its closure. *Book:* the proof of Lemma 2.5.12 (preprint 2.40), p. 59 (preprint p. 43) (a sketch). *Deviation (not in the book):* the exposition's Lemmas 4.A–4.C; (b) uses tameness condition (2) through Proposition 2.5.3 (preprint 2.35).

**V2.46 (Lemma 2.5.12 (preprint 2.40); `Polyfold.ScRetraction.IsRetractFace`, `Polyfold.lemma_2_40`, `Polyfold.lemma_2_40_injective`, `Polyfold.closure_connectedComponentIn_faceStratum`, `Polyfold.connectedComponentIn_coordHyper_eq`, `Polyfold.LocalModel.isFace_toMPolyfold_iff`, `Polyfold.subset_of_relClopen_of_isPreconnected`).** Let $r$ be tame for $T$. A *face of the retract* is $\overline{\hat{F}} \cap O$ for a connected component $\hat{F}$ of $S$. Then $G$ is a face of the retract iff $G$ is the connected component of $f_i \cap O$ through some $e \in O$ with $\lambda_i(e) = 0$; and for $x \in O$ with $\lambda_i(x) = 0$, equal components of $f_i \cap O$ and $f_j \cap O$ through $x$ force $i = j$. The faces of the M-polyfold $O$ are exactly the faces of the retract. *Book:* Lemma 2.5.12 (preprint 2.40), p. 59 (preprint p. 43). *Hidden hypotheses:* the face preservation of V2.45, which the text's sketch does not address. *Deviation (hypotheses explicit):* extended faces are the coordinate hyperplanes of a presentation.

**V2.47 (Corollary 2.5.13 (preprint 2.41); `Polyfold.corollary_2_41`, `Polyfold.corollary_2_41_retract`, `Polyfold.LocalModel.isFaceStructured_toMPolyfold`).** For a nonempty tame retract $O$, the M-polyfold $O$ is face-structured: every $x \in O$ lies in exactly $d_O(x)$ faces. *Book:* Corollary 2.5.13 (preprint 2.41), p. 59 (preprint p. 44). *Hidden hypotheses:* none.

**V2.48 (Proposition 2.5.14 (preprint 2.42); `Polyfold.proposition_2_42`, `Polyfold.MPolyfold.faces_encard_le`, `Polyfold.MPolyfold.IsTame.exists_chart`).** Let $X$ be tame and $x \in X$. There is a chart $\varphi : V \to O$ of the structure around $x$ such that for all $y \in V$, $d_X(y) = d_C(\varphi(y))$ and $d_O(\varphi(y)) = d_X(y)$, and the M-polyfold $O$ is face-structured. Globally, $x$ lies in at most $d_X(x)$ faces of $X$. *Book:* Proposition 2.5.14 (preprint 2.42), p. 60 (preprint p. 44). *Hidden hypotheses:* "$d_V = d_X$" needs that a tame chart realizes $d_X$ (V2.42). *Deviation (restated):* the neighbourhood $V$ is represented by the M-polyfold $O$.

**V2.49 (Proposition 2.5.15 (preprint 2.43); `Polyfold.proposition_2_43`, `Polyfold.killCoord`, `Polyfold.killCoord_mem`, `Polyfold.killCoord_eq_self`, `Polyfold.LocalModel.isSubMPolyfold_of_isFace`, `Polyfold.faceIncl`, `Polyfold.faceProj`, `Polyfold.degIndexWith_eq_quadrantDeg_faceProj_add_one`, `Polyfold.proposition_2_43_tameModel`, `Polyfold.MPolyfold.faceChart`, `Polyfold.MPolyfold.faceChart_c`, `Polyfold.LocalModel.proposition_2_43_isTame`).** Let $r$ be tame for $T$ and $F$ a face of the retract. (a) $F \subseteq f_i$ for some $i$, and there is an sc-smooth retraction $s$ on a set $U^*$ with $F \subseteq U^* \subseteq U$, $U^*$ relatively open in $C$, with retract $F$ and $s = r \circ P_i$ on $U^*$, where $P_i$ is the sc-projection onto $f_i$ killing the $i$-th coordinate; every face of the M-polyfold $O$ is a sub-M-polyfold of $O$. (b) With the sc-operators $\iota : \mathbb{R}^k \oplus W \to E$ (insert a zero $i$-th coordinate, onto $f_i$) and $\pi : E \to \mathbb{R}^k \oplus W$ (delete it), $\pi \circ \iota = 1$, $\iota \circ \pi = 1$ on $f_i$ and $d_C(y) = d_{C'}(\pi y) + 1$ on $f_i$ ($C' = [0,\infty)^k \oplus W$); $r' = \pi \circ r \circ \iota$ is a tame retraction of $C'$ onto $\pi(F)$, and $d_{C'}(\pi y) + 1 = d_C(y)$ for $y \in F$. (c) For a face $F$ of a nonempty tame retract $O$, with its sub-M-polyfold structure, $F$ is a tame M-polyfold and $d_F(x) + 1 = d_O(x)$ for $x \in F$; the chart $\pi : F \to \pi(F)$ is compatible with the sub-charts of Proposition 2.3.19 (preprint 2.21). *Book:* Proposition 2.5.15 (preprint 2.43), pp. 61–62 (preprint pp. 44–45). *Hidden hypotheses:* the face preservation of V2.45; the paracompactness of $F$ (faces are closed). *Deviation (restated):* the tame model lives on $\mathbb{R}^k \oplus W$ through $\iota$, $\pi$ instead of the sc-subspace $f_1$; tameness (2) of $r'$ is proved with the complement $(1 - Dr'(x))E'$ by Proposition 2.5.3 (preprint 2.35), not by intersecting with $f_1$. $U^*$ is a union of balls intersected with their $P_i$-preimages, since $P_i$ need not be norm-decreasing. *No hypothesis is added.*

**V2.50 (Proposition 2.5.16 (preprint 2.44); `Polyfold.MPolyfold.proposition_2_44`, `Polyfold.proposition_2_4`, `Polyfold.LocalModel.proposition_2_43_isTame`, `Polyfold.MPolyfold.proposition_2_44_corner`, `Polyfold.corner_retraction`, `Polyfold.corner_tameModel`, `Polyfold.cornerIncl`, `Polyfold.cornerProj`, `Polyfold.cornerKill`, `Polyfold.degIndexWith_eq_quadrantDeg_cornerProj_add_card`).** Let $X$ be tame and $x \in X$. There is a chart $\varphi : V \to O$ of the structure around $x$, with a tame retraction $r$ onto $O$ (tame for a presentation $T$), such that $\varphi(x)$ lies in exactly $d_X(x)$ faces of the M-polyfold $O$ (the local faces), and every face $F$ of $O$ is a sub-M-polyfold of $O$ which, with the induced structure, is tame with $d_F(\varphi(x)) + 1 = d_X(x)$ when $\varphi(x) \in F$. For every $\sigma \subseteq I(\varphi(x))$ (a set of $\#\sigma$ face germs at $x$) there are pairwise distinct faces $F_i \ni \varphi(x)$ of the retract, $F_i \subseteq f_i$ ($i \in \sigma$), and an open $B \ni \varphi(x)$ such that $A = \{y \in V : \varphi(y) \in B,\ \varphi(y) \in F_i\ \forall i \in \sigma\}$ is a sub-M-polyfold of $X$, tame with its induced structure, with $d_A(y) + \#\sigma = d_X(y)$ for all $y \in A$. The ingredients: the corner $f_\sigma = \bigcap_{i\in\sigma} f_i$ is modelled on $\mathbb{R}^{n-\#\sigma} \oplus W$ through sc-operators $\iota_\sigma$, $\pi_\sigma$, with $d_C(y) = d_{C'}(\pi_\sigma y) + \#\sigma$ on $f_\sigma$; near a point $o \in O$ with $\sigma \subseteq I(o)$, $r \circ P_\sigma$ ($P_\sigma = \iota_\sigma\pi_\sigma$) is an sc-smooth retraction onto $O \cap V \cap f_\sigma = O \cap V \cap \bigcap_{i\in\sigma} F_i$; and $\pi_\sigma \circ r \circ \iota_\sigma$ is a tame retraction of $[0,\infty)^{n-\#\sigma} \oplus W$ onto its image under $\pi_\sigma$. *Book:* Proposition 2.5.16 (preprint 2.44), p. 62 (preprint p. 46) (stated without proof). *Hidden hypotheses:* none. *Deviation (restated):* local faces are the faces of the M-polyfold $O \cong V$, face germs at $x$ are the faces $F_i$, $i \in I(\varphi(x))$, "the germ $F_x^\sigma$" is represented by the set $A$, and "codimension $\#\sigma$" is the number of quadrant coordinates removed in the tame model; the index identity holds on all of $A$. *Proof:* not by the exposition's induction on $\#\sigma$ but at once: Lemma 4.C for each $i \in \sigma$ shows that $r$ maps points of $C \cap f_\sigma$ near $\varphi(x)$ into $f_\sigma$;

then as in Proposition 2.5.15 (preprint 2.43) and Theorem 2.5.17 (preprint 2.45).

**V2.51 (Theorem 2.5.17 (preprint 2.45);** `Polyfold.MPolyfold.theorem_2_45_interior, Polyfold.MPolyfold.isSubMPolyfold_of_local, Polyfold.MPolyfold.theorem_2_45_isSubMPolyfold_inter Polyfold.MPolyfold.exists_localFace_of_isFaceStructured, Polyfold.MPolyfold.theorem_2_45_isSubMP Polyfold.MPolyfold.theorem_2_45_isTame_face, Polyfold.MPolyfold.isTame_of_forall_exists_chart`**).** Let $X$ be tame, $G$ the connected component of $\{d_X = 1\}$ through a point and $F = \overline{G}$ the face. (a) $F \cap \{d_X = 1\} = G$, and $G$ is a sub-M-polyfold of $X$. (b) If $X$ is face-structured, every face $F$ agrees near each of its points, in a tame chart, with a face of the model retract; $F$ is a sub-M-polyfold of $X$, tame with its induced structure, and $d_X(x) = d_F(x) + 1$ for $x \in F$. (A subset $A$ is a sub-M-polyfold as soon as near each point $\varphi(A)$ agrees on an open set with an sc-smooth retract contained in $O$; $X$ is tame as soon as every point lies in a chart of the structure with a tame model.) *Book:* Theorem 2.5.17 (preprint 2.45), p. 62 (preprint p. 46) (stated without proof). *Hidden hypotheses:* the meaning of "the interior of a face", here $F \cap \{d_X = 1\}$. *Deviation (hypotheses explicit):* the counting argument behind "each global face through $x$ contains exactly one local face" (onto by Lemma 4.C, injective by face-structuredness) is made explicit. The theorem's face counts are V2.48 and the definition of face-structured (V2.44).

**V2.52 (products of M-polyfolds, exposition Lemma 4.D;** `Polyfold.IsPartialQuadrant.prod, Polyfold.prod_eq_prodPresentation, Polyfold.degIndex_prod, Polyfold.minimalSubspaceWith_prodPrese Polyfold.ScRetraction.prod, Polyfold.IsScRetract.prod, Polyfold.ScRetraction.IsTameWith.prod, Polyfold.MChart.prod, Polyfold.MChart.Compatible.prod, Polyfold.MPolyfold.paracompactSpace_prod, Polyfold.MPolyfold.prod, Polyfold.MPolyfold.degIndex_prod_le, Polyfold.MPolyfold.isTame_prod, Polyfold.MPolyfold.isScSmoothMap_fst, Polyfold.MPolyfold.isScSmoothMap_snd, Polyfold.MPolyfold.Is`**).** For partial quadrants $C \subseteq E$, $C' \subseteq E'$, $C \times C'$ is a partial quadrant of $E \oplus E'$, presented by $E \oplus E' \cong \mathbb{R}^{n+n'} \oplus (W \oplus W')$, with $d_{C\times C'}(x, x') = d_C(x) + d_{C'}(x')$ and minimal subspace $E_x \oplus E'_{x'}$; products of sc-smooth (tame) retractions are sc-smooth (tame) retractions. For M-polyfolds $X$, $Y$ the space $X \times Y$ is paracompact and the product charts are an atlas of an M-polyfold structure; $d_{X\times Y}(x, y) \le d_X(x) + d_Y(y)$, and if $X$, $Y$ are tame, $X \times Y$ is tame with equality. The projections are sc-smooth and the pairing of sc-smooth maps is sc-smooth. *Book:* used without statement in the proof of Proposition 2.5.20 (preprint 2.46), preprint p. 47 ("$d_{X\times Y} = d_X + d_Y$"). *Hidden hypotheses:* the additivity needs tameness; in general only the inequality holds. Paracompactness of $X \times Y$ is Theorem 2.3.10 (preprint 2.16). *Deviation (extra):* not stated in the text; the exposition's Lemma 4.D.

**V2.53 (Proposition 2.5.20 (preprint 2.46);** `Polyfold.MPolyfold.IsLocalScDiffeo, Polyfold.MPolyfold.fibred Polyfold.MPolyfold.fibredSubDataL, Polyfold.MPolyfold.isSubMPolyfold_fibredProduct, Polyfold.MPolyfold.isScDiffeoOn_snd_fibredProduct, Polyfold.MPolyfold.isScDiffeoOn_fst_fibredPro Polyfold.MPolyfold.degIndex_fibredProduct_eq_snd, Polyfold.MPolyfold.degIndex_fibredProduct_eq_f Polyfold.MPolyfold.isTame_fibredProduct, Polyfold.MPolyfold.proposition_2_46, Polyfold.MPolyfold.proposition_2_46_of_isLocalScDiffeo`**).** Let $f : X \to Z$ be a local sc-diffeomorphism (every point has an open neighbourhood that $f$ maps sc-diffeomorphically onto an open subset of $Z$) and $g : Y \to Z$ sc-smooth. Then $X_f \times_g Y = \{(x, y) : f(x) = g(y)\}$ is closed in $X \times Y$, hence paracompact, and a sub-M-polyfold of $X \times Y$, near each point the retract of $R(x', y') = ((f|_U)^{-1}(g(y')), y')$; $d_{X_f\times_g Y}(x, y) = d_Y(y)$; if $Y$ is tame, so is the fibred product. If $g$ is also a local sc-diffeomorphism, then $d_{X_f\times_g Y}(x, y) = d_X(x) = d_Y(y)$, and if moreover $X$ and $Y$ are tame, $2\, d_{X_f\times_g Y}(x, y) = d_{X\times Y}(x, y)$. *Book:* Proposition 2.5.20 (preprint 2.46), pp. 63–64 (preprint pp. 46–47). *Hidden hypotheses:* the meaning of "local sc-diffeomorphism";

the paracompactness of the fibred product. The text's $\pi_2 : X_f \times_g Y \to X$ has target $Y$. *Deviation (stronger):* $d = d_Y$, and $d = d_X = d_Y$ in the last statement, hold without tameness (Proposition 2.4.4 (preprint 2.24) along $\pi_2$, $\pi_1$); tameness is used only for $d_{X\times Y} = d_X + d_Y$ (V2.52). The inverse of $\pi_2$ near $(x, y)$ is $y' \mapsto ((f|_U)^{-1}(g(y')), y')$; the exposition's factorization through the constant-in-$x$ map $y' \mapsto (x, y')$ is not sc-smooth unless $x$ is a smooth point. *No hypothesis is added.*

**V2.54 (Definition 2.6.1 (preprint 2.47);** `Polyfold.doubleLevel`, `Polyfold.shiftFiber`, `Polyfold.IsStrongBundleMap`, `Polyfold.IsStrongBundleRetraction`, `Polyfold.IsPartialQuadrant.prod_un` `Polyfold.IsRelOpenInQuadrant.prod_univ`, `Polyfold.IsStrongBundleRetraction.baseRetraction`, `Polyfold.IsStrongBundleRetraction.fiber_idem`, `Polyfold.IsStrongBundleRetraction.toScRetraction`, `Polyfold.IsStrongBundleRetraction.toScRetractionOne`**).** $C \times F_0$ is a partial quadrant in $E \oplus F$ and $U \times F_0$ is relatively open in it. A *strong bundle map* on $U \lhd F$ is $\Phi : E_0 \times F_0 \to E_0' \times F_0'$ mapping $U \times F_0$ into $U' \times F_0'$, whose first component does not depend on $h$ and second is linear in $h$ for $u \in U$, mapping $U_m \oplus F_k$ into $U_m' \oplus F_k'$ for $k \le m + 1$, sc-smooth on $U \times F_0$ for $E \oplus F$, and whose induced map on $U \times F_1$ (fibre component lifted to $F_1'$) is sc-smooth for $E \oplus F^1$. A *strong bundle retraction* is one with $R \circ R = R$ on $U \times F_0$. Then $r(u) = R(u, 0)_1$ is an sc-smooth retraction on $U$, $\Gamma(u, \cdot)$ is a projection for $u \in O$, and $R$ is an sc-smooth retraction on $(U \lhd F)[0]$ and $[1]$. *Book:* Definition 2.6.1 (preprint 2.47) and the remarks after it, pp. 65–66 (preprint pp. 48–49). *Hidden hypotheses:* $U \lhd F$ relatively open in a partial quadrant (for "sc-smooth" to make sense). *Deviation (restated):* strong bundle isomorphisms are not defined separately.

**V2.55 (Definition 2.6.2 (preprint 2.48) and $K[i]$;** `Polyfold.IsStrongBundleRetract`, `Polyfold.strongLevel`, `Polyfold.strongLevelShift`, `Polyfold.strongShiftOne`, `Polyfold.IsStrongBundleRetraction.mem_image_i` `Polyfold.IsStrongBundleRetraction.toScRetractionOne_retract`, `Polyfold.IsStrongBundleRetract.isSc` `Polyfold.IsStrongBundleRetract.isScRetract_one`, `Polyfold.IsStrongBundleRetract.isScSmoothRetract` `Polyfold.IsStrongBundleRetract.isScSmoothRetractMap_fst_one`, `Polyfold.IsStrongBundleRetract.isSc`**).** $K$ is a *local strong bundle retract* over $C$ if $K = R(U \times F_0)$ for a strong bundle retraction on a relatively open $U \subseteq C$; then $K = \{(u, h) : u \in O, \Gamma(u, h) = h\}$. $K_{m,k}$ and $K[1] = \{(u, h) \in E_0 \times F_1 : (u, \iota_1 h) \in K\}$ are defined level by level; $K$ is a retract in $(C \times F_0, E \oplus F)$, $K[1]$ (the image of $R$ on $(U \lhd F)[1]$) a retract in $(C \times F_0, E \oplus F^1)$, and the projections $K[i] \to O$ and the inclusion $K[1] \to K[0]$ are sc-smooth. *Book:* Definition 2.6.2 (preprint 2.48) and pp. 65–67 (preprint pp. 49–50). *Hidden hypotheses:* that $R$ preserves the double filtration at $(0, 1)$, for $K[1] = R((U \lhd F)[1])$. *Deviation (restated):* none in substance.

**V2.56 (Definition 2.6.3 (preprint 2.49);** `Polyfold.IsScSmoothSection`, `Polyfold.IsScPlusSection`, `Polyfold.IsScPlusSection.mem`, `Polyfold.IsScPlusSection.isScSmoothSection`**).** An *sc-smooth section* of $p : K \to O$ is $f$ with $f(x)_1 = x$ on $O$, sc-smooth from $(O, C, E)$ to $(K, C \times F_0, E \oplus F)$; an *sc⁺-section* is $f : E_0 \to E_0 \times F_1$ with $f(x)_1 = x$ on $O$, sc-smooth into $(K[1], C \times F_0, E \oplus F^1)$. An sc⁺-section is an sc-smooth section. *Book:* Definition 2.6.3 (preprint 2.49), p. 66 (preprint p. 50) ($K(0)$, $K(1)$ for $K[0]$, $K[1]$). *Deviation (restated):* the category $\mathcal{SBR}$ is not formalized.

**V2.57 (Definition 2.6.4 (preprint 2.50);** `Polyfold.IsBanachable`, `Polyfold.StrongBundleChart`, `Polyfold.StrongBundleChart.K`, `Polyfold.StrongBundleChart.continuous_fiber`, `Polyfold.StrongBundleC` `Polyfold.StrongBundleChart.Compatible.symm`, `Polyfold.StrongBundleAtlas`, `Polyfold.StrongBundleAtla` `Polyfold.StrongBundleChart.compatible_of_locally`, `Polyfold.StrongBundleAtlas.equiv_iff`, `Polyfold.StrongBundleAtlas.Equiv.trans`, `Polyfold.StrongBundleAtlas.equivalence`, `Polyfold.IsStrongBundleRetraction.restrict`, `Polyfold.StrongBundleChart.restrict`, `Polyfold.StrongBundleChart.Compatible.restrict`**).** For $P : Y \to X$ with linear structures on the fibres: a *strong bundle chart* is a chart $\varphi$ of the M-polyfold $X$, an sc-Banach space $F$, a strong

bundle retraction $R$ on a relatively open $U \subseteq C$ with base retract $O$, and a homeomorphism $\Phi : P^{-1}(V) \to K = R(U \times F_0)$ with $\Phi(y)_1 = \varphi(P(y))$ and linear on each fibre (hence continuous on fibres). Two charts are *compatible* if $\Phi' \circ \Phi^{-1}$ and $\Phi \circ \Phi'^{-1}$ are sc-smooth between the overlaps on $[0]$ and, through the lift to $F_1$, on $[1]$, and carry fibre components in $F_1$ to fibre components in $F_1'$. Atlases and their equivalence as in V2.11; two atlases are equivalent iff every chart of one is compatible with every chart of the other, and the relation is an equivalence relation (it rests on the locality of compatibility: near a point of the overlap a transition factors through a chart of the middle atlas, on $[0]$ and on $[1]$). A strong bundle chart restricts to every open $s \subseteq V$ (retraction domain $U \cap r^{-1}(W)$ for $\varphi(s) = O \cap W$, $K_s = \{(u,h) \in K \mid u \in \varphi(s)\}$, $\Phi$ unchanged), and restrictions of compatible charts are compatible. A space is *Banachable* if it is linearly homeomorphic to a Banach space. *Book:* Definition 2.6.4 (preprint 2.50), pp. 67–68 (preprint pp. 51–52). *Hidden hypotheses:* $\varphi$ a chart of $X$ (the text: a homeomorphism onto $O$); the $[1]$-transition condition. *Deviation (hypotheses explicit):* both stated. The transitivity of the equivalence, which "an equivalence class of strong bundle atlases" (Definition 2.6.5 (preprint 2.51)) presupposes, and the restriction of charts, which the text uses silently (Lemma 5.1.2 (preprint 5.2)), are proved (Findings, Chapter 2, items 40, 42); a restriction whose retraction domain lies inside a given open $W \ni 0$ of $E$ is not, since $U \cap W$ need not be invariant under $r$.

**V2.58 (Definition 2.6.5 (preprint 2.51), strong bundles, $Y[0]$, $Y[1]$, $P[0]$, $P[1]$; `Polyfold.StrongBundle`, `Polyfold.StrongBundleChart.toMChart`, `Polyfold.StrongBundleChart.Compatible.toMChart`, `Polyfold.StrongBundle.atlas0`, `Polyfold.StrongBundle.toMPolyfold0`, `Polyfold.StrongBundle.isScSmoot` `Polyfold.StrongBundle.doubleLevel`, `Polyfold.StrongBundle.mem_doubleLevel_iff`, `Polyfold.StrongBundle.oneChart`, `Polyfold.StrongBundle.oneChart_compatible`, `Polyfold.StrongBundle.` `Polyfold.StrongBundle.toMPolyfold1`, `Polyfold.StrongBundle.isScSmoothMap_proj1`, `Polyfold.StrongBundle.maximalAtlas`, `Polyfold.StrongBundle.maximalAtlas_equiv`, `Polyfold.StrongBundle.isChart_iff_exists_equiv`, `Polyfold.StrongBundle.isChart_congr`, `Polyfold.StrongBundle.IsChart.compatible`, `Polyfold.StrongBundle.IsChart.restrict`).** A *strong bundle* is a continuous surjection $P : Y \to X$, $Y$ Hausdorff paracompact, with Banachable fibres and a strong bundle atlas. The charts $\Phi$ form an sc-smooth atlas of $Y$ (the M-polyfold $Y[0]$), and $P : Y[0] \to X$ is sc-smooth. $Y_{m,k}$ ($k \le m+1$) is the set of $y$ with $\Phi(y) \in E_m \times F_k$ in the charts of the atlas around $y$; one chart around $y$ suffices. The charts $\Phi[1] : y \mapsto (\Phi(y)_1, \Phi(y)_2$ lifted to $F_1)$, defined on $Y_{0,1}$ with the topology they generate and onto the retracts $K[1]$, are pairwise compatible, their transition maps being the $[1]$-transitions of Definition 2.6.4 (preprint 2.50), and make $Y[1]$ an M-polyfold; $P[1] : Y[1] \to X$ is sc-smooth. The charts of the strong bundle (compatible with the atlas) are those of the maximal atlas, which is equivalent to the given one: a chart is one iff it belongs to some atlas of the class, any two are compatible, restrictions of charts are charts, and equivalent atlases have the same charts. *Book:* Definition 2.6.5 (preprint 2.51) and the remarks after it, preprint p. 52. *Hidden hypotheses:* the paracompactness of $Y[1]$ (proved, V2.64); for the chart-independence of $Y_{m,m+1}$ and of $Y_{0,1}$, the $F_1$-condition on the $[1]$-transition (V2.57). *Deviation (restated):* $Y[1]$ carries the topology generated by the charts $\Phi[1]$, as $X_m$ in §2.3 (preprint §2.2); the charts are those induced by the given atlas. The notions defined through the charts compatible with the atlas depend only on the class; those defined with the charts of the given atlas itself ($Y_{m,k}$, $Y[1]$ and its atlas, sc$^+$-sections) are not shown invariant (Findings, Chapter 2, item 43).

**V2.59 (Definition 2.6.6 (preprint 2.52); `Polyfold.IsSection`, `Polyfold.StrongBundle.IsScSmoothSection`, `Polyfold.StrongBundle.localRepOne`, `Polyfold.StrongBundle.IsScPlusSection`).** A *section* is $f$ with $P \circ f = \mathrm{id}$; it is *sc-smooth* if it is sc-smooth $X \to Y[0]$, and an *sc$^+$-section* if in every chart of the atlas its fibre values lie in $F_1$ and its local representative, lifted to $E_0 \times F_1$, is

sc-smooth from $O$ into $K[1]$. *Book:* Definition 2.6.6 (preprint 2.52), p. 68 (preprint p. 53). *Deviation (restated):* sc$^+$ is defined chart by chart; this is equivalent to the text's definition: $f$ is an sc$^+$-section iff $P \circ f = \mathrm{id}$ and $f = \iota \circ g$ for an sc-smooth $g : X \to Y[1]$, $\iota : Y[1] \to Y$ the inclusion (`Polyfold.StrongBundle.isScPlusSection_iff`).

**V2.60 (Definition 2.6.7 (preprint 2.53); `Polyfold.pullbackSpace`, `Polyfold.pullbackProj`, `Polyfold.pullbackProj'`, `Polyfold.pullback_comm`, `Polyfold.continuous_pullbackProj`, `Polyfold.isClosed_pullbackSpace`).** $f^*Y = \{(z, y) : P(y) = f(z)\}$ with the projections $P_f$, $P'$; $P \circ P' = f \circ P_f$; $P_f$ is continuous, and $f^*Y$ is closed in $Z \times Y$ when $X$ is Hausdorff and $P$, $f$ are continuous. *Book:* Definition 2.6.7 (preprint 2.53), p. 69 (preprint p. 53). *Hidden hypotheses:* none.

**V2.61 (Proposition 2.6.8 (preprint 2.54); `Polyfold.proposition_2_54`, `Polyfold.pullbackAtlas`, `Polyfold.PullbackChartData`, `Polyfold.PullbackChartData.toChart`, `Polyfold.PullbackChartData.transi` `Polyfold.PullbackChartData.transition_isScSmooth`, `Polyfold.PullbackChartData.transitionOne_isScS` `Polyfold.PullbackChartData.compatible_toChart`, `Polyfold.paracompactSpace_pullbackSpace`, `Polyfold.pullbackFiberCLE`, `Polyfold.pullbackRetraction`, `Polyfold.isStrongBundleRetraction_pullbac` `Polyfold.pullback_image`, `Polyfold.isStrongBundleRetract_pullback`).** Let $P : Y \to X$ be a strong bundle and $f : Z \to X$ sc-smooth. Then $f^*Y$ is paracompact, and $P_f : f^*Y \to Z$ is a strong bundle whose atlas consists of the pulled-back charts: for a strong bundle chart $\Phi$ over $\varphi$ and a chart $\psi$ of $Z$ with $f(\mathrm{dom}\,\psi) \subseteq \mathrm{dom}\,\varphi$ and $g = \varphi \circ f \circ \psi^{-1}$, the chart $\Phi_f(z, y) = (\psi(z), \mathrm{pr}_F \Phi(y))$ onto $K_g = \{(w, h) : w \in O_Z, (g(w), h) \in K\}$, the image of the strong bundle retraction $R_g(w, h) = (r_Z(w), \Gamma(g(r_Z(w)), h))$. Its fibres are $P_f^{-1}(z) \cong P^{-1}(f(z))$ (a linear homeomorphism). Transition maps of pulled-back charts are $(w, h) \mapsto (\psi_2\psi_1^{-1}(w), \mathrm{pr}_F(\Phi_2\Phi_1^{-1})(g_1(w), h))$ and are sc-smooth on $[0]$ and $[1]$. *Book:* Proposition 2.6.8 (preprint 2.54), p. 69 (preprint p. 53) ("the easy proof is left to the reader"). *Hidden hypotheses:* $f^*Y$ paracompact (required by Definition 2.6.5 (preprint 2.51), not mentioned); proved from Theorem 2.3.10 (preprint 2.16) for $Z$ and $Y[0]$. *Deviation (hypotheses explicit):* "natural structure" is the atlas of all pulled-back charts. That the projection $P' : f^*Y \to Y$ is a strong bundle map (sc-smooth on $[0]$ and $[1]$), which "natural" presumably also says, is not formalized (Findings, Chapter 2, item 51).

**V2.62 (Theorem 2.7.1 (preprint 2.55), Smirnov's theorem; `Polyfold.metrizableSpace_of_locally`, `Polyfold.metrizableSpace_of_partition`, `Polyfold.metrizableSpace_of_pullback`, `Polyfold.cone_triangle`, `Polyfold.truncDist`, `Polyfold.paracompactSpace_of_metrizableSpace`).** A paracompact Hausdorff space in which every point has a metrizable open neighbourhood is metrizable. More generally, a space $Y$ with open metrizable subspaces $V_i$ and a locally finite family of continuous $\rho_i \geq 0$ with $\mathrm{supp}\,\rho_i \subseteq V_i$, not all vanishing at any point, is metrizable, with the metric $\sum_i |\rho_i(x) - \rho_i(y)| + \min(\rho_i(x), \rho_i(y)) \min(d_i(x, y), 1)$; and if $\pi : Y \to X$ is continuous into a paracompact Hausdorff $X$ whose points have open neighbourhoods $U$ with $\pi^{-1}(U)$ metrizable, $Y$ is metrizable. Metrizable spaces are paracompact. *Book:* Theorem 2.7.1 (preprint 2.55), p. 72 (preprint p. 56), quoted from Smirnov. *Deviation (stronger):* proved, in the pull-back form used for $X_m$ and $TX$.

**V2.63 (restriction and locality of sc^k maps; `Polyfold.IsSck.of_subset`, `Polyfold.IsScSmooth.of_subset`, `Polyfold.IsSc1.tangentMap_eqOn_of_subset`, `Polyfold.IsSc1.of_locally`, `Polyfold.IsSck.of_locally`, `Polyfold.IsScSmooth.of_locally`, `Polyfold.IsScSmooth.congr`).** On sc-domains: an sc$^k$ map restricts to sc$^k$ on every smaller sc-domain, with the same tangent map there; a map that is sc$^1$ (resp. sc$^k$, sc-smooth) on $U \cap W$ for an open $W$ around each point of $U$ is sc$^1$ (resp. sc$^k$, sc-smooth) on $U$; sc-smoothness depends only on the values on $U$. *Book:* used without statement in §§2.1, 2.3 (preprint §2.1–§2.2) (Propositions 2.1.11, 2.3.1 (preprint 2.4, 2.10), Definitions 2.3.4–2.3.8 (preprint

2.13–2.14)). *Deviation (not in the book):* infrastructure the text uses silently.

**V2.64 (the space $Y[1]$ and the inclusion $Y[1] \to Y[0]$; `Polyfold.StrongBundle.oneSet`, `Polyfold.StrongBundle.mem_oneSet_iff`, `Polyfold.StrongBundle.One`, `Polyfold.StrongBundle.continuous` `Polyfold.StrongBundle.metrizableSpace_one`, `Polyfold.StrongBundle.paracompactSpace_one`, `Polyfold.StrongBundle.isScSmoothMap_oneVal`).** For a strong bundle $P : Y \to X$, $Y_{0,1}$ is the set of points whose fibre component lies in $F_1$ in every chart of the atlas around them, and one such chart suffices. With the topology generated by the charts $\Phi[1]$ (V2.58), the inclusion $Y[1] \to Y$ is continuous, so $Y[1]$ is Hausdorff; $Y[1]$ is metrizable, hence paracompact; and the inclusion $Y[1] \to Y[0]$ is sc-smooth (in the charts $\Phi[1]$, $\Phi$ it is $(u, h) \mapsto (u, \iota h)$). *Book:* not stated; the text takes $Y[1]$ to be an M-polyfold on $Y_{0,1}$ (preprint p. 52). *Hidden hypotheses:* paracompactness of $Y[1]$, required by Definition 2.3.4 (preprint 2.13). *Deviation (not in the book):* proved; metrizability by the pull-back form of Smirnov's theorem over the paracompact base (V2.62), as in Theorem 2.3.10 (preprint 2.16).

**V2.65 (Proposition 2.1.13 (preprint 2.6), the 2021 statement; `Polyfold.proposition_2_1_13_2021`).** Let $r : U \to U$ and $s : V \to V$ be sc-smooth retractions onto the same retract $O$, $\rho$ an sc-smooth retraction onto $O'$, and $f$ a map with $f(O) \subseteq O'$. Then (1) $f \circ r$ is sc-smooth on $U$ iff $f \circ s$ is sc-smooth on $V$; (2) if $f \circ r$ is sc-smooth, $T(f \circ r) = T(f \circ s)$ on $TO = Tr(TU)$; (3) if $f \circ r$ is sc-smooth, $T(f \circ r)$ maps $TO$ into $TO'$. *Book:* Proposition 2.1.13, p. 27 (preprint 2.6, p. 20); the 2021 statement of (3) has the hypothesis "$f \circ r$ sc-smooth", which the preprint and 2017 statements lack and their proofs use. *Hidden hypotheses:* $f(O) \subseteq O'$, contained in "$f : O \to O'$". *Deviation:* none. The Lean statement of (3) in V2.5 already carried the 2021 hypothesis, so the 2021 change confirms the formalization's reading. *Proof:* the conjunction of `Polyfold.proposition_2_6_1` (both ways), `Polyfold.proposition_2_6_2` and `Polyfold.proposition_2_6_3` (V2.5).

**V2.66 (Proposition 2.1.4, complete closed neighbourhoods in a retract; `Polyfold.ScRetraction.exists_isClos` `Polyfold.IsScRetract.exists_isComplete_closure`, `Polyfold.proposition_2_1_4_2021`).** Let $O$ be an sc-smooth retract in $E$, with the metric $d_O(x, y) = |x - y|_0$ of $E_0$. Every $x \in O$ has open neighbourhoods $V \subseteq W$ in $O$ with $\mathrm{cl}_O(V) \subseteq W$ and $\mathrm{cl}_O(V)$ complete for $d_O$; $W$ can be chosen inside any given neighbourhood of $x$. For a retraction $r$ onto $O$, $O \cap \bar{B}_\varepsilon(x)$ is closed in $E_0$ for all small $\varepsilon > 0$. *Book:* Proposition 2.1.4, p. 22, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (stronger):* $W$ arbitrarily small. *Proof:* with $O \cap \bar{B}_\varepsilon(x)$ in place of the text's $V'$ with $r(V') \subseteq V'$: for $\bar{B}_\varepsilon(x) \cap C \subseteq U$, a limit of points of $O \cap \bar{B}_\varepsilon(x)$ lies in $U$ and is fixed by $r$; a closed subset of the Banach space $E_0$ is complete.

**V2.67 (Definition 2.1.5, locally complete spaces; `Polyfold.AdmitsCompleteMetricOn`, `Polyfold.admitsCompleteMetricOn_of_isComplete`, `Polyfold.AdmitsCompleteMetricOn.metrizableSpace`, `Polyfold.AdmitsCompleteMetricOn.of_isEmbedding`, `Polyfold.AdmitsCompleteMetricOn.mono`, `Polyfold.isComplete_inter_of_isClosed`, `Polyfold.IsLocallyComplete`, `Polyfold.IsLocallyComplete.exi` `Polyfold.IsLocallyComplete.regularSpace`, `Polyfold.IsLocallyComplete.t1Space`, `Polyfold.IsLocallyComplete.t2Space`, `Polyfold.isLocallyComplete_of_completeSpace`, `Polyfold.isLocallyComplete_real`, `Polyfold.isLocallyComplete_of_metric`).** A topological space $X$ is locally complete if every $x$ has open $V \ni x$ and $W$ with $\mathrm{cl}_X(V) \subseteq W$ such that $W$ carries a metric $d$ inducing its subspace topology for which $\mathrm{cl}_X(V)$ is complete. Then $W$ can be chosen inside any neighbourhood of $x$, and $X$ is regular, $T_1$ and Hausdorff. A complete metric space is locally complete ($\mathbb{R}$ in particular), and so is a metric space in which every point has open $V \subseteq W$, $\mathrm{cl}(V) \subseteq W$, with $\mathrm{cl}(V)$ complete. *Book:* Definition 2.1.5, p. 23, new in the 2021 edition, for a Hausdorff space $X$, with (1) the metric on $W(x)$ and (2) $V(x) \subseteq W(x)$, $\mathrm{cl}_X(V(x)) \subseteq W(x)$, $\mathrm{cl}_X(V(x))$ complete. *Hidden hypotheses:* none. *Deviation (stronger):* the Hausdorff assumption

is not part of the definition; it follows ($T_1$ from the metrizable $W$, regularity from the global condition $\mathrm{cl}_X(V) \subseteq W$), so V2.70 and V2.71 need no Hausdorff hypothesis except for the conclusion $T_{3\frac{1}{2}}$. $V \subseteq W$ is a consequence, not a separate condition.

**V2.68 (Remark 2.1.6; `Polyfold.remark_2_1_6_2021`, `Polyfold.IsLocallyCompleteRel`, `Polyfold.IsLocallyComplete.isLocallyCompleteRel`).** If $\mathrm{cl}_X(V) \subseteq W$, the closure of $V$ in the subspace $W$ is $\mathrm{cl}_X(V)$. The modified notion, in which $\mathrm{cl}_X(V) \subseteq W$ is replaced by $V \subseteq W$ and completeness is asked of $\mathrm{cl}_W(V)$, is implied by local completeness. *Book:* Remark 2.1.6, p. 23, new in the 2021 edition: the condition $\mathrm{cl}_X(V(x)) \subseteq W(x)$ is global, and the modified definition is weaker. *Hidden hypotheses:* none. *Deviation (narrower):* "weaker" is proved here as the implication; that the modified notion is strictly weaker is V2.91, an example (gap G.76, closed).

**V2.69 (Corollary 2.1.7; `Polyfold.corollary_2_1_7_2021`, `Polyfold.isLocallyComplete_univ_retract`).** An sc-smooth retract $O$, with the topology of $E_0$, is locally complete; so is $E_0$ itself. *Book:* Corollary 2.1.7, p. 23, new in the 2021 edition. *Hidden hypotheses:* none. *Proof:* V2.66, with the metric $d_O$ restricted to $W$ (`Polyfold.isLocallyComplete_of_metric`).

**V2.70 (Proposition 2.1.8; `Polyfold.IsLocallyComplete.of_isEmbedding`, `Polyfold.proposition_2_1_8_1_202`, `Polyfold.proposition_2_1_8_2_2021`, `Polyfold.proposition_2_1_8_3_2021`, `Polyfold.proposition_2_1_8_`).** Let $X$ be locally complete. (1) A locally closed subspace $A$ (open $\cap$ closed) is locally complete; (2) a space homeomorphic to $X$ is locally complete; (3) an open subspace is locally complete. (4) A space $Y$ with an open cover $(Y_i)$ by locally complete subspaces such that every $z \in Y_i$ has an open $U \ni z$, $U \subseteq Y_i$, with $\mathrm{cl}_Y(U) \subseteq Y_i$, is locally complete. More generally, a space embedded in $X$ with locally closed image is locally complete. *Book:* Proposition 2.1.8, p. 24, new in the 2021 edition, for Hausdorff spaces. *Hidden hypotheses:* none. *Deviation (stronger):* no Hausdorff hypothesis (V2.67). "Locally closed" is Mathlib's notion, equivalent to the text's use ("$U \cap A$ closed in $U$ for an open $U \ni a$"). *Proof:* (1)–(3) from the embedding statement; (4) by transporting the metric of $W \subseteq Y_i$ to $W \subseteq Y$.

**V2.71 (Lemma 2.1.9 and Proposition 2.1.10; `Polyfold.lemma_2_1_9_2021`, `Polyfold.IsLocallyComplete.compl`, `Polyfold.proposition_2_1_10_2021`).** A locally complete space is first countable and completely regular; if it is Hausdorff it is $T_{3\frac{1}{2}}$ (Tychonoff). *Book:* Lemma 2.1.9 (countable neighbourhood bases) and Proposition 2.1.10 (Hausdorff locally complete spaces are completely regular), p. 25, both new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (stronger):* no Hausdorff hypothesis for complete regularity. *Proof:* without sequences: the neighbourhood filter of $x$ is the image of that of the metrizable open $W$, and the bump function is built in $W$ and extended by a constant outside $\mathrm{cl}_X(V)$. Mathlib's complete regularity separates with $f(x) = 0$ and $f = 1$ on the closed set (the text's $\beta$ is $1 - f$).

**V2.72 (Lemma 2.2.1, the levels of a connected relatively open set; `Polyfold.ScBanach.lvlSetInfty`, `Polyfold.ScBanach.smoothSeq`, `Polyfold.ScBanach.toBase_smoothSeq`, `Polyfold.ScBanach.smoothSeq_mem`, `Polyfold.ScBanach.smoothSeq_zero`, `Polyfold.ScBanach.image_lvlSetInfty`, `Polyfold.ScBanach.joinedIn`, `Polyfold.IsPartialQuadrant.exists_smooth_near`, `Polyfold.IsRelOpenInQuadrant.exists_ball_inter_su`, `Polyfold.lemma_2_2_1_2021`, `Polyfold.lemma_2_2_1_infty_2021`).** Let $U$ be relatively open in a partial quadrant $C$ and connected. For every $k \in \mathbb{N}$ the level $U_k = U \cap E_k$, with the topology of $E_k$, is path-connected; and $U_\infty$, represented as the compatible sequences $(v_m) \in \prod_m E_m$ with $v_0 \in U$ (with the product topology, i.e. the topology of all the norms $|\cdot|_m$), is path-connected. Smooth points are dense in every level $C_m$, and the image of this $U_\infty$ in $E_0$ is $U \cap E_\infty$. *Book:* Lemma 2.2.1, p. 30, new in the 2021 edition, for $k \in \mathbb{N} \cup \{\infty\}$. *Hidden hypotheses:* the topology of $E_\infty$, which the text does not specify. *Deviation (explicit):* $U_\infty$ carries the inverse-limit topology.

*Proof:* not the text's polygon along a path $\sigma$ in $U_0$: the set of points of $U$ near which every point of the level is joined to a fixed one is open and closed in the connected $U$; segments in convex pieces $B_\varepsilon \cap C$ join smooth points, and smooth points are dense.

**V2.73 (Lemma 2.2.2, Definition 2.2.3 and Theorem 2.2.4; `Polyfold.lemma_2_2_2_2021`, `Polyfold.definition_2_2_3_2021`, `Polyfold.theorem_2_2_4_2021`, `Polyfold.IsScRetract.locallyPathConn`**
For a retraction $r : U \to U$ onto $O$, $x \in O$ and $V = V' \cap U$ with $V' \ni x$ open in $E_0$, the set $W = V \cap r^{-1}(V)$ is relatively open in $C$, contains $x$, and satisfies $W \cap O = V \cap O$, $r(W) \subseteq W$, $r(W) = O \cap V$. Mathlib's local path-connectedness is the text's (every open neighbourhood of $z$ contains an open path-connected neighbourhood). A retract is locally path-connected. *Book:* Lemma 2.2.2, p. 31, Definition 2.2.3 and Theorem 2.2.4, p. 32, all new in the 2021 edition. *Hidden hypotheses:* none. *Proof:* Theorem 2.2.4 was already proved for Chapter 7 (`Polyfold.IsScRetract.locallyPathConnectedSpace`), joining nearby points by $\tau \mapsto r(o + \tau(p - o))$ instead of through Lemma 2.2.2 and the components of $W(x)$.

**V2.74 (Definition 2.2.5 and Proposition 2.2.6; `Polyfold.IsAlmostLocallyContractible`, `Polyfold.proposition_2_2_6_2021`, `Polyfold.isAlmostLocallyContractible_univ`).** A space $Z$ is almost locally contractible if for every $z$ and open $V \ni z$ there are an open $W$, $z \in W \subseteq V$, and a continuous $\Phi : [0,1] \times W \to Z$ with values in $V$, $\Phi(0, y) = y$, $\Phi(t, z) = z$ and $\Phi(1, y) = z$. An sc-smooth retract is almost locally contractible; so is $E_0$. *Book:* Definition 2.2.5 and Proposition 2.2.6, p. 33, new in the 2021 edition. *Hidden hypotheses:* none. *Proof:* the text's $\Phi(t, y) = r((1-t)y + tx)$ on $W = B_\varepsilon(x) \cap O$.

**V2.75 (the footnote to Definition 2.3.4 (preprint 2.13), paracompactness, and the convention of Appendix 2.7.2; `Polyfold.IsLocallyFiniteSetFamily`, `Polyfold.IsParacompactStar`, `Polyfold.isParacompactStar_iff`, `Polyfold.definition_2_3_4_2021`, `Polyfold.isParacompactStar_real`, `Polyfold.IsParacompactBySubcover`, `Polyfold.not_isParacompactBySubcover_real`).** A space is paracompact* if every open cover has a locally finite open cover refining it, with no separation axiom; this is Mathlib's paracompactness, so "Hausdorff paracompact" *is exactly the hypothesis of the M-polyfold structure (V2.12);* $\mathbb{R}$ *is paracompact*. Under the reading"every open cover has a locally finite open **sub**cover", $\mathbb{R}$ is not paracompact (the cover by the intervals $(-n-1, n+1)$). *Book:* Definition 2.3.4, p. 35 (preprint 2.13, p. 23), with the footnote, new in the 2021 edition, that paracompactness does not include the Hausdorff property; Appendix 2.7.2, p. 72 (and 2017 Appendix 2.6.2, p. 51), states the convention with a locally finite open subcovering. *Hidden hypotheses:* none. *Deviation (restated):* the footnote changes nothing for M-polyfolds, whose Hausdorff property is required separately. The subcover wording of Appendix 2.7.2 does not hold as we read it for $\mathbb{R}$; it is read as "refinement" (possible misprint), as in the footnote.

**V2.76 (Lemma 2.3.5 and Lemma 2.7.2; `Polyfold.lemma_2_7_2_2021`, `Polyfold.exists_isOpen_closure_subset`, `Polyfold.lemma_2_3_5_2021`).** In a regular space every open $U \ni x$ contains an open $V \ni x$ with $\mathrm{cl}(V) \subseteq U$. In a Hausdorff paracompact space, and so in an M-polyfold, every open $W \ni x$ contains such a $V$ inside any given neighbourhood of $x$. *Book:* Lemma 2.3.5, p. 35, and Lemma 2.7.2, p. 72, new in the 2021 edition; Lemma 2.7.2 is stated for regular Hausdorff spaces. *Hidden hypotheses:* none. *Deviation (stronger):* Lemma 2.7.2 needs only regularity. *Proof:* Mathlib (regular spaces; Hausdorff paracompact spaces are normal).

**V2.77 (Propositions 2.3.6 and 2.3.7; `Polyfold.MChart.isLocallyComplete_source`, `Polyfold.MPolyfold.isLoc`, `Polyfold.proposition_2_3_6_2021`, `Polyfold.proposition_2_3_7_2021`, `Polyfold.MPolyfold.locallyPathC`**
A chart domain of an M-polyfold is locally complete. An M-polyfold $X$ is locally complete, with $W(x)$ inside any given neighbourhood of $x$: there are open $V \subseteq W$ around $x$ with $\mathrm{cl}_X(V) \subseteq W$

and a metric on $W$ inducing its topology for which $\mathrm{cl}_X(V)$ is complete. An M-polyfold is locally path-connected. *Book:* Propositions 2.3.6 and 2.3.7, p. 36, new in the 2021 edition. *Hidden hypotheses:* none. *Proof:* Proposition 2.3.6 by the open-cover part of V2.70, the cover by chart domains ($V \cong O$, V2.69 and V2.70 (2)) with condition (b) from V2.76, instead of the text's direct argument in a chart; Proposition 2.3.7 was already proved for Chapter 7 (`Polyfold.MPolyfold.locallyPathConnectedSpace`).

**V2.78 (Definitions 2.5.18 and 2.5.19, tame sub-M-polyfolds; `Polyfold.MPolyfold.HasTameInducedStructure`, `Polyfold.MPolyfold.HasReducedComplementAt`, `Polyfold.MPolyfold.hasReducedComplementAt_self`, `Polyfold.MPolyfold.TameSubData`, `Polyfold.MPolyfold.IsTameSubMPolyfold`, `Polyfold.MPolyfold.IsTameS` `Polyfold.MPolyfold.hasTameInducedStructure_univ`, `Polyfold.MPolyfold.isTameSubMPolyfold_univ`).** For a sub-M-polyfold $Y$ of $X$, the induced structure is tame if the M-polyfold $Y$ (V2.24) is tame. For a tame $X$, $Y$ is a tame sub-M-polyfold if every $y \in Y$ lies in an open $V$ with an sc-smooth retraction $r$ of the sub-M-polyfold data ($r(V) = Y \cap V$) such that $d_X(r(x)) = d_X(x)$ on $V$ and, at every smooth $y \in Y \cap V$, in some chart around $y$ with a presentation of the partial quadrant, the tangent space of the chart image $Q$ of $Y$ has an sc-complement $A$ in $T_zO$ inside the reduced tangent space $T_zO \cap E_z$ ($(T_zO)_m = (T_zQ)_m \oplus A_m$ on every level). A tame sub-M-polyfold is a sub-M-polyfold; in a tame $X$, $Y = X$ is a tame sub-M-polyfold with tame induced structure. *Book:* Definitions 2.5.18, p. 62, and 2.5.19, p. 63, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (narrower):* condition (3) is stated in a chart, for some chart around $y$ ($T_yX$ has no linear structure here, gap G.7); its independence of the chart is V2.99, the remark that a tame sub-M-polyfold is tame V2.100, and the examples in $[0,\infty)^2 \times \mathbb{R}$ V2.101 (gap G.77, closed).

**V2.79 (faces of open sets and the corner axioms; `Polyfold.TameBoundary.faceAt`, `Polyfold.TameBoundary.IsF` `Polyfold.TameBoundary.facesThrough`, `Polyfold.TameBoundary.LocalPieces`, `Polyfold.TameBoundary.IsCo` `Polyfold.TameBoundary.faceAt_inter_eq`, `Polyfold.TameBoundary.faceExt`, `Polyfold.TameBoundary.faceE` `Polyfold.TameBoundary.exists_faceExt_eq`, `Polyfold.TameBoundary.isCornerIndex_zero`, `Polyfold.MPolyfold.exists_localPieces_chart`, `Polyfold.MPolyfold.isCornerIndex_degIndex`, `Polyfold.MPolyfold.isFaceIn_univ_iff`).** For a map $d : X \to \mathbb{N}$ on a space $X$ and an open $V$, a face of $V$ is the closure in $V$ of a connected component of $V \cap \{d = 1\}$; the faces of $X$ itself are those of V2.44. The corner axioms ask that $d$ be upper semicontinuous and that every $x$ have an open $D$ such that every $y \in D$ has, in arbitrarily small neighbourhoods, $d(y)$ preconnected pieces of $\{d = 1\}$ accumulating at $y$ and covering $\{d = 1\}$ near $y$, no preconnected subset of $D \cap \{d = 1\}$ meeting two of them. For open $U \subseteq V$, a face $\theta$ of $U$ lies in a unique face $\theta'$ of $V$, the one containing it. The degeneracy index of a tame M-polyfold satisfies the corner axioms; so does $d \equiv 0$. *Book:* not stated; §2.7.4, pp. 78–82, new in the 2021 edition, uses the faces of open subsets $V$ (Definition 2.5.10 for $V$ with its induced structure, whose degeneracy index is $d_X|V$ by Proposition 2.4.4) and the local structure of a tame chart. *Hidden hypotheses:* none. *Deviation (not in the book):* all of §2.7.4 is proved from the corner axioms, so that it applies to the object spaces of tame ep-groupoids (2021 §7.6.5); the local pieces in a tame chart come from the local face structure of V2.45.

**V2.80 (Proposition 2.7.5; `Polyfold.TameBoundary.exists_nbhd_faces_mem`, `Polyfold.TameBoundary.proposit` `Polyfold.MPolyfold.proposition_2_7_5_2021`).** In a tame M-polyfold (more generally, for $d$ with the corner axioms) every $x$ has an open $V \ni x$ with (1) $d(y) \le d(x)$ on $V$, (2) every $y \in V$ in exactly $d(y)$ faces of $V$, (3) an open $U(x)$, $x \in U(x) \subseteq V$, such that every face of $V$ meeting $U(x)$ contains $x$; (3) holds for every open $V \ni x$. *Book:* Proposition 2.7.5, p. 78, new in the 2021 edition. *Hidden hypotheses:* none.

**V2.81 (Definition 2.7.6 and Proposition 2.7.7; `Polyfold.TameBoundary.IsFaceStructuredOpen`,**

`Polyfold.TameBoundary.faceStructuredOpens`, `Polyfold.TameBoundary.faceStructuredNbhds`, `Polyfold.TameBoundary.faceStructuredNbhdsAt`, `Polyfold.TameBoundary.faceStructuredNbhds_eq_iUnion` `Polyfold.TameBoundary.faceStructuredNbhdsAt_nonempty`, `Polyfold.TameBoundary.sUnion_faceStructure` `Polyfold.TameBoundary.mem_faceStructuredNbhds_of_subset`, `Polyfold.TameBoundary.inter_mem_faceStr` `Polyfold.TameBoundary.inter_mem_faceStructuredNbhdsAt`, `Polyfold.MPolyfold.proposition_2_7_7_2021` `Polyfold.TameBoundaryExample.isFaceStructuredOpen_quadrant`). An open $V$ is face-structured if every $y \in V$ lies in exactly $d(y)$ faces of $V$; $N_X$ is the set of nonempty face-structured $V$ with $d|V$ bounded, $N_{X,x}$ the set of $V \in N_X$ with $x \in V$ and $d \leq d(x)$ on $V$, and $N_X = \bigcup_x N_{X,x}$. For a tame $X$: every $N_{X,x}$ is nonempty (so $N_X$ covers $X$); a nonempty open subset of a member of $N_X$ is in $N_X$; nonempty intersections of members of $N_X$ are in $N_X$, and $N_{X,x}$ is closed under intersections. Every open subset of $[0,\infty)^n \oplus W$ is face-structured. *Book:* Definition 2.7.6, p. 79, and Proposition 2.7.7, p. 80, new in the 2021 edition. *Hidden hypotheses:* none.

**V2.82 (Definitions 2.7.8 and 2.7.9, the category $L_X$ and the boundary structure functor;** `Polyfold.TameBoundary.IsMor`, `Polyfold.TameBoundary.IsMor.trans`, `Polyfold.TameBoundary.LObj`, `Polyfold.TameBoundary.boundaryFunctor`, `Polyfold.TameBoundary.boundaryFunctor_map_injective`, `Polyfold.TameBoundary.boundaryFunctor_map_bijective`, `Polyfold.MPolyfold.LocalBoundaryCategory`, `Polyfold.MPolyfold.boundaryStructureFunctor`). The objects of $L_X$ are the pairs $(x, V)$ with $d(x) \geq 1$ and $V \in N_{X,x}$; there is a morphism $(x, V) \to (y, W)$, and then only one, iff $V \subseteq W$, $x$ lies in $d(x)$ faces of $W$ and each of them contains $y$; morphisms compose. The functor $F_X$ sends $(x, V)$ to the set of faces of $V$ through $x$ and a morphism to $\theta \mapsto \theta'$, the face of $W$ containing $\theta$; $F_X(\varphi)$ is injective, and bijective for a morphism $(x, V) \to (x, W)$. *Book:* Definitions 2.7.8, p. 80, and 2.7.9, p. 81, new in the 2021 edition, with the text's argument for the composition and the injection (2.43). *Hidden hypotheses:* none. *Deviation (restated):* $L_X$ is a preorder (a thin category) and $F_X$ takes values in types; $\theta'$ is defined as the union of the faces of $W$ containing $\theta$, which is the text's "the face of $W$ through $x$ and $y$ with $\theta \cap \theta'$ open in $\theta$" (`Polyfold.TameBoundary.faceExt_unique`).

**V2.83 (Definition 2.7.10, the set $bX$ and the map $B$;** `Polyfold.TameBoundary.BTriple`, `Polyfold.TameBoundary.BRel`, `Polyfold.TameBoundary.BRel.refl`, `Polyfold.TameBoundary.BRel.symm`, `Polyfold.TameBoundary.BRel.trans`, `Polyfold.TameBoundary.bSetoid`, `Polyfold.TameBoundary.BSpace`, `Polyfold.TameBoundary.bMap`, `Polyfold.TameBoundary.range_bMap`, `Polyfold.TameBoundary.encard_bMap_p` `Polyfold.MPolyfold.TameBoundarySpace`, `Polyfold.MPolyfold.tameBoundaryMap`, `Polyfold.MPolyfold.rang` `Polyfold.MPolyfold.encard_tameBoundaryMap_preimage`). An admissible triple is $(x, V, \theta)$ with $(x, V)$ an object of $L_X$ and $\theta$ a face of $V$ through $x$; $(x, V, \theta) \sim (x', V', \theta')$ iff $x = x'$ and some $V'' \in N_{X,x}$, $V'' \subseteq V \cap V'$, has a face $\theta''$ through $x$ sent to $\theta$ and to $\theta'$. This is an equivalence relation; $bX$ is the set of classes and $B[x, V, \theta] = x$. The image of $B$ is $\partial X = \{d \geq 1\}$, and $B^{-1}(x)$ has exactly $d(x)$ points. *Book:* Definition 2.7.10, p. 81, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (stronger):* the fibre count, the set-level content of Theorem 2.7.13 (4).

**V2.84 (Definition 2.7.11 and (2.44), the sets $U(x, V, \theta)$ and the maps $\Phi$;** `Polyfold.TameBoundary.uSet`, `Polyfold.TameBoundary.phi`, `Polyfold.TameBoundary.phi_injective`, `Polyfold.TameBoundary.reach`, `Polyfold.TameBoundary.IsCenteredAt`, `Polyfold.TameBoundary.isOpen_reach`, `Polyfold.TameBoundary.bMa` `Polyfold.TameBoundary.phi_bijective_of_isCenteredAt`, `Polyfold.TameBoundary.isCenteredAt_reach`, `Polyfold.TameBoundary.reach_mem_faceStructuredNbhdsAt`). $U(x, V, \theta)$ is the set of classes with a representative $(y, V', \theta')$ such that $y \in \theta$, there is a morphism $(y, V') \to (x, V)$, and it sends $\theta'$ to $\theta$; $\Phi_{(x,V,\theta)} : U(x, V, \theta) \to \theta$ is $[y, V', \theta'] \mapsto y$. $\Phi$ is injective, and its image is $\theta \cap V^x$, where $V^x = \{y \in V : \text{every face of } V \text{ through } y \text{ contains } x\}$ is open. If every face of $V$ contains $x$ ($V$ centred at $x$), $\Phi$ is a bijection onto $\theta$; $V^x$ is centred at $x$, and $V^x \in N_{X,x}$ for

$V \in N_{X,x}$. *Book:* Definition 2.7.11 and (2.44), p. 82, new in the 2021 edition: $\Phi_{(x,V,\theta)}$ is a canonical bijection onto $\theta$. *Hidden hypotheses:* that $V$ be small enough; the text says, for the topology of $bX$ (p. 85), that the main point is to take $V$ sufficiently small. *Deviation (narrower):* the bijection is proved for $V$ centred at $x$; for every $V \in N_{X,x}$ it does not hold as we read it (V2.85). The centred neighbourhoods are cofinal in $N_{X,x}$, which is what "$V$ sufficiently small" provides.

**V2.85 ((2.44) for every $V \in N_{X,x}$: an example in $[0,\infty)^3$; `Polyfold.TameBoundaryExample.dq`, `Polyfold.TameBoundaryExample.isCornerIndex_dq`, `Polyfold.TameBoundaryExample.dq_eq`, `Polyfold.TameBoundaryExample.part`, `Polyfold.TameBoundaryExample.dq_eq_one_iff`, `Polyfold.TameBoundaryExample.subset_part`, `Polyfold.TameBoundaryExample.faceAt_subset_coord_eq_ze` `Polyfold.TameBoundaryExample.exists_part_mem_faceAt`, `Polyfold.TameBoundaryExample.slab`, `Polyfold.TameBoundaryExample.slab_mem_faceStructuredNbhdsAt`, `Polyfold.TameBoundaryExample.slabFa` `Polyfold.TameBoundaryExample.counterTriple`, `Polyfold.TameBoundaryExample.exists_face_not_mem`, `Polyfold.TameBoundaryExample.pt_010_not_mem_image`, `Polyfold.TameBoundaryExample.phi_not_surjecti` `Polyfold.TameBoundaryExample.not_forall_bijective_phi_2021`).** In the tame M-polyfold $X = [0,\infty)^3 \oplus W$ ($W$ any sc-Banach space), let $x = (0,0,1)$, $V = \{p_0 < 1/2,\ 1/2 < p_1 + p_2 < 3/2\}$ (a convex open slab around the segment from $x$ to $(0,1,0)$), and $\theta$ the face of $V$ through that segment. Then $V \in N_{X,x}$, $d_X(x) = 2$, $(x,V,\theta)$ is an admissible triple, $(0,1,0) \in \theta$ also lies on the face of $V$ in $\{p_2 = 0\}$, which misses $x$, and so $(0,1,0)$ is not in the image of $\Phi_{(x,V,\theta)}$: $\Phi$ is not onto $\theta$, and not every $\Phi_{(x,V,\theta)}$ is a bijection. *Book:* Definition 2.7.11, (2.44), p. 82, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (not in the book):* the example shows that (2.44) does not hold as we read it for every $V \in N_{X,x}$, and that some restriction such as "$V$ centred at $x$" (V2.84) is needed. *Proof:* $d_X(p)$ is the number of vanishing half-line coordinates, a face of an open $V$ through a point with only $p_k = 0$ lies in $\{p_k = 0\}$.

**V2.86 (Lemma 2.7.15, the topology of $bX$; `Polyfold.TameBoundary.IsBOpen`, `Polyfold.TameBoundary.BTriple.` `Polyfold.TameBoundary.BRel.exists_common`, `Polyfold.TameBoundary.eq_of_mem_uSet`, `Polyfold.TameBoundary.isBOpen_iff`, `Polyfold.TameBoundary.bTopology`, `Polyfold.TameBoundary.isOpen_` `Polyfold.TameBoundary.isOpen_iff_forall_uSet`, `Polyfold.TameBoundary.lemma_2_7_15`, `Polyfold.MPolyfold.lemma_2_7_15_2021`).** A set $\mathcal{U} \subseteq bX$ is in $T_{bX}$ if every class $p \in \mathcal{U}$ has a representative $(x,V,\theta)$ and a set $O \ni p$, $O \subseteq \mathcal{U} \cap U(x,V,\theta)$, with $\Phi_{(x,V,\theta)}(O)$ open in $\theta$; equivalently, if every $p \in \mathcal{U}$ has a representative with $U(x,V,\theta) \subseteq \mathcal{U}$. These sets are the open sets of a topology. *Book:* the description of $T_{bX}$ in §2.7.4.3 and Lemma 2.7.15, p. 85, new in the 2021 edition. *Hidden hypotheses:* $p \in O$, without which $O = \emptyset$ would make every set open. *Deviation (explicit):* the first description is read with $O$ containing the class. *Proof:* from the second description and a common refinement of equivalent triples, without the sc-smoothness of the transition maps that the text's proof of the intersection axiom uses.

**V2.87 (Proposition 2.7.12, topological part; `Polyfold.TameBoundary.isOpen_uSet`, `Polyfold.TameBoundary.is` `Polyfold.TameBoundary.continuous_bMap`, `Polyfold.TameBoundary.exists_nbhd_faceExt_inter_subset`, `Polyfold.TameBoundary.exists_isOpen_bMap_image`, `Polyfold.TameBoundary.isOpenEmbedding_phi`, `Polyfold.TameBoundary.phiHomeomorph`, `Polyfold.TameBoundary.proposition_2_7_12`, `Polyfold.MPolyfold.proposition_2_7_12_2021`).** The sets $U(x,V,\theta)$ are open and form a basis of $T_{bX}$; $B$ is continuous; every $\Phi_{(x,V,\theta)}$ is an open embedding into $\theta$, and a homeomorphism onto $\theta$ for $V$ centred at $x$. For two triples, the image of $U(x,V,\theta) \cap U(x',V',\theta')$ is open in $\theta$ and in $\theta'$, and the transition map $\Phi_{(x',V',\theta')} \circ \Phi^{-1}_{(x,V,\theta)}$ is the identity on underlying points of $X$. *Book:* Proposition 2.7.12, p. 83, new in the 2021 edition: the images are open and the transition map is an sc-diffeomorphism. *Hidden hypotheses:* none. *Deviation (narrower):* the sc-diffeomorphism statement is not part of this item: for representatives with $V$ centred at $x$ it is proved with the

M-polyfold structure of $bX$ (V2.103), and for every representative in V2.104 (gap G.78, closed).

**V2.88 (Proposition 2.7.16; `Polyfold.TameBoundary.tripleOf`, `Polyfold.TameBoundary.exists_mem_uSet_trip` `Polyfold.TameBoundary.metrizableSpace_preimage`, `Polyfold.TameBoundary.proposition_2_7_16`, `Polyfold.MPolyfold.proposition_2_7_16_2021`).** For $d$ with the corner axioms on a Hausdorff $X$, $bX$ is Hausdorff; on a metrizable $X$, $bX$ is metrizable. For a tame M-polyfold, $T_{bX}$ is Hausdorff and metrizable. *Book:* Proposition 2.7.16, pp. 85–87, new in the 2021 edition. *Hidden hypotheses:* none. *Proof:* not the text's paracompactness argument (a locally finite refinement built from one of the cover $(V_x)$ of $\partial X$): the pull-back form of Smirnov's theorem (V2.62) along $B$, each point having an open neighbourhood $B^{-1}(W) \cap \bigcup_\theta U(x, W, \theta)$ embedded into the metrizable $X \times \mathbb{R}^{\text{faces}}$ by $B$ and the indicators of the clopen sets $U(x, W, \theta)$.

**V2.89 (Theorem 2.7.13, topological part, and Definition 2.7.14; `Polyfold.TameBoundary.eq_of_isOpen_uSet`, `Polyfold.TameBoundary.theorem_2_7_13_topology`, `Polyfold.MPolyfold.theorem_2_7_13_2021`, `Polyfold.MPolyfold.tameBoundary`).** For a tame M-polyfold $X$: (1) $bX$ carries the metrizable topology $T_{bX}$; every $U(x, V, \theta)$ is open and every $\Phi_{(x,V,\theta)}$ is an open embedding into $\theta$; $T_{bX}$ is the only topology in which every $U(x, V, \theta)$ is open and every $\Phi$ maps open sets to relatively open subsets of $\theta$; $B : bX \to X$ is continuous, with image $\partial X$, and every $x$ has exactly $d_X(x)$ preimages. The tame boundary is $B$ as a continuous map. *Book:* Theorem 2.7.13, pp. 83–84, and Definition 2.7.14, p. 84, new in the 2021 edition: (1) the metrizable topology, (2) a natural tame M-polyfold structure on $bX$, (3) $B$ sc-smooth, (4) $d_X(B(p)) - 1 = d_{bX}(p)$, characterized by: every $U(x, V, \theta)$ open and every $\Phi$ a homeomorphism and an sc-diffeomorphism onto $\theta$. *Hidden hypotheses:* none. *Deviation (narrower):* this item is (1) and the topological half of the characterization; "homeomorphism onto $\theta$" is proved for $V$ centred at $x$ (V2.84, V2.87), the open embedding for every $V$. Parts (2)–(4) beyond the fibre count are V2.102 and V2.106; the sc-diffeomorphism half of the characterization is V2.103 for centred representatives and V2.104 for every representative, with image $\theta \cap V^x$ (V2.105) (gap G.78, closed).

**V2.90 (the tame quadrant as a witness; `Polyfold.MPolyfold.quadrantPolyfold`, `Polyfold.MPolyfold.quadran` `Polyfold.MPolyfold.quadrantPolyfold_encard_preimage`, `Polyfold.MPolyfold.quadrantPolyfold_tameBou` `Polyfold.MPolyfold.quadrantPolyfold_univ`).** The quadrant $[0, \infty)^n \oplus W$ is a tame M-polyfold; its corner 0 has exactly $n$ preimages under $B$, so $bX \neq \emptyset$ for $n \geq 1$; and $Y = X$ has a tame induced structure and is a tame sub-M-polyfold. *Book:* not stated. *Hidden hypotheses:* none. *Deviation (not in the book):* satisfiability of the hypotheses of V2.78–V2.89.

**V2.91 (Remark 2.1.6, strictness: the K-topology on $\mathbb{R}$; `Polyfold.remark_2_1_6_strict_2021`, `Polyfold.isLocallyCompleteRel_of_forall_isCompletelyMetrizable`, `Polyfold.isCompletelyMetrizableS` `Polyfold.kSeq`, `Polyfold.kTop`, `Polyfold.RealKTop`, `Polyfold.RealKTop.not_regularSpace`, `Polyfold.RealKTop.not_isLocallyComplete`, `Polyfold.RealKTop.isLocallyCompleteRel`).** A space in which every point has an open, completely metrizable neighbourhood satisfies the modified definition of Remark 2.1.6 (with $V = W$). A space embedded in $\mathbb{R}$ onto the complement of a sequence is completely metrizable. Let $\mathbb{R}_K$ be $\mathbb{R}$ with the K-topology, generated by the usual open sets and $\mathbb{R} \setminus K$, $K = \{1/(n+1) : n \in \mathbb{N}\}$. It is Hausdorff; it satisfies the modified definition, the points outside $K$ lying in $\mathbb{R} \setminus K$ and those of $K$ in $\mathbb{R} \setminus \{0\}$, both with the usual subspace topology; and it is not regular, so not locally complete (V2.67). So some Hausdorff space satisfies the modified definition and not Definition 2.1.5. *Book:* Remark 2.1.6, p. 23, new in the 2021 edition: the modified definition would be weaker. *Hidden hypotheses:* none. *Deviation (not in the book):* the example; with V2.68 the remark holds as we read it, and "weaker" is strict (gap G.76, closed). *Proof:* every closed neighbourhood of 0 in $\mathbb{R}_K$ meets $K$, through irrational points near $1/(n+1)$.

The complement of a sequence $(a_n)$ embeds as a closed set into $\mathbb{R} \times \mathbb{R}^{\mathbb{N}}$ by $y \mapsto (y, ((y-a_n)^{-1})_n)$, in place of the theorem that $G_\delta$ subsets of complete metric spaces are completely metrizable, which Mathlib lacks.

**V2.92 (Definition 2.1.16 (preprint 2.9), bundle retractions and bundle maps; `Polyfold.ScRetraction.IsBundleR` `Polyfold.ScRetraction.tangentRetraction_isBundleRetractionOver`, `Polyfold.IsBundleMapOn`, `Polyfold.IsBundleMapOn.fst_eq`, `Polyfold.IsScSmoothRetractMap.retractTangentMap_isBundleMapOn`).** A retraction $R$ of $E \oplus F$ (V2.1) is a bundle retraction covering the retraction $r$ of $E$ if its domain is $U \oplus F$, $U$ the domain of $r$, and $R(u,v) = (r(u), \rho(u)v)$ there, with $\rho(u)$ a continuous linear map of $F_0$. A map $\Phi$ from $E \oplus F$ to $E' \oplus F'$ is a bundle map on $B$ covering $a$ if $\Phi(u,v) = (a(u), \phi(u)v)$ on $B$ with $\phi(u)$ continuous linear; then $p' \circ \Phi = a \circ p$ on $B$. The tangent retraction $Tr$ is a bundle retraction covering the level-1 retraction $r^1$, and for an sc-smooth $f$ between retracts $Tf$ is a bundle map on $TO$ covering the level-1 map $f^1$. *Book:* Definition 2.1.16, p. 29 (preprint 2.9, p. 22). *Hidden hypotheses:* none. *Deviation (restated):* the definition fixes the shape of the maps; sc-smoothness and $R \circ R = R$ are part of the data (a retraction of V2.1, an sc-smooth map of V2.6), and $\rho(u)$ is continuous on level 0, which the text's linear $\rho(u,\cdot)$ is as a restriction of the continuous $R$. "$p' \circ \Phi = p$" is read as $p' \circ \Phi = a \circ p$ (possible misprint; Findings, Chapter 2, item 76). The categories $\mathcal{R}$, $\mathcal{BR}$ and the forgetful functor are not formalized; they name these objects and maps and state nothing further (gap G.5, formalized since apart from them).

**V2.93 (the levels of $X^1$, gap G.6; `Polyfold.MPolyfold.levelLift_transition`, `Polyfold.MPolyfold.levelLift` `Polyfold.MPolyfold.shiftLevelMap`, `Polyfold.MPolyfold.shiftLevelInv`, `Polyfold.MPolyfold.continuous` `Polyfold.MPolyfold.continuous_shiftLevelInv`, `Polyfold.MPolyfold.shiftLevelHomeomorph`, `Polyfold.MPolyfold.shift_level_eq`).** For every $m$, the identity of the underlying points of $X$ is a homeomorphism $(X^1)_m \cong X_{m+1}$; in a chart $\varphi$ of $X$, the level-$m$ lift of the level chart $\varphi^1$ of $X^1$ is the level-$(m+1)$ lift of $\varphi$ ($(E^1)_m = E_{m+1}$), and level lifts in two charts differ by the level map of the transition map. So the iterate $X^m = (X^1)^1 \cdots$ is, for each $m$, the space $X_m$ with its topology and the charts $(V_m, \varphi, (O_m, C_m, E_m))$, by composing these homeomorphisms. *Book:* the paragraph after Definition 2.3.8, p. 37 (preprint after Definition 2.14, p. 25): these charts form an atlas of $X_m$, which with it is the M-polyfold $X^m$. *Hidden hypotheses:* none. *Deviation (restated):* $X^m$ is the iterate, identified with $X_m$ for each fixed $m$; one homeomorphism for the family indexed by $m$ is not stated. This closes gap G.6 (V2.16).

**V2.94 (Theorem 2.3.16 as a bundle statement; `Polyfold.theorem_2_3_16_2021`, `Polyfold.tangentMap_levelMap` `Polyfold.toBase_tangentMap_levelMap_one`, `Polyfold.retractTangent_shift_of_isScRetract`, `Polyfold.toBase_retractTangentMap_shift`).** In a local model, for an sc¹ map $f$ on $U$, the tangent map of $f^1 : U^1 \to F^1$ is the level-1 map of $Tf$ on $T(U^1) = (TU)^1$; $T(O^1) = (TO)^1$ for a retract $O$; and $T(f^1)$ is the level-1 map of $Tf$ for an sc-smooth map between retracts. For an M-polyfold $X$: the tangent charts $T\varphi$ belong to the structure of $TX$; $p : TX \to X^1$ is sc-smooth; in a tangent chart, $p$ is the first projection onto the level chart $\varphi^1$ of $X^1$; for every retraction $r$ onto a model $O$, the model $(TO, TC, TE)$ is a bundle retract ($Tr$ covers $r^1$, V2.92); and the transition map of two tangent charts is a bundle map on $T(\varphi(V \cap V'))$ covering the level-1 map of the transition map of the charts. *Book:* Theorem 2.3.16, p. 40 (2017: running text after Proposition 2.5; preprint: after Proposition 2.19, p. 27): the construction $X \mapsto (p : TX \to X^1)$; the text before it says that $p$ is locally built on the bundle retractions $TO \to O^1$ and that the transition maps are sc-smooth bundle maps. *Hidden hypotheses:* the paracompactness of $TX$ and $X_1$ (instance arguments, supplied by V2.20 and V2.17). *Deviation (restated):* "sc-smooth M-polyfold bundle" is read through the models and the transition maps, with $TX$ the M-polyfold of V2.21; the fibres $T_xX$ carry no linear structure of their own (gap G.7).

**V2.95 (Proposition 2.3.17 for $i = 1$, $T(X^1) \equiv (TX)^1$; `Polyfold.MPolyfold.someChart`, `Polyfold.MPolyfold.TangentPre.lower`, `Polyfold.MPolyfold.TangentPre.transport_lower`, `Polyfold.MPolyfold.tangentLower`, `Polyfold.MPolyfold.tangentShiftMap`, `Polyfold.MPolyfold.tangentSh` `Polyfold.MPolyfold.tangentShiftHomeomorph`, `Polyfold.MPolyfold.tangentProjShift`, `Polyfold.MPolyfold.tangentShiftEquiv_isScDiffeoOn`, `Polyfold.MPolyfold.tangentChartMap_tangentLow` `Polyfold.MPolyfold.levelChart_tangentShiftMap`, `Polyfold.proposition_2_3_17_2021`).** The inclusion $T(X^1) \to TX$, defined in a chosen chart and independent of it, takes values in $(TX)_1$ and gives a bijection $\Phi : T(X^1) \to (TX)^1$ with: (1) $\Phi$ is an sc-diffeomorphism of the M-polyfolds $T(X^1)$ and $(TX)^1$, in particular a homeomorphism; (2) $p^1 \circ \Phi = p_{X^1}$, where $p^1 : (TX)^1 \to (X^1)^1$ is the level-1 map of $p : TX \to X^1$; (3) $p^1$ is sc-smooth; (4) in every chart $\psi$ of $X$ around the base point, $T\psi(\Phi(\alpha))$ is the inclusion of $T(\psi^1)(\alpha)$, so $\Phi$ does not depend on the chart used to define it; (5) in the charts $T(\varphi^1)$ of $T(X^1)$ and $(T\varphi)^1$ of $(TX)^1$, $\Phi$ is the identity of $T(O^1) = (TO)^1$, in particular linear on the fibres. *Book:* Proposition 2.3.17, p. 42, new in the 2021 edition, with pp. 40–42 ($(TO)^i = T(O^i)$ in a local model, compatible with the tangent charts): natural identifications of $(TX)^i$ with $T(X^i)$ and of the bundles $(TX)^i \to (X^1)^i$ and $T(X^i) \to (X^i)^1$, covering $(X^1)^i \equiv (X^i)^1$. *Hidden hypotheses:* none. *Deviation (restated):* "naturally identified, also as bundles" is read as (1)–(5); for $i = 1$ the identification $(X^1)^1 \equiv (X^1)^1$ is the identity.

**V2.96 (Proposition 2.3.17 for $i = 2$, and raising the index of an sc-diffeomorphism; `Polyfold.MPolyfold.levelOneEquiv`, `Polyfold.MPolyfold.levelOneEquiv_isScDiffeoOn`, `Polyfold.MPolyfold.tangentShiftEquivTwo`, `Polyfold.MPolyfold.tangentProjShiftTwo`, `Polyfold.proposition_2_3_17_two_2021`).** A bijection $e : Y \to Z$ of M-polyfolds that is sc-smooth in both directions induces an sc-diffeomorphism $e^1 : Y^1 \to Z^1$. Composing $\Phi$ of V2.95 for $X^1$ with the index raise of $\Phi^{-1}$ for $X$ gives an sc-diffeomorphism $\Phi_2 : T(X^2) \to (TX)^2$, $X^2 = (X^1)^1$, $(TX)^2 = ((TX)^1)^1$, covering the identity of $(X^1)^2 = (X^2)^1$: $p^2 \circ \Phi_2 = p_{X^2}$. *Book:* Proposition 2.3.17, p. 42, for $i = 2$. *Hidden hypotheses:* none. *Deviation (narrower):* $i = 2$; every $i$ is V2.97.

**V2.97 (Proposition 2.3.17 for every $i$; `Polyfold.MPolyfold.ScEquiv`, `Polyfold.MPolyfold.ScEquiv.shift`, `Polyfold.MPolyfold.ScMap`, `Polyfold.MPolyfold.tangentShiftScEquiv`, `Polyfold.MPolyfold.tangentProjS` `Polyfold.MPolyfold.tangentLower_tangentMap`, `Polyfold.BundledMPolyfold`, `Polyfold.BundledMPolyfold.` `Polyfold.BundledMPolyfold.lift`, `Polyfold.BundledMPolyfold.tangent`, `Polyfold.BundledMPolyfold.tang` `Polyfold.BundledMPolyfold.shiftLiftEquiv`, `Polyfold.BundledMPolyfold.tangentProjLift`, `Polyfold.BundledMPolyfold.incl`, `Polyfold.BundledMPolyfold.tangentIncl`, `Polyfold.BundledMPolyfold.` `Polyfold.BundledMPolyfold.tangentLiftEquiv_unique`, `Polyfold.BundledMPolyfold.tangentLiftEquiv_na` `Polyfold.proposition_2_3_17_all_2021`).** Bundle an M-polyfold $X$ with its space, so that the lifts $X^0 = X$, $X^{i+1} = (X^i)^1$ form a family indexed by $i$. For every $i$: (1) $\Phi_i : T(X^i) \to (TX)^i$, with $\Phi_0 = \mathrm{id}$ and $\Phi_{i+1} = (\Phi_i)^1 \circ \Phi$, and $J_i : (X^1)^i \to (X^i)^1$ are sc-diffeomorphisms; (2) $p^i : (TX)^i \to (X^1)^i$, the $i$-th index raise of $p$, and $p : T(X^i) \to (X^i)^1$ are sc-smooth; (3) $J_i \circ p^i \circ \Phi_i = p$; (4) $\Phi_i$ commutes with the inclusions $T(X^i) \to TX$ and $(TX)^i \to TX$, (5) and with the inclusions $T(X^{i+1}) \to T(X^i)$ and $(TX)^{i+1} \to (TX)^i$ of the two towers of the text; (6) $J_i$ is the identity on the points of $X$; (7) $\Phi_i$ is the only map with (4). $\Phi_1$ is the $\Phi$ of V2.95, and $\Phi_i \circ T(f^i) = (Tf)^i \circ \Phi_i$ for every sc-smooth $f : X \to Y$, from the naturality $\iota \circ T(f^1) = Tf \circ \iota$ of the inclusion $\iota : T(X^1) \to TX$. A bundled M-polyfold exists (the half-line). *Book:* Proposition 2.3.17, p. 42, with the two towers of p. 41. *Hidden hypotheses:* none. *Deviation (restated):* "natural identification" is read as (1)–(7) and the naturality in $X$. Linearity on the fibres is not part of the statement, since $T_xX$ has no linear structure (gap G.7); $\Phi_i$ is a composite of index raises of maps that are the identity in charts (V2.95 (5)).

**V2.98 (Remark 2.4.2, the diagonal half-line in $[0,\infty)^3$; `Polyfold.ScOperator.ofConst`, `Polyfold.diagonalAverage`, `Polyfold.diagonalRetraction`, `Polyfold.mem_diagonalRetraction_retract_if` `Polyfold.diagonalModel`, `Polyfold.diagonalPolyfold`, `Polyfold.diagonalEmbed`, `Polyfold.diagonalProjec` `Polyfold.diagonalEmbed_isScDiffeoOn`, `Polyfold.degIndex_diagonal_quadrant`, `Polyfold.diagonalPolyfo` `Polyfold.diagonalHalfLineChart`, `Polyfold.diagonalHalfLineChart_spec`, `Polyfold.remark_2_4_2_2021`).** In $C = [0,\infty)^3 \oplus W$, $W$ any sc-Banach space, replacing the three coordinates by their average is an sc-smooth retraction onto $\Delta \oplus W$, $\Delta = \{(s,s,s) : s \geq 0\}$. With the M-polyfold structure of this retract: (1) $(s,w) \mapsto ((s,s,s),w)$ is an sc-diffeomorphism from the half-line M-polyfold $[0,\infty) \oplus W$ onto $\Delta \oplus W$; (2) $d_C((s,s,s),w)$ is 3 for $s = 0$ and 0 otherwise, while $d_\Delta$ is 1 for $s = 0$ and 0 otherwise, and in the defining chart (the identity of the retract) the chart index is $d_C$; (3) one chart, defined on all of $\Delta \oplus W$, has as model the relatively open subset $[0,\infty) \oplus W$ of a partial quadrant, and in it the chart index is $d_\Delta$. *Book:* Remark 2.4.2, pp. 44–45, new in the 2021 edition, for $\Delta \subseteq [0,\infty)^3 \subseteq \mathbb{R}^3$. *Hidden hypotheses:* none. *Deviation (stronger):* any $W$ ($W = 0$ is the text's case); "classically smooth" is read as sc-smooth for the constant sc-structure, and the retraction, which the text does not give, is the orthogonal projection onto the diagonal. *Proof:* $d_\Delta$ by Proposition 2.4.4 (preprint 2.24) (V2.28) applied to the sc-diffeomorphism (1).

**V2.99 (Definition 2.5.19 (3) does not depend on the chart; `Polyfold.MPolyfold.mem_smoothSet_iff_of_isChart` `Polyfold.MPolyfold.TameSubData.hasReducedComplementAt_of_isChart_2021`, `Polyfold.MPolyfold.TameSu`** A point of an M-polyfold is smooth iff its image in one chart is a smooth point of the model. Let $Y$ have tame sub-M-polyfold data $(V,r)$ (V2.78), condition (3) being asked in one chart around each smooth point, for one presentation of its partial quadrant. Then at every smooth $y \in Y \cap V$ condition (3) holds in every chart $\varphi \ni y$, for every presentation $T$ of the partial quadrant of $\varphi$ and every level-1 representative $z$ of $\varphi(y)$, for the chart image $\varphi(W' \cap Y)$ of any sub-M-polyfold retraction $r'$ onto $Y$ near $y$, in particular for $r' = r$. *Book:* Definition 2.5.19 (3), p. 63, new in the 2021 edition, stated on $T_yX$. *Hidden hypotheses:* none. *Deviation (restated):* (3) remains a statement in charts, since $T_yX$ has no linear structure here (gap G.7); the chart, the presentation and the retraction do not matter (gap G.77, closed). *Proof:* by Chapter 4's chart change for the general-position condition (`Polyfold.MPolyfold.isGeneralPositionRetractAt_chart_change`, V4.45), which is (3) word for word.

**V2.100 (a tame sub-M-polyfold is tame; `Polyfold.ScBanach.levelSub_sup_of_le_of_isScSplitting`, `Polyfold.ScBanach.isScSplitting_sup_of_complementIn`, `Polyfold.MPolyfold.TameSubData.isTameRetrac` `Polyfold.MPolyfold.IsTameSubMPolyfold.hasTameInducedStructure_2021`).** If $E = M \oplus B$ is an sc-splitting and $N, A \subseteq M$ are sc-subspaces with $M_m = N_m \oplus A_m$ on every level, then $E = N \oplus (A \oplus B)$ is an sc-splitting. For tame sub-M-polyfold data $(V,r)$ and a chart $\varphi$ of $X$ with a tame model, the chart image $\varphi(W \cap Y)$ of $Y$ is a tame retract, of the retraction $(\varphi r \varphi^{-1}) \circ t$ with $t$ the tame retraction of the model. Hence a tame sub-M-polyfold $Y$ of a tame $X$ has a tame induced structure (Definition 2.5.18): the sub-charts built on tame charts of $X$ have tame models. *Book:* the text after Definition 2.5.19, p. 63, new in the 2021 edition ("not difficult to see"). *Hidden hypotheses:* none. *Proof:* the first condition of tameness from (2) of Definition 2.5.19 and $d_X = d_C \circ \varphi$ in a tame chart; the sc-complement of the tangent space of $\varphi(W \cap Y)$ inside $E_x$ is $A \oplus B$, with $A$ from (3) in the chart $\varphi$ (V2.99) and $B$ the complement of $T_xO$ given by the tameness of $t$.

**V2.101 (the examples $A$ and $B$ in $[0,\infty)^2 \times \mathbb{R}$; `Polyfold.TameSubExample.fixSet`, `Polyfold.TameSubExample.is` `Polyfold.TameSubExample.isTameSubMPolyfold_fixSet`, `Polyfold.TameSubExample.setA`, `Polyfold.TameSubExample.setB`, `Polyfold.TameSubExample.isTameSubMPolyfold_setA_2021`, `Polyfold.TameSubExample.hasTameInducedStructure_setA_2021`, `Polyfold.TameSubExample.isSubMPolyfol` `Polyfold.TameSubExample.not_mem_tameSubData_setB_2021`, `Polyfold.TameSubExample.not_isTameSubMPol`**

`Polyfold.TameSubExample.halfLineToB_isScDiffeoOn_2021`, `Polyfold.TameSubExample.hasTameInducedStr`
`Polyfold.TameSubExample.examples_A_B_2021`). In $X = [0,\infty)^n \oplus W$, $W$ any sc-Banach space, the fixed set of an idempotent sc-operator $L$ mapping the quadrant into itself is a sub-M-polyfold, and a tame sub-M-polyfold if $L$ does not change the $\mathbb{R}^n$-component. For $n = 2$: $A = \{(x, y, 0)\}$ (with $L(a, w) = (a, 0)$) is a tame sub-M-polyfold with tame induced structure; $B = \{(x, x, 0)\}$ is a sub-M-polyfold, but no retraction onto $B$ with conditions (1) and (2) of Definition 2.5.19 is defined near the corner 0, so $B$ is not a tame sub-M-polyfold; $t \mapsto (t, t, 0)$ is an sc-diffeomorphism from the half-line $[0,\infty)$ onto $B$ with its induced structure, which is tame. For $W = \mathbb{R}$ these are the text's statements. *Book:* the text after Definition 2.5.19, p. 63, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (stronger):* any $W$. *Proof:* for small $t > 0$ the point $(t, 0, 0)$ of $V$ has $d_X = 1$, while $d_X$ takes only the values 0 and 2 on $B$; (3) is not needed.

**V2.102 (Theorem 2.7.13 (2)–(4), the tame M-polyfold $bX$;** `Polyfold.MPolyfold.exists_tame_chart_subset`, `Polyfold.MPolyfold.exists_localFace_of_faceIn`, `Polyfold.MPolyfold.BChartData`, `Polyfold.MPolyfold.BChartData.chart`, `Polyfold.MPolyfold.BChartData.chart_transition_isScSmooth`, `Polyfold.MPolyfold.BChartData.chart_compatible`, `Polyfold.MPolyfold.exists_bChartData`, `Polyfold.MPolyfold.paracompactSpace_tameBoundarySpace`, `Polyfold.MPolyfold.bAtlas`, `Polyfold.MPolyfold.tameBoundaryPolyfold`, `Polyfold.MPolyfold.tameBoundaryPolyfold_isTame`, `Polyfold.MPolyfold.tameBoundaryMap_isScSmooth`, `Polyfold.MPolyfold.degIndex_tameBoundaryMap`, `Polyfold.MPolyfold.degIndex_tameBoundaryMap_sub_one`). Let $X$ be tame. A chart datum of $bX$ consists of a triple $(x, V, \theta)$ with $V$ centred at $x$ (V2.84), a chart $\varphi$ of $X$, an open $V_1 \subseteq \operatorname{dom}\varphi$, an sc-isomorphism $T$ of the model space of $\varphi$ with $\mathbb{R}^{k+1} \oplus W$ and a coordinate $i$ vanishing on $T\varphi(\theta \cap V_1)$, such that the image of $\varphi(\theta \cap V_1)$ under the face projection $\pi$ (Proposition 2.5.15 (preprint 2.43)) is a tame retract of $[0,\infty)^k \oplus W$ and $d_X(y) = d(\pi\varphi(y)) + 1$ on $\theta \cap V_1$. Its chart is $p \mapsto \pi(\varphi(B(p)))$ on $U(x, V, \theta) \cap B^{-1}(V_1)$; two such charts are compatible, with transition map $\pi' \circ (\varphi' \circ \varphi^{-1}) \circ \iota$ ($\iota$ the face inclusion); every point of $\theta$ lies in the $V_1$ of a chart datum with the triple $(x, V, \theta)$, so the charts cover $bX$. With the topology $T_{bX}$, paracompact by V2.88, they make $bX$ a tame M-polyfold (2); $B : bX \to X$, which is $\iota$ in these charts, is sc-smooth (3); and $d_X(B(p)) = d_{bX}(p) + 1$, hence $d_X(B(p)) - 1 = d_{bX}(p)$ (4). *Book:* Theorem 2.7.13 (2)–(4), pp. 83–84, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation:* none. *Proof:* the charts are the charts $\pi \circ \varphi$ of the face $\theta$ composed with $\Phi_{(x,V,\theta)}$; the local faces of $\theta$ in a tame chart inside $V$ are found as in Theorem 2.5.17 (preprint 2.45).

**V2.103 (the characterization after Theorem 2.7.13, and Proposition 2.7.12, for centred representatives;** `Polyfold.TameBoundary.BSpace.exists_centred_rep`, `Polyfold.MPolyfold.isSubMPolyfold_of_isFac`
`Polyfold.MPolyfold.facePolyfold`, `Polyfold.MPolyfold.phiTot`, `Polyfold.MPolyfold.phi_isScDiffeoOn`, `Polyfold.MPolyfold.tameBoundaryPolyfold_unique`, `Polyfold.MPolyfold.tameBoundaryPolyfold_satisfie`
`Polyfold.MPolyfold.proposition_2_7_12_scDiffeo_2021`, `Polyfold.MPolyfold.theorem_2_7_13_structure`
Let $X$ be tame. Every class of $bX$ has a representative $(x, V, \theta)$ with $V$ centred at $x$. A face $\theta$ of a face-structured open $V$ is a sub-M-polyfold of $X$ and carries the induced structure. For $V$ centred at $x$, $\Phi_{(x,V,\theta)}$ is an sc-diffeomorphism of the open set $U(x, V, \theta)$ onto $\theta$, with inverse $\Phi^{-1}$; an M-polyfold structure on $bX$ (with the topology $T_{bX}$) for which all these $\Phi$ are sc-diffeomorphisms is equivalent to that of V2.102, which has the property. For two triples with centred neighbourhoods, the transition map $\Phi_{(x',V',\theta')} \circ \Phi^{-1}_{(x,V,\theta)}$ is an sc-diffeomorphism between the open subsets $\Phi_{(x,V,\theta)}(U \cap U')$ of $\theta$ and $\Phi_{(x',V',\theta')}(U \cap U')$ of $\theta'$, and the identity on points of $X$. One statement collects (2)–(4) with this characterization and the uniqueness. *Book:* Proposition 2.7.12, p. 83, and the characterization after Theorem 2.7.13, p. 84, new in the 2021 edition: for every class and every representative, $U(x, V, \theta)$ is open and $\Phi_{(x,V,\theta)}$ a homeomorphism and an sc-diffeomorphism

onto $\theta$, with the structure induced from $V$. *Hidden hypotheses:* none. *Deviation (narrower):* centred representatives; every representative is V2.104. "The structure induced from $V$" is read as the sub-M-polyfold structure of $\theta$ in $X$, and "uniquely characterized" as: every structure with the property has an equivalent atlas.

**V2.104 (Proposition 2.7.12 and the characterization for every representative;** `Polyfold.TameBoundary.BTriple.` `Polyfold.TameBoundary.phiInv`, `Polyfold.TameBoundary.phiInv_spec`, `Polyfold.MPolyfold.reachSet`, `Polyfold.MPolyfold.isOpen_reachSet`, `Polyfold.MPolyfold.reachSet_eq_univ_of_isCenteredAt`, `Polyfold.MPolyfold.phiTot_isScSmoothOn`, `Polyfold.MPolyfold.phiInv_isScSmoothOn`, `Polyfold.MPolyfold.phi_isScDiffeoOn_nonCentred_2021`, `Polyfold.MPolyfold.tameBoundaryPolyfold_uni` `Polyfold.MPolyfold.tameBoundaryPolyfold_satisfies_allReps_2021`, `Polyfold.MPolyfold.proposition_2`
Every triple $s$ has an equivalent triple $s'$ centred at its point with $U(s') \subseteq U(s)$. For every representative $(x, V, \theta)$, $\Phi_{(x,V,\theta)}$ is sc-smooth on $U(x, V, \theta)$, its inverse is sc-smooth on the image $\theta \cap V^x$, and $\Phi$ is an sc-diffeomorphism of $U(x, V, \theta)$ onto the open subset $\theta \cap V^x$ of $\theta$, which is $\theta$ when $V$ is centred at $x$. An M-polyfold structure on $bX$ with this property for every representative is equivalent to that of V2.102, which has it. For every pair of representatives, $\Phi_{(x,V,\theta)}(U \cap U')$ and $\Phi_{(x',V',\theta')}(U \cap U')$ are open in $\theta$ and $\theta'$, and the transition map between them is an sc-diffeomorphism, with inverse $\Phi_{(x,V,\theta)} \circ \Phi^{-1}_{(x',V',\theta')}$, the identity on points of $X$. *Book:* as V2.103. *Hidden hypotheses:* none. *Deviation (narrower):* Proposition 2.7.12 holds as printed for all representatives; in the characterization the image of $\Phi$ is $\theta \cap V^x$, and "onto $\theta$" does not hold as we read it for every representative (V2.105) (gap G.78, closed). *Proof:* near each point, $\Phi^{-1}$ is $\Phi^{-1}_{s'}$ for a centred refinement $s'$, composed with the inclusion of an open piece of $\theta$ into the face of $s'$; the sub-M-polyfold calculus (a map into a sub-M-polyfold is sc-smooth if it is sc-smooth into $X$) replaces the text's open sc-embeddings of faces.

**V2.105 ("$\Phi$ is an sc-diffeomorphism onto $\theta$" for every representative: the slab of V2.85;** `Polyfold.MPolyfold.not_surjOn_phiTot_2021`**).** For the triple $(x, V, \theta)$ of V2.85 in $[0, \infty)^3 \oplus W$, $\Phi_{(x,V,\theta)}$ does not map $U(x, V, \theta)$ onto $\theta$. *Book:* the characterization after Theorem 2.7.13, p. 84, for "any representative". *Hidden hypotheses:* none. *Deviation (not in the book):* read for every representative, the characterization does not hold as we read it; the strongest version that holds is V2.104, onto $\theta \cap V^x$, which for centred $V$ is the printed statement (V2.103). *Proof:* V2.85.

**V2.106 (Definition 2.7.14 and the iterated boundaries (2.45)–(2.47);** `Polyfold.MPolyfold.TameMPolyfoldBundle`, `Polyfold.MPolyfold.TameMPolyfoldBundle.b`, `Polyfold.MPolyfold.TameMPolyfoldBundle.iter`, `Polyfold.MPolyfold.TameMPolyfoldBundle.iterMap`, `Polyfold.MPolyfold.TameMPolyfoldBundle.iterMap_i` `Polyfold.MPolyfold.TameMPolyfoldBundle.degIndex_iterMap`, `Polyfold.MPolyfold.TameMPolyfoldBundle.` `Polyfold.MPolyfold.definition_2_7_14_2021`**).** The tame boundary of a tame $X$ is $bX$ with the tame structure of V2.102 and the sc-smooth $B$, whose image is $\partial X$. For a tame M-polyfold bundled with its space, $b^0X = X$ and $b^kX = b(b^{k-1}X)$ are tame M-polyfolds; $B^0 = \mathrm{id}$ and $B^{k+1} = B_X \circ B^k_{bX}$, so $B^k = B_X \circ B_{bX} \circ \cdots \circ B_{b^{k-1}X}$; $B^k : b^kX \to X$ is sc-smooth, $d_X(B^k(p)) = d_{b^kX}(p) + k$, and the image of $B^k$ is $\{d_X \geq k\}$. *Book:* Definition 2.7.14 and (2.45)–(2.47), p. 84, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (stronger):* the index formula, Theorem 2.7.13 (4) iterated. The bundling makes $b$ a map that can be iterated.

**V2.107 (witnesses for Theorem 2.7.13 (2)–(4);** `Polyfold.MPolyfold.quadrantPolyfold_bChartData_nonempty`, `Polyfold.MPolyfold.quadrantPolyfold_degIndex_tameBoundary`**).** For the tame quadrant $[0, \infty)^n \oplus W$, $n \geq 1$, chart data of $bX$ exist, and the $n$ points of $bX$ over the corner 0 have $d_{bX} = n - 1$. *Book:* not stated. *Hidden hypotheses:* none. *Deviation (not in the book):* the hypotheses of V2.102–V2.106 are satisfiable, and (4) is not vacuous.

**What is not verified in this chapter**

- p. 22 (preprint p. 18), Proposition 2.1.2 (preprint 2.2) (classical $C^\infty$ retractions have submanifold images): motivation only; not attempted (the exposition proves it, §8.1).
- p. 29 (preprint p. 22), Definition 2.1.16 (preprint 2.9) (sc-smooth bundle retractions, the categories $\mathcal{R}$, $\mathcal{BR}$): not formalized as a definition; its instances ($Tr$, strong bundle retractions) are. Formalized since as a definition: bundle retractions and bundle maps, with $Tr$ and $Tf$ as instances (V2.92); the categories $\mathcal{R}$, $\mathcal{BR}$ and their forgetful functors are still not formalized (gap G.5, formalized since apart from the categories; Findings, item 76).
- preprint p. 25, $X_m$ as an M-polyfold for $m \geq 2$: only $X^1$ is constructed (V2.16); the iterate $(X^1)^1 \cdots$ has the points of $X_m$, its topology is not compared with that of $X_m$; Theorem 2.3.10 (preprint 2.16) is proved for every $m$. Verified since: $(X^1)_m \cong X_{m+1}$ through the identity, with the charts of $X_{m+1}$, for every $m$, so each iterate is $X_m$ with its topology and charts (V2.93) (gap G.6, closed; Findings, item 77).
- pp. 38–44 (preprint pp. 26–30), the vector space, Banach and sc-Banach structures of $T_xX$, and Proposition 2.3.19 (3),
  (4) (preprint 2.21) in $T_aX$ rather than in a chart: not formalized.
- p. 39 (preprint p. 27), Proposition 2.3.15 (2) (preprint 2.19), openness of $p : TX \to X_1$: not proved (not used).
- pp. 48–49 (preprint pp. 33–35), Lemma 2.4.9 (preprint 2.29) in its path formulation, and $T_xO = C_xO - C_xO$: not formalized; the sign arguments are V2.34.
- preprint p. 39, the intrinsic description of extended faces ($k$ components of $\{d_C = 1\}$, minimal subspaces, independence of the presentation): the formalization works in a presentation.
- p. 57 (preprint p. 42), Definition 2.5.8 (preprint 2.38) ($T^R_xX$, $C_xX$ in $T_xX$): only the local form, its invariance and the partial quadrant $C_oO$ (in a tame chart V2.42, in any chart V2.33); not the transport to $T_xX$ or $d_{C_xX}(0_x) = d_X(x)$.
- pp. 65–67 (preprint pp. 48–51), strong bundle isomorphisms, the category $\mathcal{SBR}$ and the functors forget$[i]$: not formalized.
- p. 69 (preprint p. 53), Proposition 2.6.8 (preprint 2.54): that the projection $P' : f^*Y \to Y$ of the pull-back is a strong bundle map; the pulled-back atlas is V2.61.
- p. 72 (preprint p. 56), Lemma 2.7.3 (preprint 2.56) and Proposition 2.7.4 (preprint 2.57): not needed, since the proofs of Theorem 2.3.10 (preprint 2.16) and Proposition 2.3.15 (3) (preprint 2.19) avoid them.
- p. 23, Remark 2.1.6: that the modified notion of local completeness is strictly weaker; only the implication is proved (V2.68). Gap G.76. Verified since: $\mathbb{R}$ with the K-topology is Hausdorff, satisfies the modified notion and is not locally complete (V2.91) (gap G.76, closed; Findings, item 75).
- p. 30, Remark 2.1.17: it fixes the words "tangent" and "tangent bundle" of a retract and has no further content.
- pp. 62–63, Definition 2.5.19 (3): its independence of the chart, the remark that a tame sub-M-polyfold is tame, and the examples in $[0,\infty)^2 \times \mathbb{R}$ (V2.78). Gap G.77. Verified since, (3) still in charts (gap G.7) (V2.99, V2.100, V2.101) (gap G.77, closed; Findings, items 81, 82).
- pp. 83–84, Proposition 2.7.12 and the characterization after Theorem 2.7.13 for representatives $(x, V, \theta)$ whose $V$ is not centred at $x$: $\Phi_{(x,V,\theta)}$ is an open embedding there (V2.87), but the sc-smoothness of the transition maps is not formalized; every class has a centred representative. Gap G.78. Verified since: Proposition 2.7.12 for every representative, and the characterization with $\Phi$ an sc-diffeomorphism onto the open set $\theta \cap V^x$, which is $\theta$ for centred $V$ (V2.104); onto $\theta$ it does not hold as we read it for every representative (V2.105) (gap G.78, closed; Findings,

items 83, 84).

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## 4. Chapter 3 (HWZ §§3.1–3.8): partly verified

Chapter 3 of Part I (pp. 89–174 (preprint pp. 61–146)) develops sc-Fredholm theory. §3.1 defines germs, sections of strong bundles, fillers, basic germs, sc-Fredholm germs and sections, proves local compactness (Theorem 3.1.13 (preprint 3.12)) and the stability under sc$^+$-perturbations (Theorem 3.1.18 (preprint 3.15)), and states the implicit function theorems (Theorems 3.1.22, 3.1.26 (preprint 3.19, 3.23)), with good position (Definition 3.1.24 (preprint 3.21), Proposition 3.1.25 (preprint 3.22)). §3.2 introduces the tangent germ property and shows that it makes a subset a sub-M-polyfold (Definition 3.2.1 (preprint 3.25), Proposition 3.2.2 (preprint 3.26), Theorem 3.2.3 (preprint 3.27)). §3.3 proves the germ implicit function theorem (Definition 3.3.1 (preprint 3.28), Theorems 3.3.2–3.3.7 (preprint 3.29–3.34) and Theorem 3.35 (preprint; not in the 2021 edition)). §3.4 proves that basic germs are stable under sc$^+$-perturbations (Theorem 3.4.2 (preprint 3.38)). §3.5 proves local regularity and compactness (Theorem 3.5.1 (preprint 3.40)) and local solvability (Corollary 3.5.3 (preprint 3.41), Theorem 3.5.6 (preprint 3.43), Theorem 3.5.11 (preprint Corollary 3.47)). §3.6 states and proves the implicit function theorem for sc-Fredholm sections (Proposition 3.6.5 (preprint 3.52), Definition 3.6.6 (preprint 3.53), Theorem 3.6.8 (preprint 3.54), Lemma 3.6.9 (preprint 3.55)) and its global forms (Theorems 3.6.10, 3.6.11 (preprint 3.56, 3.57)). §3.7 gives a criterion for being conjugate to a basic germ (Theorem 3.7.1 (preprint 3.58), Lemmas 3.7.2–3.7.4 (preprint 3.59–3.61)). The appendix §3.8 proves Proposition 3.1.25 (preprint 3.22), Lemma 3.5.10 (preprint 3.46), Lemma 3.6.9 (preprint 3.55), and the inverse and implicit function theorems in partial quadrants (Theorems 3.8.15, 3.8.22 (preprint 3.72, 3.78)).

What is verified. **The germ implicit function theorem (Theorem 3.3.6 (preprint 3.33)) holds with the text's hypotheses**, and so do Corollary 3.3.7 (preprint 3.34) and Theorem 3.35 (preprint; not in the 2021 edition): the solution germ is sc^k on the germ of neighbourhoods shrunk by $2^k-1$ levels, and sc-smooth on one diagonal germ, which shrinks fast with the level (V3.31, V3.32). **The implicit function theorems also hold with the text's hypotheses, local and global**: Theorems 3.6.8, 3.1.26 and 3.1.22 (preprint 3.54, 3.23 and 3.19) (V3.94, V3.95), Theorem 3.6.11 (preprint 3.57), Corollary 3.1.27 (preprint 3.24) and Theorem 3.6.10 (preprint 3.56) (V3.87, V3.70, V3.86), once three steps of the text's proof are replaced. The surjectivity of $f'(y)$ at the zeros $y$ near a given zero, which the text derives from Corollary 3.5.3 (preprint 3.41) and which does not follow from it (V3.82; the same step stands in the book's arXiv version), is proved by a new argument, an implicit function theorem at the nearby zero for a perturbed equation (V3.93); the regularity of the solution germ away from the centre is proved by re-centring (V3.73, V3.74); and the relative openness of the set $\Sigma$ in the proof of Theorem 3.2.3 (preprint 3.27) by the choice of the good complement (V3.75, V3.77). With them: the dimension of the solution set is the Fredholm index (V3.96, V3.97), the index is locally constant along the solution set (V3.98; also without good position, with the text's hypotheses: tame germ data and no bump functions), and so, with the text's hypotheses, is the index of Definition 6.2.1 (preprint 6.1) (read with the text's locally defined sc$^+$-sections) along sc-smooth paths $\mathbb{R} \to X$, the text's Proposition 6.2.2 (preprint 6.2) for paths on $\mathbb{R}$ (V3.98), $\ker f'(y)$ is the tangent space of the solution set in a chart (V3.99), and good position is the surjectivity of $f'(y)$ at zeros with $d_X(y) = 0$, so that Theorems 3.1.22 and 3.6.10 (preprint 3.19 and 3.56) hold with the text's hypothesis "$f'(x)$ surjective" (V3.100, V3.101). The forms first proved under an added filler condition at the nearby zeros (V3.68, V3.69, V3.82, V3.85) are kept; that condition follows from good position (V3.102), and with Definition 3.1.4 (preprint 3.4) as stated the text's argument

gives the kernel half of Theorem 3.6.8 (1) (preprint 3.54) (V3.81) but not the surjectivity. All these statements are the repaired ones. They were first proved under the hypothesis that the base retraction of every strong bundle chart compatible with the atlas is tame, which no strong bundle over a nonempty base satisfies (V3.104, where only the half of the argument that concerns retractions is machine-checked), so those first statements were vacuous; they now assume what the text's Definitions 3.6.3, 3.6.4 and 3.6.6 (preprint 3.50, 3.51 and 3.53) ask, germ data and good position in a chart whose base retraction is tame, and conclude good position in such charts (V3.103, V3.107). Also verified: every definition of §§3.1–3.6 (Definitions 3.1.1–3.1.4, 3.1.6, 3.1.7, 3.1.11, 3.1.16, 3.1.24, 3.2.1, 3.3.1, 3.4.1, 3.6.1, 3.6.3, 3.6.4, 3.6.6 (preprint 3.1–3.4, 3.6, 3.7, 3.11, 3.14, 3.21, 3.25, 3.28, 3.37, 3.48, 3.50, 3.51, 3.53); since the switch of Definition 3.1.11 (preprint 3.11) to the text's germ reading, the fillings, the local sc$^+$-sections and every notion built on them are germs whose levels may shrink with $m$, as in the text, V3.7, V3.9, V3.13, and so, since a fourth addendum, is Definition 3.1.3 (preprint 3.3), V3.5); Proposition 3.1.5 (preprint 3.5), also in sc form (V3.8); Lemma 3.1.9 (preprint 3.9) as stated (V3.11); Proposition 3.1.10 (preprint 3.10), both statements, the second on every level (V3.12); Theorems 3.1.13 (preprint 3.12; with fewer hypotheses), 3.1.18 (preprint 3.15; without tameness), Corollary 3.1.15 (preprint 3.13); Theorem 3.1.21 (preprint 3.18) (without tameness), with the linearization of the perturbed section (V3.91, V3.92); Lemmas 3.1.19, 3.1.20 (preprint 3.16, 3.17); Proposition 3.1.25 (preprint 3.22) (without the extreme-ray lemmas); Proposition 3.2.2 (preprint 3.26); Theorem 3.2.3 (preprint 3.27) under two added regularity conditions, each with a counterexample to the step of the text's proof it replaces (for solution sets both are theorems); Theorems 3.3.2, 3.3.3 (preprint 3.29, 3.30), Lemmas 3.3.4, 3.3.5 (preprint 3.31, 3.32) (as one induction), Theorem 3.4.2 (preprint 3.38) (with a linear coordinate change), Corollary 3.4.3 (preprint 3.39), Theorem 3.5.1 (preprint 3.40) (with compact closures), Corollary 3.5.3 (preprint 3.41), Lemmas 3.5.5, 3.5.8–3.5.10 (preprint 3.42, 3.44–3.46); Theorem 3.5.6 (preprint 3.43) and Theorem 3.5.11 (preprint Corollary 3.47) (their clause (1), "the solution map is an sc-smooth germ", in the level-wise sense: $C^k$ into level $k$ on a neighbourhood depending on $k$); Remark 3.6.2 (preprint 3.49); Proposition 3.6.5 (preprint 3.52) in charts, as an sc-Fredholm operator with the index $n - N$ of the conjugating basic germ on every level, in every chart (V3.46, V3.96); Lemma 3.8.25 (preprint 3.81) as stated (V3.66); Lemma 3.6.9 (preprint 3.55) (with the added hypothesis that the good complement is closed); Theorem 3.6.8 (preprint 3.54) as described above, with its graph description; Theorem 3.7.1 (preprint 3.58) in the coordinates of its proof; Lemmas 3.7.2 and 3.7.3 (preprint 3.59 and 3.60) (in an abstract form, with their hypotheses derived in Theorem 3.7.1 (preprint 3.58)) and Lemma 3.7.4 (preprint 3.61); and the appendix except Lemmas 3.8.1–3.8.3, 3.8.9–3.8.11, 3.8.20 (preprint 3.62–3.64, 3.66–3.68, 3.76), which the formal proofs avoid. Not verified: Proposition 3.6.5 (preprint 3.52) on $T_xX$ itself, which carries no linear structure here; $\ker f'(y) = T_yS$ with $T_yS$ a subspace of $T_yX$ (it is verified in a chart); Proposition 6.2.2 (preprint 6.2) for paths sc-smooth on $[0,1]$ only (the project has no M-polyfold $[0,1]$); the passage from an atlas of tame charts to tame germ data (Definition 3.1.11 (preprint 3.11) $\Rightarrow$ Definition 3.6.3 (preprint 3.50)), which is open. Remarks 3.1.8 and 3.1.23 (preprint 3.8 and 3.20) compare the definitions with the earlier paper [21] and make no claim to verify. Items V3.68–V3.90 were added after the first edition of this section (the Lean formalized after it); the earlier items whose content changed with them (V3.13, V3.23, V3.24, V3.34, V3.38, V3.42, V3.46–V3.54, V3.59, V3.62) are updated in place. Items V3.91–V3.102 were added in a second addendum (Theorem 3.1.21 (preprint 3.18), Theorem 3.6.8 (1) (preprint 3.54) with the text's hypotheses, the index of the solution set, the kernel as tangent space, good position at interior zeros); the items whose content changed with them (V3.46, V3.47, V3.68, V3.69, V3.81–V3.87) are updated in place. Items V3.103–V3.107 were added after the repair of the tameness hypothesis (Findings, Chapter 3, items 190–199: the tame germ notions, the non-tame retraction behind the vacuity, Theorems 3.1.18 and 3.1.21 (preprint 3.15 and 3.18) for tame germ data, good position in any chart

of the germ data); the items whose statements changed with it (V3.45, V3.47, V3.48, V3.68–V3.70, V3.80–V3.82, V3.84–V3.87, V3.94, V3.95, V3.97–V3.102) are updated in place. A third addendum (sc-Fredholm theory on all levels, the one-germ form of Theorem 3.35 (preprint), Lemmas 3.1.9 and 3.8.25 (preprint 3.9 and 3.81) as stated, good position with the text's sc-complement, the index without good position, and the difference between the text's germs and the formal definitions on the whole chart) updated V3.5, V3.7–V3.9, V3.11–V3.13, V3.16, V3.19, V3.20, V3.22, V3.28, V3.31, V3.32, V3.43, V3.45–V3.47, V3.66, V3.68, V3.70, V3.86, V3.87, V3.94–V3.98 and V3.103 in place. The switch of Definition 3.1.11 (preprint 3.11) and of everything built on it to the text's germ reading (phase 2: the canonical notions and theorems are the germ reading, the project's earlier whole-chart ones are kept as `…WholeChart`, `…_wholeChart`; Findings, Chapter 3, items 330–334) updated V3.5, V3.7–V3.9, V3.13, V3.16, V3.17, V3.19, V3.43, V3.45–V3.47, V3.68, V3.78, V3.79, V3.81, V3.82, V3.91, V3.93, V3.96, V3.98, V3.102, V3.103, V3.105 and V3.107 in place. A fourth addendum (2026-11-02: Definition 3.1.3 (preprint 3.3) in the germ reading, Definition 6.2.1 (preprint 6.1) and Proposition 6.2.2 (preprint 6.2), germ data and the charts of a tame atlas; Findings, Chapter 3, items 340–355) updated V3.5, V3.13, V3.98 and V3.103 in place. A fifth addendum (2026-11-02: the restriction of a strong bundle to an open set, the index along the zero set and Proposition 6.2.2 (preprint 6.2) without bump functions, Definition 6.2.1 (preprint 6.1) with the text's local sections; Findings, Chapter 3, items 349, 360–367) updated V3.68 and V3.98 in place.

The setting. Germs are fixed sequences of sets (Definition 3.1.1 (preprint 3.1)), and the calculus of germs — restriction, chain rule, pairing, Propositions 1.2.1–1.2.3 (preprint 1.16–1.18) for germs — is proved, not assumed. The finite-dimensional parameter spaces $\mathbb{R}^n$, $\mathbb{R}^N$ are arbitrary finite-dimensional normed spaces $P$, $Q$, and the parameter set $\widetilde{C}$ a convex set containing $0$ with nonempty interior, which is closed or a partial quadrant where a theorem needs it. A "strong bundle chart" in Definitions 3.1.3, 3.1.6, 3.1.11 (preprint 3.3, 3.6, 3.11) is a chart compatible with the atlas. The fillings and the local sc$^+$-sections of Definition 3.1.11 (preprint 3.11) are, as in the text, germs on sc-germs of neighbourhoods whose levels may shrink with $m$ (the text's germ reading, since phase 2; V3.7, V3.9, V3.13); the project's earlier notions, with these data on the whole $U$ of the chart (levels $U \cap E_m$), are stricter and are kept under the names `…WholeChart`. The sc-smooth section germs of Definition 3.1.3 (preprint 3.3), which no theorem assumes, are germs in the same sense since a fourth addendum (V3.5; Findings, Chapter 3, items 220–221, 330). Tameness of the strong bundle is nowhere used in §3.1. It enters §3.6 as the text's Definitions 3.6.3, 3.6.4 and 3.6.6 (preprint 3.50, 3.51 and 3.53) put it: the germ data of $f$ at a point, and its good position, lie in a strong bundle chart whose base retraction is tame (V3.103; "tame germ data" and "good position in a tame chart" below). It is used, for the chart at the zero $x_0$, through Proposition 2.5.3 (preprint 2.35), $(1-Dr(0))E \subseteq E_0$, and, for the tangent germ property of the solution set, through the tameness of the model of the chart $\psi \circ \varphi$ (V3.71). The reading first formalized, that the base retractions of all charts compatible with the atlas are tame, cannot be met (V3.104, the retraction-level half machine-checked). Classical regularity in the appendix is Mathlib's $C^k$ within the set, which on relatively open subsets of partial quadrants is the text's Definition 3.8.14 (preprint 3.71). Findings referred to below are those of `Exposition/Findings.md`, "Chapter 3".

**V3.1 (Definition 3.1.1 (preprint 3.1), sc-germs of neighbourhoods; `Polyfold.ScNbhdGerm`, `Polyfold.ScNbhdGerm.tangent`, `Polyfold.ScNbhdGerm.ofSet`, `Polyfold.IsQuadrantLike`, `Polyfold.IsPartialQuadrant.isQuadrantLike`).** An sc-germ of neighbourhoods around $0$ in $C \subseteq E_0$ is a family $U_m \subseteq E_m$, each of the form $O \cap C_m$ with $O$ open in $E_m$, containing $0$, with $\iota U_{m+1} \subseteq U_m$. Its tangent germ has levels $U_{m+1} \times E_m$. A relatively open $U \ni 0$ in $C$ gives the germ $U_m = U \cap E_m$. $C$ is quadrant-like if $0 \in C$ and every $C_m$ is convex with nonempty interior;

partial quadrants are. *Book:* Definition 3.1.1 (preprint 3.1), p. 89 (preprint p. 62). *Hidden hypotheses:* none. *Deviation (restated):* a germ is a fixed sequence, not a class up to shrinking; the text's "the size of the sets $U_m$ does not matter" is not built in, and the theorems say on which germ they hold. $(U_{m+1})_m$ is a germ only when all levels coincide (finite-dimensional $E$).

**V3.2 (Definition 3.1.2 (preprint 3.2), sc^k-germs; `Polyfold.IsSc0Germ`, `Polyfold.IsSc1GermWith`, `Polyfold.IsSc1Germ`, `Polyfold.IsSckGerm`, `Polyfold.IsScSmoothGerm`, `Polyfold.IsSck.isSckGerm`).** $f$ is an sc$^0$-germ on $(U_m)$ if $f(U_m) \subseteq F_m$ and $f : U_m \to F_m$ is continuous for every $m$; sc$^1$ if moreover, for a family of bounded $Df(x) : E_0 \to F_0$, $x \in U_1$, $f$ is differentiable within $U_1$ at $x$ (increments in $E_1$, remainder in $F_0$) with derivative $Df(x)$, and the tangent map $(x,h) \mapsto (f(x), Df(x)h)$ is an sc$^0$-germ on the tangent germ; sc^{k+1} if sc$^1$ with sc^k tangent map; sc-smooth if sc^k for all $k$. An sc^k map on a relatively open $U \ni 0$ is an sc^k-germ on $(U \cap E_m)$. *Book:* Definition 3.1.2 (preprint 3.2), pp. 90–91 (preprint pp. 62–63). *Hidden hypotheses:* for quadrant-like $C$ the derivative is unique on $U_1$. *Deviation:* none.

**V3.3 (the calculus of germs; `Polyfold.IsSckGerm.congr`, `Polyfold.IsSckGerm.mono`, `Polyfold.ScNbhdGerm.MapsToGerm`, `Polyfold.IsSckGerm.comp`, `Polyfold.IsSc1Germ.comp`, `Polyfold.ScOperator.isScSmoothGerm`, `Polyfold.IsSckGerm.pair`, `Polyfold.ScNbhdGerm.preimage`).** For quadrant-like sets: sc^k-germs depend only on their values on $U_0$ and restrict to smaller germs; if $f$ is an sc^k-germ on $(U_m)$, $g$ an sc^k-germ on $(V_m)$ and $f(U_m) \subseteq V_m$ for every $m$, then $g \circ f$ is an sc^k-germ, with $T(g \circ f) = Tg \circ Tf$ on the tangent germ (k = 1); sc-operators are sc-smooth germs on every germ; pairings of sc^k-germs are sc^k-germs; for an sc$^0$-germ $f$ with $f(0) = 0$ and a germ $G'$ around 0 there is a germ inside $(U_m)$ on which $f$ maps into $G'$ level by level. *Book:* not stated; used throughout §§3.3–3.6. *Hidden hypotheses:* the level-wise inclusion $f(U_m) \subseteq V_m$ for compositions. *Deviation (not in the book).*

**V3.4 (Propositions 1.2.1–1.2.3 (preprint 1.16–1.18) for germs; `Polyfold.IsSc1GermWith.exists_levelDeriv`, `Polyfold.IsSc1GermWith.continuousOn_levelDeriv_clm`, `Polyfold.IsSc1GermWith.hasFDerivWithinAt_lev` `Polyfold.ScNbhdGerm.shift`, `Polyfold.IsSckGerm.shift`, `Polyfold.IsSckGerm.contDiffOn_levels`).** For an sc$^1$-germ with derivative family $D$ and $x \in U_{m+1}$, $D(x)$ restricts to bounded operators $E_m \to F_m$, continuous in $x$ in the operator norm on $U_{m+1}$, and $f : U_{m+1} \to F_m$ is differentiable within $U_{m+1}$ with these derivatives. $(U_{m+1})_m$ is a germ in $E^1$ around $0 \in C^1$, and an sc^k-germ on $(U_m)$ is an sc^k-germ $(U_{m+1})_m \to F^1$. An sc^k-germ is $C^k$ from $U_{m+k}$ to $F_m$. *Book:* not stated for germs; the text applies Propositions 1.2.1–1.2.3 (preprint 1.16–1.18) to germs (Lemma 3.1.9 (preprint 3.9): "recalling Proposition 1.17", which is 1.16; Lemma 3.3.4 (preprint 3.31); Corollary 3.3.7 (preprint 3.34)). *Hidden hypotheses:* none. *Deviation (not in the book):* the germ versions, with the proofs of Chapter 1.

**V3.5 (Definition 3.1.3 (preprint 3.3), sc-smooth section germs; `Polyfold.StrongBundle.IsScSmoothSectionGerm` `Polyfold.StrongBundle.IsChart`, `Polyfold.StrongBundle.IsScSmoothSection.isScSmoothSectionGerm`, `Polyfold.StrongBundle.isChart_congr`).** $(f, x_0)$ is an sc-smooth section germ of the strong bundle $P : Y \to X$ if $x_0$ is smooth and in some strong bundle chart compatible with the atlas, centred at $x_0$, the principal part $\mathbf{f}$ of $f$ satisfies: $\mathbf{f} \circ r$ is an sc-smooth germ on some sc-germ of neighbourhoods $G$ of 0 with $G_0 \subseteq U$. An sc-smooth section is an sc-smooth section germ at every smooth point that has a centred chart (with the germ $U \cap E_m$). *Book:* Definition 3.1.3 (preprint 3.3), p. 91 (preprint p. 63). *Hidden hypotheses:* "a strong bundle chart" means a chart compatible with the atlas (a chart of the atlas need not be centred); the notion does not depend on the atlas within its class (`Polyfold.StrongBundle.isChart_congr`, V2.58). *Deviation (hypotheses explicit):* the chart, as just said. Since a fourth addendum the germ is the text's: $\mathbf{f} \circ r$ is an sc-smooth

germ on some sc-germ of neighbourhoods $G$ with $G_0 \subseteq U$, whose levels may shrink with $m$, as for Definitions 3.1.4, 3.1.6 and 3.1.11 (preprint 3.4, 3.6 and 3.11) since phase 2. Until then it was required on the whole $U$ of the chart, which is stricter (a function $C^k$ on $(-1/k, 1/k)$ for every $k$ is an sc-smooth germ in the constant scale $\mathbb{R}$ and sc-smooth on no neighbourhood of $0$, `Polyfold.exists_isScSmoothGerm_not_isScSmooth`), and the item was "narrower". No theorem assumes the definition (Findings, Chapter 3, items 220–221, 260).

**V3.6 (the section algebra and the sum rule; `Polyfold.IsScSmooth.add`, `Polyfold.sectionAdd`, `Polyfold.StrongBundleChart.principal_sectionAdd`, `Polyfold.StrongBundleChart.isZeroAt_iff`, `Polyfold.StrongBundleChart.add_mem_K`, `Polyfold.StrongBundle.IsScSmoothSection.isScSmoothRetractM` `Polyfold.StrongBundle.IsScPlusSection.isScSmoothRetractMap_one`).** Sums, differences and negatives of sc-smooth maps on relatively open subsets of partial quadrants are sc-smooth. The fibrewise sum $f + s$ of sections is a section, with principal part $\mathbf{f} + \mathbf{s}$ in every chart; $f(x) = 0$ iff $\mathbf{f}(\varphi(x)) = 0$; the fibres of $K$ over $O$ are linear. In every strong bundle chart compatible with the atlas, the representative $z \mapsto (z, \mathbf{f}(z))$ of an sc-smooth section is sc-smooth $O \to K$, and the representative $z \mapsto (z, \mathbf{s}_1(z))$ of an sc⁺-section is sc-smooth $O \to K[1]$. *Book:* not stated; used in Theorems 3.1.13, 3.1.18 (preprint 3.12, 3.15). *Hidden hypotheses:* the compatibility on $[1]$ of Definition 2.6.4 (preprint 2.50) (its first use). *Deviation (not in the book).*

**V3.7 (Definition 3.1.4 (preprint 3.4), fillers; `Polyfold.IsFillingGerm`, `Polyfold.fillDefect`, `Polyfold.fillDefect_isScSmoothGerm`, `Polyfold.IsFillerOf`, `Polyfold.StrongRetr.r`, `Polyfold.StrongRetr.`, `Polyfold.StrongRetr.CLM`).** For a strong bundle retraction $R(u, h) = (r(u), \rho(u)h)$ on $U \lhd F$ with $O = r(U)$, $0 \in U$: $g$ is a filling of $f$ if $g$ is an sc-smooth germ on an sc-germ of neighbourhoods $G$ with $G_0 \subseteq U$ and (1) $f = g$ on $O$ near 0; (2) a point $y \in U$ near 0 with $g(y) = \rho(r(y))g(y)$ lies in $O$; (3) $y \mapsto [1 - \rho(r(y))]g(y)$ is an sc¹-germ on $G$ with a derivative family $Dk$, and $Dk(0)$ restricts to a topological linear isomorphism $\ker Dr(0) \to \ker \rho(0)$. ($\rho(u)$ is a bounded operator, $\rho(x)$ a projection for $x \in O$.) *Book:* Definition 3.1.4 (preprint 3.4), p. 91 (preprint p. 64). *Hidden hypotheses:* the map in (3) is differentiable at 0 (true for every sc-smooth germ $g$, and $Dk(0)$ is unique). *Deviation (hypotheses explicit):* stated for the principal parts; tameness unused. Since the switch of Definition 3.1.11 (preprint 3.11) to the text's germ reading (phase 2) $g$ and the map in (3) are germs whose levels may shrink with $m$; the stricter whole-chart filler (`Polyfold.IsFillerOf`, the map in (3) sc¹ on all of $U$, the sc-smoothness of $g$ on $U$ added in Definition 3.1.6 (preprint 3.6)) is kept for the whole-chart notions (Findings, Chapter 3, items 220–221, 261, 330). Formerly "narrower", for the whole-chart reading.

**V3.8 (Proposition 3.1.5 (preprint 3.5) and the matrix of $Dg(0)$; `Polyfold.IsFillerOf.proposition_3_5`, `Polyfold.proposition_3_5`, `Polyfold.fillerRestrict`, `Polyfold.fillerRestrict_kerEquiv`, `Polyfold.fillerRestrict_cokerEquiv`, `Polyfold.StrongRetr.scDeriv_fiber_zero`, `Polyfold.IsFillerOf.d` `Polyfold.IsFillerOf.scDeriv_section_comp_r`, `Polyfold.IsScFredholm.restrict`, `Polyfold.IsFillerOf.p` `Polyfold.IsFillerOf.filler_condition_three_sc`, `Polyfold.IsFillerOf.proposition_3_5_sc_converse`, `Polyfold.IsFillerOf.proposition_3_5_sc_iff`).** Let $g$ be sc-smooth on $U$ and fill the section $f$ of $K \to O$, with $g(0) = 0$. Then $D(\rho(r(\cdot))g)(0) = \rho(0)Dg(0)$, $(1 - \rho(0))Dg(0)Dr(0) = 0$, $D(f \circ r)(0) = Dg(0)Dr(0)$; and $Tf(0) := Dg(0)$ restricted to $Dr(0)E \to \rho(0)F$ is surjective iff $Dg(0)$ is, Fredholm iff $Dg(0)$ is, and has the same index ($\dim\ker - \dim\operatorname{coker}$). The linear algebra holds for any idempotents $p$, $q$ and $L$ with $(1 - q)Lp = 0$ and $(1 - q)L : \ker p \to \ker q$ a topological isomorphism. In sc form: if $Dg(0)$ is an sc-Fredholm sc-operator, then $Dr(0)E$, $\rho(0)F$, $\ker Dr(0)$ and $\ker \rho(0)$ are sc-subspaces, $Tf(0) : Dr(0)E \to \rho(0)F$ is sc-Fredholm with the index of $Dg(0)$ on every level, and $(1 - \rho(0))Dg(0) : \ker Dr(0) \to \ker \rho(0)$ is an sc-isomorphism (condition (3) on every level); for $Dg(0)$ an sc-operator, $Dg(0)$ is sc-Fredholm iff

$Tf(0)$ is sc-Fredholm and condition (3) holds on every level. *Book:* the paragraph after Definition 3.1.4 (preprint 3.4) and Proposition 3.1.5 (preprint 3.5), pp. 91–93 (preprint pp. 64–65). *Hidden hypotheses:* $(u,h) \mapsto \rho(u)h$ differentiable at $(0,0)$; $g(0) = 0$. *Deviation (stronger):* the index equality without any finiteness assumption, via isomorphisms of kernels and cokernels instead of the text's factorization. The text's target "$(1 - D\rho(0))F$" is $(1 - \rho(0))F$. The sc form needs condition (3) only on level 0, as the text states it (the higher levels of $Tf(0)$ come from the regularity of $Dg(0)$); its converse needs condition (3) on every level and fails with level 0 alone (an example with the shift on a weighted $\ell^2$, not formalized; Findings, Chapter 3, items 234, 237, 238). The same holds for a filling germ (V3.7), with $Dg(0)$ the value at 0 of any derivative family of $g$ (`Polyfold.IsFillingGerm.proposition_3_5`, `Polyfold.IsFillingGerm.proposition_3_5_sc`, `Polyfold.IsFillingGerm.proposition_3_5_sc_iff`; Findings, Chapter 3, items 272, 289, 314).

**V3.9 (Definition 3.1.6 (preprint 3.6), filled versions; `Polyfold.IsFillingGerm`, `Polyfold.IsFilledVersion`, `Polyfold.IsFilledVersion.isFillingGerm`, `Polyfold.IsFillingGerm.isFilledVersion`).** A filling of the principal part **f** (V3.7) is an sc-smooth germ $g$ on an sc-germ of neighbourhoods inside $U$, whose levels may shrink with $m$, filling **f** near 0; the centred chart is that of Definition 3.1.11 (preprint 3.11) (V3.13). *Book:* Definition 3.1.6 (preprint 3.6), p. 93 (preprint p. 66). *Hidden hypotheses:* none. *Deviation (restated):* the datum in a given chart, for the principal parts; the chart is quantified in Definition 3.1.11 (preprint 3.11). The project's earlier filled version (`Polyfold.IsFilledVersion`), $g$ sc-smooth on all of $U$, is stricter for germs whose levels shrink with $m$ (round-trip items V3.9, V3.43); it is exactly a filling sc-smooth on all of $U$ (`IsFilledVersion.isFillingGerm`, `IsFillingGerm.isFilledVersion`; Findings, Chapter 3, items 220–221, 260–262, 330). Formerly "narrower", for the whole-chart reading.

**V3.10 (Definition 3.1.7 (preprint 3.7), basic germs; `Polyfold.IsBasicGerm`, `Polyfold.IsSc0ContractionGerm`).** For a finite-dimensional $P$, a set $\widetilde{C} \subseteq P$, a finite-dimensional $Q$ and a germ $G$ around 0 in $\widetilde{C} \oplus W$: $f$ is a basic germ with contraction map $B$ if $f$ is an sc-smooth germ on $G$ with $f(0) = 0$, the $W$-component of $f(a,w)$ is $w - B(a,w)$ on $U_0$, and $B$ is an sc$^0$-contraction germ (V3.25). *Book:* Definition 3.1.7 (preprint 3.7), p. 94 (preprint p. 66) (Remark 3.1.8 (preprint 3.8): historical). *Hidden hypotheses:* the points of the contraction estimate lie in $U_m$. *Deviation (restated):* $\mathbb{R}^n$, $\mathbb{R}^N$ replaced by finite-dimensional spaces.

**V3.11 (Lemma 3.1.9 (preprint 3.9); `Polyfold.ContractionGerm.norm_levelDeriv_snd_le`, `Polyfold.IsSc0ContractionGerm.deriv_zero_snd`, `Polyfold.IsSc0ContractionGerm.exists_norm_deriv_sm`, `Polyfold.IsSc0ContractionGerm.exists_norm_levelDeriv_snd_le`).** For an sc$^0$-contraction germ $B$ that is an sc$^1$-germ: (a) in the working form, $\|D_2B(a,w)\|_{L(W_m)} \leq \frac{1}{2}$ at points $(a,w) \in U_{m+1}$ with $a \in N(m+1) \cap \widetilde{C}$, $w \in O(m+1)$, $\iota w \in O(m)$; (b) $D_2B(0,0) = 0$; (c) for every $\varepsilon > 0$ there is $r > 0$ with $|D_2B(x)\zeta|_0 \leq \varepsilon|\zeta|_0$ for $x \in U_1$, $|x|_1 < r$, $\zeta \in W_0$; (d) as stated, for $\widetilde{C}$ quadrant-like: for every $\varepsilon > 0$ and $m$ there is $r > 0$ with $|D_2B(x)\zeta|_m \leq \varepsilon|\zeta|_m$ for $\zeta \in W_m$ at every $x \in U_{m+1}$ with $|x|_m < r$. *Book:* Lemma 3.1.9 (preprint 3.9), p. 94 (preprint p. 67) (every $\varepsilon$, every level, $(a,w) \in E_{m+1}$ small in $E_m$; $B$ sc-smooth). *Hidden hypotheses:* differentiability of $B : U_{m+1} \to W_m$ (Proposition 1.2.1 (preprint 1.16) for germs, V3.4). *Deviation (stronger):* sc$^1$ suffices, and the general statement is proved as stated (d); smallness on level $m+1$ is not needed, since the contraction estimate of Definition 3.3.1 (preprint 3.28) is applied at points small on level 0 (Findings, Chapter 3, item 215). Formerly "narrower", when only (a)–(c) were proved.

**V3.12 (Proposition 3.1.10 (preprint 3.10); `Polyfold.proposition_3_10`, `Polyfold.proposition_3_10_scPlus`, `Polyfold.IsSc1GermWith.isScOperator_deriv_zero`, `Polyfold.ScBanach.scIndex_projWOp`, `Polyfold.StabilityData.exists_isFredholm_Dfs'`, `Polyfold.proposition_3_10_level`,**

`Polyfold.proposition_3_10_full`, `Polyfold.IsBasicGerm.levelDeriv_add_eq`, `Polyfold.IsBasicGerm.snd_`
Let $\widetilde{C}$ be a partial quadrant, $h$ a basic germ, $s$ an sc¹-germ into $(Q \oplus W)^1$ and $f = h + \iota s$ an sc¹-germ with derivative family $Df$. Then $Df(0)$ is an sc-operator, sc-Fredholm of index $\dim P - \dim Q$: it is $T_0 + S$ with $T_0(\delta a, \delta w) = (0, \delta w)$ and $S$ an sc⁺-operator. For the data of Theorem 3.4.2 (preprint 3.38), $D(h+s)(x) : E_0 \to F_0$ is Fredholm of index $\dim P - \dim Q$ for every $x \in U_1$ small on level 1. On every level (the second statement): for every $m$ there is $\delta > 0$ such that at every $x \in U_{m+1}$ with $|x|_m < \delta$ the level-$m$ operator $Df(x)_m : P \oplus W_m \to Q \oplus W_m$ is Fredholm of index $\dim P - \dim Q$ ($\widetilde{C}$ quadrant-like); $f(0) = 0$ is not used. *Book:* Proposition 3.1.10 (preprint 3.10), p. 95 (preprint p. 68). *Hidden hypotheses:* the finite-rank terms are sc⁺ because $Dh(0)$ is an sc-operator at the smooth point 0. *Deviation (stronger):* both statements as in the text, with $s$ and $f$ only sc¹ and $f(0) = 0$ unused. The second statement by the text's proof on level $m$: $Df(x)_m = T_0 + S(x) + K(x)$ with $\|S(x)\| \le \varepsilon$ by Lemma 3.1.9 (preprint 3.9) on level $m$ (V3.11) and $K(x)$ compact; the $W$-component of $Df(x)_m$, which the text uses silently, follows from the uniqueness of derivatives within $U_1$ (Findings, Chapter 3, item 230). Formerly "narrower", the second statement on level 0 only.

**V3.13 (Definition 3.1.11 (preprint 3.11), sc-Fredholm germs;** `Polyfold.StrongBundle.IsScFredholmGerm`, `Polyfold.IsFillingGerm`, `Polyfold.IsScPlusGerm`, `Polyfold.IsConjugateToBasicGerm`, `Polyfold.IsStrongBundleIso`, `Polyfold.StrongBundle.IsScFredholmGermWholeChart`, `Polyfold.StrongBundle.IsScFredholmGermWholeChart.isScFredholmGerm`, `Polyfold.StrongBundle.isScFre`
$(f, x_0)$ is an sc-Fredholm germ if $x_0$ is smooth and in a strong bundle chart of $P$ centred at $x_0$ the principal part has a filling $g$ (V3.9: an sc-smooth germ on a germ of neighbourhoods inside $U$), there is an sc⁺-germ $s$ (`IsScPlusGerm`: $s : E_0 \to F_1$ an sc-smooth germ into $F^1$ on a germ inside $U$) with $\iota s(0) = g(0)$, and $k = g - \iota s$ is conjugated to a basic germ: strong bundle maps $\Psi : U \lhd F \to U' \lhd F'$ and $\Psi'$ back, inverse to each other, with base map $\psi$, $\psi(0) = 0$, $U'$ relatively open in $\widetilde{C} \oplus W$ ($\widetilde{C}$ a partial quadrant of a finite-dimensional space), a basic germ $h$ on a germ inside $U'$, and $(\Psi(u, k(u)))_2 = h(\psi(u))$ for $u \in U$ near 0. *Book:* Definition 3.1.11 (preprint 3.11), p. 96 (preprint p. 69). *Hidden hypotheses:* none. *Deviation (restated):* the identity is required for $u \in U$; tameness unused; $C' = T\psi(0)C$ not imposed (it is proved, V3.48). The witnesses of the conjugation are also available as data (`Polyfold.ConjugationData`, V3.46). Since the switch to the text's germ reading (phase 2) the filling and the sc⁺-section are germs whose levels may shrink with $m$; $\Psi$ on all of $U$ is no restriction (on a relatively open $U_0 \subseteq U$ it is the same after restricting the chart on level 0). The project's earlier notion, with $g$, $s$ sc-smooth on the whole $U$ (`IsScFredholmGermWholeChart`), implies this one and is exactly this one with $g$, $s$ sc-smooth on $U$; whether the two agree for sections sc-smooth on $X$ is open, and no theorem depends on it (Findings, Chapter 3, items 220–221, 261–264, 330). The existential over charts is essential: a filling in a chart gives a topological isomorphism $\ker Dr(0) \cong \ker \rho(0)$, a condition on the chart (`Polyfold.IsFillingGerm.nonempty_kerEquiv`), so a filling in one chart gives none in another unless these normal spaces match; a compatible tame chart of the trivial bundle carries no filling of any section (`Polyfold.TrivialBundle.not_isFillingGerm_evenBundleChart`; Findings, Chapter 3, items 350–352).

**V3.14 (Theorem 3.1.13 (preprint 3.12);** `Polyfold.theorem_3_12`, `Polyfold.theorem_3_12_local_subset`, `Polyfold.MPolyfold.continuousOn_levelChartSymm`**).** Let $f$ be a section of a strong bundle $P : Y \to X$ ($P \circ f = \mathrm{id}$) with $(f, x_0)$ an sc-Fredholm germ and $f(x_0) = 0$. There are open sets $O(0) \supseteq O(1) \supseteq \cdots$ of $X$ containing $x_0$ such that for every $i$ the closure in $X$ of $\{x \in O(i) \mid f(x) = 0\}$ is the image of a compact subset of $X_i$, with the level topology, under the inclusion $X_i \to X$. *Book:* Theorem 3.1.13 (preprint 3.12), p. 97 (preprint p. 70) (sc-smooth section

of a tame strong bundle). *Hidden hypotheses:* none beyond those used. *Deviation (stronger):* only $P \circ f = \mathrm{id}$ of the section is used; no tameness. *Proof:* the local form (Theorem 3.5.1 (preprint 3.40) for any subset of the solution set), transported through chart, filler and conjugation, with the inverse chart $O_i \to X_i$ continuous for the level topology.

**V3.15 (Corollary 3.1.15 (preprint 3.13); `Polyfold.corollary_3_13`, `Polyfold.corollary_3_13_section`, `Polyfold.tendsto_of_isCompact_of_tendsto_map`).** (a) Under the hypotheses of V3.14, if $f(x_k) = 0$ and $x_k \to x_0$ in $X$, then for every $m$ there are $y_k \in X_m$ with $\iota y_k = x_k$ for $k$ large and $y_k \to y_0$ in $X_m$, $\iota y_0 = x_0$. (b) If $f$ is an sc-Fredholm section and $x_k \to x$ are zeros, then $x$ is a zero and a smooth point, and the same holds at $x$. *Book:* Corollary 3.1.15 (preprint 3.13), p. 97 (preprint p. 70). *Hidden hypotheses:* the germ property at the limit. *Deviation (restated):* the text's arbitrary limit $x$ is covered for $x = x_0$ and, for sc-Fredholm sections, for every $x$.

**V3.16 (Definition 3.1.16 (preprint 3.14), sc-Fredholm sections; `Polyfold.StrongBundle.IsScFredholmSection`, `Polyfold.StrongBundle.IsScFredholmSectionWholeChart`, `Polyfold.StrongBundle.IsScFredholmSectionWh`**
$f$ is an sc-Fredholm section of the strong bundle $P$ if it is sc-smooth, regularizing ($x \in X_m$, $f(x) \in Y_{m,m+1}$ imply $x \in X_{m+1}$) and an sc-Fredholm germ (V3.13) at every smooth point. *Book:* Definition 3.1.16 (preprint 3.14), p. 97 (preprint p. 70). *Hidden hypotheses:* none. *Deviation:* none; the bundle need not be tame. Clause (3) is the text's germ reading since phase 2; the project's earlier whole-chart notion (`IsScFredholmSectionWholeChart`), stricter, implies it (Findings, Chapter 3, items 220–221, 330). Formerly "restated", for the whole-chart reading.

**V3.17 (Theorem 3.1.18 (preprint 3.15); `Polyfold.theorem_3_15`, `Polyfold.StrongBundle.IsScSmoothSection.s`, `Polyfold.StrongBundle.regularizing_sectionAdd`, `Polyfold.StrongBundle.IsScFredholmGerm.sectionAdd`**
If $f$ is an sc-Fredholm section and $s$ an sc$^+$-section of a strong bundle, the fibrewise sum $f + s$ is an sc-Fredholm section. *Book:* Theorem 3.1.18 (preprint 3.15), p. 98 (preprint p. 71) (tame bundle). *Hidden hypotheses:* the sum rule; the [1]-compatibility of charts (V3.6). *Deviation (stronger):* no tameness. In the germ reading of Definition 3.1.11 (preprint 3.11) (V3.13), with the text's proof: $\tilde{t} = \hat{s} \circ r$ is sc-smooth on the whole $U$, and a germ plus a whole-chart map is a germ on the same germ (V3.19; Findings, Chapter 3, item 300); the whole-chart form is `Polyfold.theorem_3_15_wholeChart`.

**V3.18 (Lemma 3.1.19 (preprint 3.16); `Polyfold.lemma_3_16`).** For a section $s$ of $K \to O$: $s(r(x)) = s(x)$ for $x \in O$, and $\rho(r(u))s(r(u)) = s(r(u))$, $R(r(u), s(r(u))) = (r(u), s(r(u)))$ for $u \in U$. *Book:* Lemma 3.1.19 (preprint 3.16), p. 99 (preprint p. 72). *Hidden hypotheses:* $s$ takes values in $K$. *Deviation (restated):* the sc$^+$-regularity of $s \circ r$ is proved inside Theorem 3.1.18 (preprint 3.15) (V3.17).

**V3.19 (Lemma 3.1.20 (preprint 3.17); `Polyfold.IsFillingGerm.add_section_germ`, `Polyfold.IsFillingGerm.a`, `Polyfold.lemma_3_17`).** If $g$ is a filling of $\mathbf{f}$ (V3.7) and $s$ a section of $K \to O$, then $g + s \circ r$ is a filling of $\mathbf{f} + s$, with the same derivative family in (3): for $s$ sc-smooth on $U$, on the same germ; for $s$ an sc-smooth germ on $G_s$, on $G \cap r^{-1}(G_s)$. *Book:* Lemma 3.1.20 (preprint 3.17), p. 99 (preprint p. 72). *Hidden hypotheses:* none. *Deviation (stronger):* the regularity of $g + s \circ r$ is part of the conclusion, and of $s$ only the sc-smoothness is used, not that it is sc$^+$. This is the text's statement for fillings that are germs, proved in the project since the switch to the germ reading (phase 2); the whole-chart `Polyfold.lemma_3_17` (fillers in the sense of `IsFillerOf`, any section $s$, the regularity left to Theorem 3.1.18 (preprint 3.15)) neither implies it nor follows from it (round-trip item V3.19; Findings, Chapter 3, items 220–221, 273). Formerly "restated".

**V3.20 (Definition 3.1.24 (preprint 3.21), good position; `Polyfold.IsGoodComplement`, `Polyfold.IsInGoodPosition`, `Polyfold.IsScGoodComplement`, `Polyfold.IsInGoodPositionSc`,**

`Polyfold.IsInGoodPosition.isInGoodPositionSc`, `Polyfold.isScGoodComplement_iff`, `Polyfold.isInGoodPositionSc_iff`). A subspace $P$ is a good complement of $N$ with respect to $C$ if $E_0 = N \oplus P$ (algebraically) and for some $\varepsilon > 0$, $n \in N$, $p \in P$, $|p|_0 \le \varepsilon |n|_0$ imply $(n \in C \iff n + p \in C)$. $N$ is in good position to $C$ if $N \cap C$ has nonempty interior in $N$ and $N$ has a good complement. Definition 3.1.24 (preprint 3.21) as written (`IsScGoodComplement`: $E = N \oplus P$ an sc-splitting with the constant $\varepsilon$; `IsInGoodPositionSc`) implies this; on a partial quadrant a smooth finite-dimensional $N$ in good position is in good position with an sc-complement, also inside any sc-subspace $T \supseteq N$, and for smooth $N$ the text's good complements are the project's closed ones. *Book:* Definition 3.1.24 (preprint 3.21), pp. 102–103 (preprint pp. 76–77) (an sc-complement, or a topological one). *Hidden hypotheses:* none. *Deviation (stronger):* only the algebraic splitting is required, as a hypothesis. As a conclusion the algebraic form is weaker (an algebraic good complement need not be closed, round-trip item V3.20), so the text's form is formalized as well and proved where the theorems conclude good position (V3.47, V3.94; Findings, Chapter 3, items 210, 211).

**V3.21 (Proposition 3.1.25 (preprint 3.22);** `Polyfold.proposition_3_22`, `Polyfold.exists_quadrantNormalForm`**).** If the finite-dimensional subspace $N$ is in good position to the partial quadrant $C$, then $N \cap C$ is a partial quadrant in $N$. More precisely, for $C = \{\ell_i \ge 0\}$ with continuous functionals jointly onto $\mathbb{R}^n$, there is a linear isomorphism $e : N \to \mathbb{R}^k \times N_0$ carrying $N \cap C$ onto $[0,\infty)^k \times N_0$, with $k = \dim N - \dim(N \cap W)$ and the degeneracy counts preserved off $N \cap W$. *Book:* Proposition 3.1.25 (preprint 3.22), p. 103 (preprint p. 77), proof pp. 143–153 (preprint pp. 120–126). *Hidden hypotheses:* none. *Deviation:* none in the statement. *Proof:* a segment argument and Lemma 3.8.8 (preprint 3.65) instead of the extreme-ray Lemmas 3.8.1–3.8.3, 3.8.9–3.8.11 (preprint 3.62–3.64, 3.66–3.68).

**V3.22 (Definition 3.2.1 (preprint 3.25), the tangent germ property;** `Polyfold.MPolyfold.TangentGermChart`, `Polyfold.MPolyfold.HasTangentGermProperty`, `Polyfold.IsScInftyGermAt`**).** $M \subseteq X$ has the $n$-dimensional tangent germ property if $M \subseteq X_\infty$ and every $x \in M$ has: a chart $\varphi : (U, x) \to (O, 0)$ of $X$ onto a tame retract $(O, C, E)$; an $n$-dimensional smooth $N \subseteq E$ in good position to $C$ with a good complement $Y$ that is **closed**; a relatively open $V \ni 0$ in $N \cap C$; and a continuous $\delta : V \to E$ with values in $Y$, such that $\varphi(M \cap U) = \{v + \delta(v) \mid v \in V\}$, $\delta(0) = 0$, $D\delta(0) = 0$, and for every $m$, $k$ there is a relatively open $V_{m,k} \ni 0$ in $N \cap C$ with $\delta(V_{m,k}) \subseteq E_m$ and $\delta : V_{m,k} \to E_m$ of class $C^k$. *Book:* Definition 3.2.1 (preprint 3.25), p. 104 (preprint p. 79). *Hidden hypotheses:* $Y$ closed (the text's sc-complement is). *Deviation (narrower):* the added closedness; "sc^∞-germ" read with $k$-dependent neighbourhoods, which the fixed-germ and the set-sense readings imply (`Polyfold.IsScSmoothGerm.isScInftyGermAt`, `Polyfold.IsScSmooth.isScInftyGermAt`). The germ implicit function theorem gives the fixed-germ reading as well (V3.31, V3.32); Theorem 3.5.6 (1) (preprint 3.43) is proved in the $k$-dependent one (V3.38).

**V3.23 (Proposition 3.2.2 (preprint 3.26);** `Polyfold.proposition_3_26`, `Polyfold.proposition_3_26_image`, `Polyfold.MPolyfold.TangentGermChart.transport`**).** Let $\psi : X \to X'$ be an sc-diffeomorphism of M-polyfolds (sc-smooth on all of $X$, with an sc-smooth inverse $\psi'$ on all of $X'$) and $\psi(M) = M'$. Then $M'$ has the $n$-dimensional tangent germ property (V3.22) if and only if $M$ has it; the data at $\psi(x)$ are the chart $\varphi \circ \psi^{-1}$ with the same $N$, $Y$, $V$, $\delta$. *Book:* Proposition 3.2.2 (preprint 3.26), p. 105 (preprint p. 80). *Hidden hypotheses:* none. *Deviation:* none (the tameness of $X$ is not used). The text's "$X' = \psi(X)$ is tame" is not needed here; that tameness is carried by sc-diffeomorphisms is V3.71.

**V3.24 (Theorem 3.2.3 (preprint 3.27); `Polyfold.theorem_3_27`, `Polyfold.theorem_3_27_isSubMPolyfold`, `Polyfold.MPolyfold.TangentGermChart.IsRegular`, `Polyfold.MPolyfold.theorem_3_27_isChart`, `Polyfold.MPolyfold.theorem_3_27_isTame`, `Polyfold.MPolyfold.theorem_3_27_isManifoldWithCorners`, `Polyfold.MPolyfold.theorem_3_27_degIndex`).** Let $M \neq \emptyset$ be a subset of an M-polyfold $X$ such that every point of $M$ has data of Definition 3.2.1 (preprint 3.25) (V3.22) that are *regular*: (a) $\delta$ is sc-smooth on $V$ as a map on a set, and (b) the projection $\pi$ along $Y$ maps $O$ into $C$. Then $M$ is a sub-M-polyfold of $X$; its induced M-polyfold structure (Proposition 2.3.19 (preprint 2.21)) is tame and has an equivalent atlas of charts on relatively open subsets of $n$-dimensional partial quadrants; each map $y \mapsto \pi\varphi(y)$ on $M \cap \varphi^{-1}(\Sigma \cap O)$, modelled on $(V, N \cap C, N)$, is a chart of the induced structure containing $x$; and $d_M(y) = d_{N\cap C}(\pi\varphi(y))$ there. *Book:* Theorem 3.2.3 (preprint 3.27), p. 106 (preprint p. 81), proof pp. 106–109 (preprint pp. 81–84). *Hidden hypotheses:* (a) and (b); the text's proof asserts both and each fails in an example (a smooth curve with a vertical tangent written over the horizontal axis; $C = [0,\infty)^3$, $N = \mathbb{R}(1,1,1)$, $Y = \{x + y - z = 0\}$). *Deviation (narrower):* the two added conditions; $X$ need not be tame; the relation $d_{N\cap C} = d_X$ via Lemma 3.8.13 (preprint 3.70) is not transported to a general presentation of $C$. For the solution set of an sc-Fredholm section both conditions are theorems (V3.74, V3.77, V3.85); for any set with the tangent germ property, (a) follows from (b) after shrinking the data (V3.74). With a dimension that may vary from point to point: V3.72.

**V3.25 (Definition 3.3.1 (preprint 3.28), sc⁰-contraction germs; `Polyfold.IsSc0ContractionGerm`, `Polyfold.ContractionGerm`, `Polyfold.IsSc0ContractionGerm.toContractionGerm`).** For a germ $G$ around 0 in $\widetilde{C} \oplus W$: $B$ is an sc⁰-contraction germ if $B$ is an sc⁰-germ, $B(0,0) = 0$, and for every $\varepsilon > 0$ and $m$ there is $r > 0$ such that the points $(a, w)$ with $a \in \widetilde{C}$, $|a| < r$, $|w|_m < r$ lie in $U_m$ and $|B(a,w) - B(a,w')|_m \leq \varepsilon |w - w'|_m$ for such $(a,w)$, $(a,w')$. Every such $B$ has a working form: open $N(0) \supseteq N(1) \supseteq \cdots \ni 0$ in $P$ and $0 \in K(m) \subseteq O(m) \subseteq W_m$ ($K$ closed, $O$ open, $\iota K(m+1) \subseteq K(m)$) with $\{a\} \times O(m) \subseteq U_m$, $B_m(a,\cdot)$ a $\frac{1}{2}$-contraction on $O(m)$ mapping $K(m)$ into itself, for $a \in N(m) \cap \widetilde{C}$. *Book:* Definition 3.3.1 (preprint 3.28), p. 109 (preprint p. 85). *Hidden hypotheses:* the points lie in $U_m$. *Deviation (restated):* explicit regions, needed for Lemma 3.3.4 (preprint 3.31).

**V3.26 (Theorem 3.3.2 (preprint 3.29); `Polyfold.theorem_3_29`, `Polyfold.theorem_3_29_of_isSc0ContractionG`, `Polyfold.exists_fixedPoint_continuousOn`).** For a contraction germ in working form with quadrant-like $\widetilde{C}$: the level-0 fixed point $\delta(a)$ of $B(a,\cdot)$ is an sc⁰-germ on $V_m = N(m) \cap \widetilde{C}$ with $\delta(0) = 0$, $(a, \delta_m(a)) \in U_m$ for $a \in V_m$, $B(a, \delta(a)) = \delta(a)$ on $V_0$, and every $w \in O(0)$ with $B(a,w) = w$, $a \in V_0$, equals $\delta(a)$. From Definition 3.3.1 (preprint 3.28) directly: the same, with uniqueness among small solutions. The parametrized Banach fixed point theorem: for $\frac{1}{2}$-contractions $g_l$ of a nonempty closed $K$ in a complete metric space, continuous in $l$ pointwise, the fixed point is continuous in $l$ and unique in $K$. *Book:* Theorem 3.3.2 (preprint 3.29), p. 110 (preprint p. 86). *Hidden hypotheses:* the uniqueness region. *Deviation (hypotheses explicit).*

**V3.27 (Theorem 3.3.3 (preprint 3.30); `Polyfold.theorem_3_30`, `Polyfold.ContractionGerm.isSc1GermWith_sol`)** If $B$ is an sc¹-germ, the solution germ $\delta$ is a solution germ and an sc¹-germ on $V_m = N(m+1) \cap \widetilde{C}$ (the germ shifted by one level), with $D\delta(a) = (1 - D_2 B(a, \delta(a)))^{-1} D_1 B(a, \delta(a))$. *Book:* Theorem 3.3.3 (preprint 3.30), p. 110 (preprint p. 86), quoted from [21] without proof. *Hidden hypotheses:* none. *Deviation (restated):* sc¹ on the shifted germ; a germ only because $\widetilde{C}$ is finite-dimensional. *Proof:* supplied; it needs $(a_0, \delta(a_0)) \in U_2$, hence the shift.

**V3.28 (Lemma 3.3.4 (preprint 3.31); `Polyfold.ContractionGerm.lemma_3_31`, `Polyfold.ContractionGerm.next`**

`Polyfold.ContractionGerm.isSc0ContractionGerm_next`**).** If $B$ is sc^{k+1} with derivative family $D$ and $\delta$ is sc^j on $(N(m+s)\cap\widetilde{C})_m$, then $f^{(1)}(v,b,u,w) = (u - B(v,u), w - DB(v,\delta(v))(b,w))$, in working form with parameter set $\widetilde{C}\times P$, fibre $TW$ and regions $N(m+2+s)\times P$, $O(m+1)\times W_m$, $K(m+1)\times W_m$, is a contraction germ of class sc^{min(k, j)} whose solution germ is $(v,b)\mapsto(\delta(v), D\delta(v)b)$ on $(N(2+s)\cap\widetilde{C})\times P$. If the working form also satisfies Definition 3.3.1 (preprint 3.28) and $B$ is sc¹, then $f^{(1)}$ satisfies Definition 3.3.1 (preprint 3.28) on its germ, for every $s$ (an $\varepsilon$-contraction near 0 for every $\varepsilon>0$ on every level), by Lemma 3.1.9 (preprint 3.9) on level $m$ (V3.11). *Book:* Lemma 3.3.4 (preprint 3.31), pp. 111 ff. (preprint pp. 86–87). *Hidden hypotheses:* uniqueness of the fixed point only in a region unbounded in the tangent directions; the shifted solution germ (Proposition 1.2.2 (preprint 1.17) for germs). *Deviation (restated):* the working form records a $\frac{1}{2}$-contraction on explicit regions, all the proof of Theorem 3.3.6 (preprint 3.33) uses; the text's "$f^{(1)}$ is an sc⁰-contraction germ" in the sense of Definition 3.3.1 (preprint 3.28) is proved separately (Findings, Chapter 3, item 252).

**V3.29 (Lemma 3.3.5 (preprint 3.32), as an induction;** `Polyfold.ContractionGerm.sol_isSckGerm`**,** `Polyfold.ContractionGerm.isSckGerm_succ_of_next`**).** For every contraction germ over any finite-dimensional parameter space: if $B$ is sc^k and $j\le k$, then $\delta$ is an sc^j-germ on $(N(m+s)\cap\widetilde{C})_m$ for some $s$; the step from $j$ to $j+1$ turns shifts $s$ (for $\delta$) and $s'$ (for $T\delta$) into $s+s'+1$. *Book:* Lemma 3.3.5 (preprint 3.32), p. 111 (preprint p. 87) (the iterated germs $f^{(j)}$ with solution $T^j\delta$). *Hidden hypotheses:* none. *Deviation (restated):* the $f^{(j)}$ are not constructed; one induction quantified over all contraction germs.

**V3.30 (Theorem 3.3.6 (preprint 3.33), the germ implicit function theorem;** `Polyfold.theorem_3_33`**,** `Polyfold.theorem_3_33_smooth`**,** `Polyfold.theorem_3_33_of_isSc0ContractionGerm`**).** If $B$ is an sc^k-germ, there is $s$ such that the solution germ $\delta$ is a solution germ and an sc^k-germ on $(N(m+s)\cap\widetilde{C})_m$; the proof gives $s=2^k-1$. If $B$ is sc-smooth, this holds for every $k$ (with $s$ depending on $k$). From Definition 3.3.1 (preprint 3.28) with balls: there is an sc^k solution germ. *Book:* Theorem 3.3.6 (preprint 3.33), p. 112 (preprint p. 88). *Hidden hypotheses:* **none; the theorem holds with the text's hypotheses**. *Deviation (restated):* the germ on which $\delta$ is sc^k shrinks with $k$; in the text's up-to-shrinking reading this is the theorem.

**V3.31 (Corollary 3.3.7 (preprint 3.34);** `Polyfold.corollary_3_34`**,** `Polyfold.corollary_3_34_isScSmoothGerm`**,** `Polyfold.isSckGerm_of_levels`**).** If $B$ is an sc-smooth germ, then for all $m$, $k$ there is a relatively open $V\ni 0$ in $\widetilde{C}$ with $\delta(V)\subseteq W_m$ and $\delta: V\to W_m$ of class $C^k$. The last sentence on one germ: $\delta$ is an sc-smooth germ (sc^k for every $k$ on the same sequence of sets) on a diagonal germ $V_m = N(m+t_m)\cap\widetilde{C}$, $t_m=\sum_{j\le m}s_j$, and a solution germ there. *Book:* Corollary 3.3.7 (preprint 3.34), p. 113 (preprint p. 88). *Hidden hypotheses:* Proposition 1.2.3 (preprint 1.18) for germs. *Deviation:* none; its last sentence ("$\delta$ is sc-smooth at 0") holds on one germ: sc^j involves only the levels $\ge j$ of a germ (`Polyfold.isSckGerm_of_levels`), so the $k$-dependent germs of Theorem 3.3.6 (preprint 3.33) can be replaced by one diagonal germ, which shrinks fast with the level (Findings, Chapter 3, item 250).

**V3.32 (Theorem 3.35 (preprint);** `Polyfold.theorem_3_35`**,** `Polyfold.theorem_3_35_smooth`**,** `Polyfold.theorem_3_35_isScSmoothGerm`**,** `Polyfold.theorem_3_35_of_isSc0ContractionGerm`**,** `Polyfold.isSckGerm_contractionMap_iff`**).** For $\widetilde{C}$ a partial quadrant in a finite-dimensional $P$: if $B$ is sc^k, $\delta$ is sc^k on $(N(m+s)\cap\widetilde{C})_m$ for some $s$; if $B$ is sc-smooth, then for every $k$ there is such an $s$, and $\delta$ is an sc-smooth germ on one germ $N(m+t_m)\cap\widetilde{C}$. In the text's form: for an

sc⁰-contraction germ $f = w - B(a,w)$ of Definition 3.3.1 (preprint 3.28) on a quadrant-like $\widetilde{C}$ and any representative $(V,\delta)$ of its solution germ, if $f$ is sc^k then $\delta$ is sc^k on a germ $V' \subseteq V$, and if $f$ is sc-smooth then $\delta$ is an sc-smooth germ on one germ $V' \subseteq V$. *Book:* Theorem 3.35 (preprint), preprint p. 89. *Hidden hypotheses:* none. *Deviation (restated):* the conclusion holds on a smaller germ $V' \subseteq V$, as for Theorem 3.3.6 (preprint 3.33) (V3.30); the hypotheses are on $f$ ($f$ is sc^k iff $B$ is); $\mathbb{R}^n$ is generalized to a quadrant-like $\widetilde{C}$ in a finite-dimensional $P$ (Findings, Chapter 3, items 250, 251). Before the third addendum "sc-smooth" held only for every $k$ on a germ depending on $k$.

**V3.33 (Definition 3.4.1 (preprint 3.37), the basic class; `Polyfold.IsBasicGerm`).** As V3.10. *Book:* Definition 3.4.1 (preprint 3.37), p. 114 (preprint p. 90). *Hidden hypotheses:* none. *Deviation:* none.

**V3.34 (Theorem 3.4.2 (preprint 3.38); `Polyfold.theorem_3_38`, `Polyfold.StabilityData`, `Polyfold.StabilityData.exists_of_scPlus`, `Polyfold.StabilityData.pushMap`, `Polyfold.StabilityData.p` `Polyfold.scPlusRemainder_contraction`, `Polyfold.StabilityData.pushMap_eq_zero_iff`, `Polyfold.StabilityData.finrank_Cs_eq_finrank_Zs`).** Let $f$ be a basic germ on $G$ with contraction $B$, $s$ an sc-smooth germ into $W^1$ with $s(0) = 0$ (the $W$-component of an sc⁺-germ), and $W = C \oplus X = R \oplus Z$ an sc-Fredholm splitting of $1 + D_2 s(0)$ (it exists, Proposition 1.1.11 (preprint 1.10)). With $P_1 : W \to X$, $P_2 : W \to R$ the projections and $L = (1 + D_2 s(0))|_X$, the linear maps $\psi(a,w) = ((a,(1-P_1)w), P_1 w)$ and $\Psi(q,w) = ((q,(1-P_2)w), L^{-1}P_2 w)$ carry $f + \iota s$ to the germ $\Psi \circ (f + \iota s) \circ \psi^{-1}$ on the index-raised spaces $(P \oplus C) \oplus X^1 \to (Q \oplus Z) \oplus X^1$, with germ $\psi(U_{m+1})$, which is a basic germ with contraction map $L^{-1}P_2(B - S)$, $S = s - D_2 s(0)$; its zeros correspond to those of $f + \iota s$; and $\dim C = \dim Z$. *Book:* Theorem 3.4.2 (preprint 3.38), pp. 114–116 (preprint pp. 90–93). *Hidden hypotheses:* the operator-norm continuity of the level derivatives of $s$ (for "$S$ is an arbitrarily small contraction"). *Deviation (restated):* a linear coordinate change; only the $W$-component of $s$; the text's $\tau : Z \to C$ not applied. The coordinate change is a strong bundle isomorphism of the index-raised structures (`Polyfold.StabilityData.linIso_isStrongBundleIso`, V3.76).

**V3.35 (Theorem 3.5.1 (preprint 3.40); `Polyfold.theorem_3_40_compactness`, `Polyfold.theorem_3_40_regulari` `Polyfold.IsSc0ContractionGerm.exists_lift`, `Polyfold.theorem_3_12_local`).** Let $B$ be an sc⁰-contraction germ on $G$ around 0 in $\widetilde{C} \oplus W$ with $\widetilde{C}$ **closed**, $\mathbf{s}$ an sc⁰-germ into $W^1$ with $\mathbf{s}(0) = 0$, and $S$ any set of solutions in $U_0$ of $w = B(a,w) - \iota\mathbf{s}(a,w)$. There are open $O(0) \supseteq O(1) \supseteq \cdots \ni 0$ in $E_0$ and compact $K_i \subseteq E_i$ with $\mathrm{cl}_{E_0}(S \cap O(i)) = \iota_i(K_i)$; in particular these closures lie in $E_i$. For a basic germ $h$ and an sc⁰-germ $s$ into $(Q \oplus W)^1$ with $s(0) = 0$, the closures of $\{h + \iota s = 0\} \cap O(i)$ are compact subsets of $E_i$. *Book:* Theorem 3.5.1 (preprint 3.40), p. 118 (preprint p. 94), proof pp. 118–123 (preprint pp. 95–98). *Hidden hypotheses:* $\widetilde{C}$ closed (automatic for a partial quadrant). *Deviation (stronger):* compact closures; $B$ only sc⁰ with the contraction estimate, $\mathbf{s}$ only sc⁰ into $W^1$.

**V3.36 (Corollary 3.5.3 (preprint 3.41); `Polyfold.StabilityData.corollary_3_41`, `Polyfold.StabilityData.ex` `Polyfold.StabilityData.exists_ker_Dfs'_inf_eq_bot`, `Polyfold.isFredholm_add_of_norm_lt`, `Polyfold.HasFiniteIndex.surjective_of_finrank_ker_le`).** For the data of V3.34 with $D(f + \iota s)(0)$ surjective: there is $r > 0$ such that every zero $x$ of $f + \iota s$ in $U_0$ with $|x|_0 < r$ is $\iota(x_1)$ for some $x_1 \in U_1$, and $D(f + \iota s)(x_1) : E_0 \to F_0$ is a surjective Fredholm operator of index $\dim P - \dim Q$. *Book:* Corollary 3.5.3 (preprint 3.41), p. 122 (preprint p. 98). *Hidden hypotheses:* none. *Deviation (restated):* the $\mathbb{R}^N$-component of the sc⁺-map is absorbed into the basic germ. *Proof:* the Fredholm property holds at every $x \in U_1$ small on level 1 (small plus compact perturbation of $D(f + \iota s)(0)$); surjectivity from $\dim\ker \le \mathrm{ind}$.

**V3.37 (Lemma 3.5.5 (preprint 3.42); `Polyfold.StabilityData.lemma_3_42`).** For the push-forward $g = \Phi_*(f + \iota s)$ of V3.34 (on level 0): for every $\varepsilon > 0$ there is $r > 0$ such that every zero $x'$ of $g$ in $\psi(U_0)$ with $|x'|_0 < r$ has its $X$-component $\iota(w_1)$ with $w_1 \in X_1$, $|w_1|_1 < \varepsilon$. *Book:* Lemma 3.5.5 (preprint 3.42), p. 126 (preprint p. 101). *Hidden hypotheses:* none. *Deviation (restated):* the form Theorem 3.5.6 (preprint 3.43) uses; $w' = \delta(a')$ is derived there.

**V3.38 (Theorem 3.5.6 (preprint 3.43); `Polyfold.theorem_3_43_sc_smooth`, `Polyfold.theorem_3_43`, `Polyfold.ShiftedBasicData`, `Polyfold.ShiftedBasicData.Sol.exists_contDiffOn_level`, `Polyfold.exists_proj_mem_of_eq_zero`).** Let $\widetilde{C}$ be a partial quadrant, $g$ an sc-smooth germ on $G$ with $g(0) = 0$ whose index-raised germ is a basic germ, with $Dg(0)$ surjective, $K' = \ker Dg(0)$ in good position to $\widetilde{C} \oplus W$, and such that for every $\varepsilon > 0$ there is $r > 0$ with every zero $x$ of $g$ in $U_0$, $|x|_0 < r$, having $W$-component $\iota(w_1)$, $|w_1|_1 < \varepsilon$. Then there are a good complement $Y$ of $K'$, a relatively open $V \ni 0$ in $K' \cap (\widetilde{C} \oplus W)$ and $\tau : K' \to P \times W_0$ with values in $Y$ and $W$-components on level 1, $C^1$ on $V$, $\tau(0) = 0$, $D\tau(0) = 0$, such that $y + \tau(y) \in U_0$ and $g(y + \tau(y)) = 0$ for $y \in V$; for every $k \geq 1$ a relatively open $V_k \subseteq V$ containing 0 and a $C^k$ map $\tau_k : V_k \to E_k$ with $\iota_k \tau_k = \tau$ on $V_k$; and an open $U'' \ni 0$ in which every zero of $g$ in $U_0$ is $y + \tau(y)$ with $y \in V$. *Book:* Theorem 3.5.6 (preprint 3.43), p. 126 (preprint p. 102). *Hidden hypotheses:* the regularity of Lemma 3.5.5 (preprint 3.42) (made a hypothesis); the projection of a solution to $\widetilde{K}$ lies in $\widetilde{C}$ (a gap in the text, proved); that the solutions given by Theorem 3.8.22 (preprint 3.78) at the different regularities $k$ coincide near 0 (uniqueness clause). *Deviation (restated):* clause (1), "$\tau$ is an sc-smooth germ", in the level-wise sense with $k$-dependent neighbourhoods. With $Y$ closed and its projection onto $K'$ mapping $\widetilde{C} \oplus W$ into itself: V3.77.

**V3.39 (Lemma 3.5.8 (preprint 3.44); `Polyfold.ShiftedBasicData.Deriv.ker_Dg_eq`).** In the setting of V3.38, $\ker Dg(0) = \{(\alpha, D\delta(0)\alpha) \mid \alpha \in \ker DH(0)\}$, $H(a) = \operatorname{pr}_Q g(a, \delta(a))$. *Book:* Lemma 3.5.8 (preprint 3.44), p. 127 (preprint p. 103). *Hidden hypotheses:* $D_2 B(0) = 0$ on level 0, by density from level 1. *Deviation:* none.

**V3.40 (Lemma 3.5.9 (preprint 3.45); `Polyfold.IsGoodComplement.map`, `Polyfold.IsInGoodPosition.map`).** A continuous linear isomorphism $S : E_0 \to E_0'$ with $x \in C \iff Sx \in C'$ carries a good complement $P$ of $N$ to a good complement $S(P)$ of $S(N)$ (constant $\varepsilon/(\|S\|\|S^{-1}\| + 1)$), and good position to good position. *Book:* Lemma 3.5.9 (preprint 3.45), p. 128 (preprint p. 104) (for $T\varphi(0)$). *Hidden hypotheses:* none. *Deviation (stronger):* any such $S$.

**V3.41 (Lemma 3.5.10 (preprint 3.46); `Polyfold.lemma_3_46`, `Polyfold.lemma_3_46_isInGoodPosition`).** If $K' = \{(\alpha, L\alpha) \mid \alpha \in \widetilde{K}\}$ has a good complement with respect to $\widetilde{C} \oplus W$, then $\widetilde{K}$ has a good complement $Z$ with respect to $\widetilde{C}$ such that $Z \times W$ is a good complement of $K'$; good position of $K'$ gives good position of $\widetilde{K}$. *Book:* Lemma 3.5.10 (preprint 3.46), p. 129 (preprint p. 104), proof pp. 155–161 (preprint pp. 129–132). *Hidden hypotheses:* none. *Deviation:* none (the proof avoids the intermediate complement $Y_0$).

**V3.42 (Theorem 3.5.11 (preprint Corollary 3.47); `Polyfold.theorem_3_47_sc_smooth`, `Polyfold.theorem_3_47`).** For $\widetilde{C}$ a partial quadrant, a basic germ $f$ and the data of V3.34 with $D(f + \iota s)(0)$ surjective and kernel $K$ in good position to $\widetilde{C} \oplus W$: there are a good complement $Y$ of $K$, a relatively open $V \ni 0$ in $K \cap (\widetilde{C} \oplus W)$, and $\sigma$ with values in $Y$ and $W$-components on level 1, $C^1$ on $V$, $\sigma(0) = 0$, $D\sigma(0) = 0$, with $v + \sigma(v) \in U_0$ a zero for $v \in V$; for every $k \geq 1$ a $C^k$ lift $\sigma_k : V_k \to E_k$ of $\sigma$ on a relatively open $V_k \subseteq V$ containing 0; and an open $U'' \ni 0$ in which every zero in $U_0$ is of this form. *Book:* Theorem 3.5.11 (preprint Corollary 3.47),

p. 131 (preprint p. 106). *Hidden hypotheses:* none. *Deviation (restated):* (1) level-wise, as in V3.38. *Proof:* Theorem 3.5.6 (preprint 3.43) for the push-forward, transported back linearly; Lemma 3.8.13 (preprint 3.70) and Theorem 3.8.15 (preprint 3.72) are not needed. With $Y$ closed and its projection onto $K$ mapping $\widetilde{C} \oplus W$ into itself: V3.77.

**V3.43 (Definition 3.6.1 (preprint 3.48); `Polyfold.IsFillingGerm`, `Polyfold.IsFillerOf`).** As V3.7. *Book:* Definition 3.6.1 (preprint 3.48), p. 134 (preprint p. 109). *Hidden hypotheses:* none. *Deviation:* none. Since the switch to the text's germ reading (phase 2) the filling is the germ notion of V3.7; the stricter whole-chart filler (`IsFillerOf`, with `IsFilledVersion`), for which "filled version = filling" on a given $U$ is false (round-trip item V3.43), is kept (Findings, Chapter 3, items 220–221, 330). Formerly "narrower", for the whole-chart reading.

**V3.44 (Remark 3.6.2 (preprint 3.49); `Polyfold.IsFillerOf.remark_3_49`).** For a filler there is an open $W \ni 0$ with $f = g$ on $O \cap W$ and, for $y \in U \cap W$, $g(y) = \rho(r(y))g(y)$ only if $y \in O$. *Book:* Remark 3.6.2 (preprint 3.49), p. 135 (preprint p. 109). *Hidden hypotheses:* none. *Deviation:* none.

**V3.45 (Definitions 3.6.3, 3.6.4 (preprint 3.50, 3.51); `Polyfold.StrongBundle.IsScFredholmGerm`, `Polyfold.StrongBundle.IsScFredholmSection`, `Polyfold.StrongBundle.IsScFredholmGermTame`, `Polyfold.StrongBundle.IsScFredholmSectionTame`, `Polyfold.StrongBundle.IsScFredholmGermTameWholeCh`** As V3.13 and V3.16, with the chart of the germ data tame: V3.103. *Book:* Definitions 3.6.3, 3.6.4 (preprint 3.50, 3.51), p. 135 (preprint p. 110) (for a tame strong bundle; the chart of Definition 3.6.3 (preprint 3.50) has a tame local strong bundle $K \to O$). *Hidden hypotheses:* none. *Deviation:* none. The forms with a tame chart (V3.103) are the ones the theorems of §3.6 use; since the switch to the text's germ reading (phase 2) they take the filling and the sc$^+$-section as germs, as the text does, and the earlier whole-chart forms (`…WholeChart`), which imply them (round-trip item V3.45), are kept (Findings, Chapter 3, items 220–221, 330). The forms of V3.13 and V3.16 drop the tameness, which §3.1 does not need. Formerly "narrower", for the whole-chart reading.

**V3.46 (Proposition 3.6.5 (preprint 3.52); `Polyfold.proposition_3_52_index`, `Polyfold.proposition_3_52_loc`** `Polyfold.ConjugationData`, `Polyfold.ConjugationData.scDeriv_isFredholm_index`, `Polyfold.ConjugationData.index_eq`, `Polyfold.proposition_3_52`, `Polyfold.proposition_3_52_local`, `Polyfold.proposition_3_52_sc`, `Polyfold.proposition_3_52_local_sc`, `Polyfold.ConjugationData.scDeri` `Polyfold.StrongBundle.linAt_isScFredholm`**).** Let $f$ be a section of a strong bundle with $(f, x_0)$ an sc-Fredholm germ and $f(x_0) = 0$. In the strong bundle chart of Definition 3.1.11 (preprint 3.11) centred at $x_0$, with the filling germ $g$ (a derivative family $Dg$ on a germ inside $U$) and the local sc$^+$-germ $s$ for which $g - s$ is conjugated to a basic germ, $Dg(0) : E_0 \to F_0$ and $Tf(x_0) := Dg(0)|_{Dr(0)E_0} : Dr(0)E_0 \to \rho(0)F_0$ are Fredholm operators, and for **every** conjugation of $g - s$ to a basic germ $h : O(\widetilde{C} \oplus W, 0) \to \mathbb{R}^N \oplus W$, $\widetilde{C} \subseteq \mathbb{R}^n$, both have index $n - N$ (`ConjugationData`: the witnesses of a conjugation, Definition 3.1.11 (preprint 3.11), as data). In particular $n - N$ is the same for all conjugations of one germ: for $k$ sc-smooth with $k(0) = 0$ and any conjugation of $k$ to a basic germ, $Dk(0)$ is Fredholm of index $n - N$. They are sc-Fredholm operators: $Dg(0) : E \to F$ is an sc-operator, $T_0O = Dr(0)E$ and $K_0 = \rho(0)F$ are sc-subspaces, and $Tf(x_0) : T_0O \to K_0$ is sc-Fredholm, both of index $n - N$ on every level; for an sc-Fredholm section the same holds in every strong bundle chart around a zero $y$, at the point $z$ over $y$, for $\mathrm{Lin}(z) : T_zO \to K_z$. *Book:* Proposition 3.6.5 (preprint 3.52), p. 135 (preprint p. 110) ($Tf(x) : T_xX \to Y_x$ is sc-Fredholm); the index $n - N$ is used in the proof of Theorem 3.6.8 (preprint 3.54). *Hidden hypotheses:* none. *Deviation (restated):* in charts ($T_xX$ carries no linear structure here, Chapter 2). The hypothesis is the text's since the switch to the germ reading (phase 2): the filling and the sc$^+$-section are germs (V3.13), and $Dg(0)$ is the value at 0

of any derivative family of the filling germ; the forms for germ data on the whole chart are the `_wholeChart` declarations of the same names (Findings, Chapter 3, items 220–221, 289, 310, 331). Formerly "narrower", for that hypothesis. The sc-Fredholm property, first proved on level 0 only, follows the text's argument with two steps it leaves out: $D(g-s)(0) = \Gamma(0)^{-1}Dh(0)D\psi(0)$ is sc-Fredholm by the criterion of V1.8, and $T_0O$, $K_0$ are sc-subspaces (Definition 2.6.1 (preprint 2.47) at the smooth point 0, the closed graph theorem); in other charts the levels are transported by the chart change at smooth points (Findings, Chapter 3, items 235, 239). The first version (`Polyfold.proposition_3_52`, kept for its consumers) states the index as "$n-N$ for some natural numbers $n$, $N$", which carries no information; `proposition_3_52_index` identifies $n$ and $N$ with the dimensions of the conjugating basic germ (Findings, Chapter 3, item 120). In every other strong bundle chart around $x_0$ the linearization is Fredholm of the same index (V3.96).

**V3.47 (Definition 3.6.6 (preprint 3.53); `Polyfold.IsGoodPositionAtGerm`, `Polyfold.IsGoodPositionAt`, `Polyfold.IsInGoodPositionIn`, `Polyfold.StrongBundle.IsGoodPositionGerm`, `Polyfold.StrongBundle.IsGo` `Polyfold.IsScGoodComplementIn`, `Polyfold.IsInGoodPositionScIn`, `Polyfold.IsGoodPositionAtScGerm`, `Polyfold.IsGoodPositionAtSc`, `Polyfold.StrongBundle.IsGoodPositionGermSc`, `Polyfold.StrongBundle.Is` `Polyfold.StrongBundle.isGoodPositionGermSc_iff`).** In a chart centred at the zero, with the filling germ $g$ and $Dg(0)$ the value at 0 of a derivative family of $g$ (the same for all): the germ is in good position if $Tf(0) = Dg(0)|_{T_0O} : T_0O \to K_0$ is onto and $\ker Dg(0)$ is in good position to $T_0O \cap C$ inside $T_0O$ (interior of $\ker Dg(0) \cap C$ in $\ker Dg(0)$ nonempty, a good complement inside $T_0O$). For a section, $(f, x_0)$ is in good position if it has the data of Definition 3.1.11 (preprint 3.11) at $x_0$ with the germ in good position in that chart. With the text's sc-complement (`IsGoodPositionAtSc`; for a section `IsGoodPositionGermSc`, in a tame chart) the kernel has a good sc-complement inside $T_0O$ ($T_m = N_m \oplus P_m$ on every level); at a zero of an sc-Fredholm section with tame germ data the two forms agree. *Book:* Definition 3.6.6 (preprint 3.53), p. 136 (preprint p. 111). *Hidden hypotheses:* none. *Deviation (restated):* in a chart rather than in $T_xX$; no case distinction $d_X(x) \geq 1$, which is recovered as a theorem for tame germ data: at a zero with $d_X = 0$ good position in a tame chart is the surjectivity of the linearization (V3.100). At a zero the linearization and the two conditions do not depend on the chart (V3.78); Definition 3.6.6 (preprint 3.53) at a zero from a chart not centred there is V3.79, and in any chart of the germ data V3.107. The form in a tame chart, which the theorems of §3.6 assume and conclude, is V3.103. The germ data are the text's since the switch to the germ reading (phase 2, V3.13); the whole-chart forms are `IsGoodPositionGermWholeChart`, `IsGoodPositionGermScWholeChart` (Findings, Chapter 3, items 220–221, 280, 320, 330); that "an sc-splitting of $T_0O$", written in $E$, is an sc-splitting of the sc-Banach space $T_0O$ is not formalized (item 214).

**V3.48 (Theorem 3.6.8 (preprint 3.54), the graph description of the solution set; `Polyfold.theorem_3_54_graph`, `Polyfold.theorem_3_54_linear`, `Polyfold.theorem_3_54_pushforward`, `Polyfold.scDeriv_mem_iff_of_inve`** Let $f$ be a section of a strong bundle, with $f(x_0) = 0$ and $(f, x_0)$ in good position in a tame chart (V3.103). Then there are: a tame strong bundle chart $\varphi$ centred at $x_0$ (compatible with the atlas, with tame base retraction); an sc-smooth $\psi$ on its $U$ with $\psi(0) = 0$ into a relatively open $U'$ of $\widetilde{C} \oplus W$ ($\widetilde{C}$ a partial quadrant), with an sc-smooth inverse $\psi'$ on $U'$; the data of Theorem 3.4.2 (preprint 3.38) and a basic germ $h'$ with $D(h' + \iota t)(0)$ surjective and kernel $K$ in good position, $K$ a smooth subspace; a closed good complement $Y$ of $K$; a relatively open $V' \ni 0$ in $K \cap (\widetilde{C} \oplus W)$; $\sigma$ with values in $Y$ and $W$-components on level 1, $C^1$ on $V'$, $\sigma(0) = 0$, $D\sigma(0) = 0$, such that every $v + \sigma(v)$, $v \in V'$, is a zero of $h' + \iota t$ in its germ, and for every $k \geq 1$ a $C^k$-lift $\sigma_k : V_k \to E'_k$ of $\sigma$ on a relatively open $V_k \ni 0$, $V_k \subseteq V'$ (Theorem 3.5.11 (1) (preprint Corollary 3.47)); and an open $V_x \ni x_0$ in the chart domain on which $\psi\varphi$ is injective, $f(y) = 0 \iff (h' + \iota t)(\psi\varphi(y)) = 0$,

and $\psi\varphi(\{y \in V_x \mid f(y) = 0\}) = \{v + \sigma(v) \mid v \in V'\}$. On the way: in the chart, $Dg(0)$ is surjective with kernel in good position to $C$ in $E$, $x \in C \iff D\psi(0)x \in C'$, and $D(\Psi_* g)(0) = \Gamma(0)Dg(0)D\psi(0)^{-1}$ is surjective Fredholm with kernel in good position. *Book:* Theorem 3.6.8 (preprint 3.54), p. 136 (preprint p. 111), the step (3) of its proof, pp. 137–139 (preprint pp. 113–114). *Hidden hypotheses:* the tameness of the chart at $x_0$, used only as $(1 - Dr(0))E \subseteq E_0$; the neighbourhood $V$ made explicit; the kernel is smooth (the text: without comment). *Deviation (narrower):* only the graph description (3) of the proof; conclusions (1) and (2) of the theorem are V3.82 and V3.85, the choice of $Y$ with the projection property V3.77.

**V3.49 (Lemma 3.6.9 (preprint 3.55); `Polyfold.lemma_3_55`, `Polyfold.lemma_3_55_core`).** In $E = \mathbb{R}^n \oplus W$, $C = [0,\infty)^n \oplus W$: let $N$ be finite-dimensional in good position with good complement $Y$, $Y$ **closed**, and $\tau : N \to E$ with values in $Y$, $C^1$ on $N \cap C$, $\tau(0) = 0$, $D\tau(0) = 0$. There is $\rho > 0$ such that for $v \in N \cap C$ with $|v| < \rho$: $v + \tau(v) \in C$, and $Y$ is a good complement of $N_v = \{n + D\tau(v)n \mid n \in N\}$ with respect to the partial quadrant $C_{v+\tau(v)}$ (so $E_0 = N_v \oplus Y$, both closed, with the good-complement constant independent of $v$). The core holds for any $C$ cut out by coordinate functionals. *Book:* Lemma 3.6.9 (preprint 3.55), p. 139 (preprint p. 114), proof pp. 160–164 (preprint pp. 132–135). *Hidden hypotheses:* $Y$ closed (needed for $D\tau(v)n \in Y$). *Deviation (narrower):* the added closedness; $\tau$ given on all of $N \cap C$. The form used in Theorem 3.6.8 (preprint 3.54) — for a relatively open $V \subseteq N \cap C$ and any presentation of $C$, with the good position of $N_v$ — is `Polyfold.lemma_3_55_relOpen` (V3.80).

**V3.50 (Theorem 3.6.10 (preprint 3.56) from the tangent germ property; `Polyfold.theorem_3_56_of_tangentGer`**
If $X$ has no boundary and the nonempty zero set $S$ of $f$ has at every point regular data of Definition 3.2.1 (preprint 3.25) (V3.24), then $S$ is a sub-M-polyfold whose induced structure is an $n$-dimensional manifold with corners without boundary. *Book:* Theorem 3.6.10 (preprint 3.56), p. 140 (preprint p. 115). *Hidden hypotheses:* the regularity conditions of V3.24. *Deviation (narrower):* assumes the tangent germ property of $S$, which the text derives from Theorem 3.6.8 (preprint 3.54). The theorem with the text's hypotheses is V3.86.

**V3.51 (Theorem 3.6.11 (preprint 3.57) from the tangent germ property; `Polyfold.theorem_3_57_of_tangentGer`**
If the nonempty zero set $S$ of $f$ has at every point regular data of Definition 3.2.1 (preprint 3.25), then $S$ is a sub-M-polyfold whose induced structure is tame and an $n$-dimensional manifold with boundary with corners. *Book:* Theorem 3.6.11 (preprint 3.57), p. 140 (preprint p. 115). *Hidden hypotheses:* as V3.50. *Deviation (narrower):* as V3.50. The theorem with the text's hypotheses is V3.87.

**V3.52 (Lemma 3.7.2 (preprint 3.59), abstract form; `Polyfold.lemma_3_59`, `Polyfold.FamilyCompactness`).** Let $b \mapsto L(b)$ be a family of sc-operators $X \to Y$, $b$ in a finite-dimensional space, with $(b,h) \mapsto L(b)h$ continuous on $O \times X_0$ for a set $O \ni 0$, $L(0)$ injective on $X_0$, and the compactness property on level 0: if $b_j \to b$ in $O$, $(h_j)$ is bounded in $X_0$ and $L(b_j)h_j \to 0$, then $(h_j)$ has a convergent subsequence. Then $L(b)$ is injective on $X_0$ for $b \in O$ near 0. *Book:* Lemma 3.7.2 (preprint 3.59), p. 141 (preprint p. 117) (for $L(b) = P_Y \circ Df(b)|_X$ in the setting of Theorem 3.7.1 (preprint 3.58)). *Hidden hypotheses:* none. *Deviation (restated):* for an abstract family; in the proof of Theorem 3.7.1 (preprint 3.58) its hypotheses are derived from those of the theorem for $L(b) = P_Y \circ Df(b,0)|_X$ (`Polyfold.familyCompactness_derivFamOp`, `Polyfold.continuousOn_derivFamOp`, V3.89).

**V3.53 (Lemma 3.7.3 (preprint 3.60), abstract form; `Polyfold.lemma_3_60`).** For a compact set $K$ of parameters, a family of sc-operators $L(b)$ with $(b,h) \mapsto L(b)_m h$ continuous on $K \times X_m$, every $L(b)_m$, $b \in K$, injective, and the compactness property on level $m$: there is $c_m > 0$ with $c_m|h|_m \le |L(b)h|_m$ for all $b \in K$, $h \in X_m$. *Book:* Lemma 3.7.3 (preprint 3.60), p. 142 (preprint p. 118).

*Hidden hypotheses:* none. *Deviation (restated):* abstract form; the injectivity on level $m$ is a hypothesis, which in Theorem 3.7.1 (preprint 3.58) comes from level 0 (`Polyfold.ScOperator.injective_lvl`, V3.89).

**V3.54 (Lemma 3.7.4 (preprint 3.61); `Polyfold.lemma_3_61`).** Let $V$ be a germ of relatively open neighbourhoods of 0 in a partial quadrant $\widetilde{C}$ of a finite-dimensional $G$, and $L(v) : E \to F$ sc-isomorphisms such that $(v,h) \mapsto L(v)h$ is an sc-smooth germ on $V \oplus E$ (levels $V_m \times E_m$) and for every $m$ there is $c_m > 0$ with $c_m|h|_m \le |L(v)h|_m$ for $v \in V_0$, $h \in E_m$. Then $(v,k) \mapsto L(v)^{-1}k$ is an sc-smooth germ on $V \oplus F$. *Book:* Lemma 3.7.4 (preprint 3.61), p. 142 (preprint p. 118), quoted from [25] without proof. *Hidden hypotheses:* none (no operator-norm continuity of $v \mapsto L(v)$). *Deviation (restated):* for germs. *Proof:* supplied; the differentiability uses the compactness of $F_1 \to F_0$. Used in Theorem 3.7.1 (preprint 3.58) in set form (`Polyfold.famInv_isScSmooth_of`, V3.88).

**V3.55 (Lemma 3.8.8 (preprint 3.65); `Polyfold.lemma_3_65`).** For functionals $\ell_i$ cutting out $C$, a good complement $P$ of $N$ (constant $\varepsilon$) satisfies $\ell_i(p) = 0$ for $p \in P$ and every $i$ vanishing at some nonzero point of $N \cap C$. *Book:* Lemma 3.8.8 (preprint 3.65), p. 150 (preprint p. 123). *Hidden hypotheses:* none. *Deviation:* none.

**V3.56 (Proposition 3.8.12 (preprint 3.69); `Polyfold.proposition_3_69`).** For $N$ finite-dimensional in good position to $C = [0,\infty)^n \oplus W$ in $\mathbb{R}^n \oplus W$ and $x \in N \cap C$: $d_{N\cap C}(x) = d_C(x)$ if $x \notin N \cap W$, and $d_{N\cap C}(x) + \dim(N \cap W) = \dim N$ if $x \in N \cap W$. *Book:* Proposition 3.8.12 (preprint 3.69), p. 153 (preprint p. 126). *Hidden hypotheses:* none. *Deviation (restated):* (2) additively.

**V3.57 (Lemma 3.8.13 (preprint 3.70); `Polyfold.lemma_3_70`).** For $N$ in good position to $C = [0,\infty)^n \oplus W$ with good complement $Y$ and $\tau : N \to E$ with values in $Y$, $C^1$ on $N \cap C$, $\tau(0) = 0$, $D\tau(0) = 0$: there is $\rho > 0$ such that for $v \in N \cap C$, $|v| < \rho$, $v + \tau(v) \in C$, and $d_{N\cap C}(v) = d_C(v + \tau(v))$ if $v \notin N \cap W$. *Book:* Lemma 3.8.13 (preprint 3.70), p. 154 (preprint p. 127). *Hidden hypotheses:* none. *Deviation (hypotheses explicit):* "close to 0" made explicit.

**V3.58 (Theorem 3.8.15 (preprint 3.72); `Polyfold.theorem_3_72`).** For $C$ a partial quadrant in a finite-dimensional $V$, $U$ a relatively open neighbourhood of 0 in $C$ and $f$ of class $C^j$, $j \ge 1$, on $U$ with $f(U) \subseteq C$, $f(0) = 0$, $Df(0)C = C$ and $d_C(x) = d_C(f(x))$ on $U$: there are relatively open $U' \subseteq U$ and $V'$ containing 0 with $f(U') = V'$, $f$ injective on $U'$, and an inverse $g$ of class $C^j$ on $V'$. *Book:* Theorem 3.8.15 (preprint 3.72), p. 165 (preprint p. 136) (smooth). *Hidden hypotheses:* none. *Deviation (stronger):* $C^j$; the degeneracy hypothesis used only as "interior iff interior".

**V3.59 (Proposition 3.8.17 (preprint 3.73); `Polyfold.proposition_3_73`, `Polyfold.InvData`).** For $U$ relatively open and convex in $C$, $f : U \to C$ of class $C^j$, $j \ge 1$, $\|Df - 1\| \le \frac{1}{2}$ on $U$, and $x$ interior iff $f(x)$ interior: $f(U)$ is relatively open in $C$, $f$ is injective on $U$, with an inverse of class $C^j$ on $f(U)$. *Book:* Proposition 3.8.17 (preprint 3.73), p. 165 (preprint p. 137). *Hidden hypotheses:* none. *Deviation (stronger):* $0 \in U$, $f(0) = 0$ not needed; (3) weakened. With the hypotheses its proof uses, without the partial quadrant: V3.90.

**V3.60 (Lemma 3.8.18 (preprint 3.74); `Polyfold.InvData.injOn`, `Polyfold.InvData.ball_inter_subset_image`, `Polyfold.InvData.image_relOpen`).** Under the hypotheses of V3.59: $\frac{1}{2}|x-x'| \le |f(x)-f(x')| \le \frac{3}{2}|x-x'|$, $f$ is injective, $B(x',3r)\cap C \subseteq U$ implies $B(f(x'),r)\cap C \subseteq f(U)$, and $f(U)$ is relatively open in $C$. *Book:* Lemma 3.8.18 (preprint 3.74), p. 165 (preprint p. 137). *Hidden hypotheses:* the interior starting point with $f(x_0) \in B(y',r)$. *Deviation:* none in the statement (openness of $f$ on subsets not stated separately).

**V3.61 (Corollary 3.8.19 (preprint 3.75); `Polyfold.corollary_3_75`).** Under the hypotheses of

V3.59, $f : U \to f(U)$ is a homeomorphism. *Book:* Corollary 3.8.19 (preprint 3.75), p. 167 (preprint p. 139). *Hidden hypotheses:* none. *Deviation:* none.

**V3.62 (Lemma 3.8.21 (preprint 3.77); `Polyfold.InvData.hasFDerivWithinAt_inv`).** Under the hypotheses of V3.59 the inverse is differentiable within $f(U)$ with $Dg(y) = [Df(g(y))]^{-1}$. *Book:* Lemma 3.8.21 (preprint 3.77), p. 168 (preprint p. 140) (for any homeomorphism $f : U \to V$ with $V$ relatively open and $Df$ invertible). *Hidden hypotheses:* none. *Deviation (narrower):* under the hypotheses of Proposition 3.8.17 (preprint 3.73), which imply the lemma's. With the hypotheses its proof uses, without the partial quadrant: V3.90.

**V3.63 (Theorem 3.8.22 (preprint 3.78); `Polyfold.theorem_3_78`, `Polyfold.ContractionData.exists_solution`).** For a partial quadrant $C$ in a finite-dimensional $V$, $U$ a relatively open neighbourhood of 0 in $C$, $f : U \to F$ of class $C^j$, $j \geq 1$, $f(0) = 0$, $Df(0)$ onto, $K = \ker Df(0)$ in good position with good complement $Y$: there are a relatively open $V' \ni 0$ in $K \cap C$, $\tau : K \to Y$ of class $C^j$ on $V'$, $\varepsilon, \sigma > 0$ with (1) $\tau(0) = 0$, $D\tau(0) = 0$, $k + \tau(k) \in U$ and $f(k + \tau(k)) = 0$ for $k \in V'$; (2) $|\tau(k)| \leq \sigma$ for $k \in V'$, $|k| \leq \varepsilon$; (3) $k \in V'$, $|k| \leq \varepsilon$, $y \in Y$, $|y| \leq \sigma$, $k + y \in U$, $f(k + y) = 0$ imply $y = \tau(k)$. *Book:* Theorem 3.8.22 (preprint 3.78), p. 169 (preprint p. 141). *Hidden hypotheses:* $f(k + \tau(k)) = 0$ and $k + y \in U$, omitted in the text. *Deviation (stronger):* the two clauses; $F$ any normed space; only convexity and interior of $C$ used.

**V3.64 (Lemma 3.8.23 (preprint 3.79); `Polyfold.ContractionData.exists_radii`, `Polyfold.Radii.eq_`).** In the normalized data there are $\varepsilon, a, b > 0$, $\varepsilon a \leq b$, with the region $R = \{|k| \leq a, |y| \leq b\} \cap C \subseteq U$ and $\|DB\| \leq \varepsilon/(2(1+\varepsilon))$ on $R$; a fixed point $y$ of $B(k, \cdot)$ with $(k, y) \in R$ is $\tau(k)$. *Book:* Lemma 3.8.23 (preprint 3.79), p. 170 (preprint p. 142). *Hidden hypotheses:* none. *Deviation:* none.

**V3.65 (Lemma 3.8.24 (preprint 3.80); `Polyfold.Radii.mapsTo_ball`).** $B(k, \cdot)$ maps $\{|y| \leq \varepsilon|k|\}$ into itself for $k \in K \cap C$, $|k| < a$, and has a fixed point there. *Book:* Lemma 3.8.24 (preprint 3.80), p. 170 (preprint p. 143). *Hidden hypotheses:* the maximum is over $B \cap C$. *Deviation (hypotheses explicit).*

**V3.66 (Lemma 3.8.25 (preprint 3.81); `Polyfold.Radii.norm_ _sub_le`, `Polyfold.Radii.norm_ _sub_le_global`, `Polyfold.Radii.norm_ _sub_le_eps`, `Polyfold.Radii.lipschitzOnWith_`).** $|\tau(k)| \leq \varepsilon|k|/2$, $\tau$ is continuous, and $|\tau(k) - \tau(k_0)| \leq \frac{\varepsilon}{1+\varepsilon}|k - k_0| \leq \varepsilon|k - k_0|$ for all $k, k_0 \in V_0$ (an older form with the constant $1 + 2\varepsilon$ is kept for the proofs that use it). *Book:* Lemma 3.8.25 (preprint 3.81), p. 171 (preprint p. 144) (constant $\varepsilon$). *Hidden hypotheses:* none. *Deviation (stronger):* the sharper constant $\varepsilon/(1+\varepsilon)$. The text's argument uses the point $(k, \tau(k_0))$, which lies in the region of the contraction when $|k_0| \leq 2|k|$; otherwise $|k| < 2|k_0|$ and the same argument with $k$ and $k_0$ exchanged applies, a case distinction the text omits (Findings, Chapter 3, item 216). Formerly "narrower", with the constant $1 + 2\varepsilon$.

**V3.67 (Lemma 3.8.26 (preprint 3.82); `Polyfold.Radii.contDiffOn_`).** $\tau$ is differentiable on $V_0 = \{k \in K \cap C : |k| < a\}$ with $D\tau(k) = (1 - D_2B(k, \tau(k)))^{-1} D_1B(k, \tau(k))$ and of class $C^j$ on $V_0$. *Book:* Lemma 3.8.26 (preprint 3.82), p. 172 (preprint p. 144) ("on a possibly smaller neighbourhood"). *Hidden hypotheses:* none. *Deviation (stronger):* no shrinking.

**V3.68 (Theorem 3.1.22 (preprint 3.19), the implicit function theorem, interior case, first form; `Polyfold.theorem_3_19`, `Polyfold.theorem_3_19_linearization`, `Polyfold.MPolyfold.IsWithoutBoundary`, `Polyfold.theorem_3_19_linearization_sc`, `Polyfold.StrongBundle.linAt_isScFredholm`).** Let $X$ be an M-polyfold with $d_X \equiv 0$, $f$ an sc-Fredholm section (V3.16) of a strong bundle $P : Y \to X$ with tame germ data (V3.103), $f(x_0) = 0$, and let $(f, x_0)$ be in good position in a tame chart (V3.47, V3.103) **with a filler that satisfies condition (3) of Definition 3.1.4**

**(preprint 3.4) at the zeros of the filler near** 0 (V3.82). Then there are an open $U \ni x_0$ and $d \in \mathbb{N}$ such that every zero $y \in U$ is a smooth point with $(f, y)$ in good position in a tame chart (in particular $f'(y)$ is onto), and $S(f, U) = \{y \in U \mid f(y) = 0\}$ is a sub-M-polyfold of $X$ whose induced structure (Proposition 2.3.19 (preprint 2.21)) is equivalent to that of a $d$-dimensional manifold with boundary with corners and has degeneracy index 0 at every point. The first clause: for an sc-Fredholm section and a zero $x_0$ in good position, in the chart of Definition 3.1.11 (preprint 3.11) centred at $x_0$, $Dg(0)$ and $f'(x_0) = Tf(x_0) = Dg(0)|_{T_0O} : T_0O \to K_0$ are sc-Fredholm operators with the same index $n - N$ on every level, and $Tf(x_0)$ is onto on every level (`Polyfold.theorem_3_19_linearization_sc`; Proposition 3.6.5 (preprint 3.52) in sc form, V3.46); with no chart singled out, in every strong bundle chart around $x_0$ the linearization $\mathrm{Lin}(z) : T_zO \to K_z$ is an sc-Fredholm operator with the same index on every level (`Polyfold.StrongBundle.linAt_isScFredholm`, V3.96). *Book:* Theorem 3.1.22 (preprint 3.19), p. 100 (preprint p. 74) ("$f'(x_0)$ surjective"; the local solution set is a manifold of dimension the Fredholm index, $f'(y)$ onto and $\ker f'(y) = T_yS(f, U)$). *Hidden hypotheses:* tameness of the bundle, read as Definitions 3.6.3, 3.6.4 and 3.6.6 (preprint 3.50, 3.51 and 3.53) read it: germ data and good position in charts with tame base retraction (V3.103; the reading "every chart compatible with the atlas has a tame base retraction", under which this item was first proved, cannot be met, V3.104). *Deviation (narrower):* (i) the filler condition at the nearby zeros, through Theorem 3.6.8 (1) (preprint 3.54) (V3.82), which was an added hypothesis when this form was proved and is now a theorem: it holds whenever $(f, x_0)$ is in good position in a tame chart (V3.102), so this form has the text's hypotheses after all, and `Polyfold.theorem_3_19'` states it without the condition (V3.95); (ii) the hypothesis "$f'(x_0)$ surjective" is read as good position (Definition 3.6.6 (preprint 3.53) in the chart form), which at $d_X = 0$ is the same (V3.100; the form with "$f'(x_0)$ surjective" is V3.101); (iii) $d$ is existentially quantified here; "the dimension agrees with the Fredholm index" is V3.97; (iv) "$\ker f'(y) = T_yS$" in the chart is V3.99; (v) $f'(x_0)$ in charts (V3.46); its sc-Fredholm property, first proved only on level 0, holds on every level. *Proof:* Theorem 3.1.26 (preprint 3.23) (V3.69) and Lemma 2.4.6 (preprint 2.26) ($d_S \leq d_X = 0$).

**V3.69 (Theorem 3.1.26 (preprint 3.23), the implicit function theorem, boundary case; `Polyfold.theorem_3_23_of_nearby`, `Polyfold.theorem_3_23`, `Polyfold.MPolyfold.subPolyfold`, `Polyfold.MPolyfold.IsManifoldWithCorners`).** Let $f$ be an sc-Fredholm section of a strong bundle, $f(x) = 0$, $(f, x)$ in good position in a tame chart (V3.47, V3.103), and let every zero in an open $V_{x_0} \ni x$ be in good position in a tame chart. Then there are an open $U$ with $x \in U \subseteq V_{x_0}$ and $d \in \mathbb{N}$ such that every zero in $U$ is a smooth point and $S(f, U) = \{y \in U \mid f(y) = 0\}$ is a sub-M-polyfold of $X$ whose induced structure (Proposition 2.3.19 (preprint 2.21)) is tame and equivalent to that of a $d$-dimensional manifold with boundary with corners (an equivalent atlas of charts on relatively open subsets of $d$-dimensional partial quadrants). For $f$ with tame germ data (V3.103) and the filler condition of V3.82 at $x$ in place of the nearby good position (`Polyfold.theorem_3_23`): the same, and every zero in $U$ is in good position in a tame chart. *Book:* Theorem 3.1.26 (preprint 3.23), p. 103 (preprint p. 77) ($f'(x)$ surjective with kernel in good position; $S(f, U)$ consists of smooth points and is a tame sub-M-polyfold with a uniquely determined structure of a manifold with boundary with corners, the induced one). *Hidden hypotheses:* tameness as in V3.68. *Deviation (narrower):* the good position of the nearby zeros (Theorem 3.6.8 (1) (preprint 3.54)) is a hypothesis of the first form, the filler condition at the nearby zeros (V3.82) of the second; both are now theorems (V3.94, V3.102), and the form with the text's hypotheses is V3.95, with the dimension as the Fredholm index V3.97; $d$ is existentially quantified here; "uniquely determined" is the equivalence of atlases, not stated separately. *Proof:* regular data of Definition 3.2.1 (preprint 3.25) at every point of $S(f, U)$, of the same dimension by the

local constancy (V3.85, V3.74), then Theorem 3.2.3 (preprint 3.27) (V3.24).

**V3.70 (Corollary 3.1.27 (preprint 3.24); `Polyfold.corollary_3_24`, `Polyfold.corollary_3_24_sc`, `Polyfold.forall_isGoodPositionGermSc_iff`).** Let $f$ be an sc-Fredholm section of a strong bundle such that $(f, x)$ is in good position in a tame chart (V3.103) at every zero $x$, and let $S = \{x \in X \mid f(x) = 0\}$ be nonempty. Then $S$ is a sub-M-polyfold of $X$ whose induced structure is tame and equivalent to that of a manifold with boundary with corners of locally constant dimension (V3.72). The same with good position in the text's sense (sc-complement, V3.47) at every zero, which for an sc-Fredholm section with tame germ data is the same hypothesis (`corollary_3_24_sc`). *Book:* Corollary 3.1.27 (preprint 3.24), p. 104 (preprint p. 78). *Hidden hypotheses:* tameness as in V3.68; $S \neq \emptyset$ (for $S = \emptyset$ the formal charts have no values; the statement is then empty). *Deviation (hypotheses explicit):* **no hypothesis added**; the text names no dimension, the formal one is locally constant. It is Theorem 3.6.11 (preprint 3.57) (V3.87) without its last clause.

**V3.71 (tameness is carried by sc-diffeomorphisms and by open subsets; `Polyfold.IsScDiffeoQuadrant`, `Polyfold.IsScDiffeoQuadrant.transport`, `Polyfold.IsScDiffeoQuadrant.scDeriv_transport`, `Polyfold.IsScSubspace.map`, `Polyfold.IsScSplitting.map`, `Polyfold.IsScDiffeoQuadrant.isTame_transpo`, `Polyfold.IsScDiffeoQuadrant.isTameRetract_image`, `Polyfold.ScRetraction.restrict_isTame`, `Polyfold.IsTameRetract.inter_isOpen`, `Polyfold.MChart.isTameRetract_image`).** Let $\psi : U \to U'$ be an sc-diffeomorphism between relatively open subsets of partial quadrants ($\psi$ sc-smooth on $U$, $\psi'$ sc-smooth on $U'$, mapping $U$ and $U'$ into each other, inverse to each other) and $r : U \to U$ a tame sc-smooth retraction onto $O$ whose domain is $U$. Then $r' = \psi \circ r \circ \psi'$ is a tame retraction of $U'$ onto $\psi(O)$, so $(\psi(O), C', E')$ is a tame retract; on the way $d_{C'}(\psi(x)) = d_C(x)$ (Theorem 1.3.5 (preprint 1.25)), $Dr'(y) = D\psi(x)Dr(x)D\psi'(y)$ at the level-1 points $y = \psi(x)$ with $x \in O$, and an sc-operator with an sc-operator inverse carries sc-subspaces and sc-splittings. The restriction of a tame retraction to $U \cap r^{-1}(W)$, $W$ open, is tame; an open subset $O \cap W$ of a tame retract is a tame retract; the model $\varphi(s)$ of a chart restricted to an open $s$ of its domain is tame if the model of $\varphi$ is. *Book:* used without statement in the proof of Proposition 3.2.2 (preprint 3.26) ("$X' = \psi(X)$ is tame", preprint p. 80) and in the proof of Theorem 3.6.8 (preprint 3.54) (the chart $\psi \circ \varphi$ of a tame M-polyfold, preprint p. 114). *Hidden hypotheses:* condition (2) of Definition 2.5.2 (preprint 2.34) is transported at smooth points, where $D\psi(x)$ is an sc-isomorphism, with $D\psi(x)E_x \subseteq E'_{\psi(x)}$ (the one-sided sign argument). *Deviation (not in the book).*

**V3.72 (Theorem 3.2.3 (preprint 3.27) with locally constant dimension; `Polyfold.theorem_3_27_loc`, `Polyfold.MPolyfold.IsManifoldWithCornersLoc`, `Polyfold.MPolyfold.theorem_3_27_loc_isSubMPolyfold`, `Polyfold.MPolyfold.theorem_3_27_loc_isTame`, `Polyfold.MPolyfold.theorem_3_27_loc_isManifoldWithCo`, `Polyfold.MPolyfold.theorem_3_27_loc_atlas_equiv`).** Let $M \neq \emptyset$ be a subset of an M-polyfold $X$ such that every point of $M$ has regular data of Definition 3.2.1 (preprint 3.25) (V3.22, conditions (a), (b) of V3.24) of some dimension, which may depend on the point. Then $M$ is a sub-M-polyfold of $X$, its induced structure is tame, and the charts $y \mapsto \pi\varphi(y)$ of all these data form an atlas equivalent to the induced one, consisting of charts on relatively open subsets of partial quadrants of finite-dimensional spaces (`IsManifoldWithCornersLoc`: a manifold with boundary with corners of locally constant dimension); each of these charts is a chart of the induced structure containing its point. *Book:* Theorem 3.2.3 (preprint 3.27), p. 106 (preprint p. 81) (for a fixed dimension $n$). *Hidden hypotheses:* (a), (b) of V3.24. *Deviation (narrower):* the conditions (a), (b); the dimension may vary from point to point (same proof as V3.24).

**V3.73 (the re-centring lemma; `Polyfold.recentre_contDiffOn_levels`, `Polyfold.finrank_le_of_interior_i`**

`Polyfold.exists_contDiffOn_inverse`**).** Let $\chi_1$, $\chi_2$ be charts of an M-polyfold $X$ and $S \subseteq X$. Suppose that $\chi_1(S \cap V_{x_1}) = \{v + \sigma_1(v) \mid v \in V_1\}$ for an open $V_{x_1}$ in the domain of $\chi_1$, where $K_1$ is a finite-dimensional smooth subspace in good position to $C_1$ with a closed complement $Y_1$, $V_1$ is relatively open in $K_1 \cap C_1$, and $\sigma_1$ has values in $Y_1$ and is of class $C^1$ on all of $V_1$; that $\chi_2$ is centred at a point $y \in V_{x_1} \cap V_{x_2}$ and $\chi_2(S \cap V_{x_2})$ is the graph of $\sigma_2$ over a relatively open $V_2 \ni 0$ in $K_2 \cap C_2$ (the same conditions on $K_2$, $Y_2$), with $\sigma_2$ continuous, $\sigma_2(0) = 0$, $D\sigma_2(0) = 0$ and $C^k$-lifts $\sigma_{2,k} : V_{2,k} \to E_{2,k}$ on relatively open neighbourhoods of 0 for every $k \geq 1$; and that $\chi_1(y) = v_0 + \sigma_1(v_0)$, $v_0 \in V_1$. Then $\dim K_1 = \dim K_2$, and for all $m$, $k$ there is a relatively open $B' \ni v_0$ in $K_1 \cap C_1$, $B' \subseteq V_1$, on which $\sigma_1$ takes values in $E_{1,m}$ and is of class $C^k$ into $E_{1,m}$. On the way: a map differentiable on a set whose image has nonempty interior does not lower the dimension (Sard's lemma in fixed dimension); and a $C^k$ map between relatively open subsets of convex sets with nonempty interior in finite-dimensional spaces, with a continuous left inverse on its relatively open image and an invertible derivative at a point, has a $C^k$ inverse near the image of that point. *Book:* the step "$\delta$ is an sc$^\infty$-germ near every point $v_0 \in V$" of the proof of Theorem 3.2.3 (preprint 3.27), preprint p. 84. *Hidden hypotheses:* $\sigma_1$ differentiable at $v_0$ (here $C^1$ on $V_1$; without it the step fails, V3.24); $Y_1$, $Y_2$ closed; the lifts at 0 (Theorem 3.5.11 (1) (preprint Corollary 3.47), V3.42). *Deviation (narrower):* for two graph descriptions of one set, with $\sigma_1$ of class $C^1$ on $V_1$. *Proof:* the linearization of $\beta = \pi_1 \circ \chi_1\chi_2^{-1} \circ (1 + \sigma_2)$ at 0 is injective by the chain rule for the transition and its inverse on level 1 and the tangency of the graph of $\sigma_2$ to the retract; it is onto by the dimension count, not by the symmetric chain rule (which would need $\sigma_1$ differentiable into level 1 at $v_0$); the inverse is $C^k$ by V3.90, not by Theorem 3.8.15 (preprint 3.72) (which would need $d_C(x) = d_C(\beta(x))$).

**V3.74 (condition (a) of `IsRegular` by re-centring;** `Polyfold.MPolyfold.TangentGermChart.exists_isRegular_` `Polyfold.MPolyfold.TangentGermChart.shrink`, `Polyfold.MPolyfold.TangentGermChart.exists_shrinkDat` `Polyfold.MPolyfold.TangentGermChart.shrink_isRegular`, `Polyfold.MPolyfold.TangentGermChart.exists`
Let $D$ be data of Definition 3.2.1 (preprint 3.25) (V3.22) of dimension $n$ for $M \subseteq X$ at $x$ with $\pi(O) \subseteq C$ (condition (b) of V3.24), and let every point of $M$ in an open $U \ni x$ have data of Definition 3.2.1 (preprint 3.25) of some dimension. Then $M$ has at $x$ regular data (conditions (a) and (b) of V3.24) of dimension $n$ whose chart domain lies in $U$, and every datum of Definition 3.2.1 (preprint 3.25) for $M$ at a point of $M$ in that domain has dimension $n$. (The proof shrinks $D$ — the chart restricted to an open subset, $V$ to a relatively open $V'$ inside the neighbourhood of 0 on which $\delta$ is $C^1$, the same $N$, $Y$, $\delta$ — and applies V3.73 at every $v_0 \in V'$; the statement exports only the existence of the regular data.) *Book:* the proof of Theorem 3.2.3 (preprint 3.27), preprint p. 84; the proof of Theorem 3.6.8 (preprint 3.54), p. 139 (preprint p. 115) ("proceeding as above"). *Hidden hypotheses:* the data at the nearby points. *Deviation (narrower):* condition (b) is a hypothesis. Read off the statement (not a separate Lean theorem): every point of a set with the $n$-dimensional tangent germ property has data, so *for such a set, data at $x$ that satisfy (b) can be shrunk to regular data*; condition (a) of V3.24 follows from (b). For the solution set of an sc-Fredholm section (b) holds by the choice of the good complement (V3.75, V3.84).

**V3.75 (good complements whose projection maps the partial quadrant into itself;** `Polyfold.ProjectsInto`, `Polyfold.exists_goodComplement_projectsInto`, `Polyfold.exists_proj_goodComplement`, `Polyfold.exists_proj_quadCone`, `Polyfold.ProjectsInto.mapL`**).** `ProjectsInto C N Y` means: $z \in C$, $k \in N$, $z - k \in Y$ imply $k \in C$; for $E_0 = N \oplus Y$ the projection onto $N$ along $Y$ maps $C$ into $N \cap C$. If the finite-dimensional subspace $N$ is in good position to the partial quadrant $C$ of the sc-Banach space $E$, there is a continuous projection $\pi : E_0 \to N$ with $\pi(C) \subseteq N \cap C$ whose kernel $Y$ is a good complement of $N$; $Y$ is closed and `ProjectsInto C N Y` holds. The property

is carried by continuous linear isomorphisms $S$ with $x \in C \iff Sx \in C'$. *Book:* not stated; the proof of Theorem 3.2.3 (preprint 3.27) (preprint p. 83) asserts that $\Sigma = \{v+y \in C \mid v \in V, y \in Y\}$ is relatively open in $C$, which fails for an arbitrary good complement (V3.24) and holds when $\pi(O) \subseteq C$. *Hidden hypotheses:* none. *Deviation (not in the book):* the construction keeps the coordinates of a set $S$ of indices; when no coordinate vanishes at a nonzero point of $N \cap C$ and $N \cap C \neq N$, one coordinate is kept (the natural choice $S = \Sigma$ fails there: for $C = [0,\infty)^2$, $N = \mathbb{R}(1,1)$, the projection along the good complement $\mathbb{R}(1,2)$ maps $(0,1)$ to $-(1,1)$).

**V3.76 (Corollary 3.4.3 (preprint 3.39);** `Polyfold.corollary_3_39`, `Polyfold.corollary_3_39_isConjugate`, `Polyfold.ConjugationData.compIso_eq_strongShift`, `Polyfold.IsStrongBundleIso.shift`, `Polyfold.IsStrongBundleIso.comp`, `Polyfold.IsStrongBundleIso.pushforward_comp`, `Polyfold.StabilityData.linIso_isStrongBundleIso`, `Polyfold.ScOperator.isStrongBundleIso_prodMap`**).** Let $U$ be relatively open in a partial quadrant of $E$ with $0 \in U$, $g : U \to F$ sc-smooth and conjugated to a basic germ (V3.13), and $s : U \to F_1$ sc-smooth into $F^1$ with $s(0) = 0$. Then the level-1 map $(g + \iota s)^1 : U_1 \to F_1$ is conjugated to a basic germ on the shifted structures $E^1 \lhd F^1$. Explicitly, for a conjugation $\Phi$ of $g$ to the basic germ $h$, with $t = \Phi[1]_* s$ ($t(0) = 0$) and $h' = h + (t_{\mathbb{R}^N}, 0)$, there are the data of Theorem 3.4.2 (preprint 3.38) (V3.34) for $t_W$ such that: $h'$ is a basic germ and $\Phi_*(g+s) = h + t$ near 0; the composite $\Gamma = \Lambda \circ \Phi^1$ of the index raise of $\Phi$ and the linear coordinate change $\Lambda$ of Theorem 3.4.2 (preprint 3.38) is a strong bundle isomorphism of the shifted structures onto a relatively open subset of $(\widetilde{C} \oplus C) \oplus X^1$, with base map vanishing at 0 and $\Gamma_* = \Lambda_* \circ \Phi^1_*$; and $(\Gamma(u, (g+s)^1(u)))_2 = (\Lambda_*(h+t))^1(\gamma(u))$ for $u \in U_1$ near 0, a basic germ. The text's form "$\Gamma = \Psi \circ \Phi$, then raise the index" is the same map (`compIso_eq_strongShift`). On the way: a pair of sc-isomorphisms, the index raise of a strong bundle isomorphism and a composition of strong bundle isomorphisms are strong bundle isomorphisms, and push-forwards compose. *Book:* Corollary 3.4.3 (preprint 3.39), p. 116 (preprint p. 93). *Hidden hypotheses:* the index raise of a strong bundle isomorphism is one (used silently). *Deviation (restated):* "sc-germ" read as sc-smooth on $U$, $s$ as an sc$^+$-map on all of $U$ with $s(0) = 0$; the conclusion is a conjugation of $(g+s)^1$ on the shifted structures; the text's "strong bundle map $\Psi$" is an isomorphism; the $\mathbb{R}^N$-component of $t$ is absorbed into $h$; the text's $\tau : Z \to C$ is not applied.

**V3.77 (Theorems 3.5.6, 3.5.11 and 3.6.8 (3) (preprint 3.43, 3.47 and 3.54) with a closed good complement whose projection maps the quadrant into itself;** `Polyfold.theorem_3_43_closed`, `Polyfold.theorem_3_43_proj`, `Polyfold.theorem_3_47_proj`, `Polyfold.theorem_3_47_closed`, `Polyfold.theorem_3_54_graph_proj`, `Polyfold.IsGoodComplement.prod_top_graph`, `Polyfold.ShiftedBasic` The conclusions of V3.38, V3.42 and V3.48 hold with a good complement $Y$ that is closed and satisfies `ProjectsInto` (V3.75) for the partial quadrant ($\widetilde{C} \oplus W$, resp. $C' = \widetilde{C} \oplus W$): the projection onto the kernel along $Y$ maps the partial quadrant into the kernel's cone. In Theorem 3.5.6 (preprint 3.43), $Y = Z \oplus W$ for a good complement $Z$ of $\widetilde{K}$ chosen by V3.75: every good complement $Z$ of $\widetilde{K}$ gives the good complement $Z \oplus W$ of $K' = \{(\alpha, L\alpha)\}$ (constant $\varepsilon/(\|L\|+1)$), the data of the proof exist for a prescribed $Z$, and the projection property passes from $(\widetilde{K}, Z)$ to $(K', Z \oplus W)$; in Theorem 3.5.11 (preprint Corollary 3.47) both properties are carried by the linear homeomorphism of Theorem 3.4.2 (preprint 3.38); in Theorem 3.6.8 (3) (preprint 3.54) the shrinking of $V'$ leaves $Y$ unchanged. *Book:* Theorem 3.5.6 (preprint 3.43), p. 126 (preprint p. 102); Theorem 3.5.11 (preprint Corollary 3.47), p. 131 (preprint p. 106); Theorem 3.6.8 (preprint 3.54), step (3) of the proof, preprint p. 114. *Hidden hypotheses:* none. *Deviation (stronger):* two additional conclusions; the closedness is the text's (its $Y'$ is an sc-complement), the projection property is not in the text and is what makes condition (b) of Theorem 3.2.3 (preprint 3.27) hold for the solution set (V3.84).

**V3.78 (the linearization at a zero in charts, and its chart independence; `Polyfold.linAt`, `Polyfold.rho_scDeriv_comp_r`, `Polyfold.linAt_eq_scDeriv`, `Polyfold.IsFilledVersion.fillerRestrict_e` `Polyfold.linAt_eq_germDeriv`, `Polyfold.IsFillingGerm.fillerRestrict_eq_linAt`, `Polyfold.linAt_chart_change`, `Polyfold.StrongBundle.linAt_chart_independence`, `Polyfold.IsInGoodPositionIn.transport`).** In a strong bundle chart (retraction $r$ on $U$ onto $O$, fibre projections $\rho$, principal part **f**), $\mathrm{Lin}(z) = D(\mathbf{f}\circ r)(z)|_{T_zO}$, $T_zO = Dr(z)E$, at every level-1 point $z$ of the retract. At a zero its values lie in $K_z = \rho(z)F$; it equals $Dg(z)$ on $T_zO$ for every $g$ sc¹ on $U$ agreeing with **f** on $O$ near $z$, and for every sc¹-germ $g$ agreeing with **f** near $z$ with $z$ in its germ ($Dg(z)$ the value of its derivative family; Findings, Chapter 3, item 284); at a centred zero it is the $Tf(0)$ of Proposition 3.1.5 (preprint 3.5) (V3.8), also for a filling germ. For strong bundle charts $c$, $c'$ whose transition is sc-smooth on $[0]$, a section with sc-smooth principal parts in both, and a zero $y$ in both domains with level-1 points $z$, $z'$ over $\varphi(y)$, $\varphi'(y)$: there are topological linear isomorphisms $A : T_zO \to T_{z'}O'$ and $\Gamma : K_z \to K'_{z'}$ with $\Gamma\circ\mathrm{Lin}(z) = \mathrm{Lin}'(z')\circ A$ and, on $T_zO$, $x \in C_z \iff Ax \in C'_{z'}$ (the partial quadrants at the two points). Hence for two strong bundle charts of $P$ and an sc-smooth section, $\mathrm{Lin}(z)$ and $\mathrm{Lin}'(z')$ are simultaneously onto, simultaneously Fredholm with the same index, and their kernels are simultaneously in good position to the partial quadrants inside $T_zO$, $T_{z'}O'$ (good position inside a subspace is carried by isomorphisms respecting the cones). *Book:* Definition 3.6.6 (preprint 3.53) (p. 136 (preprint p. 111)) presupposes it; Definition 2.5.8 (preprint 2.38) (p. 57 (preprint p. 42)) asserts the chart independence of $T^R_xX$, $C_xX$ by Proposition 2.4.10 (preprint 2.30); nothing is said about $f'(x)$. *Hidden hypotheses:* the point is a zero (away from zeros the linearization is not chart independent). *Deviation (not in the book):* the chart form of what gap G.13 leaves open; of the compatibility of Definition 2.6.4 (preprint 2.50) only the sc-smoothness of the transition on $[0]$ is used; level-1 points suffice; no tameness.

**V3.79 (Definition 3.6.6 (preprint 3.53) at a zero, from a chart not centred there; `Polyfold.StrongBundle.isGoodP` `Polyfold.StrongBundle.isGoodPositionGerm_of_chart`, `Polyfold.StrongBundle.isGoodPositionGerm_of_l` `Polyfold.StrongBundle.isGoodPositionAtGerm_of_chart_dense`).** Let $f$ be an sc-smooth section, $y$ a zero with tame germ data (V3.103), $c$ a strong bundle chart of $P$ with $y$ in its domain (not necessarily centred), $z$ a level-1 point over $\varphi(y)$, and $g$ an sc¹-germ on a germ $G$ of $c$ with derivative family $Dg$ and $z \in G_1$, agreeing with **f** on $O$ near $\varphi(y)$. If the range of $Dg(z)|_{T_zO}$ contains $\rho(z)k$ for every smooth $k \in F_\infty$, and $\ker Dg(z)\cap T_zO$ is in good position to $C_{\varphi(y)}$ inside $T_zO$, then $(f, y)$ is in good position in a tame chart (V3.47, V3.103). Variants: with all of $K_z$ in the range, for an sc-Fredholm germ, in the chart centred at $y$ of Definition 3.1.11 (preprint 3.11); with the hypotheses on $\mathrm{Lin}(z)$ (V3.78). *Book:* the proof of Theorem 3.6.8 (preprint 3.54), p. 139 (preprint p. 115) ("proceeding as above", at a nearby zero). *Hidden hypotheses:* the kernel hypothesis is on $\ker Dg(z)\cap T_zO$, not on $\ker Dg(z)$ (condition (3) of the filler holds at its centre only); $z$ lies in the germ of $g$. *Deviation (not in the book):* a dense part of $K_z$ in the range suffices, since the range of the linearization in the chart of the germ data is closed (Proposition 3.6.5 (preprint 3.52), V3.46), which is why the dense form is stated for tame germ data. In the germ reading since phase 2; the whole-chart forms are `Polyfold.StrongBundle.isGoodPositionGerm_of_chart_dense`, `Polyfold.StrongBundle.isGoodPositionGerm_of_linAt` and the `_wholeChart` declarations (Findings, Chapter 3, items 285, 301, 313).

**V3.80 (property (4) of the proof of Theorem 3.6.8 (preprint 3.54) in the chart at $x_0$: Corollary 3.5.3 (preprint 3.41) and Lemma 3.6.9 (preprint 3.55) on the graph; `Polyfold.theorem_3_54_nearby`, `Polyfold.lemma_3_55_relOpen`, `Polyfold.StabilityData.exists_surjective_Dfs'_graph`).** In the data of V3.48, for a section $f$ of a strong bundle with $f(x_0) = 0$ and $(f, x_0)$ in good position

in a tame chart (V3.103), there are a presentation $T$ of $C' = \widetilde{C} \oplus W$ and an open $V_{x,1} \ni x_0$ such that for every zero $y \in V_{x,1}$, with $\psi\varphi(y) = v + \sigma(v)$, $v \in V'$: $v + \sigma(v)$ lies on level 1 and $D(h+t)$ there is a surjective Fredholm operator of index $n - N$ ($n = \dim \mathbb{R}^n$, $N = \dim \mathbb{R}^N$ of the basic germ); $v + \sigma(v) \in C'$; and $Y$ is a good complement of $N_v = \{k + D\sigma(v)k \mid k \in K\}$ with respect to the partial quadrant $C'_{v+\sigma(v)}$, and $N_v$ is in good position to it. Lemma 3.6.9 (preprint 3.55) is proved in the form used: for $N$ in good position to $C$ (any presentation) with closed good complement $Y$, $V$ relatively open in $N \cap C$, $\sigma : V \to Y$ of class $C^1$ on $V$ with $\sigma(0) = 0$, $D\sigma(0) = 0$, for $v \in V$ near 0: $v + \sigma(v) \in C$, $Y$ is a good complement of $N_v$ with respect to $C_{v+\sigma(v)}$, and $N_v$ is in good position to it. *Book:* the proof of Theorem 3.6.8 (preprint 3.54), property (4), preprint p. 114; Lemma 3.6.9 (preprint 3.55), p. 139 (preprint p. 114). *Hidden hypotheses:* $Y$ closed (V3.49); that $N_v \cap C_z$ has interior in $N_v$, which Lemma 3.6.9 (preprint 3.55) does not state. *Deviation (narrower):* in the chart at $x_0$, for the tangent space $N_v$ of the graph; its identification with the kernel at $y$ is V3.81.

**V3.81 (Theorem 3.6.8 (1) (preprint 3.54) at the nearby zeros, the kernel;** `Polyfold.theorem_3_54_nearby_kernel` `Polyfold.theorem_3_54_chart_kernel`, `Polyfold.StabilityData.graphSpace_eq_ker_Dfs'`, `Polyfold.IsStrongBundleIso.ker_scDeriv_eq_map`, `Polyfold.scDeriv_mem_partialQuadrantAtWith_iff`, `Polyfold.IsInGoodPosition.isInGoodPositionIn_of_le`, `Polyfold.IsStrongBundleIso.ker_deriv_eq_map`**.** Let $f$ be a section of a strong bundle, $f(x_0) = 0$, $(f, x_0)$ in good position in a tame chart (V3.103). In a tame chart centred at $x_0$ (filling germ $g$ with a derivative family $Dg$, conjugation covering $\psi$ with inverse $\psi'$, the germ $h + t$ with kernel $K$, closed good complement, graph $\sigma : V' \to Y$) there is an open $V_{x,2} \ni x_0$ such that for every zero $y \in V_{x,2}$, with $z_0 = \varphi(y)$, $\psi(z_0) = v + \sigma(v)$ and level-1 points $z_{0,1}$ over $z_0$, which lies in the germ of $g$, $x_1$ over $v + \sigma(v)$: $\mathbf{f} = g$ on $O$ near $z_0$ and $g(z_0) = 0$; $D(h + t)(x_1)$ is a surjective Fredholm operator of index $n - N$; $N_v = \ker D(h + t)(x_1)$; $D(\Psi_* g)(x_1) = D(h + t)(x_1)$ and $\ker Dg(z_{0,1}) = D\psi'(x_1)(N_v)$; $Dg(z_{0,1})$ is onto; $\ker Dg(z_{0,1}) \subseteq T_{z_0}O$; and $\ker Dg(z_{0,1})$ is in good position to $C_{z_0}$ inside $T_{z_0}O$. *Book:* Theorem 3.6.8 (1) (preprint 3.54), the kernel half, "from Corollary 3.41 and Lemma 3.55", preprint p. 114. *Hidden hypotheses:* the $C^1$-lift of $\sigma$ into level 1 (Theorem 3.5.11 (1) (preprint Corollary 3.47)) for $N_v \subseteq \ker D(h + t)$; $\ker Dg(z_0) \subseteq T_{z_0}O$ comes from the tangency of the solution set to $O$, not from the filler (whose condition (3) holds at 0 only). *Deviation (narrower):* in the chart at $x_0$; **no added hypothesis**; the surjectivity of $f'(y)$ is not part of it (V3.93, V3.94). In the germ reading $z_{0,1}$ lies in the germ of the filling because the basic germ is taken on the points that $\psi'$ maps into the germs of $g$ and $s$ (Findings, Chapter 3, items 276–278, 295).

**V3.82 (Theorem 3.6.8 (1) (preprint 3.54) under the filler condition at the nearby zeros;** `Polyfold.theorem_3_54_one`, `Polyfold.StrongBundle.IsGoodPositionGermFilled`, `Polyfold.exists_range_deriv_eq_ _of_normal_injective_germ`**).** `IsGoodPositionGermFilled` at $x_0$: $x_0$ is smooth and there are the data of Definition 3.6.6 (preprint 3.53) at $x_0$ in a tame chart (V3.47, V3.103: a centred chart with tame base retraction, a filling germ $g$ with a derivative family $Dg$ on a germ $G$, a local sc$^+$-germ $s$ with $g - s$ conjugated to a basic germ, the germ in good position) and an open $N \ni 0$ such that at every level-1 point $z_1 \in G_1$ with $z_1 \in O \cap N$ and $g(z_1) = 0$, $Dr(z_1)b = 0$ and $\rho(z_1)Dg(z_1)b = Dg(z_1)b$ imply $b = 0$: condition (3) of Definition 3.1.4 (preprint 3.4), the injectivity of $(1-\rho)Dg$ on $\ker Dr$, at the zeros of $g$ near 0. Theorem: if $f$ is an sc-Fredholm section with tame germ data (V3.103), $f(x_0) = 0$, and $(f, x_0)$ is in good position in this strengthened sense, there is an open $V_x \ni x_0$ such that $(f, y)$ is in good position in a tame chart (V3.47, V3.103) at every zero $y \in V_x$. The step it needs: at a zero $z$ of a filler with $Dg(z)$ onto and condition (3) at $z$, $\rho(z)F \subseteq Dg(z)(T_zO)$ (Proposition 3.1.5 (1) (preprint 3.5) at $z$). *Book:* Theorem 3.6.8 (1) (preprint 3.54), p. 136 (preprint p. 111), proof preprint p. 114. *Hidden hypotheses:*

the filler condition at the nearby zeros, added when this form was proved; it is a consequence of good position (V3.102), so the form has the text's hypotheses after all. *Deviation (narrower):* the condition is a hypothesis of the statement. Definition 3.1.4 (preprint 3.4) imposes (3) at 0 only (so does the 2017 version, Definition 3.13 there), and then the text's argument does not give the surjectivity of $f'(y)$: with respect to $E = T_{z_0}O \oplus \ker Dr(z_0)$, $F = K_{z_0} \oplus \ker \rho(z_0)$, $Dg(z_0)$ is block triangular and $Tf(z_0)$ is onto iff the block $(1-\rho(z_0))Dg(z_0)|_{\ker Dr(z_0)}$ is injective, which is the local constancy of the index along the solution set (Findings, Chapter 3, items 100, 157); both directions are formalized (V3.102). The surjectivity holds with Definition 3.1.4 (preprint 3.4) as stated, by an implicit function theorem at the nearby zero (V3.93), and Theorem 3.6.8 (1) (preprint 3.54) with the text's hypotheses is V3.94.

**V3.83 (the parameter extension of the proof of Proposition 6.2.2 (preprint 6.2): the obstruction and the tilted kernel; `Polyfold.mem_lineality_of_goodComplement_prod`, `Polyfold.sup_lineality_eq_top_`, `Polyfold.isInGoodPosition_extendedKernel`).** (a) Let $C \subseteq V$ be closed under nonnegative scaling with $0 \in C$ and $N \subseteq V$ a subspace. For every good complement of $\mathbb{R} \times N$ with respect to $\mathbb{R} \times C$, the $V$-components of its vectors lie in the lineality space $C \cap (-C)$; so $\mathbb{R} \times N$ is in good position only if $N + (C \cap -C) = V$. (b) Let $K, Y'$ be subspaces of $\mathbb{R}^n \oplus W$ and $j : \Lambda \to \mathbb{R}^n \oplus W$ linear, $\Lambda$ finite-dimensional, with $K \cap j(\Lambda) = 0$, $(K + j(\Lambda)) \oplus Y' = \mathbb{R}^n \oplus W$, $Y'$ in the lineality space of $\widetilde{C} \oplus W$ ($v + y \in \widetilde{C} \oplus W \iff v \in \widetilde{C} \oplus W$ for $y \in Y'$), and $K + j(\Lambda)$ meeting the interior of $\widetilde{C} \oplus W$. Then $\{((\lambda, a), w) \mid (a, w) + j(\lambda) \in K\}$ is in good position to $(\Lambda \times \widetilde{C}) \oplus W$. *Book:* not in Chapter 3; the linear algebra of the perturbation argument of the text's Proposition 6.2.2 (preprint 6.2), tried for the surjectivity of V3.82. *Hidden hypotheses:* none. *Deviation (not in the book):* the rest of the argument is not formalized; it would give the surjectivity only on a neighbourhood depending on $\Lambda$ (Findings, Chapter 3, item 106). The surjectivity is proved by another argument, with one perturbation direction at a time and the implicit function theorem at the nearby zero (V3.93).

**V3.84 (Theorem 3.6.8 (2) (preprint 3.54) at the centre: the data of Definition 3.2.1 (preprint 3.25) for the solution set at $x_0$; `Polyfold.theorem_3_54_tangentGermChart_proj`, `Polyfold.theorem_3_54_tangentG`, `Polyfold.MChart.compDiffeo`, `Polyfold.MChart.compDiffeo_isChart`, `Polyfold.isScInftyGermAt_of_lifts`, `Polyfold.StabilityData.ker_Dfs_le_smooth`, `Polyfold.MPolyfold.TangentGermChart._mem_of_projectsI`.** Let $f$ be a section of a strong bundle, $f(x_0) = 0$, $(f, x_0)$ in good position in a tame chart (V3.103). Then there are an open $V_x \ni x_0$ and $d \in \mathbb{N}$ such that $\{y \in V_x \mid f(y) = 0\}$ has at $x_0$ data of Definition 3.2.1 (preprint 3.25) of dimension $d$ (V3.22) with $\pi(O) \subseteq C$ (condition (b) of V3.24). The data: the chart $\psi \circ \varphi$ onto $\psi(O)$ (for a chart $\varphi$ of $X$ and an sc-diffeomorphism $\psi$ between relatively open subsets of partial quadrants whose domain contains the retract $O$ of $\varphi$, $\psi \circ \varphi$ is a chart of $X$ when $\psi(O)$ is a retract), whose model is a tame retract by V3.71; $N = \ker D(h+t)(0)$, a smooth subspace; the closed good complement with the projection property and the map $\sigma$ of V3.77; $\sigma$ is an sc$^\infty$-germ in the sense of V3.22 because it has $C^k$-lifts into level $k$ for every $k$. *Book:* the proof of Theorem 3.6.8 (preprint 3.54), p. 139 (preprint p. 115) ("$S$ possesses the $d$-dimensional tangent germ property", $d = \dim\ker f'(x)$). *Hidden hypotheses:* the kernel is a smooth subspace (the text takes it so without comment); $\psi(O)$ is tame (V3.71). *Deviation (narrower):* at $x_0$ only (the other points: V3.85); $d$ is existentially quantified here (the proof takes $d = \dim\ker D(h+t)(0)$); that $d$ is the Fredholm index of $f'(x_0)$ is V3.97.

**V3.85 (Theorem 3.6.8 (2) (preprint 3.54), and Theorem 3.6.8 (preprint 3.54); `Polyfold.theorem_3_54_two`, `Polyfold.theorem_3_54`, `Polyfold.theorem_3_54_tangentGermChart_solutionSet`).** Let $f$ be a section of a strong bundle, $f(x_0) = 0$, $(f, x_0)$ in good position in a tame chart (V3.103). (i) The

solution set $S = \{f = 0\}$ has at $x_0$ data of Definition 3.2.1 (preprint 3.25) with $\pi(O) \subseteq C$. (ii) If moreover every zero in an open $V_{x_0} \ni x_0$ is in good position in a tame chart, there are $d \in \mathbb{N}$ and regular data (V3.24) of dimension $d$ for $S$ at $x_0$ whose chart domain lies in $V_{x_0}$, and every datum of Definition 3.2.1 (preprint 3.25) for $S$ at a point of $S$ in that domain has dimension $d$. (iii) Theorem 3.6.8 (preprint 3.54), first form: if $f$ is an sc-Fredholm section with tame germ data and $(f, x_0)$ is in good position with the filler condition of V3.82 (in a tame chart), there is an open $V_x \ni x_0$ such that every zero in $V_x$ is in good position in a tame chart, and (ii) holds with $V_{x_0} = V_x$. *Book:* Theorem 3.6.8 (1), (2) (preprint 3.54), p. 136 (preprint p. 111). *Hidden hypotheses:* tameness as in V3.68. *Deviation (narrower):* (2) is proved from (1), with the nearby good position as a hypothesis; (1) under the filler condition of V3.82, which follows from good position (V3.102). With the text's hypotheses: V3.94; with the dimension as the index and its local constancy: V3.97, V3.98. "$S$ is a sub-M-polyfold with the structure of a manifold with boundary with corners" follows from (ii) by Theorem 3.2.3 (preprint 3.27) (V3.69). The regularity conditions (a), (b) of Theorem 3.2.3 (preprint 3.27), hypotheses for general sets (V3.24), are theorems for the solution set (V3.74, V3.77).

**V3.86 (Theorem 3.6.10 (preprint 3.56); `Polyfold.theorem_3_56`, `Polyfold.theorem_3_56_sc`).** Let $X$ be an M-polyfold with $d_X \equiv 0$, $f$ an sc-Fredholm section of a strong bundle, with $(f, x)$ in good position in a tame chart (V3.103) at every zero $x$, and $S = \{f = 0\}$ nonempty. Then $S$ is a sub-M-polyfold of $X$ whose induced structure is equivalent to that of a manifold with boundary with corners of locally constant dimension and has degeneracy index 0 at every point (a manifold without boundary). The same with good position in the text's sense (sc-complement, V3.47) at every zero, which for an sc-Fredholm section with tame germ data is the same hypothesis (`theorem_3_56_sc`). *Book:* Theorem 3.6.10 (preprint 3.56), p. 140 (preprint p. 115) ($f'(x)$ surjective at every zero). *Hidden hypotheses:* as V3.70. *Deviation (restated):* the hypothesis "$f'(x)$ surjective" is read as good position, which by the text's Definition 3.6.6 (preprint 3.53) is surjectivity at points with $d_X(x) = 0$; in the chart form the kernel condition is stated for the chart's partial quadrant, and that it is void when $d_X(x) = 0$ is V3.100; the form with "$f'(x)$ surjective" is V3.101. No other hypothesis is added. The tameness of the induced structure is in V3.87 and V3.101.

**V3.87 (Theorem 3.6.11 (preprint 3.57); `Polyfold.theorem_3_57`, `Polyfold.theorem_3_57_dim`, `Polyfold.solutionSetDim`, `Polyfold.theorem_3_57_sc`).** Let $f$ be an sc-Fredholm section of a strong bundle, with $(f, x)$ in good position in a tame chart (V3.103) at every zero $x$. If $S = \{f = 0\}$ is nonempty, $S$ is a sub-M-polyfold of $X$, its induced structure is tame and equivalent to that of a manifold with boundary with corners of locally constant dimension, and at every $x \in S$ there are regular data of Definition 3.2.1 (preprint 3.25) of a dimension $d(x)$ such that every datum at a point of $S$ in their chart domain has dimension $d(x)$. Moreover $S$ is the disjoint union of the sets $S_n$ of points at which $S$ has data of dimension $n$, each $S_n$ is open and closed in $S$, and each nonempty $S_n$ is a sub-M-polyfold whose induced structure is tame and equivalent to that of an $n$-dimensional manifold with boundary with corners. The same with good position in the text's sense (sc-complement, V3.47) at every zero, which for an sc-Fredholm section with tame germ data is the same hypothesis (`theorem_3_57_sc`). *Book:* Theorem 3.6.11 (preprint 3.57), p. 140 (preprint p. 115). *Hidden hypotheses:* as V3.70. *Deviation (stronger):* **the text's hypotheses**; the local constancy of the dimension and the parts $S_n$ are not in the text. Two steps of the text's proof are replaced: the regularity of the solution germ away from the centre (V3.73, V3.74) and the relative openness of $\Sigma$ (V3.75, V3.77). The dimension $d(x)$ is the Fredholm index of $f'(y)$ at the zeros near $x$ (V3.97).

**V3.88 (Theorem 3.7.1 (preprint 3.58), given the family of sc-isomorphisms; `Polyfold.theorem_3_58_of_family`, `Polyfold.DerivCondition`, `Polyfold.contrMap358_contraction`, `Polyfold.norm_sub_sub_le_of_hasFDerivW`, `Polyfold.fiberIsoMap_isStrongBundleIso`, `Polyfold.famInv_isScSmooth_of`).** In the coor-

dinates of the text's proof — $E = \mathbb{R}^n \oplus X$ ($\mathbb{R}^n$ any finite-dimensional space with a partial quadrant $\widetilde{C}$), $C = \widetilde{C} \oplus X$, $F = \mathbb{R}^N \oplus Y$ — let $U$ be relatively open in $C$ with $0 \in U$ and $f : U \to F$ sc-smooth; let $V_o \ni 0$ be open in $\mathbb{R}^n$ with $(b,0) \in U$ for $b \in V_o \cap \widetilde{C}$, and $L(b) : X \to Y$ sc-isomorphisms with $L(b)h = (Df(b,0)(0,h))_Y$ on level 0 and lower bounds $c_m|h|_m \le |L(b)h|_m$ for $b \in V_o \cap \widetilde{C}$, $h \in X_m$, every $m$; and let condition (3) hold (`DerivCondition`: for all $m$ and $\varepsilon > 0$ there is $r > 0$ with $|[D_2 f(b,0) - D_2 f(b,x)]h|_m \le \varepsilon |h|_m$ for $b \in \widetilde{C}$, $|b| < r$, $x \in X_{m+1}$, $|x|_m < r$, $(b,0), (b,x) \in U$, $h \in X_m$). Then on $U' = U \cap (V_o \oplus X)$: $k(b,x) = ((f(b,x) - f(0))_{\mathbb{R}^N}, L(b)^{-1}(f(b,x) - f(0))_Y)$ is a basic germ with contraction map $H(b,x) = x - L(b)^{-1}(f(b,x) - f(0))_Y$; $\Phi((b,x),(c,y)) = ((b,x),(c, L(b)^{-1}y))$ is a strong bundle isomorphism $U' \lhd (\mathbb{R}^N \oplus Y) \to U' \lhd (\mathbb{R}^N \oplus X)$ covering the identity; and $f - f(0)$ is conjugated to a basic germ. *Book:* the second half of the proof of Theorem 3.7.1 (preprint 3.58), pp. 142–145 (preprint pp. 118–120). *Hidden hypotheses:* the mean value inequality along a segment with the derivative bound in the weaker norm of $X_m$ (the text's integral; Findings, Chapter 3, item 73). *Deviation (restated):* in the coordinates of the proof; the sc-smoothness of $(b,y) \mapsto L(b)^{-1}y$ is Lemma 3.7.4 (preprint 3.61) (V3.54) in set form.

**V3.89 (Theorem 3.7.1 (preprint 3.58), the basic germ criterion; `Polyfold.theorem_3_58`, `Polyfold.CompactnessCondition`, `Polyfold.familyCompactness_derivFamOp`, `Polyfold.fredholmIndex_bloc` `Polyfold.surjective_of_fredholmIndex_block`, `Polyfold.derivFamOp`).** In the coordinates of V3.88, let $U$ be relatively open in $\widetilde{C} \oplus X$ with $0 \in U$ and $f : U \to \mathbb{R}^N \oplus Y$ sc-smooth with: (1) at every $(b,0) \in U$, $b \in \widetilde{C}$, $Df(b,0)$ is sc-Fredholm with the sc-index of $Df(0)$; (2) $h \mapsto (Df(0)(0,h))_Y$ is bijective on level 0, and the compactness condition holds (`CompactnessCondition`: there is $r > 0$ such that for every $m$, if $b_j \to b_0$ in $\widetilde{C}$ with $(b_j,0) \in U$ and $|b_0| < r$, $(\eta_j)$ is bounded in $X_m$ and $Df(b_j,0)(0,\eta_j) = y_j + z_j$ with $y_j \to 0$ in $F_m$ and $(z_j)$ bounded in $F_{m+1}$, then $(\eta_j)$ has a convergent subsequence in $X_m$); (3) `DerivCondition` (V3.88). Then there are an open $V_o \ni 0$ in $\mathbb{R}^n$ and sc-isomorphisms $L(b) : X \to Y$ with $L(b)h = (Df(b,0)(0,h))_Y$ for $b \in V_o \cap \widetilde{C}$, for which the conclusions of V3.88 hold; in particular $f - f(0)$ is conjugated near 0 to a basic germ. On the way: Lemmas 3.7.2, 3.7.3 (preprint 3.59, 3.60) for this family, with their hypotheses derived from (2) (V3.52, V3.53); the index of a block operator, $\operatorname{ind} T = \dim \mathbb{R}^n + \operatorname{ind} L - \dim \mathbb{R}^N$ for $T : \mathbb{R}^n \oplus X \to \mathbb{R}^N \oplus Y$ Fredholm with block $L$ of finite index, and an injective block is onto when $\operatorname{ind} T = \dim \mathbb{R}^n - \dim \mathbb{R}^N$. *Book:* Theorem 3.7.1 (preprint 3.58), p. 140 (preprint p. 116), proof pp. 140–145 (preprint pp. 116–120). *Hidden hypotheses:* the smallness of $b$ in (2) uniform in the level $m$. *Deviation (narrower):* in the coordinates of the proof; the reduction of a general sc-splitting $E = B \oplus X$, $F = A \oplus Y$ to them is not formalized; (1) is needed only at the points $(b,0)$; the text's "$B \supseteq \ker Df(0)$ and $Y = Df(0)X$" is replaced by the bijectivity it implies.

**V3.90 (Proposition 3.8.17 (preprint 3.73) and Lemma 3.8.21 (preprint 3.77) with the hypotheses their proofs use; `Polyfold.InvDataW`, `Polyfold.InvDataW.injOn`, `Polyfold.InvDataW.hasFDerivWithinAt_inv` `Polyfold.InvDataW.contDiffOn_inv`).** Let $U$ be a convex subset of a finite-dimensional normed space $V$ with unique differentiability at its points, $f : V \to V$ of class $C^j$ ($j \ge 1$) on $U$ with $\|Df(x) - 1\| \le \frac{1}{2}$ for $x \in U$, and let $f(U)$ have unique differentiability. Then $f$ is injective on $U$ ($\frac{1}{2}|x - x'| \le |f(x) - f(x')|$), its inverse $g$ is differentiable within $f(U)$ with $Dg(y) = [Df(g(y))]^{-1}$, and $g$ is of class $C^j$ on $f(U)$. *Book:* Proposition 3.8.17 (preprint 3.73), p. 165 (preprint p. 137); Lemma 3.8.21 (preprint 3.77), p. 168 (preprint p. 140). *Hidden hypotheses:* none. *Deviation (restated):* no partial quadrant; the hypothesis "$x$ interior iff $f(x)$ interior" of V3.59 is replaced by the unique differentiability of $f(U)$, which is what the bootstrapping uses;

the relative openness of $f(U)$ is not concluded. Used in the re-centring (V3.73), where $f(U)$ is relatively open in a cone for another reason.

**V3.91 (Theorem 3.1.21 (preprint 3.18), parametrized perturbations; `Polyfold.theorem_3_18`, `Polyfold.theorem_3_18'`, `Polyfold.paramSection`, `Polyfold.paramSection_eq_sectionAdd`, `Polyfold.StrongBundle.pullback`, `Polyfold.StrongBundle.IsScFredholmSection.param`, `Polyfold.StrongBundle.isScPlusSection_paramComb`, `Polyfold.IsBasicGerm.paramExtend`, `Polyfold.IsConjugateToBasicGerm.paramExtend`, `Polyfold.rnPolyfold`, `Polyfold.prod_rn_isTame`).** Let $P : Y \to X$ be a strong bundle over an M-polyfold ($X$, $Y$ Hausdorff and paracompact, as throughout), $f$ an sc-Fredholm section (V3.16) and $s_1, \dots, s_n$ sc$^+$-sections of $P$ (Definition 2.6.6 (preprint 2.52)). Let $\mathbb{R}^n$ carry its M-polyfold structure with the constant sc-structure and the charts $r \mapsto r - r_0$, $\mathbb{R}^n \times X$ the product M-polyfold, $\pi(r, x) = x$, and $\pi^*P : \pi^*Y \to \mathbb{R}^n \times X$ the pull-back strong bundle with the natural structure of Proposition 2.6.8 (preprint 2.54) (the atlas of all charts pulled back from charts of $P$ through charts of $\mathbb{R}^n \times X$). Then $F(r, x) = f(x) + \sum_i r_i s_i(x)$, computed in the fibre $P^{-1}(x)$ and read in the fibre of $\pi^*Y$ over $(r, x)$, is an sc-Fredholm section of $\pi^*P$. The paracompactness of $\mathbb{R}^n \times X$ and of $\pi^*Y$, instance arguments of `theorem_3_18`, is supplied by Theorem 2.3.10 (preprint 2.16) in `theorem_3_18'`. On the way: $\tilde{f}(r, x) = f(x)$ is an sc-Fredholm section of $\pi^*P$ — in the product chart $(r \mapsto r - r_0) \times \varphi$, centred at $(r_0, x)$, the filling germ is $(a, u) \mapsto g(u)$ on $\mathbb{R}^n \times G$ (filler conditions (1)–(3) from those of $g$), the sc$^+$-germ $(a, u) \mapsto t_0(u)$, and the conjugation is the product of the given one with the identity of $\mathbb{R}^n$, onto the $\mathbb{R}^n$-extension $((a, \lambda), w) \mapsto h(a, w)$ of the basic germ, again a basic germ; $\tilde{s}(r, x) = \sum_i r_i s_i(x)$ is an sc$^+$-section (the product rule with a scalar factor, from Proposition 1.2.4 (preprint 1.19)); $F = \tilde{f} + \tilde{s}$ fibrewise, and Theorem 3.1.18 (preprint 3.15) (V3.17) applies; $\mathbb{R}^n \times X$ is tame when $X$ is. *Book:* Theorem 3.1.21 (preprint 3.18), p. 100 (preprint p. 74) (for a tame strong bundle; "$\tilde{f}$ … is a sc-Fredholm section as is readily verified", preprint p. 73). *Hidden hypotheses:* the translation charts of $\mathbb{R}^n$ (a chart centred at $(r_0, x)$ is needed for the germ there), the strong bundle structure of $\pi^*P$, the scalar product rule. *Deviation (stronger):* tameness is not used, as for Theorem 3.1.18 (preprint 3.15).

**V3.92 (the linearization of the parametrized section; `Polyfold.linAt_paramSection_zero`, `Polyfold.linAt_paramSection`, `Polyfold.scDeriv_paramSection`, `Polyfold.PullbackChartData.ofProd`).** In the setting of V3.91, let $c$ be a strong bundle chart of $P$ compatible with the atlas, with 0 in its retract $O$, and consider the product chart of $\pi^*P$ through $(r \mapsto r - r_0) \times c$, with the retraction $1 \times r$. At its centre: $D(\hat{F} \circ (1 \times r))(0)(\delta a, \delta u) = D(\hat{G} \circ r)(0)\delta u + \sum_i \delta a_i \hat{s}_i(0)$, $\hat{G} = \hat{f} + \sum_i r_{0,i}\hat{s}_i$; in the chart form of the linearization (V3.78), on $T_0(\mathbb{R}^n \times O) \subseteq \mathbb{R}^n \times T_0O$: $\mathrm{Lin}_F(0)(\delta a, \delta u) = \mathrm{Lin}_{f+\sum r_{0,i}s_i}(0)\delta u + \sum_i \delta a_i \hat{s}_i(0)$; and at $r_0 = 0$: $\mathrm{Lin}_F(0)(\delta r, \xi) = \mathrm{Lin}_f(0)\xi + \sum_i \delta r_i \hat{s}_i(0)$. *Book:* after Theorem 3.1.22 (preprint 3.19), p. 101 (preprint p. 75) ("the linearization of $\tilde{f}$ at $(0, x_0)$ is $(h, u) \mapsto f'(x_0)u + \sum h_i e_i$"). *Hidden hypotheses:* none (the formula does not use $F = 0$ at the point). *Deviation (restated):* in the chart form (Definition 3.6.6 (preprint 3.53) is formalized in charts); at $(r_0, x)$ with $r_0 \neq 0$ the first term is the linearization of $f + \sum r_{0,i}s_i$; the text's "sc$^1$-sections $s_i$" are sc$^+$ here, as Theorem 3.1.21 (preprint 3.18) needs.

**V3.93 (Theorem 3.6.8 (1) (preprint 3.54), the surjectivity at the nearby zeros, in the chart at $x_0$; `Polyfold.theorem_3_54_chart_onto`, `Polyfold.StrongRetr.mem_0_of_perturbed_zero`, `Polyfold.exists_inward_of_isInGoodPosition`, `Polyfold.inward_add_smul`, `Polyfold.exists_curve_of_in`
`Polyfold.ShiftedBasicData.exists_radii`, `Polyfold.ShiftedBasicData.perturbContraction`,
`Polyfold.ShiftedBasicData.perturbContraction_isSc1GermWith`, `Polyfold.xiMap`, `Polyfold.exists_xi_ra`
`Polyfold.ShiftedBasicData.surjective_DH_and_ker_eq`, `Polyfold.ShiftedBasicData.exists_perturbed_`**

In the data of V3.48 at a zero $x_0$ in good position — the strong bundle chart centred at $x_0$ with retraction $(r, \rho)$ on $U$ onto $O$, the filling germ $g$, the conjugation $\Psi$ covering $\psi$ of $g - s$ to a basic germ, $h' + \iota t$ with kernel $K = \ker D(h' + \iota t)(0)$ in good position, the closed good complement, the $C^1$ map $\sigma$ on $V'$ with its lifts, and an open $N_2 \ni 0$ on which $\psi$ maps the zeros of $g$ in $O$ onto the graph of $\sigma$ — assume condition (2) of Definition 3.1.4 (preprint 3.4) on $N_2$ and $K \neq 0$. Then there is an open $N_4 \subseteq N_2$ containing 0 such that for every zero $z_0 \in O \cap N_4$ of $g$, every level-1 point over $z_0$ and every smooth $k \in F_\infty$, $\rho(z_0)k = Dg(z_0)x$ for some $x \in T_{z_0}O = Dr(z_0)E$. On the way: (a) a zero of $g + \mu\rho(r(\cdot))k$ at which condition (2) holds lies in $O$; (b) a nonzero finite-dimensional subspace in good position to the partial quadrant $C_z$ at $z \in C$ contains a vector $v$ with $z + t(v + \zeta) \in C$ for $t \in [0, t_0]$, $|\zeta| < \eta$, and such inward vectors stay inward after adding any vector and rescaling; (c) for $\widetilde{C}$ convex and $H$ of class $C^1$ on $\Omega \cap (\mathbb{R} \times \widetilde{C})$ with $H(0, a_0) = 0$, $D_a H(0, a_0)$ onto and an inward $\alpha_0$ with $DH(0, a_0)(1, \alpha_0) = 0$, there is a curve $a(\mu)$, $0 \leq \mu < \mu_0$, $a(0) = a_0$, with right derivative $\alpha_0$ at 0 and $H(\mu, a(\mu)) = 0$; (d) for the normal form of Theorem 3.4.2 (preprint 3.38) (a basic germ $(G_1, w - B)$ on the shifted structure) there are $\frac{1}{4}$-contraction regions of $B$ below any prescribed radii, and $B - \mu P\hat{e}$ is a contraction germ in the working form on $(-\mu_m, \mu_m) \times \{|a| < \nu_m\}$ with the same regions in $a$ for every perturbation $\hat{e}$ with given bounds, its solution germ sc¹ (Theorem 3.3.3 (preprint 3.30)); (e) the push-forward $\hat{e}_k = \Xi(\cdot, k)$ of $u \mapsto \rho(r(u))k$ to the normal form, with $\Xi$ sc-smooth into the index-raised target and linear in $k$, is bounded with its Lipschitz constant on the contraction regions by $c_m = 2|k|_{m+2}(M_m + 1)/\delta_m$, $\delta_m$, $M_m$ independent of $k$; (f) at a zero $(a_0, w_0)$ of the normal form with $|a_0| < \nu_3$, $|w_0|_1 < \rho_0$ and a kernel of dimension $n - N$, $D_a H(0, a_0)$ is onto for $H$ the $\mathbb{R}^N$-component of $(G + \mu\hat{e})(a, \delta_\mu(\mu, a))$, the kernel is the graph of $D_a\delta_\mu(0, a_0)$ over $\ker D_a H(0, a_0)$, and there is a curve of zeros of $G + \mu\hat{e}$ through it, right differentiable at 0 into level 2. *Book:* the proof of Theorem 3.6.8 (preprint 3.54), property (4), preprint p. 114 (from Corollary 3.5.3 (preprint 3.41) and Lemma 3.6.9 (preprint 3.55)). *Hidden hypotheses:* none added. *Deviation (not in the book):* a new argument (Findings, Chapter 3, items 180–187): the text's argument gives the surjectivity of the filled $Dg(z_0)$ only (V3.82). The neighbourhood $N_4$ is determined by $g$, $r$, $\rho$, $\Psi$ and the normal form, not by $k$; Corollary 3.4.3 (preprint 3.39) is not used. In the germ reading the curve of zeros lies in the germ of the filling, so $g$ is differentiated within its germ (Findings, Chapter 3, items 276–278, 298).

**V3.94 (Theorem 3.6.8 (1) (preprint 3.54), and Theorem 3.6.8 (preprint 3.54), with the text's hypotheses; `Polyfold.theorem_3_54_one'`, `Polyfold.theorem_3_54'`, `Polyfold.theorem_3_54_one_sc_of_tame`, `Polyfold.theorem_3_54_one_sc`, `Polyfold.theorem_3_54_sc`).** Let $f$ be an sc-Fredholm section of a strong bundle with tame germ data (V3.103), $f(x_0) = 0$, and $(f, x_0)$ in good position in a tame chart (V3.47, V3.103). Then there is an open $V_x \ni x_0$ such that $(f, y)$ is in good position in a tame chart at every zero $y \in V_x$; in particular $f'(y)$ is onto (`theorem_3_54_one'`). Moreover (`theorem_3_54'`) there are $d \in \mathbb{N}$ and regular data of Definition 3.2.1 (preprint 3.25) (V3.24) of dimension $d$ for $S = \{f = 0\}$ at $x_0$ whose chart domain lies in $V_x$, and every datum of Definition 3.2.1 (preprint 3.25) for $S$ at a point of $S$ in that domain has dimension $d$. With the text's notion of good position (sc-complement, V3.47) in the conclusion, from either form at $x_0$: every zero $y \in V_x$ is in good position with an sc-complement. *Book:* Theorem 3.6.8 (1), (2) (preprint 3.54), p. 136 (preprint p. 111). *Hidden hypotheses:* tameness as in V3.68. *Deviation (restated):* no hypothesis added; (2) as regular tangent germ data with locally constant dimension (the manifold structure of the local solution set is V3.95); the proof of the surjectivity in (1) is V3.93, through Definition 3.6.6 (preprint 3.53) from a chart not centred at $y$ (V3.79), read in the tame chart of the germ data at $y$ (V3.107); for $\dim K = 0$ the only zero near $x_0$ is $x_0$. The project's good position asks only for an algebraic good complement (V3.20), less than the text's as a conclusion; the sc form follows at

each zero (Findings, Chapter 3, items 210, 213).

**V3.95 (Theorems 3.1.26 and 3.1.22 (preprint 3.23 and 3.19) with the text's hypotheses; `Polyfold.theorem_3_23'`, `Polyfold.theorem_3_19'`, `Polyfold.theorem_3_23_sc`).** Let $f$ be an sc-Fredholm section of a strong bundle with tame germ data (V3.103), $f(x) = 0$ and $(f, x)$ in good position in a tame chart. Then there are an open $U \ni x$ and $d \in \mathbb{N}$ such that every zero in $U$ is in good position in a tame chart and a smooth point, and $S(f, U) = \{y \in U \mid f(y) = 0\}$ is a sub-M-polyfold whose induced structure is tame and equivalent to that of a $d$-dimensional manifold with boundary with corners (`theorem_3_23'`). If moreover $d_X \equiv 0$, the induced structure also has degeneracy index 0 everywhere (`theorem_3_19'`). Theorem 3.1.26 (preprint 3.23) with the text's notion of good position (sc-complement, V3.47) at $x$ and at every zero in $U$: `theorem_3_23_sc`. *Book:* Theorem 3.1.26 (preprint 3.23), p. 103 (preprint p. 77); Theorem 3.1.22 (preprint 3.19), p. 100 (preprint p. 74). *Hidden hypotheses:* tameness as in V3.68. *Deviation (hypotheses explicit):* no hypothesis added. For Theorem 3.1.22 (preprint 3.19) the hypothesis "$f'(x_0)$ surjective" is read as good position, the same at $d_X = 0$ (V3.100, V3.101); "the dimension agrees with the Fredholm index" is V3.97, "$\ker f'(y) = T_y S$" V3.99; "uniquely determined" is the equivalence of atlases.

**V3.96 (the index of the linearization at a zero, in every chart; `Polyfold.StrongBundle.IsSurjLinOfIndex`, `Polyfold.StrongBundle.linAt_index_of_conjugationData`, `Polyfold.StrongBundle.isSurjLinOfIndex_of_`, `Polyfold.StrongBundle.IsGoodPositionGerm.exists_isSurjLinOfIndex`, `Polyfold.theorem_3_19_lineariz`, `Polyfold.StrongBundle.linAt_isScFredholm`, `Polyfold.StrongBundle.linAt_isScFredholm_of_isSurjLin`, `Polyfold.proposition_3_52_sc`, `Polyfold.theorem_3_19_linearization_sc`).** `IsSurjLinOfIndex f y d`: in every strong bundle chart compatible with the atlas whose domain contains $y$, at the level-1 point $z$ over $\varphi(y)$, the linearization $\mathrm{Lin}(z) : T_z O \to K_z$ (V3.78) is surjective and Fredholm with $\mathrm{ind}\,\mathrm{Lin}(z) = \dim\ker \mathrm{Lin}(z) = d$. Theorem: let $f$ be an sc-smooth section, $x_0$ a zero, $c$ the strong bundle chart of Definition 3.1.11 (preprint 3.11) centred at $x_0$ with filling germ $g$, sc$^+$-germ $s$, $s(0) = g(0)$, and conjugation data of $g - s$ to a basic germ $h : O(\widetilde{C} \oplus W, 0) \to \mathbb{R}^N \oplus W$, $\widetilde{C} \subseteq \mathbb{R}^n$. Then in every strong bundle chart around $x_0$ the linearization at the point over $x_0$ is Fredholm of index $n - N$, and it is onto if the germ is in good position in $c$. Hence at a zero in good position `IsSurjLinOfIndex f x_0 d` holds with $d = n - N$ for every conjugation in the chart of Definition 3.6.6 (preprint 3.53). For an sc-Fredholm section and a zero $x_0$ in good position, in the chart of Definition 3.6.6 (preprint 3.53), $Dg(0)$ and $Tf(x_0) = Dg(0)|_{T_0 O}$ are Fredholm, $Tf(x_0)$ is onto, and for every conjugation $\mathrm{ind}\, Dg(0) = \mathrm{ind}\, Tf(x_0) = \dim\ker Tf(x_0) = n - N$. In sc form: for an sc-Fredholm section, a zero $y$ and any strong bundle chart around $y$, at the point $z$ over $\varphi(y)$ (a smooth point), $T_z O = Dr(z)E$ and $K_z = \rho(z)F$ are sc-subspaces and $\mathrm{Lin}(z) : T_z O \to K_z$ is an sc-Fredholm operator of index $n - N$ on every level; under `IsSurjLinOfIndex f y d` it is onto on every level, of index $d$ on every level; in the chart of Definition 3.6.6 (preprint 3.53), $Dg(0)$ and $f'(x_0)$ are sc-Fredholm and $f'(x_0)$ is onto on every level. *Book:* Proposition 3.6.5 (preprint 3.52), p. 135 (preprint p. 110), and the first clause of Theorem 3.1.22 (preprint 3.19), p. 100 (preprint p. 74). *Hidden hypotheses:* none. *Deviation (stronger):* in charts ($T_x X$ carries no linear structure here, Chapter 2); in every chart by the chart independence at zeros (V3.78), which the text does not state, realised on all levels by the chart change at smooth points (Findings, Chapter 3, items 236, 239); the index on every level, and the surjectivity on every level at a zero in good position, go beyond the text's statement. The hypothesis is the text's since the switch to the germ reading (phase 2): Proposition 3.6.5 (preprint 3.52) enters at the centre of the chart of the germ datum, where $\mathrm{Lin}(0) = Dg(0)$ on $T_0 O$ for the filling germ (Findings, Chapter 3, items 284, 310, 314, 330). `theorem_3_19_linearization_sc`

gives the conclusions in the chart at $x_0$, and `linAt_isScFredholm_of_isSurjLinOfIndex` with `IsGoodPositionGerm.exists_isSurjLinOfIndex` in every chart. The sc-Fredholm property was first proved on level 0 only; formerly "narrower", for the whole-chart hypothesis.

**V3.97 (the dimension of the solution set is the Fredholm index; `Polyfold.theorem_3_54_tangentGermChart_in` `Polyfold.theorem_3_54_two_index`, `Polyfold.theorem_3_23_index`, `Polyfold.theorem_3_19_index`, `Polyfold.theorem_3_57_index`, `Polyfold.theorem_3_54_graph_proj_conj`, `Polyfold.theorem_3_54_tangent` `Polyfold.StrongBundle.linAt_isScFredholm_of_isSurjLinOfIndex`).** Let $f$ be an sc-smooth section of a strong bundle, $f(x_0) = 0$, $(f, x_0)$ in good position in a tame chart (V3.103). (i) The data of Definition 3.2.1 (preprint 3.25) at $x_0$ of V3.84 have a dimension $d$ with `IsSurjLinOfIndex f x_0 d` (V3.96): $d = n - N$ for the conjugation in the chart from which $\psi\varphi$ is built. (ii) The regular data of V3.85 (ii) have such a dimension. (iii) For $f$ sc-Fredholm with tame germ data there are an open $U \ni x_0$ and $d$ such that every zero $y \in U$ is in good position in a tame chart with `IsSurjLinOfIndex f y d`, the zeros in $U$ are smooth points, and $S(f, U)$ is a sub-M-polyfold whose induced structure is tame and equivalent to that of a $d$-dimensional manifold with boundary with corners (`theorem_3_23_index`); if $d_X \equiv 0$, also of degeneracy index 0 everywhere (`theorem_3_19_index`). (iv) Under the hypotheses of V3.87 (good position in a tame chart at every zero, $S \neq \emptyset$): $S$ is a sub-M-polyfold with tame induced structure, a manifold with boundary with corners of locally constant dimension, and at every $x \in S$ there are regular data of a dimension $d$ such that every datum at a point of $S$ in their chart domain has dimension $d$ and `IsSurjLinOfIndex f y d` holds at every zero $y$ of that domain (`theorem_3_57_index`). For an sc-Fredholm section, at a zero with `IsSurjLinOfIndex f y d`, $\mathrm{Lin}(z)$ is moreover sc-Fredholm and onto on every level, of index $d$ on every level, in every chart (V3.96). *Book:* Theorem 3.1.22 (preprint 3.19), p. 100 (preprint p. 74) ("whose dimension agrees with the Fredholm index"); the proof of Theorem 3.6.8 (preprint 3.54), preprint p. 115 ($d = \dim\ker f'(x)$). *Hidden hypotheses:* tameness as in V3.68. *Deviation (stronger):* the index at every zero near the point and in every chart; the text states no dimension in Theorems 3.1.26, 3.6.11 (preprint 3.23, 3.57).

**V3.98 (the Fredholm index along the solution set and along paths, Definition 6.2.1 (preprint 6.1), Proposition 6.2.2 (preprint 6.2); `Polyfold.theorem_3_54_index_locally_constant`, `Polyfold.theorem_3_54_index`, `Polyfold.isSurjLinOfIndex_of_forall_dim`, `Polyfold.theorem_3_54_index` `Polyfold.StrongBundle.linAt_isScFredholm`, `Polyfold.index_locally_constant_tame`, `Polyfold.theorem_3_54_index_locally_constant_general_tame`, `Polyfold.index_locally_constant_of_ac` `Polyfold.theorem_3_54_index_locally_constant_general`, `Polyfold.StrongBundle.IsLinOfIndex`, `Polyfold.StrongBundle.restrictOpen`, `Polyfold.StrongBundle.IsIndexAtLocal`, `Polyfold.StrongBundle.I` `Polyfold.StrongBundle.exists_isIndexAtLocal`, `Polyfold.StrongBundle.IsIndexAt`, `Polyfold.StrongBundle.IsIndexAt.unique`, `Polyfold.StrongBundle.exists_isIndexAt`, `Polyfold.StrongBundle.isIndexAtLocal_iff_isIndexAt`, `Polyfold.exists_isScPlusSection_pullback_of_` `Polyfold.StrongBundle.isIndexAt_of_pullback_isLinOfIndex`, `Polyfold.proposition_6_2_local`, `Polyfold.proposition_6_2_local'`, `Polyfold.proposition_6_2`).** Let $f$ be an sc-Fredholm section of a strong bundle with tame germ data (V3.103), $f(x_0) = 0$, $(f, x_0)$ in good position in a tame chart. Then there are an open $V_x \ni x_0$ and $d$ such that `IsSurjLinOfIndex f y d` holds at every zero $y \in V_x$: $f'(y)$ is onto with $\mathrm{ind}\, f'(y) = \dim\ker f'(y) = d = \mathrm{ind}\, f'(x_0)$ (`theorem_3_54_index_locally_constant`). Theorem 3.6.8 (preprint 3.54) with the index (`theorem_3_54_index`): moreover every zero in $V_x$ is in good position in a tame chart, and $S$ has at $x_0$ regular data of dimension $d$ whose chart domain is $V_x$, every datum at a point of $S$ in $V_x$ having dimension $d$. On the way: at a zero in good position in a tame chart at which every datum of Definition 3.2.1 (preprint 3.25) for $S$ has dimension $d$, `IsSurjLinOfIndex f y d` ($f$ sc-smooth). In sc form, at every zero in $V_x$ and in every chart, $\mathrm{Lin}(z)$ is sc-Fredholm,

onto on every level, of index $d$ on every level (`theorem_3_54_index_locally_constant_sc`); at any zero of an sc-Fredholm section, in every chart, $\mathrm{Lin}(z)$ is sc-Fredholm with the same index on every level (`linAt_isScFredholm`). Without good position, with the text's hypotheses: if $f$ is an sc-Fredholm section with tame germ data and $f(x_0) = 0$, there are an open $V \ni x_0$ and $i \in \mathbb{Z}$ such that at every zero $y \in V$, in every strong bundle chart, $f'(y)$ is Fredholm of index $i$ (`IsLinOfIndex f y i; index_locally_constant_tame`) and $\mathrm{Lin}(z)$ is sc-Fredholm of index $i$ on every level (`theorem_3_54_index_locally_constant_general_tame`); no bump functions are assumed (the first forms, for $X$ admitting sc-smooth bump functions, Definition 5.3.1 (preprint 5.9): `index_locally_constant_of_admitsBumpFunctions`, `theorem_3_54_index_locally_constant_general`). **Definition 6.2.1 (preprint 6.1)**, with the text's locally defined section: an sc-Fredholm section $f$ has index $i$ at a smooth point $x$ (`IsIndexAtLocal f x i`) if for some open $V \ni x$ and some sc$^+$-section $s$ of the restriction $P|V$ of $P$ to $V$ (`StrongBundle.restrictOpen`: the strong bundle over the M-polyfold $V$ whose atlas consists of the charts of $P$ with base domain inside $V$), $(f|_V + s)(x) = 0$ and $(f|_V + s)'(x)$ is Fredholm of index $i$ in every strong bundle chart of $P|V$; the index is unique, at a zero it is the index of $f'(x)$, and it exists at every smooth point. With an sc$^+$-section on all of $X$ in place of $s$ (`IsIndexAt f x i`) the index is unique as well and implies the local one; if $X$ admits sc-smooth bump functions it exists at every smooth point and the two readings agree (`isIndexAtLocal_iff_isIndexAt`). **Proposition 6.2.2 (preprint 6.2)**: if $f$ is an sc-Fredholm section with tame germ data, then for every sc-smooth path $\varphi : \mathbb{R} \to X$ the index $\mathrm{ind}(f, \varphi(t))$ (local reading) is the same for all $t$ (`proposition_6_2_local`), and any two values of the index at $\varphi(0)$ and $\varphi(1)$ agree (`proposition_6_2_local'`); if $X$ admits sc-smooth bump functions, the same holds for the global reading (`proposition_6_2`). *Book:* Definition 6.2.1 (preprint 6.1), pp. 237–240 (preprint pp. 200–201); Proposition 6.2.2 (preprint 6.2), p. 240 (preprint p. 201) (over a tame M-polyfold, the index is constant along sc-smooth paths $[0,1] \to X$ of smooth points; the proof: "we do not need a sc-smooth partition of unity", preprint p. 202), whose proof uses the step of V3.82; in Chapter 3 the statement along the solution set is implicit in Theorem 3.6.8 (1) (preprint 3.54). *Hidden hypotheses:* tameness as in V3.68; for Definition 6.2.1 (preprint 6.1), the text's local sc$^+$-section $s$ with $s(x) = f(x)$ is a section of the restriction of the bundle to an open neighbourhood of $x$ ($-s$ here), a construction the text uses without comment. *Deviation (narrower):* the path is defined on $\mathbb{R}$, not on $[0,1]$ ($\mathbb{R} \times X$ is the product of Theorem 3.1.21 (preprint 3.18) and has no boundary, and the text's tame M-polyfold $[0,1] \times X$ is not used): a path on $[0,1]$ that extends sc-smoothly to $\mathbb{R}$ is covered, a path sc-smooth on $[0,1]$ only is not, and the project has no M-polyfold $[0,1]$. This is the only narrowing left; the hypotheses on $f$ are the text's, in the reading of tameness of V3.103 (as in V3.94), and no bump functions are assumed. *Proof:* Proposition 6.2.2 (preprint 6.2) is proved by the text's device, run on the restriction $P|V$ to the domain $V$ of a chart around $\varphi(t_0)$, where the text's sections need no cut-off — for each $t_0$ an sc$^+$-section $S$ of the pull-back to $\mathbb{R} \times V$ with $\tilde{f} + S = 0$ along the graph of the path near $t_0$ (the text's $R(x, f(\varphi(t)))$ in the chart), the local constancy of the index of $\tilde{f} + S$ along its zero set, the count $\mathrm{ind}(\tilde{f} + S)'(t,x) = \mathrm{ind}(f,x) + 1$ through the slice $S(t,\cdot)$, and the connectedness of $\mathbb{R}$; the earlier form for M-polyfolds with bump functions cuts the sections off instead (Findings, Chapter 3, items 340–349, 360–367). The local constancy along the zero set without good position is the same device: sc$^+$-sections $R(\cdot, e_i)$ on $P|V$ put the parametrized section of Theorem 3.1.21 (preprint 3.18) into general position at $(0, x_0)$, its index is constant along its solution set, at $(0,y)$ it is $k + \mathrm{ind}\, f'(y)$, and the index at a zero is the same for $P$ and $P|V$, through the germ datum (items 240–242, 360–363). Definition 6.2.1 (preprint 6.1) is well defined because two local sections are compared on the intersection of their domains; its independence of $s$ is proved from the germ

data and Theorem 3.1.18 (preprint 3.15), not by the text's appeal to Proposition 1.1.11 (preprint 1.10); and it exists at every smooth point through the section $-R(\cdot, f(x))$ on a chart domain (items 340, 364). Under bump functions a local section is cut off to a global one with the same index (item 367). The candidate counterexample of items 188 and 243 is excluded by Theorem 3.6.8 (1) (preprint 3.54). Without tame germ data the local constancy is open: it amounts to the index 0 of the block $(1-\rho(z))Dg(z)|_{\ker Dr(z)}$ of the filling at the nearby zeros, which condition (3) of Definition 3.1.4 (preprint 3.4) gives at $z=0$ only (items 243, 343).

**V3.99 (**$\ker f'(y) = T_yS$ **in a chart;** `Polyfold.theorem_3_54_ker_eq_tangent`**).** Let $f$ be an sc-smooth section of a strong bundle, $f(x_0)=0$, $(f,x_0)$ in good position in a tame chart (V3.103). Then there are a tame strong bundle chart $c$ centred at $x_0$ (as in Definition 3.6.6 (preprint 3.53)), the conjugating sc-diffeomorphism $\psi : U \to U' \subseteq C'$ (inverse $\psi'$, $\psi(0)=0$), the finite-dimensional $K = \ker D(h+t)(0)$, the solution graph $\sigma : V' \to E'$ over a relatively open $V' \ni 0$ of $K \cap C'$ ($C^1$, $\sigma(0)=0$, $D\sigma(0)=0$), an open $V_x \ni x_0$ with $\psi\varphi(\{y \in V_x \mid f(y)=0\}) = \{v+\sigma(v) \mid v \in V'\}$, and an open $V_{x,2} \subseteq V_x$ containing $x_0$, such that for every zero $y \in V_{x,2}$, with $\psi(\varphi(y)) = v+\sigma(v)$, $v \in V'$, and every level-1 point $z$ over $\varphi(y)$: $D\psi(z)$ maps the kernel of $\mathrm{Lin}(z) : T_zO \to K_z$ onto $N_v = \{n + D\sigma(v)n \mid n \in K\}$, the tangent space at $v$ of the graph. *Book:* Theorem 3.1.22 (preprint 3.19), p. 100 (preprint p. 74) ("$\ker f'(y) = T_yS(f,U)$ is the tangent space"). *Hidden hypotheses:* tameness as in V3.68. *Deviation (restated):* in the chart $\psi\varphi$ of the tangent germ data, the kernel of the linearization in the chart $c$; $T_yS$ as a subspace of $T_yX$ is not formalized (Chapter 2, gap G.13).

**V3.100 (good position is surjectivity at a zero with** $d_X = 0$**;** `Polyfold.StrongBundle.IsLinSurjAt`**,** `Polyfold.isInGoodPositionIn_of_subset`**,** `Polyfold.StrongBundle.isGoodPositionGerm_of_linAt_surj_of`
`Polyfold.StrongBundle.isGoodPositionGerm_of_isLinSurjAt`**,** `Polyfold.StrongBundle.IsGoodPositionGer`
`Polyfold.StrongBundle.isGoodPositionGerm_iff_isLinSurjAt`**,** `Polyfold.StrongBundle.IsLinSurjAt.linA`
`IsLinSurjAt f y` ("$f'(y)$ is surjective"): in some strong bundle chart compatible with the atlas whose domain contains $y$, at a level-1 point $z$ over $\varphi(y)$, $\mathrm{Lin}(z)$ maps onto the fibre $K_z = \{k \mid (\varphi(y),k) \in K\}$; then, at a zero, it does so in every such chart and at every level-1 point over $\varphi(y)$ (V3.78). Let $f$ be an sc-Fredholm section of a strong bundle with tame germ data (V3.103) and $y$ a zero with $d_X(y)=0$. Then $(f,y)$ is in good position in a tame chart if and only if `IsLinSurjAt f y`. On the way: a subspace $N \subseteq T$ is in good position inside $T$ to every cone containing $T$; in a chart with tame base retraction around a zero $y$ with $d_X(y)=0$ at which $(f,y)$ is an sc-Fredholm germ, if the linearization is onto there, $(f,y)$ is in good position, since $d_C(\varphi(y)) = d_X(y) = 0$ makes the partial quadrant at $\varphi(y)$ the whole space; the surjectivity is carried to the tame chart of the germ data at $y$, and the good position read there (V3.107). *Book:* the remark after Theorem 3.6.8 (preprint 3.54), p. 136 (preprint p. 111) (for $\partial X = \emptyset$ good position just means surjectivity), and Corollary 3.1.27 (preprint 3.24) ("an empty condition if $d_X(x)=0$", p. 104 (preprint p. 79)). *Hidden hypotheses:* the chart is tame, for $d_C = d_X$ (in a non-tame chart $d_C(\varphi(y))$ can exceed $d_X(y)$). *Deviation (hypotheses explicit).*

**V3.101 (Theorems 3.1.22 and 3.6.10 (preprint 3.19 and 3.56) with "**$f'(x)$ **surjective", and Theorem 3.6.10 (preprint 3.56) with tameness;** `Polyfold.theorem_3_19_of_surjective`**,** `Polyfold.theorem_3_56_of_surjective`**,** `Polyfold.theorem_3_56_tame`**).** Let $X$ be an M-polyfold with $d_X \equiv 0$ and $f$ an sc-Fredholm section of a strong bundle. (i) If $f$ has tame germ data (V3.103), $f(x_0)=0$ and `IsLinSurjAt f x_0` (V3.100), the conclusions of `theorem_3_19_index` hold (V3.97 (iii)): an open $U \ni x_0$ and $d$ with every zero $y \in U$ in good position in a tame chart, `IsSurjLinOfIndex f y d`, the zeros in $U$ smooth, and $S(f,U)$ a sub-M-polyfold whose induced structure is tame, a $d$-dimensional manifold with degeneracy index 0 everywhere. (ii) If $(f,x)$ is in good position in a tame chart at every zero (`theorem_3_56_tame`), or $f$ has tame germ data

and `IsLinSurjAt f x` at every zero (`theorem_3_56_of_surjective`), and $S \neq \emptyset$, then $S$ is a sub-M-polyfold whose induced structure is tame, a manifold of locally constant dimension with degeneracy index 0 everywhere, with regular data of Definition 3.2.1 (preprint 3.25) at every zero of a dimension $d$ constant near it and equal to the Fredholm index at the zeros of the chart domain. *Book:* Theorem 3.1.22 (preprint 3.19), p. 100 (preprint p. 74); Theorem 3.6.10 (preprint 3.56), p. 140 (preprint p. 115). *Hidden hypotheses:* tameness as in V3.68; $S \neq \emptyset$ as in V3.70. *Deviation (stronger):* the text's hypotheses; the tameness of the induced structure and the index are added conclusions.

**V3.102 (the filler condition at the nearby zeros follows from good position;** `Polyfold.normal_injective_of_ex` `Polyfold.exists_range_deriv_eq_ _iff_normal_injective_germ`, `Polyfold.isGoodPositionGermFilled_of`
(a) Let $U$ be relatively open in a partial quadrant, $R$ a strong bundle retraction on $U \lhd F$, $g$ an sc¹-germ with derivative family $Dg$, and $z$ a level-1 point in its germ over a point $z_0$ of the retract $O$ with $\ker Dg(z) \subseteq T_{z_0}O$. If for every $k \in F$ there is $x \in T_{z_0}O$ with $Dg(z)x = \rho(z_0)k$, then $Dr(z)b = 0$ and $\rho(z_0)Dg(z)b = Dg(z)b$ imply $b = 0$: condition (3) of Definition 3.1.4 (preprint 3.4) at $z_0$. (b) If moreover $\mathbf{f} \circ r$ is sc¹, $\mathbf{f}$ is a section of $K$ on $O$ with $\mathbf{f}(z_0) = 0$, $\mathbf{f} = g$ on $O$ near $z_0$, and $Dg(z)$ is onto, the two conditions are equivalent. (c) For an sc-Fredholm section with tame germ data (V3.103) and a zero $x_0$ in good position in a tame chart, $(f, x_0)$ is in good position in a tame chart with the filler condition at the nearby zeros (`IsGoodPositionGermFilled`, V3.82). *Book:* not in the text (Definition 3.1.4 (3) (preprint 3.4), p. 91 (preprint p. 64), at 0 only). *Hidden hypotheses:* none. *Deviation (not in the book):* (c) uses V3.94 and the tangency $\ker Dg(z_0) \subseteq T_{z_0}O$ of V3.81; it shows that the added hypothesis of V3.68, V3.69, V3.82, V3.85 is satisfied whenever the text's hypotheses are.

**V3.103 (Definitions 3.6.3, 3.6.4 and 3.6.6 (preprint 3.50, 3.51 and 3.53) in a tame chart;** `Polyfold.StrongBundle.IsTameChart`, `Polyfold.StrongBundle.IsScFredholmGermTame`, `Polyfold.StrongBundle.IsGoodPositionGermTame`, `Polyfold.StrongBundle.IsScFredholmSectionTame`, `Polyfold.StrongBundle.IsScFredholmGermTame.isScFredholmGerm`, `Polyfold.StrongBundle.IsGoodPositio` `Polyfold.StrongBundle.IsGoodPositionGermTame.isScFredholmGermTame`, `Polyfold.StrongBundle.IsGoodP` `Polyfold.StrongBundle.isTameChart_of_eqOn_id`, `Polyfold.ScRetraction.IsTame.congr`, `Polyfold.IsFillingGerm.nonempty_kerEquiv`, `Polyfold.TrivialBundle.tame_atlas_no_germData_in_atlas`
A strong bundle chart of $P$ is *tame* (`IsTameChart`) if it is compatible with the atlas and its base retraction $r : U \to U$ is tame (Definition 2.5.2 (preprint 2.34)). `IsScFredholmGermTame f x` (Definition 3.6.3 (preprint 3.50), "tame germ data at $x$"): $x$ is a smooth point and there are the data of Definition 3.1.11 (preprint 3.11) (V3.13: a strong bundle chart centred at $x$, a filling germ $g$ of the principal part, a local sc⁺-germ $s$ with $s(0) = g(0)$, and $g - s$ conjugated near 0 to a basic germ; the text's germ reading) in a tame chart. `IsGoodPositionGermTame f x` ("good position in a tame chart"): the same data with the germ in good position in that chart (Definition 3.6.6 (preprint 3.53) in the chart form of V3.47). `IsScFredholmSectionTame f` (Definition 3.6.4 (preprint 3.51), "tame germ data"): $f$ is an sc-Fredholm section (V3.16) with tame germ data at every smooth point. Each of these implies the form without tameness (V3.13, V3.47), good position in a tame chart gives tame germ data, and the strengthened good position of V3.82 is stated in a tame chart and implies good position in it. A retraction agreeing on its domain with a tame retraction is tame; hence a compatible chart whose base retraction is the identity of $U$ (the retract $O = U$ relatively open in the partial quadrant) is tame, so the hypotheses can be met. *Book:* Definition 3.6.3 (preprint 3.50), p. 135 (preprint p. 110) (a strong bundle chart around $x$ "where $K \to O$ is a tame strong local bundle", as in Definition 3.1.6 (preprint 3.6), p. 93 (preprint p. 66)); Definition 3.6.4 (preprint 3.51), p. 135 (preprint p. 110), and Definition 3.6.6 (preprint 3.53), p. 136 (preprint p. 111) (for a tame strong bundle); a strong bundle retraction is tame when its $r$ is (preprint p. 49); the proof

of Proposition 3.6.5 (preprint 3.52) starts from "a tame strong bundle chart" (p. 135 (preprint p. 110)). *Hidden hypotheses:* none. *Deviation (hypotheses explicit):* these are the hypotheses of the implicit function theorems of §3.6 (V3.48, V3.68–V3.70, V3.80–V3.82, V3.84–V3.87, V3.94–V3.102), in place of the reading of "tame strong bundle" as the tameness of the base retraction of every compatible chart, under which they were first proved and which cannot be met (V3.104). Whether germ data in some chart (V3.13) can be moved to some compatible tame chart when the atlas consists of tame charts is open; it cannot be done by re-centring a chart of the tame atlas, since whether a filling exists in a chart depends on the chart: the trivial bundle with a one-chart tame atlas has sc-Fredholm sections with tame germ data in compatible charts and no germ data in the chart of the atlas (Findings, Chapter 3, items 350–353). The text builds the tame chart into Definition 3.6.3 (preprint 3.50), and so do the theorems. Since the switch to the text's germ reading (phase 2) the three notions carry the text's germ data (V3.13, V3.45); the earlier whole-chart forms are `IsScFredholmGermTameWholeChart`, `IsGoodPositionGermTameWholeChart`, `IsScFredholmSectionTameWholeChart`, and the non-vacuity witnesses satisfy both readings (Findings, Chapter 3, items 220–221, 304, 330). For Definition 3.6.6 (preprint 3.53) the comparison is that of V3.47. Formerly "restated", for the whole-chart reading (round-trip item V3.103).

**V3.104 (a retraction onto a product with a point that is not tame: the earlier tameness hypothesis is unsatisfiable; `Polyfold.ScRetraction.halfLineConst_prod_not_isTame`, `Polyfold.ScRetraction.halfLineConst`, `Polyfold.ScRetraction.halfLineConst_retract`, `Polyfold.halfLineSc`).** Let $W$ be an sc-Banach space and $c$ the constant retraction $(a, w) \mapsto (1, 0)$ of the partial quadrant $[0, \infty) \oplus W$ of $\mathbb{R} \oplus W$ (sc-smooth, $(1, 0)$ being a smooth point), with retract $\{(1, 0)\}$. For every sc-smooth retraction $r$ of a nonempty relatively open subset $U$ of a partial quadrant $C$, the product retraction $c \times r$ of $([0, \infty) \oplus W) \times U$, whose retract is $\{(1, 0)\} \times r(U)$, is not tame: at $((0, 0), u)$ with $u \in r(U)$ the degeneracy index is $1 + d_C(u)$, at its image $((1, 0), u)$ it is $d_C(u)$, against condition (1) of Definition 2.5.2 (preprint 2.34). *Book:* not in the text (Definition 2.5.2 (preprint 2.34), pp. 53–54 (preprint pp. 39–40); Definition 2.6.4 (preprint 2.50), pp. 67–68 (preprint pp. 51–52)). *Hidden hypotheses:* none. *Deviation (not in the book):* the retraction-level half of the argument (Findings, Chapter 3, item 190) that the hypothesis under which the theorems of §3.6 were first proved, "the base retraction of every strong bundle chart compatible with the atlas is tame", fails for every strong bundle over a nonempty base: from a compatible chart $(\varphi, U, R, \Phi)$, the chart $(1, \varphi)$ onto $\{1\} \times O$ with base retraction $(a, u) \mapsto (1, r(u))$ on $[0, \infty) \times U$ (the retraction above, with $W = 0$) has the transition maps of the first chart composed with affine maps, hence is compatible with every chart the first one is. That chart and its compatibility are not formalized (see "What is not verified"); so the earlier statements, vacuous by this argument, are replaced by those with tame germ data (V3.103).

**V3.105 (Theorem 3.1.18 (preprint 3.15) for tame germ data; `Polyfold.theorem_3_15_tame`, `Polyfold.StrongBundle.IsScFredholmGermTame.sectionAdd`).** If $f$ is an sc-Fredholm section with tame germ data (V3.103) and $s$ an sc$^+$-section, then $f + s$ is an sc-Fredholm section with tame germ data: at every smooth point the germ data of $f + s$ are taken in the tame chart of those of $f$ (filling germ $g + \hat{s} \circ r$, sc$^+$-germ $\hat{s}_1 \circ r + t_0$, the same conjugation, as in V3.17). *Book:* Theorem 3.1.18 (preprint 3.15), p. 98 (preprint p. 71), recalled after Definition 3.6.4 (preprint 3.51), p. 135 (preprint p. 110), for the tame strong bundle of §3.6. *Hidden hypotheses:* none. *Deviation (hypotheses explicit):* Theorem 3.1.18 (preprint 3.15) for the tame reading of Definitions 3.6.3, 3.6.4 (preprint 3.50, 3.51); the proof of V3.17 does not change the chart.

**V3.106 (Theorem 3.1.21 (preprint 3.18) for tame germ data; `Polyfold.theorem_3_18_tame`, `Polyfold.PullbackChartData.ofProd_isTame`, `Polyfold.StrongBundle.IsScFredholmGermTame.param`,**

`Polyfold.StrongBundle.IsScFredholmSectionTame.param`**).** In the setting of V3.91, if $f$ is an sc-Fredholm section with tame germ data (V3.103) and $s_1, \dots, s_n$ are sc$^+$-sections, then $F(r,x) = f(x) + \sum_i r_i s_i(x)$ is an sc-Fredholm section of $\pi^*P$ with tame germ data. On the way: the product chart $(r \mapsto r - r_0) \times c$ of a strong bundle chart $c$ of $P$ (V3.92) has the base retraction $1 \times r$ on $\mathbb{R}^n \times U$, which is tame when $r$ is ($\mathbb{R}^n$ carries the identity retraction of the whole space, so $d(a,u) = d_C(u)$, and the minimal subspaces and tangent spaces are products); so $\tilde{f}(r,x) = f(x)$ has germ data at every smooth $(r_0, x)$ in the tame product chart, and V3.105 adds $\tilde{s}$. *Book:* Theorem 3.1.21 (preprint 3.18), p. 100 (preprint p. 74) (for a tame strong bundle). *Hidden hypotheses:* none. *Deviation (hypotheses explicit):* the tame form of V3.91, which is what the implicit function theorems of §3.6 need when applied to $F$; the tameness of the base $\mathbb{R}^n \times X$ (`prod_rn_isTame`, V3.91) is a different statement.

**V3.107 (good position in any chart of the germ data;** `Polyfold.StrongBundle.isGoodPositionAtGerm_of_chart` `Polyfold.StrongBundle.isGoodPositionAtGerm_of_chart_dense`, `Polyfold.StrongBundle.IsGoodPosition`**).** (a) Let $f$ be an sc-smooth section, $y$ a zero, $c$ a strong bundle chart of $P$ around $y$ (not necessarily centred), $z$ a level-1 point over $\varphi(y)$, and $g$ an sc$^1$-germ agreeing with **f** on $O$ near $\varphi(y)$, with $z$ in its germ, such that $Dg(z)$ maps $T_zO$ onto $K_z$ and $\ker Dg(z) \cap T_zO$ is in good position to $C_{\varphi(y)}$ inside $T_zO$. Then for **every** datum of Definition 3.1.11 (preprint 3.11) at $y$ (a strong bundle chart $c'$ compatible with the atlas and centred at $y$, a filling germ $g'$ with a derivative family $Dg'$) the germ is in good position in $c'$ at $Dg'(0)$; with only $\rho(z)k$ in the range for every smooth $k \in F_\infty$ (the hypotheses of V3.79), the same when $Dg'(0)$ is Fredholm. (b) If $(f,y)$, $f(y) = 0$, is in good position in some chart (V3.47) and has tame germ data at $y$ (V3.103), it is in good position in that tame chart. *Book:* Definition 3.6.6 (preprint 3.53), p. 136 (preprint p. 111), presupposes that good position does not depend on the chart; the proof of Theorem 3.6.8 (preprint 3.54), p. 139 (preprint p. 115) ("proceeding as above"). *Hidden hypotheses:* none. *Deviation (not in the book):* V3.79 gives the good position in the chart that Definition 3.1.11 (preprint 3.11) provides at $y$; (a) gives it in every chart carrying germ data at $y$. (b) applies (a) at $z = 0$ in the chart of the given good position, centred at $y$, where 0 lies in every germ, $Tf(0)$ maps onto $K_0$, $\ker Dg(0) \subseteq T_0O$ and $C_0 = C$; in the germ reading no Fredholm input is needed (Findings, Chapter 3, item 301). This is how the good position of the nearby zeros in Theorem 3.6.8 (1) (preprint 3.54) (V3.94) and the good position from surjectivity (V3.100) are obtained in tame charts. The whole-chart form of (a) is `Polyfold.StrongBundle.isGoodPositionAt_of_chart_dense`.

**V3.108 (Definitions 3.1.4 and 3.6.1 (preprint 3.4 and 3.48), the 2021 filling;** `Polyfold.IsFilling2021`, `Polyfold.HasFilling2021`, `Polyfold.IsFilling2017`, `Polyfold.isFilling2021_iff`, `Polyfold.IsFilling202` `Polyfold.IsFillingGerm.isFilling2021`, `Polyfold.IsFilledVersion.isFilling2021`, `Polyfold.IsFilling2017.isFilling2021`, `Polyfold.isFilling2017_iff`, `Polyfold.isFilling2017_id`, `Polyfold.isFilling2017_rn`**).** Let $K = R(U \lhd F) \to O$ be a strong local bundle, $U$ relatively open in $C$ with $0 \in U$, and $R(u,h) = (r(u), \rho(u)h)$ covering the sc-smooth retraction $r : U \to U$ onto $O = r(U)$. A germ $g$, sc-smooth on an sc-germ of neighbourhoods of 0 in $C$ inside $U$, is a 2021 filling of the section with principal part $\hat{f}$ if $r(0) = 0$ and (1) $\hat{f} = g$ on $O$ near 0; (2) every $y \in U$ near 0 with $g(y) = \rho(r(y))g(y)$ lies in $O$; (3) the derivative at 0 of $k(y) = [1 - \rho(r(y))]g(y)$, restricted to $\ker Dr(0)$, is a topological linear isomorphism onto $\ker \rho(0)$. The 2017 filling is the same with $r$ tame. A 2021 filling is exactly a filling of V3.7 with $r(0) = 0$, and a filling of V3.7 is a 2021 filling as soon as $0 \in O$; a 2017 filling is a 2021 filling; with the identity strong bundle retraction (tame) every sc-smooth $g$ fills itself. *Book:* Definition 3.1.4, p. 91 (preprint 3.4, p. 64), and Definition 3.6.1, p. 134 (preprint 3.48, p. 109): the 2021 text asks $r$ to be an sc-smooth retraction, where the 2017 text and the preprint ask for a tame one, and assumes $r(0) = 0$. *Hidden hypotheses:*

none. *Deviation (restated):* the germ reading of V3.7, which never asked for tameness; the 2021 change brings the text to the formalization's definition. $r(0) = 0$ is a condition of the definition.

**V3.109 (a 2021 filling over a retraction that is not tame; `Polyfold.CornerFilling.Rc`, `Polyfold.CornerFilling.gc`, `Polyfold.CornerFilling.isFilledVersion_gc`, `Polyfold.CornerFilling.isFi` `Polyfold.CornerFilling.not_isTame_r`, `Polyfold.CornerFilling.not_isFilling2017_gc`, `Polyfold.exists_isFilling2021_not_isFilling2017`).** In $E = \mathbb{R} \oplus \mathbb{R}$, $C = U = [0, \infty) \times \mathbb{R}$, $F = \mathbb{R}$, let $R((a, b), h) = ((0, b), 0)$, so that $r(a, b) = (0, b)$, $O = \{0\} \times \mathbb{R}$ and $\rho \equiv 0$. Then $g(a, b) = a$ is a 2021 filling of the zero section, and $r$ is not tame ($d_C(1, 0) = 0$, $d_C(r(1, 0)) = 1$), so it is not a 2017 filling: the 2021 filling is strictly more general than the 2017 one. *Book:* not stated. *Hidden hypotheses:* none. *Deviation (not in the book):* compares Definition 3.1.4 with its 2017 form.

**V3.110 (Definitions 3.1.11, 3.1.16 = 3.6.3, 3.6.4 (preprint 3.11, 3.14 = 3.50, 3.51) with the 2021 filling; `Polyfold.StrongBundle.IsScFredholmGerm2021`, `Polyfold.StrongBundle.IsScFredholmGerm2017`, `Polyfold.StrongBundle.IsScFredholmSection2021`, `Polyfold.StrongBundle.zero_mem_0_of_chart`, `Polyfold.StrongBundle.isScFredholmGerm2021_iff`, `Polyfold.StrongBundle.isScFredholmGerm2017_iff`, `Polyfold.StrongBundle.IsScFredholmGerm2017.isScFredholmGerm2021`, `Polyfold.StrongBundle.isScFredh` `Polyfold.isScFredholmSection2021_graphSection`, `Polyfold.isScFredholmGerm2017_graphSection`).** $(f, x_0)$ is a 2021 sc-Fredholm germ if $x_0$ is smooth and some compatible strong bundle chart centred at $x_0$ carries a 2021 filling $g$ of the principal part and an sc$^+$-germ $s$ with $s(0) = g(0)$ such that $g - s$ is conjugate near 0 to a basic germ; the 2017 germ uses the 2017 filling. $f$ is a 2021 sc-Fredholm section if it is sc-smooth, regularizing, and a 2021 sc-Fredholm germ at every smooth point. The 2021 germ is the project's sc-Fredholm germ (V3.13, V3.45), the 2021 section the project's sc-Fredholm section (V3.16), and the 2017 germ the project's germ with tame germ data (V3.103); 2017 implies 2021. The graph section of a $C^\infty$ map $\mathbb{R}^n \to \mathbb{R}^k$ is a 2021 sc-Fredholm section and a 2017 sc-Fredholm germ at every point. *Book:* Definitions 3.1.11, p. 96 (preprint 3.11, p. 69), 3.1.16, p. 97 (preprint 3.14, p. 70), 3.6.3 and 3.6.4, p. 135 (preprint 3.50 and 3.51, p. 110). *Hidden hypotheses:* none. *Deviation (restated):* as V3.13; $r(0) = 0$ holds in a chart centred at $x_0$. Whether every 2021 germ is a 2017 germ is open (gap G.79).

**V3.111 (Proposition 3.1.5 (preprint 3.5) for the 2021 filling; `Polyfold.proposition_3_1_5_2021`).** If $g$ is a 2021 filling of $\hat{f}$ with $\hat{f}(0) = 0$ and $Dg$ its derivative germ, then $Dg(0)$ maps $T_0O = \operatorname{range} Dr(0)$ into $K_0 = \operatorname{range} \rho(0)$, so that its restriction $T\hat{f}(0) : T_0O \to K_0$ is defined, and (1) $T\hat{f}(0)$ is onto iff $Dg(0)$ is, (2) one is Fredholm iff the other is, with the same index. *Book:* Proposition 3.1.5, p. 93 (preprint 3.5, p. 65). *Hidden hypotheses:* as V3.8. *Deviation (restated):* V3.8 for the 2021 definition; $0 \in O$ from $r(0) = 0$.

**V3.112 (Theorem 3.1.13 (preprint 3.12) and Corollary 3.1.15 (preprint 3.13) for the 2021 definitions; `Polyfold.theorem_3_1_13_2021`, `Polyfold.corollary_3_1_15_2021`).** For a 2021 sc-Fredholm germ $(f, x_0)$ at a zero: nested open neighbourhoods $O(i)$ of $x_0$ such that the closure of the zeros in $O(i)$ is compact on level $i$; and a sequence of zeros converging to $x_0$ converges on every level. *Book:* Theorem 3.1.13 and Corollary 3.1.15, p. 97 (preprint 3.12 and 3.13, p. 70). *Hidden hypotheses:* none. *Deviation (restated):* V3.14 and V3.15 through V3.110; their proofs do not use tameness.

**V3.113 (Theorems 3.1.18 (preprint 3.15) and 3.1.21 (preprint 3.18) for the 2021 definitions; `Polyfold.theorem_3_1_18_2021`, `Polyfold.theorem_3_1_21_2021`).** If $f$ is a 2021 sc-Fredholm section and $s$ an sc$^+$-section, $f + s$ is a 2021 sc-Fredholm section; for sc$^+$-sections $s_1, \dots, s_n$, $(\lambda, x) \mapsto f(x) + \sum_i \lambda_i s_i(x)$ is a 2021 sc-Fredholm section of the pull-back bundle over $\mathbb{R}^n \times X$. *Book:* Theorem 3.1.18, p. 98 (preprint 3.15, p. 71), and Theorem 3.1.21, p. 100 (preprint 3.18, p. 74).

*Hidden hypotheses:* none. *Deviation (restated):* V3.17 and V3.91 through V3.110.

**V3.114 (Proposition 3.6.5 (preprint 3.52) for the 2021 definitions; `Polyfold.proposition_3_6_5_2021`).** At a zero $x_0$ of a 2021 sc-Fredholm germ, in a chart centred at $x_0$ with a filling $g$, $Dg(0)$ and the restriction $T_0O \to K_0$ are Fredholm of index $n - N$. *Book:* Proposition 3.6.5, p. 135 (preprint 3.52, p. 110). *Hidden hypotheses:* none. *Deviation (restated):* V3.46 through V3.110.

**V3.115 (Theorem 3.6.8 (preprint 3.54), the proof step on p. 138 without a tame chart; `Polyfold.CornerFilling.isGoodPositionAtGerm_gc`, `Polyfold.CornerFilling.not_isInGoodPosition_gc`, `Polyfold.CornerFilling.not_range_le_minimalSubspace`, `Polyfold.theorem_3_6_8_tame_step_2021`).** In the example of V3.109: $g$ is a 2021 filling of $\hat{f} = 0$, $r$ is not tame, $0 \in O$, $\hat{f}(0) = 0$, $g$ is sc$^1$ with $Dg(0)$ onto, and $\ker T\hat{f}(0)$ is in good position to the partial cone in $T_0O$ (V3.47); but the range of $1 - Dr(0)$ does not lie in the subspace $E_0$ of $C$ at 0, and $\ker Dg(0) = 0 \oplus \mathbb{R}$ is not in good position to $C$ in $E$. *Book:* Theorem 3.6.8, p. 136 (preprint 3.54, p. 111), proof pp. 137–138: the passage from good position in $T_0O$ to good position of $\ker Dg(0)$ in $E$ uses the sc-complement $\ker Dr(0) \subseteq E_0$ of $T_0O$, given by Proposition 2.5.3 (preprint 2.35) for a tame retract; the 2021 text calls the retract tame "by assumption". *Hidden hypotheses:* a tame chart. *Deviation (not in the book):* with the untamed 2021 definitions this step does not hold as we read it; the example's conclusions hold (its solution set $O \cong \mathbb{R}$ is a manifold), so the theorem itself is not refuted. Theorem 3.6.8 and what rests on it are verified with tame germ data (V3.85, V3.94, V3.103); for untamed data they are open (gap G.79).

**V3.116 (Theorem 3.5.1 (preprint 3.40) with the 2021 sequential conclusion; `Polyfold.IsBasicGerm.solves`, `Polyfold.theorem_3_5_1_2021`).** Let $h$ be a basic germ on an sc-germ of neighbourhoods $G$ of 0 in $C = \tilde{C} \times W$ ($\tilde{C}$ closed), $U = G_0$, $s$ an sc$^+$-germ on $G$ with $s(0) = 0$, and $S = \{x \in U : h(x) + s(x) = 0\}$. Then there are nested neighbourhoods $O(0) \supseteq O(1) \supseteq \cdots$ of 0, relatively open in $C$ and inside $U$, with $\mathrm{cl}(S \cap O(m)) \subseteq C_m$; and if $x_k \in S$ converges to 0 in $E_0$, then for every $m$ there are $y_k \in E_m$ with $y_k = x_k$ for $k$ large and $y_k \to 0$ in $E_m$. *Book:* Theorem 3.5.1, p. 118 (preprint 3.40, p. 94); the sequential conclusion is new in the 2021 edition. *Hidden hypotheses:* as V3.35. *Deviation (restated):* the closure is taken in $E_0$ (it lies in $C$, which is closed); "converges on level $m$" is stated with level-$m$ lifts. *Proof:* from V3.14 and V3.15 in the local form (`Polyfold.theorem_3_12_local`, `Polyfold.corollary_3_13_local`).

**V3.117 (closed convex cones and extreme rays; `Polyfold.IsClosedConvexCone`, `Polyfold.IsExtremeRayGen`, `Polyfold.IsClosedConvexCone.eq_zero`, `Polyfold.isClosedConvexCone_Ici`, `Polyfold.isExtremeRayGen_or`, `Polyfold.isExtremeRayGen_symm_single`, `Polyfold.exists_sub_single_mem`, `Polyfold.exists_quadrantEqu`).** A closed convex cone $K$ is a closed convex set with $K \cap (-K) = \{0\}$ and $tK \subseteq K$ for $t \geq 0$; $a$ generates an extreme ray if $a \in K \setminus \{0\}$ and $y \in K$, $a - y \in K$ imply $y \in \mathbb{R}_+ a$. $[0, \infty)$ is a closed convex cone generated by the extreme ray of 1. If $N$ is in good position to $C = T^{-1}([0,\infty)^n \oplus W)$ and $N \cap C$ is pointed, a linear isomorphism $e : N \cong \mathbb{R}^{\dim N}$ carries $N \cap C$ onto $[0,\infty)^{\dim N}$, and the $e^{-1}(e_j)$ generate extreme rays. *Book:* p. 145, new in the 2021 edition, for finite-dimensional spaces. *Hidden hypotheses:* none. *Deviation (stronger):* the quadrant normal form, from V3.21, is added; it replaces the Krein–Milman-type Lemmas 3.8.1–3.8.3 in the proofs of V3.118–V3.121.

**V3.118 (Lemma 3.8.4; `Polyfold.lemma_3_8_4_2021`, `Polyfold.lemma_3_65`).** Let $N$ be in good position to $C = T^{-1}([0,\infty)^n \oplus W)$ with good complement $N^\perp$, $N \cap C$ a closed convex cone, $\sigma_a$ the set of $i$ with $a_i = 0$, and $\Gamma = \bigcup_{0 \neq a \in N \cap C} \sigma_a$. Then (1) $y_i = 0$ for $y \in N^\perp$ and $i \in \sigma_a$, $a$ a generator of an extreme ray; (2) $\Gamma$ is the union of the $\sigma_a$ over generators of extreme rays; (3) $y_i = 0$ for $y \in N^\perp$ and $i \in \Gamma$. *Book:* Lemma 3.8.4, p. 147, new in the 2021 edition. *Hidden hypotheses:* none.

*Deviation (stronger):* (1) and (3) hold without the cone hypothesis (Lemma 3.8.8, V3.55); (2) uses only that $N \cap C$ is pointed. *Proof:* (2) not by writing $a$ as a sum of extreme-ray generators (Krein–Milman, Lemma 3.8.1): for $0 \neq a \in N \cap C$ with $a_i = 0$ and $e(a)_j > 0$, $a - e(a)_j e^{-1}(e_j) \in N \cap C$ forces $e^{-1}(e_j)_i = 0$ (V3.117).

**V3.119 (Lemma 3.8.5; `Polyfold.lemma_3_8_5_2021`, `Polyfold.lemma_3_8_5_not_literal_2021`).** For $N$ in good position to $C$ with $N \cap C$ a closed convex cone, either $N \cap (\mathbb{R}^{\Gamma^c} \oplus W) = \{0\}$, or $N \subseteq \mathbb{R}^{\Gamma^c} \oplus W$, $\dim N = 1$ and $\Gamma = \emptyset$. In $E = \mathbb{R}$, $C = [0, \infty)$, the subspace $N = 0$ is in good position, $N \cap C = \{0\}$ is a closed convex cone and $N \subseteq \mathbb{R}^{\Gamma^c} \oplus W$, but $\dim N = 0$. *Book:* Lemma 3.8.5, p. 147, new in the 2021 edition: either $N \subseteq \mathbb{R}^{\Gamma^c} \oplus W$, in which case $\dim N = 1$ and $\Gamma = \emptyset$, or $N \cap (\mathbb{R}^{\Gamma^c} \oplus W) = \{0\}$. *Hidden hypotheses:* $N \neq 0$ for the first case. *Deviation (restated):* stated as the dichotomy that the proof shows (it assumes $N \cap (\mathbb{R}^{\Gamma^c} \oplus W) \neq \{0\}$); read as "$N \subseteq \mathbb{R}^{\Gamma^c} \oplus W$ implies $\dim N = 1$" it does not hold as we read it for $N = 0$. *Proof:* if $\Gamma \neq \emptyset$, a point of $N$ with $x_i = 0$ for $i \in \Gamma$ lies in $N \cap C$ with $-x$; if $\Gamma = \emptyset$ and $\dim N \geq 2$, two generators with all coordinates positive give $b_0 - t b_1 \in N \cap C$ for small $t > 0$, against V3.117.

**V3.120 (Lemma 3.8.6; `Polyfold.lemma_3_8_6_2021`, `Polyfold.lemma_3_8_6_two_2021`, `Polyfold.halfLine_witness_2021`).** Let $N$ be in good position to $C$ and $P : N \to \mathbb{R}^\Gamma$ the coordinate projection. (1) If $N \cap (\mathbb{R}^{\Gamma^c} \oplus W) = \{0\}$, $P$ is a linear isomorphism carrying $N \cap C$ onto $[0, \infty)^\Gamma$. (2) If $N \cap C$ is a closed convex cone, $N \subseteq \mathbb{R}^{\Gamma^c} \oplus W$ and $N \neq 0$, there is a linear isomorphism $N \cong \mathbb{R}$ carrying $N \cap C$ onto $[0, \infty)$. The hypotheses of (2) are satisfiable ($N = E = \mathbb{R}$, $C = [0, \infty)$). *Book:* Lemma 3.8.6, p. 148, new in the 2021 edition, both parts under the cone hypothesis. *Hidden hypotheses:* $N \neq 0$ in (2). *Deviation (narrower):* (2) assumes $N \neq 0$; as printed it does not hold as we read it for $N = 0$ (V3.119). (1) holds without the cone hypothesis.

**V3.121 (Corollary 3.8.7; `Polyfold.corollary_3_8_7_2021`).** If $N$ is in good position to $C$ and $N \cap C$ is a closed convex cone, then $N \cap C$ is a partial quadrant in $N$, and a linear isomorphism $N \cong \mathbb{R}^{\dim N}$ carries $N \cap C$ onto $[0, \infty)^{\dim N}$. *Book:* Corollary 3.8.7, p. 149, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (stronger):* the quadrant form, which the text also derives on p. 149; the partial quadrant is V3.21 (Proposition 3.1.25 (preprint 3.22)) without the cone hypothesis.

**What is not verified in this chapter**

- pp. 136–139 (preprint pp. 111–114), Theorem 3.6.8 (1) (preprint 3.54): nothing of the statement; the proof is not the text's (V3.93), and the first forms under the filler condition at the nearby zeros (V3.68, V3.69, V3.82, V3.85) are kept beside the forms with the text's hypotheses (V3.94, V3.95). Of the perturbation argument of the text's Proposition 6.2.2 (preprint 6.2), used for the surjectivity at the nearby zeros, only the linear algebra is formalized (V3.83); for the index it is formalized (V3.98).
- preprint p. 201 (Chapter 6), Proposition 6.2.2 (preprint 6.2) for paths sc-smooth on $[0, 1]$ only: it is verified with the text's hypotheses (tame germ data, no bump functions, Definition 6.2.1 (preprint 6.1) with local sections) for paths on $\mathbb{R}$ (V3.98); a path on $[0, 1]$ that does not extend sc-smoothly to $\mathbb{R}$ is not covered, the project having no M-polyfold $[0, 1]$. Without tame germ data the local constancy of the index at the zeros is open (Findings, Chapter 3, items 243, 343).
- p. 100 (preprint p. 74), Theorem 3.1.22 (preprint 3.19): "$\ker f'(y) = T_y S(f, U)$" with $T_y S$ a subspace of $T_y X$; it is verified in the chart of the tangent germ data (V3.99).
- p. 113 (preprint p. 89), Remark 3.3.8 (preprint 3.36) in quantitative form: its content is V3.26 and V3.31.
- p. 135 (preprint p. 110), Proposition 3.6.5 (preprint 3.52) on $T_x X \to Y_x$ itself: verified in

charts, as an sc-Fredholm operator $T_zO \to K_z$ in every chart (V3.46, V3.96); $T_xX$ carries no linear structure here (Chapter 2).

- pp. 140–145 (preprint pp. 116–120), Theorem 3.7.1 (preprint 3.58) for a general sc-splitting $E = B \oplus X$, $F = A \oplus Y$: verified in the coordinates of the text's proof (V3.89).
- pp. 145–152 (preprint pp. 121–125), Lemmas 3.8.1–3.8.3, 3.8.9–3.8.11 (preprint 3.62–3.64, 3.66–3.68) and p. 167 (preprint p. 139), Lemma 3.8.20 (preprint 3.76): superseded by the formal proofs of Proposition 3.1.25 (preprint 3.22) and Proposition 3.8.17 (preprint 3.73), not formalized.
- p. 164 (preprint p. 135), Definition 3.8.14 (preprint 3.71): Mathlib's $C^k$ within the set is used, equal to it on these sets.
- Lemma 3.8.13 (preprint 3.70) in a general presentation of $C$ (hence $d_M = d_X$ in Theorem 3.2.3 (preprint 3.27)); Lemma 3.6.9 (preprint 3.55) is verified in the model, in coordinates, and for a general presentation of $C$ (V3.49, V3.80).
- Theorem 3.2.3 (preprint 3.27) for a set whose data do not satisfy condition (b) of V3.24 (it would need a re-graphing along another complement); for solution sets (b) holds by the choice of the complement (V3.77, V3.84).
- The linearization $f'(x) : T_xX \to Y_x$ and good position in $T_xX$ (Definition 2.5.8 (preprint 2.38), gap G.13 of Chapter 2): Definition 3.6.6 (preprint 3.53) is verified in charts, with its chart independence at zeros (V3.78).
- The chart-level half of the argument that the earlier hypothesis "the base retraction of every compatible strong bundle chart is tame" cannot be met (Findings, Chapter 3, items 190, 199): that the strong bundle chart $(1, \varphi)$ with base retraction $(a, u) \mapsto (1, r(u))$ on $[0, \infty) \times U$ is a chart compatible with every chart $(\varphi, U, R, \Phi)$ is (sc-calculus for affine maps, not done). Only the non-tameness of its base retraction is verified (V3.104); the theorems no longer use the hypothesis.
- Definition 3.1.11 (preprint 3.11) ⇒ Definition 3.6.3 (preprint 3.50) (Findings, Chapter 3, items 199, 353): whether in a strong bundle whose atlas consists of tame charts every sc-Fredholm germ has germ data in some compatible tame chart is open. Re-centring a tame chart of the atlas does not do it: a filling needs $\ker Dr(0) \cong \ker \rho(0)$ in its chart, and a one-chart tame atlas of the trivial bundle carries no germ data at all (V3.103). The text builds the tame chart into Definition 3.6.3 (preprint 3.50), and the theorems of §3.6 assume tame germ data.
- §3.1: Definitions 3.1.4, 3.1.6, 3.1.11, 3.1.16 (preprint 3.4, 3.6, 3.11, 3.14) (and 3.48, 3.50, 3.51, 3.53) are in the text's germ reading since phase 2 (V3.7, V3.9, V3.13, V3.16, V3.43, V3.45, V3.47), and Definition 3.1.3 (preprint 3.3) since the fourth addendum (V3.5). Whether the text's germ reading and the project's earlier whole-chart reading of Definition 3.1.11 (preprint 3.11) agree for sections sc-smooth on $X$ is open; no theorem depends on it (Findings, Chapter 3, items 220–221, 264, 330).
- pp. 91, 134–138, the 2021 (untamed) definitions: whether every 2021 sc-Fredholm germ has a filling over a tame retraction (Findings, Chapter 3, item 353), and Theorem 3.6.8 for untamed germ data, with what rests on it (Theorems 3.1.22, 3.1.26, Corollary 3.1.27, Theorems 3.6.10, 3.6.11): its proof step on p. 138 needs a tame chart (V3.115). Gap G.79.

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## 5. Chapter 4 (HWZ §§4.1–4.4): verified except Lemma 4.1.11 (preprint 4.10) and §4.4

Chapter 4 of Part I (pp. 175–204 (preprint pp. 147–168)) asks when an M-polyfold is an ordinary manifold with boundary with corners. §4.1 defines a compatible manifold structure with corners

(Definition 4.1.1 (preprint 4.1)), states the characterization (Theorem 4.1.2 (preprint 4.2): $X_0 = X_\infty$, $1_0^1 : X \to X^1$ sc-smooth, and centred tame charts whose tangent space at the centre is finite-dimensional and in good position), and prepares its proof: Remark 4.1.3 (preprint 4.3), sc$^+$-retractions and sc$^+$-retracts (Definitions 4.1.4, 4.1.6 (preprint 4.4, 4.6), Lemmas 4.1.5, 4.1.7 (preprint 4.5, 4.7), Proposition 4.1.8 (preprint 4.8)), and the description of an sc$^+$-retract near a point as a graph over its tangent space (Proposition 4.1.9 (preprint 4.9), with Lemmas 4.1.11–4.1.13 (preprint 4.10–4.12)). §4.2 defines smooth finite-dimensional submanifolds (Definition 4.2.1 (preprint 4.13), Remark 4.2.2 (preprint 4.14), Proposition 4.2.3 (preprint 4.15), an open question), good position and tameness of submanifolds (Definitions 4.2.5, 4.2.6 (preprint 4.16, 4.17)), proves that a submanifold in good position is a tame manifold with corners (Theorem 4.2.7 (preprint 4.18), Lemma 4.2.8 (preprint 4.19), Remark 4.2.9 (preprint 4.20)), defines general position (Definition 4.2.10 (preprint 4.21)) and describes a submanifold near a point of general position (Theorem 4.2.11 (preprint 4.22)). §4.3 proves, with Sard's theorem, that the fibres of a family $A \subseteq \mathbb{R}^m \times X$ over almost every parameter are compact manifolds with corners in general position (Theorem 4.3.1 (preprint 4.23), Remark 4.3.2 (preprint 4.24)). §4.4 sketches sc-differential forms, de Rham cohomology and Stokes' theorem (Definitions 4.4.1, 4.4.6 (preprint 4.25, 4.27), Proposition 4.4.5 (preprint 4.26), Theorem 4.4.7 (preprint 4.28)).

What is verified. **Theorems 4.1.2, 4.2.7, 4.2.11 and 4.3.1 (preprint 4.2, 4.18, 4.22 and 4.23) hold with the text's hypotheses**, and Theorem 4.1.2 (preprint 4.2) holds without its tameness hypothesis (V4.5). Every numbered statement of §§4.1–4.3 is verified except Lemma 4.1.11 (preprint 4.10), which is no longer used: Definition 4.1.1 (preprint 4.1) (V4.1–V4.4), Theorem 4.1.2 (preprint 4.2) (V4.5, V4.27, V4.28), Remark 4.1.3 (preprint 4.3), with an example for "$X^1 \neq X$ is possible", which the text does not give (V4.6), Definitions 4.1.4, 4.1.6 (preprint 4.4, 4.6) and Lemmas 4.1.5, 4.1.7 (preprint 4.5, 4.7) (V4.9–V4.12), Proposition 4.1.8 (preprint 4.8) with its chart form (V4.13, V4.14), Proposition 4.1.9 (preprint 4.9) with Lemmas 4.1.12 and 4.1.13 (preprint 4.11 and 4.12) (V4.20–V4.25), Definition 4.2.1 (preprint 4.13) (V4.29), Remark 4.2.2 (preprint 4.14) in a chart (V4.31, V4.32), Proposition 4.2.3 (preprint 4.15) (V4.33–V4.36), Definitions 4.2.5, 4.2.6, 4.2.10 (preprint 4.16, 4.17, 4.21) in charts with their chart independence (V4.38–V4.40, V4.44, V4.45), Theorem 4.2.7 (preprint 4.18), Lemma 4.2.8 (preprint 4.19), Remark 4.2.9 (preprint 4.20) (V4.41–V4.43), Theorem 4.2.11 (1)–(4) (preprint 4.22) (V4.49–V4.52), Theorem 4.3.1 (preprint 4.23) (V4.54, V4.63, V4.64) and Remark 4.3.2 (preprint 4.24) (V4.55). Four places of the text needed more than a reading. (a) Proposition 4.1.9 (preprint 4.9) is proved with two hypotheses the text leaves implicit — the good complement $Y$ is closed, and $p$ is the projection along $Y$, not $Dt(0)$ — and by a direct argument, because the text's route through Lemma 4.1.11 (preprint 4.10) and Theorem 3.5.11 (preprint Corollary 3.47) yields the graph map only as a germ over a complement of its own choosing (V4.20); where Proposition 4.1.9 (preprint 4.9) is applied the closed complement is chosen, so Theorems 4.1.2, 4.2.7, 4.2.11, 4.3.1 (preprint 4.2, 4.18, 4.22, 4.23) need nothing extra. (b) One step of Lemma 4.2.8 (preprint 4.19) is false ("$z \in Z \subset C$" does not give $n + y \in C$) and is repaired (V4.42). (c) Theorem 4.2.11 (4) (preprint 4.22) has no proof in the text and is proved (V4.51, V4.52). (d) In Theorem 4.3.1 (preprint 4.23) "regular value" is the stratified notion of the proof — every stratified regular value is a regular value, and with ordinary regular values the conclusion fails at the value — and the finite cover of the proof does not exist in general; a countable one is used (V4.53, V4.54, V4.64). Proved although the text does not construct or state it: charts and strong bundle charts centred at a smooth point (V4.15–V4.18), the sc$^+$-retraction onto the fibre $A_\lambda$ and its general position (V4.61, V4.62), Theorem 3.8.22 (preprint 3.78) in $C^\infty$ form (V4.58), and Sard's theorem in all dimensions (V4.56, vendored from our Milnor project, `math_faithful`). Not verified: Lemma 4.1.11 (preprint 4.10), §4.4, the open question after Proposition 4.2.3 (preprint 4.15), and the chart independence of stratified reg-

ularity (see the list at the end of this section). A fourth addendum (2026-11-02; Findings, items 140–146) added the example of Remark 4.1.3 (preprint 4.3) to V4.6. A fifth addendum (2026-11-08; Findings, items 150–159) covers the Springer 2021 edition: the 2021 forms of Theorems 4.2.11 and 4.3.1 (V4.65, V4.66), the sc-smooth vector fields of §4.4 (Definition 4.4.3, Lemma 4.4.4) and their Lie bracket (Proposition 4.4.5, Appendix 4.5) in a local model (V4.67–V4.72), Definition 4.4.3 on an M-polyfold (V4.68), and an example in which the bracket at a point of $O_2$ does not lie in $E_1$ (V4.73). It continues (Findings, items 160–163) with Lemma 4.5.1 in a local model and the independence of the chart (V4.74, V4.75), and on an M-polyfold Proposition 4.4.5 (V4.76, closing gap G.38), Lemma 4.5.1 for every sc-smooth map (V4.77), Lemma 4.4.4 for $i = 1$ (V4.78) and for every $i$ (V4.79).

The setting. An sc$^+$-map is a map whose values lie in $F_1$ and whose lift into $F^1$ is sc-smooth (V4.7); the text uses the word for maps and defines it only for retractions. $T_xX$ carries no linear structure in the formalization (gap G.7 of Chapter 2), so every condition the text states in $T_aX$ — the tangent space of a submanifold, good and general position, stratified regularity — is stated in a chart and, except for stratified regularity, proved independent of the chart. A chart "with $\psi(x) = 0$" is a chart re-centred at the smooth point $x$, modelled in the partial quadrant $C_{\varphi(x)}$ of Definition 2.5.1 (preprint 2.33) (V4.17). The map $1^1_0 : X \to X^1$ is defined on $X_1$, with an arbitrary value off $X_1$ that plays no role. Tameness of $X$ enters §4.2 only through Lemma 4.2.8 (preprint 4.19) (Theorem 4.2.7 (preprint 4.18), Remark 4.2.9 (preprint 4.20)), the tame centred chart of Theorem 4.2.11 (preprint 4.22) and Proposition 2.5.6 (preprint 2.36) (general position implies good position, V4.48). Findings referred to below are those of `Exposition/Findings.md`, "Chapter 4".

**V4.1 (Definition 4.1.1 (preprint 4.1), corner charts and corner structures; `Polyfold.MChart.IsCornerChart`, `Polyfold.MPolyfold.HasCornerStructure`, `Polyfold.MPolyfold.hasCornerStructure_iff`, `Polyfold.MPolyfold.HasCornerStructure.isTame`, `Polyfold.MChart.IsCornerChart.isTameRetract`, `Polyfold.MChart.IsCornerChart.isScPlusRetract`).** A chart of $X$ is a *corner chart* if its local model is $(U, C, V)$ with $V$ a finite-dimensional real normed space carrying the constant sc-structure, $C$ a partial quadrant in $V$, $U$ relatively open in $C$, and the identity retraction. $X$ has a *corner structure* if it has an atlas of corner charts equivalent to its atlas; equivalently, every point of $X$ lies in a corner chart that is a chart of the M-polyfold structure. The dimension of $V$ may vary from chart to chart. The model of a corner chart is a tame retract and an sc$^+$-retract, and an M-polyfold with a corner structure is tame. *Book:* Definition 4.1.1 (preprint 4.1), p. 175 (preprint p. 147). *Hidden hypotheses:* "sc-smoothly compatible charts" with "$\varphi$ an sc-diffeomorphism" is read as compatibility with the atlas of $X$ (equivalence of atlases, as for tameness). *Deviation (restated):* the classical manifold structure with corners is represented by the atlas of corner charts; local constancy of the dimension is neither stated by the text nor formalized.

**V4.2 (Definition 4.1.1 (preprint 4.1), the transition maps are $C^\infty$; `Polyfold.MChart.IsCornerChart.contDiffOn`, `Polyfold.MChart.isRelOpenInQuadrant_overlap`).** For compatible corner charts $\chi$, $\chi'$ the overlap $\chi(V \cap V')$ is relatively open in the partial quadrant of $\chi$, and the transition map $\chi' \circ \chi^{-1}$ is of class $C^\infty$ on it (derivatives within the set, one-sided at the boundary). *Book:* Definition 4.1.1 (preprint 4.1), p. 175 (preprint p. 147), the paragraph after the definition. *Hidden hypotheses:* none. *Deviation:* "the atlas defines the structure of a smooth manifold with boundary with corners" is not formalized beyond this (Mathlib's manifolds with corners have a single model space).

**V4.3 (finite-dimensional sc-Banach spaces; `Polyfold.ScBanach.finiteDimensional_lvl`, `Polyfold.ScBanach.toBase_surjective`, `Polyfold.ScBanach.scIsoConst`, `Polyfold.ScBanach.mem_smooth_c`** If $E_0$ is finite-dimensional, every level $E_m$ is finite-dimensional, every inclusion $E_m \to E_0$ is onto, $E$ is sc-isomorphic to $E_0$ with the constant structure (the level maps being the inclusions), and

every point of $E_0$ is smooth. *Book:* §1.1, p. 6 (preprint p. 6), after Definition 1.1.1 (preprint 1.1) ("a finite-dimensional Banach space has precisely one sc-structure"), used in Definition 4.1.1 (preprint 4.1). *Hidden hypotheses:* the density of $E_\infty$ in every level (Definition 1.1.1 (2) (preprint 1.1)) is what makes the inclusions onto; compact injective inclusions alone do not ($E_0 = \mathbb{R}$, $E_m = 0$ for $m \geq 1$). *Deviation (restated):* "one sc-structure" as an sc-isomorphism onto the constant structure.

**V4.4 (Definition 4.1.1 (preprint 4.1) with arbitrary finite-dimensional models, and the manifolds with corners of Chapter 3; `Polyfold.MChart.exists_isCornerChart_of_finiteDimensional`, `Polyfold.IsRelOpenInQuadrant.isScDiffeoQuadrant_toConst`, `Polyfold.MPolyfold.hasCornerStructure_i`, `Polyfold.MPolyfold.isManifoldWithCorners_iff`, `Polyfold.MPolyfold.IsManifoldWithCorners.hasCorner`**
A chart of $X$ onto a relatively open subset of a partial quadrant in an sc-Banach space with finite-dimensional $E_0$ is, with the same map and domain, a corner chart for the constant structure on $E_0$, of the same dimension. Hence $X$ has a corner structure iff it has an equivalent atlas of charts onto relatively open subsets of partial quadrants in sc-Banach spaces with finite-dimensional $E_0$ (the manifolds with corners of locally constant dimension of Theorem 3.2.3 (preprint 3.27)). The manifolds with corners of dimension $n$ of Theorem 3.2.3 (preprint 3.27) are exactly the corner structures all of whose corner charts have dimension $n$. *Book:* Definition 4.1.1 (preprint 4.1), p. 175 (preprint p. 147) ("or in a finite-dimensional vector space"); the relation to Chapter 3 is not stated. *Hidden hypotheses:* none. *Deviation (not in the book).*

**V4.5 (Theorem 4.1.2 (preprint 4.2), the characterization; `Polyfold.MPolyfold.theorem_4_2`, `Polyfold.MPolyfold.theorem_4_2_general`).** Let $X$ be an M-polyfold and $x_0 \in X_1$ the junk value of $1^1_0$ off $X_1$. The following are equivalent. (i) $X$ has a corner structure (V4.1). (ii) Every point of $X$ is smooth, $1^1_0 : X \to X^1$ is sc-smooth, and every $x \in X$ lies in a chart $\psi$ of $X$ with $\psi(x) = 0$ whose model $(O, C, E)$ is a tame retract and whose tangent space $N = T_0O$ (Definition 2.3.12 (preprint 2.17)) is finite-dimensional and in good position to $C$ (Definition 3.1.24 (preprint 3.21)). `theorem_4_2` is the text's form, with the hypothesis "$X$ tame". *Book:* Theorem 4.1.2 (preprint 4.2), p. 176 (preprint p. 147); proof pp. 183–186 (preprint pp. 156–157). *Hidden hypotheses:* the chart with $\psi(x) = 0$ is a re-centred chart (V4.17). *Deviation (stronger):* the tameness of $X$ is not used, and (i) implies it (V4.1). *Proof:* V4.27, V4.28.

**V4.6 (Remark 4.1.3 (preprint 4.3); `Polyfold.ScBanach.liftOne`, `Polyfold.MPolyfold.toLevelOne`, `Polyfold.MPolyfold.remark_4_3_levelVal`, `Polyfold.MPolyfold.remark_4_3_isScDiffeo`, `Polyfold.remark_4_3_const`, `Polyfold.remark_4_3_example`, `Polyfold.remark_4_3_retract_example`, `Polyfold.IsLossFamily.isScSmooth`, `Polyfold.exists_retract_smooth_not_scPlus_of_data`, `Polyfold.MPolyfold.mem_shift_levelSet_iff`).** The inclusion $1^0_1 : X^1 \to X$ is always sc-smooth; in the charts $(V_1, \varphi, (O_1, C_1, E^1))$ and $\varphi$ it is the sc-operator $\iota : E^1 \to E$. If every point of $X$ lies in $X_1$ and $1^1_0 : X \to X^1$ is sc-smooth, then $1^1_0$ is an sc-diffeomorphism with inverse $1^0_1$. For a finite-dimensional $V$ with the constant structure, $V^1 = V$ as sc-Banach spaces, the lift $V \to V^1$ is the identity, and every sc-smooth retract in $V$ is an sc$^+$-retract. The levels of $X^1$ are those of $X$ shifted by one, as sets: $(X^1)_m = X_{m+1}$. **The example:** there is an M-polyfold all of whose points are smooth ($X_0 = X_\infty$) for which $1^1_0 : X \to X^1$ is not sc-smooth, so $X^1 \neq X$ as M-polyfolds. It is a retract $O \subseteq \mathbb{R} \oplus X$, $X$ the weighted sequence space of V1.13 ($|x|_m = \sup_k (k+1)^m |x_k|$), with a single chart: the image of $r(t, z) = (t, L(t)z)$, $L(t)$ the projection onto $\cos\theta\, e_{n_k} + \sin\theta\, e_{n_{k+1}}$ ($n_k = 2^k$), the angle $\theta$ turning from 0 to $\pi/2$ with flat ends while $t$ runs through $[1/(k+2), 1/(k+1)]$, and $L(t) = 0$ for $t \leq 0$. Its points are smooth, and the lift $O \to E_1$ is discontinuous at 0, so $O$ is not an sc$^+$-retract (Proposition 4.1.8 (preprint 4.8)). The sc-smoothness of $r$ comes from a criterion proved for it: for level operators $L(p)_m : Z_m \to W_m$ depending on a finite-dimensional parameter $p$, compatible with

the inclusions, with $(p,z) \mapsto L(p)_m z$ continuous and $p \mapsto L(p)_m$ of class $C^\infty$ as a map into the operators $Z_{m+1} \to W_m$ (one level of loss), $(p,z) \mapsto L(p)z$ is sc-smooth. *Book:* Remark 4.1.3 (preprint 4.3), p. 176 (preprint p. 148). *Hidden hypotheses:* $1^1_0$ is defined on $X_1$ (the text: when $X_0 = X_\infty$). *Deviation (stronger):* the sc-diffeomorphism clause needs only $X = X_1$, not $X_0 = X_\infty$; "$X^1 \neq X$ is possible" has an example, which the text does not give, and in it $X$ and $X^1$ even have different topologies on the same set. "$X^1 \neq X$" is read as "$1^1_0$ is not sc-smooth", equivalently "the identity is not an sc-diffeomorphism". An example needs a retraction that loses a level in its first derivative, as the splicings of HWZ do (a retraction with $C^1$ level maps gives none: an argument, not formalized), so its sc-smoothness does not follow from Proposition 1.2.4 (preprint 1.19). "$(X^1)_m = X_{m+1} = X_i$" misprints $X_0$; the topologies of the iterated levels agree (`Polyfold.MPolyfold.shiftLevelHomeomorph`; gap G.6, closed) (Findings, items 5, 125, 140–146). Formerly "narrower", for the missing example (gap G.31, now closed).

**V4.7 (sc⁺-maps; `Polyfold.IsScPlus`, `Polyfold.IsScPlus.mem_range_succ`, `Polyfold.IsScPlus.iff_exists`, `Polyfold.IsScPlus.isScSmooth`, `Polyfold.IsScPlus.comp_isScSmooth`, `Polyfold.IsScSmooth.comp_isScPlu` `Polyfold.ScPlus.isScPlus`).** A map $f$ on $U \subseteq E_0$ is *sc⁺* if $f(U) \subseteq F_1$ and the lift $U \to F^1$, $x \mapsto f(x) \in F_1$, is sc-smooth. Then $f(U_m) \subseteq F_{m+1}$ for every $m$. On an sc-domain $U$: $f$ is sc⁺ iff some sc-smooth $g : U \to F^1$ has $\iota \circ g = f$ on $U$; an sc⁺-map is sc-smooth; $g \circ f$ is sc⁺ if $g$ is sc⁺ on an sc-domain $V$ and $f$ is sc-smooth with $f(U) \subseteq V$; $g \circ f$ is sc⁺ if $f$ is sc⁺ with $f(U) \subseteq V$ and $g$ is sc-smooth on $V$ (through Proposition 1.2.2 (preprint 1.17) for $g$); an sc⁺-operator is an sc⁺-map. *Book:* not defined (Definition 4.1.4 (preprint 4.4) is for retractions; the proof of Lemma 4.1.7 (preprint 4.7) says "$r$ is a sc⁺-map"). *Hidden hypotheses:* $U$, $V$ sc-domains for the composition rules. *Deviation (not in the book).*

**V4.8 (the derivative of an sc⁺-map; `Polyfold.IsScPlus.exists_scPlus_deriv`, `Polyfold.IsScPlus.isCompact` `Polyfold.finiteDimensional_range_of_isCompactOperator_of_idem`).** Let $f$ be sc⁺ on an sc-domain $U$ and $x \in U_1$ a smooth point. There is an sc⁺-operator $S : E \to F$ whose level-0 operator is $Df(x)$; $Df(x) : E_0 \to F_0$ is compact, and so is every level operator of $S$. A compact continuous linear projection $P = P \circ P$ of a normed space has finite-dimensional range. *Book:* proof of Lemma 4.1.7 (ii) (preprint 4.7), p. 177 (preprint p. 149) ("a sc⁺-operator, since $r$ is a sc⁺-map … must be finite-dimensional"). *Hidden hypotheses:* $x$ smooth; the range of a compact projection is finite-dimensional by Riesz's theorem. *Deviation (hypotheses explicit).*

**V4.9 (Definition 4.1.4 (preprint 4.4), sc⁺-retractions; `Polyfold.ScRetraction.IsScPlus`, `Polyfold.ScRetraction.IsScPlus.mem_lvlSet_succ`, `Polyfold.MPolyfold.IsScPlusRetractionOn`).** An sc-smooth retraction $r : U \to U$ is an *sc⁺-retraction* if it is an sc⁺-map on $U$ (V4.7); then $r(U_m) \subseteq U_{m+1}$ for every $m$. On an open $V$ of an M-polyfold: $s$ is an sc⁺-retraction on $V$ if $s$ is sc-smooth on $V$, $s(V) \subseteq V$, $s \circ s = s$ on $V$, $s(V) \subseteq X_1$, and $1^1_0 \circ s : V \to X^1$ is sc-smooth (for every junk value). *Book:* Definition 4.1.4 (preprint 4.4), p. 176 (preprint p. 148). *Hidden hypotheses:* in the M-polyfold form, $s(V) \subseteq X_1$, so that "$r : V \to V^1$" makes sense. *Deviation (restated):* $r(U_m) \subseteq U_{m+1}$ is derived, not assumed.

**V4.10 (Lemma 4.1.5 (preprint 4.5); `Polyfold.lemma_4_5`).** If $t$ is an sc⁺-retraction and $s$ an sc-smooth retraction (possibly of another relatively open subset of $C$) with the same retract, then $s$ is an sc⁺-retraction. *Book:* Lemma 4.1.5 (preprint 4.5), p. 177 (preprint p. 148). *Hidden hypotheses:* none. *Proof:* the text's, $s = t \circ s$ and sc⁺ □ sc-smooth is sc⁺.

**V4.11 (Definition 4.1.6 (preprint 4.6), sc⁺-retracts; `Polyfold.IsScPlusRetract`, `Polyfold.IsScPlusRetract.isS` `Polyfold.isScPlusRetract_iff_forall`).** $(O, C, E)$ is an *sc⁺-retract* if $O$ is the retract of an sc⁺-retraction of a relatively open $U \subseteq C$; then every sc-smooth retraction onto $O$ is sc⁺, and $O$ is an

sc⁺-retract iff it is a retract all of whose retractions are sc⁺. *Book:* Definition 4.1.6 (preprint 4.6) and the remark after it, pp. 176–178 (preprint pp. 148–149). *Hidden hypotheses:* none.

**V4.12 (Lemma 4.1.7 (preprint 4.7); `Polyfold.lemma_4_7_i`, `Polyfold.lemma_4_7_ii`, `Polyfold.lemma_4_7_ii_t`** Let $r$ be an sc⁺-retraction onto $O$. (i) $O \subseteq E_\infty$. (ii) For every level-1 point $o$ with $\iota(o) \in O$ the range of $Dr(o)$ is finite-dimensional; every $o \in O$ has a smooth lift $o_1 \in E_1$, and $T_{o_1}O = Dr(o_1)E_0$ is a finite-dimensional linear subspace. *Book:* Lemma 4.1.7 (preprint 4.7), p. 177 (preprint p. 149). *Hidden hypotheses:* none. *Deviation (hypotheses explicit):* the tangent space of Definition 2.3.12 (preprint 2.17) is indexed by level-1 points; the step "$Dr(o)$ is an sc⁺-operator" is V4.8.

**V4.13 (Proposition 4.1.8 (preprint 4.8); `Polyfold.proposition_4_8`).** An sc-smooth retract $(O, C, E)$ is an sc⁺-retract iff $O \subseteq E_\infty$ and the lift $1^1_0$ is sc-smooth from $(O, C, E)$ to $(O_1, C_1, E^1)$ in the sense of Definition 2.1.14 (preprint 2.7). *Book:* Proposition 4.1.8 (preprint 4.8), p. 177 (preprint p. 149). *Hidden hypotheses:* none. *Deviation:* none (the proof ends "the lemma is proved").

**V4.14 (Proposition 4.1.8 (preprint 4.8) in charts; `Polyfold.MPolyfold.levelChart_toLevelOne`, `Polyfold.MPolyfold.proposition_4_8_chart`, `Polyfold.MPolyfold.isScSmoothMap_toLevelOne_iff`, `Polyfold.MPolyfold.isScSmoothMap_toLevelOne_of_cover`).** In the charts $\varphi$ of $X$ and $(V_1, \varphi, (O_1, C_1, E^1))$ of $X^1$ the map $1^1_0$ is the lift $O \to O_1$. For a chart $\varphi : V \to O$ of $X$: the model is an sc⁺-retract iff $V \subseteq X_\infty$ and $1^1_0$ is sc-smooth on $V$. Hence "every point smooth and $1^1_0$ sc-smooth" holds iff every chart of $X$ is modelled on an sc⁺-retract, iff every point lies in some chart so modelled. *Book:* not stated; used in the proofs of Theorem 4.1.2 (preprint 4.2) and Proposition 4.2.3 (4) (preprint 4.15). *Hidden hypotheses:* none. *Deviation (not in the book).*

**V4.15 (the partial quadrant $C_e$ and the neighbourhood $G(e)$; `Polyfold.recentreNbhd`, `Polyfold.recentreIso`, `Polyfold.isPartialQuadrant_partialQuadrantAtWith`, `Polyfold.image_sub_partia`, `Polyfold.recentreNbhd_inter_eq`, `Polyfold.degIndex_sub_recentre`, `Polyfold.minimalSubspaceWith_rece`, `Polyfold.partialQuadrantAtWith_recentreIso`).** Let $T$ present $C$ with coordinates $\lambda_1, \dots, \lambda_n$ and let $e \in E_0$. $G(e) = \{v : \lambda_i(v) > 0 \text{ whenever } \lambda_i(e) \neq 0\}$ is open and, for $e \in C$, contains $e$. $C_e$ is a partial quadrant, presented by reordering the coordinates as $\mathbb{R}^{I(e)} \oplus (\mathbb{R}^{\text{rest}} \oplus W)$; $C_e - e = C_e$; $G(e) \cap C = G(e) \cap C_e$; and for $v \in G(e)$ the vanishing coordinates of $v - e$ for this presentation are those of $v$, so that $d_{C_e}(v - e) = d_C(v)$ ($v \in C$) and the minimal subspace and the partial quadrant of $v - e$ are $E_v$ and $C_v$. *Book:* Definition 2.5.1 (preprint 2.33), p. 53 (preprint p. 39), asserts without proof that $C_e$ is a partial quadrant; the rest is not in the text. *Hidden hypotheses:* none. *Deviation (not in the book).*

**V4.16 (translations and re-centred retracts; `Polyfold.isScSmooth_add_const`, `Polyfold.isScSmooth_sub_const`, `Polyfold.relOpen_translate`, `Polyfold.isScDiffeoQuadrant_translate`, `Polyfold.ScRetraction.restrict`, `Polyfold.ScRetraction.restrictIn_isTame`, `Polyfold.isScRetract_recentre`, `Polyfold.isTameRetract_re`** Translation by a smooth vector is sc-smooth on every sc-domain. For $e$ smooth and $U$ relatively open in $C$ with $U \subseteq G(e)$, the translate $U - e$ is relatively open in $C_e$ and $v \mapsto v - e$ is an sc-diffeomorphism $U \to U - e$. The restriction of a retraction $r$ to $U \cap r^{-1}(W) \cap W$ ($W$ open) is a retraction onto $r(U) \cap W$, tame if $r$ is. For a retract (tame retract) $(O, C, E)$ and a smooth $e \in E_0$, $((O \cap G(e)) - e, C_e, E)$ is a retract (tame retract), with the retraction $u \mapsto r(u + e) - e$. *Book:* not in the text. *Hidden hypotheses:* $e$ smooth — translation by $e$ preserves every level only then. *Deviation (not in the book).*

**V4.17 (charts centred at a smooth point; `Polyfold.MChart.recentre`, `Polyfold.MChart.recentre_isChart`, `Polyfold.MChart.recentre_apply_self`, `Polyfold.MChart.isTameRetract_recentre`, `Polyfold.MChart.chartDegIndex_recentre`, `Polyfold.MPolyfold.exists_centredChart`,**

`Polyfold.MPolyfold.IsTame.exists_centredChart`**).** For a chart $\varphi : V \to O \subseteq C$ of $X$, a presentation of $C$ and a smooth $e$, the chart $\varphi_e = \varphi - e$ on $V \cap \varphi^{-1}(G(e))$, modelled on $((O \cap G(e)) - e, C_e, E)$, is a chart of $X$, with $\varphi_e(x) = 0$ when $\varphi(x) = e$, the same degeneracy index as $\varphi$ on its domain, and a tame model when $\varphi$ has one. So every smooth point $x$ of the domain of a chart $\varphi$ lies in a chart $\chi$ of $X$ with domain inside that of $\varphi$ and $\chi(x) = 0$; in a tame M-polyfold every smooth point lies in a tame chart centred at it. *Book:* used without construction in Theorem 4.1.2 (ii) (preprint 4.2), the proofs of Theorems 4.2.7 and 4.2.11 (preprint 4.18 and 4.22), and Lemma 5.3.3 (preprint 5.11). *Hidden hypotheses:* $x$ smooth; the model lies in $C_{\varphi(x)}$, not in $C$. *Deviation (not in the book).*

**V4.18 (strong bundle charts centred at a smooth point;** `Polyfold.StrongBundleChart.recentre`**,** `Polyfold.StrongBundleChart.Compatible.recentre`**,** `Polyfold.StrongBundle.IsChart.recentre`**,** `Polyfold.StrongBundle.IsChart.exists_centred`**).** For a strong bundle chart $c$ over $\varphi$, a presentation and a smooth $e$, the chart $c_e$ over $\varphi_e$ with bundle retraction $R_e(u,h) = (r(u+e) - e, \Gamma(u+e,h))$ and $\Phi_e(y) = (\Phi(y)_1 - e, \Phi(y)_2)$ is compatible with every chart $c$ is compatible with (the projection being continuous), and is a chart of the strong bundle. Every smooth point of the base in the domain of a strong bundle chart lies in a strong bundle chart centred at it, with smaller domains. *Book:* used in Lemma 5.3.3 (preprint 5.11). *Hidden hypotheses:* $e$ smooth; $(e,0)$ is then smooth in $E \oplus F$ and $E \oplus F^1$. *Deviation (not in the book).*

**V4.19 (the level data of an sc⁺-retraction and the mean value estimate;** `Polyfold.ScRetraction.lvlMap`**,** `Polyfold.ScRetraction.lvlDeriv`**,** `Polyfold.ScRetraction.IsScPlus.hasFDerivWithinAt_lvlMap`**,** `Polyfold.ScRetraction.IsScPlus.continuousOn_lvlDeriv`**,** `Polyfold.ScRetraction.IsScPlus.exists_mvt`**,** `Polyfold.exists_norm_sub_sub_le_of_hasFDerivWithinAt`**).** Let $t$ be an sc⁺-retraction and $k \geq 0$. Then $t : U_{k+1} \to E_{k+1}$ is differentiable within $U_{k+1}$, with derivative $Dt(x) \in \mathcal{L}(E_{k+1})$ continuous in $x$ in the operator norm. For $u_0 \in U_{k+1}$ and $\theta > 0$ there is $r > 0$ with $C_{k+1} \cap B(u_0, r) \subseteq U_{k+1}$, $\|Dt(z) - Dt(u_0)\| \leq \theta$ there, and $|t(z') - t(z) - Dt(u_0)(z' - z)|_{k+1} \leq \theta |z' - z|_{k+1}$ for $z, z' \in C_{k+1} \cap B(u_0, r)$. *Book:* (76)–(77) and the estimates in the proofs of Lemmas 4.1.12 and 4.1.13 (preprint 4.11 and 4.12), pp. 181–183 (preprint pp. 154–155). *Hidden hypotheses:* the mean value inequality is taken on the convex set $C_{k+1} \cap B(u_0, r)$, since $U_{k+1}$ need not be convex. *Deviation (restated):* the text's integrals are the mean value inequality.

**V4.20 (Proposition 4.1.9 (preprint 4.9);** `Polyfold.proposition_4_9`**,** `Polyfold.finiteDimensional_tangentSpa`
`Polyfold.GraphData`**).** Let $t : U \to U$ be an sc⁺-retraction onto $O \ni 0$, $N = T_0 O$ (finite-dimensional by Lemma 4.1.7 (preprint 4.7)), in good position to $C$, $Y$ a good complement of $N$, $Y$ **closed**, and $p : E_0 \to N$ continuous linear with $p = 1$ on $N$ and $p = 0$ on $Y$ (**the projection along** $Y$). There are an open $W \ni 0$ in $E_0$, a relatively open $V \ni 0$ in the partial quadrant $N \cap C$ of $N$ (constant structure) and $\Gamma : N \to E_0$, sc-smooth on $V$, with $p \circ \Gamma = 1$ on $V$, $\Gamma(a) - a \in Y$, $\Gamma(V) = O \cap W$, $\Gamma \circ p = 1$ on $O \cap W$, $\Gamma(0) = 0$, and $p$, $\Gamma$ sc-smooth between the retracts $(O \cap W, C, E)$ and $(V, N \cap C, N)$: $p : O \cap W \to V$ is an sc-diffeomorphism with inverse $\Gamma$. *Book:* Proposition 4.1.9 (preprint 4.9), p. 178 (preprint p. 150); proof pp. 178–184 (preprint pp. 150–156). *Hidden hypotheses:* $Y$ closed (Definition 3.1.24 (preprint 3.21) asks for an sc-complement, which is closed; the formal good complement records only the splitting), and $p$ the projection along $Y$: the text writes $p = Dt(0)$, a projection along $\ker Dt(0)$, which nothing relates to $Y$, while Lemma 4.1.13 (preprint 4.12) needs the projection along $Y$ (Findings, items 20, 21). *Deviation (hypotheses explicit):* the conclusion is stated on fixed sets, with $\Gamma$ sc-smooth on the fixed $V$. *Proof:* not the text's. The text applies Theorem 3.5.11 (preprint Corollary 3.47) to the section $\mathbf{f}(u) = (1-p)(u - t(u))$ (Lemma 4.1.11 (preprint 4.10)), which gives a graph over a complement of its own choosing, sc-smooth only as a germ (Findings, item 22); the formal proof

is direct (V4.21–V4.25).

**V4.21 (Lemma 4.1.12 (preprint 4.11), uniqueness; `Polyfold.GraphData.qL_lvlDeriv_zero`, `Polyfold.GraphData.lvlDeriv_zero_of_mem_N`, `Polyfold.GraphData.exists_contraction_radius`, `Polyfold.GraphData.eq_of_p_eq`).** In the data of V4.20, on every level $k+1$: $(1-p)Dt(0)=0$ and $Dt(0)=1$ on $N$; for $\theta>0$ there is $r>0$ for which the estimates of V4.19 hold at $u_0=0$ and $|(1-p)(u'-u)|_{k+1}\le\|1-p\|\theta|u'-u|_{k+1}$ for $u,u'\in O_{k+1}$ with $|u|_{k+1},|u'|_{k+1}<r$; and two points $u,u'$ with $p(u)=p(u')$ and $|(1-p)(u'-u)|\le c|u'-u|$, $c<1$, coincide. *Book:* Lemma 4.1.12 (preprint 4.11), p. 181 (preprint p. 154): small solutions $a+y$ of $\mathbf{f}=0$ have $y=\delta(a)$. *Hidden hypotheses:* none. *Deviation (restated):* uniqueness among points of $O$, without the $\delta$ of Theorem 3.5.11 (preprint Corollary 3.47).

**V4.22 (Lemma 4.1.13 (preprint 4.12), existence; `Polyfold.GraphData.eps'`, `Polyfold.GraphData.exists_graph`).** Let $\varepsilon$ be the constant of the good complement $Y$ and $\varepsilon'=\varepsilon/(\|\iota_1\|\|\mathrm{lift}_N\|+1)$. If the estimates of V4.21 hold on $C_1\cap B(0,r)$ with $\theta<1$, $\|1-p\|\theta<1$ and $\|1-p\|\theta(1+\varepsilon')\le\varepsilon'$, there is $\rho>0$ such that every $a\in N\cap C$ with $|a|_1<\rho$ has a $u\in O_1$ with $p(u)=a$, $u-a\in Y$, $|u-a|_1\le\varepsilon'|a|_1$ and $|u|_1<r$. *Book:* Lemma 4.1.13 (preprint 4.12), pp. 182–183 (preprint pp. 154–155): $a+\delta(a)\in O$ for small $a$. *Hidden hypotheses:* none. *Deviation (restated):* the point is constructed as the fixed point of $y\mapsto(1-p)t(a+y)$ on $\{y\in Y_1:|y|_1\le\varepsilon'|a|_1\}$, where $a+y\in C$ by the good complement property; then the text's argument $a_1=a$ shows $t(u)=u$.

**V4.23 ($p(O\cap W)\subseteq C$ and the neighbourhoods; `Polyfold.GraphData.p_mem_C`, `Polyfold.GraphData.Nbhd`, `Polyfold.GraphData.exists_nbhd`, `Polyfold.GraphData.Nbhd.image_V`, `Polyfold.GraphData.Nbhd.injOn_p`).** (a) If $u\in O_1$ satisfies $|(1-p)u|_1\le\kappa|u|_1$ with $0\le\kappa\le1/2$ and $2\|\iota_1\|\kappa\|\mathrm{lift}_N\|\le\varepsilon$, then $p(u)\in C$. (b) For every $\theta_0>0$ there are $r>0$, an open $W=B_0(0,\delta_0)\cap p^{-1}(V')$, an open $V'=\{a:|a|_1<\rho'\}$ of $N$ and $\Gamma$ on $V=V'\cap C$ with $\Gamma(V)\subseteq O\cap W$, $p\circ\Gamma=1$, $\Gamma(a)-a\in Y$, $|\Gamma(a)|_1<r$, $p(O\cap W)\subseteq V$, $\Gamma\circ p=1$ on $O\cap W$, $\|Dt(z)-Dt(0)\|_{\mathcal{L}(E_1)}\le\theta_0$ on $C_1\cap B(0,r)$, and $|(1-p)(u'-u)|_1\le\frac12|u'-u|_1$ on $O_1\cap B_1(0,r)$; then $\Gamma(V)=O\cap W$ and $p$ is injective on $O\cap W$. *Book:* the last paragraph of the proof of Proposition 4.1.9 (preprint 4.9), preprint p. 156 ("$V=\{a\in T_0O\cap C:|a|_0<\varepsilon\}$"). *Hidden hypotheses:* $p(u)\in C$ for $u\in O$ near 0, which the text never checks (Findings, item 25). *Deviation (not in the book):* the neighbourhoods are open on level 0 and controlled on level 1 through the continuity of $t:U\to E_1$ at 0.

**V4.24 (the regularity of $\Gamma$: continuity and the frame formula; `Polyfold.GraphData.Nbhd.norm_Γ_lvl_one_sub_`, `Polyfold.GraphData.Nbhd.continuousOn_Γ_lvl`, `Polyfold.norm_sub_le_of_projection`, `Polyfold.hasFDerivWithinAt_of_projection`, `Polyfold.GraphData.A_injective`, `Polyfold.GraphData.eq_z`, `Polyfold.GraphData.scDeriv_sub_frame`, `Polyfold.GraphData.Nbhd.hasFDerivWithinAt_Γ_lvl`).** With the data of V4.23: $|\Gamma(a')-\Gamma(a)|_1\le2\|\mathrm{lift}_N\||a'-a|$, and $\Gamma$ is continuous into every level $E_m$ on $V$. If $\|Dt(u)-Dt(0)\|_{\mathcal{L}(E_1)}\le\theta_0$ with $\theta_0$ small (`FrameSmall`), then $A(u)=p\circ Dt(u)|_N$ is invertible, $T_uO\cap Y=0$ and $(1-B_up)Dt(u)=0$ for the frame $B_u=Dt(u)|_N\circ A(u)^{-1}$; and $\Gamma$ is differentiable into every level $E_{k+1}$ on $V$ with $D\Gamma(a)=Dt(\Gamma(a))|_N\circ(p\circ Dt(\Gamma(a))|_N)^{-1}$. The abstract step: if $s$ satisfies the mean value estimate at $u_0$ on a set $S$, $q\circ B=1$, $(1-Bq)Ds(u_0)=0$ and $\theta\|1-Bq\|\le1/2$, then fixed points $u,u'$ of $s$ in $S$ satisfy $|u'-u|\le2\|B\||q(u'-u)|$, and every continuous right inverse of $q$ with values in the fixed points has derivative $B$ at $q(u_0)$. *Book:* not in the text (Theorem 3.5.11 (preprint Corollary 3.47) gives $\delta$ as an sc-smooth germ). *Hidden hypotheses:* none. *Deviation (not in the book).*

**V4.25 (the $C^k$-bootstrap and the sc-diffeomorphism; `Polyfold.GraphData.contDiffOn_lvlDeriv_apply`, `Polyfold.GraphData.Nbhd.contDiffOn_Γ_lvl`, `Polyfold.GraphData.Nbhd.isScSmooth_Γ`, `Polyfold.GraphData.Nbhd.V_isRelOpenInQuadrant`, `Polyfold.GraphData.Nbhd.p_isScSmoothRetractMap`,**

`Polyfold.GraphData.Nbhd.Γ_isScSmoothRetractMap`). $(x,w) \mapsto Dt_{j+1}(x)w$ is $C^k$ on $U_{j+k+1} \times E_{j+k+1}$ (Proposition 1.2.3 (preprint 1.18) for the tangent map of the lift); $\Gamma$ is $C^k$ into every $E_{j+1}$ on $V$ for every $k$, hence $C^\infty$ into every level and sc-smooth on $V$ (Proposition 1.2.4 (preprint 1.19)); $(V, N \cap C, N)$ is a local model; $p$ and $\Gamma$ are sc-smooth between the retracts $(O \cap W, C, E)$ and $(V, N \cap C, N)$. *Book:* the last sentence of the proof of Proposition 4.1.9 (preprint 4.9), preprint p. 156 ("having the sc-smooth map $a \mapsto a + \delta(a)$ as its inverse"). *Hidden hypotheses:* none. *Deviation (not in the book):* derivatives of $t$ are applied only to vectors of the smooth finite-dimensional $N$, so the level loss of $Tt$ does not accumulate.

**V4.26 (charts composed with an sc-diffeomorphism between retracts;** `Polyfold.MChart.compRetract`, `Polyfold.MChart.compRetract_isChart`**).** For a chart $\varphi$ onto $(O, C, E)$ and mutually inverse $f$, $g$ that are sc-smooth between $(O, C, E)$ and a retract $(O', C', E')$ (Definition 2.1.14 (preprint 2.7)), $f \circ \varphi$ is a chart onto $(O', C', E')$, and a chart of the M-polyfold whenever $\varphi$ is. *Book:* not stated; "we obtain a smoothly compatible atlas", preprint p. 157. *Hidden hypotheses:* none. *Deviation (not in the book).*

**V4.27 (Theorem 4.1.2 (preprint 4.2), (i) ⇒ (ii);** `Polyfold.MChart.IsCornerChart.mem_smooth`, `Polyfold.MChart.IsCornerChart.exists_centred`, `Polyfold.MChart.IsCornerChart.exists_centred_tame`, `Polyfold.tangentSpace_relOpen`, `Polyfold.isInGoodPosition_top`**).** Every point of a corner chart is smooth for its model. A point $x$ in the domain of a corner chart $\chi$ of $X$ lies in a chart $\psi$ of $X$ with $\psi(x) = 0$, modelled on a relatively open subset of the partial quadrant $C_{\chi(x)}$ in the same finite-dimensional space; this model is a tame retract (identity retraction), its tangent space at 0 is the whole space, and the whole space is in good position to any partial quadrant (complement 0). Together with V4.14 and V4.1, (i) implies (ii). *Book:* proof of Theorem 4.1.2 (preprint 4.2), preprint p. 156. *Hidden hypotheses:* the centred chart, the tangent space at its centre and its good position, which the text's proof does not address (Findings, item 53). *Deviation (hypotheses explicit).*

**V4.28 (Theorem 4.1.2 (preprint 4.2), (ii) ⇒ (i);** `Polyfold.exists_proj_goodComplement`, `Polyfold.CornerChartData`, `Polyfold.exists_cornerChartData`, `Polyfold.CornerChartData.chart`, `Polyfold.CornerChartData.chart_isCornerChart`, `Polyfold.CornerChartData.chart_isChart`, `Polyfold.MPolyfold.theorem_4_2_chart`, `Polyfold.MPolyfold.hasCornerStructure_of_centred`**).** If every point is smooth and $1_0^1$ is sc-smooth, then for every chart $\psi$ of $X$ with $\psi(x) = 0$ whose tangent space $N = T_0O$ is finite-dimensional and in good position to $C$ there are a closed good complement $Y$ of $N$, the projection $p$ along it, and $W$, $V$, $\Gamma$ of Proposition 4.1.9 (preprint 4.9); the chart $p \circ \psi$ on $\psi^{-1}(W)$, modelled on $(V, N \cap C, N)$, is a corner chart and a chart of $X$, with $x$ in its domain and $(p \circ \psi)(x) = 0$. If every point has such a chart $\psi$ (tame or not), $X$ has a corner structure. *Book:* proof of Theorem 4.1.2 (preprint 4.2), pp. 183–186 (preprint pp. 156–157). *Hidden hypotheses:* none; the closed good complement is chosen by the normal form of Proposition 3.1.25 (preprint 3.22), and the compatibility of the charts $p \circ \psi$, not argued in the text, is V4.26 (Findings, items 54, 55). *Deviation (hypotheses explicit).*

**V4.29 (Definition 4.2.1 (preprint 4.13), smooth finite-dimensional submanifolds;** `Polyfold.MPolyfold.IsSmooth`, `Polyfold.MPolyfold.SubmanifoldData`, `Polyfold.MPolyfold.isSmoothSubmanifold_iff`, `Polyfold.MPolyfold.SubmanifoldData.toSubData`, `Polyfold.MPolyfold.IsSmoothSubmanifold.isSubMPolyf`**)** $A \subseteq X$ is a *smooth finite-dimensional submanifold* if every $a \in A$ has an open $V \ni a$ and an sc$^+$-retraction $s$ on $V$ (V4.9) with $s(V) = A \cap V$. Such an $A$ is a sub-M-polyfold (Definition 2.3.18 (preprint 2.20)) and carries the induced structure of Proposition 2.3.19 (preprint 2.21). *Book:* Definition 4.2.1 (preprint 4.13), p. 184 (preprint p. 157). *Hidden hypotheses:* none.

**V4.30 (the sc$^+$-retraction in a chart;** `Polyfold.MPolyfold.SubmanifoldData.isScSmoothRetractMap_lift`,

`Polyfold.MPolyfold.SubmanifoldData.exists_chartRetraction`, `Polyfold.MPolyfold.SubmanifoldData.is`
`Polyfold.MPolyfold.SubmanifoldData.apply_mem_smooth`). Let $(V, s)$ be data of Definition 4.2.1 (preprint 4.13), $\varphi$ a chart of $X$ and $W = V \cap \operatorname{dom}\varphi \cap s^{-1}(V \cap \operatorname{dom}\varphi)$. The lift of $\varphi \circ s \circ \varphi^{-1}$ to $E^1$ is sc-smooth from $(\varphi(W), C, E)$ to $(O_1, C_1, E^1)$. For every retraction $r$ onto $O$ there is an open $G$ with $\varphi(W) = O \cap G$ and an sc⁺-retraction $t$ on $U \cap r^{-1}(G)$, equal to $\varphi \circ s \circ \varphi^{-1} \circ r$ there, onto $\varphi(W \cap A)$. So $\varphi(W \cap A)$ is an sc⁺-retract, and $\varphi(a)$ is smooth for $a \in A \cap V \cap \operatorname{dom}\varphi$. *Book:* proof of Theorem 4.2.7 (preprint 4.18), pp. 188–190 (preprint pp. 160–161) ("taking $V$ and $U$ small, we may assume $V = \varphi^{-1}(O)$ … it is a sc⁺-retraction since $s$ is"). *Hidden hypotheses:* the set $W$ of Proposition 2.3.19 (preprint 2.21) in place of "$V = \varphi^{-1}(O)$", and the sc⁺ condition read in the charts of $X$ and $X^1$ (Findings, item 60). *Deviation (hypotheses explicit).*

**V4.31 (Remark 4.2.2 (preprint 4.14) in a chart;** `Polyfold.MPolyfold.SubmanifoldData.remark_4_14`**).** At $a \in A \cap V \cap \operatorname{dom}\varphi$ and the level-1 point $z$ over $\varphi(a)$: $z$ is smooth; the tangent map of $\rho = \varphi \circ s \circ \varphi^{-1}$ at $z$ maps $T_zO$ onto the tangent space $T_z(\varphi(W \cap A))$; this tangent space lies in $T_zO$; and it is a finite-dimensional linear subspace. *Book:* Remark 4.2.2 (preprint 4.14), p. 184 (preprint p. 157) ("$T_aA = Ts(a)(T_aX)$ … finite-dimensional"). *Hidden hypotheses:* $T_aX$ is read in a chart (gap G.7). *Deviation (restated):* the sc⁺-operator is $Dt(z)$ for the chart retraction $t$ of V4.30, not $Ts(a)$.

**V4.32 (Remark 4.2.2 (preprint 4.14) at points with $d_X = 0$;** `Polyfold.isInGoodPosition_univ`, `Polyfold.partialQuadrantAtWith_eq_univ_of_degIndex_eq_zero`, `Polyfold.MPolyfold.isGoodPositionAt_`
`Polyfold.MPolyfold.hasCornerStructure_of_degIndex_eq_zero`**).** Every subspace is in good position to the whole space; at a point $e \in C$ with $d_C(e) = 0$, $C_e$ is the whole space. If $a \in A$ and $d_X(a) = 0$, $A$ is in good position at $a$ (Definition 4.2.5 (preprint 4.16), V4.38). If $d_X = 0$ on all of $A$, the induced structure on $A$ has a corner structure; $X$ need not be tame. *Book:* Remark 4.2.2 (preprint 4.14), p. 184 (preprint p. 157) ("near points $a \in A$ satisfying $d_X(a) = 0$, necessarily smooth finite-dimensional manifolds in the classical sense"). *Hidden hypotheses:* none. *Deviation (narrower):* the local form, for the open piece $A \cap \{d_X = 0\}$ of an arbitrary $A$, is not formalized (gap G.33).

**V4.33 (Proposition 4.2.3 (1) (preprint 4.15);** `Polyfold.MPolyfold.IsSmoothSubmanifold.subset_smoothSet`, `Polyfold.MPolyfold.IsSmoothSubmanifold.mem_levelSet_subPolyfold_iff`, `Polyfold.MPolyfold.IsSmooth`
$A \subseteq X_\infty$; the levels of the induced structure are the traces of the levels of $X$; every point of $A$ is smooth for the induced structure. *Book:* Proposition 4.2.3 (1) (preprint 4.15), p. 186 (preprint p. 158). *Hidden hypotheses:* none. *Deviation (hypotheses explicit):* "$A \subseteq X_\infty$" and "$A = A_\infty$ for the induced structure" are separated; the proof goes through the chart retraction of V4.30 and Lemma 4.1.7 (i) (preprint 4.7).

**V4.34 (Proposition 4.2.3 (2) (preprint 4.15);** `Polyfold.MPolyfold.IsSmoothSubmanifold.proposition_4_15_2`**).** $A$ is locally closed in $X$: the intersection of an open and a closed set (equivalently, every $a \in A$ has an open $V$ with $V \cap \overline{A} \subseteq A$). *Book:* Proposition 4.2.3 (2) (preprint 4.15), p. 186 (preprint p. 158), with a garbled definition of "locally closed" (the letter $U$ used twice). *Hidden hypotheses:* none. *Deviation (restated):* Mathlib's notion; a filter replaces the text's sequence, so no metrizability is needed.

**V4.35 (Proposition 4.2.3 (3) (preprint 4.15);** `Polyfold.MPolyfold.IsSmoothSubmanifold.proposition_4_15_3`**).** The boundary $\{d_A \geq 1\}$ of $A$ with its induced structure lies in $\partial X = \{d_X \geq 1\}$. *Book:* Proposition 4.2.3 (3) (preprint 4.15), p. 186 (preprint p. 158). *Hidden hypotheses:* none. *Proof:* $d_A \leq d_X$ (Lemma 2.4.6 (preprint 2.26)).

**V4.36 (Proposition 4.2.3 (4) (preprint 4.15); `Polyfold.MPolyfold.IsSmoothSubmanifold.proposition_4_15_4`).** For the induced structure on $A$, every point is smooth and $1^1_0 : A \to A^1$ is sc-smooth (for any junk value). *Book:* Proposition 4.2.3 (4) (preprint 4.15), p. 186 (preprint p. 158). *Hidden hypotheses:* none. *Proof:* every point of $A$ lies in an induced chart whose model $\varphi(W \cap A)$ is an sc$^+$-retract (V4.30), and V4.14 applies; the text says only that the sc$^+$-retraction "is the identity on the retract $A$".

**V4.37 (tangent spaces of sub-retracts; `Polyfold.tangentSpace_inter_isOpen`, `Polyfold.tangentSpace_subset` `Polyfold.scDeriv_mem_tangentSpace_of_subset`).** For a retract $Q$ and an open $K$, $T_z(Q \cap K) = T_zQ$ at points of $Q \cap K$. For retracts $Q \subseteq O = r(U)$, $T_zQ \subseteq T_zO$. If $F \circ r$ is sc-smooth and $F$ maps the retract $Q \subseteq O$ into a retract $Q'$, then $D(F \circ r)(z)$ maps $T_zQ$ into $T_{F(z)}Q'$. *Book:* not stated. *Hidden hypotheses:* none. *Deviation (not in the book).*

**V4.38 (Definition 4.2.5 (preprint 4.16) in a chart; `Polyfold.IsGoodPositionRetractAt`, `Polyfold.MPolyfold.SubmanifoldData.IsGoodPositionChart`, `Polyfold.MPolyfold.IsGoodPositionAt`, `Polyfold.MPolyfold.SubmanifoldData.isGoodPositionChart_iff`).** For retracts $Q \subseteq O$, a presentation $T$ and a level-1 point $z$: $T_zQ$ is in good position to $C_z$ inside $T_zO$ ($T_zQ \subseteq T_zO$, $T_zQ \cap C_z$ has interior points in $T_zQ$, and $T_zQ$ has a complement $P \subseteq T_zO$ in $T_zO$ with a constant $\varepsilon$ as in Definition 3.1.24 (preprint 3.21)). $A$ is *in good position at* $a \in A$ if this holds for $Q = \varphi(W \cap A)$ and the model $O$ of some chart $\varphi$ at $a$, some data of Definition 4.2.1 (preprint 4.13) at $a$, some presentation and the level-1 point over $\varphi(a)$. For retractions $r$ onto $O$ and $t$ onto $\varphi(W \cap A)$ the condition says that $Dt(z)E$ is in good position to $C_z$ inside $Dr(z)E$. *Book:* Definition 4.2.5 (preprint 4.16), p. 188 (preprint p. 159). *Hidden hypotheses:* the chart form (gap G.7); the chart form needs no tameness. *Deviation (restated).*

**V4.39 (the chart independence of Definition 4.2.5 (preprint 4.16); `Polyfold.IsGoodPositionRetractAt.transpo` `Polyfold.MPolyfold.chart_change_of_transport`, `Polyfold.MPolyfold.isGoodPositionRetractAt_chart_` `Polyfold.MPolyfold.IsGoodPositionAt.isGoodPositionChart`, `Polyfold.MPolyfold.isGoodPositionAt_if`).** Good position of a sub-retract is carried by a germ of sc-diffeomorphisms between retracts that maps the sub-retracts into each other. Hence if $A$ is in good position at $a$, the chart condition holds for every data of Definition 4.2.1 (preprint 4.13) at $a$, every chart at $a$, every presentation and the level-1 point over the image of $a$; and good position at $a$ is equivalent to the condition for all of them. *Book:* implicit in Definition 4.2.5 (preprint 4.16). *Hidden hypotheses:* none. *Deviation (not in the book).*

**V4.40 (Definition 4.2.6 (preprint 4.17); `Polyfold.MPolyfold.IsTameSubmanifold`).** $A$ is a *tame* submanifold if it is a smooth finite-dimensional submanifold whose induced M-polyfold structure is tame. *Book:* Definition 4.2.6 (preprint 4.17), p. 188 (preprint p. 159). *Hidden hypotheses:* none.

**V4.41 (Theorem 4.2.7 (preprint 4.18); `Polyfold.MPolyfold.theorem_4_18`, `Polyfold.MPolyfold.hasCornerStru` `Polyfold.MPolyfold.IsGoodPositionAt.isInGoodPosition_of_tame`).** Let $X$ be tame and $A$ a smooth finite-dimensional submanifold in good position at every point. Then the induced structure on $A$ has a corner structure (Definition 4.1.1 (preprint 4.1)) and $A$ is tame. More generally, the induced structure has a corner structure as soon as every $a \in A$ lies in a chart $\psi$ of $X$ with $\psi(a) = 0$ in which $T_0(\psi(W \cap A))$ is in good position to $C$ in $E$; and good position at $a$ gives this in every tame chart centred at $a$. *Book:* Theorem 4.2.7 (preprint 4.18), p. 188 (preprint p. 159); proof pp. 188–190 (preprint pp. 160–161). *Hidden hypotheses:* none. *Proof:* the text's route through Theorem 4.1.2 (preprint 4.2), with the steps it skips: the centred chart (V4.17), the chart retraction (V4.30), the closed complement (V4.28) and the compatibility of the charts $p \circ \psi$ (V4.26); tameness of $X$ enters only through Lemma 4.2.8 (preprint 4.19).

**V4.42 (Lemma 4.2.8 (preprint 4.19); `Polyfold.isGoodComplement_sup_range`, `Polyfold.lemma_4_19`,**

`Polyfold.IsInGoodPositionIn.isInGoodPosition`**).** Let $T$ present $C$, $\pi$ a bounded idempotent with $(1-\pi)E_0 \subseteq E_0^{\min}$, the minimal subspace at 0 ($\{\lambda = 0\}$), and $N$ in good position to $C$ inside $\pi E_0$. Then $N$ is in good position to $C$ in $E_0$, and for every good complement $Y$ of $N$ inside $\pi E_0$ with constant $\varepsilon$, $Y \oplus (1-\pi)E_0$ is a good complement of $N$ in $E_0$, with constant $\varepsilon/(\|\pi\|+1)$. *Book:* Lemma 4.2.8 (preprint 4.19), p. 188 (preprint p. 160). *Hidden hypotheses:* $(1-\pi)E_0$ lies in the minimal subspace at the corner 0 of the centred chart (Proposition 2.5.3 (preprint 2.35)), not merely in $C$. *Deviation (hypotheses explicit):* the text concludes $n + y \in C$ from $n + y + z \in C$ "since $z \in Z \subset C$", which is false ($C = [0,\infty)$, $z = 1$, $n + y = -\frac{1}{2}$); it holds because $Z \subseteq C \cap (-C)$. The text's norm $|y|_0 + |z|_0$ is replaced by the constant $\varepsilon/(\|\pi\|+1)$ (Findings, item 69).

**V4.43 (Remark 4.2.9 (preprint 4.20);** `Polyfold.tangentSpace_eq_range_graph`**,** `Polyfold.lemma_3_55_relOpen`**,** `Polyfold.IsInGoodPosition.isInGoodPositionIn_of_le`**,** `Polyfold.MPolyfold.remark_4_20`**).** If the retract $Q$ is the image of an sc-diffeomorphism $\Gamma$ from $(V, N \cap C, N)$ with inverse $p$, then at $v \in V$ and the level-1 point $y$ over $\Gamma(v)$, $T_yQ = D\Gamma(v)N$ and $p \circ D\Gamma(v) = 1$. For $X$ tame and $A$ in good position at $a$, there is an open $U \ni a$ such that $A$ is in good position at every point of $U \cap A$. *Book:* Remark 4.2.9 (preprint 4.20), p. 189 (preprint p. 161), without "tame". *Hidden hypotheses:* $X$ tame (Definition 4.2.5 (preprint 4.16) is stated for tame $X$; Lemma 4.2.8 (preprint 4.19) is used). *Deviation (hypotheses explicit):* "the tangent spaces move only slowly" is $D\sigma(0) = 0$ for $\Gamma = 1 + \sigma$, and Lemma 3.6.9 (preprint 3.55) is applied with the closed complement of the normal form.

**V4.44 (Definition 4.2.10 (preprint 4.21) in a chart;** `Polyfold.IsGeneralPositionRetractAt`**,** `Polyfold.MPolyfold.SubmanifoldData.IsGeneralPositionChart`**,** `Polyfold.MPolyfold.IsGeneralPositionA`**).** For retracts $Q \subseteq O$, a presentation $T$ and a level-1 point $z$: there is an sc-subspace $B \subseteq T_zO \cap E_z$ with $(T_zO)_m = (T_zQ)_m \oplus B_m$ on every level $m$ ($T_zO \cap E_z = T_z^R O$). $A$ is *in general position at* $a \in A$ if this holds for $Q = \varphi(W \cap A)$ and the model of some chart at $a$, some data, presentation and lift. *Book:* Definition 4.2.10 (preprint 4.21), p. 190 (preprint p. 161). *Hidden hypotheses:* the sc-structure of $T_zO$ is the one induced from $E$ at the smooth point; the chart form (gap G.7). *Deviation (restated).*

**V4.45 (the chart independence of Definition 4.2.10 (preprint 4.21);** `Polyfold.IsGeneralPositionRetractAt.tra`**,** `Polyfold.MPolyfold.isGeneralPositionRetractAt_chart_change`**,** `Polyfold.MPolyfold.IsGeneralPositio`**,** `Polyfold.MPolyfold.isGeneralPositionAt_iff_forall`**).** General position of a sub-retract is carried by a germ of sc-diffeomorphisms at a smooth point; hence general position at $a$ holds in every chart, for every data, presentation and lift, and is equivalent to the condition for all of them. *Book:* implicit in Definition 4.2.10 (preprint 4.21). *Hidden hypotheses:* the point is smooth (true on $A$). *Deviation (not in the book).*

**V4.46 (linear algebra at the corner of a centred chart;** `Polyfold.mem_minimalSubspaceWith_zero`**,** `Polyfold.minimalSubspaceWith_zero_le`**,** `Polyfold.add_mem_iff_of_mem_zero`**,** `Polyfold.degIndexWith_add`**,** `Polyfold.coordsOn_surjective`**,** `Polyfold.exists_isClosed_isCompl_le`**,** `Polyfold.exists_goodComplement`**,** `Polyfold.degIndex_slice`**).** For $C = T^{-1}([0,\infty)^n \oplus W)$: the minimal subspace at 0 is $E_0^{\min} = \{\lambda = 0\}$ and lies in every $E_x$; adding a vector of $E_0^{\min}$ changes neither membership in $C$ nor the degeneracy index. For a finite-dimensional $N$ with $N + E_0^{\min} = E_0$: $\lambda|_N$ is onto $\mathbb{R}^n$, $N$ is in good position to $C$ with a closed good complement $Y \subseteq E_0^{\min}$, and $d_{N\cap C}(v) = d_C(v)$ for $v \in N \cap C$. A finite-dimensional $N$ and a closed $K$ with $N + K$ the whole space give a closed complement of $N$ inside $K$. *Book:* proof of Theorem 4.2.11 (preprint 4.22), pp. 190–191 (preprint pp. 162–163) ("$T_0A$ has in $E$ an sc-complement $Y$ contained in $W$"). *Hidden hypotheses:* $Y$ closed. *Deviation (restated):* the text's splitting $T_0A = N \oplus (T_0A \cap W)$ is replaced by the coordinates $\lambda|_{T_0A}$ (Findings, items 82, 83).

**V4.47 (Lemma 7.A of the exposition, and general position from $T_yQ + E_0^{\min} = E_0$; `Polyfold.exists_scComplement_le`, `Polyfold.isGeneralPositionRetractAt_of_sup_eq_top`).** (a) For $N \subseteq E_\infty$ finite-dimensional, an sc-subspace $R$ and $M = N + R$, there is an sc-subspace $B \subseteq R$ with $M_m = N_m \oplus B_m$ on every level. (b) For $r$ tame, a retract $Q \subseteq O$, a smooth $y \in Q_1$ with $T_yQ$ finite-dimensional, contained in $E_\infty$ and $T_yQ + E_0^{\min} = E_0$, the retract $Q$ is in general position at $y$ in $O$. *Book:* proof of Theorem 4.2.11 (3) (preprint 4.22), preprint p. 164 ("$T_aO = T_aA + T_a^RO$, implying that $T_aA$ has in $T_aO$ a sc-complement contained in $T_a^RO$"). *Hidden hypotheses:* $T_y^RO$ an sc-subspace (Proposition 2.5.6 (preprint 2.36), for tame retracts). *Deviation (hypotheses explicit).*

**V4.48 (general position implies good position; `Polyfold.IsGeneralPositionRetractAt.isGoodPositionRetra`).** For $r$ tame and a smooth $z \in O_1$, if $Q$ is in general position at $z$ in $O$, it is in good position at $z$. *Book:* not stated; needed for the tameness in Theorem 4.2.11 (4) (preprint 4.22) and for Theorem 4.3.1 (preprint 4.23). *Hidden hypotheses:* none. *Deviation (not in the book).*

**V4.49 (Theorem 4.2.11 (preprint 4.22) in a tame chart centred at the point; `Polyfold.sup_minimalSubspaceWith_`, `Polyfold.sup_minimalSubspaceWith_eq_top_graph`, `Polyfold.theorem_4_22_chart`).** Let $r$ be tame onto $O \ni 0$, $t$ an sc$^+$-retraction onto $Q \subseteq O$ with $0 \in Q$, and $N = T_0Q$, with $Q$ in general position at 0. Then $N + E_0^{\min} = E_0$, $N$ is in good position to $C$, $N$ has a closed good complement $Y \subseteq E_0^{\min}$ with a projection $p$ whose kernel is exactly $Y$, and the data $W$, $V$, $\Gamma$ of Proposition 4.1.9 (preprint 4.9) exist for this $Y$, with (2) $d_{N\cap C}(v) = d_C(\Gamma(v))$ on $V$ and (3) general position of $Q$ at every level-1 point over $Q \cap W$. *Book:* proof of Theorem 4.2.11 (preprint 4.22), pp. 190–193 (preprint pp. 162–164). *Hidden hypotheses:* none. *Deviation (restated):* $T_yQ + E_0^{\min} = E_0$ along the graph comes from $p \circ D\Gamma(v) = 1$ and $\ker p = Y$ (the text: "by means of the particular form of the map $\alpha$").

**V4.50 (Theorem 4.2.11 (preprint 4.22), local structure in general position; `Polyfold.MPolyfold.theorem_4_22`).** Let $X$ be tame, $A$ a smooth finite-dimensional submanifold, in general position at $x \in A$. There are a chart $\psi$ of $X$ with $\psi(x) = 0$, a tame retraction $r$ onto its model for a presentation $T$ with $n$ half-line coordinates, and an open $U \ni x$ inside the domain of $\psi$ with: (1) $d_X(x) = n$, and $\psi(x)$ lies in exactly $d_X(x)$ faces of the M-polyfold $O$; (2) $d_X(a) = d_A(a)$ for $a \in A \cap U$ ($d_A$ for the induced structure); (3) $A$ is in general position at every point of $A \cap U$; (4) for every $\sigma \subseteq \{1, \dots, n\}$ there are faces $F_i \ni 0$ of the retract ($i \in \sigma$, $F_i$ in the coordinate hyperplane $\lambda_i = 0$) and an open $G \ni 0$ such that $F_\sigma = \psi^{-1}(G \cap \bigcap_{i\in\sigma} F_i)$ contains $x$, lies in $U$, and is a tame sub-M-polyfold with $d_{F_\sigma} + \#\sigma = d_X$ on it, and $A \cap F_\sigma$ is a smooth finite-dimensional submanifold of $F_\sigma$, in general position in $F_\sigma$ at each of its points, and tame. *Book:* Theorem 4.2.11 (preprint 4.22), p. 190 (preprint p. 162); proof pp. 190–193 (preprint pp. 162–164). *Hidden hypotheses:* the chart is re-centred at $x$; "smooth submanifold" is finite-dimensional. *Deviation (restated):* (1) is a statement about the chart model (local faces are not identified with global faces, gap G.35); $F_\sigma$ is a germ, taken inside $U$; (4), without proof in the text, is V4.51, V4.52. (1) and (4) differ from the text's in content, not only in form: the text's local faces are faces of the neighbourhood $U$ itself (the closures in $U$ of the components of $U \cap \{d_X = 1\}$, all through $x$), the formal ones faces of the model retract of the chart; (2) and (3) are the text's. Identifying the two needs a local face theorem for small neighbourhoods, believed true for tame $X$ and not formalized (Findings, Chapter 4, items 90, 130; round-trip item V4.50).

**V4.51 (Theorem 4.2.11 (4) (preprint 4.22) in the chart; `Polyfold.killOn`, `Polyfold.exists_scPlus_retraction_`, `Polyfold.exists_cornerModel_retraction`, `Polyfold.theorem_4_22_four_chart`).** In the situation of V4.49, for $\sigma \subseteq \{1, \dots, n\}$ and an open $B \ni 0$: with $\kappa_\sigma(v) = v - s(\lambda(v)1_\sigma)$ ($s$ a linear

section of $\lambda|_N$) there are an open $G \ni 0$, $G \subseteq W \cap B$, and an sc⁺-retraction $t_\sigma = \Gamma \circ \kappa_\sigma \circ p \circ t$ onto $Q \cap G \cap f_\sigma$, $f_\sigma = \{\lambda_i = 0, i \in \sigma\}$, whose tangent spaces contain $T_y Q \cap f_\sigma$. A retraction onto a subset of $f_\sigma$ has a corner model on $[0,\infty)^{n-\#\sigma} \oplus W$, sc⁺ if the retraction is. For every open $B_0 \ni 0$ there are faces $F_i \ni 0$ ($i \in \sigma$) and an open $G \ni 0$, $G \subseteq W \cap B_0$, such that $O \cap G \cap f_\sigma$ is the retract of a retraction whose corner model is tame with $d_{C'}(\pi_\sigma y) + \#\sigma = d_C(y)$, $Q \cap G \cap f_\sigma$ and its corner model are sc⁺-retracts, and the corner model of $Q \cap G \cap f_\sigma$ is in general position at each of its level-1 points. *Book:* not in the text ("the previous arguments, but now applied to $A \cap F_\sigma$, conclude the proof"). *Hidden hypotheses:* none. *Deviation (not in the book).*

**V4.52 (sc⁺-retractions in a chart, and Theorem 4.2.11 (4) (preprint 4.22) for one $\sigma$; `Polyfold.MPolyfold.isScPlusR` `Polyfold.MChart.cornerPreimage`, `Polyfold.MPolyfold.theorem_4_22_four_of_chart`).** For a chart $\chi$ of $X$ and an sc⁺-retraction $\tau$ of its model with $\tau(U) \subseteq O$, $\chi^{-1} \circ \tau \circ \chi$ is an sc⁺-retraction on the open set $\chi^{-1}(\tau.U)$ with image $\chi^{-1}(\tau(U))$. In a tame chart $\psi$ centred at $x$ with the data of V4.49 for the chart retraction of $A$, for every $\sigma$ there are faces $F_i \ni 0$ and an open $G \ni 0$ such that $\psi^{-1}(G \cap \bigcap_{i\in\sigma} F_i)$ contains $x$, lies in the open set $W$ of the submanifold chart, and has the properties (4) of V4.50. *Book:* not in the text. *Hidden hypotheses:* none. *Deviation (not in the book).*

**V4.53 (the family $\mathbb{R}^m \times X$ and stratified regular points; `Polyfold.prod_rn_isTame`, `Polyfold.fibreSet`, `Polyfold.rnPres`, `Polyfold.IsStratRegularPoint`, `Polyfold.IsStratRegularValue`).** For tame $X$, $\mathbb{R}^m \times X$ with the product charts is tame. The fibre of $A \subseteq \mathbb{R}^m \times X$ over $\lambda$ is $A_\lambda = \{x : (\lambda, x) \in A\}$. A point $z \in A$ is a *stratified regular point* if, for some data of Definition 4.2.1 (preprint 4.13) at $z$, some $a \in \mathbb{R}^m$, some chart $\varphi$ of $X$ at $z_2$, some presentation $T$ of its partial quadrant and the level-1 point $y$ over the image of $z$ in the product chart $(r \mapsto r - a) \times \varphi$, every $\mu \in \mathbb{R}^m$ is the first component of a tangent vector $v$ of the chart model of $A$ at $y$ with $v_2 \in E_{\varphi(z_2)}$ (the projection maps $T_z A \cap E_z$, the tangent space of the stratum through $z$, onto $\mathbb{R}^m$); $\lambda$ is a *stratified regular value* if every $z \in A$ over $\lambda$ is a stratified regular point. *Book:* §4.3, preprint p. 164, and the proof of Theorem 4.3.1 (preprint 4.23), p. 194 (preprint p. 166) ("regular values of $p_\sigma$"). *Hidden hypotheses:* the chart form (gap G.7). *Deviation (restated):* the chart independence is not formalized (gap G.36).

**V4.54 (Theorem 4.3.1 (preprint 4.23); `Polyfold.theorem_4_23`).** Let $X$ be tame and $A$ a smooth finite-dimensional submanifold of $\mathbb{R}^m \times X$ with tame induced structure and compact closure, $\varepsilon > 0$, and suppose (1) for $|\lambda| \le \varepsilon$ ($|\lambda|^2 = \sum \lambda_i^2$) the set $A \cap \{z_1 = \lambda\}$ is compact and non-empty, (2) $A$ is in general position at every point. Then there is $\Sigma \subseteq \{|\lambda| < \varepsilon\}$ whose complement in $\{|\lambda| < \varepsilon\}$ is Lebesgue-null, consisting of stratified regular values, such that for $\lambda \in \Sigma$ the fibre $A_\lambda$ is compact, a smooth finite-dimensional submanifold of $X$, in general position at each of its points, and its induced structure has a corner structure and is tame. *Book:* Theorem 4.3.1 (preprint 4.23), p. 193 (preprint p. 165); proof pp. 193–194 (preprint pp. 165–166). *Hidden hypotheses:* hypothesis (2), misprinted in the text ("there exists a sc-complement $T_z A$ in $T_z Z$"), is Definition 4.2.10 (preprint 4.21). *Deviation (stronger):* $\Sigma$ consists of stratified regular values, which are regular values; the tameness of $A$, the non-emptiness in (1) and $\varepsilon > 0$ are not used, and $A_\lambda$ is moreover tame. With ordinary regular values the conclusion fails at the value: in $\mathbb{R} \times [0,\infty)$, $A = \{(x,x) : 0 \le x < 1\} \cup (-1,1) \times \{5\}$ satisfies the hypotheses, $Dp$ is onto everywhere, and $A_0 = \{0, 5\}$ meets $\partial X$ (Findings, item 102).

**V4.55 (Remark 4.3.2 (preprint 4.24); `Polyfold.quotDim_le_finrank_of_eq_sup`, `Polyfold.remark_4_24_chart`, `Polyfold.MPolyfold.remark_4_24_codim`, `Polyfold.MPolyfold.remark_4_24`, `Polyfold.MPolyfold.remark_4`** For tame $X$ and smooth $x$, in a tame chart at $x$ the reduced tangent space has codimension $d_X(x)$ in the tangent space. If $A$ is in general position at $a$, then $d_X(a) \le \dim T_a A$ (in any tame chart

at $a$); in particular a submanifold in general position with $T_aA = 0$ does not meet $\partial X$ at $a$. *Book:* Remark 4.3.2 (preprint 4.24), p. 193 (preprint p. 165). *Hidden hypotheses:* $X$ tame and $x$ smooth for the codimension. *Deviation (hypotheses explicit):* stated for any submanifold in general position, not only for the $A_\lambda$ of Theorem 4.3.1 (preprint 4.23) (to which it applies by V4.54).

**V4.56 (Sard's theorem, vendored; `Polyfold.MilnorSard.Sard.critSet`, `Polyfold.MilnorSard.Sard.sardStat` `Polyfold.MilnorSard.sard`, `Polyfold.MilnorSard.dense_compl_image_critSet`).** Let $E$, $F$ be finite-dimensional real normed spaces (in the same universe), $F$ with its Borel σ-algebra and a Haar measure $\mu$, $U \subseteq E$ open and $f$ of class $C^\infty$ on $U$. Then $\mu(f(\mathrm{Crit})) = 0$, where Crit is the set of points of $U$ at which $Df$ is not onto, and the complement of $f(\mathrm{Crit})$ is dense. *Book:* "Sard's theorem", proof of Theorem 4.3.1 (preprint 4.23), p. 194 (preprint p. 166), without statement. *Hidden hypotheses:* Mathlib has only the cases $\dim E \le \dim F$; the general case is Milnor's induction, copied from the Milnor project (`math_faithful`) with only the namespace changed (Findings, item 100). *Deviation (not in the book).*

**V4.57 (the strata and Sard's theorem on them; `Polyfold.quadC`, `Polyfold.stratSub`, `Polyfold.zeroSetOf`, `Polyfold.finrank_stratSub`, `Polyfold.stratCritSet`, `Polyfold.measure_image_strat`** Let $V$ be finite-dimensional with linear coordinates $\ell : V \to \mathbb{R}^n$ and $C = \{\ell \ge 0\}$. The stratum subspace of $\sigma$ is $\{\ell_i = 0, i \in \sigma\}$, of dimension $\dim V - \#\sigma$ when $\ell$ is onto; $\sigma(v) = \{i : \ell_i(v) = 0\}$. For $f$ of class $C^\infty$ on $U = O \cap C$ ($O$ open) into a finite-dimensional $F$, the stratified critical points are the $v \in U$ at which $Df(v)$ restricted to the stratum subspace of $\sigma(v)$ is not onto, and their image is null for every Haar measure. *Book:* proof of Theorem 4.3.1 (preprint 4.23), pp. 193–194 (preprint pp. 165–166) ("$A \cap F^\circ_\sigma$ is a smooth manifold … of dimension $\dim(A) - \#\sigma$"; "Using Sard's theorem we find a subset $\Sigma_\sigma$ … having full measure"). *Hidden hypotheses:* the dimension count, which Theorem 4.2.11 (preprint 4.22) does not state (the text cites "Theorem 4.23" for it). *Deviation (restated):* in the model space $T_0A$ of the chart.

**V4.58 (Theorem 3.8.22 (preprint 3.78) in $C^\infty$ form; `Polyfold.ContractionData.exists_solution_smooth`, `Polyfold.theorem_3_78_smooth`).** Under the hypotheses of Theorem 3.8.22 (preprint 3.78) (V3.63) with $f$ of class $C^\infty$, one solution $\tau$ is $C^\infty$ on $V'$, with the other conclusions of V3.63 (including the uniqueness clause). *Book:* Theorem 3.8.22 (preprint 3.78), p. 169 (preprint p. 141), states one $j$; the proof of Theorem 4.3.1 (preprint 4.23) needs $C^\infty$. *Hidden hypotheses:* none. *Deviation (stronger):* the contraction radii of Lemmas 3.8.23, 3.8.24 (preprint 3.79, 3.80) and the fixed point do not depend on $j$.

**V4.59 (the kernel at a stratified regular point is in good position; `Polyfold.quadAt`, `Polyfold.isPartialQuadrant_quadAt`, `Polyfold.exists_goodComplement_stratSub`).** For $\ell$ onto, $\{\ell_i \ge 0, i \in \sigma\}$ is a partial quadrant. If $K + \{\ell_i = 0, i \in \sigma\} = V$, then $K$ is in good position to $\{\ell_i \ge 0, i \in \sigma\}$ with a closed good complement inside the stratum subspace. *Book:* not in the text here; the argument of Lemma 5.26 (preprint; not in the 2021 edition). *Hidden hypotheses:* none. *Deviation (not in the book).*

**V4.60 (a retraction onto the fibre over a stratified regular point; `Polyfold.exists_fibre_retraction`).** For $\ell$ onto, $f$ of class $C^\infty$ on $O \cap C$ and $v_0 \in O \cap C$ not stratified-critical, there are an open $O' \ni v_0$, $O' \subseteq O$, and a map $\rho$, $C^\infty$ on $O' \cap C$, mapping $O' \cap C$ into itself and into $f^{-1}(f(v_0))$, equal to the identity on $O' \cap C \cap f^{-1}(f(v_0))$, with $D\rho(v_0)$ of range $\ker Df(v_0)$ and equal to the identity there. *Book:* "for $\lambda \in \Sigma$ … $A_\lambda$ is a smooth compact manifold with boundary with corners", p. 194 (preprint p. 166), in the model. *Hidden hypotheses:* none. *Deviation (not in the book):* the implicit function theorem at corner points (V4.58 with V4.59).

**V4.61 (the sc⁺-retraction onto $A_\lambda$; `Polyfold.isScSmooth_slice`, `Polyfold.scDeriv_slice`,**

`Polyfold.exists_slice_retraction`**).** In the model $\mathbb{R}^m \oplus E$ of a product chart, let $t$ be an sc$^+$-retraction described near a point as the graph $\Gamma : V \to t(U) \cap W$ with inverse $p$ (Proposition 4.1.9 (preprint 4.9)), and $\rho$ a $C^\infty$ retraction of $O' \cap (\mathbb{R}^m \times C)$ onto the points $v$ with $\Gamma(v)_1 = c$. Then $\tau(e) = (\Gamma(\rho(p(t(c,e)))))_2$ is an sc$^+$-retraction of $E$ (partial quadrant $C$) onto $\{e : (c,e) \in t(U) \cap W, p(c,e) \in O'\}$; if $D\rho(v_0) = 1$ on $\{h : (D\Gamma(v_0)h)_1 = 0\}$, the tangent space of its retract at the point over $\Gamma(v_0)_2$ contains $(D\Gamma(v_0)h)_2$ for all such $h$. The slice $e \mapsto (c,e)$ is sc-smooth with derivative $h \mapsto (0,h)$. *Book:* not constructed (Definition 4.2.1 (preprint 4.13) requires it). *Hidden hypotheses:* none. *Deviation (not in the book).*

**V4.62 (general position of $A_\lambda$;** `Polyfold.isGeneralPositionRetractAt_slice`**).** In a product chart with $r_X$ tame: if the chart model of $A$ is in general position at $z = (c,y)$, the projection maps the tangent space of the stratum $T_zA \cap E_z$ onto $\mathbb{R}^m$, and $\tau$ is an sc$^+$-retraction onto $Q \subseteq O$ whose tangent space at the smooth $y$ contains the second components of the tangent vectors of $A$ at $z$ with first component $0$, then $Q$ is in general position at $y$. *Book:* proof of Theorem 4.3.1 (preprint 4.23), p. 194 (preprint p. 166) ("in view of the assumption (2) … $T_{z'}(\{\lambda\} \times A_\lambda)$ has a sc-complement in $T^R_{z'}X$"). *Hidden hypotheses:* stratified regularity at $z$ itself. *Deviation (restated).*

**V4.63 (the local step of Theorem 4.3.1 (preprint 4.23);** `Polyfold.FibreLocal`**,** `Polyfold.exists_local_family`**).** Let $X$ be tame and $A \subseteq \mathbb{R}^m \times X$ a smooth finite-dimensional submanifold in general position at every point. Every $z_0 \in A$ has an open neighbourhood $U$ and a Lebesgue-null CV $\subseteq \mathbb{R}^m$ such that at every $z \in A \cap U$ with $z_1 \notin$ CV: $z$ is a stratified regular point, $A_{z_1}$ is near $z_2$ the image of an sc$^+$-retraction on an open subset of $X$, and $A_{z_1}$ is in general position and in good position at $z_2$. *Book:* proof of Theorem 4.3.1 (preprint 4.23), pp. 193–194 (preprint pp. 165–166), in one $U(z)$. *Hidden hypotheses:* none. *Deviation (restated):* in the product chart of a tame chart of $X$ centred at $z_{0,2}$, through V4.49, V4.57, V4.60–V4.62.

**V4.64 (the countable cover: Theorem 4.3.1 (preprint 4.23) for any parameter set;** `Polyfold.theorem_4_23_of_sub`**).** For $X$ tame, $A \subseteq \mathbb{R}^m \times X$ a smooth finite-dimensional submanifold with compact closure, in general position at every point, and any $S \subseteq \mathbb{R}^m$, there is $\Sigma \subseteq S$ with $S \setminus \Sigma$ Lebesgue-null such that every $\lambda \in \Sigma$ is a stratified regular value and the conclusions of V4.63 hold at every point of $A_\lambda$. *Book:* proof of Theorem 4.3.1 (preprint 4.23), p. 193 (preprint p. 165) ("We cover the set $\{z \in A : |p(z)| \le \varepsilon\}$ with the finitely many such neighborhoods"). *Hidden hypotheses:* none. *Deviation (stronger):* any $S$; the text's set need not be compact and need not have a finite subcover (in $\mathbb{R} \times \mathbb{R}$, $A = (-2\varepsilon, 2\varepsilon) \times \{2\} \cup \{(\varepsilon(1-x), x) : 0 < x < 1\}$); $A$ is second countable, its closure being compact in the metrizable $\mathbb{R}^m \times X$, and countably many neighbourhoods cover it (Findings, item 103).

**V4.65 (Theorem 4.2.11 (preprint 4.22) in the 2021 form of (1);** `Polyfold.MPolyfold.theorem_4_2_11_2021`**).** Let $X$ be a tame M-polyfold, $A \subseteq X$ a smooth finite-dimensional submanifold and $x \in A$ a point at which $A$ is in general position. There are a tame chart $\psi$ of $X$ with $\psi(x) = 0$, with a tame retraction $R$ onto its model $O$ and a presentation $T : E \cong \mathbb{R}^n \oplus W$ of its partial quadrant, and an open $U \ni x$ inside the chart domain, such that: (1) $d_X(x) = n$, $\psi(x)$ lies in exactly $d_X(x)$ faces of the M-polyfold $O$, and for every $i < n$ there are a face $F_i \ni 0$ of the retract inside the $i$-th coordinate hyperplane and an open $G \ni 0$ with $x \in \psi^{-1}(G \cap F_i) \subseteq U$; (2)–(4) are those of V4.50. *Book:* Theorem 4.2.11 (preprint 4.22), p. 190 (preprint p. 162); the 2021 edition asks in (1) that the $d$ local faces in $U$ also contain $x$. *Hidden hypotheses:* none. *Deviation (restated):* as in V4.50, the local faces are faces of the chart model, and their identification with faces of $U$ itself is not formalized (gap G.35). The count in (1) of V4.50 already counts the faces through $\psi(x)$, which is the 2021 statement; the germs $\psi^{-1}(G \cap F_i)$ that contain $x$ and lie in $U$ are read off from (4) for $\sigma = \{i\}$.

**V4.66 (Theorem 4.3.1 (preprint 4.23) in the 2021 form; `Polyfold.theorem_4_3_1_2021`).** Let $X$ be tame and $A \subseteq \mathbb{R}^m \times X$ a smooth finite-dimensional submanifold with tame induced structure and compact closure, $\varepsilon > 0$, such that (1) $A \cap \{z_1 = \lambda\}$ is compact for every $\lambda$ with $|\lambda| \le \varepsilon$ (possibly empty), and (2) $A$ is in general position at every point. Then there is a **measurable** $\Sigma \subseteq \{|\lambda| < \varepsilon\}$ whose complement in $\{|\lambda| < \varepsilon\}$ is Lebesgue-null, consisting of stratified regular values, such that for $\lambda \in \Sigma$ the fibre $A_\lambda$ is compact, a smooth finite-dimensional submanifold of $X$ in general position at each of its points, with a corner structure, and tame. *Book:* Theorem 4.3.1 (preprint 4.23), p. 193 (preprint p. 165); the 2021 edition drops "non-empty" from (1) and asks $\Sigma$ to be measurable. *Hidden hypotheses:* none. *Deviation (stronger):* stratified regular values, as in V4.54; the tameness of $A$ and $\varepsilon > 0$ are kept and not used. $\Sigma$ is that of V4.64 for $S = \{|\lambda| < \varepsilon\}$ with a measurable null hull of its exceptional set removed (Mathlib's `MeasureTheory.toMeasurable`), which keeps every other conclusion; the compactness of $A_\lambda$ comes from (1). Hypothesis (2) is printed as in the earlier texts and read as general position (Findings, items 101, 151).

**V4.67 (Definition 4.4.3 in a local model; `Polyfold.IsRetractVectorField`, `Polyfold.isRetractVectorField_z` `Polyfold.IsRetractVectorField.mem_retractTangent`, `Polyfold.IsRetractVectorField.isScSmooth_comp`, `Polyfold.IsRetractVectorField.scDeriv_apply`).** For a local model $(O, C, E)$, a map $A : E_1 \to E_0$ is (the principal part of) a vector field on $O$ if $x \mapsto (x, A(x))$ is an sc-smooth map from the retract $(O^1, C^1, E^1)$ to the tangent $(TO, TC, TE)$. Then $(x, A(x)) \in TO$ for $x \in O_1$; for every sc-smooth retraction $r$ onto $O$ the map $\tilde{A} = A \circ r^1$ is sc-smooth on $U^1$; and $Dr(x)A(x) = A(x)$ for $x \in O_1$. The zero map is a vector field on every sc-smooth retract. *Book:* Definition 4.4.3, p. 195, new in the 2021 edition (an sc-smooth section of $p : TX \to X^1$, defined on $X^1$, not on $X$), and formula (4.8), p. 200. *Hidden hypotheses:* none. *Deviation (restated):* the definition read in the model of a chart, where the text works in Appendix 4.5.

**V4.68 (Definition 4.4.3 on an M-polyfold; `Polyfold.MPolyfold.IsScVectorField`, `Polyfold.MPolyfold.zeroVe` `Polyfold.MPolyfold.tangentChartMap_zeroVectorField`, `Polyfold.MPolyfold.continuous_zeroVectorFiel` `Polyfold.MPolyfold.isScVectorField_zero`).** An sc-smooth vector field on the M-polyfold $X$ is a map $A : X^1 \to TX$, sc-smooth for the M-polyfold structures of $X^1$ and of the tangent bundle $TX$ (both paracompact), with $p(A(x)) = x$ for every $x \in X^1$. The zero section, which reads $(\psi(x), 0)$ in every chart $\psi$, is continuous and an sc-smooth vector field. *Book:* Definition 4.4.3, p. 195, new in the 2021 edition. *Hidden hypotheses:* none.

**V4.69 (Lemma 4.4.4 in a local model, $i = 1$; `Polyfold.retractTangent_shift`, `Polyfold.IsRetractVectorFiel`** For an sc-smooth retraction $r$ onto $O$, the tangent of the retract $(O^1, C^1, E^1)$ is the level-1 part of $TO$: $T(O^1) = (TO)^1$. If $A$ is a vector field on $(O, C, E)$, its level map $A_1 : E_2 \to E_1$ is a vector field on $(O^1, C^1, E^1)$: $x \mapsto (x, A_1(x))$ is an sc-smooth section of $T(O^1) \to O^2$. *Book:* Lemma 4.4.4, p. 196, new in the 2021 edition, said to be obvious from the identification of $(TX)^i \to (X^1)^i$ with $T(X^i) \to X^{i+1}$. *Hidden hypotheses:* none. *Deviation (narrower):* in a local model and for $i = 1$; larger $i$ follow by iterating the local statement. The lemma on an M-polyfold, with the identification of $(TX)^1$ and $T(X^1)$ of Proposition 2.3.17 (gap G.6, closed), is V4.78, and for every $i$ V4.79.

**V4.70 (the second derivative and its symmetry; `Polyfold.scSecondDeriv`, `Polyfold.scSecondDeriv_symm`, `Polyfold.fderivWithin_fderivWithin_eq`, `Polyfold.IsRelOpenInQuadrant.mem_closure_interior_lvlSet`, `Polyfold.scDeriv_tangentMap_inr`).** For an sc² map $f$ on $U \subseteq E$, $z \in U_2$ and $a, b \in E_1$, $D^2 f(z)(a, b)$ is the second component of $D(Tf)(z, a)(b, 0)$, the derivative of $z \mapsto Df(z)a$ in the direction $b$; and $D(Tf)(z, a)(0, l) = (0, Df(z)l)$. If $U$ is relatively open in a partial quadrant, $D^2 f(z)(a, b) = D^2 f(z)(b, a)$. *Book:* §4.5.1, p. 200, new in the 2021 edition ("using

the symmetry of the second derivative", for $f = r$). *Hidden hypotheses:* none ($r$ lives on a relatively open subset of a partial quadrant). *Deviation (restated):* the text does not say what $D^2r(x)(A(x), B(x))$ is when $A(x), B(x)$ lie only in $E_1$, while $r$ is twice differentiable only as a map $U_2 \to E_0$, in directions of $E_2$; it is defined through the sc²-structure, the derivative of $Tr$. The symmetry holds for $a, b \in E_2$ by Mathlib's symmetry of second derivatives within a set (`ContDiffWithinAt.isSymmSndFDerivWithinAt`; points of a partial quadrant lie in the closure of its interior, also on level 2), and for $a, b \in E_1$ by density of $E_2$ in $E_1$ and continuity of $D(Tf)$ (Findings, item 154).

**V4.71 (§4.5.1, the Lie bracket in a local model; `Polyfold.retractLieBracket`, `Polyfold.retractLieBracket_an` `Polyfold.IsRetractVectorField.diff_identity`, `Polyfold.retractLieBracket_mem_tangent`).** Let $r$ be an sc-smooth retraction onto $O$ and $A$, $B$ vector fields on $(O, C, E)$. For $x \in E_2$ put $[A, B](x) = D\tilde{A}(x)B_1(x) - D\tilde{B}(x)A_1(x)$, with $\tilde{A} = A \circ r^1$, $\tilde{B} = B \circ r^1$ differentiated as maps on $U^1$; $[A, B] = -[B, A]$. For $x \in O_2$ and $b$ with $Dr^1(x)b = b$: $D^2r(x)(A(x), b) + Dr(x)D\tilde{A}(x)b = D\tilde{A}(x)b$. Hence $Dr(x)[A, B](x) = [A, B](x)$ for $x \in O_2$: the bracket is tangent to $O$. *Book:* §4.5.1, pp. 200–201, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (restated):* $DA(x)$ is read as the derivative of $\tilde{A} = A \circ r^1$, the principal part being defined only on $O^1$; the computation is the text's, with the symmetry of V4.70 (Findings, item 154).

**V4.72 (Proposition 4.4.5 (preprint 4.26) in a local model; `Polyfold.proposition_4_4_5_local_2021`, `Polyfold.isScSmooth_scDeriv_apply`, `Polyfold.levelMap_shift_shift_eq`, `Polyfold.levelMap_comp_shift`).** For vector fields $A$, $B$ on $(O, C, E)$ and an sc-smooth retraction $r$ onto $O$: $(x, [A, B](x)) \in TO$ for every $x \in O_2$, and $x \mapsto (x, [A, B](x))$ is an sc-smooth map from the retract $(O^2, C^2, E^2)$ to $(TO, TC, TE)$, the local form of "$[A, B]$ is an sc-smooth section of $TX|X^2 \to X^2$". *Book:* Proposition 4.4.5 (preprint 4.26), p. 196 (preprint p. 167), with the proof in §§4.5.1–4.5.2, pp. 200–202, new in the 2021 edition (the earlier texts quote it from another paper, as a vector field on $X^1$). *Hidden hypotheses:* none. *Deviation (narrower):* in a local model. The statement on an M-polyfold is V4.76, with Lemma 4.5.1 (independence of the chart), whose proof the text leaves to the reader, in V4.74, V4.75 and V4.77 (gap G.38, closed); the pull-back bundle $TX|X^2 \to X^2$ is not built (gap G.92). *Proof:* the second component is $x \mapsto D\tilde{A}(r^2x)(\tilde{B})_1(x) - D\tilde{B}(r^2x)(\tilde{A})_1(x)$ on $U^2$, sc-smooth as the tangent map of $\tilde{A}$ composed with the sc-smooth pair $(r^2, (\tilde{B})_1)$, and equal to $[A, B] \circ r^2$; the first is V4.71.

**V4.73 (the level of the bracket on pp. 200–201; `Polyfold.lieBracket_level_counterexample_2021`, `Polyfold.not_forall_lieBracket_mem_level_one_2021`, `Polyfold.BracketLevel.opS`, `Polyfold.BracketLevel.opT`, `Polyfold.BracketLevel.xPt`, `Polyfold.BracketLevel.bracket_not_mem_level` `Polyfold.isRetractVectorField_scOperator`, `Polyfold.retractLieBracket_scOperator`).** There are an sc-Banach space $E$, the identity retraction of $E$ onto $O = E$ (with $C = E$), vector fields $A$, $B$ on $(O, C, E)$ and a point $x \in O_2$ with $[A, B](x) \notin E_1$; so the statement "$[A, B](x) \in E_1$ for every local model, all vector fields $A$, $B$ and all $x \in O_2$" does not hold. Every sc-operator $E^1 \to E$ is a vector field on $(E, E, E)$, and the bracket of two of them is the commutator $S_0T_1 - T_0S_1$. The example: $E$ the weighted sequence space of V1.13 ($|x|_m = \sup_k (k+1)^m |x_k|$), $(Sx)_k = (-1)^k(k+1)x_k$, $(Tx)_k = (k+1)x_{k+1}$, $x_k = (k+1)^{-3}$. *Book:* §4.5.1, pp. 200–201, new in the 2021 edition: $DA(x)B(x) - DB(x)A(x)$ "belongs to the fiber $(T_xO)_1$ whenever $x \in O_2$" (p. 200), and for $x \in O_2$ the bracket "belongs to the level 1" (p. 201). *Hidden hypotheses:* none. *Deviation (not in the book):* both sentences, read as $[A, B](x) \in E_1$, do not hold as we read them: $[S, T]x = ((-1)^k(k+1)(2k+3)x_{k+1})_k$, and

for this $x$ the level-1 coordinates have absolute value $P^3 + P^2 \geq 1/4$, $P = (k+1)/(k+2)$, so they do not tend to 0 (here $T_xO = E_0$, so $(T_xO)_1 = E_1$). Read as "a tangent vector at the point $x$ of level 1 of $X^1$", that is $x \in O_{m+2}$ gives $[A,B](x) \in E_m$, the sentence is V4.72, and Proposition 4.4.5 is not affected. A first example on a weighted $\ell^2$ space (sketched in the header of `Polyfold/Chapter4/VectorFields.lean`) is not formalized (Findings, item 156).

**V4.74 (Lemma 4.5.1 in a local model, the computation of §4.5.2; `Polyfold.IsRetractVectorField.diff_relate` `Polyfold.IsRetractVectorField.related_shift`, `Polyfold.retractLieBracket_natural`, `Polyfold.levelMap_levelMap_comp_eq`, `Polyfold.IsRetractVectorField.restrict`, `Polyfold.retractLieBr` `Polyfold.retractLieBracket_natural_inter`).** Let $r$, $r'$ be sc-smooth retractions onto $O$, $O'$, let $f$ be an sc-smooth map from the retract $O$ to the retract $O'$, $g = f \circ r$ and $Df = Dg$, and let $A$, $B$ be vector fields on $(O, C, E)$ and $A'$, $B'$ on $(O', C', F)$ with $Df(x)A(x) = A'(f(x))$ and $Df(x)B(x) = B'(f(x))$ for every $x \in O_1$. Then for $x \in O_2$: (a) the relation for $A$ differentiated in a direction $b$ with $Dr^1(x)b = b$ reads $D^2g(x)(A(x), b) + Dg(x)D\tilde{A}(x)b = D\tilde{A}'(g_2(x))Dg^1(x)b$, $\tilde{A} = A \circ r^1$, $\tilde{A}' = A' \circ r'^1$; (b) the relation for $B$ holds on level 1, $Dg^1(x)B_1(x) = B'_1(g_2(x))$, and $g_2 = f_2$ on $O_2$; (c) $Df(x)[A,B](x) = [A',B'](f(x))$. A vector field on $O$ is one on the open part $O \cap W$, $W$ open, with the restricted retraction, and at points of $(O \cap W)_2$ the bracket is the same for both retractions. For $f$ sc-smooth on $O \cap W$ with $Tf(z, A(z)) = (f(z), A'(f(z)))$ and $Tf(z, B(z)) = (f(z), B'(f(z)))$ on $(O \cap W)_1$, also $Tf(x, [A,B](x)) = (f(x), [A',B'](f(x)))$ on $(O \cap W)_2$. *Book:* Lemma 4.5.1, p. 201, new in the 2021 edition, whose proof is left to the reader, and the computation of §4.5.2, pp. 201–202, for a map $\varphi$ with $D\varphi \cdot A = A \circ \varphi$ and $D\varphi \cdot B = B \circ \varphi$. *Hidden hypotheses:* the relations on an open set of level 1 (here on all of $O_1$), since the computation differentiates them at points of $O_2$. *Deviation (restated):* the computation for an sc-smooth $f$ between two local models with two pairs of vector fields, with $D^2g$ read as in V4.70 and its symmetry; the form on $O \cap W$ is the shape of a transition map of charts (V4.75). §4.5.2 ends by calling the result "Proposition 4.5.1", a possible misprint for Lemma 4.5.1 (Findings, items 157, 162). *Proof:* the text's: (a) for $A$ with $b = B_1(x)$ and for $B$ with $b = A_1(x)$, both fixed by $Dr^1(x)$ (V4.69), (b) on both sides, the difference, and the symmetry of $D^2g$.

**V4.75 (principal parts in a chart and the independence of the chart; `Polyfold.MPolyfold.tangentChartMap_cha` `Polyfold.MPolyfold.chartRetraction`, `Polyfold.MPolyfold.chartRetraction_retract`, `Polyfold.MPolyfold.principalPart`, `Polyfold.MPolyfold.tangentChartMap_eq_principalPart`, `Polyfold.MPolyfold.IsScVectorField.isRetractVectorField_principalPart`, `Polyfold.MPolyfold.IsScVe` `Polyfold.MPolyfold.chartLieBracket`, `Polyfold.MPolyfold.IsScVectorField.chartLieBracket_spec`, `Polyfold.MPolyfold.IsScVectorField.chartLieBracket_change`).** Let $A$, $B$ be sc-smooth vector fields on the M-polyfold $X$ (V4.68) and $\varphi$, $\psi$ charts of $X$, the model of each with a chosen sc-smooth retraction. Tangent charts change by $T\psi = T(\psi \circ \varphi^{-1}) \circ T\varphi$ over the overlap (formula (2.9)). The principal part $A_\varphi$, given on $O_1$ by $T\varphi(A(y)) = (\varphi(y), A_\varphi(\varphi(y)))$ for $y$ of level 1 in the domain of $\varphi$, is a vector field on the model $(O, C, E)$ (V4.67); the principal parts in $\varphi$ and $\psi$ are related by the transition map, $T(\psi\varphi^{-1})(z, A_\varphi(z)) = (\psi\varphi^{-1}(z), A_\psi(\psi\varphi^{-1}(z)))$ on level 1 of the overlap. The bracket $[A_\varphi, B_\varphi]$ of the principal parts satisfies Proposition 4.4.5 in the model (V4.72), and the brackets in $\varphi$ and $\psi$ are related by the tangent map of the transition map on level 2 of the overlap. *Book:* p. 201, new in the 2021 edition: Lemma 4.5.1 shows that the definition of $[A,B]$ does not depend on the local coordinates. *Hidden hypotheses:* none. *Proof:* the overlap is $O \cap W$ for an open $W$; V4.74 there, with the relations of the principal parts.

**V4.76 (Proposition 4.4.5 (preprint 4.26) on an M-polyfold; `Polyfold.MPolyfold.levelLift_levelChart_spec`, `Polyfold.MPolyfold.lieBracketPre`, `Polyfold.MPolyfold.lieBracket`, `Polyfold.MPolyfold.tangentProj_l`**

**`Polyfold.MPolyfold.tangentChartMap_lieBracket`, `Polyfold.MPolyfold.tangentChartMap_lieBracket_eq`, `Polyfold.MPolyfold.continuous_lieBracket`, `Polyfold.proposition_4_4_5_2021`).** For sc-smooth vector fields $A$, $B$ on the M-polyfold $X$, the Lie bracket $[A,B] : X^2 \to TX$, $X^2 = (X^1)^1$, is the tangent vector at $x$ represented by $[A_\varphi, B_\varphi](\varphi(x))$ in a chart $\varphi$ chosen around $x$. In every chart $\psi$ around $x$, $T\psi([A,B](x)) = (\psi(x), [A_\psi, B_\psi](\psi(x)))$, so $[A,B]$ depends neither on the chart nor on the retraction chosen for its model; $p([A,B](x)) = x$; and $[A,B]$ is continuous and sc-smooth from $X^2$ to $TX$. *Book:* Proposition 4.4.5 (preprint 4.26), p. 196 (preprint p. 167), with the proof in §§4.5.1–4.5.2, pp. 200–202: the bracket, defined in local coordinates by the usual formula, is an sc-smooth section of $TX|X^2 \to X^2$, the pull-back of $TX \to X^1$ by the inclusion $X^2 \to X^1$ (the preprint and 2017 texts quote it from another paper, as a vector field on $X^1$). *Hidden hypotheses:* none. *Deviation (restated):* "an sc-smooth section of $TX|X^2 \to X^2$" is read as an sc-smooth map $X^2 \to TX$ that covers the inclusion $X^2 \to X^1$; the pull-back bundle itself is not built (gap G.92). The earlier statement, a vector field on $X^1$, would through $\Phi$ (V4.78) put $[A,B](x)$ in $E_1$ for $x \in X_2$, which the example of V4.73 excludes; as we read it, it does not hold, and the 2021 statement does. With V4.72, V4.75 and V4.77 this closes gap G.38 (Findings, item 161). *Proof:* in the chart $(\psi^1)^1$ of $X^2$ and the tangent chart $T\psi$, $[A,B]$ reads $z \mapsto (z, [A_\psi, B_\psi](z))$, sc-smooth by V4.72; the chart does not matter by V4.75.

**V4.77 (Lemma 4.5.1 on M-polyfolds; `Polyfold.MPolyfold.liftLevelOneMap`, `Polyfold.MPolyfold.levelTwoMa` `Polyfold.MPolyfold.levelMap_coord_levelLift`, `Polyfold.lemma_4_5_1_2021`).** Let $f : X \to Y$ be an sc-smooth map of M-polyfolds with level maps $f_1 : X^1 \to Y^1$ and $f_2 : X^2 \to Y^2$, $A$, $B$ sc-smooth vector fields on $X$ and $A'$, $B'$ on $Y$ with $Tf(A(x)) = A'(f_1(x))$ and $Tf(B(x)) = B'(f_1(x))$ for every $x \in X^1$. Then $Tf([A,B](x)) = [A',B'](f_2(x))$ for every $x \in X^2$. *Book:* Lemma 4.5.1, p. 201, new in the 2021 edition, whose proof is left to the reader: for a local sc-diffeomorphism $\varphi : X \to X$ and vector fields with $T\varphi(A(x)) = A(\varphi(x))$ and $T\varphi(B(x)) = B(\varphi(x))$ at points of $X_2$, $T\varphi([A,B](x)) = [A,B](\varphi(x))$ for $x \in X_2$. *Hidden hypotheses:* the relations on an open set of $X^1$, which the computation of §4.5.2 differentiates. *Deviation (restated):* more general in the map (any sc-smooth $f$, between two M-polyfolds, with a pair of vector fields on each side); the relations are assumed on all of $X^1$, not only at the points of $X_2$. A local sc-diffeomorphism between open subsets of $X$ is covered only through such an $f$: open subsets of an M-polyfold as M-polyfolds are not built (gap G.92) (Findings, item 162). *Proof:* in charts $\varphi$ around $x$ and $\psi$ around $f(x)$, $Tf$ reads $T(\psi \circ f \circ \varphi^{-1})$, the principal parts are related by $\psi f \varphi^{-1}$ on the open part of the model where it is defined, and V4.74 applies there.

**V4.78 (Lemma 4.4.4 on an M-polyfold, $i = 1$; `Polyfold.MPolyfold.vectorFieldLevelOne`, `Polyfold.MPolyfold.raiseIndex`, `Polyfold.lemma_4_4_4_2021`, `Polyfold.MPolyfold.tangentShiftEquiv`, `Polyfold.proposition_2_3_17_2021`).** For an sc-smooth vector field $A$ on $X$, let $A_1 : X^2 \to (TX)^1$ be its level-1 map and $\Phi : T(X^1) \to (TX)^1$ the identification of Proposition 2.3.17 (an sc-diffeomorphism, compatible with the projections to $X^2$). Then $A^{(1)} = \Phi^{-1} \circ A_1$ is an sc-smooth vector field on the M-polyfold $X^1$, that is, an sc-smooth section of $T(X^1) \to X^2$, and $\Phi(A^{(1)}(x)) = A(x)$ in $(TX)^1$ for every $x \in X^2$. *Book:* Lemma 4.4.4, p. 196, new in the 2021 edition: through the natural identification of $(TX)^i \to (X^1)^i$ with $T(X^i) \to X^{i+1}$ the index of $A$ can be raised by every $i \geq 0$. *Hidden hypotheses:* none. *Deviation (narrower):* $i = 1$; every fixed $i$ follows by applying the lemma to $X^1$, $X^2$, …, since $A^{(1)}$ is again an sc-smooth vector field; the statement for every $i$ is V4.79 (Findings, items 160, 164). The identification $(TX)^i \cong T(X^i)$ is Proposition 2.3.17 (`Polyfold.proposition_2_3_17_2021`, `Polyfold.proposition_2_3_17_two_2021`, and for every $i$ `Polyfold.proposition_2_3_17_all_2021`; gap G.6, closed). *Proof:* $A_1$ is sc-smooth (the level-1 map of an sc-smooth map), $\Phi^{-1}$ is sc-smooth, and $\Phi$ is compatible with the projections, so

$p \circ A^{(1)}$ is the identity of $X^2$.

**V4.79 (Lemma 4.4.4 on an M-polyfold, every $i$; `Polyfold.BundledMPolyfold`, `Polyfold.BundledMPolyfold.lift`, `Polyfold.BundledMPolyfold.raiseIndexAll`, `Polyfold.BundledMPolyfold.tangentIncl_raiseIndexAll`, `Polyfold.BundledMPolyfold.vectorFieldScMap`, `Polyfold.BundledMPolyfold.tangentLiftEquiv_raiseInde`, `Polyfold.lemma_4_4_4_all_2021`, `Polyfold.BundledMPolyfold.lemma_4_4_4_all_2021_zero`, `Polyfold.BundledMPolyfold.tangentLiftEquiv`, `Polyfold.BundledMPolyfold.shiftLiftEquiv`, `Polyfold.proposition_2_3_17_all_2021`).** Let $X$ be an M-polyfold, $X^i$ its $i$-th lift ($X^0 = X$, $X^{i+1} = (X^i)^1$), and $\Phi_i : T(X^i) \to (TX)^i$, $J_i : (X^1)^i \to (X^i)^1$ the sc-diffeomorphisms of Proposition 2.3.17 for every $i$. For an sc-smooth vector field $A$ on $X$ put $A^{(0)} = A$ and $A^{(i+1)} = (A^{(i)})^{(1)}$, the index raise of V4.78 on $X^i$. Then for every $i \geq 0$: $A^{(i)}$ is an sc-smooth vector field on $X^i$, a section of $T(X^i) \to X^{i+1}$; it is $A$, in that its image under the inclusion $T(X^i) \to TX$ at $x \in X^{i+1}$ is $A(x)$; and $\Phi_i \circ A^{(i)} \circ J_i$ is the $i$-th index raise $(X^1)^i \to (TX)^i$ of $A : X^1 \to TX$. The zero vector field gives a witness for every $i$. *Book:* Lemma 4.4.4, p. 196, new in the 2021 edition, with the diagram printed before it. *Hidden hypotheses:* none. *Deviation (restated):* the lifts $X^i$ are bundled with their spaces, so that they form one family indexed by $i$; "the natural identification" is $\Phi_i$ and $J_i$, and "$A^{(i)}$ is $A$" is read through the inclusions into $TX$ and through the diagram (Findings, item 164). *Proof:* induction on $i$, with V4.78 on $X^i$; the two identities are checked after the inclusion into $TX$, which is injective and commutes with $\Phi_i$ (Proposition 2.3.17).

**What is not verified in this chapter**

- p. 179 (preprint p. 151), Lemma 4.1.11 (preprint 4.10) (the section $u \mapsto (1-p)(u - t(u))$ of $U \lhd Y$ is sc-Fredholm): not formalized, because no longer used — Proposition 4.1.9 (preprint 4.9) is proved directly (V4.20). It would need $U \lhd Y$ as a strong bundle, $Y$ and $Y \cap F$ as sc-subspaces, and at each smooth point the re-centred strong bundle chart, filler and conjugation (gap G.32).
- pp. 195–199 (preprint pp. 167–168), §4.4 in full: Definitions 4.4.1, 4.4.6 (preprint 4.25, 4.27), the exterior derivative and the relative complex, Theorem 4.4.7 (preprint 4.28). It needs $\oplus^k TX$ as an M-polyfold and a linear $T_xX$ (gap G.7), the M-polyfolds $X^i$ for $i \geq 2$ with $(TX)^i \cong T(X^i)$ (gap G.6, now closed: Proposition 2.3.17 for every $i$), and integration on manifolds with corners of varying dimension, which Mathlib lacks; $d^2 = 0$ is asserted, and Theorem 4.4.7 (preprint 4.28) has no proof in the text (gaps G.37, G.39, G.40). Proposition 4.4.5 (preprint 4.26), quoted there from another paper, is proved in the 2021 edition and verified on an M-polyfold as V4.76 (gap G.38, closed).
- pp. 195–208, the 2021 §4.4 and Appendix 4.5: Definitions 4.4.1–4.4.2 (the Whitney sum $\oplus_k TX$, the direct limit $\Omega^*_\infty(X)$), the exterior derivative $d : \Omega^k(X^i) \to \Omega^{k+1}(X^{i+1})$ and $d \circ d = 0$, Definition 4.4.6, the relative complex, Theorem 4.4.7 and the Poincaré Lemmas 4.4.8 and 4.5.2 (gap G.80, continuing G.37–G.40). Verified: Definition 4.4.3 (V4.67, V4.68), Lemma 4.4.4 for $i = 1$ and Proposition 4.4.5 in a local model (V4.69–V4.72), and on an M-polyfold Lemma 4.5.1 (V4.74, V4.75, V4.77), Proposition 4.4.5 (V4.76) and Lemma 4.4.4 for $i = 1$ and for every $i$ (V4.78, V4.79) (Findings, items 152–164).
- pp. 196 and 201, the 2021 Proposition 4.4.5 and Lemma 4.5.1: the pull-back bundle $TX|X^2 \to X^2$ (the bracket is an sc-smooth map $X^2 \to TX$ covering the inclusion, V4.76), and Lemma 4.5.1 as printed, for a local sc-diffeomorphism between open subsets of $X$, for which open subsets of an M-polyfold would be needed as M-polyfolds (V4.77 proves it for every sc-smooth map $X \to Y$) (gap G.92) (Findings, item 163).
- preprint p. 159, the open question after Proposition 4.2.3 (preprint 4.15): a question, not a

statement. Its last sentence, for a retract without boundary, is V4.32 with $A = O$; that the "T" is not a topological manifold is not formalized (gap G.34).

- p. 176 (preprint p. 148), Remark 4.1.3 (preprint 4.3): why an example of "$X^1 \neq X$" needs a retraction whose first derivative loses a level (one whose level maps are $C^1$ gives none, by a Baire category argument): not formalized; the example itself is verified (V4.6; Findings, item 141).
- p. 176, the example that the 2021 edition adds to Remark 4.1.3 (preprint 4.3): the space of Subsection 1.4.4, a line with a half-plane attached, made an M-polyfold with $X_\infty = X$, for which $1^1_0$ is not sc-smooth by Theorem 4.1.2 (preprint 4.2), as the local dimension of $X$ is not constant. The argument holds as we read it and is not formalized; the formalized example of V4.6 is a different one (gap G.31; Findings, item 159).
- The chart independence of stratified regularity (V4.53): true by the transport argument of V4.39, since the transition maps of product charts are id $\times\, \tau$ up to translation, but not formalized; Theorem 4.3.1 (preprint 4.23) states it for some chart (gap G.36).
- p. 184 (preprint p. 157), Remark 4.2.2 (preprint 4.14), the local form "near points with $d_X(a) = 0$ $A$ is a classical manifold" for the open piece $A \cap \{d_X = 0\}$ of an arbitrary $A$: verified when $d_X = 0$ on all of $A$ (V4.32), not for the piece (gap G.33).
- p. 190 (preprint p. 162), Theorem 4.2.11 (1) and (4) (preprint 4.22) for the local faces of the neighbourhood $U$ itself: the identification of the faces of the chart model with (germs of) global faces of $X$, a local face theorem for small neighbourhoods; (1) and (4) are verified for the chart (V4.50) (gap G.35; Findings, Chapter 4, item 130).
- Definition 4.1.1 (preprint 4.1): the classical manifold structure beyond the $C^\infty$ property of the transition maps, and the local constancy of the dimension (V4.1, V4.2).
- Definitions 4.2.5, 4.2.10 (preprint 4.16, 4.21) and Remark 4.2.2 (preprint 4.14) in $T_aX$: verified in charts, with chart independence (V4.38, V4.39, V4.44, V4.45), since $T_aX$ carries no linear structure here (gap G.7 of Chapter 2).

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## 6. Chapter 5 (HWZ §§5.1–5.3, 5.5 (preprint §§5.1–5.4)): verified except Theorem 5.3.15 (preprint 5.22) and Theorem 5.27 (preprint; not in the 2021 edition) and the cited Theorems 5.5.5, 5.5.13, 5.5.16 (preprint 5.32, 5.40, 5.43)

Chapter 5 of Part I (pp. 205–236 (preprint pp. 169–199)) is the Fredholm package for M-polyfolds. §5.1 introduces auxiliary norms on the part $Y_{0,1}$ of biregularity $(0, 1)$ of a strong bundle (Definition 5.1.1 (preprint 5.1), Lemma 5.1.2 (preprint 5.2), Proposition 5.1.3 (preprint 5.3)). §5.2 defines compact solution sets, weakly proper and proper sections (Definition 5.2.1 (preprint 5.4)) and proves that the three agree for sc-Fredholm sections (Theorem 5.2.2 (preprint 5.5), proof in §5.2 (preprint Appendix 5.4.1)), that solution sets are locally compact on every level (Theorem 5.2.3 (preprint 5.6)) and compact in $X_\infty$ (Theorem 5.2.4 (preprint 5.7)), and that the shifted sections are sc-Fredholm (Corollary 5.2.5 (preprint 5.8)). §5.3 builds sc$^+$-sections from sc-smooth bump functions (Definition 5.3.1 (preprint 5.9), Proposition 5.3.2 (preprint 5.10), Lemma 5.3.3 (preprint 5.11)), introduces the allowable sections $\Gamma^{+,1}_U(P)$ (Remark 5.3.4 (preprint 5.12)), and proves the perturbation theorems: transversality when $\partial X = \emptyset$ (Theorem 5.3.5 (preprint 5.13), Definition 5.3.7 (preprint 5.15)), general position (Definition 5.3.9 (preprint 5.17), Theorem 5.3.10 (preprint 5.18)), good position relative to the boundary (Theorem 5.3.12 (preprint 5.19)), the Morse-type structure of homotopies (Definition 5.3.13 (preprint 5.20), Theorem 5.3.15 (preprint 5.22) with Lemmas 5.3.16, 5.3.17 (preprint 5.23, 5.24)), perturbation into good position (Theorem 5.25 (preprint; not in the 2021 edition), Lemma

5.26 (preprint; not in the 2021 edition)) and cobordisms (Theorem 5.27 (preprint)). §5.5 (preprint Appendix 5.4.2) discusses sc-smooth bump functions and partitions of unity (Definitions 5.5.1–5.5.14 (preprint 5.28–5.41), Theorems 5.5.5–5.5.16 (preprint 5.32–5.43), Corollary 5.5.17 (preprint 5.44)).

What is verified. **Theorems 5.2.2, 5.2.3, 5.2.4, 5.3.5, 5.3.10, 5.3.12, 5.5.7, 5.5.10 (preprint 5.5, 5.6, 5.7, 5.13, 5.18, 5.19, 5.34, 5.37) and Theorem 5.25 (preprint) hold**, with the readings recorded below, and so do Lemmas 5.1.2, 5.3.3, 5.3.16, 5.3.17 (preprint 5.2, 5.11, 5.23, 5.24) and Lemma 5.26 (preprint), Propositions 5.1.3, 5.3.2, 5.5.3, 5.5.8, 5.5.11, 5.5.15 (preprint 5.3, 5.10, 5.30, 5.35, 5.38, 5.42), Remark 5.3.4 (preprint 5.12), Corollary 5.2.5 (preprint 5.8) (for each fixed $i$) and Corollary 5.5.17 (preprint 5.44) (with Theorem 5.5.16 (preprint 5.43) as an explicit hypothesis). Every definition of the chapter is formalized: Definitions 5.1.1 (preprint 5.1; V5.3), 5.2.1 (preprint 5.4; V5.7), 5.3.1 and 5.5.2 (preprint 5.9 and 5.29; V5.16), 5.3.7 (preprint 5.15; V5.23), 5.3.9 (preprint 5.17; V5.31), 5.3.13 (preprint 5.20; V5.41), 5.5.1 (preprint 5.28; V5.47), 5.5.4 (preprint 5.31; V5.50), 5.5.6 (preprint 5.33; V5.51), 5.5.9 (preprint 5.36; V5.54), 5.5.12 (preprint 5.39; V5.59), 5.5.14 (preprint 5.41; V5.60). Corollary 5.2.5 (preprint 5.8) is verified on the shifted strong bundle $P^1 : Y^1 \to X^1$, which the text uses without defining it, and by iteration for every fixed $i$, with the filler condition on level 1 that its "it is a priori clear" hides (V5.13, V5.14). Theorems 5.2.2–5.2.4 (preprint 5.5–5.7), Corollary 5.2.5 (preprint 5.8) and Lemma 5.3.3 (preprint 5.11) need no tameness (V5.10–V5.14, V5.20). Several places of the text needed more than a reading. (a) *Tameness.* The §3.6 implicit function theorems are proved for sections whose germ data lie in a tame strong bundle chart (Definitions 3.6.3, 3.6.4 (preprint 3.50, 3.51); Chapter 3, Findings 190–199); Theorems 5.3.5, 5.3.10, 5.3.12 (preprint 5.13, 5.18, 5.19) and Theorem 5.25 (preprint) take $f$ with tame germ data, the reading of the tame strong bundle with which §5.3 opens, and nothing else (Findings 74–78). Theorem 5.3.10 (preprint 5.18) also assumes $X$ tame, which its proof uses through Theorems 4.2.11 and 4.3.1 (preprint 4.22 and 4.23) (V5.38). (b) *"Generic" and "dense".* The perturbations are $s_0 + \sum \lambda_i t_i$ for $\lambda$ outside a Lebesgue null set of a small ball (V5.29, V5.38), and density is the bound $\rho(s, s') = \sup_x N(s(x) - s'(x)) \leq \varepsilon$ in the metric space of $\rho$ (V5.21, V5.30, V5.39). (c) *Omitted steps proved.* The spanning sections (V5.24), the compactness that keeps the zeros of small perturbations near $\{0\} \times S_{f+s_0}$ (V5.27), the parametrized solution set as a smooth submanifold in the sense of Definition 4.2.1 (preprint 4.13) (V5.32), $T_y S = \ker f'(y)$ and with it the chart independence of Definition 5.3.9 (preprint 5.17) (V5.33, V5.34), the openness of general position (V5.35), and the properties (i)–(ii) of the proof of Theorem 5.3.10 (preprint 5.18) (V5.36). (d) *Repairs.* Definition 5.3.1 (preprint 5.9) is read with $f(x) \neq 0$ (V5.16), which the weak reading gives at the smooth points of an sc-Banach space (V5.16); "$(\Gamma^{+,1}_U(P), \rho)$ is not complete" is false without a hypothesis and is proved in general, by an example (V5.21); the function $\sigma$ of Proposition 5.5.3 (preprint 5.30) is misprinted (V5.17); the extension by 0 in Theorem 5.5.7 (preprint 5.34) and both directions of Theorem 5.5.10 (preprint 5.37) need smaller charts (V5.52, V5.55, V5.56); Proposition 5.5.11 (preprint 5.38) is not "an immediate consequence of Theorem 5.37" (V5.58). (e) *Theorem 5.3.12 (preprint 5.19)* is proved without applying the implicit function theorem to the parametrized section at the boundary zeros, where it is in general not in good position (V5.40; Finding 88, on paper). (f) *Lemma 5.3.16 (preprint 5.23)* exposes a hidden hypothesis of Theorem 5.3.15 (preprint 5.22), $dt(x) \neq 0$ (V5.42). Not verified: Theorem 5.3.15 (preprint 5.22) and Theorem 5.27 (preprint), Remarks 5.3.6, 5.3.8, 5.3.14 (preprint 5.14, 5.16, 5.21), the cited Theorems 5.5.5, 5.5.13, 5.5.16 (preprint 5.32, 5.40, 5.43), the Sobolev remark and a dense form of Theorem 5.3.12 (preprint 5.19) (see the list at the end of this section). A fourth addendum (2026-11-02: the shifted strong bundle and Corollary 5.2.5 (preprint 5.8), the metric $\rho$, weak against pointwise bump functions; Findings, items 140–164) updated V5.13, V5.14, V5.16, V5.21, V5.22 and V5.30 in place. A fifth addendum (2026-11-08; Findings, items 170–179) covers the Springer 2021

edition: Definition 5.2.1 with the new properties weakly proper$_{<\infty}$ and proper$_{<\infty}$, its a priori dependencies and Theorem 5.2.2 for them (V5.64–V5.66), the compact sub-M-polyfold of Theorem 5.3.5 (V5.67), and Definition 5.3.13 with its new condition (3), which supplies the hidden hypothesis of V5.42 (V5.68–V5.70).

The setting. A section of $P : Y \to X$ is a map $s : X \to Y$ with $P \circ s = \mathrm{id}$; its support is the closure of $\{s \neq 0\}$. $X$ and $Y$ are paracompact Hausdorff spaces, as for every M-polyfold of the project. $Y_{0,1}$ is the M-polyfold $Y[1]$ of §2.6 (preprint §2.5), an auxiliary norm is a function on it, and "$N(y) \leq r$" for a point $y$ of $Y$ includes $y \in Y_{0,1}$ (the text's convention $N(y) = \infty$ off $Y_{0,1}$). The linearization $f'(x) : T_xX \to Y_x$ is read in a strong bundle chart, as the operator $\mathrm{Lin}(z) : T_zO \to K_z$ of Chapter 3, because $T_xX$ carries no linear structure here (gap G.7 of Chapter 2); transversality is chart independent (Chapter 3), and general position is shown chart independent at the zeros near a zero in good position (V5.34). In the perturbation theorems $|\lambda|$ is the maximum norm on $\mathbb{R}^m$, and "the solution set is a manifold" is stated for a non-empty solution set (the empty set is trivially one). Findings referred to below are those of `Exposition/Findings.md`, "Chapter 5".

**V5.1 (the charts $\Phi[1]$ for every strong bundle chart; `Polyfold.StrongBundle.oneChartOf`, `Polyfold.StrongBundle.isChart_oneChartOf`, `Polyfold.StrongBundle.oneChartOf_compatible`, `Polyfold.StrongBundle.mem_oneSet_iff_of_isChart`).** For every strong bundle chart $\Phi$ of $P$ compatible with the atlas (not only those of the atlas), a point $y$ in its domain lies in $Y_{0,1}$ iff the fibre component of $\Phi(y)$ lies in $F_1$; the map $\Phi[1] : P^{-1}(V) \cap Y_{0,1} \to K[1]$ is a chart of the M-polyfold $Y[1]$, and the charts $\Phi[1]$ of a compatible chart and of a chart of the atlas are compatible. *Book:* §5.1, preprint p. 169, and Lemma 5.1.2 (ii) (preprint 5.2), which quantifies over strong bundle isomorphisms $\Phi : Y|V \to K$. *Hidden hypotheses:* none. *Deviation:* none; Chapter 2 built $\Phi[1]$ only for the charts of the atlas (Findings, item 2).

**V5.2 (the fibre $(Y_{0,1})_x$ and its chart norm $|\cdot|_1$; `Polyfold.StrongBundle.oneFiber`, `Polyfold.StrongBundle.toOn`
`Polyfold.StrongBundle.zeroOne`, `Polyfold.StrongBundle.liftOneFiber`, `Polyfold.StrongBundle.liftOneF`
`Polyfold.StrongBundle.mem_range_liftOneFiber`, `Polyfold.StrongBundle.isClosed_slice`,
`Polyfold.StrongBundle.complete_liftOneFiber`, `Polyfold.StrongBundle.exists_norm_liftOne_le_mul`).** The points of the fibre $P^{-1}(x)$ that lie in $Y_{0,1}$ form a linear subspace $(Y_{0,1})_x$; its zero is the point $0_x$ of $Y[1]$. In a strong bundle chart around $x$ the fibre component of $\Phi[1]$ is an injective linear map $(Y_{0,1})_x \to F_1$ onto the slice $\{h \in F_1 \mid (\varphi(x), h) \in K[1]\}$, which is closed in $F_1$, so $(Y_{0,1})_x$ is complete for $|h|_1$. The norms $|\cdot|_1$ of two charts around $x$ are comparable, with one constant, on the fibres over a neighbourhood of $x$. *Book:* §5.1, preprint p. 169 ("the fibers $Y_x$ have the structure of Banach spaces"; $|h|_1$ in Lemma 5.1.2 (preprint 5.2)). *Hidden hypotheses:* none. *Deviation (restated):* the text takes the linear and Banach structure of the fibres of $Y_{0,1}$ for granted; it is constructed, and $|h|_1$ depends on the chart up to equivalence (Findings, item 1).

**V5.3 (Definition 5.1.1 (preprint 5.1), auxiliary norms; `Polyfold.StrongBundle.IsCompleteFiberNorm`,
`Polyfold.StrongBundle.IsAuxNorm`, `Polyfold.StrongBundle.IsAuxNorm.nonneg`, `Polyfold.StrongBundle.Is`
An auxiliary norm is a continuous $N : Y[1] \to \mathbb{R}$ such that (1) on every fibre $(Y_{0,1})_x$, $N$ is a norm ($N(v) = 0 \Rightarrow v = 0$, $N(tv) = |t|N(v)$, the triangle inequality) for which every Cauchy sequence converges, and (2) if $P(w_k) \to x$ in $X$ and $N(w_k) \to 0$, then $w_k \to 0_x$ in $Y[1]$. It is non-negative and homogeneous. *Book:* Definition 5.1.1 (preprint 5.1), p. 205 (preprint p. 169). *Hidden hypotheses:* none. *Deviation (restated):* "complete norm" is read as a norm on $(Y_{0,1})_x$ with complete normed space; the topology of $Y_{0,1}$ is that of $Y[1]$ (Findings, item 1).

**V5.4 (Lemma 5.1.2 (preprint 5.2); `Polyfold.lemma_5_2`, `Polyfold.StrongBundle.exists_le_mul_norm_liftOne`**

`Polyfold.StrongBundle.IsAuxNorm.exists_norm_liftOne_le`, `Polyfold.StrongBundle.IsAuxNorm.exists_b` `Polyfold.lemma_5_2_literal`**).** Let $N : Y[1] \to \mathbb{R}$ be continuous and a complete norm on every fibre. Then $N$ is an auxiliary norm iff every $x \in X$ lies in the domain of a strong bundle chart $\Phi$ with an open neighbourhood $V \subseteq \operatorname{dom}\varphi$ of $x$ and constants $0 < a < A$ such that $a\,N(w) \le |h|_1 \le A\,N(w)$ for all $w \in Y_{0,1}$ with $P(w) \in V$, $\Phi(w) = (p,h)$. For an auxiliary norm such bounds hold in *every* strong bundle chart around $x$; in the text's literal form (`lemma_5_2_literal`) the bounds hold for all $w \in Y_{0,1}$ over the whole domain of the chart, obtained by restricting the chart to the neighbourhood (V2.57); the upper bound $g(w) \le C|h|_1$ holds for every $g$ continuous at $0_x$ and homogeneous on the fibres near $x$. *Book:* Lemma 5.1.2 (preprint 5.2), p. 205 (preprint p. 169); proof pp. 205–206 (preprint pp. 169–170). *Hidden hypotheses:* the proof gives the bounds near $x$ only; on all of $P^{-1}(V)$ for the domain $V$ of $\Phi$ they hold after restricting the chart, which the text does silently. *Deviation:* none for `lemma_5_2_literal`; the first form has $V$ a neighbourhood of $x$ on which the bounds hold; the contradiction argument needs no metrizability (Findings, items 2, 3; Chapter 2, items 42, 44). Formerly "restated", before the literal form was proved.

**V5.5 (Proposition 5.1.3 (preprint 5.3), existence;** `Polyfold.proposition_5_3_exists`, `Polyfold.StrongBundle.gluedNorm`, `Polyfold.StrongBundle.continuous_gluedNorm`, `Polyfold.StrongBundle.isCompleteFiberNorm_gluedNorm`, `Polyfold.StrongBundle.gluedNorm_bounds`**).** Every strong bundle has an auxiliary norm: for the charts $\Phi_i$ of the atlas and a partition of unity $(\rho_i)$ subordinate to their domains, $N(w) = \sum_i \rho_i(P(w))\,|\Phi_i[1](w)_2|_1$ is continuous on $Y[1]$, a complete norm on every fibre, and comparable near each point with the norm $|\cdot|_1$ of any chart there. *Book:* Proposition 5.1.3 (preprint 5.3), p. 206 (preprint p. 171). *Hidden hypotheses:* none. *Deviation:* none; the text's proof is one sentence (Findings, item 4).

**V5.6 (Proposition 5.1.3 (preprint 5.3), comparison;** `Polyfold.proposition_5_3_compare`**).** For auxiliary norms $N_0$, $N_1$ there is a continuous $f : X \to (0,\infty)$ with $f(x)N_0(w) \le N_1(w) \le f(x)^{-1}N_0(w)$ for every $w \in Y_{0,1}$ over $x$. *Book:* Proposition 5.1.3 (preprint 5.3), p. 206 (preprint p. 171). *Hidden hypotheses:* none. *Deviation:* none; the statement's $N_0(h)$ is $N_0(w)$ (Findings, item 5).

**V5.7 (Definition 5.2.1 (preprint 5.4);** `Polyfold.StrongBundle.AuxLe`, `Polyfold.StrongBundle.IsWeaklyProper`, `Polyfold.StrongBundle.IsProper`, `Polyfold.StrongBundle.IsProper.isWeaklyProper`, `Polyfold.StrongBundle.IsWeaklyProper.isCompact_solutionSet`**).** A section $f$ has a compact solution set if $f^{-1}(0)$ is compact in $X$; it is weakly proper if moreover for every auxiliary norm $N$ there is an open $U \supseteq f^{-1}(0)$ such that $\{f = s\}$ is compact for every sc$^+$-section $s$ with support in $U$ and $N(s(y)) \le 1$ for all $y$; it is proper if moreover for every $N$ there is an open $U \supseteq f^{-1}(0)$ with $\mathrm{cl}_X\{x \in U \mid N(f(x)) \le 1\}$ compact, where $N(f(x)) \le 1$ includes $f(x) \in Y_{0,1}$. A continuous proper section is weakly proper, and a weakly proper section has a compact solution set. *Book:* Definition 5.2.1 (preprint 5.4), p. 209 (preprint p. 171). *Hidden hypotheses:* "proper implies weakly proper" needs $f$ continuous (so that $\{f = s\}$ is closed). *Deviation:* none (Findings, item 26).

**V5.8 (Theorem 5.2.2 (preprint 5.5), the local form for basic germs;** `Polyfold.IsSc0ContractionGerm.exists_isC` `Polyfold.IsBasicGerm.exists_isCompact_bdd`**).** Let $f$ be a basic germ on $\tilde{C} \oplus W$ with $P \circ f(a,w) = w - B(a,w)$, $\tilde{C}$ closed. There is an open $O \ni 0$, independent of $c$, such that for every $c$ the points $x$ of $U_0 \cap O$ with $f(x) \in \mathbb{R}^N \oplus W_1$ and $|f(x)|_1 \le c$ lie in a compact subset of $E_0$, namely the image of $\{(a,z) \mid a \in \tilde{C}, |a|_0 \le \rho, z \in K_c, |z|_0 \le r/8\}$ under $(a,z) \mapsto (a, \delta(a,z))$, with $K_c$ a compact set containing the level-1 ball of radius $c$ and $\delta$ the parametrized fixed point of $w \mapsto B(a,w) - z$. *Book:* §5.2 (preprint Appendix 5.4.1), preprint pp. 189–192, (82)–(90). *Hidden*

*hypotheses:* $\tilde{C}$ closed, as in Theorem 3.5.1 (preprint 3.40). *Deviation (restated):* only the level-0 contraction is used, and the neighbourhood does not depend on the bound (Findings, items 23, 25).

**V5.9 (Theorem 5.2.2 (preprint 5.5), the local step; `Polyfold.IsStrongBundleMap.exists_fiber_bound_one`, `Polyfold.exists_nbhd_isCompact_closure_bdd`).** A strong bundle map $\Psi$ has a uniform $(0,1)$ fibre bound near each point, $|\Psi[1](u,k)_2|_1 \le M|k|_1$ for $|u-u_0|_0 < \delta$. If $(f,x_0)$ is an sc-Fredholm germ, then in the strong bundle chart $\Phi$ of the germ there is an open $V \ni x_0$ such that for every $C$ the closure in $X$ of $\{x \in V \mid \Phi(f(x))_2 \in F_1, |\Phi(f(x))_2|_1 \le C\}$ is compact. *Book:* §5.2 (preprint Appendix 5.4.1), preprint pp. 189–192 ("without loss of generality $f = h + t$", "it suffices to find in local coordinates"). *Hidden hypotheses:* the uniform fibre bound of the conjugating isomorphism. *Deviation (stronger):* $x_0$ need not be a zero; subtracting the sc$^+$-part before the conjugation replaces the text's $t$ and its estimates (83), (85), and the closure is pulled back to $X$ because the retract is the fixed point set of the retraction (Findings, items 23, 24).

**V5.10 (Theorem 5.2.2 (preprint 5.5); `Polyfold.theorem_5_5`, `Polyfold.theorem_5_5_iff`).** Let $f$ be an sc-Fredholm section of a strong bundle over an M-polyfold $X$. If $f$ has a compact solution set, $f$ is proper; and for $f$ the properties proper, weakly proper and having a compact solution set are equivalent. *Book:* Theorem 5.2.2 (preprint 5.5), p. 210 (preprint p. 172); proof §5.2 (preprint Appendix 5.4.1), preprint pp. 189–192. *Hidden hypotheses:* none. *Deviation (stronger):* $X$ need not be tame (the appendix states the theorem without tameness); the finite cover of $f^{-1}(0)$ and the closure of a finite union are made explicit (Findings, items 24, 25).

**V5.11 (Theorem 5.2.3 (preprint 5.6); `Polyfold.theorem_5_6`).** Let $f$ be an sc-Fredholm section and $f(x) = 0$. There are open neighbourhoods $x \in U(i+1) \subseteq U(i)$ of $x$ in $X$ such that for every $i$ the closure in $X$ of $\{y \in U(i) \mid f(y) = 0\}$ is the image of a compact subset of $X_i$ (with its level topology). *Book:* Theorem 5.2.3 (preprint 5.6), p. 213 (preprint p. 172). *Hidden hypotheses:* none. *Deviation:* none; it is Theorem 3.1.13 (preprint 3.12) at a zero, which is a smooth point where $(f,x)$ is an sc-Fredholm germ.

**V5.12 (Theorem 5.2.4 (preprint 5.7); `Polyfold.theorem_5_7`, `Polyfold.MPolyfold.tendsto_smoothLevel_iff`, `Polyfold.MPolyfold.smoothLevel_topology_eq`, `Polyfold.MPolyfold.metrizableSpace_smoothLevel`).** $X_\infty$ carries the initial topology of the inclusions $X_\infty \to X_m$; it is metrizable, and a sequence converges in it iff it converges on every level. If $f$ is an sc-Fredholm section whose solution set $S$ is compact in $X$, then $S \subseteq X_\infty$ and $S$ is a compact subset of $X_\infty$. *Book:* Theorem 5.2.4 (preprint 5.7), p. 213 (preprint p. 172). *Hidden hypotheses:* the topology of $X_\infty$, which the text does not define. *Deviation (stronger):* $X$ need not be tame. The convergence on every level of a sequence of zeros is Corollary 3.1.15 (preprint 3.13) at its limit, not Theorem 5.2.3 (preprint 5.6) as the text says (Findings, item 20).

**V5.13 (Corollary 5.2.5 (preprint 5.8); `Polyfold.corollary_5_8`, `Polyfold.corollary_5_8_two`, `Polyfold.StrongBundle.solutionSet_sectionShift`, `Polyfold.isCompact_solutionSet_level`).** Let $f$ be an sc-Fredholm section of $P : Y \to X$ (Definition 3.1.16 (preprint 3.14), in the text's germ reading of Definition 3.1.11 (preprint 3.11)) whose solution set is compact. Then $f^1 = f|_{X_1}$ is an sc-Fredholm section of the shifted strong bundle $P^1 : Y^1 \to X^1$ (V5.14) and its solution set is compact in $X^1$; $f^2 = (f^1)^1$ is an sc-Fredholm section of $P^2 = (P^1)^1$ with compact solution set; and for every $i$ the zeros of $f$ in $X_i$ form a compact subset of $X_i$. *Book:* Corollary 5.2.5 (preprint 5.8), p. 213 (preprint p. 173) (over a tame $X$, for every $i$). *Hidden hypotheses:* the bundle $P^i$, which the text does not define, is the iterated shift $P^i = (P^{i-1})^1$. *Deviation:* none; the first statement holds for every strong bundle, so every fixed $i$ follows by iterating it ($i = 2$ is stated);

one statement for a family indexed by $i$ is not formalized, and the base $(X^1)^1$ of $P^2$ has the points of $X_2$, and its topology is that of $X_2$ (V2.93; gap G.6, closed). Tameness is not used; the conclusion is Definition 3.1.16 (preprint 3.14) for $f^i$, not tame germ data, which would need tame charts of $P^1$ (Findings, items 140–146). Formerly "narrower": only the compactness half was verified (gap G.41, now closed for each fixed $i$).

**V5.14 (Corollary 5.2.5 (preprint 5.8): the shifted strong bundle, and $f^1$ sc-Fredholm on it;** `Polyfold.StrongBundle.shift`, `Polyfold.StrongBundleChart.shift`, `Polyfold.StrongBundleChart.Compati` `Polyfold.StrongBundle.mem_shift_doubleLevel_iff`, `Polyfold.StrongBundle.sectionShift`, `Polyfold.StrongBundle.regularizing_sectionShift`, `Polyfold.ScNbhdGerm.shiftParam`, `Polyfold.IsSc0ContractionGerm.shift`, `Polyfold.IsBasicGerm.shift`, `Polyfold.IsConjugateToBasicGerm.` `Polyfold.IsConjugateToBasicGerm.shift_germ`, `Polyfold.IsFillingGerm.mem_range_toBase_of_fillDefec` `Polyfold.IsFillingGerm.germ_shift`, `Polyfold.isScFredholmGerm_sectionShift`, `Polyfold.corollary_5_8` `Polyfold.corollary_5_8_chart`)**.** *The shifted strong bundle.* $P^1 : Y^1 \to X^1$ is a strong bundle over the M-polyfold $X^1$ (Definition 2.6.5 (preprint 2.51)): its total space is $Y_{1,1}$, the level 1 of the M-polyfold $Y[0]$ with its level topology; $P^1$ is the restriction of $P$; the fibre over $x$ is $P^{-1}(x) \cap Y_{1,1}$, a linear subspace, carrying the topology of $F_1$; and the atlas consists of the index raises $\Phi^1$ of the strong bundle charts $\Phi$ of $P$ (base chart the level chart $\varphi^1$, fibre model $F^1$, retraction $R$ restricted to $U^1 \lhd F^1$), compatible charts having compatible index raises. Its double levels are $(Y^1)_{m,k} = Y_{m+1,k+1}$. *$f^1$ is sc-Fredholm.* For every sc-Fredholm section $f$ (no compactness needed), $f^1 = f|_{X_1}$ is an sc-Fredholm section of $P^1$: it is sc-smooth, regularizing, and at every smooth point of $X^1$ it has the data of Definition 3.1.11 (preprint 3.11) in the index raise $\Phi^1$ of the chart $\Phi$ of a datum $(g, s)$ of $f$ there. The principal part of $f^1$ is the level-1 map of that of $f$; $g^1$ is a filling germ of it, including condition (3) of Definition 3.1.4 (preprint 3.4) on level 1, which comes from a regularity statement at the centre of every datum of Definition 3.1.11 (preprint 3.11): $v \in \ker Dr(0)$ and $Dk(0)v \in F_m$ imply $v \in E_m$, $k$ the fill defect. *In a chart.* The level shift $f^1$ of a basic germ $f$ on $\tilde{C} \oplus W$, $\tilde{C}$ a partial quadrant, is a basic germ on $\tilde{C} \oplus W^1$ with $B^1$ in place of $B$; if a principal part $k$ is conjugated to a basic germ $h$ by $\Psi$, then $k^1$ is conjugated to $h^1$ by the index raise $\Psi^1$. Hence for an sc-Fredholm section with compact solution set: the zeros in $X_1$ form a compact subset of $X_1$, and at every smooth point $x$, in the chart $\Phi$ centred at $x$ of the germ $(f, x)$ with filling germ $g$ and sc$^+$-germ $s$ (the text's germ reading of Definition 3.1.11 (preprint 3.11)), the level-1 germ $(g - s)^1$ is conjugated to a basic germ (by the germ form of the index raise, which uses $g - s$ only near 0) and $s^1$ is an sc$^+$-germ on the shifted germ of $s$. *Book:* Corollary 5.2.5 (preprint 5.8), p. 213 (preprint p. 173) ("it is a priori clear that $f^i$ is sc-Fredholm", as a section of $P^i : Y^i \to X^i$). *Hidden hypotheses:* the bundle $Y^1 \to X^1$, which the text does not define ($Y[i]$ is defined for $i = 0, 1$ only): $Y^1 = Y_{1,1}$, the convention of $E^1$, forced by "$f^i$ is obtained by raising the index" (an sc-smooth section maps $X_1$ into $Y_{1,1}$, in general not into $Y_{1,2}$); $\tilde{C}$ a partial quadrant. *Deviation:* none; "it is a priori clear" hides condition (3) of the filling on level 1, which does not follow from level 0 alone (an sc-operator can be bijective on level 0 and not onto on level 1); it is proved for the data of Definition 3.1.11 (preprint 3.11) by reducing to $g(0) = 0$ with Lemma 3.1.20 (preprint 3.17) (subtracting the section $z \mapsto \Gamma(z, -g(0))$, which leaves the fill defect unchanged) and applying Proposition 3.6.5 (preprint 3.52) in sc form. No hypothesis beyond the text's is added; tameness is not used, and the tame form (germ data in tame charts of $P^1$) is not stated (Findings, items 22, 140–146). Formerly "narrower": the bundle was not built and the filler on level 1 not verified (gap G.41, now closed for each fixed $i$). In the text's germ reading of Definition 3.1.11 (preprint 3.11) (phase 2; Findings, Chapter 3, item 333) $s^1$ is an sc$^+$-germ on the shifted germ; in the whole-chart reading it was sc$^+$ on all of $U^1$.

**V5.15 ($\mathbb{R}$ as an M-polyfold; sc-smooth functions; `Polyfold.realPolyfold`, `Polyfold.realChart`, `Polyfold.isChart_realChart`, `Polyfold.MPolyfold.isScSmoothMap_of_local`, `Polyfold.MPolyfold.IsScSmo` `Polyfold.isScSmoothMap_real_of_local`, `Polyfold.isScSmoothMap_real_const`, `Polyfold.isScSmoothMap_r` `Polyfold.MPolyfold.IsScSmoothMap.comp_contDiff`).** $\mathbb{R}$ with the constant sc-structure and the single chart id is an M-polyfold, and an sc-smooth function on $X$ is an sc-smooth map $X \to \mathbb{R}$ (Definition 2.3.8 (preprint 2.14)). Sc-smoothness of a map between M-polyfolds can be checked in one pair of charts around each point (continuity included); an sc-smooth function read in a chart is sc-smooth on the retract; constants are sc-smooth, and $\sigma \circ f$ is sc-smooth for $\sigma : \mathbb{R} \to \mathbb{R}$ of class $C^\infty$. *Book:* Definitions 5.3.1, 5.5.2, 5.5.6 (preprint 5.9, 5.29, 5.33) ("sc-smooth map $f : X \to \mathbb{R}$"). *Hidden hypotheses:* none. *Deviation (not in the book):* the M-polyfold $\mathbb{R}$ is presupposed by the text (Findings, item 40).

**V5.16 (Definitions 5.3.1 and 5.5.2 (preprint 5.9 and 5.29); `Polyfold.MPolyfold.AdmitsBumpFunctions`, `Polyfold.MPolyfold.AdmitsBumpFunctionsWeak`, `Polyfold.MPolyfold.AdmitsBumpFunctions.weak`, `Polyfold.MPolyfold.HasBumpAt`, `Polyfold.MPolyfold.admitsBumpFunctionsWeak_of_dense`, `Polyfold.MPolyfold.AdmitsBumpFunctionsWeak.hasBumpAt_of_mem_smooth`, `Polyfold.MPolyfold.admitsBum`** $X$ admits sc-smooth bump functions if for every $x$ and every open $U \ni x$ there is an sc-smooth $f : X \to \mathbb{R}$ with $f(x) \neq 0$ and $\operatorname{supp} f \subseteq U$. Definition 5.5.2 (preprint 5.29) as written ("$f$ not identically zero") is a separate notion, implied by the first. A point $x$ is a *bump point* if every open $U \ni x$ carries an sc-smooth $f$ with $f(x) \neq 0$ and $\operatorname{supp} f \subseteq U$: the first notion says that every point is a bump point, and the second follows if the bump points are dense. On an sc-Banach space $E$ (as an M-polyfold with the identity chart) the second notion makes every smooth point a bump point; on M-polyfolds built on sc-Hilbert spaces the first notion holds. *Book:* Definition 5.3.1 (preprint 5.9), p. 216 (preprint p. 173); Definition 5.5.2 (preprint 5.29), p. 229 (preprint p. 192). *Hidden hypotheses:* none. *Deviation (restated):* "$f \neq 0$" is read as $f(x) \neq 0$, the reading Proposition 5.5.3's (preprint 5.30) proof uses. Whether the weak reading implies it is open only at non-smooth points: at a smooth point of an sc-Banach space a bump that is non-zero near $x$ is translated to $x$ by a smooth vector, and translation by a non-smooth vector does not preserve the levels; that the weak reading is exactly the density of the bump points is argued, one direction formalized (Findings, items 41, 160–163).

**V5.17 (Propositions 5.3.2 and 5.5.3 (preprint 5.10 and 5.30); `Polyfold.proposition_5_30`, `Polyfold.proposition_5_10`).** If $X$ admits sc-smooth bump functions, then for every $x$ and open $U \ni x$ there is an sc-smooth $f : X \to [0,1]$ with $f(x) = 1$, $\operatorname{supp} f \subseteq U$, and $f \equiv 1$ on a neighbourhood of $x$. *Book:* Proposition 5.3.2 (preprint 5.10), p. 216 (preprint p. 173); Proposition 5.5.3 (preprint 5.30), p. 230 (preprint p. 193). *Hidden hypotheses:* Definition 5.3.1 (preprint 5.9) in the reading of V5.16. *Deviation:* none; the text's $\sigma$ ("0 for $s \leq 1$ and 1 for $s \geq 1$") is misprinted and $g(x) = 1$ is not given; the proof uses $f = \tau(3g/g(x) - 1)$ with a smooth transition $\tau$ (Findings, item 42).

**V5.18 (the sc-calculus of sections: $\Gamma^+(P)$, supports, $f \cdot s$ and sums; `Polyfold.sectionSupport`, `Polyfold.sectionSmul`, `Polyfold.sectionSum`, `Polyfold.StrongBundle.scPlusSections`, `Polyfold.StrongBundle.IsScPlusSection.mem_oneSet`, `Polyfold.StrongBundle.isScPlusSection_fiberZer` `Polyfold.StrongBundle.IsScPlusSection.smul_left`, `Polyfold.StrongBundle.IsScPlusSection.add`, `Polyfold.StrongBundle.isScPlusSection_sectionSum`).** $\Gamma^+(P)$ is the set of sc$^+$-sections (Definition 2.6.6 (preprint 2.52)); their values lie in $Y_{0,1}$. The zero section is an sc$^+$-section; for an sc-smooth function $f$ and an sc$^+$-section $s$, $f \cdot s$ (fibrewise) is an sc$^+$-section; sums and finite sums of sc$^+$-sections are sc$^+$-sections. *Book:* §5.3, preprint p. 173 ($\Gamma^+(P)$); used in the proof of Lemma 5.3.3 (preprint 5.11) ("$s(y) = \beta(y)t(y)$ is a sc$^+$-section") and in the finite combinations $\sum \lambda_i s_i$ of

Theorems 5.3.5, 5.3.10 (preprint 5.13, 5.18). *Hidden hypotheses:* none. *Deviation (not in the book):* the text uses these rules without stating them; in a chart $(f \cdot s)^\wedge = (f \circ \varphi^{-1}) \cdot \hat{s}$ and the product rule of the sc-calculus applies (Findings, item 45).

**V5.19 (local sc⁺-sections cut off and extended by $0$; `Polyfold.StrongBundle.cutoffSection`, `Polyfold.StrongBundle.isScPlusSection_cutoffSection`).** Let $\Phi$ be a strong bundle chart of the atlas, $\tau : O \to F_1$ such that $z \mapsto (z, \tau(z))$ is sc-smooth $O \to K[1]$, and $\beta : X \to \mathbb{R}$ sc-smooth with $\operatorname{supp}\beta \subseteq \operatorname{dom}\varphi$. Then $y \mapsto \Phi^{-1}(\varphi(y), \beta(y)\tau(\varphi(y)))$ on $\operatorname{dom}\varphi$, $0_y$ outside, is an sc⁺-section. *Book:* proof of Lemma 5.3.3 (preprint 5.11), preprint p. 174 ("if we construct $s$ with support in $Q'$ we can extend it by 0"). *Hidden hypotheses:* none. *Deviation (not in the book):* the extension by 0 is proved (Findings, item 46 (c)).

**V5.20 (Lemma 5.3.3 (preprint 5.11); `Polyfold.lemma_5_11`).** Let $X$ admit sc-smooth bump functions and $N$ be an auxiliary norm. For a smooth point $x$, a smooth point $e$ of $Y$ with $P(e) = x$, $\varepsilon > 0$ and an open $U \ni x$, there is an sc⁺-section $s$ with $s(x) = e$, $\operatorname{supp} s \subseteq U$ and $N(s(y)) < N(e) + \varepsilon$ **for all** $y$. *Book:* Lemma 5.3.3 (preprint 5.11), p. 217 (preprint p. 173); proof preprint p. 174. *Hidden hypotheses:* Definition 5.3.1 (preprint 5.9) in the reading of V5.16. *Deviation (stronger):* the bound holds at every point (the text's bound at $x$ is trivial; the proof gives the uniform one, which §5.3 needs); tameness of $P$ and $X$, normality of $X$ and a chart centred at $x$ are not needed (Findings, item 46).

**V5.21 ($\Gamma^{+,1}_U(P)$ and the metric $\rho$; `Polyfold.StrongBundle.allowableSections`, `Polyfold.StrongBundle.secti`
`Polyfold.StrongBundle.allowableDist`, `Polyfold.StrongBundle.allowableMetric`, `Polyfold.StrongBundle`
`Polyfold.TrivialBundle.not_completeSpace_allowableSections`, `Polyfold.StrongBundle.completeSpace_`
`Polyfold.TrivialBundle.completeSpace_allowableSections_point`).** For an auxiliary norm $N$ and an open $U$, $\Gamma^{+,1}_U(P)$ is the set of sc⁺-sections $s$ with $\operatorname{supp} s \subseteq U$ and $N(s(x)) \le 1$ for all $x$ (the allowable sections). For two sc⁺-sections $N(s(x) - s'(x))$ is the norm of the difference in the fibre $(Y_{0,1})_x$. The uniform distance $\rho(s, s') = \sup_x N(s(x) - s'(x))$ is finite ($\rho \le 2$) and, for an auxiliary norm, a metric on $\Gamma^{+,1}_U(P)$; $\rho(s, s') \le \varepsilon$ iff $N(s(x) - s'(x)) \le \varepsilon$ for all $x$. For the trivial bundle $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$ with $N(x, v) = |v|$ and $U = \mathbb{R}$ the metric space is not complete; for $U = \emptyset$ (any bundle) and over a point with fibre $\mathbb{R}^k$ (any $U$) it is complete. *Book:* §5.3, preprint p. 174 ("a metric space … not complete"). *Hidden hypotheses:* none for the metric. *Deviation (restated):* the text's "not complete" is false without a hypothesis (the cases $U = \emptyset$ and a point base) and is read as "not complete in general", proved by the example: $s_j(x) = (x, \sqrt{x^2 + 1/(j+1)^2}/(2\sqrt{1+x^2}))$ is a $\rho$-Cauchy sequence of allowable sections whose pointwise limit has the fibre part $|x|/(2\sqrt{1+x^2})$, not differentiable at 0, while the fibre part of an sc⁺-section of the trivial bundle is $C^1$. Non-completeness in the text's setting ($U$ containing a non-isolated point over which $Y_{0,1} \neq 0$) is expected by the same mechanism and not formalized (Findings, items 47, 150–153).

**V5.22 (Remark 5.3.4 (preprint 5.12); `Polyfold.remark_5_12`, `Polyfold.StrongBundle.IsAuxNorm.auxLe_of_sec`
`Polyfold.StrongBundle.IsInAllowableCompletion`, `Polyfold.remark_5_12_completion`,
`Polyfold.remark_5_12_of_isScFredholmSection`).** Let $f$ have its solution set inside the open $U$ with $\operatorname{cl}\{x \in U \mid N(f(x)) \le 1\}$ compact. If a section $s$ vanishes outside $U$ and has $N(s(x)) \le 1$ everywhere — as the uniform limits of allowable sections do — then the solutions of $f + s = 0$ lie in $\{x \in U \mid N(f(x)) \le 1\}$, and their closure is compact. As stated (`remark_5_12_completion`): with the points of the completion represented by sections $s$ with values in $Y_{0,1}$ that are uniform $N$-limits of allowable sections (`IsInAllowableCompletion`), for $f$ continuous the solution set of $f + s = 0$ is closed, hence compact; for $f$ sc-Fredholm with compact solution set, with the $U$ of Theorem 5.2.2 (preprint 5.5) (`remark_5_12_of_isScFredholmSection`). *Book:* Remark

5.3.4 (preprint 5.12), p. 217 (preprint p. 174). *Hidden hypotheses:* none. *Deviation (restated):* the completion of $\Gamma_U^{+,1}$ is represented by sections, uniform $N$-limits of allowable sections; the metric space itself is V5.21, and that every $\rho$-Cauchy sequence has such a limit section is not formalized (Findings, items 131, 153). The closedness of the solution set, which the remark does not discuss, is proved by a chart argument: Lemma 5.1.2's (preprint 5.2) bound turns $N(s-s') \le \varepsilon$ into a level-0 bound in a chart, and $f+s'$ is continuous for allowable $s'$; "adapted to $N$" is read as the properness of V5.7 (Findings, items 27, 130, 131).

**V5.23 (Definition 5.3.7 (preprint 5.15); `Polyfold.StrongBundle.IsTransversal`, `Polyfold.StrongBundle.IsLi`).** A section $g$ is transversal to the zero section if at every zero $x$ the linearization is onto, in a strong bundle chart around $x$; it is then onto in every such chart. *Book:* Definition 5.3.7 (preprint 5.15), p. 218 (preprint p. 175). *Hidden hypotheses:* none. *Deviation (restated):* $g'(x) : T_xX \to Y_x$ is read in charts (gap G.7 of Chapter 2) (Findings, item 70).

**V5.24 (spanning sections at a zero; `Polyfold.exists_fin_mem_of_dense_of_closedComplemented`, `Polyfold.exists_spanning_sections`).** If $R$ is a closed, complemented subspace of finite codimension of a Banach space $K$ and $D$ a dense subspace, finitely many vectors of $D$ span a complement of $R$. Hence, if $X$ admits sc-smooth bump functions, $N$ is an auxiliary norm, $g$ is sc-Fredholm and $g(y) = 0$, then for every open $U \ni y$ there are finitely many allowable sc$^+$-sections $s_1, \dots, s_k$ supported in $U$ and a strong bundle chart centred at $y$ in which $\mathrm{Lin}(0)(T_0O) + \mathrm{span}\{\hat{s}_i(0)\} = K_0$. *Book:* proof of Theorem 5.3.5 (preprint 5.13), p. 218 (preprint p. 175) ("we find finitely many allowable sc$^+$-sections such that the range of $(f+s_0)'(x)$ together with the sections span $Y_x$"). *Hidden hypotheses:* none. *Deviation (not in the book):* the text does not say how; the density of the smooth vectors in $K_0$ is the point (Findings, item 66).

**V5.25 (the parametrized section and its slices; `Polyfold.paramSlice`, `Polyfold.isZeroAt_paramSection_iff`, `Polyfold.isScFredholmSection_paramSlice`, `Polyfold.isScPlusSection_paramSlice`, `Polyfold.sectionSupport_paramSlice_subset`, `Polyfold.auxLe_paramSlice`).** For a section $g$ and sections $t_1, \dots, t_m$, the slice $g + \sum \lambda_i t_i$ of $F(\lambda, y) = g(y) + \sum \lambda_i t_i(y)$ (Theorem 3.1.21 (preprint 3.18)) is a section of $P$, and $(\lambda, y)$ is a zero of $F$ iff $y$ is a zero of the slice. For $g$ sc-Fredholm and $t_i$ sc$^+$ the slices are sc-Fredholm; slices of sc$^+$-sections are sc$^+$ with support in the union of the supports; and if $N(s_0) \le 1-\varepsilon$, $N(t_i) \le 1$ and $m|\lambda| \le \varepsilon$, then $N(s_0 + \sum \lambda_i t_i) \le 1$. *Book:* proof of Theorem 5.3.5 (preprint 5.13), preprint p. 176 ("there is a number $\delta_0 > 0$ such that $|\lambda| < \delta_0$ implies $N(\sum \lambda_j t_j(y)) < \varepsilon/2$"). *Hidden hypotheses:* none. *Deviation (not in the book):* bookkeeping the text leaves implicit.

**V5.26 ("regular value $\Leftrightarrow$ transversal" and transversality of $F$ at $(0,y)$; `Polyfold.surjective_of_forall_exists`, `Polyfold.forall_exists_ker_fst_of_surjective`, `Polyfold.isLinSurjAt_paramSlice_of_ker_fst`, `Polyfold.isLinSurjAt_paramSection_zero_of_spanning`, `Polyfold.isLinSurjAt_paramSection_zero_of_co`).** For $M(a,u) = Lu + \sum a_i e_i$ onto, the projection $\ker M \to \mathbb{R}^m$ is onto iff $L$ is onto. Consequently, if the linearization of $F$ at a zero $(\lambda, y)$ is onto and the kernel of $\mathrm{Lin}_F(0)$ in the product chart $(r \mapsto r-\lambda) \times \Phi$ projects onto $\mathbb{R}^m$, the slice $g + \sum \lambda_i t_i$ is transversal at $y$. Conversely, if the range of $\mathrm{Lin}_g(0)$ and the $\widehat{t}_i(0)$ span $K_0$ in a chart centred at the zero $y$ of $g$, $F$ is transversal at $(0,y)$, and adding sections keeps this. *Book:* proof of Theorem 5.3.5 (preprint 5.13), pp. 218–219 (preprint pp. 175–176) ("the linearization at $(0,x)$ is surjective"; "it is easily verified that $f + s_0 + s(\lambda_0, \cdot)$ is transversal"). *Hidden hypotheses:* none. *Deviation (not in the book):* only "regular value $\Rightarrow$ transversal" is needed and proved for slices; the text's "$\Leftrightarrow$" for $\tilde{S}$ is not formalized (Findings, items 64, 67).

**V5.27 (the openness steps and the omitted compactness step; `Polyfold.isWithoutBoundary_prod_rn`,**

`Polyfold.isClosed_solutionSet`, `Polyfold.exists_isOpen_isLinSurjAt_zero_near`, `Polyfold.exists_isOpen_goodPosition_of_isLinSurjAt`, `Polyfold.exists_pos_forall_mem_of_isCompact`) If $\partial X = \emptyset$ then $\partial(\mathbb{R}^m \times X) = \emptyset$; the solution set of an sc-smooth section is closed. For $\partial X = \emptyset$ and $g$ with tame germ data: if $F$ is transversal at $(0, x)$, it is transversal at $(0, y)$ for the zeros $y$ of $g$ near $x$; if $F$ is transversal at every $(0, y)$, $y \in S_g$, there is an open $V \supseteq \{0\} \times S_g$ all of whose zeros of $F$ are in good position in a tame chart. If $Z \subseteq \mathbb{R}^m \times X$ is compact and the open $V$ contains $Z \cap \{\lambda = 0\}$, then $V$ contains $Z \cap \{|\lambda| < \delta\}$ for some $\delta > 0$. *Book:* proof of Theorem 5.3.5 (preprint 5.13), pp. 218–219 (preprint pp. 175–176) ("in view of the interior case of the implicit function theorem, Theorem 3.19"; "$U \subset \{|\lambda| < \delta_0\} \oplus X$"). *Hidden hypotheses:* tame germ data, the reading of the tame strong bundle of §5.3 (Findings, item 75). *Deviation (not in the book):* the passage to $|\lambda| < \delta_0$ is the omitted step, proved with Remark 5.3.4 (preprint 5.12) (Findings, items 63, 68).

**V5.28 (the Sard step; `Polyfold.exists_null_isLinSurjAt_paramSlice_near`).** Let $g$ have tame germ data, $t_i$ be sc$^+$-sections, and $z_0$ a zero of $F(\lambda, y) = g(y) + \sum \lambda_i t_i(y)$ in good position in a tame chart. There are an open $W \ni z_0$ and a Lebesgue null set $CV \subseteq \mathbb{R}^m$ such that for every zero $(\lambda, y) \in W$ of $F$ with $\lambda \notin CV$ the slice $g + \sum \lambda_i t_i$ is transversal at $y$. *Book:* proof of Theorem 5.3.5 (preprint 5.13), preprint p. 176 ("taking a regular value $\lambda_0$ for the projection $\tilde{S} \to \mathbb{R}^m$"). *Hidden hypotheses:* none. *Deviation (not in the book):* $\tilde{S}$ is not built as a manifold; Sard's theorem is applied in its stratified form on the model of the regular tangent germ chart of Theorem 3.6.8 (2) (preprint 3.54), which avoids showing that the points over the zeros are interior (Findings, items 64, 65).

**V5.29 (Theorem 5.3.5 (preprint 5.13), generic form; `Polyfold.theorem_5_13_generic`, `Polyfold.StrongBundle.SolutionSetIsManifold`).** Let $\partial X = \emptyset$, $X$ admit sc-smooth bump functions, $f$ be an sc-Fredholm section with tame germ data and compact solution set, and $N$ an auxiliary norm. There is an open $U \supseteq S_f$ such that (1) $S_{f+s}$ is compact for every $s \in \Gamma^{+,1}_U(P)$, and (2) for every sc$^+$-section $s_0$ with $\operatorname{supp} s_0 \subseteq U$ and $N(s_0(x)) \leq 1 - \varepsilon$ ($\varepsilon > 0$) there are allowable $t_1, \dots, t_m \in \Gamma^{+,1}_U(P)$, $\delta > 0$ and a Lebesgue null set $CV \subseteq \mathbb{R}^m$ such that for every $\lambda$ with $|\lambda| < \delta$, $\lambda \notin CV$: $s_\lambda = s_0 + \sum \lambda_i t_i$ is allowable, $f + s_\lambda$ is transversal, $S_{f+s_\lambda}$ is compact and, if non-empty, a sub-M-polyfold whose induced structure is tame and that of a manifold without boundary of locally constant dimension, with regular tangent germ data at every zero of dimension the Fredholm index of $(f + s_\lambda)'(y)$. *Book:* Theorem 5.3.5 (preprint 5.13), pp. 217–218 (preprint pp. 174–175); proof pp. 218–219 (preprint pp. 175–176). *Hidden hypotheses:* $f$ with tame germ data (Definition 3.6.4 (preprint 3.51)), the reading of §5.3's tame strong bundle; the statement's "$N(f(x)) \leq 1$" is $N(s(x)) \leq 1$ and "$(f, x)$ in good position" is $(f + s, x)$. *Deviation (restated):* the dense set of the text is described as the parameters off a null set of a ball (Findings, items 60, 62, 74, 75).

**V5.30 (Theorem 5.3.5 (preprint 5.13), the text's dense form; `Polyfold.theorem_5_13`, `Polyfold.StrongBundle.allowableDist_le_iff`).** Under the hypotheses of V5.29 there is an open $U \supseteq S_f$ such that $S_{f+s}$ is compact for every $s \in \Gamma^{+,1}_U(P)$, and for every allowable $s'$ and $\varepsilon > 0$ there is an allowable $s$ with $N(s(x) - s'(x)) \leq \varepsilon$ for all $x$ such that $f + s$ is transversal and $S_{f+s}$ is compact and, if non-empty, a manifold without boundary as in V5.29. *Book:* Theorem 5.3.5 (preprint 5.13), pp. 217–218 (preprint pp. 174–175). *Hidden hypotheses:* as in V5.29. *Deviation (restated):* density in $(\Gamma^{+,1}_U(P), \rho)$ (V5.21) is the bound $\rho(s, s') \leq \varepsilon$, stated pointwise and equivalent to it; the text's "$N(s_0(y) + s(\lambda_0, y)) \leq 1 - \varepsilon/2$" is $\leq 1$ (Findings, items 60, 62).

**V5.31 (Definition 5.3.9 (preprint 5.17), in a tame chart; `Polyfold.StrongBundle.IsGeneralPositionGerm`,**

`Polyfold.StrongBundleChart.linKer`, `Polyfold.StrongBundle.IsGeneralPositionGerm.isLinSurjAt`**).** $(f, y)$ with $f(y) = 0$ is in general position if in some tame strong bundle chart around $y$, at the level-1 point $z$ over $\varphi(y)$, $\mathrm{Lin}(z) : T_zO \to K_z$ is onto, and for every presentation $T$ of the partial quadrant the kernel of $\mathrm{Lin}(z)$ has an sc-complement in $T_zO$ contained in $T_z^R O = T_zO \cap E_z$: an sc-subspace $B$ with $(T_zO)_m = (\ker \mathrm{Lin}(z))_m \oplus B_m$ on every level. General position includes transversality at $y$. *Book:* Definition 5.3.9 (preprint 5.17), p. 220 (preprint p. 177). *Hidden hypotheses:* the chart is tame (the reduced tangent space of Definition 2.5.8 (preprint 2.38) is that of a tame M-polyfold). *Deviation (restated):* stated in a chart (gap G.7 of Chapter 2); chart independence at the zeros near a zero in good position is V5.34 (Findings, items 71, 76, 82).

**V5.32 (the bridge lemma: solution sets as smooth submanifolds;** `Polyfold.isScPlus_of_isScSmooth_const`, `Polyfold.MPolyfold.TangentGermChart.exists_scPlusRetraction`, `Polyfold.MPolyfold.TangentGermChart` `Polyfold.MPolyfold.isSmoothSubmanifold_of_isRegular`, `Polyfold.StrongBundle.isSmoothSubmanifold_s`
An sc-smooth map on a finite-dimensional space with the constant structure is sc$^+$. For regular tangent germ data of Definition 3.2.1 (2) (preprint 3.25), the retraction $\tau = \Gamma \circ \pi \circ r$ of the proof of Theorem 3.2.3 (preprint 3.27) is an sc$^+$-retraction, so a set with regular tangent germ data at every point is a smooth finite-dimensional submanifold (Definition 4.2.1 (preprint 4.13)). In particular, if every zero of a section in the open $W$ is in good position in a tame chart, $S_f \cap W$ is a smooth finite-dimensional submanifold. *Book:* proof of Theorem 5.3.10 (preprint 5.18), p. 221 (preprint p. 179) ("hence we can apply Theorem 4.23"). *Hidden hypotheses:* none. *Deviation (not in the book):* the text never verifies Definition 4.2.1 (preprint 4.13) for solution sets (Findings, item 81).

**V5.33 (**$T_yS = \ker f'(y)$**;** `Polyfold.tangentSpace_subset_linKer`, `Polyfold.MPolyfold.TangentGermChart.ex` `Polyfold.HasIndepTangent.transport`, `Polyfold.exists_tangentSpace_eq_linKer`**).** In a strong bundle chart the tangent space of a retract on which the principal part vanishes lies in $\ker \mathrm{Lin}(z)$. For $f$ with tame germ data and $(f, x_0)$ in good position in a tame chart there is an open $V \ni x_0$ whose zeros are all in good position such that at every zero $y \in V$, in every strong bundle chart around $y$ and for every open $G \ni y$ in its domain with $\varphi(G \cap S_f \cap V)$ a retract, the tangent space of that retract at the point over $\varphi(y)$ equals $\ker \mathrm{Lin}(z)$. *Book:* used in the proofs of Theorem 5.3.10 (preprint 5.18) and Theorem 5.25 (preprint) ("$T_zS_\lambda$ has a sc-complement contained in $T_x^R X$"); Lemma 5.3.17 (preprint 5.24) ("$T_xS = \ker f'(x)$"). *Hidden hypotheses:* none. *Deviation (not in the book):* "$\supseteq$" is a dimension count, $d$ independent tangent vectors of the bridge retraction carried to every chart (Findings, item 82).

**V5.34 (Definition 5.3.9 (preprint 5.17) is Definition 4.2.10 (preprint 4.21) for the solution set;** `Polyfold.exists_isGeneralPositionGerm_iff`, `Polyfold.isGoodPositionGermTame_of_isGeneralPosition`
For $f$ with tame germ data and $(f, x_0)$ a zero in good position in a tame chart there is an open $V \ni x_0$ such that every zero in $V$ is in good position, $S_f \cap V$ is a smooth finite-dimensional submanifold, and at every zero $y \in V$, $(f, y)$ is in general position (Definition 5.3.9 (preprint 5.17)) iff $S_f \cap V$ is in general position at $y$ (Definition 4.2.10 (preprint 4.21)); both are then independent of the chart. If $f'(y)$ is onto and a smooth submanifold $A \subseteq S_f$ is in general position at $y$, then $(f, y)$ is in good position in a tame chart. *Book:* Definition 5.3.9 (preprint 5.17), p. 220 (preprint p. 177), and the proofs of Theorem 5.3.10 (preprint 5.18) and Theorem 5.25 (preprint). *Hidden hypotheses:* none. *Deviation (not in the book):* the chart independence of Definition 5.3.9 (preprint 5.17) holds near zeros in good position (Findings, items 82, 85).

**V5.35 (openness of general position;** `Polyfold.exists_isOpen_isGeneralPositionGerm`, `Polyfold.MPolyfold.IsGeneralPositionAt.congr_nhds`**).** Definition 4.2.10 (preprint 4.21) depends only on the germ of the submanifold. For $X$ tame, $f$ with tame germ data and $(f, x_0)$ in general position at the zero $x_0$, there is an open $W \ni x_0$ such that $S_f \cap W$ is a smooth

finite-dimensional submanifold and at every zero $y \in W$, $(f, y)$ is in good and in general position and $S_f \cap W$ is in general position at $y$. *Book:* proof of Theorem 5.3.10 (preprint 5.18), preprint p. 178 ("they will also hold at $(0, y)$ for $y \in S_f$ close to $x$"; "the properties (i)–(ii) hold for all $(\lambda, y) \in \tilde{S}$"). *Hidden hypotheses:* $X$ tame. *Deviation (not in the book):* used without proof in the text; proved through V5.34 and Theorem 4.2.11 (3) (preprint 4.22) (Findings, item 83).

**V5.36 (properties (i)–(ii) of the proof of Theorem 5.3.10 (preprint 5.18);** `Polyfold.StrongBundleChart.IsParamG` `Polyfold.StrongBundleChart.IsParamGP.comp`, `Polyfold.isGeneralPositionGerm_paramSection_of_isPara` `Polyfold.isParamGP_of_isGeneralPositionGerm`, `Polyfold.exists_sections_isParamGP`**).** In a chart centred at $y$, (i) is $\mathrm{Lin}_g(0)(T_0O) + \mathrm{span}\{\hat{t}_i(0)\} = K_0$ and (ii) is: every $w \in T_0O$ is $u + e$ with $e \in E_0$ and $\mathrm{Lin}_g(0)u + \sum a_i \hat{t}_i(0) = 0$ for some $a$. Adding sections keeps (i)–(ii). For $g$ with tame germ data and a tame chart centred at the zero $y$, (i)–(ii) hold iff $F$ is in general position at $(0, y)$ (Definition 5.3.9 (preprint 5.17)). If $X$ admits sc-smooth bump functions and $N$ is an auxiliary norm, finitely many allowable sc$^+$-sections supported in any given open $U \ni y$ achieve (i)–(ii). *Book:* proof of Theorem 5.3.10 (preprint 5.18), preprint p. 178. *Hidden hypotheses:* none. *Deviation (restated):* the complement $L$ of $T_x^R X$ is spanned by smooth vectors, so that Lemma 5.3.3 (preprint 5.11) applies; (i)–(ii) give good position of $F$ first, and then, with $\ker F' = T\tilde{S}$, general position (Findings, item 84).

**V5.37 (Theorem 5.3.10 (preprint 5.18) from a family;** `Polyfold.theorem_5_18_of_family`, `Polyfold.StrongBundle.SolutionSetInGeneralPosition`**).** Let $X$ be tame, $f$ an sc-Fredholm section with tame germ data and compact solution set, $N$ an auxiliary norm. There is an open $U \supseteq S_f$ such that $S_{f+s}$ is compact for every $s \in \Gamma_U^{+,1}(P)$, and for every sc$^+$-section $s_0$ with $\mathrm{supp}\, s_0 \subseteq U$, $N(s_0) \leq 1 - \varepsilon$, and every family of allowable $t_1, \dots, t_m$ for which $F(\lambda, y) = f(y) + s_0(y) + \sum \lambda_i t_i(y)$ is in general position at every $(0, x)$, $x \in S_{f+s_0}$, there are $\delta > 0$ and a null set $CV$ such that for $|\lambda| < \delta$, $\lambda \notin CV$, $s_\lambda = s_0 + \sum \lambda_i t_i$ is allowable and $f + s_\lambda$ is transversal, in general position at every zero, and $S_{f+s_\lambda}$ is compact, a smooth finite-dimensional submanifold in general position at each point, whose induced structure has a corner structure and is tame, with $d_{S_{f+s_\lambda}} = d_X$ on it. *Book:* Theorem 5.3.10 (preprint 5.18), p. 220 (preprint p. 177); proof pp. 220–222 (preprint pp. 178–179). *Hidden hypotheses:* as in V5.38. *Deviation (not in the book):* the theorem with the family as a hypothesis; Theorem 4.3.1 (preprint 4.23) is used in its countable-cover form (Findings, item 85).

**V5.38 (Theorem 5.3.10 (preprint 5.18), generic form;** `Polyfold.theorem_5_18`**).** Let $X$ be tame and admit sc-smooth bump functions, $f$ an sc-Fredholm section with tame germ data and compact solution set, $N$ an auxiliary norm. There is an open $U \supseteq S_f$ such that (1) $S_{f+s}$ is compact for every $s \in \Gamma_U^{+,1}(P)$, and (2) for every sc$^+$-section $s_0$ with $\mathrm{supp}\, s_0 \subseteq U$ and $N(s_0(x)) \leq 1 - \varepsilon$ there are allowable $t_1, \dots, t_m$, $\delta > 0$ and a Lebesgue null set $CV$ such that for every $\lambda$ with $|\lambda| < \delta$, $\lambda \notin CV$, $s_\lambda = s_0 + \sum \lambda_i t_i$ is allowable and $f + s_\lambda$ has all the conclusions of V5.37. *Book:* Theorem 5.3.10 (preprint 5.18), p. 220 (preprint p. 177); proof pp. 219–222 (preprint pp. 177–179). *Hidden hypotheses:* $X$ tame (the proof uses Theorems 4.2.11 and 4.3.1 (preprint 4.22 and 4.23); the opening of §5.3 places the discussion over a tame strong bundle); $f$ with tame germ data. *Deviation (restated):* the dense set is described as the parameters off a null set, as in V5.29; "replacing $f + s_0$ by $f$" is not needed (Findings, items 80, 85).

**V5.39 (Theorem 5.3.10 (preprint 5.18), the text's dense form;** `Polyfold.theorem_5_18_dense`**).** Under the hypotheses of V5.38 there is an open $U \supseteq S_f$ such that $S_{f+s}$ is compact for every $s \in \Gamma_U^{+,1}(P)$, and for every allowable $s'$ and $\varepsilon > 0$ an allowable $s$ with $N(s(x) - s'(x)) \leq \varepsilon$ for

all $x$ such that $f+s$ is transversal, $(f+s,x)$ is in general position at every $x \in S_{f+s}$, and $S_{f+s}$ is a compact smooth submanifold in general position with a tame corner structure and $d_{S_{f+s}} = d_X$. *Book:* Theorem 5.3.10 (preprint 5.18), p. 220 (preprint p. 177). *Hidden hypotheses:* as in V5.38. *Deviation (restated):* density as in V5.30 (Findings, item 80).

**V5.40 (Theorem 5.3.12 (preprint 5.19), generic form; `Polyfold.theorem_5_19`, `Polyfold.StrongBundle.IsGoodP`).** Good position depends only on the germ of the section. Let $X$ admit sc-smooth bump functions, $f$ be an sc-Fredholm section with tame germ data and compact solution set, $N$ an auxiliary norm, and $(f,x)$ in good position in a tame chart at every zero $x \in \partial X$. There is an open $U \supseteq S_f$ with $S_{f+s}$ compact for every $s \in \Gamma^{+,1}_U(P)$, and there are allowable $t_1, \dots, t_m$ with supports disjoint from $\partial X$, $\delta > 0$ and a null set $CV$ such that for $|\lambda| < \delta$, $\lambda \notin CV$, $s_\lambda = \sum \lambda_i t_i$ is allowable, vanishes on the neighbourhood $X \setminus \operatorname{supp} s_\lambda$ of $\partial X$, $f + s_\lambda$ is transversal and in good position at every zero, and $S_{f+s_\lambda}$ is compact and, if non-empty, a sub-M-polyfold whose induced structure is tame and a manifold with boundary with corners. *Book:* Theorem 5.3.12 (preprint 5.19), p. 222 (preprint p. 180). *Hidden hypotheses:* none. *Deviation (stronger):* $X$ need not be tame; "arbitrarily small" is $\lambda \to 0$; "$(f,x)$ in good position" is $(f+s,x)$. The text's "as before" would apply the implicit function theorem to $F$ at the boundary zeros, where $F$ is in general not in good position (gap G.46, on paper); the proof uses it only at interior points and keeps $f+s = f$ near the boundary zeros. No dense form (gap G.45) (Findings, items 87, 88).

**V5.41 (Definition 5.3.13 (preprint 5.20); `Polyfold.restrictSection`, `Polyfold.StrongBundle.IsGenCompactHo`r `Polyfold.StrongBundle.IsGenCompactHomotopy.isCompact_solutionSet_restrict`).** $(f,t)$ is a generalized compact homotopy if $X$ is tame, $f$ is an sc-Fredholm section with tame germ data and compact solution set, $t : X \to \mathbb{R}$ is sc-smooth with $t(X) = [0,1]$, and for every $\tau \in [0,1]$ the fibre $X_\tau = t^{-1}(\tau)$ is a sub-M-polyfold with tame induced structure on which the restriction $f|X_\tau$ — the section $a \mapsto (a, f(a))$ of the pull-back of $P$ along the inclusion — is an sc-Fredholm section with tame germ data. The restrictions then have compact solution sets. *Book:* Definition 5.3.13 (preprint 5.20), pp. 223 ff. (preprint pp. 180–181). *Hidden hypotheses:* none. *Deviation (restated):* $Y|X_\tau$ is the pull-back bundle (Proposition 2.6.8 (preprint 2.54)); the end problems are the restrictions to $t^{-1}(0)$, $t^{-1}(1)$, and the identification up to isomorphism (Remark 5.3.14 (preprint 5.21)) is not formalized (gap G.47; Findings, item 100).

**V5.42 (Lemma 5.3.16 (preprint 5.23); `Polyfold.lemma_5_23`, `Polyfold.ne_zero_of_isCompl_ker`, `Polyfold.exists_isCompl_ker_of_ne_zero`).** Let $\ell : T \to \mathbb{R}$ be linear and non-zero and $Z$ a complement of $\ker \ell$. For every linear $\tau : S \to \mathbb{R}$ there is a unique linear $b : S \to Z$ with $\ell \circ b = \tau$, and $\ell \circ (b+a) = \tau$ whenever $\ell \circ b = \tau$ and $a$ takes values in $\ker \ell$. A non-zero complement of $\ker \ell$ exists only if $\ell \neq 0$, and then it can be spanned by a vector of any spanning set. *Book:* Lemma 5.3.16 (preprint 5.23), p. 224 (preprint p. 182), and the sentence before it ("since $x$ is a smooth point, we find a one-dimensional smooth linear sc-subspace $Z$ with $T_xX = Z \oplus \ker dt(x)$"). *Hidden hypotheses:* $dt(x) \neq 0$, which Theorem 5.3.15 (preprint 5.22) does not assume. *Deviation (explicit):* linear algebra; the hidden hypothesis is exposed (Findings, item 101).

**V5.43 (the product rule at a level-$1$ point; `Polyfold.scDeriv_smul_apply`, `Polyfold.scDeriv_clm_sub_apply`, `Polyfold.chartLinCutoff`, `Polyfold.isScSmoothMap_chartLinCutoff`).** For sc-smooth $a : U \to \mathbb{R}$ and $w : U \to F$, $D(a \cdot w)(z)v = Da(z)v \cdot w(z) + a(z) \cdot Dw(z)v$; $D(\ell(\rho(\cdot) - p))(z) = \ell \circ D\rho(z)$; and $g = \beta \cdot \ell(\varphi(\cdot) - \varphi(x))$ on $\operatorname{dom}\varphi$, $0$ outside, is sc-smooth when $\beta$ is and $\operatorname{supp}\beta \subseteq \operatorname{dom}\varphi$. *Book:* proof of Lemma 5.3.17 (preprint 5.24), p. 225 (preprint pp. 182–183). *Hidden hypotheses:* none. *Deviation (not in the book):* tools for V5.44.

**V5.44 (Lemma 5.3.17 (preprint 5.24); `Polyfold.exists_scPlusSection_linAt`, `Polyfold.lemma_5_24`).**

Let $X$ admit sc-smooth bump functions, $\Phi$ a strong bundle chart around the smooth point $x$, $z$ the level-1 point over $\varphi(x)$, and $e$ a smooth point of the fibre over $x$. For every continuous linear $\ell$ on the model space and open $U \ni x$ there is an sc$^+$-section $s$ with $s(x) = 0$, $\operatorname{supp} s \subseteq U$ and $\operatorname{Lin}_s(z)v = \ell(v)\Phi(e)_2$ on $T_zO$. Hence, for a section $f$, a finite-dimensional $K \subseteq T_zO$ and a linear $b : K \to T_zO$ with $\operatorname{Lin}_f(z)(bv) \in \mathbb{R}\,\Phi(e)_2$ for all $v$, there is an sc$^+$-section $s$ with $s(x) = 0$, $\operatorname{supp} s \subseteq U$ and $\operatorname{Lin}_f(z)(bv) + \operatorname{Lin}_s(z)v = 0$ for $v \in K$. *Book:* Lemma 5.3.17 (preprint 5.24), p. 225 (preprint pp. 182–183). *Hidden hypotheses:* that $f'(x) \circ b_\tau$ has its image in the line of a smooth point $e$ is a hypothesis (the text's justification, "the image of any vector in $T_xS$ belongs to $Y_\infty$", is not formalized). *Deviation (restated):* $s'(x)$ is read in a chart, $T_xS$ as a finite-dimensional $K$; the graph of $S$ over $T_xS$ and the case distinction are not needed (Findings, item 102).

**V5.45 (Theorem 5.25 (preprint); `Polyfold.theorem_5_25`).** Let $X$ be tame and admit sc-smooth bump functions, $f$ a proper sc-Fredholm section with tame germ data, $N$ an auxiliary norm. There is an open $U \supseteq S_f$ with $S_{f+s}$ compact for every $s \in \Gamma^{+,1}_U(P)$, and an $s \in \Gamma^{+,1}_U(P)$ such that $(f+s, x)$ is in good position in a tame chart at every $x \in S_{f+s}$, and $f+s$ has the conclusions of V5.37; in particular $S_{f+s}$ has a corner structure. *Book:* Theorem 5.25 (preprint), preprint pp. 186–187; proof preprint pp. 187–188. *Hidden hypotheses:* $f$ with tame germ data. *Deviation (restated):* $U$ is the neighbourhood the proof produces (that of Theorem 5.3.10 (preprint 5.18)), not an arbitrary one with the compactness property; the text's "in view of Theorem 3.54 … in general position" is V5.35, and "$x \in S_f$" is $x \in S_{f+s_0}$ (Findings, item 86).

**V5.46 (Lemma 5.26 (preprint); `Polyfold.isInGoodPositionIn_of_scComplement_le_reduced`, `Polyfold.StrongBundle.IsGeneralPositionGerm.isGoodPositionGerm`, `Polyfold.StrongBundle.IsGeneralP`**
In a tame retract, a subspace $N$ of $T_zO$ with an algebraic complement $B \subseteq T^R_zO$ is in good position to the partial cone, with good complement $B$ and $N \cap C_z$ of non-empty interior in $N$. Hence for an sc-Fredholm section $f$, a zero $y$ with $(f, y)$ in general position (Definition 5.3.9 (preprint 5.17)) is in good position (Definition 3.6.6 (preprint 3.53)), and in the tame chart of the germ data when $f$ has tame germ data. *Book:* Lemma 5.26 (preprint), preprint p. 187; "clearly general position implies good position", p. 222 (preprint p. 179). *Hidden hypotheses:* none beyond the tame chart of Definition 5.3.9 (preprint 5.17). *Deviation (stronger):* $X$ need not be tame; the interior condition of Definition 3.1.24 (preprint 3.21), which the text omits, is proved (Findings, items 72, 76).

**V5.47 (Definition 5.5.1 (preprint 5.28); `Polyfold.MPolyfold.IsScSmoothPartitionOfUnity`, `Polyfold.MPolyfold.AdmitsScPartitionsOfUnity`).** $X$ admits sc-smooth partitions of unity if for every open cover $(U_i)_{i\in\iota}$ (with $\iota$ in the universe of $X$) there is a partition of unity $(\rho_i)$ — continuous, $\rho_i \geq 0$, locally finite supports, $\sum \rho_i = 1$ — with $\operatorname{supp}\rho_i \subseteq U_i$ and every $\rho_i$ sc-smooth. *Book:* Definition 5.5.1 (preprint 5.28), p. 229 (preprint p. 192). *Hidden hypotheses:* none. *Deviation (restated):* the index type is restricted to the universe of $X$, which loses nothing (every cover can be indexed by subsets of $X$) (Findings, item 110).

**V5.48 (sc-smooth functions: algebra, local finiteness, extension by $0$; `Polyfold.MPolyfold.IsScSmoothMap.rea`
`Polyfold.MPolyfold.IsScSmoothMap.real_sub`, `Polyfold.MPolyfold.IsScSmoothMap.real_mul`,
`Polyfold.isScSmoothMap_real_finsetSum`, `Polyfold.isScSmoothMap_real_finsetProd`,
`Polyfold.isScSmoothMap_real_finsum`, `Polyfold.isScSmoothMap_real_of_eventuallyEq`,
`Polyfold.isScSmoothMap_real_of_chart`).** Sums, differences, products, finite sums and finite products of sc-smooth functions are sc-smooth; so is a locally finite sum; a function that near every point agrees with an sc-smooth function is sc-smooth; a function supported in the domain of a chart whose representative is sc-smooth on the retract is sc-smooth on $X$. *Book:* proof of Theorem

5.5.10 (preprint 5.37), pp. 232–234 (preprint pp. 195–197). *Hidden hypotheses:* none. *Deviation (not in the book):* used and not stated (Findings, item 112).

**V5.49 (partitions of unity give bump functions; `Polyfold.MPolyfold.AdmitsScPartitionsOfUnity.admitsBum`** If $X$ admits sc-smooth partitions of unity, it admits sc-smooth bump functions (Definition 5.3.1 (preprint 5.9) in the reading of V5.16). *Book:* §5.5 (preprint §5.4.2), preprint p. 192 (the open problem whether the converse holds). *Hidden hypotheses:* none. *Deviation (not in the book):* the easy direction, which the text implies; the converse is not addressed (Findings, item 119).

**V5.50 (Definition 5.5.4 (preprint 5.31); `Polyfold.HasContDiffBump`, `Polyfold.AdmitsContDiffPartitionsOfUni`** A Banach space admits a $C^k$-bump function ($k \in \mathbb{N} \cup \{\infty\}$) if some $C^k$ function $V \to \mathbb{R}$, not identically zero, has bounded support; it admits $C^k$ partitions of unity if every open cover has a subordinate partition of unity of $C^k$ functions. *Book:* Definition 5.5.4 (preprint 5.31), p. 230 (preprint p. 193); partitions of unity of Banach spaces, preprint pp. 192, 197–198. *Hidden hypotheses:* none. *Deviation:* none (Findings, item 114).

**V5.51 (Definition 5.5.6 (preprint 5.33); `Polyfold.LocalModel.HasBumpProperty`, `Polyfold.closure_inter_bal`** A local model $(O, C, E)$ has the sc-smooth bump function property if for every $x \in O$ and open $W \ni x$ of $E_0$ with $\mathrm{cl}_E(O \cap W) \subseteq O$ there is an sc-smooth $f : O \to \mathbb{R}$ with $f(x) \neq 0$ and support (in $O$) inside $W$. A retract is closed near each of its points: $O \cap \mathrm{cl}\, B(z, r/2) \subseteq O$ when $B(z, r)$ lies in the open set of the domain of a retraction onto $O$. *Book:* Definition 5.5.6 (preprint 5.33), p. 231 (preprint p. 194). *Hidden hypotheses:* none. *Deviation (restated):* "$f \neq 0$" is read as $f(x) \neq 0$, as in V5.16 (Findings, items 41, 43).

**V5.52 (Theorem 5.5.7 (preprint 5.34); `Polyfold.theorem_5_34`).** $X$ admits sc-smooth bump functions iff some family of charts of $X$ covering $X$ has local models with the sc-smooth bump function property. *Book:* Theorem 5.5.7 (preprint 5.34), p. 231 (preprint p. 194) ("clearly"). *Hidden hypotheses:* none. *Deviation (explicit):* the extension by 0 of a local bump can fail to be continuous (a chart of the unit ball onto the whole space); the proof takes the local bump in a small ball whose preimage has closure inside the chart domain; "atlas" is read as a covering family of charts (Findings, item 43).

**V5.53 (Proposition 5.5.8 (preprint 5.35) and M-polyfolds on sc-Hilbert spaces; `Polyfold.LocalModel.hasBumpPr` `Polyfold.proposition_5_35`, `Polyfold.MPolyfold.admitsBumpFunctions_of_hilbert`).** If $\kappa\|v\|^2 \leq b(v, v)$ for a continuous bilinear $b$ on $E_0$ and some $\kappa > 0$ — in particular if $E_0$ is a Hilbert space, $\|v\|^2 = b(v, v)$ — the local model $(O, C, E)$ has the sc-smooth bump function property. An M-polyfold covered by charts on sc-Hilbert spaces admits sc-smooth bump functions. *Book:* Proposition 5.5.8 (preprint 5.35), p. 231 (preprint p. 194); §5.3, p. 216 (preprint p. 173) ("for example, the case if $X$ is built on sc-Hilbert spaces"). *Hidden hypotheses:* none. *Deviation (restated):* "Hilbert space" is a norm given by a continuous bilinear form, not a second normed structure; translation by a non-smooth point is legitimate because the bump is $C^\infty$ on $E_0$ (Findings, item 44).

**V5.54 (Definition 5.5.9 (preprint 5.36); `Polyfold.LocalModel.HasApproxProperty`).** A local model $(O, C, E)$ has the sc-smooth approximation property if for every open $W$ of $E_0$ with $\mathrm{cl}(O \cap W) \subseteq O$, every continuous $f : O \to [0, 1]$ with support (in $O$) inside $W$ and every $\varepsilon > 0$ there is an sc-smooth $g : O \to [0, 1]$ with support inside $W$ and $|f - g| \leq \varepsilon$ on $O$. *Book:* Definition 5.5.9 (preprint 5.36), p. 231 (preprint p. 195). *Hidden hypotheses:* none. *Deviation (restated):* $V = O \cap W$ and the closure is taken in $E_0$ ($C$ is closed) (Findings, item 110).

**V5.55 (Theorem 5.5.10 (preprint 5.37) "(i) $\Rightarrow$ (ii)"; `Polyfold.hasApproxProperty_restrict`, `Polyfold.MChart.isClosed_inter_preimage`).** If $\mathrm{cl}_X(s)$ lies in the domain of a chart $\varphi$, then

$\varphi^{-1}(A)$ is closed in $X$ for closed $A \subseteq \varphi(s)$. If $X$ admits sc-smooth partitions of unity, every chart restricted to an open $s$ with $\mathrm{cl}_X(s) \subseteq \operatorname{dom}\varphi$ has a local model with the sc-smooth approximation property. *Book:* proof of Theorem 5.5.10 (preprint 5.37), pp. 232–233 (preprint pp. 195–196). *Hidden hypotheses:* none. *Deviation (narrower):* the text claims the property for the charts of any atlas; its step "$\mathrm{cl}_X(\varphi^{-1}(V)) \subseteq U$" fails (the chart of the unit ball onto the whole space), so the restricted charts are used; whether every chart has the property is not settled (Findings, item 111).

**V5.56 (Theorem 5.5.10 (preprint 5.37) "(ii) ⇒ (i)"; `Polyfold.MChart.exists_small_nhds`, `Polyfold.LocalModel.HasApproxProperty.exists_approx`, `Polyfold.exists_isScSmoothMap_urysohn`, `Polyfold.admitsScPartitionsOfUnity_of_approx`).** Every point of a chart domain has arbitrarily small open neighbourhoods $N$ with $\mathrm{cl}_X(N) \subseteq \operatorname{dom}\varphi$ and $\mathrm{cl}(\varphi(N)) \subseteq O$; the approximation property transports to continuous functions supported in such $N$. If a family of charts covering $X$ has local models with the approximation property, then disjoint closed sets of $X$ are separated by sc-smooth Urysohn functions $X \to [0,1]$, and $X$ admits sc-smooth partitions of unity. *Book:* proof of Theorem 5.5.10 (preprint 5.37), pp. 232–234 (preprint pp. 196–197). *Hidden hypotheses:* none. *Deviation (restated):* the refinement "with the same index set" does not exist in general and the closure condition is not arranged in the text; the proof goes through Urysohn functions and Mathlib's bump coverings; metrizability is not used (Findings, item 112).

**V5.57 (Theorem 5.5.10 (preprint 5.37); `Polyfold.theorem_5_37`).** $X$ admits sc-smooth partitions of unity subordinate to any open cover iff some family of charts of $X$ covering $X$ has local models with the sc-smooth approximation property. *Book:* Theorem 5.5.10 (preprint 5.37), p. 232 (preprint p. 195). *Hidden hypotheses:* none. *Deviation (restated):* "atlas" is a covering family of charts; for "(i) ⇒ (ii)" it is the family of V5.55 (Findings, items 111, 112).

**V5.58 (Proposition 5.5.11 (preprint 5.38); `Polyfold.proposition_5_38`).** If $E_0$ admits $C^\infty$ partitions of unity, every local model $(O, C, E)$ has the sc-smooth approximation property. *Book:* Proposition 5.5.11 (preprint 5.38), p. 234 (preprint p. 197). *Hidden hypotheses:* "the sc-Banach space admits smooth partitions of unity" is read for $E_0$. *Deviation (explicit):* not "an immediate consequence of Theorem 5.37" (which yields some atlas, and would need partitions of unity on an open subset of $C$); the argument of "(i) ⇒ (ii)" is repeated in $E_0$ (Findings, item 113).

**V5.59 (Definition 5.5.12 (preprint 5.39) and the statement of Theorem 5.5.13 (preprint 5.40); `Polyfold.cZero`, `Polyfold.mem_cZero_iff`, `Polyfold.norm_cZero`, `Polyfold.IsCoordinatewiseContDiffEmbe`
`Polyfold.ToruncykStatement`).** $c_0(\Gamma)$ is the closed subspace of $\ell^\infty(\Gamma)$ of the functions tending to 0 along the cofinite filter, equivalently those with $\{\gamma \mid |f(\gamma)| > \varepsilon\}$ finite for every $\varepsilon > 0$, with the supremum norm; a homeomorphic embedding into $c_0(\Gamma)$ is coordinate-wise $C^k$ if every coordinate is $C^k$. Theorem 5.5.13 (preprint 5.40) is recorded as a statement, not proved and not used. *Book:* Definition 5.5.12 (preprint 5.39), Theorem 5.5.13 (preprint 5.40), p. 234 (preprint p. 198). *Hidden hypotheses:* none. *Deviation (restated):* the text's "max" norm is the supremum (attained for $f \neq 0$) (gap G.52; Findings, items 114, 117).

**V5.60 (Definition 5.5.14 (preprint 5.41); `Polyfold.IsWCG`, `Polyfold.IsReflexive`).** A Banach space is weakly compactly generated if some set compact in the weak topology has dense linear span; it is reflexive if the canonical map into the bidual is onto. *Book:* Definition 5.5.14 (preprint 5.41), p. 234 (preprint p. 198) (reflexivity is used in Proposition 5.5.15 (preprint 5.42), not defined). *Hidden hypotheses:* none. *Deviation:* none (Findings, item 114).

**V5.61 (Proposition 5.5.15 (preprint 5.42); `Polyfold.proposition_5_42_reflexive`, `Polyfold.proposition_5_4`
`Polyfold.span_closedBall_eq_top`).** (i) A reflexive Banach space is WCG; (ii) a separable Banach space is WCG. *Book:* Proposition 5.5.15 (preprint 5.42), p. 235 (preprint p. 198). *Hidden hypotheses:*

none. *Deviation:* none; Kakutani's theorem is Mathlib's Banach–Alaoglu argument, and in (ii) a dense sequence of the space, suitably scaled, replaces the text's $\{0\} \cup \{x_n/n\}$ (Findings, item 115).

**V5.62 (the statement of Theorem 5.5.16 (preprint 5.43), and Hilbert spaces; `Polyfold.GodefroyTroyanskiWhitfi` `Polyfold.hasContDiffBump_of_hilbert`, `Polyfold.isReflexive_of_hilbert`).** Theorem 5.5.16 (preprint 5.43) is recorded as a statement: a WCG space with a $C^k$-bump function admits $C^k$ partitions of unity. A Banach space whose norm comes from a continuous bilinear form has a $C^\infty$-bump function and, if complete, is reflexive. *Book:* Theorem 5.5.16 (preprint 5.43), p. 235 (preprint p. 198); proof of Corollary 5.5.17 (preprint 5.44), p. 235 (preprint p. 199) ("a Hilbert space is reflexive"). *Hidden hypotheses:* none. *Deviation:* Theorem 5.5.16 (preprint 5.43) is not proved (gap G.53) (Findings, items 116, 117).

**V5.63 (Corollary 5.5.17 (preprint 5.44); `Polyfold.corollary_5_44`, `Polyfold.MPolyfold.admitsScPartitionsO`** If $E_0$ is a Hilbert space ($\|v\|^2 = b(v,v)$) and Theorem 5.5.16 (preprint 5.43) holds for $E_0$ and $k = \infty$, the local model $(O, C, E)$ has the sc-smooth approximation property. An M-polyfold covered by charts on such spaces admits sc-smooth partitions of unity. *Book:* Corollary 5.5.17 (preprint 5.44), p. 235 (preprint p. 199). *Hidden hypotheses:* Theorem 5.5.16 (preprint 5.43) for $E_0$ (cited, not proved). *Deviation (narrower):* Theorem 5.5.16 (preprint 5.43) is an explicit hypothesis; the proof's "in view of Theorem 5.37" is Proposition 5.5.11 (preprint 5.38), and the bump Theorem 5.5.16 (preprint 5.43) needs is the classical $C^\infty$ one (Findings, item 116).

**V5.64 (Definition 5.2.1 (preprint 5.4) in the 2021 form; `Polyfold.StrongBundle.IsWeaklyProperOnU`, `Polyfold.StrongBundle.IsWeaklyProperLtInfty`, `Polyfold.StrongBundle.IsProperLtInfty`, `Polyfold.StrongBundle.isWeaklyProperOnU_iff`, `Polyfold.StrongBundle.IsAuxNorm.auxLe_fiberZero`, `Polyfold.StrongBundle.AuxLe.mono`, `Polyfold.StrongBundle.isZeroAt_of_not_mem_of_support_subset`).** Let $f$ be a section of a strong bundle. (2) $f$ is weakly proper if its solution set is compact and for every auxiliary norm $N$ there is an open $U \supseteq f^{-1}(0)$ such that $\{f = s\}$ is compact for every sc$^+$-section $s$ supported in $U$ with $N(s(y)) \le 1$ for $y \in U$; (4) $f$ is weakly proper$_{<\infty}$ if the same holds for every sc$^+$-section $s$ supported in $U$ with $N(s(y)) \le C$ for $y \in U$ and some constant $C$; (5) $f$ is proper$_{<\infty}$ if its solution set is compact and for every $N$ there is an open $U \supseteq f^{-1}(0)$ such that $\mathrm{cl}_X\{x \in U \mid N(f(x)) \le C\}$ is compact for every $C > 0$, where $N(f(x)) \le C$ includes $f(x) \in Y_{0,1}$. The form (2), with the bound on $U$ only, is equivalent to weakly proper in the sense of V5.7 (bound everywhere). *Book:* Definition 5.2.1 (preprint 5.4), p. 209 (preprint p. 171); (4) and (5) are new in the 2021 edition, (1) and (3) are those of V5.7. *Hidden hypotheses:* none. *Deviation (explicit):* "$\sup_{y\in U} N(s(y)) < \infty$" is a constant bound on $U$, and the convention $N = \infty$ off $Y_{0,1}$ is `AuxLe`, as in V5.7; the bound on $U$ is no weaker than a bound everywhere, since $s$ vanishes off $U$ and $N(0_x) = 0$ (Findings, item 170).

**V5.65 (the a priori dependencies after Definition 5.2.1 (preprint 5.4); `Polyfold.StrongBundle.IsProperLtInfty` `Polyfold.StrongBundle.IsWeaklyProperLtInfty.isWeaklyProperOnU`, `Polyfold.StrongBundle.IsProper.is` `Polyfold.StrongBundle.IsProperLtInfty.isWeaklyProperLtInfty`, `Polyfold.StrongBundle.IsWeaklyPrope`** For a section $f$: proper$_{<\infty}$ implies proper, weakly proper$_{<\infty}$ implies weakly proper, and weakly proper implies a compact solution set; if $f$ is continuous, proper implies weakly proper and proper$_{<\infty}$ implies weakly proper$_{<\infty}$. *Book:* the paragraph and diagram after Definition 5.2.1 (preprint 5.4), p. 209; the arrows from (4) and (5) are new in the 2021 edition. *Hidden hypotheses:* $f$ continuous for the two arrows from a properness to a weak properness, so that $\{f = s\}$ is closed, as in V5.7. *Deviation (explicit):* the text gives no argument; a solution of $f = s$ lies in $U$ (outside $U$, $s$ vanishes, so it would be a zero of $f$), where $N(f(x)) = N(s(x)) \le C$ (Findings, item 171).

**V5.66 (Theorem 5.2.2 (preprint 5.5) in the 2021 form; `Polyfold.theorem_5_2_2_2021`,**

`Polyfold.theorem_5_2_2_2021_tfae`**).** Let $f$ be an sc-Fredholm section of a strong bundle over an M-polyfold $X$. If $f$ has a compact solution set, $f$ is proper$_{<\infty}$; and for $f$ the five properties of V5.64 — compact solution set, weakly proper, proper, weakly proper$_{<\infty}$ and proper$_{<\infty}$ — are equivalent. *Book:* Theorem 5.2.2 (preprint 5.5), p. 210 (preprint p. 172); proof pp. 210–213. *Hidden hypotheses:* none. *Deviation:* none; as in V5.10, $X$ need not be tame and the finite cover of $f^{-1}(0)$ is explicit. Near a zero, the closure of $\{N(f) \leq C\}$ lies in the compact set of V5.9 for the bound $A \cdot C$, where $A$ compares $N$ with the chart norm, so the neighbourhood does not depend on $C$ (Findings, item 172).

**V5.67 (Theorem 5.3.5 (preprint 5.13) in the 2021 form;** `Polyfold.theorem_5_3_5_2021`**,** `Polyfold.isSubMPolyfold_solutionSet_of_manifold`**).** Under the hypotheses of V5.29 ($\partial X = \emptyset$, sc-smooth bump functions, $f$ sc-Fredholm with tame germ data and compact solution set, $N$ an auxiliary norm) there is an open $U \supseteq S_f$ such that $S_{f+s}$ is compact for every $s \in \Gamma^{+,1}_U(P)$, and for every allowable $s'$ and $\varepsilon > 0$ there is an allowable $s$ with $N(s(x) - s'(x)) \leq \varepsilon$ for all $x$ such that $f + s$ is transversal and $S_{f+s}$ is a compact sub-M-polyfold, which, if non-empty, has the structure of a manifold without boundary as in V5.29. A solution set that is such a manifold whenever it is non-empty is a sub-M-polyfold. *Book:* Theorem 5.3.5 (preprint 5.13), pp. 217–218 (preprint pp. 174–175); the 2021 edition concludes that $S_{f+s}$ is a compact sub-M-polyfold with the structure of a compact manifold without boundary. *Hidden hypotheses:* as in V5.29; the printed bound "$N(f(x)) \leq 1$ for all $x \in X$" is read as $N(s(x)) \leq 1$ (possible misprint; the proof and $\Gamma^{+,1}_U(P)$ bound $s$). *Deviation (restated):* density is pointwise, as in V5.30; the empty solution set counts as a sub-M-polyfold (Findings, items 173, 174).

**V5.68 (Definition 5.3.13 (preprint 5.20) in the 2021 form;** `Polyfold.MPolyfold.HasSurjDerivAt`**,**
`Polyfold.StrongBundle.IsCompactFredholmRestriction`**,** `Polyfold.StrongBundle.IsSubmersiveGenCompact`
`Polyfold.StrongBundle.IsGenCompactHomotopy.isSubmersiveGenCompactHomotopy`**,** `Polyfold.StrongBundle`
`Polyfold.StrongBundle.isCompactFredholmRestriction_of_eq_empty`**,** `Polyfold.isScSmoothRetractMap_of`
$(f, t)$ is a generalized compact homotopy in the 2021 sense if $X$ is tame, $f$ is an sc-Fredholm section with tame germ data and compact solution set, $t : X \to \mathbb{R}$ is sc-smooth with values in $[0, 1]$, for $i = 0, 1$ the end $t^{-1}(i)$ is a sub-M-polyfold with tame induced structure over which the restriction of $f$ (a section of the pull-back) has tame germ data, and (3) $dt(x)$ is onto at every smooth $x$: in some chart $\varphi$ of the atlas around $x$, with a retraction $R$ onto its model and a level-1 point $z$ over $\varphi(x)$, every $a \in \mathbb{R}$ is the derivative of $t \circ \varphi^{-1} \circ R$ at $z$ on some tangent vector $DR(z)v \in T_zO$. The data of V5.41 together with (3) give these data; every fibre $t^{-1}(\tau)$ has a compact solution set; and the end conditions hold over an empty end. *Book:* Definition 5.3.13 (preprint 5.20), p. 223 (preprint pp. 180–181); the 2021 edition asks $t : X \to [0, 1]$ without surjectivity, drops the conditions on the fibres $X_\tau$, $0 < \tau < 1$, and adds (3). *Hidden hypotheses:* none. *Deviation (restated):* as in V5.41, $Y|X_i$ is the pull-back and the end problems are the restrictions (Remark 5.3.14 (preprint 5.21), gap G.47); $dt(x)$ is read in a chart, since $T_xX$ carries no linear structure here (gap G.7 of Chapter 2) (Findings, items 175, 176).

**V5.69 (Lemma 5.3.16 (preprint 5.23) under the 2021 Definition 5.3.13 (3);** `Polyfold.lemma_5_3_16_2021`**).** If $dt(x)$ is onto in the sense of V5.68, then in a chart around $x$ the derivative $\ell$ of $t \circ \varphi^{-1} \circ R$ at $z$, restricted to $T_zO = DR(z)(E_0)$, is non-zero, $\ker \ell$ has a one-dimensional complement $Z$ in $T_zO$, and for every linear $\tau : S \to \mathbb{R}$ there is a unique linear $b : S \to Z$ with $\ell \circ b = \tau$, while $\ell \circ (b + a) = \tau$ whenever $\ell \circ b = \tau$ and $a$ takes values in $\ker \ell$. *Book:* Lemma 5.3.16 (preprint 5.23), p. 224 (preprint p. 182), with Definition 5.3.13 (3), p. 223. *Hidden hypotheses:* none: the hypothesis $dt(x) \neq 0$ of V5.42 now follows from (3). *Deviation (explicit):* $Z$ is an algebraic complement, spanned by a vector on which $\ell$ is non-zero; that it can be chosen smooth, as the proof of Theorem

5.3.15 (preprint 5.22) asks, is not part of the statement (Findings, items 101, 178).

**V5.70 (a generalized compact homotopy in the 2021 sense that is not one in the earlier sense; `Polyfold.GenCompactHomotopyWitness.arctanTime`, `Polyfold.GenCompactHomotopyWitness.arctanTime_mem` `Polyfold.GenCompactHomotopyWitness.hasSurjDerivAt_arctanTime`, `Polyfold.GenCompactHomotopyWitness` `Polyfold.GenCompactHomotopyWitness.arctanTime_preimage_eq_empty`, `Polyfold.GenCompactHomotopyWitn` `Polyfold.GenCompactHomotopyWitness.not_isGenCompactHomotopy_witness`).** On the trivial bundle $\mathbb{R}\times\mathbb{R}\to\mathbb{R}$, the graph section of the identity (solution set $\{0\}$) and $t(x)=1/2+\arctan(x)/\pi$ satisfy Definition 5.3.13 in the 2021 sense (V5.68): $t$ takes values in $(0,1)$, so both ends are empty, and $dt(x)=1/(\pi(1+x^2))\neq 0$ at every $x$. They are not a generalized compact homotopy in the sense of V5.41, since $t$ does not take the value $0$. *Book:* Definition 5.3.13 (preprint 5.20), p. 223; not in the book. *Hidden hypotheses:* none. *Deviation (not in the book):* satisfiability of the 2021 definition, and the 2021 data do not imply the earlier ones (Findings, item 177).

**What is not verified in this chapter**

- p. 213 (preprint p. 173), Corollary 5.2.5 (preprint 5.8) as one statement for all $i$: it is verified for each fixed $i$ by iterating the shift ($i=1,2$ stated; V5.13, V5.14); a family $P^i$ indexed by $i$ and the tame form of the conclusion (the tameness of the index raises of tame charts) are not formalized (Findings, items 145, 146). The identification of the iterated base $(X^1)^1$ with $X_2$ as a topological space, listed here before, is verified since (V2.93; gap G.6, closed).
- preprint p. 174, the non-completeness of $(\Gamma_U^{+,1}(P),\rho)$ in the text's setting: it is proved for the trivial bundle over $\mathbb{R}$, and it fails for $U=\emptyset$ and over a point (V5.21); a general sufficient condition is not formalized, and neither is the identification of the completion of Remark 5.3.4 (preprint 5.12) with the uniform $N$-limits of allowable sections (V5.22) (Findings, items 152, 153).
- preprint p. 173 and preprint p. 192, Definitions 5.3.1 and 5.5.2 (preprint 5.9 and 5.29): whether the weak reading gives a bump function non-zero at a non-smooth point (V5.16; Findings, item 163).
- p. 218 (preprint p. 175), Remark 5.3.6 (preprint 5.14), and p. 219 (preprint p. 176), Remark 5.3.8 (preprint 5.16): comments on [21] and a sketch of the homotopy argument "once we have finished our general discussion of the boundary case"; not formalized (gaps G.43, G.44).
- preprint p. 180, a dense form of Theorem 5.3.12 (preprint 5.19): the generic form gives perturbations $\sum\lambda_i t_i$ with $\lambda$ arbitrarily small (V5.40), not density around a given allowable $s'$ (gap G.45).
- preprint p. 180, Finding 88: at a zero $(0,x)$, $x\in\partial X$, where all $t_i$ vanish, the parametrized section is in good position only if $\ker f'(x)+E_x\supseteq T_xX$, a general position condition that good position does not imply; checked on paper only. The proof of Theorem 5.3.12 (preprint 5.19) avoids the point (V5.40) (gap G.46).
- p. 223 (preprint p. 181), Remark 5.3.14 (preprint 5.21) (end problems up to isomorphism): not formalized; Definition 5.3.13 (preprint 5.20) takes the restrictions as the end problems (V5.41) (gap G.47).
- p. 223 (preprint p. 181), Theorem 5.3.15 (preprint 5.22) (Morse-type structure): needs finite-dimensional Morse theory on manifolds with boundary, the cotangent bundle of the solution set, a parametric Sard theorem and a trivialization $\Phi_\lambda : S\to S_\lambda$ of the family, and the hidden hypothesis $dt(x)\neq 0$ of V5.42; Lemmas 5.3.16 and 5.3.17 (preprint 5.23 and 5.24) are verified (V5.42–V5.44) (gap G.48).
- preprint p. 188, Theorem 5.27 (preprint) (cobordism between good position perturbations):

its proof is left to the reader (gap G.49).

- p. 230 (preprint p. 193), Theorem 5.5.5 (preprint 5.32) (Bonic–Frampton, cited from [9]): not stated (gap G.50).
- pp. 230–231 (preprint pp. 193–194), the $L^p$ and Sobolev remark: the norms of $W^{k,p}(\Omega)$, the bump function of $L^p$, $W^{1,4}(\Omega)$ against $W^{1,3/2}(\Omega)$ and the question it raises; it needs Sobolev spaces and their sc-structures (gap G.1) and Theorem 5.5.5 (preprint 5.32) (gap G.51).
- preprint p. 198, Theorems 5.5.13 (preprint 5.40; Toruńczyk) and 5.5.16 (preprint 5.43; Godefroy–Troyanski–Whitfield–Zizler), cited from [9], [12]: recorded as statements (V5.59, V5.62), not proved; Theorem 5.5.16 (preprint 5.43) is an explicit hypothesis of Corollary 5.5.17 (preprint 5.44) (V5.63) (gaps G.52, G.53).
- Definitions 5.3.7 and 5.3.9 (preprint 5.15 and 5.17) in $T_xX$: verified in charts (V5.23, V5.31), with chart independence for Definition 5.3.9 (preprint 5.17) at the zeros near a zero in good position (V5.34), since $T_xX$ carries no linear structure here (gap G.7 of Chapter 2).
- p. 209, the remark after Definition 5.2.1 (2021) that for sc-smooth sections the five compactness properties are in general not equivalent: no example is formalized (Findings, item 171).
- p. 223, Theorem 5.3.15 (preprint 5.22) under the 2021 Definition 5.3.13: condition (3) supplies the hypothesis $dt(x) \neq 0$ of V5.42 (V5.69); the theorem itself is still not formalized (gap G.48; Findings, item 178).
- p. 230, Theorem 5.5.5 (preprint 5.32): the 2017 and 2021 statements print the exponent $p$ that the preprint omits in (1) and (2); the theorem is still cited (2021: from [25]) and not stated (gap G.50; Findings, item 179).

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## 7. Chapter 6 (HWZ §§6.2–6.8 (preprint §§6.1–6.7)): verified except Proposition 6.6.3 (1) (preprint 6.28), the global parts of Theorem 6.6.18 (preprint 6.41) and Remark 6.6.19 (preprint 6.42), and Theorem 6.7.1 (preprint 6.43)

Chapter 6 of Part I (pp. 237–286 (preprint pp. 200–237)) builds the orientation theory of sc-Fredholm sections. §6.2 (preprint §6.1) defines the space $\mathrm{Lin}(f,x)$ of linearizations at a smooth point (Definition 6.2.1 (preprint 6.1)) and the index; Proposition 6.2.2 (preprint 6.2), the stability of the index along paths, was verified with Chapter 3 (V3.98). §6.3 (preprint §6.2) fixes the conventions for determinant lines $\lambda(E) = \Lambda^{\max}E$: the isomorphism $\iota : \lambda(E^*) \to \lambda^*(E)$ (Definition 6.3.1 (preprint 6.3), Proposition 6.3.2 (preprint 6.4)) and the isomorphism $\Phi_E$ of a four-term exact sequence, formula (91) (Lemma 6.3.3 (preprint 6.5), Proposition 6.3.5 (preprint 6.6)). §6.4 (preprint §6.3) defines the determinant $\det(T) = \lambda(\ker T)\otimes\lambda^*(\operatorname{coker} T)$ of a Fredholm operator and its orientations (Definitions 6.4.1, 6.4.2 (preprint 6.7, 6.8)), the good left-projections $\Pi_T$ (Definition 6.4.3 (preprint 6.9), Lemma 6.10 (preprint; not in the 2021 edition)), the isomorphisms $\gamma_T^P : \det(T) \to \det(PT)$ (Proposition 6.4.10 (preprint Lemma 6.11), Proposition 6.4.11 (preprint 6.12), Corollary 6.4.12 (preprint 6.13)), the stabilization $T_\varphi$ with formula (94) (Lemma 6.4.14 (preprint 6.14), Proposition 6.4.15 (preprint 6.15)) and the direct sum convention (Definition 6.4.16 (preprint 6.16)). §6.5 (preprint §6.4) builds the local determinant bundles $\mathrm{DET}(T,P,\varepsilon)$ (Lemmas 6.5.1–6.5.3 (preprint 6.17–6.19), Proposition 6.5.4 (preprint 6.20), Lemma 6.5.5 (preprint 6.21)) and the determinant line bundle $\mathrm{DET}(E,F)$ over the Fredholm operators (Definition 6.5.6 (preprint 6.22), Propositions 6.5.7, 6.5.8 (preprint 6.23, 6.24), Lemma 6.5.9 (preprint 6.25)). §6.6 (preprint §6.5) restricts it to $\mathrm{Lin}(f,x)$ (Proposition 6.6.1 (preprint 6.26), Definition 6.6.2 (preprint 6.27)), propagates orientations along sc-smooth paths in small neighbourhoods (Proposition 6.6.3 (preprint 6.28), Lemmas 6.6.4–6.6.6 (preprint 6.29–6.31), Proposition 6.6.7 (preprint 6.32) with Lemmas 6.6.8, 6.6.10 (preprint 6.33, 6.34),

Theorem 6.6.11 (preprint 6.35), Definition 6.6.12 (preprint 6.36)), globalizes the propagation to a functor $\Gamma_f$ on the path category (Theorem 6.6.13 (preprint 6.37)), defines orientability and orientations of sc-Fredholm sections (Definitions 6.6.14–6.6.16 (preprint 6.38–6.40)), and orients the solution set (Theorem 6.6.18 (preprint 6.41)) and the sections $f + s$ over $X \times [0,1]$ (Remark 6.6.19 (preprint 6.42)). §6.7 (preprint §6.6) states the invariant $\Psi_{(f,o)}$ (Theorem 6.7.1 (preprint 6.43)). Appendix 6.8 (preprint 6.7) proves Lemma 6.3.3 (preprint 6.5) (§6.8.1 (preprint §6.7.1)) and Proposition 6.4.11 (preprint 6.12) (§6.8.2 (preprint §6.7.2)).

What is verified. **Theorems 6.6.11 and 6.6.13 (preprint 6.35 and 6.37) hold**, with the readings recorded below, and so do Propositions 6.3.2, 6.3.5, 6.4.11, 6.4.15, 6.5.4, 6.5.7, 6.5.8, 6.6.1, 6.6.7 (preprint 6.4, 6.6, 6.12, 6.15, 6.20, 6.23, 6.24, 6.26, 6.32), Proposition 6.6.3 (preprint 6.28) with (1′) in place of (1), Lemma 6.3.3, Proposition 6.4.10, Lemma 6.4.14, Lemma 6.5.1, Lemma 6.5.2, Lemma 6.5.5, Lemma 6.5.9, Lemma 6.6.6, Lemma 6.6.8, Lemma 6.6.10 (preprint Lemmas 6.5, 6.11, 6.14, 6.17, 6.18, 6.21, 6.25, 6.31, 6.33, 6.34) and Lemma 6.10 (preprint), Lemma 6.6.4 (preprint 6.29) in the form the text uses afterwards, the kernel statement of Lemma 6.6.5 (preprint 6.30), Corollary 6.4.12 (preprint 6.13), the equivalence of Definitions 6.6.14 and 6.6.15 (preprint 6.38 and 6.39), Theorem 6.6.18 (preprint 6.41) in the form of tangent orientations that vary continuously along the paths of zeros in a chart, and Remark 6.6.19 (preprint 6.42) at a point. Every definition of the chapter is formalized: Definitions 6.2.1 (preprint 6.1; V6.1), 6.3.1 (preprint 6.3; V6.3), 6.4.1 (preprint 6.7; V6.8), 6.4.2 (preprint 6.8; V6.9), 6.4.3 (preprint 6.9; V6.10), 6.4.16 (preprint 6.16; V6.19), 6.5.6 (preprint 6.22; V6.25), 6.6.2 (preprint 6.27; V6.31), 6.6.12 (preprint 6.36; V6.50), 6.6.14 (preprint 6.38; V6.58), 6.6.15 (preprint 6.39; V6.59), 6.6.16 (preprint 6.40; V6.60). Several places of the text needed more than a reading. (a) *Lemma 6.10 (preprint).* The projection of the text's proof does not lie below the two given projections, as we read it (an example on $\mathbb{R}^2$); the lemma holds by another construction, without the hypothesis that $T$ is Fredholm (V6.11). The 2017 version drops the lemma and remarks that a common lower bound need not exist; with the order of Definition 6.4.3 (preprint 6.9) it always exists, so the preprint's route to Lemma 6.5.9 (preprint 6.25) is the one formalized. (b) *Signs.* The isomorphism $\Phi_E$ of formula (94) does not vary continuously in the topology of DET where $\dim \operatorname{coker} T$ changes parity, as we read it, so the orientations it carries along a family need not be continuous (an example, V6.40); the propagation uses $(-1)^{\dim \operatorname{coker} T}\Phi_E$ at both end points (V6.41). The map $h \mapsto \tilde{h}$ of Remark 6.6.19 (preprint 6.42) has the same property, and the same sign is used there (V6.62). (c) *Contractibility.* Proposition 6.6.3 (1) (preprint 6.28), the sc-smooth contractibility of the neighbourhood, is not established by its proof as we read it; what Theorem 6.6.11 (preprint 6.35) uses is (1′), two paths with the same end points are homotopic in a larger region where (3) holds, and that is proved (V6.32, V6.48). (d) *The chart-free objects.* $T_xX$, $Y_x$, $\operatorname{Lin}(f,x)$ and $\operatorname{Or}_x^f$ are made in a chart point and identified between charts by explicit transition maps, and the local propagation is shown to be independent of the chart (V6.53, V6.54). (e) *Omitted proofs.* The orientability of a line bundle over a convex set (V6.29), the general position of Lemma 6.6.4 (preprint 6.29) (V6.33), the "simple computation" of Lemma 6.6.8 (preprint 6.33) (V6.45), the covering argument of Theorem 6.6.13 (preprint 6.37) (V6.56) and the equivalence of Definitions 6.6.14 and 6.6.15 (preprint 6.38 and 6.39), stated without proof (V6.59), are proved. Not verified: Theorem 6.7.1 (preprint 6.43) and the notion of an admissible section of §6.7 (preprint §6.6), Proposition 6.6.3 (1) (preprint 6.28), the orientation of the solution set as an abstract manifold and the perturbation paragraph after Theorem 6.6.18 (preprint 6.41), the global part of Remark 6.6.19 (preprint 6.42), the general Lemma 6.5.3 (preprint 6.19), several remarks, and the facts on paths recalled on preprint p. 201 (see the list at the end of this section). A fifth addendum (2026-11-08; Findings, items 110–115) covers the Springer 2021 edition: Lemma 6.4.4 (2017: 6.3) with its new index equalities (V6.65), Proposition 6.4.13 (2017: 6.6) without its last sentence (V6.66, V6.67), and Lemmas 6.6.8 and 6.6.9,

the two halves of preprint Lemma 6.33 (V6.68). A sixth addendum (2026-11-08; Findings, items 116–121) adds the last sentence of Proposition 6.4.13, with a closed common complement (V6.69, V6.70; gap G.81, closed), and Theorem 6.6.17, the topology of the orientation space $O_f$ (V6.71–V6.75): exactly one Hausdorff topology with properties (1) and (3), and with (2) when small open sets have connected smooth parts (gap G.90); the remarks after the theorem are gap G.91.

The setting. $X$ and $Y$ are paracompact Hausdorff spaces, as for every M-polyfold of the project, and from §6.6 (preprint §6.5) on $X$ is tame and $f$ is an sc-Fredholm section with tame germ data (Definition 3.6.4 (preprint 3.51)), the reading of Chapter 5. The linearization $(f-s)'(x) : T_xX \to Y_x$ is read in a strong bundle chart, at a *chart point* $(c,z)$ of $x$, as the operator $T_zO = Dr(z)E \to K_z = \rho(z)F$ on level 0, because $T_xX$ carries no linear structure here (gaps G.6, G.7, G.13); the text's $f-s$ with $s(x)=f(x)$ is $f+s$ with $(f+s)(x)=0$. The sc⁺-sections are global, made with sc-smooth bump functions: the text assumes sc-smooth partitions of unity from §6.3 (preprint §6.2) on (preprint p. 203), and these give bump functions (V5.49) (Findings, item 43). Paths, homotopies and families are maps on $\mathbb{R}$, $\mathbb{R}^2$, $\mathbb{R}^n$ of which only $[0,1]$, $[0,1]^2$, $[0,1]^n$ matters (decision D2, as for Proposition 6.2.2 (preprint 6.2) in V3.98), and $X \times [0,1]$ is read as $\mathbb{R}^n \times X$ with the pull-back $\pi^*P$ of the bundle along the projection. An orientation of a line is a ray (Mathlib's `Module.Ray`), and an orientation of a topological line bundle is a choice of rays given near every point by a continuous nowhere-zero section. Findings referred to below are those of `Exposition/Findings.md`, "Chapter 6".

**V6.1 (Definition 6.2.1 (preprint 6.1), the space $\mathrm{Lin}(f,x)$; `Polyfold.ChartPoint`, `Polyfold.ChartPoint.T`, `Polyfold.ChartPoint.K`, `Polyfold.ChartPoint.IsScPlusOp`, `Polyfold.ChartPoint.linSet`, `Polyfold.ChartPoint.mem_linSet_iff`, `Polyfold.ChartPoint.exists_isScPlusOp_linAt_eq`, `Polyfold.ChartPoint.linSet_nonempty`, `Polyfold.ChartPoint.isScPlusOp_sub_of_mem_linSet`, `Polyfold.ChartPoint.isFredholm_isIndexAt_of_mem_linSet`, `Polyfold.ChartPoint.exists_index_linSet`, `Polyfold.ChartPoint.fredholmIndex_of_mem_linSet`, `Polyfold.ChartPoint.convex_linSet`, `Polyfold.MPolyfold.AdmitsBumpFunctions.hasScPlusSectionsThrough`).** At a chart point $p=(c,z)$ of a smooth point $x$, $\mathrm{Lin}(f,x)$ is the set of operators $(f+s)'(x)+a : T_zO \to K_z$ for an sc⁺-section $s$ with $(f+s)(x)=0$ and an sc⁺-operator $a$; for one such $s$ it is $(f+s)'(x)$ plus the sc⁺-operators. Two such sections give linearizations that differ by an sc⁺-operator; the members of $\mathrm{Lin}(f,x)$ are Fredholm on level 0, all of the index $\mathrm{ind}(f,x)$ of V3.98; $\mathrm{Lin}(f,x)$ is convex, and it is not empty when $P$ has sc⁺-sections through the smooth points of its fibres, which bump functions give. *Book:* §6.2 (preprint §6.1), pp. 237–240 (preprint pp. 200–201) (the sections $s$ and the paragraph before the definition), Definition 6.2.1 (preprint 6.1), p. 239 (preprint p. 201), and preprint p. 219 (convexity); the index and Proposition 6.2.2 (preprint 6.2) are V3.98. *Hidden hypotheses:* an sc⁺-section through $f(x)$; the text constructs a local one in a chart, the project uses global ones (`HasScPlusSectionsThrough`), which exist under the standing bump functions. *Deviation (restated):* made in a chart point; global sections; "sc-Fredholm" is proved on level 0, which is what the determinant lines use, and not on every level (gap G.64). The paragraph before the definition compares $(f-s)'(x)$ with itself where $(f-t)'(x)$ is meant (possible misprint) (Findings, items 40, 41).

**V6.2 (the determinant lines $\lambda(E)$, $\lambda^*(E)$ and the maps $\lambda(\Phi)$; `Polyfold.maxPow`, `Polyfold.maxPowDual`, `Polyfold.finrank_maxPow`, `Polyfold.wedge`, `Polyfold.wedge_eq_det_smul`, `Polyfold.wedge_append_basis_l`, `Polyfold.wedge_append_basis_right`, `Polyfold.wedge_append_add_left`, `Polyfold.wedge_append_add_righ`, `Polyfold.dualVec`, `Polyfold.dualVec_self`, `Polyfold.eq_dualVec_of_apply_eq_one`, `Polyfold.maxPowMap`, `Polyfold.maxPowMap_wedge`, `Polyfold.maxPowMap_comp`, `Polyfold.maxPowMap_ne_zero_`, `Polyfold.maxPowDualMap`).** $\lambda(E)$ is the top exterior power of a finite-dimensional real space and

$\lambda^*(E)$ its dual, both lines; a change of basis multiplies the wedge by the determinant, also in one block of a two-block wedge, and adding to one block vectors of the span of the other does not change the wedge. For $v \neq 0$ in a line, $v^*$ is the functional with $v^*(v) = 1$. For $\dim E = \dim F$, $\lambda(\Phi)(a_1 \wedge \cdots \wedge a_n) = \Phi a_1 \wedge \cdots \wedge \Phi a_n$, with $\lambda(\Psi\Phi) = \lambda(\Psi)\lambda(\Phi)$, nonzero iff $\Phi$ is bijective; $\lambda^*(\Phi)$ is its dual. *Book:* §6.3 (preprint §6.2), pp. 241–243 (preprint pp. 203–204). *Hidden hypotheses:* none. *Deviation:* none.

**V6.3 (Definition 6.3.1 (preprint 6.3); `Polyfold.iotaDual`, `Polyfold.iotaDual_wedge_wedge`, `Polyfold.iotaDual_wedge_dualBasis`, `Polyfold.iotaDual_ne_zero`).** $\iota : \lambda(E^*) \to \lambda^*(E)$, $\iota(e_1^* \wedge \cdots \wedge e_n^*)(a_1 \wedge \cdots \wedge a_n) = \det(e_i^*(a_j))$, is a nonzero map of lines, and for a basis and its dual basis $\iota(e_1^* \wedge \cdots \wedge e_n^*) = (e_1 \wedge \cdots \wedge e_n)^*$. *Book:* Definition 6.3.1 (preprint 6.3), p. 242 (preprint p. 204). *Hidden hypotheses:* none. *Deviation:* none; the convention $\lambda(\{0\}^*) = \mathbb{R}$ is the zeroth exterior power, and the text's matrix is the transpose of Mathlib's, with the same determinant.

**V6.4 (Proposition 6.3.2 (preprint 6.4); `Polyfold.proposition_6_4`).** For an isomorphism $\Phi : E \to F$ of finite-dimensional spaces, $\lambda^*(\Phi) \circ \iota = \iota \circ \lambda(\Phi^*)$. *Book:* Proposition 6.3.2 (preprint 6.4), p. 243 (preprint p. 205). *Hidden hypotheses:* none. *Deviation:* none.

**V6.5 (four-term exact sequences and the isomorphism $\Phi_E$ of (91); `Polyfold.FourTermExact`, `Polyfold.FourTermExact.Choice`, `Polyfold.FourTermExact.Choice.value`, `Polyfold.FourTermExact.canoni` `Polyfold.FourTermExact.phi`, `Polyfold.FourTermExact.phi_ne_zero`, `Polyfold.FourTermExact.phiEquiv`).** For an exact sequence $0 \to A \to B \to C \to D \to 0$ of finite-dimensional spaces, a *choice* is the data of (91): complements $Z$ of $\alpha(A)$ in $B$ and $V$ of $\beta(B)$ in $C$, bases of $A$, $D$, $Z$, and the $\gamma$-preimages in $V$ of the basis of $D$; $\Phi_E : \lambda(A) \otimes \lambda^*(D) \to \lambda(B) \otimes \lambda^*(C)$ is the linear map given by (91) for one choice, an isomorphism of lines. *Book:* §6.3 (preprint §6.2), preprint p. 206, formula (91) with its two ordering conventions. *Hidden hypotheses:* none. *Deviation:* none.

**V6.6 (Lemma 6.3.3 (preprint 6.5); `Polyfold.FourTermExact.lemma_6_5`).** Formula (91) holds for every choice: $\Phi_E$, defined from one choice, maps $(a_1 \wedge \cdots \wedge a_n) \otimes (d_1 \wedge \cdots \wedge d_l)^*$ to the right-hand side of (91) computed with any other choice. *Book:* Lemma 6.3.3 (preprint 6.5), p. 244 (preprint p. 206); proof in §6.8.1 (preprint §6.7.1) (V6.63). *Hidden hypotheses:* none. *Deviation (restated):* "does not depend on the choices" is read as this, the strongest form and the one every later computation of $\gamma_T^P$ uses (Findings, item 1).

**V6.7 (Proposition 6.3.5 (preprint 6.6); `Polyfold.contractDual`, `Polyfold.FourTermExact.psi`, `Polyfold.FourTermExact.proposition_6_6`).** With $\bar{\iota}(v \otimes b \otimes \varphi) = \varphi(v)\, b$ and $\Psi_E(c \otimes h) = \bar{\iota}(c \otimes \Phi_E(h))$, $\Psi_E$ maps the wedge of the basis of $C$ of (91), tensored with $(a_1 \wedge \cdots \wedge a_n) \otimes (d_1 \wedge \cdots \wedge d_l)^*$, to $\alpha(a_1) \wedge \cdots \wedge \alpha(a_n) \wedge b_1' \wedge \cdots \wedge b_{m-n}'$. *Book:* (92) and Proposition 6.3.5 (preprint 6.6), p. 246 (preprint p. 207). *Hidden hypotheses:* none. *Deviation (restated):* the wedge in $C$ is taken in the order of (91); the statement exchanges the names of its two blocks (possible misprint) (Findings, item 2).

**V6.8 (Definition 6.4.1 (preprint 6.7); `Polyfold.linDet`, `Polyfold.fredholmDet`, `Polyfold.finrank_fredholmDet` `Polyfold.linDetEquivOfEq`).** $\det(T) = \lambda(\ker T) \otimes \lambda^*(F/R(T))$, a line for $T$ Fredholm; equal operators have identified determinants. *Book:* Definition 6.4.1 (preprint 6.7), p. 246 (preprint p. 208). *Hidden hypotheses:* none. *Deviation:* none; the alternative definition $\lambda(\ker T) \otimes \lambda(\operatorname{coker}(T)^*)$ mentioned after it is not formalized (gap G.57).

**V6.9 (Definition 6.4.2 (preprint 6.8); `Polyfold.FredholmOrientation`, `Polyfold.nonempty_fredholmOrientati` `Polyfold.exists_two_fredholmOrientation`).** An orientation of $T$ is a ray of $\det(T)$; a Fredholm operator has exactly two. *Book:* Definition 6.4.2 (preprint 6.8), p. 246 (preprint p. 208). *Hidden hypotheses:* none. *Deviation:* none.

**V6.10 (Definition 6.4.3 (preprint 6.9) and the facts after it; `Polyfold.IsGoodLeftProj`, `Polyfold.goodLeftProjs`, `Polyfold.ProjLE`, `Polyfold.ProjLE.refl`, `Polyfold.ProjLE.antisymm`, `Polyfold.ProjLE.trans`, `Polyfold.IsGoodLeftProj.isFredholm`, `Polyfold.IsGoodLeftProj.fredholmIndex_`, `Polyfold.IsGoodLeftProj.isFredholm_comp`, `Polyfold.IsGoodLeftProj.fredholmIndex_comp`, `Polyfold.IsGoodLeftProj.finiteDimensional_quotient_inf`, `Polyfold.exists_isGoodLeftProj`).** A good left-projection for $T$ is a bounded projection $P$ with $\dim F/R(P) < \infty$ and $R(PT) = R(P)$; $P \le Q$ iff $P = PQ = QP$ is a partial order. A good left-projection is Fredholm of index 0, $PT$ is Fredholm with $\mathrm{ind}(PT) = \mathrm{ind}(T)$, $R(P) \cap R(P')$ has finite codimension, and $\Pi_T$ is not empty for $T$ Fredholm. *Book:* Definition 6.4.3 (preprint 6.9) and the paragraphs after it, preprint p. 208. *Hidden hypotheses:* none; that $\Pi_T \neq \emptyset$ is used from Proposition 6.4.10 (preprint Lemma 6.11) on and is not stated. *Deviation:* none (Findings, item 3).

**V6.11 (Lemma 6.10 (preprint); `Polyfold.lemma_6_10`, `Polyfold.lemma_6_10_text_reading`).** For $P, P' \in \Pi_T$ there is $P'' \in \Pi_T$ with $P'' \le P$ and $P'' \le P'$, for any bounded $T$. On $\mathbb{R}^2$ with $T = 1$ there are $P, P' \in \Pi_T$ such that no projection with range $R(P) \cap R(P')$ lies below both. *Book:* Lemma 6.10 (preprint), preprint p. 208, proof preprint pp. 208–209. *Hidden hypotheses:* none. *Deviation (stronger):* $T$ need not be Fredholm, and the construction differs from the text's: the text's $P''$ has range $R(P) \cap R(P')$, and as we read it $P'' = P''P$ does not hold for it in general; the formalized $P''$ projects onto $R(P) \cap R(P') \cap N$, for a closed finite-codimensional $N$ meeting $\ker P + \ker P'$ only in 0, along a complement containing $\ker P + \ker P'$ (Findings, item 4).

**V6.12 (Proposition 6.4.10 (preprint Lemma 6.11); `Polyfold.projSeqJ`, `Polyfold.projSeqPhi`, `Polyfold.projSeqPi`, `Polyfold.lemma_6_11`, `Polyfold.projSeq`).** For $P \in \Pi_T$ the sequence $0 \to \ker T \to \ker(PT) \to F/R(P) \to \mathrm{coker}\, T \to 0$, with the inclusion, $x \mapsto Tx + R(P)$ and $y + R(P) \mapsto (I - P)y + R(T)$, is exact. *Book:* Proposition 6.4.10 (preprint Lemma 6.11), p. 249 (preprint p. 209). *Hidden hypotheses:* none. *Deviation:* none.

**V6.13 (the isomorphism $\gamma_T^P$; `Polyfold.gammaProj`, `Polyfold.gammaProj_tmul`, `Polyfold.gammaProj_wedge_tmu`).** $\gamma_T^P : \det(T) \to \det(PT)$ is $\Phi_{E(T,P)}$ followed by the identification of $\lambda^*(F/R(P))$ with $\lambda^*(F/R(PT))$; on wedges it is given by (91), for any choice. *Book:* preprint p. 210. *Hidden hypotheses:* none. *Deviation:* none; "recalling $R(P) = R(PT)$" is read as this identification (Findings, item 5).

**V6.14 (Proposition 6.4.11 (preprint 6.12); `Polyfold.proposition_6_12`, `Polyfold.IsGoodLeftProj.of_projLE`, `Polyfold.comp_comp_eq_of_projLE`).** For $P \le Q$ in $\Pi_T$, $QT$ is Fredholm, $P \in \Pi_{QT}$, and $\gamma_{QT}^P \circ \gamma_T^Q = \gamma_T^P$, up to the identification $\det(P(QT)) = \det(PT)$ of the equal operators $PQT = PT$. *Book:* Proposition 6.4.11 (preprint 6.12), p. 251 (preprint p. 211); proof in §6.8.2 (preprint §6.7.2) (V6.64). *Hidden hypotheses:* none. *Deviation:* none.

**V6.15 (Corollary 6.4.12 (preprint 6.13); `Polyfold.corollary_6_13`).** For $P \le S$ and $P \le R$ in $\Pi_T$, $(\gamma_{RT}^P)^{-1} \circ \gamma_{ST}^P = \gamma_T^R \circ (\gamma_T^S)^{-1}$, so the left side does not depend on $P$. *Book:* Corollary 6.4.12 (preprint 6.13), p. 251 (preprint p. 211). *Hidden hypotheses:* none. *Deviation:* none.

**V6.16 (the stabilization and Lemma 6.4.14 (preprint 6.14); `Polyfold.stabJ`, `Polyfold.stabPi`, `Polyfold.stabClass`, `Polyfold.lemma_6_14`, `Polyfold.stabSeq`, `Polyfold.exists_surjective_coprod`).** For $T$ Fredholm and $\varphi : \mathbb{R}^n \to F$ with $T_\varphi(e, r) = Te + \varphi r$ onto, the sequence $0 \to \ker T \to \ker T_\varphi \to \mathbb{R}^n \to F/R(T) \to 0$, with $j(e) = (e, 0)$, $\pi(e, r) = r$ and $[\varphi](r) = \varphi(r) + R(T)$, is exact; a stabilizing $\varphi$ exists for every Fredholm $T$. *Book:* pp. 251–256 (preprint pp. 211–212) and Lemma 6.4.14 (preprint 6.14), p. 256 (preprint p. 212). *Hidden hypotheses:* none. *Deviation:* none; the proof writes $j(e) = (0, e)$ for $(e, 0)$ (possible misprint) (Findings, item 8).

**V6.17 (formula (94); `Polyfold.stdWedge`, `Polyfold.stabOne`, `Polyfold.stabPsi`, `Polyfold.stabPsi_dualVec`).** $\psi : \lambda^*(\mathbb{R}^n) \to \lambda^*(\{0\})$, $(e_1 \wedge \cdots \wedge e_n)^* \mapsto 1^*$ for the standard basis, satisfies $\psi((c_1 \wedge \cdots \wedge c_n)^*) = \det[c_1, \dots, c_n]^{-1}\, 1^*$ for every basis. *Book:* formula (94), p. 256 (preprint p. 212). *Hidden hypotheses:* none. *Deviation:* none.

**V6.18 (Proposition 6.4.15 (preprint 6.15); `Polyfold.stabIota`, `Polyfold.stabIota_bijective`, `Polyfold.stab_card_eq`, `Polyfold.isCompl_range_stabPi_span`, `Polyfold.proposition_6_15`).** $\iota_\varphi = (1 \otimes \psi) \circ \Phi_E : \det(T) \to \det(T_\varphi)$ is an isomorphism, and for a basis $a$ of $\ker T$, $\bar{c}_1, \dots, \bar{c}_m$ with $[\varphi](\bar{c}_i)$ a basis of $F/R(T)$ and $\bar{b}_i = (b_i, r_i)$ completing $(a, 0)$ to a basis of $\ker T_\varphi$, $\iota_\varphi(h) = \det(\bar{c}, r)^{-1}\, ((a_1, 0) \wedge \cdots \wedge (a_k, 0) \wedge \bar{b}_1 \wedge \cdots \wedge \bar{b}_l) \otimes 1^*$; the $r_i$ and $\bar{c}_j$ form a basis of $\mathbb{R}^n$. *Book:* Proposition 6.4.15 (preprint 6.15), p. 257 (preprint p. 213). *Hidden hypotheses:* none. *Deviation (restated):* the identity $m + l = n$, needed to form $\det(\bar{c}, r)$, is a hypothesis of the statement and is proved to hold always; that the $r_i$ and $\bar{c}_j$ form a basis, which the text notices, is proved; the statement calls $\psi$ "$\varphi$" (possible misprint). The identification (95) and the maps $V \otimes \lambda(\mathbb{R}^n) \to V$, $V \otimes \lambda^*(\mathbb{R}^n) \to V$ after the proposition are not formalized (gap G.58) (Findings, items 9, 10).

**V6.19 (Definition 6.4.16 (preprint 6.16); `Polyfold.kerInl`, `Polyfold.kerInr`, `Polyfold.cokerInl`, `Polyfold.cokerInr`, `Polyfold.kerProdEquiv`, `Polyfold.cokerProdEquiv`, `Polyfold.finrank_ker_prodMap`, `Polyfold.finrank_coker_prodMap`, `Polyfold.isFredholm_prodMap`, `Polyfold.directSumVec`, `Polyfold.directSumDet`, `Polyfold.directSumVec_eq_smul`, `Polyfold.definition_6_16`, `Polyfold.directSumDet_bijective`).** $T \oplus S$ is Fredholm, with $\ker(T \oplus S) \cong \ker T \times \ker S$ and $\operatorname{coker}(T \oplus S) \cong \operatorname{coker} T \times \operatorname{coker} S$, and $\det(T) \otimes \det(S) \to \det(T \oplus S)$, $h \otimes h' \mapsto (a \wedge a') \otimes (d \wedge d')^*$, is an isomorphism satisfying the formula for every choice of bases. *Book:* Definition 6.4.16 (preprint 6.16), p. 258 (preprint p. 214). *Hidden hypotheses:* none. *Deviation:* none; "natural" is read as the formula for every choice, as in Lemma 6.3.3 (preprint 6.5) (Findings, item 11).

**V6.20 (Lemma 6.5.1 (preprint 6.17); `Polyfold.FredholmOp`, `Polyfold.projRestrict`, `Polyfold.lemma_6_17`).** For a bounded $T$, $P \in \Pi_T$ and a topological complement $Y$ of $\ker(PT)$, there is $\varepsilon > 0$ such that for $\|S - T\| < \varepsilon$, $P \in \Pi_S$ and $PS|_Y : Y \to R(P)$ is a topological isomorphism. *Book:* Lemma 6.5.1 (preprint 6.17), p. 258 (preprint p. 214). *Hidden hypotheses:* none. *Deviation (stronger):* the hypothesis that $T$ is Fredholm is not used; a topological complement is an algebraic complement with a continuous projection (Findings, item 20).

**V6.21 (Lemma 6.5.2 (preprint 6.18); `Polyfold.LocalDetChart`, `Polyfold.LocalDetChart.ySol`, `Polyfold.LocalDetChart.kerMap`, `Polyfold.LocalDetChart.kerProj`, `Polyfold.LocalDetChart.lemma_6_18`).** A *local determinant chart* is the data $(T, P, Y, \varepsilon)$ of Lemma 6.5.1 (preprint 6.17) with its conclusions. The solution $y(k, S) = -(PS|_Y)^{-1} PSk$ is continuous in $(k, S)$, and $(S, k) \mapsto (S, k + y(k, S))$ is a continuous map onto $\bigcup_{\|S-T\|<\varepsilon} \{S\} \times \ker(PS)$, inverted by the continuous projection onto $\ker(PT)$ along $Y$. *Book:* Lemma 6.5.2 (preprint 6.18), p. 259 (preprint p. 215). *Hidden hypotheses:* none. *Deviation (restated):* "is a trivial vector bundle" is stated as this trivialization and its inverse (Findings, item 21).

**V6.22 (Lemma 6.5.3 (preprint 6.19) in the case used, and Proposition 6.5.4 (preprint 6.20); `Polyfold.LocalDetChart.kerFrame`, `Polyfold.LocalDetChart.frame`, `Polyfold.LocalDetChart.frame_ne_ze` `Polyfold.proposition_6_20`).** Over the ball of a local determinant chart, $\omega(S) = (k_1(S) \wedge \cdots \wedge k_n(S)) \otimes (e_1 \wedge \cdots \wedge e_l)^*$, with $k_i(S) = k_i + y(k_i, S)$ and a constant basis $e$ of $F/R(P) = F/R(PS)$, is a nowhere-zero frame of $\mathrm{DET}(T, P, \varepsilon)$, and the frames of two charts with the same $P$ differ by a continuous nowhere-zero factor. *Book:* Lemma 6.5.3 (preprint 6.19) and Proposition 6.5.4 (preprint 6.20), p. 260 (preprint p. 216). *Hidden hypotheses:* none. *Deviation (narrower):* Lemma 6.5.3

(preprint 6.19) (exterior powers, duals, sums and tensor products of topological vector bundles) is not formalized in general (gap G.59); it is applied only to trivial bundles, and Proposition 6.5.4 (preprint 6.20) is proved with the explicit frames; "in a natural way" is read as the relation between frames (Findings, item 22).

**V6.23 (Lemma 6.5.5 (preprint 6.21); `Polyfold.LocalDetChart.gammaProj_frame_eq`, `Polyfold.lemma_6_21`).** For $Q \le P$, $\gamma^Q_{PS}$ maps the $P$-frame at $S$ to a continuous nowhere-zero multiple of the $Q$-frame, on the common ball of any two local determinant charts. *Book:* Lemma 6.5.5 (preprint 6.21), p. 260 (preprint p. 216), sketch pp. 259–261 (preprint pp. 216–217). *Hidden hypotheses:* none. *Deviation (stronger):* two charts with different centres and radii, the form Lemma 6.5.9 (preprint 6.25) uses; the text's sketch (extend the kernel sections of $PS$ to sections spanning $\ker(QS)$) is the proof (Findings, item 23).

**V6.24 ($\mathrm{DET}(E,F)$ and the charts $\gamma_{(T,P,\varepsilon)}$; `Polyfold.DetFiber`, `Polyfold.LocalDetChart.pretriv`, `Polyfold.LocalDetChart.coord`, `Polyfold.exists_localDetChart`, `Polyfold.detPrebundle`).** The fibre over a Fredholm $S$ is $\det(S)$; a local determinant chart gives the pretrivialization $(S,h) \mapsto (S, \omega(S)^*(\gamma^P_S h))$ with model fibre $\mathbb{R}$; every Fredholm operator is the centre of one; together they form a Mathlib vector prebundle, whose topology and bundle structure are those of $\mathrm{DET}(E,F)$. *Book:* preprint p. 217. *Hidden hypotheses:* none. *Deviation (restated):* $\gamma_{(T,P,\varepsilon)}$ is read in the frame of Proposition 6.5.4 (preprint 6.20) (Findings, item 24).

**V6.25 (Definition 6.5.6 (preprint 6.22); `Polyfold.detBasis`).** $\mathcal{B}$ is the collection of sets lying over the ball of some chart whose image in ball $\times\,\mathbb{R}$ is open. *Book:* Definition 6.5.6 (preprint 6.22), p. 261 (preprint p. 218). *Hidden hypotheses:* none. *Deviation:* none.

**V6.26 (Proposition 6.5.7 (preprint 6.23); `Polyfold.proposition_6_23`).** $\mathcal{B}$ is a basis of the topology of the prebundle, so the topology it generates is that one. *Book:* Proposition 6.5.7 (preprint 6.23), p. 261 (preprint p. 218). *Hidden hypotheses:* none. *Deviation:* none.

**V6.27 (Proposition 6.5.8 (preprint 6.24); `Polyfold.proposition_6_24`).** $\mathrm{DET}(E,F)$ is a topological vector bundle with model fibre $\mathbb{R}$, and every $\gamma_{(T,P,\varepsilon)}$, read in its frame, is a trivialization. *Book:* Proposition 6.5.8 (preprint 6.24), p. 261 (preprint p. 218). *Hidden hypotheses:* none. *Deviation:* none.

**V6.28 (Lemma 6.5.9 (preprint 6.25); `Polyfold.lemma_6_25`, `Polyfold.lemma_6_25_local`).** The transition between two charts is $(S,t) \mapsto (S, g(S)t)$ with $g$ continuous and nowhere zero on the common ball. *Book:* Lemma 6.5.9 (preprint 6.25), p. 262 (preprint p. 218). *Hidden hypotheses:* none. *Deviation:* none; the proof is the text's, a common lower bound $Q \le P, P'$ (Lemma 6.10 (preprint), V6.11), Corollary 6.4.12 (preprint 6.13) and Lemma 6.5.5 (preprint 6.21).

**V6.29 (orientations of line bundles; `Polyfold.IsLineOrientation`, `Polyfold.LineOrientation`, `Polyfold.LineOrientation.eq_of_apply_eq`, `Polyfold.OrCover`, `Polyfold.isCoveringMap_orCover`, `Polyfold.exists_lineOrientation_apply_eq`, `Polyfold.exists_two_lineOrientation`).** For a topological line bundle, two orientations that agree on one fibre agree over a preconnected base; over a simply connected, locally path-connected base every orientation of one fibre extends, and over a nonempty such base there are exactly two orientations; the rays of the fibres form a two-sheeted covering. *Book:* the reason given in Proposition 6.6.1 (preprint 6.26) ("since the base space is contractible", p. 263 (preprint p. 219)) and the opening of §6.6 (preprint §6.5), p. 263 (preprint p. 219). *Hidden hypotheses:* none. *Deviation (not in the book):* the text cites contractibility without the theorem it needs; the proof is the lifting criterion for the orientation double cover, and nonempty convex sets of normed spaces are simply connected and locally path-connected (Findings, item 25).

**V6.30 (Proposition 6.6.1 (preprint 6.26); `Polyfold.fredholmIncl`, `Polyfold.DetOn`, `Polyfold.proposition_6_2`**
Over a nonempty convex set $C$ of Fredholm operators, the restriction $\mathrm{DET}(E,F)|_C$ is a topological line bundle with exactly two orientations. *Book:* Proposition 6.6.1 (preprint 6.26), p. 263 (preprint p. 219), for $C = \mathrm{Lin}(f,x)$. *Hidden hypotheses:* none. *Deviation (stronger):* any nonempty convex $C$; $\mathrm{Lin}(f,x)$ is one (V6.1) (Findings, item 25).

**V6.31 (Definition 6.6.2 (preprint 6.27); `Polyfold.ChartPoint.isFredholm_of_mem_linSet`, `Polyfold.ChartPoint.detLin`, `Polyfold.ChartPoint.Or`, `Polyfold.ChartPoint.exists_two_or`, `Polyfold.ChartPoint.orEquivRay`, `Polyfold.nonempty_equiv_of_two`, `Polyfold.equiv_eq_of_apply_eq_of_`**
At a chart point, $\mathrm{DET}(f,x)$ is the pull-back of $\mathrm{DET}(T_zO, K_z)$ to $\mathrm{Lin}(f,x)$ and $\mathrm{Or}_x^f$ is the set of its orientations, with exactly two elements; for each $L \in \mathrm{Lin}(f,x)$, $o \mapsto o(L)$ is a bijection onto the two orientations of $\det(L)$. Between two two-element sets there are bijections, and one value determines a bijection. *Book:* Definition 6.6.2 (preprint 6.27), p. 263 (preprint pp. 219–220). *Hidden hypotheses:* none. *Deviation (restated):* made in a chart point (V6.53 compares charts); the category $\mathcal{O}_f$ is read through its objects, with bijections as morphisms, and $\{o_x, -o_x\}$ as "exactly two elements" (Findings, item 44).

**V6.32 (Proposition 6.6.3 (preprint 6.28); `Polyfold.IsScHomotopyRel`, `Polyfold.pt1`, `Polyfold.pt2`, `Polyfold.exists_contDiff_eq_id_near_Icc_mapsTo`, `Polyfold.exists_isScSmoothMap_eq_one_nhdsSet`, `Polyfold.exists_nhds_isCompact_zeros`, `Polyfold.exists_isScPlusSection_pullback_of_mapsTo_of_bump` `Polyfold.exists_isScPlusSection_pullback_of_family`, `Polyfold.hasScPlusSectionsThrough_pullback_` `Polyfold.proposition_6_28`, `Polyfold.MPolyfold.paracompactSpace_prod`, `Polyfold.paracompactSpace_pu`**
With sc-smooth bump functions, every smooth $x$ has open $U \subseteq W$ in the domain of one chart of the atlas with (1′) any two sc-smooth paths in $U$ with the same end points are sc-smoothly homotopic in $W$ with the end points fixed; (2) $\{y \in \mathrm{cl}(U) \mid f(y) = 0\}$ is compact; (3) for every sc-smooth family $\Phi : \mathbb{R}^n \to X$ with $\Phi([0,1]^n) \subseteq W$ there is an sc$^+$-section $S$ of $\pi^*P \to \mathbb{R}^n \times X$ with $(f+S)(q,\Phi(q)) = 0$ for $q$ near $[0,1]^n$, depending on $q \in [0,1]^n$ only through $\Phi(q)$; (4) $\pi^*P$ has sc$^+$-sections through all smooth fibre points. *Book:* Proposition 6.6.3 (preprint 6.28), p. 263 (preprint p. 220), proof pp. 263–264 (preprint pp. 220–221). *Hidden hypotheses:* bump functions, for the global sections of (3) and (4) (the text assumes partitions of unity from §6.3 (preprint §6.2) on and works in a chart after "assuming $X = U$"); the paracompactness of $\mathbb{R}^n \times X$ and of $\pi^*P$, which holds. *Deviation (narrower):* (1′) in place of (1): the contraction $r(ty)$ of the proof contracts $r(V)$ for a convex $V$, but for $U = r(V')$ it needs $r(V') \subseteq V'$, which is not stated, and as we read it (1) is not established for $U$ (gap G.61); (3) is stated for families, as Theorem 6.6.11 (preprint 6.35) needs it for homotopies; the smoothness of $x$ is not used (Findings, items 45, 46).

**V6.33 (Lemma 6.6.4 (preprint 6.29); `Polyfold.isZeroAt_paramSection_zero`, `Polyfold.exists_sections_isGe` `Polyfold.lemma_6_29`).** Over a tame $X$ with bump functions, for $f$ with tame germ data, an sc-smooth $\Phi : \mathbb{R}^n \to X$ and an sc$^+$-section $S$ of $\pi^*P$ with $(f+S)(q,\Phi(q)) = 0$ near $[0,1]^n$, there are finitely many sc$^+$-sections $s_1, \dots, s_k$ of $\pi^*P$ such that $F(\lambda,(q,y)) = f(y) + S(q,y) + \sum \lambda_i s_i(q,y)$ is in general position at $(0,(q,\Phi(q)))$ for all $q$ near $[0,1]^n$. *Book:* the section $F$, preprint p. 221, Lemma 6.6.4 (preprint 6.29) and the paragraph after it, preprint p. 222. *Hidden hypotheses:* $X$ tame, bump functions, tame germ data (the standing hypotheses of §6.6 (preprint §6.5) and of Theorem 5.3.10 (preprint 5.18)). *Deviation (narrower):* the text gives no proof ("using the results from the transversality theory"); the one formalized is that of Theorem 5.3.10 (preprint 5.18). The conclusion is the general position of $F$ at the points $(t,\varphi(t),0)$, which the text derives from (P) and which Lemmas 6.6.5, 6.6.6 (preprint 6.30, 6.31) use; (P) itself, the general position of each slice $F_t$, is not stated (gap G.63) (Findings, item 47).

**V6.34 (the kernels of Lemma 6.6.5 (preprint 6.30); `Polyfold.StrongBundle.IsSurjLinOfIndex.finrank_linKer` `Polyfold.exists_local_frame_linKer`, `Polyfold.exists_subspaceFamily_linKer`, `Polyfold.hasFDerivAt_s` `Polyfold.fderiv_mem_linKer`).** For an sc-Fredholm section with tame germ data and a continuous family $y_q$ of zeros in good position, read in one chart at continuously varying level-1 points, the kernels of the linearizations form a continuous family of subspaces of the model space, of dimension the index; the derivative of an sc-smooth family of zeros $\mathbb{R}^n \to X$, read in a chart, lies in the kernel of the linearization. *Book:* Lemma 6.6.5 (preprint 6.30), p. 265 (preprint p. 222). *Hidden hypotheses:* none beyond the standing ones. *Deviation (narrower):* the bundle $L$ is a continuous family of subspaces; that the solution set is a manifold with boundary with corners near $(t, \varphi(t), 0)$, the map $\pi$, and the smooth structure of $L$ are not formalized (gap G.62) (Findings, item 48).

**V6.35 (continuous families of subspaces and their determinant lines; `Polyfold.SubspaceFamily`, `Polyfold.SubspaceFamily.det`, `Polyfold.SubspaceFamily.Frame.pretriv`, `Polyfold.SubspaceFamily.Frame` `Polyfold.SubspaceFamily.comp`, `Polyfold.LineOrientation.comp`, `Polyfold.SubspaceFamily.transport`, `Polyfold.SubspaceFamily.transport_eq`, `Polyfold.SubspaceFamily.transport_injective`, `Polyfold.SubspaceFamily.transport_comp`).** A family of $n$-dimensional subspaces of a normed space, spanned near every point by a continuous frame, has the line bundle $b \mapsto \lambda(L_b)$ with the frames as trivializations; families and orientations restrict along continuous maps, and over $[0,1]$ an orientation of $\lambda(L_0)$ is continued to $\lambda(L_1)$, injectively, and over a simply connected, locally path-connected base the continuation along a path depends only on its end points. *Book:* pp. 264–266 (preprint pp. 222–223) (the bundles $L$, $\lambda(L)$ and "an orientation of the lines … determines by continuation an orientation of all the other lines"), and the bundles $\tilde{L}$ of pp. 266–273 (preprint pp. 224–227). *Hidden hypotheses:* none. *Deviation (restated):* "vector bundle" is read as a family of subspaces of one model space with continuous local frames.

**V6.36 (continuous families of kernels and their images; `Polyfold.exists_subspaceFamily_ker`, `Polyfold.SubspaceFamily.exists_map`, `Polyfold.SubspaceFamily.isLineOrientation_map`, `Polyfold.SubspaceFamily.exists_inf_ker`).** The kernels of a norm-continuous family of surjective Fredholm operators over a connected base form a continuous family of subspaces; so do the images of a continuous family under a jointly continuous family of linear maps injective on its members, and orientations are carried along; cutting a continuous family by the kernel of a fixed $\ell$ onto on every member gives one. *Book:* the continuity statements of pp. 265–272 (preprint pp. 223–226) ("as $\tau$ varies, the bundle varies continuously"). *Hidden hypotheses:* none. *Deviation (not in the book):* "varies continuously" is read in the operator norm, and for maps into a model space as joint continuity.

**V6.37 (the stabilized section in a product chart; `Polyfold.PullbackChartData.ofProd_scDeriv_r_apply`, `Polyfold.PullbackChartData.ofProd_mem_tangent_iff`, `Polyfold.PullbackChartData.mem_K_ofProd_iff`, `Polyfold.StrongBundle.scDeriv_ofProd_slice_apply`, `Polyfold.StrongBundle.scDeriv_paramSection_app` `Polyfold.StrongBundleChart.paramDerivAt`, `Polyfold.StrongBundle.scDeriv_paramSection_ofProd_ofPro` `Polyfold.sliceProj`, `Polyfold.sliceIncl`, `Polyfold.linKer_inf_ker_sliceProj_eq_map`, `Polyfold.exists_sliceOps`, `Polyfold.mem_linSet_sectionAdd_pullbackSlice`, `Polyfold.surjective_copro` `Polyfold.exists_sliceKernel`).** In the product chart $(\lambda \mapsto \lambda) \times ((q \mapsto q) \times \varphi)$ at $(0,(q,z))$, the linearization of $F$ is $(\lambda,(\sigma,v)) \mapsto D(\hat{g}_q \circ r)(z)v + \sum \lambda_i \hat{s}_i(q,\cdot)(z) + D_q\sigma$, with $g_q = f + S(q,\cdot)$; its kernel cut by $\sigma = 0$ is the image of $\ker(T \oplus \varphi)$, with $T = g_q'(x) \in \mathrm{Lin}(f,x)$ and $\varphi(\lambda) = \sum \lambda_i s_i(q,x)$; and $T \oplus \varphi$ is onto when $F$ is in general position at $(0,(q,\Phi(q)))$. *Book:* $F$, preprint p. 221, $F_t$, $L_t$ and (96), pp. 264–266 (preprint pp. 222–223). *Hidden hypotheses:* none. *Deviation (restated):* $\ker F_t'(\varphi(t),0)$ is read as the kernel of the linearization of the whole $F$ cut by

$\sigma = 0$ (Findings, item 49).

**V6.38 (Lemma 6.6.6 (preprint 6.31); `Polyfold.lemma_6_31`).** With the operators of V6.37, the sequence (96) $0 \to \ker T \to \ker(T \oplus \varphi) \to \mathbb{R}^k \to K_z/R(T) \to 0$ is the stabilization sequence of $T$ and $\varphi$ (V6.16), hence exact. *Book:* (96) and Lemma 6.6.6 (preprint 6.31), p. 266 (preprint p. 223). *Hidden hypotheses:* none. *Deviation (restated):* the text has $c$ onto by (P); here it is onto by the general position of $F$ (V6.33) and the derivative of the family of zeros, which absorbs the $\mathbb{R}^n$-directions (`Polyfold.fderiv_mem_linKer`) (Findings, item 49).

**V6.39 (continuation and the isomorphisms $\Phi_{E_t}$; `Polyfold.SubspaceFamily.cont`, `Polyfold.SubspaceFamily.c` `Polyfold.SubspaceFamily.cont_self`, `Polyfold.SubspaceFamily.cont_trans`, `Polyfold.SubspaceFamily.co` `Polyfold.SubspaceFamily.cont_comp`, `Polyfold.SubspaceFamily.cont_eq_transport`, `Polyfold.detEquivOfSurjective`, `Polyfold.stabRayEquiv`, `Polyfold.propagate`, `Polyfold.propagate_self`, `Polyfold.propagate_trans`, `Polyfold.propagate_comp`).** Over a simply connected, locally path-connected base the continuation of a ray of $\lambda(L_{q_0})$ to $\lambda(L_{q_1})$ is a bijection, compatible with composition and pull-back, and along $[0,1]$ it is the continuation of V6.35. For a stabilization $T \oplus \varphi$ onto and an identification of its kernel with $L$, the rays of $\det(T)$ and of $\lambda(L)$ correspond ($\iota_\varphi$ of V6.18 and $\det(T \oplus \varphi) \cong \lambda(\ker(T \oplus \varphi))$); $\Phi_{E_0}$, continuation and $\Phi_{E_1}^{-1}$ compose to a bijection of the rays of $\det(T_0)$ and $\det(T_1)$. *Book:* pp. 265–269 (preprint pp. 223–224), after Lemma 6.6.6 (preprint 6.31). *Hidden hypotheses:* none. *Deviation (restated):* over any simply connected, locally path-connected base, so that one statement serves the paths and the squares; the fixed identification $\lambda^*(\mathbb{R}^k) \cong \lambda^*(\{0\})$ of (94) occurs at both ends.

**V6.40 (the stabilization sign; `Polyfold.coord_stabIota`, `Polyfold.stabD`, `Polyfold.continuousOn_stabD`, `Polyfold.stabD_ne_zero`, `Polyfold.stabSign`, `Polyfold.isLineOrientation_stabRay`, `Polyfold.not_isLineOrientation_stabRay_uncorrected`).** In a local determinant chart with the projection $P$, the chart coordinate of $\iota_\psi(x) \in \det(S \oplus \psi)$ is $(-1)^{k \dim F/R(P) + \dim \operatorname{coker} S} D(S,\psi)$ times that of $x$, with $D$ continuous and nowhere zero near the centre. For a continuous orientation $o$ of DET, a norm-continuous family $T_q$ and a continuous stabilization $\varphi_q$ with $T_q \oplus \varphi_q$ onto, the rays $(-1)^{\dim \operatorname{coker} T_q}\, \Phi(o(T_q))$ of $\lambda(\ker(T_q \oplus \varphi_q))$ form an orientation; without the sign they need not ($E = F = \mathbb{R}$, $k = 1$, $T_q = q/(1+|q|)$, $\varphi(r) = r_0$). *Book:* formula (94), p. 256 (preprint p. 212), as it is used in the continuity statements of pp. 265–272 (preprint pp. 223–226). *Hidden hypotheses:* none. *Deviation (restated):* as we read it, the orientations carried by $\Phi_E$ of (94) along a continuous family are not continuous where $\dim \operatorname{coker} T$ changes parity; the propagation uses the sign-normalized $(-1)^{\dim \operatorname{coker} T}\Phi_E$, which on each operator carries the text's orientation or its opposite (Findings, item 50).

**V6.41 (the push-forward along stabilized kernels; `Polyfold.chartPt`, `Polyfold.ChartPoint.stabT`, `Polyfold.ChartPoint.stabT_mem_linSet`, `Polyfold.ChartPoint.stabPhi`, `Polyfold.ChartPoint.stabPhi_fi` `Polyfold.stabSign_inv`, `Polyfold.toPerm_stabSign_symm`, `Polyfold.isFredholm_stabCoprod`, `Polyfold.IsStabKernels`, `Polyfold.IsStabKernels.comp`, `Polyfold.IsStabKernels.push`, `Polyfold.IsStabKernels.push_congr`, `Polyfold.IsStabKernels.orRay`, `Polyfold.IsStabKernels.push_eq_c` `Polyfold.IsStabKernels.cont_orRay_path`, `Polyfold.IsStabKernels.push_comp`, `Polyfold.IsStabKernels.`** At a chart point, $T = (f+S)'(x) \in \operatorname{Lin}(f,x)$ and $\varphi(\lambda) = \sum \lambda_i \hat{s}_i(x)$. A family of *stabilized kernels* is a continuous family $\Lambda$, over a parameter space, of the subspaces $\ker(T_q \oplus \varphi_q)$ of $E_0 \times \mathbb{R}^k$ at points $y(q)$, read in one chart; its push-forward $\varphi_* : \operatorname{Or}^f_{x_0} \to \operatorname{Or}^f_{x_1}$ between the end points is $o \mapsto o(T_a) \mapsto (-1)^{\dim \operatorname{coker} T_a} \Phi_{E_a}(o(T_a))$, continued to $\lambda(\Lambda_b)$, and carried back in the same way at $T_b$. It depends only on the data, is compatible with pull-back and with concatenation, and along a path with continuously varying data continuation carries the rays read at one parameter

to those read at the next. *Book:* the push-forward $o_1 = \varphi_* o_0$, preprint p. 224. *Hidden hypotheses:* none. *Deviation (restated):* the sign of V6.40 at both end points; for given end-point operators the result differs from the text's by $(-1)^{\dim\operatorname{coker} T_a + \dim\operatorname{coker} T_b}$; the sections enter pointwise, and the continuity in the parameter is that of $\Lambda$ (Findings, item 50).

**V6.42 (Lemma 6.6.5 (preprint 6.30) for the push-forward: the stabilized kernels exist; `Polyfold.exists_continuousOn_lift_chart`, `Polyfold.exists_isStabKernels`).** For an sc-smooth $\Phi : \mathbb{R}^n \to X$ with $\Phi([0,1]^n)$ in the domain of a chart, an sc$^+$-section $S$ with $f + S = 0$ along $\Phi$ near $[0,1]^n$ and sc$^+$-sections $s_1, \dots, s_k$ as in V6.33, the stabilized kernels of the slices form a continuous family over $[0,1]^n$. *Book:* Lemma 6.6.5 (preprint 6.30), p. 265 (preprint p. 222), and the bundle $L$ after it. *Hidden hypotheses:* none beyond the standing ones. *Deviation (narrower):* as V6.34 (gap G.62); the family is the tangent spaces of the solution set (V6.34) cut by $\{dt = 0\}$ (V6.36) and identified with $\ker(T_q \oplus \varphi_q)$ (V6.37).

**V6.43 (push-forward data along paths; `Polyfold.pt1_mem_Icc`, `Polyfold.exists_lift_of_smooth`, `Polyfold.chartPtOf`, `Polyfold.isZeroAt_sectionAdd_reparamSection`, `Polyfold.IsStabKernels.push_eq_c` `Polyfold.IsPush`, `Polyfold.isPush_reparam`, `Polyfold.IsPush.split`, `Polyfold.exists_isPush`).** $e$ is a push-forward along an sc-smooth path $\gamma$ in a chart if there are global sc$^+$-sections $S, s_1, \dots, s_k$ of $\pi^*P \to \mathbb{R} \times X$ with $(f + S)(t, \gamma(t)) = 0$ near $[0,1]$ and a family of stabilized kernels of their slices along $\gamma$ whose push-forward is $e$. A push-forward along $\gamma$ is the composite of push-forwards along its two halves, and along every sc-smooth path in a set with property (3) of Proposition 6.6.3 (preprint 6.28) there is one. *Book:* preprint p. 224 (the choices of $s(\cdot, t)$ and $s_1, \dots, s_k$ along the path). *Hidden hypotheses:* none. *Deviation (restated):* global sections of the pull-back in place of sections over $U \times [0,1]$; property (P) is used only to construct a push-forward.

**V6.44 (Proposition 6.6.7 (preprint 6.32); `Polyfold.IsStabKernels.congr_data`, `Polyfold.IsStabKernels.push` `Polyfold.IsStabKernels.push_eq_of_prefix`, `Polyfold.isZeroAt_sectionAdd_combo`, `Polyfold.linAt_principal_combo`, `Polyfold.continuous_principal_pullbackSlice`, `Polyfold.proposition_6_32`).** Two push-forwards along the same sc-smooth path in the domain of a chart agree. *Book:* Proposition 6.6.7 (preprint 6.32), p. 267 (preprint p. 224), proof pp. 266–272 (preprint pp. 224–226). *Hidden hypotheses:* the standing ones of §6.6 (preprint §6.5) ($X$ tame, bump functions, tame germ data). *Deviation (restated):* "does not depend on the choices" is read as the agreement of any two data; the homotopy of Lemma 6.6.10 (preprint 6.34) and the added sections of Lemma 6.6.8 (preprint 6.33) are combined over one square, $S^\tau = (1-\tau)S^0 + \tau S^1$ with the two families of sections interpolated and the sections of Lemma 6.6.4 (preprint 6.29) on the square appended, so that Lemma 6.6.8 (preprint 6.33) applies on both edges without reordering (Findings, item 52).

**V6.45 (Lemma 6.6.8 (preprint 6.33); `Polyfold.wedgeExt`, `Polyfold.wedgeExt_wedge`, `Polyfold.wedgeExt_congr` `Polyfold.wedgeExt_bijective`, `Polyfold.appendStab`, `Polyfold.kerAppend`, `Polyfold.detEquiv_stabIota_a` `Polyfold.SubspaceFamily.exists_continuous_lifts`, `Polyfold.isLineOrientation_wedgeExt`, `Polyfold.kerEquivMap`, `Polyfold.stabRayEquiv_append`, `Polyfold.propagate_append`, `Polyfold.IsStabKernels.push_append`).** Appending sc$^+$-sections $\bar{s}_1, \dots, \bar{s}_d$ after $s_1, \dots, s_k$ does not change the push-forward: the stabilization isomorphism of $(\varphi, \psi)$ is that of $\varphi$ followed by $w \mapsto (w, 0) \wedge \ell_1 \wedge \cdots \wedge \ell_d$, for any $\ell_j \in \ker(T \oplus (\varphi, \psi))$ with $\psi$-coordinates the standard basis, and this map carries continuous orientations along the family. *Book:* Lemma 6.6.8 (preprint 6.33), p. 267 (preprint p. 224), proof preprint p. 225 ("one verifies by a simple computation"). *Hidden hypotheses:* none. *Deviation (restated):* the homotopy $\tilde{F}(\tau, \cdot)$ of the text's proof is not needed; with the appended sections last the two isomorphisms agree exactly, with no sign (Findings, item 51).

**V6.46 (sections over parameter families; `Polyfold.reparamSection`, `Polyfold.pullbackSlice_reparamSectio`
`Polyfold.rnProdMapOp`, `Polyfold.StrongBundle.isScPlusSection_of_local`, `Polyfold.StrongBundle.IsScP`
`Polyfold.StrongBundle.isGeneralPositionGerm_paramSection_of_comp`).** For $g$ affine, $(q, y) \mapsto S(g(q), y)$ is again an sc$^+$-section of the pull-back; sc$^+$-sections may be checked in one chart around each point; adding sections keeps the general position of $F$ at $(0, y)$. *Book:* pp. 266–272 (preprint pp. 224–226) (the sections $s(y, \tau, t)$ over a square, and "adding more sc$^+$-sections", Lemma 6.6.8 (preprint 6.33), p. 267 (preprint p. 224); the remark that after "adding more sections" the properties still hold is in the proof of Theorem 5.3.10 (preprint 5.18), p. 221 (preprint p. 178)). *Hidden hypotheses:* none. *Deviation:* none.

**V6.47 (Lemma 6.6.10 (preprint 6.34); `Polyfold.IsStabKernels.push_square`, `Polyfold.sqEdge`, `Polyfold.sqPath`).** For a family of stabilized kernels over the square $[0, 1]^2$ with the end points $x_0, x_1$ fixed and the end-point operators varying continuously, the push-forwards along the edges $\tau = 0$ and $\tau = 1$ agree. *Book:* Lemma 6.6.10 (preprint 6.34), p. 271 (preprint p. 226). *Hidden hypotheses:* none. *Deviation (restated):* the convex homotopy $s^\tau$ of the lemma is part of the square of V6.44; the continuity of the end-point orientations along $\tau$ is that of V6.40, with the sign.

**V6.48 (Theorem 6.6.11 (preprint 6.35); `Polyfold.push_eq_of_homotopy`, `Polyfold.HasLocalPropagation`, `Polyfold.theorem_6_35`).** Over a tame $X$ with bump functions, for $f$ with tame germ data, every smooth point $x$ has an open $U \ni x$ and a chart $c$ such that push-forwards exist along the sc-smooth paths in $U$ and depend only on their end points. *Book:* the homotopy argument, pp. 271–273 (preprint pp. 226–227), and Theorem 6.6.11 (preprint 6.35), p. 272 (preprint p. 227). *Hidden hypotheses:* the section $f$, which the statement does not introduce (an sc-Fredholm section, with tame germ data); bump functions. *Deviation (narrower):* $U$ is not shown to be sc-smoothly contractible (gap G.61); what the proof uses is (1′) of V6.32. Since the homotopy of (1′) meets its edge conditions only on $[0, 1]$ while a push-forward datum needs $f + S = 0$ near $[0, 1]$, the proof passes through the reparametrized paths $\gamma_i \circ \kappa$, with three homotopies whose edges are exact (Findings, item 53).

**V6.49 (sc-smooth paths in a chart and their concatenation; `Polyfold.MPolyfold.isScSmoothMap_of_eventuall`
`Polyfold.pt1_apply`, `Polyfold.contDiff_pt1_comp`, `Polyfold.exists_nhds_scPath`, `Polyfold.concatPath`,
`Polyfold.concatPath_of_le`, `Polyfold.concatPath_of_gt`, `Polyfold.isScSmoothMap_concatPath`,
`Polyfold.concatPath_zero`, `Polyfold.concatPath_one`, `Polyfold.concatPath_half`,
`Polyfold.concatPath_half'`, `Polyfold.isScSmoothMap_reparam`, `Polyfold.HasLocalPropagation.eq_trans`).**
For an open $U$ in the domain of a chart and $x \in U$ there is an open $V \ni x$ in $U$ whose smooth points are pairwise joined by sc-smooth paths in $U$. The concatenation, $\gamma_1(\kappa(3s))$ for $s \le 1/2$ and $\gamma_2(\kappa(3s - 2))$ for $s > 1/2$ with $\kappa$ the smooth transition function, is sc-smooth, with the right end points and halves; reparametrizing by a $C^\infty$ map keeps sc-smoothness; push-forwards compose along concatenations. *Book:* the paths after Theorem 6.6.11 (preprint 6.35), p. 272 (preprint p. 227), and the concatenation, preprint p. 228. *Hidden hypotheses:* none. *Deviation (narrower):* paths are joined only locally, inside a chart; the global facts recalled on preprint p. 201 (local path connectedness, sc$^+$-smooth paths within a path component) are not formalized (gap G.56).

**V6.50 (Definition 6.6.12 (preprint 6.36); `Polyfold.IsContinuousOrientation`, `Polyfold.exists_two_isContinu`
`Polyfold.exists_orientableNbhd`, `Polyfold.IsOrientableNbhd`, `Polyfold.exists_isOrientableNbhd`).**
An orientable neighbourhood of $(f, x)$ is a pair $V \subseteq U$: $U$ open with the local propagation property in a chart, $V \ni x$ open with any two smooth points joined by an sc-smooth path in $U$; a continuous orientation is a choice $y \mapsto o_y$ on $V_\infty$ carried into itself by the push-forwards along paths in $U$; there are exactly two, and every smooth point has an orientable neighbourhood. *Book:* the paragraph after Theorem 6.6.11 (preprint 6.35) and Definition 6.6.12 (preprint 6.36), p. 273

(preprint p. 227). *Hidden hypotheses:* the standing ones of §6.6 (preprint §6.5). *Deviation (restated):* the pair $V \subseteq U$ in place of an open contractible $U$, since Proposition 6.6.3 (preprint 6.28) gives (1′) (gap G.61) (Findings, item 54).

**V6.51 (reparametrization homotopies;** `Polyfold.mem_smoothSet_of_isScSmoothMap`, `Polyfold.contDiff_pt2_`
`Polyfold.isScHomotopyRel_reparam`, `Polyfold.reparam_mem`, `Polyfold.leftHalf`, `Polyfold.rightHalf`,
`Polyfold.halvesReparam`, `Polyfold.concatPath_halves`**).** For a $C^\infty$ $\theta$ with $\theta(0) = 0$, $\theta(1) = 1$, $\gamma \circ \theta$ is sc-smoothly homotopic to $\gamma$ with the end points fixed, inside every set containing $\gamma([0,1])$ when $\theta([0,1]) \subseteq [0,1]$; the concatenation of the two halves of $\gamma$ is such a reparametrization. *Book:* the homotopy classes $[\varphi]$, preprint p. 227, and their composition, preprint p. 228. *Hidden hypotheses:* none. *Deviation:* none; these are tools for V6.52 and V6.56.

**V6.52 (the path category $PX$;** `Polyfold.smoothTransition_half`, `Polyfold.isScSmoothMap_const_of_mem_smo`
`Polyfold.isScSmoothMap_ite`, `Polyfold.isScSmoothMap_ite` , `Polyfold.concatPath_of_le_two_thirds`,
`Polyfold.concatPath_of_one_third_le`, `Polyfold.ScHomotopic`, `Polyfold.ScHomotopic.refl`,
`Polyfold.ScHomotopic.symm`, `Polyfold.ScHomotopic.trans`, `Polyfold.isScHomotopyRel_triHomotopy`,
`Polyfold.assocReparam`, `Polyfold.ScPath`, `Polyfold.ScPath.source_mem_smoothSet`,
`Polyfold.ScPath.target_mem_smoothSet`, `Polyfold.ScPath.const`, `Polyfold.ScPath.concat`,
`Polyfold.ScPath.reverse`, `Polyfold.ScPath.concat_congr`, `Polyfold.ScPath.const_concat`,
`Polyfold.ScPath.concat_const`, `Polyfold.ScPath.concat_assoc`, `Polyfold.ScPath.concat_reverse`,
`Polyfold.ScPath.reverse_concat`, `Polyfold.ScPathClass`, `Polyfold.ScPathClass.comp`,
`Polyfold.PathCat`, `Polyfold.PathCat.category`, `Polyfold.PathCat.groupoid`, `Polyfold.PathCat.homMk`,
`Polyfold.PathCat.comp_mk`, `Polyfold.PathCat.id_eq_mk`, `Polyfold.PathCat.mk_eq_mk_iff`,
`Polyfold.PathCat.exists_eq_mk`, `Polyfold.PathCat.inv_mk`**).** Sc-smooth homotopy with the end points fixed is an equivalence relation on the sc-smooth paths between two points; concatenation is well defined on classes, associative, with the classes of the constant paths as units and the reversed paths as inverses; the smooth points with these classes form a category, indeed a groupoid. *Book:* the classes $[\varphi]$, their composition and the category $PX$, pp. 272–274 (preprint pp. 227–228). *Hidden hypotheses:* none. *Deviation (restated):* the edge conditions of a homotopy hold only on $[0,1]$, so transitivity is proved after reparametrizing by $\kappa$; the identities, which the text does not name, are the classes of the constant paths; the composition's second piece $\psi(\beta(s))$ is read as $\psi(\beta(s) - 1)$ (possible misprint); Mathlib's $[\varphi] \gg [\psi]$ is the text's $[\psi] * [\varphi]$ (Findings, items 55, 71).

**V6.53 (transferring orientations between chart points;** `Polyfold.conjKerEquiv`, `Polyfold.conjCokerEquiv`,
`Polyfold.conjDet`, `Polyfold.conjDet_wedge_tmul`, `Polyfold.conjDet_trans`, `Polyfold.conjDet_congr`,
`Polyfold.conjDet_refl`, `Polyfold.IsGoodLeftProj.conj`, `Polyfold.gammaProj_conjDet`,
`Polyfold.coord_conjDet_coordInv`, `Polyfold.isLineOrientation_conj`, `Polyfold.LineOrientation.conj`,
`Polyfold.LineOrientation.conj_apply`, `Polyfold.LineOrientation.conj_refl`, `Polyfold.LineOrientation`
`Polyfold.LineOrientation.conj_congr`, `Polyfold.ChartPoint.baseDeriv`, `Polyfold.ChartPoint.baseDeriv`
`Polyfold.ChartPoint.baseDeriv_trans`, `Polyfold.ChartPoint.isScOperator_baseDeriv`,
`Polyfold.ChartPoint.fiberMap`, `Polyfold.ChartPoint.fiberMap_self`, `Polyfold.ChartPoint.fiberMap_tra`
`Polyfold.ChartPoint.isScOperator_fiberMap`, `Polyfold.ChartPoint.transferT`, `Polyfold.ChartPoint.tra`
`Polyfold.ChartPoint.transferT_self_apply`, `Polyfold.ChartPoint.transferT_trans_apply`,
`Polyfold.ChartPoint.scDeriv_principal_baseDeriv`, `Polyfold.ChartPoint.IsScPlusOp.conj`,
`Polyfold.ChartPoint.mem_linSet_transfer_iff`, `Polyfold.ChartPoint.orTransfer`,
`Polyfold.ChartPoint.orTransfer_apply`, `Polyfold.ChartPoint.orTransfer_self`, `Polyfold.ChartPoint.or`
For isomorphisms $A$, $\Gamma$ with $S'A = \Gamma S$, $\det(S) \cong \det(S')$ by $A$ on kernels and $\Gamma$ on cokernels, compatible with composition and with the $\gamma^P$, and continuous orientations of DET are carried

along. For two chart points at $x$, $A$ is the derivative at $z$ of the base transition and $\Gamma$ the fibre part of the strong bundle transition, with the cocycle rules; $L \in \mathrm{Lin}_p(f,x)$ iff $\Gamma L A^{-1} \in \mathrm{Lin}_{p'}(f,x)$, and the induced bijection $\mathrm{Or}_x^f(p) \cong \mathrm{Or}_x^f(p')$ satisfies the cocycle rules. *Book:* preprint p. 219 ($T_xX$, $Y_x$ and $\mathrm{Lin}(f,x)$ as intrinsic objects) and the proof of Theorem 6.6.13 (preprint 6.37), p. 273 (preprint p. 228). *Hidden hypotheses:* none. *Deviation (not in the book):* the text treats these objects as chart-free; the identification between charts is made explicit (Findings, item 70).

**V6.54 (the push-forward does not depend on the chart;** `Polyfold.stabSign_conj`, `Polyfold.coprod_conj`, `Polyfold.ker_coprod_conj`, `Polyfold.surjective_coprod_conj`, `Polyfold.conj_stab_key`, `Polyfold.stabRayEquiv_conjDet`, `Polyfold.SubspaceFamily.cont_map`, `Polyfold.ChartPoint.fiberMap_pri` `Polyfold.ChartPoint.conj_of_linAt`, `Polyfold.ChartPoint.conj_of_principal`, `Polyfold.continuous_bas` `Polyfold.IsStabKernels.exists_transfer`, `Polyfold.IsStabKernels.orRay_transfer`, `Polyfold.IsPush.transfer`, `Polyfold.HasLocalPropagation.push_compat`**).** The stabilization isomorphism and the sign are natural under conjugation, and continuation is natural under jointly continuous families of injective linear maps. A family of stabilized kernels in one chart is carried by the derivative of the base transition to one in a second chart, with the same sections; a push-forward in one chart, conjugated by the transfer of V6.53 at the end points, is a push-forward in the other; so the local propagations of two charts agree. *Book:* pp. 264–274 (preprint pp. 222–228) (the push-forward, and the orientable neighbourhoods of the proof of Theorem 6.6.13 (preprint 6.37)). *Hidden hypotheses:* none. *Deviation (not in the book):* the text does not compare charts (Findings, item 70).

**V6.55 (**$\mathrm{Or}_x^f$ **in a chosen chart;** `Polyfold.orChart`, `Polyfold.isChart_orChart`, `Polyfold.mem_orChart`, `Polyfold.Orx`, `Polyfold.toChartOr`, `Polyfold.toChartOr_orChart`, `Polyfold.toChartOr_trans`, `Polyfold.exists_two_orx`, `Polyfold.restrictOr`**).** $\mathrm{Or}_x^f$ is the set of orientations read in a chosen chart of the atlas at $x$; it has two elements, and it is identified with the orientations read in any other chart at $x$ by the transfer of V6.53, compatibly with composition. *Book:* Definition 6.6.2 (preprint 6.27) and the objects of Theorem 6.6.13 (preprint 6.37), preprint pp. 220, 228. *Hidden hypotheses:* none. *Deviation (restated):* a chosen chart stands for the text's chart-free $\mathrm{Or}_x^f$ (Findings, item 70).

**V6.56 (transport along paths from local push-forwards;** `Polyfold.LocalTransport`, `Polyfold.LocalTransport.` `Polyfold.LocalTransport.push_eq_push`, `Polyfold.LocalTransport.push_congr`, `Polyfold.LocalTransport` `Polyfold.LocalTransport.transportN_eq_push`, `Polyfold.LocalTransport.IsFine`, `Polyfold.LocalTranspo` `Polyfold.LocalTransport.IsLebesgueNumber`, `Polyfold.LocalTransport.exists_isLebesgueNumber`, `Polyfold.LocalTransport.isFine_of_isLebesgueNumber`, `Polyfold.LocalTransport.transport`, `Polyfold.LocalTransport.transport_eq_push`, `Polyfold.LocalTransport.transport_congr`, `Polyfold.LocalTransport.transport_const`, `Polyfold.LocalTransport.transport_halves`, `Polyfold.LocalTransport.transportN_square`, `Polyfold.LocalTransport.exists_isSqLebesgueNumber`, `Polyfold.LocalTransport.transport_eq_of_isScHomotopyRel`, `Polyfold.LocalTransport.transport_repar` `Polyfold.LocalTransport.transport_concatPath`, `Polyfold.LocalTransport.transport_unique`, `Polyfold.equiv_paste_squares`, `Polyfold.vSeg`, `Polyfold.hSeg`, `Polyfold.dSeg`**).** From local data (good open sets covering $X_\infty$ and push-forwards along paths in a good set, compatible with concatenation and independent of the good set) the transport along an sc-smooth path is the composite of the push-forwards along its $2^N$ dyadic pieces, for any $N$ at which every piece lies in a good set (such $N$ exist by the Lebesgue number lemma). It is the push-forward on a path in a good set, homotopy invariant, compatible with concatenation and reparametrization, the identity on constant paths, and the only family with these properties. *Book:* the proof of Theorem 6.6.13

(preprint 6.37), p. 273 (preprint p. 228) ("using the compactness of $[0,1]$ and a finite covering", "by a covering argument", "it is obvious that $\Gamma_f$ is unique"). *Hidden hypotheses:* none. *Deviation (restated):* dyadic subdivisions of $[0,1]$ and of $[0,1]^2$ in place of arbitrary finite coverings (Findings, item 72).

**V6.57 (Theorem 6.6.13 (preprint 6.37); `Polyfold.IsPropagationNbhd`, `Polyfold.orPush`, `Polyfold.orPush_eq`, `Polyfold.orPush_compat`, `Polyfold.orPush_trans`, `Polyfold.IsStarPropertyOn`, `Polyfold.IsStarProperty`, `Polyfold.orLocalTransport`, `Polyfold.orLocalTransport_transport_eq`, `Polyfold.LocalTransport.restrict`, `Polyfold.LocalTransport.transport_apply_eq_of_local`, `Polyfold.orFunctorMap`, `Polyfold.orFunctorMap_mk`, `Polyfold.orFunctorMap_id`, `Polyfold.orFunctorMap_c` `Polyfold.orFunctor`, `Polyfold.orFunctorMap_isStarProperty`, `Polyfold.orFunctorMap_unique_of_cover`, `Polyfold.orFunctorMap_unique`, `Polyfold.theorem_6_37`).** Over a tame $X$ with bump functions, for $f$ with tame germ data and $P$ with sc$^+$-sections through the smooth fibre points, there is exactly one family of bijections $\Gamma_f([\varphi]) : \mathrm{Or}^f_x \to \mathrm{Or}^f_y$ on the morphisms of $PX$ that takes identities to identities, compositions to composites, and has property (□ ): for every chart $c$, every open $U$ with the local propagation property in $c$ and every sc-smooth path $\varphi$ in $U$, $\Gamma_f([\varphi])$ is every push-forward along $\varphi$ in $c$, read through $\mathrm{Or}^f$ of the chosen charts. It is a functor $PX \to$ Type, $x \mapsto \mathrm{Or}^f_x$, and the uniqueness needs (□ ) only on a family of such open sets covering $X_\infty$. *Book:* Theorem 6.6.13 (preprint 6.37) and its proof, preprint p. 228. *Hidden hypotheses:* the standing ones of §6.6 (preprint §6.5), and sc$^+$-sections through every smooth fibre point, so that every $\mathrm{Or}^f_x$ has two elements (implied by bump functions). *Deviation (restated):* (□ ) is asked for every open set with the local propagation property, not only for sc-smoothly contractible ones (gap G.61); $\mathcal{O}_f$ is read as types and bijections; "uniquely determined" is read for the maps on morphisms (Findings, item 72).

**V6.58 (Definition 6.6.14 (preprint 6.38); `Polyfold.scPathSetoid`, `Polyfold.IsOrientableSection`, `Polyfold.isOrientableSection_iff_exists`).** $f$ is orientable if $\Gamma_f([\varphi]) = \Gamma_f([\psi])$ for any two morphisms $x \to y$ of $PX$; equivalently there is $y \mapsto o_y$ with $\Gamma_f([\varphi])o_x = o_y$ for every $[\varphi] : x \to y$. *Book:* Definition 6.6.14 (preprint 6.38), p. 274 (preprint p. 229). *Hidden hypotheses:* none. *Deviation (restated):* "smooth points in the same path component" is read as smooth points joined by an sc-smooth path (Findings, item 73).

**V6.59 (Definition 6.6.15 (preprint 6.39) and its equivalence with Definition 6.6.14 (preprint 6.38); `Polyfold.IsOrientableSection'`, `Polyfold.isOrientableSection'_iff_exists`, `Polyfold.isOrientableSection_iff`).** $f$ is orientable in the sense of Definition 6.6.15 (preprint 6.39) if there is $y \mapsto o_y$ on $X_\infty$ such that every smooth point has an orientable neighbourhood over which it is a continuous orientation; this holds iff there is $y \mapsto o_y$ with $\Gamma_f([\varphi])o_x = o_y$ for all $[\varphi]$, so Definitions 6.6.14 and 6.6.15 (preprint 6.38 and 6.39) are equivalent. *Book:* Definition 6.6.15 (preprint 6.39), p. 274 (preprint p. 229) ("an equivalent version"). *Hidden hypotheses:* none. *Deviation (stronger):* the equivalence, stated without proof, is proved; the definition's "$y \in U_\infty$", before any $U$ is introduced, is read as $y \in X_\infty$ (possible misprint) (Findings, items 73, 74).

**V6.60 (Definition 6.6.16 (preprint 6.40); `Polyfold.IsSectionOrientation`, `Polyfold.isSectionOrientation_i` `Polyfold.exists_isSectionOrientation_iff`, `Polyfold.IsSectionOrientation.apply_eq_iff`, `Polyfold.IsSectionOrientation.exists_two`).** An orientation of $f$ is a map $y \mapsto o_y$ on $X_\infty$ whose restriction to every orientable neighbourhood of every smooth point is a continuous orientation; equivalently $\Gamma_f([\varphi])o_x = o_y$ for all $[\varphi] : x \to y$. $f$ has an orientation iff it is orientable; two orientations that agree at a point agree on the smooth points joined to it by sc-smooth paths, and there are exactly two on each such class. *Book:* Definition 6.6.16 (preprint 6.40), p. 274 (preprint

p. 229). *Hidden hypotheses:* none. *Deviation (restated):* an orientable neighbourhood is the pair of V6.50 (Findings, item 75).

**V6.61 (Theorem 6.6.18 (preprint 6.41), the natural orientation of the solution set;** `Polyfold.theorem_6_41_solut` `Polyfold.MPolyfold.IsGeneralPositionAt.isGoodPositionAt`, `Polyfold.tangentSpace_solutionSet_eq_li` `Polyfold.ChartPoint.linZero`, `Polyfold.ChartPoint.linZero_mem_linSet`, `Polyfold.ChartPoint.linZero_` `Polyfold.ChartPoint.map_ker_linZero`, `Polyfold.ChartPoint.tangentOr`, `Polyfold.tangentOrAt`, `Polyfold.stabSign_of_surjective`, `Polyfold.stabRayEquiv_zero`, `Polyfold.ChartPoint.stabRayEquiv_eq_` `Polyfold.IsStabKernels.orRay_eq_tangentOr`, `Polyfold.isStabKernels_zero`, `Polyfold.SubspaceFamily.c` `Polyfold.exists_tangentFamily`, `Polyfold.tangentFamily`, `Polyfold.tangentFamily_L`, `Polyfold.cont_tangentOr`, `Polyfold.isLineOrientation_tangentOr`, `Polyfold.isLineOrientation_tangent` `Polyfold.theorem_6_41`**).** Over a tame $X$, for a proper sc-Fredholm section $f$ with tame germ data in general position (Definition 5.3.9 (preprint 5.17)) at every zero, the solution set $S$ is compact, a smooth submanifold in general position with the induced structure of a tame manifold with boundary with corners (the conclusions of Theorem 5.3.10 (preprint 5.18)), and $T_xS = \ker f'(x)$, read in a chart. At a zero, $f'(x)$ lies in $\mathrm{Lin}(f,x)$ and is onto, so an element $o_x$ of $\mathrm{Or}_x^f$ gives a ray of $\det f'(x) \cong \lambda(\ker f'(x)) = \lambda(T_xS)$, the *tangent orientation*. For an orientation $o$ of $f$ (Definition 6.6.16 (preprint 6.40)) and every sc-smooth path $\gamma$ of zeros in an orientable neighbourhood $V \subseteq U$, the kernels $\ker f'(\gamma_q)$ form a continuous family of subspaces, and the tangent orientations of $o$ form an orientation of its line bundle. The Fredholm side is the push-forward for the datum $S = 0$, $k = 0$, which along a path of zeros is the continuation between tangent orientations. *Book:* Theorem 6.6.18 (preprint 6.41) and its proof, preprint p. 229; the paragraph after it on $S_{f+s}$, pp. 274–276 (preprint pp. 229–230). *Hidden hypotheses:* tame germ data (the standing hypothesis of Chapter 5); the orientation $o$ exists iff $f$ is orientable (V6.60). "Proper" is the text's hypothesis (`StrongBundle.IsProper`); the proof uses only the compactness of $S$, which for sc-Fredholm sections is equivalent to it (V5.10). *Deviation (narrower):* "a natural orientation" is read as the tangent orientations with their continuity along the sc-smooth paths of zeros in a chart, and "the local prolongation of the orientation on $S$ is the Fredholm one" as the identity of continuation and push-forward along such paths. Not formalized: an orientation of $S$ as an abstract manifold (gap G.67), the compatibility of the tangent orientations read in two charts at one point (gap G.68), and the perturbation paragraph after the theorem, with $S_{f+s}$ and $\mathrm{Lin}(f+s,x) = \mathrm{Lin}(f,x)$ (gap G.69) (Findings, items 90–93).

**V6.62 (Remark 6.6.19 (preprint 6.42), at a point;** `Polyfold.extOp`, `Polyfold.isFredholm_extOp`, `Polyfold.finrank_ker_extOp`, `Polyfold.extE`, `Polyfold.range_extOp`, `Polyfold.extCoker`, `Polyfold.extDet`, `Polyfold.extDet_wedge_tmul`, `Polyfold.extDetN`, `Polyfold.extDetN_eq_of_surjective`, `Polyfold.coord_extDet`, `Polyfold.coord_extDetN`, `Polyfold.isLineOrientation_extPull`, `Polyfold.LineOrientation.extPull`, `Polyfold.extConjOp`, `Polyfold.extConjDet`, `Polyfold.existsUnique_e` `Polyfold.ChartPoint.prodPt`, `Polyfold.ChartPoint.mem_prodPt_T_iff`, `Polyfold.ChartPoint.mem_prodPt_` `Polyfold.ChartPoint.prodT`, `Polyfold.ChartPoint.prodK`, `Polyfold.ChartPoint.prodLin`, `Polyfold.ChartPoint.IsScPlusOp.prodPt`, `Polyfold.constLift`, `Polyfold.ChartPoint.prodLin_mem_linSet` `Polyfold.ChartPoint.existsUnique_prodOr`, `Polyfold.ChartPoint.prodOr`, `Polyfold.ChartPoint.prodOr_a` `Polyfold.ChartPoint.prodOr_injective`**).** For Fredholm $L$, $\tilde{L} = L \oplus 0 : E \times \mathbb{R} \to F$ is Fredholm with $\ker \tilde{L} = \ker L \times \mathbb{R}$ and $\operatorname{coker} \tilde{L} = \operatorname{coker} L$, and $h \mapsto \tilde{h}$, $(a_1 \wedge \cdots \wedge a_n) \otimes d^* \mapsto ((a_1,0) \wedge \cdots \wedge (a_n,0) \wedge e) \otimes d^*$ with $e = (0,1)$, is an isomorphism $\det(L) \cong \det(\tilde{L})$; in local determinant charts its coordinate is $(-1)^{\dim F/R(P) + \dim \operatorname{coker} L}$ times a continuous nowhere-zero function, and $(-1)^{\dim \operatorname{coker} L}\, h \mapsto \tilde{h}$ pulls continuous orientations back along $L \mapsto \tilde{L}$. In charts: for an sc-Fredholm $f$, an sc$^+$-section $S$ of $\pi^*P \to \mathbb{R} \times X$, $\hat{f} = \tilde{f} + S$ with $\tilde{f}(t,y) = f(y)$, and the

product chart point $(t, z)$, the operator $\hat{L}(\sigma, v) = Lv$ lies in $\mathrm{Lin}(\hat{f}, (t, x))$ for every $L \in \mathrm{Lin}(f, x)$, and an orientation $o$ of $\mathrm{DET}(f, x)$ gives exactly one orientation $\hat{o}$ of $\mathrm{DET}(\hat{f}, (t, x))$ with $\hat{o}(\hat{L})$ the image of $o(L)$ for **every** $L$; $o \mapsto \hat{o}$ is injective. *Book:* Remark 6.6.19 (preprint 6.42), p. 276 (preprint p. 230). *Hidden hypotheses:* $x$ smooth and $\mathrm{Lin}(f, x)$ nonempty (implied by bump functions, V6.1); only the orientation of $\mathrm{DET}(f, x)$ at the one point is used. *Deviation (narrower):* at a point. As we read it, the literal $h \mapsto \tilde{h}$ is not continuous in $L$ where the parity of $\dim \operatorname{coker} L$ changes, so the orientation of $\mathrm{DET}(\hat{f}, (t, x))$ it defines could depend on the choice of $L$; the sign $(-1)^{\dim \operatorname{coker} L}$ (the sign of V6.40) is used, which agrees with the text's map when $L$ is onto. $X \times [0, 1]$ is read as $\mathbb{R} \times X$ (decision D2), with the parameter first, so $\tilde{L}(a, b) = La$ is conjugated by the swap of the factors. That $(t, y) \mapsto \hat{o}$ is an orientation of $\hat{f}$ in the sense of Definition 6.6.16 (preprint 6.40), built from an orientation of $f$, is not formalized (gap G.66) (Findings, items 95–97).

**V6.63 (Appendix 6.8.1 (preprint 6.7.1), the proof of Lemma 6.3.3 (preprint 6.5); `Polyfold.FourTermExact.Choice` `Polyfold.FourTermExact.Choice.wedgeC`).** For two choices of (91) the right-hand sides are related as the sources are: a change of the basis of $Z$ multiplies both wedges by its determinant, a change of the bases of $A$ and $D$ multiplies both sides by the same factor, and a change of complements adds to the basis of $Z$ vectors of $\alpha(A)$ and to the $\gamma$-preimages vectors of $\beta(B)$, which leaves the wedges unchanged. *Book:* §6.8.1 (preprint §6.7.1), preprint pp. 234–235. *Hidden hypotheses:* none. *Deviation (restated):* the step "independent of the choices of the basis if and only if $\det(\tau) = \det(\varepsilon)$" is read as the homogeneity of (91) in the bases of $A$ and $D$, $\tau$ being the $\sigma$ just introduced; the corrections $p_i$ lie in $\beta(B)$ (written $\beta(C)$) and are combinations of the $\beta$-images, whose names are exchanged with those of the $\gamma$-preimages in the last step (possible misprints) (Findings, item 1).

**V6.64 (Appendix 6.8.2 (preprint 6.7.2), the proof of Proposition 6.4.11 (preprint 6.12); `Polyfold.mem_span_append_of_ker`, `Polyfold.gammaProj_wedge_tmul`, `Polyfold.comp_comp_eq_of_projLE`).** For $P \leq Q$ in $\Pi_T$: if $Qg = 0$, then $g$ is a combination of the vectors $d_i$ plus $Tx$ with $x \in \ker(QT)$, so the classes of $d_i$ and $Tb_j$ span what the text calls $\ker \tilde{Q}$; the classes $[T\tilde{b}]$ and $[QT\tilde{b}]$ in $F/R(P)$ differ by elements of that span, which does not change the wedge. *Book:* §6.8.2 (preprint §6.7.2), pp. 280–286 (preprint pp. 235–237). *Hidden hypotheses:* none. *Deviation (restated):* the spanning is proved directly in $F$; the name $\alpha_q$ in the last part is read as $\alpha^P$ (possible misprint) (Findings, items 6, 7).

**V6.65 (Lemma 6.4.4 (2017: 6.3); `Polyfold.projIntertwiner`, `Polyfold.projIntertwiner_self`, `Polyfold.projIntertwiner_mul`, `Polyfold.continuous_projIntertwiner`, `Polyfold.lemma_6_4_4_2021`).** Let $E$, $F$ be Banach spaces, $T_0 : E \to F$ Fredholm and $P_0 \in \Pi_{T_0}$. There is $\varepsilon > 0$ such that for every $T$ and every projection $P$ of $F$ with $\|T - T_0\| < \varepsilon$ and $\|P - P_0\| < \varepsilon$: $T$ is Fredholm, $P \in \Pi_T$, and $\mathrm{ind}(P \circ T) = \mathrm{ind}(T) = \mathrm{ind}(T_0) = \mathrm{ind}(P_0 \circ T_0)$. For projections $P_0$, $P$ the intertwiner $J(P) = PP_0 + (1 - P)(1 - P_0)$ has $J(P_0) = 1$ and $J(P)P_0 = PJ(P)$, and depends continuously on $P$. *Book:* Lemma 6.4.4 (2017: 6.3), p. 247 (2017 p. 189); no preprint counterpart; the index equalities are new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (explicit):* $\mathrm{ind}(P \circ T)$ is the index of $P \circ T$ as a map $E \to F$ (as a map onto $R(P)$ its index would be $\dim \ker(PT)$); $E$ and $F$ are taken in the universe `Type`. The proof is not the text's: $J(P)$ is invertible near $P_0$ and $P = JP_0J^{-1}$, so $R(P) = JR(P_0)$ has finite codimension, and Lemma 6.5.1 (preprint 6.17) for $J^{-1}T$ near $T_0$ gives $R(PT) = JR(P_0J^{-1}T) = JR(P_0) = R(P)$ (Findings, items 110, 111).

**V6.66 (a common complement; `Polyfold.finrank_sup_span_singleton_of_notMem`, `Polyfold.isCompl_sup_s`**

`Polyfold.exists_isCompl_isCompl_of_finrank_eq_aux`, `Polyfold.exists_isCompl_isCompl_of_finrank_eq`
Two finite-dimensional subspaces $K_1$, $K_2$ of the same dimension of a real vector space have a common algebraic complement $X$. *Book:* the proof of Proposition 6.4.13 (2017: 6.6), p. 253 (2017 p. 195), takes a closed subspace $X$ complementing both kernels. *Hidden hypotheses:* none. *Deviation (not in the book):* by induction on $\dim(K_1 + K_2) - \dim K_1$: for $a \in K_2 \setminus K_1$ and $b \in K_1 \setminus K_2$, $a + b$ lies in neither, and a common complement of $K_1 + \mathbb{R}(a + b)$ and $K_2 + \mathbb{R}(a + b)$ plus $\mathbb{R}(a + b)$ is one of $K_1$ and $K_2$. That $X$ can be taken closed is V6.69 (Findings, items 112, 116).

**V6.67 (Proposition 6.4.13 (2017: 6.6), the algebraic part; `Polyfold.proposition_6_4_13_2021`).** Let $T : E \to F$ be Fredholm and $P, Q \in \Pi_T$ with $R(P) = R(Q) = H$. There are linear isomorphisms $A_T : \ker(PT) \to \ker(QT)$ and $B_T : F/R(P) \to F/R(Q)$ with $A_T \circ j_P = j_Q$, $\Phi_Q \circ A_T = B_T \circ \Phi_P$ and $\pi_Q \circ B_T = \pi_P$, for the maps of the exact sequences $E(T, P)$ and $E(T, Q)$ of V6.12. They are the text's: $A_T(e) = e - x(e)$, $x(e) \in X$ with $QTx(e) = QTe$, and $B_T(f + H) = \tilde{f} + H$ with $Tx + \tilde{f} = (1 - P)f$, $x \in X$, $\tilde{f} \in (1 - Q)F$, for a common complement $X$ of the kernels (V6.66), which exists because $\dim \ker(PT) = \dim \ker(QT)$ by the indices. *Book:* Proposition 6.4.13 (2017: 6.6), p. 253 (2017 p. 195), proof pp. 253–255; no preprint counterpart; the 2021 edition makes $\Pi_T \cap \mathcal{P}_H \neq \emptyset$ a hypothesis. *Hidden hypotheses:* none. *Deviation (narrower):* the last sentence, that $A_T$ and $B_T$ depend continuously on $T$, is not part of this item: it is V6.70, with a closed $X$ (V6.69) (gap G.81, closed); here $X$ is algebraic, and $F/H$ appears as $F/R(P)$ and $F/R(Q)$ (Findings, items 112, 113, 117).

**V6.68 (Lemmas 6.6.8 and 6.6.9; `Polyfold.lemma_6_6_8_2021`, `Polyfold.lemma_6_6_9_2021`).** For stabilized kernels of the data $(f, s_1, \dots, s_k)$ along a family, and of the same data with $d$ sections appended, the push-forwards of orientations between two smooth points agree, both when the appended sections are zero (Lemma 6.6.8: the new parameters $\mu$ enter as dummy variables) and when they are arbitrary sections $\bar{s}_1, \dots, \bar{s}_d$ (Lemma 6.6.9). *Book:* Lemma 6.6.8 (preprint 6.33), p. 267 (preprint p. 224), and Lemma 6.6.9, p. 269, new in the 2021 edition as a separate lemma: the 2021 text splits preprint Lemma 6.33 (2017: 6.15), whose statement is that of Lemma 6.6.9 and whose proof begins with the step of Lemma 6.6.8. *Hidden hypotheses:* the stabilized kernels of the appended data are given (`IsStabKernels`), as in V6.45. *Deviation (restated):* both are V6.45 (`Polyfold.IsStabKernels.push_append`), Lemma 6.6.8 for $\bar{s}_j = 0$; the homotopy of the text's proof is not needed (Findings, item 114).

**V6.69 (a common closed complement; `Polyfold.isCompl_sup_inf_of_isCompl`, `Polyfold.exists_isClosed_i`**
Two finite-dimensional subspaces $K_1$, $K_2$ of the same dimension of a real normed space $E$ have a common closed complement $X$, and $E = K_1 \oplus X = K_2 \oplus X$ topologically (with continuous projections). *Book:* the proof of Proposition 6.4.13 (2017: 6.6), p. 253 (2017 p. 195), takes a closed subspace $X$ that is a topological complement of both kernels, without argument. *Hidden hypotheses:* none. *Deviation (not in the book):* the argument. $X = Z + (X_0 \cap (K_1 + K_2))$, where $X_0$ is the algebraic common complement of V6.66 and $Z$ a closed complement of the finite-dimensional $K_1 + K_2$; it complements $K_1$ and $K_2$ (if $X_0 \oplus K' = E$, $K' \subseteq K$ and $K \oplus Z = E$, then $(Z + X_0 \cap K) \oplus K' = E$), it is closed as the sum of a closed and a finite-dimensional subspace, and a closed complement of a finite-dimensional subspace is a topological one (Findings, items 112, 116).

**V6.70 (Proposition 6.4.13 (2017: 6.6), the last sentence; `Polyfold.projCod`, `Polyfold.solveOp`, `Polyfold.kerTransition`, `Polyfold.cokerTransition`, `Polyfold.solveOp_mem`, `Polyfold.kerTransition_app` `Polyfold.cokerTransition_apply`, `Polyfold.apply_solveOp_of_projRestrict_eq`, `Polyfold.disjoint_ker_` `Polyfold.isTopCompl_ker_comp_of_projRestrict_eq`, `Polyfold.diagram_of_solveOp`,**

`Polyfold.continuousAt_solveOp`, `Polyfold.proposition_6_4_13_continuity_2021`, `Polyfold.proposition_`
Let $E$, $F$ be Banach spaces, $T : E \to F$ Fredholm and $P, Q \in \Pi_T$ with $R(P) = R(Q) = H$. For $S : E \to F$ and a subspace $X$ write $\xi_S = (QS|_X)^{-1} \circ Q : F \to X$ (the solution $x \in X$ of $QSx = Qg$), $A_S(e) = e - \xi_S(Se)$ on $E$ and $B_S(f) = (1-P)f - S\xi_S((1-P)f)$ on $F$. There are a closed subspace $X \subseteq E$ and $\varepsilon > 0$ such that (a) $S \mapsto A_S$ and $S \mapsto B_S$ are continuous on $\|S - T\| < \varepsilon$ in the operator norm, and $(S, e) \mapsto A_S e$, $(S, f) \mapsto B_S f$ are jointly continuous there; (b) for every such $S$: $P, Q \in \Pi_S$; $E = \ker(PS) \oplus X = \ker(QS) \oplus X$ topologically; $e - A_S e \in X$ and $QSA_S e = 0$ ((6.15), (6.16)); $QB_S f = 0$ and $Sx + B_S f = (1-P)f$ for some $x \in X$ ((6.18), (6.19)); $A_S$ restricts to a linear isomorphism $\ker(PS) \to \ker(QS)$ and $B_S$ induces a linear isomorphism $F/R(P) \to F/R(Q)$, with $A \circ j_P = j_Q$, $\Phi_Q \circ A = B \circ \Phi_P$ and $\pi_Q \circ B = \pi_P$ for the exact sequences $E(S, P)$, $E(S, Q)$ (V6.12). The hypotheses are satisfiable ($T = 1$ on $\mathbb{R}$, $P = Q$). *Book:* Proposition 6.4.13 (2017: 6.6), p. 253 (2017 p. 195), last sentence: for $T'$ near $T$ with $P, Q \in \Pi_{T'}$ the constructions of $A_{T'}$ and $B_{T'}$ depend continuously on $T'$; argument pp. 253, 255 (with $X$ fixed, the equation (6.15) for $T'$ has a unique solution depending continuously on $T'$); no preprint counterpart. *Hidden hypotheses:* none; $P, Q \in \Pi_S$ for $S$ near $T$ follows from Lemma 6.5.1 (preprint 6.17) (V6.20) and is part of the conclusion. *Deviation (explicit):* "depend continuously" is read as the continuity of $A_S$ extended to $E$ and of $B_S$ lifted to $F$, which determine the maps $\ker(PS) \to \ker(QS)$ and $F/H \to F/H$ between the varying kernels; the joint continuity says that $(S, e) \mapsto (S, A_S e)$ is continuous on the kernel bundle $\bigcup_S \{S\} \times \ker(PS)$ of Remark 6.4.5 (V6.21) with its subspace topology. *Proof:* the text's. $X$ is V6.69 for the kernels of $PT$ and $QT$, of equal dimension by the indices; by Lemma 6.5.1 (preprint 6.17) for $(T, P, X)$ and $(T, Q, X)$, $PS|_X$ and $QS|_X$ are isomorphisms onto $R(P)$, $R(Q)$ near $T$, and inversion is continuous at isomorphisms, so $S \mapsto \xi_S$ is continuous; the diagram is the construction of the algebraic part (V6.67) for the solution operator $\xi_S$ (Findings, items 113, 117).

**V6.71 (the orientation space and the properties of Theorem 6.6.17; `Polyfold.OrSpace`, `Polyfold.orProj`, `Polyfold.segPath`, `Polyfold.isScSmoothMap_segPath`, `Polyfold.OrPullback`, `Polyfold.IsContSectionOver`, `Polyfold.IsPropagationContinuous`, `Polyfold.HasTwoContSections`, `Polyfold.IsOrientationTopology`).** For an sc-Fredholm section $f$ of a strong bundle $P : Y \to X$, $O_f$ is the set of pairs $(x, o)$ with $x \in X_\infty$ and $o \in \mathrm{Or}_x^f$, an orientation of $\mathrm{DET}(f, x)$ read in the chosen chart (V6.55), and $\pi_f(x, o) = x$; $X_\infty$ carries the metrizable topology of Theorem 2.3.10 (preprint 2.16) (V2.17). For an sc-smooth path $\varphi$ (an sc-smooth map $\mathbb{R} \to X$), $\varphi^* O_f$ is the subspace $\{(t, p) : \varphi(t) = \pi_f(p)\}$ of $[0,1] \times O_f$, and the segment $s \mapsto \varphi(a + s(b-a))$ from $\varphi(a)$ to $\varphi(b)$ is sc-smooth. For a topology on $O_f$: a continuous section over $U \cap X_\infty$ is a continuous $\sigma : U_\infty \to O_f$ with $\pi_f \circ \sigma = \mathrm{id}$, $U_\infty$ with the topology of $X_\infty$; property (2) asks that every $x \in X_\infty$ have an open $U \ni x$ in $X$ with exactly two continuous sections over $U \cap X_\infty$; property (3) asks that for every sc-smooth path $\varphi$, every continuous section $\alpha(t) = (t, (\varphi(t), o_t))$ of $\varphi^* O_f$ and every $t_0 \in [0,1]$, for all $t$ near $t_0$ an open $U$ with the local propagation property in some chart contains the segment of $\varphi$ from $t_0$ to $t$, and the push-forward along it in $U$ maps $o_{t_0}$ to $o_t$; an orientation topology is a Hausdorff topology with (1) $\pi_f$ a local homeomorphism, (2) and (3). *Book:* pp. 274–275, before Theorem 6.6.17, and the three properties of Theorem 6.6.17, p. 275, new in the 2021 edition (the set $O_f$ appears without a topology in the preprint after Definition 6.27, p. 219, and in the 2017 text after Definition 6.9, p. 205). *Hidden hypotheses:* the topology of $X_\infty$ ("the natural metrizable topology"). *Deviation (explicit):* "$o_t$ is the local propagation of $o_{t_0}$" is read as the push-forward `orPush` (V6.57) along the segment of $\varphi$ in a set with the local propagation property, which does not depend on that set (`Polyfold.orPush_compat`); paths are the sc-smooth maps $\mathbb{R} \to X$ of decision D2, restricted to $[0,1]$ (Findings, item 118).

**V6.72 (sc-smooth paths in $X_\infty$; `Polyfold.rnChart_toBase`, `Polyfold.continuousOn_levelLift_comp`, `Polyfold.continuous_smoothMk_comp`, `Polyfold.chartSegPath`, `Polyfold.isScSmoothMap_chartSegPath`, `Polyfold.eventually_exists_scPath_smoothLevel`, `Polyfold.exists_nhds_scPath_of_isOpen`).** An sc-smooth map $\Gamma : \mathbb{R}^n \to X$ takes values in $X_\infty$ and is continuous into $X_\infty$, its level-$m$ coordinates in every chart being continuous. For every neighbourhood $N$ of a smooth point $x$ in $X_\infty$, every $y \in X_\infty$ close enough to $x$ in $X_\infty$ is joined to $x$ by an sc-smooth path all of whose points lie in $N$: in a chart, $s \mapsto r(u_0 + \kappa(s)(u_1 - u_0))$ with $\kappa$ the smooth transition function. For an open $W \ni x$ in $X$ there is an open $V$, $x \in V \subseteq W$, whose smooth points are pairwise joined by sc-smooth paths in $W$. *Book:* not stated; needed for the uniqueness in Theorem 6.6.17, p. 275, whose proof the text leaves to the reader. *Hidden hypotheses:* none. *Deviation (not in the book):* tools; the second statement makes small neighbourhoods in $X_\infty$ reachable by paths that stay in them, which is how property (3) controls the topology (Findings, item 119).

**V6.73 (the topology of $O_f$, property (1) and Hausdorffness; `Polyfold.IsContOrOn`, `Polyfold.orSec`, `Polyfold.orBasic`, `Polyfold.orTopology`, `Polyfold.IsContOrOn.eventually_iff`, `Polyfold.isOpen_orBasic`, `Polyfold.continuousOn_orSec`, `Polyfold.continuous_orProj`, `Polyfold.eq_orSec_of_mem_orBasic`, `Polyfold.orSec_mem_orBasic`, `Polyfold.exists_isContOrOn`, `Polyfold.orChartHomeo`, `Polyfold.isLocalHomeomorph_orProj`, `Polyfold.t2Space_orTopology`).** Let $L$ be local transport data for the sets $\mathrm{Or}^f_x$ (good open sets covering $X_\infty$ with push-forwards, V6.56). A continuous orientation over an open $U$ is a family $y \mapsto o_y$ carried along by the transport of $L$ along every sc-smooth path in $U$. The topology $\mathcal{T}_L$ on $O_f$: $W$ is open iff $\{y \in U_\infty : (y, o_y) \in W\}$ is open in $X_\infty$ for every continuous orientation $o$ over an open $U$. For every $L$: two continuous orientations agree near a smooth point iff they agree at it; the sets $\{(y, o_y) : y \in U_\infty\}$ are open, $y \mapsto (y, o_y)$ is continuous on $U_\infty$ and $\pi_f$ is continuous; every point of $O_f$ lies on a continuous orientation over an open set inside a good set of $L$; $\pi_f$ restricted to $\{(y, o_y) : y \in U_\infty\}$ is a homeomorphism onto $U_\infty$, so (1) $\pi_f$ is a local homeomorphism; and $\mathcal{T}_L$ is Hausdorff. *Book:* Theorem 6.6.17, p. 275, new in the 2021 edition: "a uniquely defined Hausdorff topology" with (1); the text leaves the proof to the reader, in view of Proposition 6.6.3 and Theorem 6.6.11 (preprint 6.28, 6.35). *Hidden hypotheses:* none (every $L$). *Deviation (not in the book):* the construction of the topology. Points over different smooth points are separated through $\pi_f$ ($X_\infty$ is metrizable); the two points over one $y$ lie on two continuous orientations that differ at $y$, hence near $y$ (Findings, item 118).

**V6.74 (property (2) under connected smooth neighbourhoods; `Polyfold.MPolyfold.HasConnectedSmoothNbhds`, `Polyfold.MPolyfold.hasConnectedSmoothNbhds_of_subsingleton`, `Polyfold.rnPolyfold_hasConnectedSmoothNbhds`, `Polyfold.eq_orSec_or_eq_orSec`, `Polyfold.hasTwoContSections_orTopology`).** $X$ has connected smooth neighbourhoods if every smooth point $x$ has, inside every open $W \ni x$, an open $U \ni x$ of $X$ with $U \cap X_\infty$ preconnected in $X_\infty$; $\mathbb{R}^n$ and every M-polyfold on a subsingleton have them. If every $\mathrm{Or}^f_y$ has exactly two elements and $X$ has connected smooth neighbourhoods, then for every $L$ the topology $\mathcal{T}_L$ has property (2): for such a $U$ inside the domain of a continuous orientation $o$, the continuous sections over $U \cap X_\infty$ are exactly $y \mapsto (y, o_y)$ and $y \mapsto (y, -o_y)$. *Book:* Theorem 6.6.17 (2), p. 275. *Hidden hypotheses:* connected smooth neighbourhoods (`HasConnectedSmoothNbhds`). *Deviation (narrower):* (2) is proved under this hypothesis only; Part I does not state it, it holds for $\mathbb{R}^n$ and is open in general (gap G.90). The proofs of §7.4 use that $U \cap X_\infty$ is connected for connected open $U$ (`MPolyfold.SmoothConnected`, gap G.85); as we read it, that fact gives the hypothesis, M-polyfolds being locally path connected (Proposition 2.3.7). *Proof:* the set where a continuous section equals $(y, o_y)$ is open, and so is the set where it equals $(y, -o_y)$, its complement; $U_\infty$ is preconnected (Findings, item 120).

**V6.75 (Theorem 6.6.17; `Polyfold.isPropagationContinuous_orTopology`, `Polyfold.eventually_eq_orSec_of`**

`Polyfold.eq_orTopology_of,Polyfold.existsUnique_orientationTopology_2021,Polyfold.theorem_6_6_1`
Let $X$ be a tame M-polyfold with sc-smooth bump functions, $P : Y \to X$ a strong bundle with sc$^+$-sections through every smooth fibre point, $f$ an sc-Fredholm section with tame germ data (the hypotheses of Theorem 6.6.13 (preprint 6.37), V6.57, with its paracompactness instances for $\mathbb{R}^n \times X$, $\mathbb{R}^k \times \mathbb{R}^n \times X$ and their pull-back bundles), and $\mathcal{T}$ the topology of V6.73 for the local transport data of $\Gamma_f$. Then $\mathcal{T}$ has property (3); every topology on $O_f$ with (1) and (3) is $\mathcal{T}$; there is exactly one Hausdorff topology on $O_f$ with (1) and (3); and if $X$ has connected smooth neighbourhoods (V6.74), exactly one Hausdorff topology with (1), (2) and (3). *Book:* Theorem 6.6.17, p. 275, new in the 2021 edition (the preprint and the 2017 text define the set $O_f$, V6.71, but no topology on it): for a strong bundle over a tame $X$ and an sc-Fredholm $f$ there is a uniquely defined Hausdorff topology on $O_f$ characterized by (1)–(3); "the easy proof is left to the reader". *Hidden hypotheses:* bump functions, tame germ data and sc$^+$-sections through the smooth fibre points (those of V6.57); connected smooth neighbourhoods for (2). *Deviation (narrower):* existence with (2) needs `HasConnectedSmoothNbhds` (gap G.90); the uniqueness is stronger than stated, from (1) and (3) alone, without Hausdorffness or (2). The remarks after the theorem (the orientation bundle, orientability iff a global continuous section, the continuous $\mathbb{Z}_2$-action, $\det_f = \mathbb{R} \otimes_{\mathbb{Z}_2} O_f$) are not formalized (gap G.91). *Proof:* (3): near $t_0$ a continuous section of $\varphi^* O_f$ stays in one open set $\{(y, o_y)\}$, and $o$ is carried along the segments of $\varphi$ by the transport, which on a good set is the push-forward. Uniqueness: for a topology with (3), a continuous local section through $(y_0, o_{y_0})$ equals $y \mapsto (y, o_y)$ near $y_0$, since $y$ is joined to $y_0$ by a small sc-smooth path (V6.72), along which (3) and the connectedness of $[0, 1]$ carry the equality; the local homeomorphisms of (1) then match the open sets of both topologies (Findings, items 119, 121).

**What is not verified in this chapter**

- preprint p. 201, the facts recalled before Proposition 6.2.2 (preprint 6.2): an M-polyfold is locally path connected, and two smooth points in one path component are joined by an sc$^+$-smooth path. Only the local form inside a chart is formalized (V6.49), and "in the same path component" (Definitions 6.6.2, 6.6.14 (preprint 6.27, 6.38)) is read as "joined by an sc-smooth path" (V6.58) (gap G.56; Findings, item 42).
- preprint p. 201, the members of $\mathrm{Lin}(f, x)$ as sc-Fredholm operators on every level: they are shown Fredholm on level 0, with the index $\mathrm{ind}(f, x)$, which is what the determinant lines use (V6.1) (gap G.64).
- preprint p. 208, the alternative definition $\det(T) = \lambda(\ker T) \otimes \lambda(\mathrm{coker}(T)^*)$ and its identification with Definition 6.4.1 (preprint 6.7) (gap G.57).
- preprint p. 213, the identification (95) and the maps $V \otimes \lambda(\mathbb{R}^n) \to V$, $V \otimes \lambda^*(\mathbb{R}^n) \to V$ after Proposition 6.4.15 (preprint 6.15); the formalized statements use only (94) (V6.17, V6.18) (gap G.58).
- p. 259 (preprint p. 216), Lemma 6.5.3 (preprint 6.19) in general (exterior powers, duals, sums and tensor products of finite-dimensional topological vector bundles); it is applied only to trivial bundles, and Proposition 6.5.4 (preprint 6.20) is proved with explicit frames (V6.22) (gap G.59).
- pp. 261–263 (preprint pp. 218–219), the remark that $\mathrm{DET}(E, F)$ is not orientable, and the classical case at the opening of §6.6 (preprint §6.5) (the determinant line bundle of a classical Fredholm section over a contractible $U$) (gap G.60; Findings, item 26).
- p. 263 (preprint p. 220), Proposition 6.6.3 (1) (preprint 6.28), the sc-smooth contractibility of $U$, and with it the word "contractible" in Theorem 6.6.11 (preprint 6.35), Definition 6.6.12 (preprint 6.36) and property (□ ) of Theorem 6.6.13 (preprint 6.37): (1′) is proved instead and

is what the proofs use (V6.32, V6.48, V6.50, V6.57) (gap G.61).

- p. 265 (preprint p. 222), Lemma 6.6.5 (preprint 6.30) as a statement about manifolds: the solution set as a manifold with boundary with corners near $(t, \varphi(t), 0)$, the map $\pi$ and the smooth vector bundle $L$; the kernels are formalized as a continuous family of subspaces (V6.34, V6.42) (gap G.62).
- preprint p. 222, property (P) of Lemma 6.6.4 (preprint 6.29) for the slices $F_t$; the general position of $F$ at the points $(t, \varphi(t), 0)$, which the text derives from it, is proved (V6.33) (gap G.63).
- §6.6 (preprint §6.5), paths, homotopies and families sc-smooth on $[0,1]$, $[0,1]^2$, $[0,1]^n$ only, and $X \times [0,1]$ as an M-polyfold with boundary: they are read on $\mathbb{R}$, $\mathbb{R}^2$, $\mathbb{R}^n$ and $\mathbb{R}^n \times X$ (decision D2), as for Proposition 6.2.2 (preprint 6.2) in Chapter 3; a path that does not extend sc-smoothly to $\mathbb{R}$ is not covered (gap G.65; Findings, item 102).
- p. 275 (preprint p. 229), Theorem 6.6.18 (preprint 6.41), the orientation of $S$ as an orientation of the abstract finite-dimensional manifold $S$ (of its tangent bundle in the charts of the induced structure); what is verified is the tangent orientation at each zero and its continuity along the sc-smooth paths of zeros in a chart (V6.61) (gap G.67).
- p. 275 (preprint p. 229), Theorem 6.6.18 (preprint 6.41), the compatibility of the tangent orientations read in two charts at the same zero (V6.61) (gap G.68).
- pp. 274–276 (preprint pp. 229–230), the paragraph after Theorem 6.6.18 (preprint 6.41): for $f$ not generic, a small sc$^+$-perturbation $s$ supported near the zero set, the orientation of $S_{f+s}$ from $o_f$, and $\mathrm{Lin}(f+s, x) = \mathrm{Lin}(f, x)$ for the orientation of $\det((f+s)'(x))$ from $\mathrm{DET}(f, x)$ (gap G.69).
- p. 276 (preprint p. 230), Remark 6.6.19 (preprint 6.42) beyond a point: that $(t, y) \mapsto \hat{o}$ is an orientation of $\hat{f}$ in the sense of Definition 6.6.16 (preprint 6.40) (continuity over the orientable neighbourhoods of $\mathbb{R} \times X$, independence of the chart of the pull-back bundle), built from an orientation of $f$; the remark is verified at a point (V6.62) (gap G.66).
- p. 276 (preprint p. 230), §6.7 (preprint §6.6), the sections "admissible for the pair $(N, U)$"; the facts recalled there are Theorem 5.3.10 (preprint 5.18) (V5.38, V5.39) and the orientation of $S_{f+s}$ (V6.61) (gap G.55).
- p. 277 (preprint p. 231), Theorem 6.7.1 (preprint 6.43) (the invariant $\Psi_{(f,o)}$ on $H^*_{dR}(X, \partial X)$), with its proof, pp. 277–279 (preprint pp. 231–233), and the case $\partial X = \emptyset$, p. 279 (preprint p. 233): it needs the sc-differential forms and their cohomology of §4.4 (gaps G.37–G.40), integration over oriented compact manifolds with boundary with corners and Stokes' theorem, and the proper sc$^+$-homotopy of Remark 5.3.8 (preprint 5.16) (gap G.44); not formalized (decision D1) (gap G.54). In the proof, the extension $p_1^*\omega \wedge p_2^* dt$ of $\omega$ is read as $p_1^*\omega$ (possible misprint; Findings, item 103).
- §§6.2 and §6.6 (preprint §6.1 and §6.5), $T_x X$, $Y_x$, $\mathrm{Lin}(f, x)$ and $\mathrm{Or}_x^f$ as chart-free objects: they are made in chart points and identified between charts (V6.53–V6.55), since $T_x X$ carries no linear structure here (gaps G.6, G.7, G.13).
- p. 253, Proposition 6.4.13 (2017: 6.6), the last sentence: that $A_T$ and $B_T$ depend continuously on $T$ near a given operator. Listed here when the 2021 additions were first reported; verified since, with a closed common complement (V6.69) and the continuity of $A_S$ on $E$ and $B_S$ on $F$, also on the kernel bundle (V6.70) (gap G.81, closed; Findings, items 113, 117).
- p. 275, Theorem 6.6.17, property (2) (two continuous sections over $U \cap X_\infty$): verified only for M-polyfolds with connected smooth neighbourhoods (small open $U$ with $U \cap X_\infty$ connected in $X_\infty$; V6.74, V6.75), which holds for $\mathbb{R}^n$; the existence and uniqueness without (2) need no such hypothesis (gap G.90; Findings, item 120).

- p. 275, the remarks after Theorem 6.6.17: $\pi_f$ as the orientation bundle of $f$, "$f$ orientable iff $\pi_f$ has a global continuous section", the continuous $\mathbb{Z}_2$-action on $O_f$ and the determinant bundle $\det_f = \mathbb{R} \otimes_{\mathbb{Z}_2} O_f$ (gap G.91; Findings, item 121).
- The 2021 changes to items verified earlier: Proposition 6.4.15 (preprint 6.15) with its hypotheses made explicit (V6.18, already with them), Proposition 6.6.3 (preprint 6.28) for a tame $X$ (V6.32, without tameness; (1) remains gap G.61), Lemma 6.6.5 (preprint 6.30) with the solution set near $(t, \varphi(t), 0)$ in the points $(t, y, \lambda)$ (gap G.62 unchanged) (Findings, item 115).

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## 8. Chapter 7 (HWZ, Part II, §§7.1–7.6 (2017 §§7.1–7.5)): verified except the tangent $T\varphi$ of Definition 7.1.4 (2017 Definition 7.4), the remark after Definition 7.1.7 (2017 Definition 7.7) in its 2017 form, the Hausdorff property of $\mathbf{X}^R$ in Proposition 7.2.10 (2017 Proposition 7.9) and Proposition 7.10 (2017; rewritten in 2021 as Theorem 7.2.14, which assumes it), and the index lifts $X^i$, $i \geq 2$, of Appendix 7.6.4

Chapter 7 (pp. 287–356 (2017 pp. 225–256); Part II has no 2014 preprint, so numbers and pages are those of the book (Springer 2021), with the 2017 text's in parentheses) introduces ep-groupoids, the sc-smooth version of étale proper Lie groupoids. §7.1 defines groupoids, isotropy groups and ep-groupoids (Definitions 7.1.1–7.1.3 (2017: 7.1–7.3)), the local sc-diffeomorphism $\widehat{\varphi}$ and the tangent $T\varphi$ of a morphism (Definition 7.1.4 (2017: 7.4)), the levels and the degeneracy index of the orbit space (Definition 7.1.5 (2017: 7.5)), tame ep-groupoids (Definition 7.1.6 (2017: 7.6)) and the category $\mathcal{EP}$ (Definition 7.1.7 (2017: 7.7)); it proves the openness of $\pi$, the countable neighbourhood bases of $|X|$, the local closure relation and the finiteness of the isotropy groups (Propositions 7.1.8–7.1.12 (2017: 7.1–7.4), Lemma 7.1.11 (2017: 7.1)), states the natural representation of $G_x$ (Theorem 7.1.13 (2017: 7.1)), discusses the translation groupoid $G_x \ltimes U(x)$ and the map $\widehat{\Phi}$ (pp. 293–297 (2017 pp. 231–233)), extends the natural representation to finite sets of isomorphic objects (Proposition 7.1.15 (2017: 7.5), Definition 7.1.18 (2017: 7.8)), identifies $|U(x)|$ with $G_x \backslash U(x)$ (Proposition 7.1.19 (2017: 7.6)), and defines local uniformizers and ep-subgroupoids (Definitions 7.1.20, 7.1.21 (2017: 7.9, 7.10), Proposition 7.1.22 (2017: 7.7)). §7.2 treats effective ep-groupoids (Definition 7.2.1 (2017: 7.11), Lemmas 7.2.2, 7.2.3 (2017: 7.2, 7.3), Proposition 7.2.4 (2017: 7.8)), the subgroups $T_x$, the reduced morphism space $\mathbf{X}^R$ and the reduced ep-groupoid $X_R$ (Definitions 7.2.7, 7.2.15 (2017: 7.12, 7.13), Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017)). §7.3 proves that $|X|$ is locally metrizable, regular and Hausdorff (Theorem 7.3.1 (2017: 7.2), Lemmas 7.3.2–7.3.5 (2017: 7.4–7.7), Corollaries 7.3.6, 7.3.8 (2017: 7.1, 7.2), Definition 7.3.7 (2017: 7.14)) and lifts ep-groupoids to level 1 (Propositions 7.3.9, 7.5.3 (2017: 7.11, 7.12)). §7.5 (2017 §7.4) defines paracompact orbit spaces and sc-smooth partitions of unity (Definitions 7.5.1, 7.5.5 (2017: 7.15, 7.16), Remarks 7.5.2, 7.5.6 (2017: 7.1, 7.2)), proves the existence of invariant partitions of unity (Theorems 7.5.4, 7.5.7 (2017: 7.3, 7.4), Corollary 7.6.6 (2017: 7.3)) and characterizes the closed subsets of $|X|$ (Definition 7.5.8 (2017: 7.17), Propositions 7.5.9, 7.5.10 (2017: 7.13, 7.14)). Appendix 7.6 (2017 7.5) proves Theorem 7.1.13 (2017: 7.1) as Theorem 7.6.1 (2017: 7.5) (Lemmas 7.6.2, 7.6.3 (2017: 7.8, 7.9)) and Theorem 7.5.7 (2017: 7.4) as Theorem 7.6.4 (2017: 7.6) (Lemma 7.6.5 (2017: 7.10)).

What is verified. **Theorems 7.1.13–7.6.4 (2017: 7.1–7.6) hold**, with the readings recorded below, and so do Propositions 7.1.8–7.2.4, 7.3.9–7.5.10 (2017: 7.1–7.8, 7.11–7.14), Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017) under the hypothesis that $\mathbf{X}^R$ is Hausdorff and paracompact, Lemmas 7.1.11–7.6.5 (2017: 7.1–7.10), and Corollaries 7.3.6–7.6.6 (2017: 7.1–7.3) (Corollary 7.6.6 (2017: 7.3) with the theorem of Godefroy–Troyanski–Whitfield–Zizler as an explicit hypothesis, as in Part I). Every definition of the chapter is formalized: Definitions 7.1.1 (2017: 7.1; V7.1), 7.1.2 (2017: 7.2;

V7.2), 7.1.3 (2017: 7.3; V7.3), 7.1.4 (2017: 7.4; V7.5, $T\varphi$ as a definition only), 7.1.5 (2017: 7.5; V7.7), 7.1.6 (2017: 7.6; V7.8), 7.1.7 (2017: 7.7; V7.9), 7.1.18 (2017: 7.8; V7.26), 7.1.20 (2017: 7.9; V7.28), 7.1.21 (2017: 7.10; V7.29), 7.2.1 (2017: 7.11; V7.33), 7.2.7 (2017: 7.12; V7.38), 7.2.15 (2017: 7.13; V7.41), 7.3.7 (2017: 7.14; V7.49), 7.5.1 (2017: 7.15; V7.53), 7.5.5 (2017: 7.16; V7.56), 7.5.8 (2017: 7.17; V7.59). Several places of the text needed more than a reading. (a) *The remark after Definition 7.1.7 (2017: 7.7).* That sc-smoothness of a functor on objects implies sc-smoothness on morphisms does not hold as we read it (a counterexample on the trivial $\mathbb{Z}/3$-action on $\mathbb{R}$); it holds for functors continuous on morphisms, which is what the text's argument uses (V7.10). (b) *The reduced morphism space.* The proof of Proposition 7.2.10 (2017: 7.9) gives the charts of $\mathbf{X}^R$ and their sc-smooth transitions, but, as we read it, neither Hausdorffness nor paracompactness; both are hypotheses of Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017), discharged for effective ep-groupoids (V7.39, V7.40; gap G.71). (c) *Connectedness.* The natural representation of Theorem 7.1.13 (2017: 7.1) is taken on a connected $U$, which the text chooses on 2017 p. 231 but does not put in the statement; it is what makes $\Phi$ a homomorphism, $\Gamma$ a functor, and Lemma 7.2.2 (2017: 7.2) true, and the local path connectedness of M-polyfolds that supplies it is proved (V7.17, V7.18, V7.34). (d) *Tameness of $X^1$.* Proposition 7.3.9's (2017: 7.11) "if $X$ is tame so is $X^1$" needs that the index lift of a tame retraction is tame, which Part I left open; it is proved (V7.52). (e) *Omitted steps.* The uniqueness of natural representations behind "adapting restrictions" (V7.24, V7.35), the conjugation $\varphi T_x \varphi^{-1} = T_y$ needed for $\mathbf{X}^R$ (V7.37), the passage from compact to component compact sets in Corollary 7.3.8 (2017: 7.2) (V7.51), the support bound that makes the sum in the proof of Theorem 7.5.7 (2017: 7.4) locally finite (V7.66, V7.67), and Michael's theorem on unions of closed paracompact sets (V7.50) are proved; Proposition 7.1.15 (2017: 7.5) is proved by a construction without induction (V7.25). Not verified: the properties of $T\varphi$ (Definition 7.1.4 (2017: 7.4)), the set $|X_\infty|$ of the filtration of $|X|$, the tangent facts (3), (4) recalled before Proposition 7.1.22 (2017: 7.7), Hausdorffness and paracompactness of $\mathbf{X}^R$ for a general ep-groupoid, and the open question of 2017 p. 249 (see the list at the end of this section).

The setting. Obj and Mor are types with M-polyfold structures `PObj`, `PMor`, Hausdorff and paracompact, as for every M-polyfold of the project (Definition 2.3.4 (2017: 2.8)). An ep-groupoid is `Polyfold.EpGroupoid`, a `Polyfold.GroupoidStr` (the maps $s, t, m, u, \iota$ with the groupoid laws on composable pairs; the composition of a non-composable pair is a junk value, never used) with the conditions (1)–(4) of Definition 7.1.3 (2017: 7.3). Properness is Mathlib's `IsProperMap` of the restriction $t : s^{-1}(\mathrm{cl}_X V) \to X$, equivalent to the sequential form the proofs use (V7.4). "An open neighbourhood $U(x)$ equipped with the natural $G_x$-action" is a datum `Polyfold.EpGroupoid.NaturalRep` (V7.17), never a hypothesis "for every such $U(x)$". $|X|$ carries the quotient topology. Part I results are cited in the same way, by their 2021 numbers with the 2017 numbers in parentheses. Findings referred to below are those of `Exposition/Findings.md`, "Chapter 7".

**V7.1 (Definition 7.1.1 (2017: 7.1), groupoids and the orbit space; `Polyfold.GroupoidStr`, `Polyfold.GroupoidStr.src`, `Polyfold.GroupoidStr.tgt`, `Polyfold.GroupoidStr.comp`, `Polyfold.GroupoidStr.unit`, `Polyfold.GroupoidStr.inv`, `Polyfold.GroupoidStr.comp_assoc`, `Polyfold.GroupoidStr.comp_unit`, `Polyfold.GroupoidStr.unit_comp`, `Polyfold.GroupoidStr.comp_inv`, `Polyfold.GroupoidStr.inv_comp`, `Polyfold.GroupoidStr.mor`, `Polyfold.GroupoidStr.morOn`, `Polyfold.GroupoidStr.Isomorphic`, `Polyfold.GroupoidStr.orbitSetoid`, `Polyfold.GroupoidStr.OrbitSpac` `Polyfold.GroupoidStr.`, `Polyfold.GroupoidStr._eq_iff`, `Polyfold.GroupoidStr.saturation`, `Polyfold.GroupoidStr.IsSaturated`, `Polyfold.GroupoidStr.isSaturated_iff`, `Polyfold.GroupoidStr.Ob`, `Polyfold.GroupoidStr.instGroupoid`, `Polyfold.GroupoidStr.orbitSetoid_eq`).** A groupoid

is a set of objects and a set of invertible morphisms with source, target, an associative composition on composable pairs, units and inverses; $|X|$ is the quotient by $x \sim y$ iff there is a morphism $x \to y$, with projection $\pi$. *Book:* Definition 7.1.1 (2017: 7.1) and the paragraph after it, pp. 287–288 (2017 pp. 225–226). *Hidden hypotheses:* none. *Deviation (restated):* a structure of maps on types, the laws required on composable pairs and the composition total with a junk value; the saturation $\pi^{-1}(\pi(A))$ of Definition 7.1.21 (2017: 7.10) is defined here; the bridge to Mathlib's `CategoryTheory.Groupoid` is added so that "full" and "faithful" are Mathlib's (Findings, item 1).

**V7.2 (Definition 7.1.2 (2017: 7.2), the isotropy group; `Polyfold.GroupoidStr.Isotropy`, `Polyfold.GroupoidStr.Isotropy.val_mul`, `Polyfold.GroupoidStr.Isotropy.val_one`, `Polyfold.GroupoidStr.Isotropy.val_inv`, `Polyfold.GroupoidStr.isotropyEquivVertexGroup`, `Polyfold.GroupoidStr.isotropyEquivAut`).** $G_x$ is the group of morphisms $x \to x$ under composition. *Book:* Definition 7.1.2 (2017: 7.2), p. 288 (2017 p. 226). *Hidden hypotheses:* none. *Deviation (restated):* the multiplication is $gh = g \circ h$, the text's; Mathlib's vertex group multiplies in the other order and is identified with the opposite group (Findings, item 2).

**V7.3 (Definition 7.1.3 (2017: 7.3), ep-groupoids; `Polyfold.EpGroupoid`, `Polyfold.EpGroupoid.PObj`, `Polyfold.EpGroupoid.PMor`, `Polyfold.EpGroupoid.isLocalScDiffeo_src`, `Polyfold.EpGroupoid.isLocalScD`, `Polyfold.EpGroupoid.isScSmoothMap_unit`, `Polyfold.EpGroupoid.isScSmoothMap_inv`, `Polyfold.EpGroupoid.exists_isProperMap`, `Polyfold.EpGroupoid.isScSmoothComp`, `Polyfold.IsScSmoothCo`, `Polyfold.IsScSmoothCompSheet`, `Polyfold.isScSmoothComp_iff_sheet`, `Polyfold.EpGroupoid.isScSmoothCo`, `Polyfold.EpGroupoid.isScSmoothMap_src`, `Polyfold.EpGroupoid.isScSmoothMap_tgt`, `Polyfold.MPolyfold.isSubMPolyfold_fibredProduct`).** An ep-groupoid is a groupoid with M-polyfold structures on objects and morphisms such that (1) $s, t$ are local sc-diffeomorphisms, (2) $u$ and $\iota$ are sc-smooth, (3) every $x$ has an open $V(x)$ with $t : s^{-1}(\mathrm{cl}_X V(x)) \to X$ proper, (4) $m$ is sc-smooth on the fibred product $\mathbf{X} \,{}_s\!\times_t \mathbf{X}$; all structure maps are then sc-smooth. *Book:* Definition 7.1.3 (2017: 7.3), pp. 288 ff. (2017 pp. 226–227). *Hidden hypotheses:* Obj and Mor Hausdorff and paracompact (the project's M-polyfolds). *Deviation (restated):* (3) existential per point, with Mathlib's proper maps; the universal form "for every open $V \ni x$" fails (the pair groupoid of $\mathbb{Z}$), and every smaller open neighbourhood works. (4) on the sub-M-polyfold of Part I's Proposition 2.5.20 (2017: 2.15), the book's statement of the text's citation "Lemma 2.8 in [38]" (an external reference, not a misprint); the proofs use the equivalent sheet form (Findings, items 3, 4, 103).

**V7.4 (properness and its sequential form; `Polyfold.isProperMap_iff_exists_tendsto_subseq`, `Polyfold.exists_tendsto_subseq_of_isProperMap`, `Polyfold.isProperMap_domRestrict_iff`, `Polyfold.isProperMap_domRestrict_iff_of_isClosed`, `Polyfold.MPolyfold.isProperMap_domRestrict_iff`, `Polyfold.MPolyfold.isProperMap_domRestrict_iff_isCompact_preimage`, `Polyfold.isProperMap_domRestr`, `Polyfold.isClosed_image_preimage_of_isProperMap_domRestrict`, `Polyfold.isClosed_of_isProperMap_do`, `Polyfold.finite_setOf_of_isProperMap_domRestrict`).** For the restriction of $t$ to a closed $s^{-1}(Q)$ between metrizable spaces, properness is equivalent to: every sequence with convergent targets has a convergent subsequence; it passes to smaller closures, and proper restrictions have closed images and finite fibres. *Book:* the notion of Definition 7.1.3 (3) (2017: 7.3), 2017 p. 227, used in sequential form in Proposition 7.1.10 (2017: 7.3), Lemma 7.1.11 (2017: 7.1) (p. 292 (2017 p. 230)), Proposition 7.1.22 (2017: 7.7) (p. 307 (2017 p. 238)), Proposition 7.3.9 (2017: 7.11) (2017 p. 248), Proposition 7.5.9 (2017: 7.13) (p. 342 (2017 p. 251)) and Lemma 7.6.2 (2017: 7.8) (2017 p. 254). *Hidden hypotheses:* metrizability (Theorem 2.3.10 (2017: 2.2)), which M-polyfolds have. *Deviation (not in the book):* the equivalence is used without comment in the text (Findings, item 3).

**V7.5 (Definition 7.1.4 (2017: 7.4), the local sc-diffeomorphism $\hat{\varphi}$ and the tangent $T\varphi$; `Polyfold.EpGroupoid.Sheet`, `Polyfold.EpGroupoid.Sheet.hat`, `Polyfold.EpGroupoid.Sheet.isScDiffeoOn_`**

`Polyfold.EpGroupoid.exists_sheet`, `Polyfold.EpGroupoid.sheetAt`, `Polyfold.EpGroupoid.mem_sheetAt`, `Polyfold.EpGroupoid.hat`, `Polyfold.EpGroupoid.isScDiffeoOn_hat`, `Polyfold.EpGroupoid.hat_src`, `Polyfold.EpGroupoid.tangentMor`). For a morphism $\varphi : x \to y$ there are open $\mathbf{U}(\varphi), U(x), U(y)$ with $s : \mathbf{U}(\varphi) \to U(x)$, $t : \mathbf{U}(\varphi) \to U(y)$ sc-diffeomorphisms, and $\widehat{\varphi} = t \circ (s|\mathbf{U}(\varphi))^{-1} : U(x) \to U(y)$ is an sc-diffeomorphism with $\widehat{\varphi}(x) = y$; $T\varphi := T\widehat{\varphi}(x)$. *Book:* 2017 p. 227 (the paragraph before the definition) and Definition 7.1.4 (2017: 7.4), p. 289 (2017 p. 227). *Hidden hypotheses:* none. *Deviation (narrower):* $\widehat{\varphi}$ is verified; $T\varphi$ is a definition only, and that it is a bounded linear isomorphism on level 0 for $\varphi \in \mathbf{X}_1$ and an sc-isomorphism for smooth $\varphi$ is not verified (gaps G.7, G.70). The display reads "$\widehat{\varphi} = t \circ (s|\mathbf{U}(\varphi))^{-1} \circ U(x) \to U(y)$", $\circ$ for : (possible misprint), and "$T\varphi : T_{s(\varphi)}X \to T_{t(\varphi)}$" lacks the $X$ (possible misprint or extraction artefact) (Findings, items 5, 110).

**V7.6 (the levels of an ep-groupoid and the filtration of $|X|$;** `Polyfold.EpGroupoid.mem_levelSet_src_iff`, `Polyfold.EpGroupoid.mem_levelSet_tgt_iff`, `Polyfold.MPolyfold.maxLevel`, `Polyfold.MPolyfold.IsLocal` `Polyfold.MPolyfold.IsLocalScDiffeo.degIndex_eq`, `Polyfold.EpGroupoid.degIndex_src`, `Polyfold.EpGroupoid.degIndex_tgt`, `Polyfold.EpGroupoid.orbitLevel`). $s$ and $t$ preserve the levels, so $ml(x) = ml(\varphi) = ml(y)$ for $\varphi : x \to y$; $d_X(s(\varphi)) = d_{\mathbf{X}}(\varphi) = d_X(t(\varphi))$, so $d_X$ is constant on classes; $|X_k| = \pi(X_k)$. *Book:* pp. 289–291 (2017 pp. 227–228), with Part I's Proposition 2.4.4 (2017: 2.7). *Hidden hypotheses:* none. *Deviation (narrower):* the set $|X_\infty|$ and $|X_\infty| = \bigcap_k |X_k|$ are not stated (gap G.72); the density of $X_\infty$ in every $X_k$, stated on 2017 p. 227 without proof, is not formalized and is not in Part I for M-polyfolds (gap G.73; Findings, items 6, 112, 116).

**V7.7 (Definition 7.1.5 (2017: 7.5), the induced degeneracy index;** `Polyfold.EpGroupoid.degIndexOrbit`, `Polyfold.EpGroupoid.degIndexOrbit_` **).** $d_{|X|}(|x|) = d_X(x)$ is well defined. *Book:* Definition 7.1.5 (2017: 7.5), p. 290 (2017 p. 228). *Hidden hypotheses:* none. *Deviation:* none (Findings, item 6).

**V7.8 (Definition 7.1.6 (2017: 7.6), tame ep-groupoids, and the tameness of X;** `Polyfold.EpGroupoid.IsTame`, `Polyfold.MPolyfold.IsTame`, `Polyfold.MPolyfold.IsScDiffeoOn.exists_tame_chart`, `Polyfold.MPolyfold.IsLocalScDiffeo.isTame`, `Polyfold.MPolyfold.IsLocalScDiffeo.isTame_iff_of_sur` $X$ is tame if the object M-polyfold is tame; then the morphism M-polyfold is tame. *Book:* Definition 7.1.6 (2017: 7.6) and the paragraph after it, 2017 p. 228. *Hidden hypotheses:* none; "tame" is the existence of an equivalent tame atlas (Definition 2.5.7 (2017: 2.19)), satisfiable (`Polyfold.EpGroupoid.isTame_trivial_rn`, `Polyfold.EpGroupoid.Examples.isTame_reflection`); "every chart tame" would be unsatisfiable. *Deviation (restated):* the tame atlas of $\mathbf{X}$ is the one transported along $s$ (Findings, items 7, 106).

**V7.9 (Definition 7.1.7 (2017: 7.7), the category $\mathcal{EP}$;** `Polyfold.GroupoidFunctor`, `Polyfold.GroupoidFunctor.mor` `Polyfold.GroupoidFunctor.id`, `Polyfold.GroupoidFunctor.comp`, `Polyfold.GroupoidFunctor.toFunctor`, `Polyfold.GroupoidFunctor.orbitMap`, `Polyfold.EpFunctor`, `Polyfold.EpFunctor.isScSmoothMap_obj`, `Polyfold.EpFunctor.isScSmoothMap_mor`, `Polyfold.EpFunctor.id`, `Polyfold.EpFunctor.comp`, `Polyfold.EpFunctor.ext`, `Polyfold.EP`). The objects of $\mathcal{EP}$ are the ep-groupoids, its morphisms the functors sc-smooth on objects and on morphisms. *Book:* Definition 7.1.7 (2017: 7.7), p. 290 (2017 p. 228). *Hidden hypotheses:* none. *Deviation (restated):* functors commute with $s, t, m, u$; the compatibility with $\iota$ is derived (Findings, item 8).

**V7.10 (the remark after Definition 7.1.7 (2017: 7.7): sc-smoothness on objects suffices;** `Polyfold.EpGroupoid.mor_eq_srcInv_obj_src`, `Polyfold.EpGroupoid.isScSmoothMap_mor_of_continuous`, `Polyfold.EpGroupoid.isScSmoothMap_mor_iff_continuous`, `Polyfold.EpFunctor.continuous_mor`, `Polyfold.EpFunctor.ofContinuous`, `Polyfold.EpGroupoid.Examples.remark_definition_7_7_counterexamp` The text claims that a functor sc-smooth on objects is sc-smooth on morphisms, by $f(\gamma) =$

$(s|\mathbf{U}(\psi))^{-1}(f(s(\gamma)))$ on $\mathbf{U}(\varphi)$. *Book:* pp. 290–292 (2017 pp. 228–229). *Hidden hypotheses:* the argument needs $f(\mathbf{U}(\varphi)) \subseteq \mathbf{U}(\psi)$, i.e. continuity of $f$ on morphisms. *Deviation (does not hold as we read it):* a functor of the trivial $\mathbb{Z}/3$-action on $\mathbb{R}$, the identity on objects, is sc-smooth on objects and not continuous on morphisms; for functors continuous on morphisms the claim is proved, with a sheet $\mathbf{U}(\psi)$ at $\psi = f(\varphi)$ and $\mathbf{U}(\varphi) = f^{-1}(\mathbf{U}(\psi))$ (Findings, items 9, 104).

**V7.11 (the quotient topology and Proposition 7.1.8 (2017: 7.1), openness of $\pi$;** `Polyfold.EpGroupoid.continuous` `Polyfold.EpGroupoid.proposition_7_1`, `Polyfold.EpGroupoid.proposition_7_1_isOpenQuotientMap`, `Polyfold.EpGroupoid.proposition_7_1_map_nhds`**).** $|X|$ carries the quotient topology and $\pi$ is open. *Book:* 2017 p. 229 and Proposition 7.1.8 (2017: 7.1), p. 291 (2017 p. 229). *Hidden hypotheses:* none. *Deviation (restated):* proved through the saturation $\pi^{-1}(\pi(U)) = t(s^{-1}(U))$ and the openness of $t$ (Findings, item 10).

**V7.12 (Proposition 7.1.9 (2017: 7.2), countable neighbourhood bases;** `Polyfold.EpGroupoid.proposition_7_2`, `Polyfold.EpGroupoid.proposition_7_2_of_hasAntitoneBasis`, `Polyfold.EpGroupoid.proposition_7_2_fi`
Every $z \in |X|$ has a monotone countable basis of open neighbourhoods. *Book:* Proposition 7.1.9 (2017: 7.2), p. 291 (2017 p. 229), with Theorem 2.3.10 (2017: 2.2). *Hidden hypotheses:* none. *Deviation (restated):* Mathlib's `HasAntitoneBasis`, radii $1/(k+1)$ (Findings, item 10).

**V7.13 (Proposition 7.1.10 (2017: 7.3), the local closure relation;** `Polyfold.EpGroupoid.proposition_7_3`, `Polyfold.EpGroupoid.proposition_7_3_of_isProperMap`**).** Every $x$ has an open $U$ with $\pi(\mathrm{cl}_X A) = \mathrm{cl}_{|X|}\pi(A)$ for all $A \subseteq U$. *Book:* Proposition 7.1.10 (2017: 7.3), pp. 292 ff. (2017 pp. 229–230). *Hidden hypotheses:* none. *Deviation (stronger):* for every $U$ with the properness property and every $A \subseteq \mathrm{cl}_X U$, proved from the closedness of the proper map instead of sequences (Findings, item 11).

**V7.14 (Lemma 7.1.11 (2017: 7.1);** `Polyfold.EpGroupoid.lemma_7_1`, `Polyfold.EpGroupoid.lemma_7_1_swap`, `Polyfold.EpGroupoid.lemma_7_1_of_isProperMap`, `Polyfold.EpGroupoid.lemma_7_1_swap_of_isProperMap`, `Polyfold.finite_setOf_mem_of_isProperMap_domRestrict`, `Polyfold.IsProperMap.finite_preimage_singl`
If $t : s^{-1}(\mathrm{cl}_X U(x_0)) \to X$ is proper, only finitely many $\psi$ have $s(\psi) = x$ and $t(\psi) \in U(x_0)$. *Book:* Lemma 7.1.11 (2017: 7.1), p. 292 (2017 p. 230). *Hidden hypotheses:* none; the openness of $U(x_0)$ and $x_0 \in U(x_0)$ are not used. *Deviation (restated):* the proof treats the inverses ($s$ and $t$ exchanged), and both forms are proved; finiteness from a compact discrete fibre. Possible misprints in the proof: "$t : V(\psi) \to V(y)$" for $V(x)$, "$s(\psi) \in s^{-1}(\mathrm{cl}_X(U(x_0)))$", and unbalanced parentheses in statement and proof (Findings, items 12, 96).

**V7.15 (Proposition 7.1.12 (2017: 7.4), finiteness of the isotropy groups;** `Polyfold.EpGroupoid.proposition_7_4`, `Polyfold.MPolyfold.finite_isotropy_of_isLocalScDiffeo`**).** $G_x$ is finite. *Book:* Proposition 7.1.12 (2017: 7.4), p. 293 (2017 p. 230). *Hidden hypotheses:* none. *Deviation:* none; from Lemma 7.1.11 (2017: 7.1) with $x_0 = x$ (Findings, item 13).

**V7.16 (the identification $G_x \to G_y$, $g \mapsto \varphi \circ g \circ \varphi^{-1}$;** `Polyfold.GroupoidStr.isotropyConj`, `Polyfold.GroupoidStr.isotropyConj_val`, `Polyfold.EpGroupoid.conjIsotropy`, `Polyfold.EpGroupoid.conj` `Polyfold.EpGroupoid.nonempty_isotropy_mulEquiv`, `Polyfold.EpGroupoid.nat_card_isotropy_eq`, `Polyfold.EpGroupoid.isotropyConj_eq_isotropyConj_conj`**).** Isomorphic objects have isomorphic isotropy groups, by conjugation with a chosen $\varphi : x \to y$; there is no canonical identification. *Book:* 2017 p. 230, after Proposition 7.1.12 (2017: 7.4). *Hidden hypotheses:* none. *Deviation (stronger):* two choices of $\varphi$ differ by an inner automorphism of $G_x$ (equivalently, of $G_y$) (Findings, item 13).

**V7.17 (Theorem 7.1.13 (2017: 7.1), the natural representation of $G_x$;** `Polyfold.EpGroupoid.NaturalRep`, `Polyfold.EpGroupoid.NaturalRep.U`, `Polyfold.EpGroupoid.NaturalRep.isOpen_U`, `Polyfold.EpGroupoid.Na`

`Polyfold.EpGroupoid.NaturalRep.mem_U`, `Polyfold.EpGroupoid.NaturalRep.Φ`, `Polyfold.EpGroupoid.Natur`
`Polyfold.EpGroupoid.NaturalRep.isScDiffeoOn_Φ`, `Polyfold.EpGroupoid.NaturalRep.Φ_mul`,
`Polyfold.EpGroupoid.NaturalRep.isScSmoothOn_Γ`, `Polyfold.EpGroupoid.NaturalRep.Γ_self`,
`Polyfold.EpGroupoid.NaturalRep.src_Γ`, `Polyfold.EpGroupoid.NaturalRep.tgt_Γ`, `Polyfold.EpGroupoid.N`
`Polyfold.EpGroupoid.NaturalRep.Φ_one`, `Polyfold.EpGroupoid.NaturalRep.mapsTo_Φ`,
`Polyfold.EpGroupoid.NaturalRep.Φ_inv_Φ`, `Polyfold.EpGroupoid.NaturalRep.bijOn_Γ`,
`Polyfold.EpGroupoid.theorem_7_1`**).** For every open $V \ni x$ there are an open $U \subseteq V$ of $x$, a homomorphism $\Phi : G_x \to \mathrm{Diff}_{sc}(U)$ and an sc-smooth $\Gamma : G_x \times U \to \mathbf{X}$ with (1) $\Gamma(g,x) = g$, (2) $s(\Gamma(g,y)) = y$, $t(\Gamma(g,y)) = \Phi(g)(y)$, (3) $\Gamma$ a bijection onto the morphisms between points of $U$. *Book:* Theorem 7.1.13 (2017: 7.1), p. 293 (2017 p. 231) (proved as Theorem 7.6.1 (2017: 7.5), V7.62). *Hidden hypotheses:* $U$ connected, which the text chooses on 2017 p. 231 and the homomorphism property needs. *Deviation (stronger):* $U$ connected; $\Phi(g)$ an sc-diffeomorphism of $U$ with inverse $\Phi(g^{-1})$; $\Gamma$ sc-smooth on each slice $\{g\} \times U$, which is sc-smoothness on $G_x \times U$ (V7.20) (Findings, item 20).

**V7.18 (local path connectedness of M-polyfolds;** `Polyfold.ScRetraction.exists_ball_joinedIn`,
`Polyfold.IsScRetract.locallyPathConnectedSpace`, `Polyfold.MChart.pathComponentIn_mem_nhds`,
`Polyfold.MPolyfold.locallyPathConnectedSpace`, `Polyfold.MPolyfold.locallyConnectedSpace`,
`Polyfold.MPolyfold.hasBasis_isOpen_isConnected`, `Polyfold.MPolyfold.isOpen_connectedComponent`,
`Polyfold.MPolyfold.isConnected_iff_isPathConnected`, `Polyfold.MPolyfold.locallyPathConnectedSpace`
`Polyfold.eqOn_sections_of_isPreconnected`**).** M-polyfolds and all their quotients are locally path connected; open connected sets are path connected; two sections of a local homeomorphism agreeing at a point agree on a connected set. *Book:* used without statement on p. 293 (2017 p. 231) ("the connected open neighborhood $U(x)$"), in Lemma 7.6.3 (2017: 7.9) (p. 345 (2017 p. 254)) and Corollary 7.3.8 (2017: 7.2) (p. 326 (2017 p. 247)). *Hidden hypotheses:* none. *Deviation (not in the book):* the first fact of gap G.56 of Part I, now formalized (Findings, items 21, 114).

**V7.19 (the splitting $\mathbf{U}(x) = \bigsqcup_g \mathbf{U}(g)$;** `Polyfold.EpGroupoid.NaturalRep.isOpen_image_Γ`,
`Polyfold.EpGroupoid.NaturalRep.Γ_mem_image`, `Polyfold.EpGroupoid.NaturalRep.pairwise_disjoint_ima`
`Polyfold.EpGroupoid.NaturalRep.iUnion_image_Γ`, `Polyfold.EpGroupoid.NaturalRep.isOpen_morOn`,
`Polyfold.EpGroupoid.NaturalRep.isConnected_image_Γ`, `Polyfold.EpGroupoid.NaturalRep.connectedComp`
`Polyfold.EpGroupoid.NaturalRep.isScDiffeoOn_src_image`, `Polyfold.EpGroupoid.NaturalRep.isScDiffeo`
`Polyfold.EpGroupoid.NaturalRep.index`, `Polyfold.EpGroupoid.NaturalRep.Γ_index`,
`Polyfold.EpGroupoid.lemma_7_8`**).** The morphisms between points of $U(x)$ form the disjoint union of the connected open sets $\mathbf{U}(g) = \Gamma(g, U)$, on which $s$ and $t$ are sc-diffeomorphisms onto $U(x)$, and $\mathbf{U}(x)$ can be put into any open neighbourhood of $G_x$. *Book:* 2017 p. 231. *Hidden hypotheses:* none. *Deviation:* none; "arbitrarily small" is Lemma 7.6.2 (2017: 7.8) for arbitrary open $N_g$ followed by Theorem 7.1.13 (2017: 7.1) (Findings, item 22).

**V7.20 (the M-polyfold $G \times U$;** `Polyfold.discretePolyfold`, `Polyfold.discreteChart`,
`Polyfold.discreteChart_compatible`, `Polyfold.isScSmoothMap_discrete_iff`, `Polyfold.isScSmoothOn_uni`
`Polyfold.isScSmoothMap_prod_discrete_iff`, `Polyfold.EpGroupoid.NaturalRep.objPolyfold`,
`Polyfold.EpGroupoid.NaturalRep.morPolyfold`, `Polyfold.EpGroupoid.NaturalRep.isScSmoothMap_smul`**).** $G \times U$ is an M-polyfold, the finite disjoint union of copies of $U$. *Book:* 2017 p. 232. *Hidden hypotheses:* none. *Deviation (restated):* the product with the discrete M-polyfold on $G$; maps on it are sc-smooth iff they are on each slice (Findings, item 23).

**V7.21 (the translation groupoid $G_x \ltimes U(x)$ is an ep-groupoid;** `Polyfold.GroupoidStr.translation`,
`Polyfold.EpGroupoid.NaturalRep.translationGroupoid`, `Polyfold.EpGroupoid.NaturalRep.shearHomeomor`
`Polyfold.EpGroupoid.NaturalRep.translation_isLocalScDiffeo_src`, `Polyfold.EpGroupoid.NaturalRep.t`

`Polyfold.EpGroupoid.NaturalRep.translation_isScSmoothCompSheet`, `Polyfold.EpGroupoid.NaturalRep.t`
`Polyfold.EpGroupoid.NaturalRep.translationEpGroupoid`**).** The category with objects $U$, morphisms $G \times U$, $s(g,y) = y$, $t(g,y) = \Phi(g)(y)$, $(g,y) \circ (h,z) = (gh,z)$ is an ep-groupoid. *Book:* p. 294 (2017 p. 232) ("using the proof of Theorem 7.1 one verifies"). *Hidden hypotheses:* none. *Deviation (stronger):* built from the data $(\Phi, \Gamma)$ alone, with the properness for every open $V$ (Findings, item 22).

**V7.22 (the functor $\Gamma : G_x \ltimes U(x) \to X$ is fully faithful;** `Polyfold.EpGroupoid.NaturalRep.Γ_mul`, `Polyfold.EpGroupoid.NaturalRep.Γ_one`, `Polyfold.EpGroupoid.NaturalRep.Γ_inv`, `Polyfold.EpGroupoid.N` `Polyfold.EpGroupoid.NaturalRep.full`, `Polyfold.EpGroupoid.NaturalRep.faithful`, `Polyfold.EpGroupoid.NaturalRep.epFunctor`, `Polyfold.GroupoidStr.fullSub`**).** $\Gamma$ is a functor, full and faithful by Theorem 7.1.13 (3) (2017: 7.1), with image the full subcategory on $U(x)$. *Book:* 2017 p. 232. *Hidden hypotheses:* $U$ connected (V7.17), for $\Gamma$ to be a functor. *Deviation (stronger):* also an ep-functor, `Polyfold.EpGroupoid.NaturalRep.epFunctor` (Findings, item 22).

**V7.23 (the map $\hat{\Phi} : G_x \to \mathrm{Diff}_{sc}(\mathbf{U}(x))$;** `Polyfold.EpGroupoid.NaturalRep.Φhat`, `Polyfold.EpGroupoid.Natur` `Polyfold.EpGroupoid.NaturalRep.Φhat_mem_image`, `Polyfold.EpGroupoid.NaturalRep.index_Φhat`, `Polyfold.EpGroupoid.NaturalRep.src_Φhat`, `Polyfold.EpGroupoid.NaturalRep.tgt_Φhat`, `Polyfold.EpGroupoid.NaturalRep.Φ_conj`, `Polyfold.EpGroupoid.NaturalRep.Φhat_one`, `Polyfold.EpGroupoid.NaturalRep.Φhat_mul`, `Polyfold.EpGroupoid.NaturalRep.isScDiffeoOn_Φhat`, `Polyfold.EpGroupoid.NaturalRep.image_Φhat`**).** $\hat{\Phi}(g)(\varphi) = \Gamma(ghg^{-1}, g * s(\varphi))$ for $\varphi \in \mathbf{U}(h)$ is a homomorphism into the sc-diffeomorphisms of $\mathbf{U}(x)$ with $\hat{\Phi}(g)(\mathbf{U}(h)) = \mathbf{U}(ghg^{-1})$ and the two commutative diagrams with $s$ and $t$. *Book:* pp. 294–297 (2017 pp. 232–233). *Hidden hypotheses:* none. *Deviation:* none; no possible misprint on pp. 293–297 (2017 pp. 231–233) (Findings, item 22).

**V7.24 (uniqueness and restriction of natural representations;** `Polyfold.EpGroupoid.NaturalRep.Γ_eqOn`, `Polyfold.EpGroupoid.NaturalRep.Φ_eqOn`, `Polyfold.EpGroupoid.NaturalRep.mapsTo_Φ_connectedComponer` `Polyfold.EpGroupoid.NaturalRep.eventually_Φ_eq_of_actsTrivially`, `Polyfold.EpGroupoid.NaturalRep.` `Polyfold.EpGroupoid.NaturalRep.restrictInter`, `Polyfold.EpGroupoid.exists_naturalRep_restrict`, `Polyfold.EpGroupoid.exists_naturalRep_shrink`**).** Two natural representations at $x$ agree on the component of $x$ in $U \cap U'$, and a natural representation restricts to invariant connected open neighbourhoods. *Book:* used without statement on pp. 307–311 (2017 pp. 239–241) ("adapting restrictions", "$U(x)$ … can be chosen arbitrarily small"). *Hidden hypotheses:* none. *Deviation (not in the book):* stated and proved (Findings, item 24).

**V7.25 (Proposition 7.1.15 (2017: 7.5), the natural representation of a finite set of isomorphic objects;** `Polyfold.EpGroupoid.NaturalRepFinite`, `Polyfold.EpGroupoid.NaturalRepFinite.F`, `Polyfold.EpGroupoid.NaturalRepFinite.isScDiffeoOn_F`, `Polyfold.EpGroupoid.NaturalRepFinite.F_comp` `Polyfold.EpGroupoid.NaturalRepFinite.F_unit`, `Polyfold.EpGroupoid.NaturalRepFinite.pairwiseDisjoi` `Polyfold.EpGroupoid.NaturalRepFinite.morOn_eq`, `Polyfold.EpGroupoid.NaturalRepFinite.isScDiffeoOn` `Polyfold.EpGroupoid.NaturalRepFinite.isScDiffeoOn_tgt`, `Polyfold.EpGroupoid.NaturalRepFinite.F_eq` `Polyfold.EpGroupoid.NaturalRepFinite.natural`, `Polyfold.EpGroupoid.NatRepFinite.transportRep`, `Polyfold.EpGroupoid.NatRepFinite.Star`, `Polyfold.EpGroupoid.NatRepFinite.exists_star`, `Polyfold.EpGroupoid.NatRepFinite.emptyNaturalRepFinite`, `Polyfold.EpGroupoid.proposition_7_5`, `Polyfold.EpGroupoid.proposition_7_5_of_mem_nhdsSet`**).** For finitely many mutually isomorphic objects $Y$ and a neighbourhood $V$ of $Y$ there is a functor $f$ assigning disjoint connected open $U(y) \subseteq V$ and sc-diffeomorphisms $f(\psi) : U(y) \to U(y')$, such that the morphisms between the $U(y)$ split into components $\mathbf{U}(\psi)$ with (1) the natural action of $G_y$ on $U(y)$, (2) $s, t : \mathbf{U}(\psi) \to U(y)$ sc-diffeomorphisms, (3) $f(\psi) = t \circ (s|\mathbf{U}(\psi))^{-1}$. *Book:* Proposition 7.1.15 (2017: 7.5), pp. 297–304

(2017 pp. 233–236). *Hidden hypotheses:* none. *Deviation (restated):* proved by transporting one natural representation along chosen morphisms from a base object, without the text's induction or shrinking; $V$ a neighbourhood of the set $Y$; $Y = \emptyset$ allowed. Twelve possible misprints in the proof, one possibly an extraction artefact (Findings, items 25, 26).

**V7.26 (Definition 7.1.18 (2017: 7.8), the natural representation of Y;** `Polyfold.EpGroupoid.NaturalRepFinite, Polyfold.EpGroupoid.NatRepFinite.mono`**).** The functor of Proposition 7.1.15 (2017: 7.5) with its properties. *Book:* Definition 7.1.18 (2017: 7.8), p. 304 (2017 p. 236). *Hidden hypotheses:* none. *Deviation:* none (Findings, item 25).

**V7.27 (Proposition 7.1.19 (2017: 7.6),** $|U(x)| \cong G_x\backslash U(x)$**;** `Polyfold.EpGroupoid.NaturalRep. _eq_ _iff_exist Polyfold.EpGroupoid.NaturalRep.orbitQuotientMap, Polyfold.EpGroupoid.NaturalRep.isOpenEmbedding_ Polyfold.EpGroupoid.NaturalRep.range_orbitQuotientMap, Polyfold.EpGroupoid.proposition_7_6, Polyfold.EpGroupoid.proposition_7_6_fullyFaithful, Polyfold.EpGroupoid.proposition_7_6_orbitMap, Polyfold.EpGroupoid.translationOrbitHomeomorph`**).** $\tau : G_x\backslash U(x) \to \pi(U(x)), G_x * y \mapsto |y|$, is a homeomorphism; the functor $G_x \ltimes U(x) \to X$ is fully faithful and induces a homeomorphism of orbit spaces. *Book:* Proposition 7.1.19 (2017: 7.6), pp. 304 ff. (2017 pp. 236–237). *Hidden hypotheses:* none. *Deviation (stronger):* $\tau$ is an open embedding into $|X|$; the openness of $\sigma$ is not needed (Findings, item 27).

**V7.28 (Definition 7.1.20 (2017: 7.9), local uniformizers, and the claims after it;** `Polyfold.EpGroupoid.LocalUnif Polyfold.EpGroupoid.LocalUniformizer.rep, Polyfold.EpGroupoid.LocalUniformizer.Ψ, Polyfold.EpGroupoid.LocalUniformizer.obj_eq, Polyfold.EpGroupoid.LocalUniformizer.full, Polyfold.EpGroupoid.LocalUniformizer.faithful, Polyfold.EpGroupoid.LocalUniformizer.isEmbedding_ Polyfold.EpGroupoid.LocalUniformizer.range_orbitMap, Polyfold.EpGroupoid.NaturalRep.toLocalUnif Polyfold.EpGroupoid.exists_localUniformizer, Polyfold.EpGroupoid.LocalUniformizer.injective_obj, Polyfold.EpGroupoid.LocalUniformizer.injective_mor, Polyfold.EpGroupoid.NaturalRep.isOpenEmbeddi Polyfold.EpGroupoid.NaturalRep.isOpenEmbedding_functor_mor`**).** A local uniformizer is a fully faithful embedding $\Psi_x : G_x \ltimes U(x) \to X$ inducing a homeomorphism onto $\pi(U(x))$; one exists around every object, and it is injective on objects and morphisms. *Book:* Definition 7.1.20 (2017: 7.9) and the two sentences after it, 2017 p. 237. *Hidden hypotheses:* none. *Deviation (restated):* "embedding" is derived, not a field; existence inside every open $V \ni x$. "$\Psi_x \circ G_x \ltimes U(x) \to X$" ($\circ$ for :) and "a a local uniformizer" (possible misprints) (Findings, item 28).

**V7.29 (Definition 7.1.21 (2017: 7.10), ep-subgroupoids;** `Polyfold.EpGroupoid.IsEpSubgroupoid, Polyfold.EpGroupoid.IsEpSubgroupoid.isSubMPolyfold, Polyfold.EpGroupoid.IsEpSubgroupoid.isSatura Polyfold.EpGroupoid.isEpSubgroupoid_univ, Polyfold.EpGroupoid.isEpSubgroupoid_of_isOpen, Polyfold.EpGroupoid.isEpSubgroupoid_saturation, Polyfold.EpGroupoid.isEpSubgroupoid_trivial`**).** An ep-subgroupoid is the full subcategory of a saturated sub-M-polyfold $A$ of $X$. *Book:* Definition 7.1.21 (2017: 7.10), p. 305 (2017 p. 237). *Hidden hypotheses:* none; open saturated sets and saturations of open sets are witnesses. *Deviation:* none (Findings, items 40, 105).

**V7.30 (the facts (1)–(4) on the sub-M-polyfold** $A$**;** `Polyfold.EpGroupoid.subObjPolyfold, Polyfold.EpGroupoid.isScSmoothMap_val_subObj, Polyfold.EpGroupoid.isScSmoothOn_retractInto_subOb Polyfold.MPolyfold.subPolyfold`**).** (1) $A \to X$ is sc-smooth and a homeomorphism onto its image, (2) $i^{-1} \circ r$ is sc-smooth for a retraction onto $A \cap V$, (3), (4) the tangent spaces of $A$ at smooth points. *Book:* 2017 pp. 237–238, from Part I's Proposition 2.3.19 (2017: 2.6). *Hidden hypotheses:* none. *Deviation (narrower):* (1) and (2) are used; the tangent facts (3) and (4) are not restated, as tangent spaces carry no linear structure here (gap G.7) (Findings, item 40).

**V7.31 (the morphism set A is a sub-M-polyfold;** `Polyfold.EpGroupoid.src_mem_of_tgt_mem_of_isSaturated,`

`Polyfold.EpGroupoid.morOn_eq_preimage_src`, `Polyfold.EpGroupoid.morOn_eq_preimage_tgt`, `Polyfold.EpGroupoid.isSubMPolyfold_preimage_src`, `Polyfold.EpGroupoid.isSubMPolyfold_preimage_tgt` `Polyfold.EpGroupoid.isSubMPolyfold_morOn`, `Polyfold.EpGroupoid.subMorPolyfold`, `Polyfold.EpGroupoid.isScSmoothMap_val_subMor`, `Polyfold.EpGroupoid.isScSmoothOn_retractInto_subMo` $\mathbf{A} = \{\varphi \mid s(\varphi), t(\varphi) \in A\}$ is a sub-M-polyfold of **X**, by the retractions $\mathbf{r} = (s|\mathbf{U}(\varphi))^{-1} \circ r \circ s$. *Book:* 2017 p. 238. *Hidden hypotheses:* none. *Deviation (restated):* $\mathbf{A} = s^{-1}(A)$ for saturated $A$, the preimage of a sub-M-polyfold under a local sc-diffeomorphism. The claim "$t(\mathbf{r}(\psi)) \in U(a)$, so that $\mathbf{r}(\psi) \in \mathbf{U}(a)$" need not hold and is not used (possible misprint or extraction artefact) (Findings, item 41).

**V7.32 (Proposition 7.1.22 (2017: 7.7), the full subcategory of a saturated sub-M-polyfold is an ep-groupoid;** `Polyfold.EpGroupoid.isLocalScDiffeo_fullSub_src`, `Polyfold.EpGroupoid.isLocalScDiffeo_fu` `Polyfold.EpGroupoid.isScSmoothMap_fullSub_unit`, `Polyfold.EpGroupoid.isScSmoothMap_fullSub_inv`, `Polyfold.EpGroupoid.exists_isProperMap_fullSub`, `Polyfold.EpGroupoid.isScSmoothComp_fullSub`, `Polyfold.EpGroupoid.instParacompactSpaceMorOn`, `Polyfold.EpGroupoid.subEpGroupoid`, `Polyfold.EpGroupoid.subEpGroupoid_src_val`, `Polyfold.EpGroupoid.subEpGroupoid_comp_val`, `Polyfold.EpGroupoid.proposition_7_7`, `Polyfold.EpGroupoid.subEpFunctor`, `Polyfold.EpGroupoid.subEpF` `Polyfold.EpGroupoid.subEpFunctor_faithful`**).** For $A$ saturated and a sub-M-polyfold, the full subcategory is an ep-groupoid with the restricted structure maps, $A$ and **A** sub-M-polyfolds. *Book:* the argument on 2017 p. 238 and Proposition 7.1.22 (2017: 7.7), pp. 307 ff. (2017 pp. 238–239). *Hidden hypotheses:* none; the paracompactness of $A$ and **A**, which the project's M-polyfolds need, is proved. *Deviation (stronger):* the steps the text leaves out (sc-smoothness into the sub-M-polyfold, the inclusion of fibred products) are proved; any $V(a)$ with the properness property works; the inclusion is a fully faithful ep-functor (Findings, item 42).

**V7.33 (Definition 7.2.1 (2017: 7.11), effective ep-groupoids, regular objects, and the sentence after it;** `Polyfold.EpGroupoid.IsEffective`, `Polyfold.GroupoidStr.IsRegular`, `Polyfold.GroupoidStr.regularSet`, `Polyfold.GroupoidStr.isRegular_tgt_iff`, `Polyfold.EpGroupoid.isRe` `Polyfold.EpGroupoid.isSaturated_regularSet`**).** $X$ is effective if every open neighbourhood of every $g \in G_x \setminus \{1_x\}$ contains a $\psi$ with $s(\psi) \neq t(\psi)$; $x$ is regular if $G_x = \{1_x\}$; if $s(\varphi)$ is regular, so is $t(\varphi)$. *Book:* Definition 7.2.1 (2017: 7.11) and the sentences after it, 2017 p. 239. *Hidden hypotheses:* none; satisfiable non-vacuously (`Polyfold.EpGroupoid.Examples.isEffective_reflection_nonvacuous`). *Deviation:* none (Findings, items 50, 101).

**V7.34 (Lemma 7.2.2 (2017: 7.2), $X_{reg}$ is open;** `Polyfold.EpGroupoid.NaturalRep.subset_regularSet`, `Polyfold.EpGroupoid.lemma_7_2`**).** The regular part is open. *Book:* Lemma 7.2.2 (2017: 7.2), p. 307 (2017 p. 239). *Hidden hypotheses:* $\Gamma(1, a) = 1_a$, i.e. $U$ connected (V7.17). *Deviation (stronger):* $U(x) \subseteq X_{reg}$ for every natural representation at a regular $x$ (Findings, item 51).

**V7.35 (Lemma 7.2.3 (2017: 7.3), effectiveness and injective natural representations;** `Polyfold.EpGroupoid.Natur` `Polyfold.EpGroupoid.NaturalRep.injective_Φ_iff`, `Polyfold.EpGroupoid.NaturalRep.kerΦ_eq_bot_iff`, `Polyfold.EpGroupoid.IsEffective.eq_one`, `Polyfold.EpGroupoid.isEffective_iff_forall_naturalRep`, `Polyfold.EpGroupoid.lemma_7_3`**).** $X$ is effective iff every $x$ has, inside every open $V(x)$, a natural representation with $\Phi$ injective. *Book:* Lemma 7.2.3 (2017: 7.3), pp. 308–309 (2017 pp. 239–240). *Hidden hypotheses:* none. *Deviation (stronger):* (1) □ (2) for every natural representation, without sequences; "adapting restrictions" read through V7.24; injectivity as trivial kernel (Findings, item 52).

**V7.36 (Proposition 7.2.4 (2017: 7.8), $X_{reg}$ is open and dense for effective $X$;** `Polyfold.EpGroupoid.NaturalRep.s` `Polyfold.EpGroupoid.proposition_7_8`, `Polyfold.EpGroupoid.isEffective_of_dense_regularSet`,

`Polyfold.EpGroupoid.isEffective_iff_dense_regularSet`**).** If $X$ is effective, $X_{reg}$ is open and dense. *Book:* Proposition 7.2.4 (2017: 7.8), pp. 309 ff. (2017 pp. 240–241). *Hidden hypotheses:* none. *Deviation (stronger):* the minimum of $\#G$ is taken over the given open set, so $U(x_0)$ is not needed; the converse also holds (Findings, item 53).

**V7.37 (the subgroup $T_x$ of elements acting trivially;** `Polyfold.EpGroupoid.ActsTrivially`**,** `Polyfold.EpGroupoid.trivialSet`**,** `Polyfold.EpGroupoid.one_mem_trivialSet`**,** `Polyfold.EpGroupoid.Natur` `Polyfold.EpGroupoid.mem_trivialSet_of_section`**,** `Polyfold.EpGroupoid.mem_trivialSet_iff_eventually` `Polyfold.EpGroupoid.exists_mem_nhds_trivialSet_iff`**,** `Polyfold.EpGroupoid.exists_naturalRep_trivia` `Polyfold.EpGroupoid.exists_naturalRep_kerΦ_eq`**,** `Polyfold.EpGroupoid.trivialSubgroup`**,** `Polyfold.EpGroupoid.trivialSubgroup_normal`**,** `Polyfold.EpGroupoid.isEffective_iff_trivialSubgroup_` `Polyfold.EpGroupoid.isotropyConj_mem_trivialSet`**,** `Polyfold.EpGroupoid.map_isotropyConj_trivialSub` `Polyfold.EpGroupoid.NaturalRep.mem_trivialSet_of_val_eq_Γ`**).** $g$ acts trivially if $\Phi(g) = 1$ for some natural representation; for $U(x)$ small the kernel of $\Phi$ is the set $T_x$ of such $g$, a normal subgroup. *Book:* 2017 p. 241. *Hidden hypotheses:* none. *Deviation (stronger):* there is a neighbourhood $N$ of $x$ with $\ker\Phi = T_x$ for every natural representation inside $N$; $g \in T_x$ iff $s = t$ near $g$; $\varphi T_x \varphi^{-1} = T_{y'}$ which Definition 7.2.7 (2017: 7.12) needs and the text does not state. "Diff$_{sc}(U(x)$" (possible misprint or extraction artefact) (Findings, item 54).

**V7.38 (Definition 7.2.7 (2017: 7.12), the non-effective part $\mathbf{X}^T$ and the reduced morphisms $\mathbf{X}^R$;** `Polyfold.EpGroupoid.nonEffectivePart`**,** `Polyfold.EpGroupoid.mem_nonEffectivePart_iff`**,** `Polyfold.EpGroupoid.nonEffectivePart_eq_interior`**,** `Polyfold.EpGroupoid.ReducedRel`**,** `Polyfold.EpGroupoid.reducedRel_iff`**,** `Polyfold.EpGroupoid.reducedSetoid`**,** `Polyfold.EpGroupoid.reduce` `Polyfold.EpGroupoid.ReducedMor`**,** `Polyfold.EpGroupoid.reducedMor_eq_quotient`**).** $\mathbf{X}^T = \bigcup_x T_x$; $\varphi \sim \varphi'$ iff $s(\varphi) = s(\varphi') = x$ and $\varphi = \varphi' \circ g$ with $g \in T_x$; $\mathbf{X}^R$ is the set of classes. *Book:* Definition 7.2.7 (2017: 7.12), p. 310 (2017 p. 241). *Hidden hypotheses:* none. *Deviation (restated):* $\sim$ is an equivalence since $T_x$ is a subgroup; $\mathbf{X}^T$ is the interior of $\{s = t\}$; "the subset $\mathbf{X}^R$" is a quotient (possible misprint) (Findings, item 55).

**V7.39 (Proposition 7.2.10 (2017: 7.9), the M-polyfold structure of $\mathbf{X}^R$;** `Polyfold.EpGroupoid.reduced`**,** `Polyfold.EpGroupoid.isQuotientMap_reduced`**,** `Polyfold.EpGroupoid.isOpenMap_reduced`**,** `Polyfold.EpGroupoid.injOn_reduced`**,** `Polyfold.EpGroupoid.reducedChart`**,** `Polyfold.EpGroupoid.reduced` `Polyfold.EpGroupoid.reducedCharts`**,** `Polyfold.EpGroupoid.reducedAtlas`**,** `Polyfold.EpGroupoid.reducedP` `Polyfold.EpGroupoid.isChart_reducedChart`**,** `Polyfold.EpGroupoid.isScDiffeoOn_reduced`**,** `Polyfold.EpGroupoid.isLocalScDiffeo_reduced`**,** `Polyfold.EpGroupoid.proposition_7_9`**,** `Polyfold.EpGroupoid.t2Space_reducedMor_of_isEffective`**,** `Polyfold.EpGroupoid.paracompactSpace_redu` `Polyfold.EpGroupoid.paracompactSpace_reducedMor_of_t3`**,** `Polyfold.EpGroupoid.Examples.exists_not_t` $\mathbf{X}^R$ has a natural M-polyfold structure. *Book:* Proposition 7.2.10 (2017: 7.9), pp. 313–319 (2017 pp. 242–243). *Hidden hypotheses:* $\mathbf{X}^R$ Hausdorff and paracompact, taken as hypotheses of `Polyfold.EpGroupoid.reducedPolyfold`, which the proof does not give as we read it; satisfiable for effective $X$ and the trivial ep-groupoid, and paracompactness follows from regularity when $\mathbf{X}$ is second countable. *Deviation (narrower):* the quotient topology, the openness of the projection, the charts $\psi \circ \sigma \circ s$ and their sc-smooth transitions are proved without hypotheses; the M-polyfold under the two hypotheses (gap G.71). The Hausdorff hypothesis cannot be dropped: for $\mathbb{Z}/2$ acting on the dimension-jumping retract of Part I's Remark 4.2.2 (2017: 4.3), $\mathbf{X}^R$ is not Hausdorff (`Polyfold.EpGroupoid.Examples.exists_not_t2Space_reducedMor`). Four possible misprints in the proof (Findings, items 56, 57, 60, 107, 111).

**V7.40 (the reduced ep-groupoid $X_R$ and Proposition 7.10 (2017);** `Polyfold.EpGroupoid.reducedSrc`**,** `Polyfold.EpGroupoid.reducedTgt`**,** `Polyfold.EpGroupoid.reducedUnit`**,** `Polyfold.EpGroupoid.reducedInv`**,**

`Polyfold.EpGroupoid.reducedComp`, `Polyfold.EpGroupoid.reducedGroupoidStr`, `Polyfold.EpGroupoid.isLo`
`Polyfold.EpGroupoid.isLocalScDiffeo_reducedTgt`, `Polyfold.EpGroupoid.isScSmoothMap_reducedUnit`,
`Polyfold.EpGroupoid.isScSmoothMap_reducedInv`, `Polyfold.EpGroupoid.exists_isProperMap_reduced`,
`Polyfold.EpGroupoid.isScSmoothCompSheet_reduced`, `Polyfold.EpGroupoid.reducedEpGroupoid`,
`Polyfold.EpGroupoid.reductionFunctor`, `Polyfold.EpGroupoid.isEffective_reducedEpGroupoid`,
`Polyfold.EpGroupoid.proposition_7_10`, `Polyfold.EpGroupoid.reductionIso`). $(X, \mathbf{X}^R)$ with the induced structure maps is an ep-groupoid $X_R$, and it is effective. *Book:* 2017 p. 243 and Proposition 7.10 (2017), 2017 p. 244. *Hidden hypotheses:* those of V7.39 (gap G.71). *Deviation (narrower):* (3) via the continuous surjection $\mathbf{X} \to \mathbf{X}^R$; (4) in sheet form, with a step the text leaves implicit; effectiveness from the intrinsic form of $T_x$; for effective $X$, $X_R \cong X$. Without the hypotheses $X_R$ is not an ep-groupoid in general, since $\mathbf{X}^R$ need not be Hausdorff (V7.39) (Findings, items 58, 60, 107).

**V7.41 (the isotropy groups of $X_R$ and Definition 7.2.15 (2017: 7.13), the effective isotropy group;** `Polyfold.EpGroupoid.reducedIsotropyHom`, `Polyfold.EpGroupoid.surjective_reducedIsotropyHom`, `Polyfold.EpGroupoid.ker_reducedIsotropyHom`, `Polyfold.EpGroupoid.quotientTrivialSubgroupEquiv`, `Polyfold.EpGroupoid.EffIsotropy`, `Polyfold.EpGroupoid.normalClosure_trivialSet`, `Polyfold.EpGroupoid.effIsotropyEquiv`, `Polyfold.EpGroupoid.effIsotropyEquivReduced`**).** The isotropy group of $x$ in $X_R$ is $G_x/T_x$, called the effective isotropy group $G_x^{\mathrm{eff}}$. *Book:* 2017 p. 244 and Definition 7.2.15 (2017: 7.13), p. 319 (2017 p. 244). *Hidden hypotheses:* the isomorphism with the isotropy of $X_R$ under those of V7.39; $G_x/T_x$ itself needs none. *Deviation:* none; "For and object" (possible misprint) (Findings, item 59).

**V7.42 (Theorem 7.3.1 (2017: 7.2), basic topological properties of $|X|$;** `Polyfold.EpGroupoid.theorem_7_2`, `Polyfold.EpGroupoid.theorem_7_2_metrizable`, `Polyfold.metrizableSpace_of_locally`**).** $|X|$ is locally metrizable, regular and Hausdorff, and metrizable if paracompact. *Book:* Theorem 7.3.1 (2017: 7.2), p. 324 (2017 p. 244). *Hidden hypotheses:* none. *Deviation (restated):* "locally metrizable" as an open metrizable neighbourhood of every point; Smirnov's theorem in Part I's form (Findings, item 70).

**V7.43 (Lemma 7.3.2 (2017: 7.4), local metrizability;** `Polyfold.EpGroupoid.metrizableSpace_orbitRel_quotien`
`Polyfold.EpGroupoid.lemma_7_4_naturalRep`, `Polyfold.EpGroupoid.lemma_7_4`**).** Every point of $|X|$ has a metrizable open neighbourhood, $|U(x)|$ for a natural representation. *Book:* Lemma 7.3.2 (2017: 7.4), pp. 324 ff. (2017 pp. 244–245). *Hidden hypotheses:* none. *Deviation (restated):* $\hat{\rho}$ computed as $\min_h \rho(y, h * y')$ (Findings, item 71).

**V7.44 (Lemma 7.3.3 (2017: 7.5), countable neighbourhood bases;** `Polyfold.EpGroupoid.lemma_7_5`**).** Every point of $|X|$ has a monotone countable basis of open neighbourhoods. *Book:* Lemma 7.3.3 (2017: 7.5), p. 324 (2017 p. 245). *Hidden hypotheses:* none. *Deviation:* none; it is Proposition 7.1.9 (2017: 7.2) (V7.12) (Findings, item 72).

**V7.45 (Lemma 7.3.4 (2017: 7.6), $|X|$ is Hausdorff;** `Polyfold.EpGroupoid.t1Space_orbitSpace`, `Polyfold.EpGroupoid.lemma_7_6`**).** *Book:* Lemma 7.3.4 (2017: 7.6), pp. 325 ff. (2017 pp. 245–246). *Hidden hypotheses:* none. *Deviation (restated):* from Lemma 7.3.5 (2017: 7.7) and the closedness of points, instead of the text's direct argument (Findings, item 72).

**V7.46 (Lemma 7.3.5 (2017: 7.7), $|X|$ is regular;** `Polyfold.EpGroupoid.lemma_7_7`**).** *Book:* Lemma 7.3.5 (2017: 7.7), p. 325 (2017 p. 246). *Hidden hypotheses:* none. *Deviation (restated):* with a closed neighbourhood basis and Proposition 7.1.10 (2017: 7.3) (Findings, item 72).

**V7.47 (Corollary 7.3.6 (2017: 7.1), the metrizability criterion;** `Polyfold.EpGroupoid.corollary_7_1`**).**

If $|X|$ is second countable, it is metrizable. *Book:* Corollary 7.3.6 (2017: 7.1), p. 326 (2017 p. 247). *Hidden hypotheses:* none. *Deviation:* none; Urysohn's theorem (Findings, item 73).

**V7.48 (subsets of $|X|$; `Polyfold.EpGroupoid.exists_metrizable_nhds_subtype`, `Polyfold.EpGroupoid.metri`** A subset of $|X|$ is again locally metrizable, regular and Hausdorff; a paracompact one is metrizable. *Book:* 2017 p. 247, before Definition 7.3.7 (2017: 7.14). *Hidden hypotheses:* none. *Deviation:* none (Findings, item 73).

**V7.49 (Definition 7.3.7 (2017: 7.14), component compact sets; `Polyfold.EpGroupoid.IsComponentCompact`).** $K \subseteq |X|$ meets every connected component of $|X|$ in a compact set. *Book:* Definition 7.3.7 (2017: 7.14), p. 326 (2017 p. 247). *Hidden hypotheses:* none. *Deviation:* none (Findings, item 74).

**V7.50 (Michael's theorem and unions of closed paracompact sets; `Polyfold.lemma_2_56`, `Polyfold.michael_tfae`, `Polyfold.proposition_2_57`, `Polyfold.paracompactSpace_of_closed_refinement`, `Polyfold.exists_closed_precise_refinement`, `Polyfold.paracompactSpace_iUnion`, `Polyfold.paracompactSpace_iUnion_of_finite`, `Polyfold.paracompactSpace_biUnion_of_finite`, `Polyfold.paracompactSpace_union`, `Polyfold.paracompactSpace_of_isClosed_subset`).** A regular space is paracompact iff every open cover has a closed locally finite refinement; a regular space with a locally finite cover by closed paracompact subspaces is paracompact. *Book:* Part I, Lemma 2.7.3 (2017: 2.7) and Proposition 2.7.4 (2017: 2.17), used for Corollary 7.3.8 (2017: 7.2) (p. 326 (2017 p. 247)) and on the text's route to Proposition 7.5.3 (2017: 7.12) (p. 340 (2017 p. 249)). *Hidden hypotheses:* none. *Deviation (stronger):* the Hausdorff property assumed by the text is not used; gap G.15 of Part I is now formalized (Findings, items 75, 114).

**V7.51 (Corollary 7.3.8 (2017: 7.2), neighbourhoods of component compact sets; `Polyfold.EpGroupoid.locallyP`, `Polyfold.EpGroupoid.exists_isOpen_metrizableSpace_closure`, `Polyfold.EpGroupoid.corollary_7_2_of_`, `Polyfold.EpGroupoid.corollary_7_2`).** A component compact $K$ has an open neighbourhood $U$ with $\mathrm{cl}_{|X|}(U)$ paracompact, hence $\mathrm{cl}(U)$ and $U$ metrizable. *Book:* Corollary 7.3.8 (2017: 7.2), p. 326 (2017 p. 247). *Hidden hypotheses:* none. *Deviation (restated):* the reduction to compact $K$ uses open components (local connectedness of $|X|$, V7.18) and a locally finite union the text does not give; Michael's theorem (V7.50) for the finite union. "$U$" names both $\bigcup \mathrm{cl}\, U(z_i)$ and $\bigcup U(z_i)$ (possible misprint) (Findings, item 74).

**V7.52 (Proposition 7.3.9 (2017: 7.11), index lifting; `Polyfold.EpGroupoid.liftSrc`, `Polyfold.EpGroupoid.liftT`, `Polyfold.EpGroupoid.liftComp`, `Polyfold.EpGroupoid.comp_mem_levelSet`, `Polyfold.EpGroupoid.liftGrou`, `Polyfold.MPolyfold.IsLocalScDiffeo.levelOne`, `Polyfold.MPolyfold.IsScSmoothMap.levelOne`, `Polyfold.EpGroupoid.isLocalScDiffeo_liftSrc`, `Polyfold.EpGroupoid.isLocalScDiffeo_liftTgt`, `Polyfold.EpGroupoid.isScSmoothMap_liftUnit`, `Polyfold.EpGroupoid.isScSmoothMap_liftInv`, `Polyfold.EpGroupoid.exists_isProperMap_lift`, `Polyfold.EpGroupoid.isScSmoothCompSheet_lift`, `Polyfold.EpGroupoid.isScSmoothComp_lift`, `Polyfold.EpGroupoid.liftOne`, `Polyfold.MPolyfold.IsTame.l`, `Polyfold.EpGroupoid.isTame_liftOne`, `Polyfold.EpGroupoid.proposition_7_11`).** $X^1 = (X^1, \mathbf{X}^1)$ is an ep-groupoid, tame if $X$ is. *Book:* Proposition 7.3.9 (2017: 7.11), pp. 327–328 (2017 pp. 247–248). *Hidden hypotheses:* none. *Deviation (restated):* (4) in sheet form, avoiding $(\mathbf{X}\,{}_s\!\times_t \mathbf{X})^1 = \mathbf{X}^1\,{}_s\!\times_t \mathbf{X}^1$; (3) with convergence in $X_1$ for the compact set of the text and without its step (b); the tameness, for which the text gives no argument and which Part I left open (Chapter 5, item 145), is proved. Display (7.5) has an unbalanced parenthesis (possible misprint) (Findings, items 80, 81, 96).

**V7.53 (Definition 7.5.1 (2017: 7.15), ep-groupoids with paracompact orbit space, the sentence before Remark 7.5.2 (2017: 7.1), and Remark 7.5.2 (2017: 7.1); `Polyfold.EpGroupoid.HasParacompactOrbitSpace`, `Polyfold.EpGroupoid.remark_7_1`, `Polyfold.EpGroupoid.hasParacompactOrbitSpace_of_secondCountableT`**

`Polyfold.EpGroupoid.hasParacompactOrbitSpace_iff_metrizableSpace`**).** (5): $|X|$ is paracompact, not required to be Hausdorff; if $X$ is second countable, (1)–(4) imply (5). Remark 7.5.2 (2017: 7.1) recalls that [38] required second countability. *Book:* Definition 7.5.1 (2017: 7.15), the paragraph after it, and Remark 7.5.2 (2017: 7.1), p. 340 (2017 p. 249). *Hidden hypotheses:* none. *Deviation (restated):* "$X$ second countable" is read for the object space; Remark 7.5.2 (2017: 7.1) has no formal content; "it is presently not known whether (5) is a consequence of (1)–(4)" is not formalized (Findings, items 82, 113).

**V7.54 (Proposition 7.5.3 (2017: 7.12), paracompactness under lifts, and the map $|X^1| \to |X|$;** `Polyfold.EpGroupoid.orbitIncl`, `Polyfold.EpGroupoid.continuous_orbitIncl`, `Polyfold.EpGroupoid.orbitIncl_injective`, `Polyfold.EpGroupoid.range_orbitIncl`, `Polyfold.EpGroupoid.NaturalRep.liftMulAction`, `Polyfold.EpGroupoid.NaturalRep.isOpenEmbedding_li` `Polyfold.EpGroupoid.NaturalRep.metrizableSpace_preimage_orbitIncl`, `Polyfold.EpGroupoid.metrizabl` `Polyfold.metrizableSpace_of_pullback`, `Polyfold.EpGroupoid.proposition_7_12`**).** If $|X|$ is paracompact, so is $|X^1|$. *Book:* Proposition 7.5.3 (2017: 7.12), pp. 340 ff. (2017 pp. 249–250). *Hidden hypotheses:* none. *Deviation (stronger):* $|X^1|$ is metrizable, by Smirnov's theorem pulled back along the continuous injection $|X^1| \to |X|$ onto $\pi(X_1)$; $U(x)_1$ carries the restricted action only (Findings, item 83).

**V7.55 (Theorem 7.5.4 (2017: 7.3), continuous partitions of unity;** `Polyfold.EpGroupoid.orbitPullback`, `Polyfold.EpGroupoid.orbitPullback_isSubordinate`, `Polyfold.EpGroupoid.exists_isSaturated_nhds_fi` `Polyfold.EpGroupoid.theorem_7_3`**).** For paracompact $|X|$, every open cover of $|X|$ has a subordinate partition of unity, and every saturated open cover of $X$ an invariant continuous one. *Book:* Theorem 7.5.4 (2017: 7.3), p. 341 (2017 p. 250). *Hidden hypotheses:* $|X|$ Hausdorff, from Lemma 7.3.4 (2017: 7.6), for normality. *Deviation:* none (Findings, item 90).

**V7.56 (Definition 7.5.5 (2017: 7.16), ep-groupoids with sc-smooth partitions of unity;** `Polyfold.EpGroupoid.HasScPartitionsOfUnity`, `Polyfold.MPolyfold.IsScSmoothPartitionOfUnity`**).** Every saturated open cover of $X$ has a subordinate sc-smooth partition of unity with (1) saturated neighbourhoods meeting finitely many supports, (2) invariance. *Book:* Definition 7.5.5 (2017: 7.16), pp. 342 ff. (2017 pp. 250–251). *Hidden hypotheses:* none. *Deviation (restated):* index types in the universe of $X$, supports as `tsupport`, (2) as $\beta_\lambda(s\varphi) = \beta_\lambda(t\varphi)$ (Findings, item 91).

**V7.57 (Remark 7.5.6 (2017: 7.2), on sc-smooth partitions of unity).** The remark (the topic is "a zoo" apart from sc-Hilbert spaces) has no formal content. *Book:* Remark 7.5.6 (2017: 7.2), p. 342 (2017 p. 251). *Hidden hypotheses:* none. *Deviation:* none; the sc-Hilbert case is Part I's `Polyfold.MPolyfold.admitsScPartitionsOfUnity_of_hilbert` (V7.68) (Findings, item 91).

**V7.58 (Theorem 7.5.7 (2017: 7.4), sc-smooth partitions of unity of an ep-groupoid;** `Polyfold.EpGroupoid.theore` If $|X|$ is paracompact and the object M-polyfold admits sc-smooth partitions of unity, $X$ has sc-smooth partitions of unity. *Book:* Theorem 7.5.7 (2017: 7.4), p. 342 (2017 p. 251) (proved as Theorem 7.6.4 (2017: 7.6), V7.65, V7.67). *Hidden hypotheses:* none; satisfiable (`Polyfold.EpGroupoid.theorem_7_4_hypotheses_rn`). *Deviation:* none (Findings, items 94, 105).

**V7.59 (Definition 7.5.8 (2017: 7.17), proper subsets and the properness property;** `Polyfold.EpGroupoid.IsPrope` `Polyfold.EpGroupoid.HasPropernessProperty`, `Polyfold.EpGroupoid.exists_hasPropernessProperty`, `Polyfold.EpGroupoid.HasPropernessProperty.mono`, `Polyfold.EpGroupoid.isProperSubset_iff_isCompact` `Polyfold.EpGroupoid.isProperSubset_iff_exists_tendsto_subseq`**).** $Q$ is proper if $t : s^{-1}(Q) \to X$ is proper; an open $U$ has the properness property if $\mathrm{cl}_X(U)$ is proper. *Book:* Definition 7.5.8 (2017: 7.17), p. 342 (2017 p. 251). *Hidden hypotheses:* none. *Deviation:* none (Findings, item 92).

**V7.60 (Proposition 7.5.9 (2017: 7.13), proper subsets are closed; `Polyfold.EpGroupoid.proposition_7_13`, `Polyfold.EpGroupoid.IsProperSubset.isClosed`).** *Book:* Proposition 7.5.9 (2017: 7.13), p. 342 (2017 p. 251). *Hidden hypotheses:* none. *Deviation:* none; by the compactness of the preimage (Findings, item 92).

**V7.61 (Proposition 7.5.10 (2017: 7.14), closed subsets of $|X|$; `Polyfold.EpGroupoid.isClosed_saturation_of_i`
`Polyfold.EpGroupoid.proposition_7_14_one`, `Polyfold.EpGroupoid.IsProperSubset.of_subset`,
`Polyfold.EpGroupoid.IsProperSubset.union`, `Polyfold.EpGroupoid.isProperSubset_iUnion_of_locallyFi`
`Polyfold.EpGroupoid.isProperSubset_of_isCompact`, `Polyfold.EpGroupoid.exists_hasPropernessPropert`
`Polyfold.EpGroupoid.proposition_7_14_two`).** (1) $\pi(Q)$ is closed for proper $Q$; (2) for paracompact $|X|$, $A \subseteq |X|$ is closed iff $A = \pi(Q)$ for a proper $Q$. *Book:* Proposition 7.5.10 (2017: 7.14), pp. 343–344 (2017 pp. 251–252). *Hidden hypotheses:* none. *Deviation (stronger):* (2) with only the paracompactness of $|X|$, without Lemma 2.7.3 (2017: 2.7) or regularity; "$\mathrm{cl}_X(U_{x_{z_i}})$" for $\mathrm{cl}_X(U(x_{z_i}))$ (possible misprint) (Findings, item 92).

**V7.62 (Theorem 7.6.1 (2017: 7.5), the natural representation, restated in Appendix 7.6.1 (2017 7.5.1); `Polyfold.EpGroupoid.theorem_7_5`, `Polyfold.EpGroupoid.NatRepConstruction.Sheets`,
`Polyfold.EpGroupoid.NatRepConstruction.exists_sheets`, `Polyfold.EpGroupoid.NatRepConstruction.She`
`Polyfold.EpGroupoid.NatRepConstruction.exists_naturalRep_sheets`, `Polyfold.EpGroupoid.NatRepConst`
Theorem 7.1.13 (2017: 7.1) with $\Phi(g) = t_g \circ s_g^{-1}$. *Book:* Theorem 7.6.1 (2017: 7.5) and its proof, pp. 344–346 (2017 pp. 253–254). *Hidden hypotheses:* as V7.17. *Deviation (restated):* `Polyfold.EpGroupoid.theorem_7_5` is `Polyfold.EpGroupoid.theorem_7_1` (V7.17); the proof starts at $U_0 := V \cap \bigcap_g s(N_g)$ so that $U \subseteq V$; the contractibility of $N_g^t$, $N_g^s$ is not needed; "$\Phi : G_x \to \mathrm{Diff}_{sc}(U)$" for $G_{x_0}$ (possible misprint) (Findings, item 29).

**V7.63 (Lemma 7.6.2 (2017: 7.8); `Polyfold.EpGroupoid.lemma_7_8`).** There is an open $U_1 \subseteq U_0$ of $x_0$ such that every morphism between points of $U_1$ lies in some $N_g$. *Book:* Lemma 7.6.2 (2017: 7.8), pp. 345 ff. (2017 pp. 253–254). *Hidden hypotheses:* none. *Deviation (stronger):* for any open $N_g \ni g$ and any neighbourhood $U_0$; "properness assumption of ep-polyfolds" (possible misprint) (Findings, item 30).

**V7.64 (Lemma 7.6.3 (2017: 7.9); `Polyfold.EpGroupoid.lemma_7_9`, `Polyfold.EpGroupoid.NatRepConstruction.`
`Polyfold.EpGroupoid.NatRepConstruction.Sheets.isConnected_unionHat`, `Polyfold.EpGroupoid.NatRepCo`
`Polyfold.EpGroupoid.NatRepConstruction.Sheets.eqOn_ _mul`, `Polyfold.EpGroupoid.NatRepConstruction`
`Polyfold.EpGroupoid.NatRepConstruction.Sheets.hat_mul`).** There is an open $U_2 \subseteq U_1$ of $x_0$ such that $U = \bigcup_g t_g \circ s_g^{-1}(U_2)$ lies in $U_1$ and is invariant under all $t_g \circ s_g^{-1}$. *Book:* Lemma 7.6.3 (2017: 7.9), p. 345 (2017 p. 254). *Hidden hypotheses:* none. *Deviation (stronger):* $U_2$, and so $U$, connected, by V7.18 (Findings, item 31).

**V7.65 (Theorem 7.6.4 (2017: 7.6), Theorem 7.5.7 (2017: 7.4) restated in Appendix 7.6.2 (2017 7.5.2); `Polyfold.EpGroupoid.theorem_7_6`).** *Book:* Theorem 7.6.4 (2017: 7.6), p. 346 (2017 p. 255). *Hidden hypotheses:* as V7.58. *Deviation:* none; `Polyfold.EpGroupoid.theorem_7_6` is `Polyfold.EpGroupoid.theorem_7_4` (V7.58) (Findings, item 95).

**V7.66 (Lemma 7.6.5 (2017: 7.10), invariant sc-smooth majorants; `Polyfold.EpGroupoid.lemma_7_10`).** For $U(x) \subseteq \mathrm{cl}_X U(x) \subseteq V(x)$, $V(x)$ with the natural action and $U(x)$ invariant, every invariant continuous $\beta : X \to [0,1]$ supported in $U(x)$ has an invariant sc-smooth majorant $f \geq 0$ supported in $V(x)$. *Book:* Lemma 7.6.5 (2017: 7.10), p. 346 (2017 p. 255). *Hidden hypotheses:* none. *Deviation (stronger):* $\mathrm{supp} f \subseteq \mathrm{cl}_X U(x)$ added (the proof gives it, the proof of Theorem 7.5.7 (2017: 7.4) needs it); "$x \in U(x)$" omitted; the partition of unity of $V(x)$ is taken for a cover of $X$

(Findings, item 93).

**V7.67 (the proof of Theorem 7.5.7 (2017: 7.4);** `Polyfold.EpGroupoid.exists_invariant_nhds_of_isOpen`, `Polyfold.EpGroupoid.mem_of_ _mem_image_of_invariant`, `Polyfold.EpGroupoid.saturation_diff_inter_e` `Polyfold.EpGroupoid.orbitExtension`, `Polyfold.EpGroupoid.isScSmoothMap_orbitExtension`, `Polyfold.EpGroupoid.exists_isScSmoothMap_orbit_urysohn`, `Polyfold.admitsScPartitionsOfUnity_of_ap` `Polyfold.EpGroupoid.theorem_7_4`**).** The neighbourhoods $V(x) \subseteq \operatorname{cl} V(x) \subseteq W(x)$ with (1)–(4), a locally finite refinement, Lemma 7.6.5 (2017: 7.10), averaging, extension to the saturation, and normalization. *Book:* pp. 346–347 (2017 pp. 255–256). *Hidden hypotheses:* none. *Deviation (restated):* (4) follows from (1); Lemma 7.6.5 (2017: 7.10) is applied with $W(x_i)$ in the role of $V(x)$, where the text reads $V(x_i)$ (possible misprint); local finiteness needs the support bound of V7.66; the saturations are closed by properness; the normalization by Mathlib's bump coverings as in Part I's Theorem 5.5.10 (2017: 5.12); "$t : s^{-1}(\mathrm{cl}_X(V(x)) \to X$" (possible misprint) (Findings, items 94, 96).

**V7.68 (Corollary 7.6.6 (2017: 7.3), sc-Hilbert atlases;** `Polyfold.EpGroupoid.corollary_7_3`, `Polyfold.MPolyfold.admitsScPartitionsOfUnity_of_hilbert`, `Polyfold.GodefroyTroyanskiWhitfieldZizl` `Polyfold.EpGroupoid.corollary_7_3_hypotheses_rn`**).** If $|X|$ is paracompact and the object M-polyfold has an atlas whose models have Hilbert spaces $E_0$, every saturated open cover has a subordinate sc-smooth partition of unity of $X$. *Book:* Corollary 7.6.6 (2017: 7.3), p. 347 (2017 p. 256), with Part I's Proposition 5.5.8 (2017: 5.6), Theorem 5.5.10 (2017: 5.12) and Corollary 5.5.17 (2017: 5.2). *Hidden hypotheses:* Part I's Theorem 5.5.16 (2017: 5.14) (Godefroy–Troyanski–Whitfield–Zizler) as an explicit hypothesis (gap G.53); satisfiable on $\mathbb{R}^1$. *Deviation (narrower):* the Hilbert atlas existential, one chart per point; "every compatible chart" would be unsatisfiable (Findings, items 95, 106).

**The additions of the 2021 edition (V7.69–V7.104)**

The Springer edition (2021) changes the remark after Definition 7.1.7 and Proposition 7.1.15, adds Proposition 7.1.14, Lemma 7.1.16 and Corollary 7.1.17, rewrites §7.2 from Definition 7.2.5 on (2017 Proposition 7.10 becomes Theorem 7.2.14, preceded by a Hausdorff criterion for the topology $T_R$ of $\mathbf{X}^R$), adds §7.4 on the Zhou condition and unique continuation, and adds Appendices 7.6.3 (Zhou's example), 7.6.4 (the reduced groupoid and index lifts) and 7.6.5 (the boundary of a tame ep-groupoid). Numbers and pages below are the 2021 text's; items new in the 2021 edition have no 2017 counterpart. Lean files: `Chapter7/Book2021Basic.lean`, `ReducedCriterion.lean`, `Zhou.lean`, `TameBoundaryEp.lean`, and, for Theorem 7.6.9 in full on top of Part I's Theorem 2.7.13 (2)–(4), `TameBoundaryEpStructure.lean` (V7.99–V7.101), and, for §7.4 on every lift $X^i$, `ZhouLevels.lean` (V7.102–V7.104).

**V7.69 (the remark after Definition 7.1.7, 2021 form;** `Polyfold.EpGroupoid.remark_7_1_7_2021`, `Polyfold.EpGroupoid.remark_7_1_7_2021_remarkFunctor`**).** Let $F : X \to Y$ be a functor between ep-groupoids which is sc-smooth on the object M-polyfolds and continuous on the morphism spaces, $\varphi \in \mathbf{X}$ and $\psi = F(\varphi)$. There are open sets $U(\varphi) \ni \varphi$, $U(\psi) \ni \psi$, $U(s(\varphi))$, $U(s(\psi))$ with $F(U(s(\varphi))) \subseteq U(s(\psi))$ and $F(U(\varphi)) \subseteq U(\psi)$ such that $s : U(\varphi) \to U(s(\varphi))$ and $s : U(\psi) \to U(s(\psi))$ are sc-diffeomorphisms and $F(\gamma) = (s|U(\psi))^{-1}(F(s(\gamma)))$ for $\gamma \in U(\varphi)$; hence $F$ is sc-smooth on the morphism spaces. The functor of the counterexample of V7.10 (on the trivial $\mathbb{Z}/3$-action on $\mathbb{R}$) is sc-smooth on objects and not continuous on morphisms. *Book:* the remark after Definition 7.1.7, p. 291 (2017: after Definition 7.7, p. 228), which in 2021 assumes continuity on morphisms. *Hidden hypotheses:* none. *Deviation (restated):* the neighbourhoods are an

existential statement; the 2021 hypothesis is the one under which V7.10 proved the 2017 remark, so the counterexample of V7.10 no longer applies (Findings, items 9, 104, 117).

**V7.70 (the data (7.3) and Proposition 7.1.14; `Polyfold.EpGroupoid.IsotropySheetData`, `Polyfold.EpGroupoid.exists_isotropySheetData`, `Polyfold.EpGroupoid.proposition_7_1_14_2021`, `Polyfold.EpGroupoid.proposition_7_1_14_2021_of_naturalRep`).** The data (7.3) at an object $x$ are an open $W(x) \ni x$ and, for every $g \in G_{x'}$ open $U^s(g), U^t(g) \ni g$ on which $s$ and $t$ are sc-diffeomorphisms onto $W(x)$, with inverses $s_g^{-1}$, $t_g^{-1}$; they exist at every object. Let $U(x)$ be an open neighbourhood of $x$, $\Phi(g)$ ($g \in G_x$) sc-diffeomorphisms of $U(x)$ with inverse $\Phi(g^{-1})$ and $\Phi(gh) = \Phi(g) \circ \Phi(h)$, and $\Gamma(g, \cdot) : U(x) \to \mathbf{X}$ sc-smooth with $\Gamma(g, x) = g$, $s(\Gamma(g, y)) = y$, $t(\Gamma(g, y)) = \Phi(g)(y)$. Then there is an open $\Phi$-invariant $V(x) \ni x$ with $V(x) \subseteq W(x) \cap U(x)$ such that, for $v \in V(x)$ and $g \in G_x$, $\Gamma(g, v) = s_g^{-1}(v)$, $\Phi(g)(v) = t(s_g^{-1}(v))$ and $\Gamma(g, v) = t_g^{-1}(\Phi(g)(v))$; the same holds for every natural representation. *Book:* (7.3) and Proposition 7.1.14, p. 296, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (stronger):* property (3) of Theorem 7.1.13 (every morphism between points of $U(x)$ is some $\Gamma(g, y)$) is assumed as printed but not used; the identity $\Gamma(g, v) = t_g^{-1}(\Phi(g)(v))$ of (7.4), from the proof, is added (Findings, item 118).

**V7.71 (Proposition 7.1.15 in the 2021 form; `Polyfold.EpGroupoid.proposition_7_1_15_2021`, `Polyfold.EpGroupoid.src_eq_tgt_of_printed_2017`, `Polyfold.EpGroupoid.not_printed_2017_reflection`).** For a finite set $Y$ of mutually isomorphic objects and a neighbourhood $V$ of $Y$ there is a natural representation of $Y$ on disjoint connected open $U(y) \subseteq V$ (Definition 7.1.18, `NaturalRepFinite`) such that, for every morphism $\psi$ between points of $Y$: $\psi \in \mathbf{U}(\psi)$, the connected component of $\psi$ in the set of morphisms between points of $U = \bigcup_y U(y)$; $s : \mathbf{U}(\psi) \to U(s(\psi))$ and $t : \mathbf{U}(\psi) \to U(t(\psi))$ are sc-diffeomorphisms; $f(\psi) = t \circ (s|\mathbf{U}(\psi))^{-1}$ on $U(s(\psi))$; and $t(\psi') = f(\psi)(s(\psi'))$ for $\psi' \in \mathbf{U}(\psi)$. The printed 2017 form of (2), $s, t : \mathbf{U}(\psi) \to U(y)$ with one $y$ for both maps, together with the other requirements forces $s(\psi) = t(\psi)$; for $\mathbb{Z}/2 \ltimes \mathbb{R}$ by reflection and $Y = \{1, -1\}$ the hypotheses hold and a natural representation in the 2021 form exists, but no data of the printed 2017 form. *Book:* Proposition 7.1.15, p. 297 (2017: 7.5, p. 233), with (2) printed as $U(s(\psi))$, $U(t(\psi))$ in 2021. *Hidden hypotheses:* none. *Deviation (same):* none for the 2021 form; the 2017 form of (2) as printed does not hold as we read it, and the structure `NaturalRepFinite` of V7.25 already had the 2021 form (Findings, items 25, 119).

**V7.72 (Lemma 7.1.16; `Polyfold.EpGroupoid.NaturalRepFinite.mem_U_of_tendsto`, `Polyfold.EpGroupoid.le`**
In the construction of V7.71, let $\psi \in \mathbf{Y}$ and $\psi'_k \in \mathbf{U}(\psi)$ converge to $\psi' \in \mathbf{X}$. If $(s(\psi'_k))$ or $(t(\psi'_k))$ converges to a point of $U(s(\psi)) \cup U(t(\psi))$, then $\psi' \in \mathbf{U}(\psi)$, $s(\psi'_k) \to s(\psi') \in U(s(\psi))$ and $t(\psi'_k) \to t(\psi') \in U(t(\psi))$. *Book:* Lemma 7.1.16, p. 300, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (same):* none; the step that a limit of points of $U(a)$ lying in $U(b)$ lies in $U(a)$ (the $U(y)$ are open and disjoint) is a separate lemma (Findings, item 120).

**V7.73 (Corollary 7.1.17 (4)–(7); `Polyfold.EpGroupoid.corollary_7_1_17_4_2021`, `Polyfold.EpGroupoid.corol`**
**`Polyfold.EpGroupoid.corollary_7_1_17_6_2021`, `Polyfold.EpGroupoid.corollary_7_1_17_7_2021`).**
In the construction of V7.71: (4) for $x \in Y$ there is a natural representation $(\Phi, \Gamma)$ of $G_x$ on $U(x)$ with $\Gamma(g, z) = (s|\mathbf{U}(g))^{-1}(z)$ and $f(g)(z) = t(\Gamma(g, z)) = \Phi(g)(z)$; (5) $f(\varphi)$ is equivariant along $\gamma_\varphi(g) = \varphi \circ g \circ \varphi^{-1}$; (6) for composable $\varphi, \psi$ the multiplication is an sc-diffeomorphism from $\mathbf{U}(\psi) \,{}_s\!\times_t \mathbf{U}(\varphi)$, an open subset of the fibred product sub-M-polyfold $\mathbf{X} \,{}_s\!\times_t \mathbf{X}$, onto $\mathbf{U}(\psi \circ \varphi)$; (7) the $U(y)$ can be taken with the properness property. *Book:* Corollary 7.1.17, p. 301, new in the 2021 edition ((1)–(3) are those of Proposition 7.1.15). *Hidden hypotheses:* none. *Deviation (restated):* in (6) the domain is read inside Part I's fibred product sub-M-polyfold (Proposition 2.5.20 (2017:

2.15)) (Findings, item 120).

**V7.74 (Definition 7.2.5 and Lemma 7.2.6; `Polyfold.EpGroupoid.NaturalRep.kerOn`, `Polyfold.EpGroupoid.Natu` `Polyfold.EpGroupoid.NaturalRep.kerOn_anti`, `Polyfold.EpGroupoid.NaturalRep.kerOn_normal`, `Polyfold.EpGroupoid.NaturalRep.kerOn_le_trivialSubgroup`, `Polyfold.EpGroupoid.definition_7_2_5_20` `Polyfold.EpGroupoid.lemma_7_2_6_2021`, `Polyfold.EpGroupoid.trivialSubgroupEquiv_2021`, `Polyfold.EpGroupoid.trivialSubgroupEquiv_2021_apply`).** For a natural representation at $x$ and $V \subseteq X$, $T_x^V$ is the subgroup of the $g$ with $\Phi(g) = \mathrm{id}$ on $V \cap U(x)$; it decreases as $V$ grows and is normal for invariant $V$. For every open $W \ni x$, $T_x^W \subseteq T_x$, and there is an invariant open $V \ni x$ in $U(x)$ with $T_x^W = T_x$ for every open $x \in W \subseteq V$; so $T_x$, the intrinsic subgroup of V7.37, is the maximal subgroup acting trivially on a small $W(x)$, for every natural representation. For $\varphi : x \to y$, $g \mapsto \varphi g \varphi^{-1}$ is injective on $T_x$ with image $T_y$, an isomorphism $T_x \cong T_y$. *Book:* Definition 7.2.5 and Lemma 7.2.6, p. 310 (Definition 7.2.5: 2017 p. 241, before Definition 7.12; Lemma 7.2.6 new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (restated):* "maximal subgroup acting trivially on some small $W(x)$" is the pair of statements above (Findings, item 121).

**V7.75 (Definition 7.2.7, Remark 7.2.8 and Lemma 7.2.9 in the 2021 form; `Polyfold.EpGroupoid.remark_7_2_8_20` `Polyfold.EpGroupoid.isOpenQuotientMap_reduced`, `Polyfold.EpGroupoid.isOpenMap_reduced`, `Polyfold.EpGroupoid.isOpenEmbedding_domRestrict_reduced`, `Polyfold.EpGroupoid.reducedTgt`).** The topology $T_R$ of $\mathbf{X}^R$ (the quotient topology) makes $\pi : \mathbf{X} \to \mathbf{X}^R$ continuous and is finer than every topology with this property; $\pi$ is an open quotient map and an open embedding on every sheet of $s$. *Book:* Definition 7.2.7, p. 310 (2017: 7.12), Remark 7.2.8, p. 311, new in the 2021 edition, and Lemma 7.2.9, p. 311. *Hidden hypotheses:* none. *Deviation (restated):* "the finest topology for which $\pi$ is continuous" is the quotient topology. In Definition 7.2.7 the target map is printed "$s([\varphi]) = t_X(\varphi)$" (possible misprint for $t([\varphi])$; `reducedTgt`); Lemma 7.2.9 is V7.39's `isOpenMap_reduced` and `isOpenEmbedding_domRestrict_reduced` (Findings, items 122, 123).

**V7.76 (Lemma 7.2.11; `Polyfold.EpGroupoid.finite_reduced _preimage`, `Polyfold.EpGroupoid.IsReducedEve` `Polyfold.EpGroupoid.exists_preimage_reduced _subset_of_t2Space`, `Polyfold.EpGroupoid.isReducedEve` `Polyfold.EpGroupoid.isClosed_nonEffectivePart_of_isReducedEvenlyCovered`, `Polyfold.EpGroupoid.isR` `Polyfold.EpGroupoid.lemma_7_2_11_2021`, `Polyfold.EpGroupoid.Examples.JumpingRetract.reduced _prei` `Polyfold.EpGroupoid.Examples.JumpingRetract.not_isReducedEvenlyCovered`).** The class $\pi^{-1}([\varphi_0])$ is finite. If $T_R$ is Hausdorff, there are open $U(\varphi) \ni \varphi$ for the members $\varphi$ of the class of $\varphi_0$ and an open $U([\varphi_0]) \ni [\varphi_0]$ such that (1) distinct $U(\varphi)$ have disjoint closures, (2) $\pi(U(\varphi)) = U([\varphi_0])$ and $\pi$ is an open embedding on $U(\varphi)$, (3) $\pi^{-1}(U([\varphi_0]))$ is covered by the $U(\varphi)$. Conversely, (1)–(3) at every unit force $\mathbf{X}^T$ closed; so (1)–(3) hold at every morphism iff $T_R$ is Hausdorff, and they fail at $(1, o_0)$ for the ep-groupoid of V7.82. *Book:* Lemma 7.2.11, p. 315, new in the 2021 edition, for every ep-groupoid. *Hidden hypotheses:* $T_R$ Hausdorff; the proof of (3) uses the uniqueness of limits in $T_R$. *Deviation (explicit):* without the hypothesis the lemma does not hold as we read it; "homeomorphism onto $U([\varphi_0])$" is read as (2) (Findings, item 124).

**V7.77 (Proposition 7.2.12; `Polyfold.EpGroupoid.paracompactSpace_of_isOpenMap`, `Polyfold.EpGroupoid.exi` `Polyfold.EpGroupoid.paracompactSpace_reducedMor`, `Polyfold.EpGroupoid.proposition_7_2_12_2021`).** For every ep-groupoid, $(\mathbf{X}^R, T_R)$ is paracompact (every open cover has a locally finite open refinement; Hausdorffness not included). *Book:* Proposition 7.2.12 and its footnote, p. 317, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (same):* none in the statement. *Proof:* not the text's, which uses Lemma 7.2.11 (1)–(3) and so, as we read it, $T_R$ Hausdorff: paracompactness descends along an open continuous surjection $p : M \to Q$ if every $q$ has a compact $K$ such that every open $N \supseteq K$ contains $p^{-1}(O)$ for an open $O \ni q$; for $\pi$, $K$ is the set of $\varphi \in s^{-1}(\mathrm{cl}\, V)$ with

$(s,t)(\varphi) = (s,t)(q)$, by properness (Findings, items 111, 125).

**V7.78 (Proposition 7.2.13 and Theorem 7.2.14;** `Polyfold.EpGroupoid.proposition_7_2_13_2021`, `Polyfold.EpGroupoid.theorem_7_2_14_2021`, `Polyfold.EpGroupoid.reducedPolyfold`, `Polyfold.EpGroupoid.reducedEpGroupoid`, `Polyfold.EpGroupoid.isEffective_reducedEpGroupoid`**).** The natural charts (`reducedAtlas`) define an M-polyfold structure on $(\mathbf{X}^R, T_R)$ — some Hausdorff paracompact M-polyfold structure on $T_R$ with the natural atlas exists — iff $T_R$ is Hausdorff. If $T_R$ is Hausdorff, $(X, \mathbf{X}^R)$ is an ep-groupoid `reducedEpGroupoid` with the groupoid structure of V7.40, the object M-polyfold of $X$, the natural atlas on $\mathbf{X}^R$, and an ep-functor $X \to X_R$; it is effective (Proposition 7.2.16, V7.40). *Book:* Proposition 7.2.13, p. 318, new in the 2021 edition, and Theorem 7.2.14, p. 318 (2017: Proposition 7.10, p. 244). *Hidden hypotheses:* none. *Deviation (restated):* Proposition 7.2.13 is read as the existence of the structure. Theorem 7.2.14 assumes exactly the hypothesis that V7.39 and V7.40 take, $T_R$ Hausdorff; paracompactness is supplied by V7.77 (gap G.71, changed) (Findings, items 56, 107, 111, 126, 143).

**V7.79 (Proposition 7.2.17;** `Polyfold.EpGroupoid.reduced _eq_iff_comp_inv`, `Polyfold.EpGroupoid.mem_nonE` `Polyfold.EpGroupoid.continuous_comp_fibredProduct`, `Polyfold.EpGroupoid.continuousOn_comp_inv`, `Polyfold.EpGroupoid.isClosed_reduced _eq_of_isClosed`, `Polyfold.EpGroupoid.isClosed_nonEffectiveP` `Polyfold.EpGroupoid.proposition_7_2_17_2021`, `Polyfold.EpGroupoid.isClosed_src_eq_tgt`, `Polyfold.EpGroupoid.closure_nonEffectivePart_subset`**).** $T_R$ is Hausdorff iff the non-effective part $\mathbf{X}^T$ is closed in $\mathbf{X}$. Also $[\varphi] = [\psi]$ iff $t(\varphi) = t(\psi)$ and $\psi^{-1} \circ \varphi \in \mathbf{X}^T$, and $\operatorname{cl} \mathbf{X}^T \subseteq \{s = t\}$. *Book:* Proposition 7.2.17, p. 320, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (same):* none. *Proof:* the relation $[\varphi] = [\psi]$ is closed in $\mathbf{X} \times \mathbf{X}$ when $\mathbf{X}^T$ is closed, and a space is Hausdorff iff this relation is closed for an open quotient map (Mathlib); properness is not used (Findings, item 127).

**V7.80 (Definition 7.2.18 (= Definition 7.4.1), Remarks 7.2.19 and 7.2.20;** `Polyfold.EpGroupoid.NaturalRep.HasL` `Polyfold.EpGroupoid.IsLUC`, `Polyfold.EpGroupoid.isLUC_iff_forall_naturalRep`, `Polyfold.EpGroupoid.N` `Polyfold.EpGroupoid.remark_7_2_20_2021`, `Polyfold.EpGroupoid.Examples.remark_7_2_19_2021`, `Polyfold.EpGroupoid.isLUC_trivial`, `Polyfold.EpGroupoid.isLUC_of_isEffective`, `Polyfold.EpGroupoid.isClosed_range_unit`, `Polyfold.EpGroupoid.nonEffectivePart_eq_range_unit_of_i` `Polyfold.EpGroupoid.Examples.isLUC_trivialZ3`, `Polyfold.EpGroupoid.Examples.isLUC_reflection`**).** $X$ has the local unique continuation property (LUC) if every object $x$ has a natural representation $U(x)$ such that $\Phi(g) = \mathrm{id}$ on a nonempty open subset of $U(x)$ forces $\Phi(g) = \mathrm{id}$ on $U(x)$. Under LUC every natural representation has this property, and there are such inside every open $O \ni x$. Not every ep-groupoid has LUC (V7.82). LUC holds for the trivial ep-groupoid of every M-polyfold, for every effective ep-groupoid ($\mathbf{X}^T$ is then the closed set of units), for $\mathbb{Z}/2 \ltimes \mathbb{R}$ by reflection, and for the non-effective trivial $\mathbb{Z}/3$-action on $\mathbb{R}$. *Book:* Definition 7.2.18, Remarks 7.2.19, 7.2.20, p. 321, new in the 2021 edition; Definition 7.4.1, p. 328, repeats Definition 7.2.18. *Hidden hypotheses:* none. *Deviation (stronger):* that the choice of $U(x)$ does not matter is proved (by V7.81, $\mathbf{X}^T$ is then closed, and a connectedness argument) (Findings, items 128, 144).

**V7.81 (Proposition 7.2.21, Theorem 7.2.22 and Proposition 7.2.23;** `Polyfold.EpGroupoid.NaturalRep.eventuall` `Polyfold.EpGroupoid.NaturalRep.Γ_mem_of_eventually`, `Polyfold.EpGroupoid.isClosed_nonEffectivePar` `Polyfold.EpGroupoid.proposition_7_2_21_2021`, `Polyfold.EpGroupoid.theorem_7_2_22_2021`, `Polyfold.EpGroupoid.t2Space_reducedMor_of_isLUC`, `Polyfold.EpGroupoid.proposition_7_2_23_2021`, `Polyfold.EpGroupoid.Examples.nonEffectivePart_trivialZ3`, `Polyfold.EpGroupoid.Examples.t2Space_re` $X$ has LUC iff $\mathbf{X}^T$ is closed. Equivalent are: $T_R$ is Hausdorff; $X$ has LUC; $\mathbf{X}^T$ is closed in $\mathbf{X}$. Under LUC, $T_R$ is Hausdorff and `reducedEpGroupoid` is an effective ep-groupoid with the structure of V7.78 and an ep-functor $X \to X_R$. For the trivial $\mathbb{Z}/3$-action, $\mathbf{X}^T = \mathbf{X}$ and $T_R$ is Hausdorff. *Book:*

Proposition 7.2.21, p. 321, Theorem 7.2.22 and Proposition 7.2.23, p. 323, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (same):* none. *Proof:* $\Gamma(g,y) \in \mathbf{X}^T$ iff $\Phi(g) = \mathrm{id}$ near $y$; if $\mathbf{X}^T$ is closed, the set of such $y$ is open and closed in the connected $U(x)$ (Findings, items 60, 128, 129).

**V7.82 (Appendix 7.6.3, Zhou's example, for an analogue;** `Polyfold.EpGroupoid.Examples.JumpingRetract.not` `Polyfold.EpGroupoid.Examples.JumpingRetract.mem_closure_nonEffectivePart,Polyfold.EpGroupoid.Ex` `Polyfold.EpGroupoid.Examples.JumpingRetract.not_isLUC,Polyfold.EpGroupoid.Examples.remark_7_2_1` For the ep-groupoid $\mathbb{Z}/2 \ltimes O$ of V7.39 ($\mathbb{Z}/2$ acting by $(t,z) \mapsto (t,\pm z)$ on the dimension-jumping retract $O$ of Part I's Remark 4.2.2 (2017: 4.3)), $(-1,o_0) \in \mathrm{cl}(\mathbf{X}^T) \setminus \mathbf{X}^T$, so $\mathbf{X}^T$ is not closed, $T_R$ is not Hausdorff, and LUC fails, consistently with Theorem 7.2.22. *Book:* Appendix 7.6.3, p. 348, new in the 2021 edition: Zhou's example, $\mathbb{Z}/2$ acting on the retract of Part I's Subsection 1.4.4 by $(s,f) \mapsto (s,-f)$, with $\mathrm{cl}(\mathbf{X}^T) \setminus \mathbf{X}^T$ one point. *Hidden hypotheses:* none. *Deviation (narrower):* our retract is not Zhou's but has the same features (the action is trivial over $t < 0$ and moves points arbitrarily close to the origin over $t > 0$); the point of $\mathrm{cl}(\mathbf{X}^T) \setminus \mathbf{X}^T$ is exhibited, $\mathbf{X}^T$ is not computed exactly (gap G.83). The round-trip counterexample of V7.39 is thus of the kind the 2021 text describes in Remark 7.2.8 (Findings, items 107, 129, 130).

**V7.83 (Appendix 7.6.4: (7.24), Theorem 7.6.7 and Corollary 7.6.8 for $i = 1$, as sets;** `Polyfold.EpGroupoid.mem_nonEffectivePart_liftOne_of_levelVal,Polyfold.EpGroupoid.NaturalRep.lev` `Polyfold.EpGroupoid.NaturalRep.continuousOn_levelOneΓ,Polyfold.EpGroupoid.NaturalRep.eventuall` `Polyfold.EpGroupoid.NaturalRep.levelOneΓ_mem_of_eventually,Polyfold.EpGroupoid.levelVal_mem_no` `Polyfold.EpGroupoid.HasConnectedDenseLevelOneReps,Polyfold.EpGroupoid.nonEffectivePart_liftOne_` `Polyfold.EpGroupoid.liftOne_reduced _eq_iff,Polyfold.EpGroupoid.reducedLevelOneMap,` `Polyfold.EpGroupoid.theorem_7_6_7_2021, Polyfold.EpGroupoid.corollary_7_6_8_2021,` `Polyfold.EpGroupoid.corollary_7_6_8_2021_of_isZhou,Polyfold.EpGroupoid.Examples.pointEpGroupoi` `Polyfold.EpGroupoid.Examples.isLUC_liftOne_point,Polyfold.EpGroupoid.Examples.hasConnectedDense` `Polyfold.EpGroupoid.Examples.theorem_7_6_7_2021_point,Polyfold.EpGroupoid.Examples.corollary_7_` A morphism of level 1 in $\mathbf{X}^T$ lies in $(\mathbf{X}^1)^T$ ((7.24)). Assume that $X^1$ has LUC and that every $x \in X_1$ has a natural representation with $U(x) \cap X_1$ connected in $X^1$ and dense in $U(x)$. Then $(\mathbf{X}^1)^T = (\mathbf{X}^T)^1$, and $[\varphi]_1 \mapsto [\varphi]$ is a well defined, injective, continuous map $(\mathbf{X}^1)^R \to \mathbf{X}^R$ onto $\pi(\mathbf{X}_1)$. The same holds under (ZHOU) for $X$ with the Part I facts of V7.86 (Corollary 7.6.8, with LUC for $X^1$ from V7.90); the hypotheses hold for the trivial ep-groupoid of a point. *Book:* (7.24), Theorem 7.6.7 and Corollary 7.6.8, pp. 348–351, new in the 2021 edition: under (SLUC), resp. (ZHOU), the identification $(X^i)_R = (X_R)^i$ is an sc-diffeomorphism on objects and morphisms, for every $i$. *Hidden hypotheses:* that $U(x) \cap X_1$ is connected in $X^1$ and dense in $U(x)$, which the proof uses (the density is that of $X_1$ in $X$, gap G.73). *Deviation (narrower):* $i = 1$ and as sets with a continuous map; the sc-diffeomorphism and $i \geq 2$ are not formalized (gap G.82); of (SLUC) only LUC for $X^1$ is used (Findings, item 131).

**V7.84 (the index lifts $X^i$, Proposition 7.4.2 and Remark 7.4.3;** `Polyfold.EP.liftOne,` `Polyfold.EP.lift,Polyfold.EP.lift_zero,Polyfold.EP.lift_succ,Polyfold.EP.lift_succ',` `Polyfold.EpGroupoid.proposition_7_4_2_2021, Polyfold.EpGroupoid.remark_7_4_3_2021`**).** $X^0 = X$ and $X^{i+1} = (X^i)^1$, also $= (X^1)^i$. Under LUC, every natural representation has the property of Definition 7.2.18, in particular every $W(x) \supseteq U(x)$, and there are such inside every open $O \ni x$. *Book:* p. 329 ($X^i$), Proposition 7.4.2, p. 328, and Remark 7.4.3, p. 329, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (stronger):* the hypotheses $U(x) \subseteq W(x)$ and LUC on $U(x)$ are not needed (Findings, item 132).

**V7.85 (Lemma 7.4.4;** `Polyfold.EpGroupoid.HasLevelOneConnectedDenseReps,Polyfold.EpGroupoid.Naturall` `Polyfold.EpGroupoid.isLUC_of_isLUC_liftOne, Polyfold.EpGroupoid.lemma_7_4_4_2021,`

`Polyfold.EpGroupoid.hasLevelOneConnectedDenseReps_of_subsingleton`, `Polyfold.EpGroupoid.Examples.`
`Polyfold.EpGroupoid.Examples.lemma_7_4_4_2021_pointEP`). If $X^i$ has LUC and, for every $k < i$, every object of $X^k$ has a natural representation $U(x)$ with $U(x) \cap X_1$ connected in $X^1$ and dense in $U(x)$ (for $X^k$), then $X$ and every $X^k$, $k \leq i$, have LUC. The hypotheses hold for all lifts of the point ep-groupoid. *Book:* Lemma 7.4.4, p. 329, new in the 2021 edition. *Hidden hypotheses:* that $U(x) \cap X_1$ is connected in $X^1$, which the proof states without proof, and dense in $U(x)$ (gaps G.85, G.73). *Deviation (explicit):* the two facts are hypotheses; in the proof, "$s : W(g) \to W(y)$" for $W(h)$ (possible misprint) (Findings, item 133).

**V7.86 (the Part I facts used in §7.4;** `Polyfold.MPolyfold.DenseSmooth`, `Polyfold.MPolyfold.SmoothConnected`, `Polyfold.MPolyfold.DenseSmoothTangent`, `Polyfold.MPolyfold.TangentDeterminedSmooth`, `Polyfold.MPolyfold.TangentDeterminedSmooth.eq_self`, `Polyfold.MPolyfold.DenseSmooth.dense`, `Polyfold.MPolyfold.DenseSmooth.subset_closure_inter`, `Polyfold.MPolyfold.DenseSmooth.exists_smoot` `Polyfold.MPolyfold.eqOn_tangentFiber_of_mem_closure`, `Polyfold.MPolyfold.SmoothRegular`, `Polyfold.MPolyfold.smoothRegular_of_subsingleton`, `Polyfold.MPolyfold.smoothRegular_discrete_unit` `Polyfold.MPolyfold.denseSmoothTangent_discrete_unit`**).** For an M-polyfold $X$: `DenseSmooth`, $X_\infty$ is dense in every $X_m$; `SmoothConnected`, $U \cap X_\infty$ is connected in $X_\infty$ for connected open $U$; `DenseSmoothTangent`, $(T_xX)_\infty$ is dense in $T_xX$ for smooth $x$; `TangentDeterminedSmooth`, two sc-smooth maps agreeing near a smooth $y$ in $X_\infty$ have the same tangent on $T_yX$. `SmoothRegular` bundles the first, the second, the last and the openness of $p : TX \to X^1$. Consequences: $X_\infty$ is dense in $X$, and, if $p$ is open, a map continuous on an open $W \subseteq TX$ and the identity over a set $A \subseteq X_\infty$ is the identity over its closure in $X_\infty$. All hold for M-polyfolds on a subsingleton with every point smooth. *Book:* used in the proofs of Lemma 7.4.6, Propositions 7.4.8, 7.4.10, 7.4.22 and Lemma 7.4.15, pp. 331–340; the density of $X_\infty$ is stated on p. 289. *Hidden hypotheses:* these Part I facts, not proved in Part I for M-polyfolds. *Deviation (explicit):* explicit hypotheses of the §7.4 results that use them (gaps G.85, G.73, G.8) (Findings, item 134).

**V7.87 (the tangent $Tg$ of an isotropy element;** `Polyfold.EpGroupoid.NaturalRep.eventually_Φ_eq_Γ`, `Polyfold.EpGroupoid.NaturalRep.isoΓ`, `Polyfold.EpGroupoid.NaturalRep.isoΓ_val`, `Polyfold.EpGroupoid.NaturalRep.isoΓ_mem_trivialSubgroup_iff`, `Polyfold.EpGroupoid.NaturalRep.tang` `Polyfold.EpGroupoid.NaturalRep.tangentMor_eq_tΦ`, `Polyfold.EpGroupoid.NaturalRep.tΦ_eq_tΦ`, `Polyfold.EpGroupoid.NaturalRep.tangentMor_isoΓ`**).** The tangent $T\varphi = Tt \circ (Ts|T_\varphi \mathbf{X})^{-1}$ of Definition 7.1.4 (defined without charts) of the morphism $\Gamma(g, y)$ is the tangent map $T\Phi(g)$ on $T_yX$; in particular $Tg = T\Phi(g)$ on $T_xX$, the same for every natural representation. For a fixed point $z$ of $\Phi(g)$, $\Gamma(g, z) \in G_z$, it lies in $T_z$ iff $\Phi(g) = \mathrm{id}$ near $z$, and it acts near $z$ as $g$ does. *Book:* p. 331 ("the linearization $Tg$") and Proposition 8.1.16's "by definition $Tg = T\hat{g}(x)$", new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (restated):* $Tg$ is a set map on the fibre $T_xX$; no linear structure is used, since (ZHOU) only asks that $Tg$ be the identity (gap G.7 not needed) (Findings, item 135).

**V7.88 (Definitions 7.4.5 and 7.4.7, the groups $T_x^\infty$;** `Polyfold.EpGroupoid.IsZhou`, `Polyfold.EpGroupoid.trivia` `Polyfold.EpGroupoid.mem_trivialSetInf_iff`, `Polyfold.EpGroupoid.trivialSubgroupInf`, `Polyfold.EpGroupoid.mem_trivialSubgroupInf_iff`, `Polyfold.EpGroupoid.trivialSubgroup_le_trivialSu` `Polyfold.EpGroupoid.IsZhou'`, `Polyfold.EpGroupoid.isZhou_of_trivialSubgroup_eq_top`, `Polyfold.EpGroupoid.isZhou'_of_trivialSubgroup_eq_top`, `Polyfold.EpGroupoid.isZhou_of_subsingleto` `Polyfold.EpGroupoid.isZhou'_of_subsingleton`, `Polyfold.EpGroupoid.isZhou_trivial`, `Polyfold.EpGroupoid.isZhou'_trivial`, `Polyfold.EpGroupoid.Examples.isZhou_trivialZ3`, `Polyfold.EpGroupoid.Examples.isZhou'_trivialZ3`**).** (ZHOU): for every smooth object $x$ and $g \in G_x$, if $Tg$ is the identity on $T_xX$ then $g \in T_x$. $T_x^\infty \supseteq T_x$ is the subgroup of the $g$ with

$\Phi(g) = \mathrm{id}$ near $x$ in $X_\infty$, for any natural representation. (ZHOU'): if $Tg$ is the identity on $(T_xX)_\infty$ then $g \in T_x^\infty$. Both hold when every $T_x = G_x$, in particular for trivial isotropy (the trivial ep-groupoid of every M-polyfold) and for the trivial $\mathbb{Z}/3$-action. *Book:* Definitions 7.4.5 and 7.4.7 and $T_x^\infty$, p. 331, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (restated):* $Tg$ as in V7.87 (Findings, items 135, 144).

**V7.89 (Lemma 7.4.6 and Proposition 7.4.8; `Polyfold.EpGroupoid.lemma_7_4_6_2021`, `Polyfold.EpGroupoid.isZhou_of_isZhou'`, `Polyfold.EpGroupoid.proposition_7_4_8_2021`, `Polyfold.EpGroupoid.NaturalRep.mem_tangentU_of_fiber`, `Polyfold.EpGroupoid.NaturalRep.tΦ_eq_self_`, `Polyfold.EpGroupoid.NaturalRep.fix_and_tangent_of_mem_closure`, `Polyfold.EpGroupoid.NaturalRep.ec`, `Polyfold.EpGroupoid.NaturalRep.eq_self_of_smooth_tangent`, `Polyfold.eq_of_mem_closure_of_continuc`, `Polyfold.EpGroupoid.Examples.proposition_7_4_8_2021_point`).** If $(T_xX)_\infty$ is dense in $T_xX$ for smooth $x$, (ZHOU) implies (ZHOU'). Given the Part I facts `SmoothRegular` of V7.86, (ZHOU') implies (ZHOU); so with both facts (ZHOU) and (ZHOU') are equivalent. The hypotheses hold for the point ep-groupoid. *Book:* Lemma 7.4.6, p. 331, and Proposition 7.4.8, p. 332, new in the 2021 edition. *Hidden hypotheses:* the Part I facts of V7.86 (gap G.85). *Deviation (explicit):* the proof of 7.4.8 uses that the smooth points of $U(x)$ near which $\hat{g} = \mathrm{id}$ in $X_\infty$ form an open, nonempty and closed subset of the connected $U(x)_\infty$, instead of sequences; "$z_k \to Y$" for $z_k \to y$ (possible misprint) (Findings, item 136).

**V7.90 (Proposition 7.4.10 and Corollary 7.4.11 for $X$ and $X^1$; `Polyfold.EpGroupoid.proposition_7_4_10_2021`, `Polyfold.EpGroupoid.IsSmoothUC`, `Polyfold.EpGroupoid.isSmoothUC_of_isZhou`, `Polyfold.EpGroupoid.isL`, `Polyfold.EpGroupoid.isLUC_liftOne_of_isSmoothUC`, `Polyfold.EpGroupoid.corollary_7_4_11_2021`, `Polyfold.EpGroupoid.corollary_7_4_11_2021_lift`).** Given `SmoothRegular`, (ZHOU) implies LUC, and more: unique continuation through $X_\infty$ (if $\hat{g} = \mathrm{id}$ on the smooth points of a nonempty open subset of $X_\infty$ inside $U(x)$, then $\hat{g} = \mathrm{id}$ on $U(x)$), which gives LUC for $X$ and for $X^1$. With (ZHOU) and `SmoothRegular` for every $X^i$, every $X^i$ has LUC. *Book:* Proposition 7.4.10, p. 333, and Corollary 7.4.11, p. 334, new in the 2021 edition (for every $X^i$, through Proposition 7.4.9). *Hidden hypotheses:* the Part I facts of V7.86 (gap G.85). *Deviation (narrower):* this form of Corollary 7.4.11 for all $i$ takes (ZHOU) for every $X^i$, the conclusion of Proposition 7.4.9, as a hypothesis; with (ZHOU) for $X$ only it is V7.102 (gap G.84, closed); the approximation of tangent vectors at the end of the proof of 7.4.10 is the consequence of the openness of $p : TX \to X^1$ in V7.86 (Findings, items 136, 137).

**V7.91 (Definition 7.4.12 and Theorem 7.4.13; `Polyfold.EP.IsSLUC`, `Polyfold.EpGroupoid.theorem_7_4_13_202`, `Polyfold.EpGroupoid.Examples.pointEP`, `Polyfold.EpGroupoid.Examples.lift_pointEP`, `Polyfold.EpGroupoid.Examples.smoothRegular_lift_pointEP`, `Polyfold.EpGroupoid.Examples.isZhou_lif`, `Polyfold.EpGroupoid.Examples.isSLUC_pointEP`, `Polyfold.EpGroupoid.Examples.theorem_7_4_13_2021_po`).** (SLUC): every $X^i$ has LUC. If every $X^i$ satisfies (ZHOU) and `SmoothRegular`, $X$ has (SLUC). The point ep-groupoid and all its lifts satisfy (ZHOU), `SmoothRegular` and (SLUC). *Book:* Definition 7.4.12 and Theorem 7.4.13, p. 334, new in the 2021 edition: (ZHOU) for $X$ implies (SLUC). *Hidden hypotheses:* the Part I facts of V7.86 for every $X^i$ (gap G.85). *Deviation (narrower):* (ZHOU) for every $X^i$ is a hypothesis here; Proposition 7.4.9 and Theorem 7.4.13 with (ZHOU) for $X$ only are V7.102 (gap G.84, closed) (Findings, items 137, 148).

**V7.92 (Definition 7.4.14, Lemma 7.4.15, Propositions 7.4.16–7.4.18; `Polyfold.EpGroupoid.IsVSLUC`, `Polyfold.EpGroupoid.isSmoothUC_of_isVSLUC`, `Polyfold.EpGroupoid.isVSLUC_of_isSmoothUC`, `Polyfold.EpGroupoid.lemma_7_4_15_2021`, `Polyfold.EpGroupoid.proposition_7_4_16_2021`, `Polyfold.EpGroupoid.proposition_7_4_17_2021`, `Polyfold.EpGroupoid.isVSLUC_of_isZhou`, `Polyfold.EpGroupoid.isSmoothUC_of_subsingleton`, `Polyfold.EpGroupoid.isVSLUC_of_subsingleton`,**

`Polyfold.EpGroupoid.isLUC_of_subsingleton_mor`, `Polyfold.EpGroupoid.isVSLUC_trivial`, `Polyfold.EpGroupoid.isSmoothUC_trivial`). (VSLUC): every smooth $x$ has a natural representation $U(x)$ and an open $V(x) \ni x$ of $X_\infty$ inside it, invariant under $G_x$, such that $\hat{g} = \mathrm{id}$ on a nonempty open subset of $V(x)$ forces $\hat{g} = \mathrm{id}$ on $V(x)$. If $X_\infty$ is dense in every $X_m$ and $U \cap X_\infty$ is connected for connected open $U$: (VSLUC) is equivalent to unique continuation through $X_\infty$, implies LUC for $X$ and for $X^1$, and (VSLUC) for every $X^i$ gives (SLUC); (ZHOU) with `SmoothRegular` implies (VSLUC). (VSLUC) holds for trivial isotropy. *Book:* Definition 7.4.14 and Lemma 7.4.15, p. 335, Propositions 7.4.16–7.4.18, p. 336, new in the 2021 edition. *Hidden hypotheses:* the Part I facts of V7.86 (gap G.85). *Deviation (narrower):* $V(x)$ is read open in $X_\infty$, inside some $U(x)$ and invariant; here Proposition 7.4.16 for $X^1$ only and Proposition 7.4.17 with (VSLUC) for every $X^i$; for every $X^i$ with (VSLUC) for $X$ only they are V7.103, and Proposition 7.4.18 in both directions is V7.104 (gap G.84, closed) (Findings, items 136, 137, 149, 150).

**V7.93 (Definitions 7.4.19 and 7.4.21, Lemma 7.4.20;** `Polyfold.EpGroupoid.IsLR`, `Polyfold.EpGroupoid.IsLR'`, `Polyfold.EpGroupoid.NaturalRep.exists_Φ_eq_of_ _eq`, `Polyfold.EpGroupoid.NaturalRep.eq_self_of_ _e` `Polyfold.EpGroupoid.lemma_7_4_20_2021`, `Polyfold.EpGroupoid.isLR_iff_isLR'`**).** (LR): inside every open $O \ni x$ there is a natural representation $U(x)$ such that every sc-diffeomorphism $\beta$ of $U(x)$ with $|\beta(y)| = |y|$ is $\hat{g}$ for some $g \in G_x$. (LR′): the same for sc-diffeomorphisms $\beta$ of $U(x)$ onto an open image with $\beta(x) = x$. If $\beta$ is an sc-diffeomorphism of $U(x)$ onto $\beta(U(x))$ with $\beta(x) = x$ and $|\beta(y)| = |y|$, then $\beta(U(x)) = U(x)$; hence (LR) and (LR′) are equivalent. *Book:* Definition 7.4.19, Lemma 7.4.20, p. 337, Definition 7.4.21, p. 338, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (restated):* "arbitrarily small" is "inside every open $O \ni x$", $|\beta(y)| = |y|$ is equality in $|X|$; Lemma 7.4.20 is proved by the finite closed cover $U(x) = \bigcup_g \{\beta = \hat{g}\}$ and connectedness, instead of paths (Findings, item 138).

**V7.94 (Proposition 7.4.22, Definition 7.4.23 and Theorem 7.4.24;** `Polyfold.EpGroupoid.NaturalRep.eqOn_Φ_of_` `Polyfold.EpGroupoid.NaturalRep.exists_eq_Φ_of_isZhou`, `Polyfold.EpGroupoid.proposition_7_4_22_202` `Polyfold.EP.IsSLR`, `Polyfold.EpGroupoid.theorem_7_4_24_2021`, `Polyfold.EpGroupoid.Examples.isSLR_po` Under (ZHOU) and LUC, $\hat{g}$ and $\hat{g}'$ agree on $U(x)$ once they agree at a smooth point together with their tangents there. Given `SmoothRegular`, (ZHOU) implies (LR). (SLR): every $X^i$ has (LR); if every $X^i$ satisfies (ZHOU) and `SmoothRegular`, $X$ has (SLR). The point ep-groupoid has (SLR). *Book:* Proposition 7.4.22, p. 338, Definition 7.4.23, p. 339, Theorem 7.4.24, p. 340, new in the 2021 edition. *Hidden hypotheses:* the Part I facts of V7.86 (gap G.85). *Deviation (narrower):* Theorem 7.4.24 takes (ZHOU) for every $X^i$ as a hypothesis here; with (ZHOU) for $X$ only it is V7.102 (gap G.84, closed). *Proof:* without passing to the reduced ep-groupoid and without the density of regular points: on the open dense union of the $X_\infty$-interiors of the sets $\{\beta = \hat{g}\}$, two pieces whose closures meet have equal tangents at a common smooth point, hence equal $\hat{g}$ (Findings, item 138).

**V7.95 (the example of Appendix 7.6.3 fails (ZHOU);** `Polyfold.EpGroupoid.Examples.JumpingRetract.not_isZ` If the object M-polyfold of the ep-groupoid of V7.82 satisfies `SmoothRegular`, the ep-groupoid does not satisfy (ZHOU): it does not have LUC, and (ZHOU) implies LUC (V7.90). *Book:* p. 331, new in the 2021 edition: the example of Appendix 7.6.3 does not satisfy the Zhou condition. *Hidden hypotheses:* the Part I facts of V7.86 for its object M-polyfold. *Deviation (narrower):* for our analogue, and conditional on these facts (gap G.83) (Findings, item 139).

**V7.96 (transport of the tame boundary along index-preserving local homeomorphisms;** `Polyfold.TameBoundary.IsDHomeoOn`, `Polyfold.TameBoundary.IsDLocalHomeo`, `Polyfold.TameBoundary.isDH` `Polyfold.TameBoundary.IsDHomeoOn.isFaceIn_image`, `Polyfold.TameBoundary.IsDHomeoOn.mem_faceStruct`

`Polyfold.TameBoundary.IsDHomeoOn.isMor_image`, `Polyfold.TameBoundary.IsDHomeoOn.image_faceExt`, `Polyfold.TameBoundary.BTriple.map`, `Polyfold.TameBoundary.map_cls_eq`, `Polyfold.TameBoundary.IsLift` `Polyfold.TameBoundary.IsLiftOf.unique`, `Polyfold.TameBoundary.IsLiftOf.comp`, `Polyfold.TameBoundary` `Polyfold.TameBoundary.bLift`, `Polyfold.TameBoundary.bLift_cls`, `Polyfold.TameBoundary.continuous_bL` `Polyfold.TameBoundary.isOpenMap_bLift`, `Polyfold.TameBoundary.isLocalHomeomorph_bLift`, `Polyfold.TameBoundary.isCompact_preimage_bMap`, `Polyfold.TameBoundary.exists_rep_centered`, `Polyfold.MPolyfold.IsScDiffeoOn.isDHomeoOn`, `Polyfold.MPolyfold.IsLocalScDiffeo.isDLocalHomeo`**).** For a corner index $d$ (the degeneracy index of a tame M-polyfold, or any `IsCornerIndex`): a map $f$ that is a homeomorphism of an open $W$ onto an open set preserving the index carries faces to faces, face-structured neighbourhoods to face-structured neighbourhoods, and admissible triples $(y, V, \theta)$ to admissible triples, with a class independent of the chosen $W$. For an index-preserving local homeomorphism $f$, $[y, V, \theta] \mapsto [f(y), f(V), f(\theta)]$ is a well defined map $bf$ of the tame boundaries, continuous, open and a local homeomorphism; preimages of compact sets under $B$ are compact. An sc-diffeomorphism between open sets preserves the degeneracy index. *Book:* §7.6.5.2, pp. 353–355, new in the 2021 edition, where the maps (7.25)–(7.28) are defined through sc-diffeomorphisms of sheets. *Hidden hypotheses:* none. *Deviation (extra):* a tool of the formalization: the topological content of those sc-diffeomorphisms (Findings, item 140).

**V7.97 ((7.25)–(7.29), the boundary groupoid;** `Polyfold.EpGroupoid.isTame_PMor`, `Polyfold.EpGroupoid.corne` `Polyfold.EpGroupoid.cornerMor`, `Polyfold.EpGroupoid.bObj`, `Polyfold.EpGroupoid.bMor`, `Polyfold.EpGroupoid.isDLocalHomeo_src`, `Polyfold.EpGroupoid.isDLocalHomeo_tgt`, `Polyfold.EpGroupoid.isScDiffeoOn_inv`, `Polyfold.EpGroupoid.isDLocalHomeo_inv`, `Polyfold.EpGroupoid.isDHomeoOn_unit`, `Polyfold.EpGroupoid.isDLocalHomeo_unit`, `Polyfold.EpGroupoid.bSrc`, `Polyfold.EpGroupoid.bTgt`, `Polyfold.EpGroupoid.bInv`, `Polyfold.EpGroupoid.bUnit`, `Polyfold.EpGroupoid.bSrc_cls`, `Polyfold.EpGroupoid.bTgt_cls`, `Polyfold.EpGroupoid.bInv_cls`, `Polyfold.EpGroupoid.bUnit_cls`, `Polyfold.EpGroupoid.eq_of_bMap_eq_of` `Polyfold.EpGroupoid.exists_bComp`, `Polyfold.EpGroupoid.bComp`, `Polyfold.EpGroupoid.bComp_spec`, `Polyfold.EpGroupoid.eventuallyEq_comp_sheet`, `Polyfold.EpGroupoid.bComp_eq`, `Polyfold.EpGroupoid.bc` `Polyfold.EpGroupoid.boundaryFunctor`**).** For a tame ep-groupoid, $\mathbf{X}$ is tame. On the tame boundaries $bX$, $b\mathbf{X}$: $s([\varphi, \mathbf{V}, \theta]) = [s(\varphi), s(\mathbf{V}), s(\theta)]$ for a sheet $\mathbf{V}$ of $s$, likewise $t$; $\iota([\varphi, \mathbf{V}, \theta]) = [\varphi^{-1}, \mathbf{V}^{-1}, \theta^{-1}]$; $u([x, V, \theta]) = [1_x, \mathbf{V}, \bar{\theta}]$ with $\bar{\theta}$ the preimage of $\theta$ under $s|\mathbf{V}$; and $m(a, b)$ is the unique class over $\psi \circ \varphi$ with source $s(b)$, which is the printed $[\psi \circ \varphi, \mathbf{W}\ {}_s\circ_t\ \mathbf{V}, \sigma\ {}_s\circ_t\ \theta]$. With these maps $(bX, b\mathbf{X})$ is a groupoid and $B$ a functor $(bX, b\mathbf{X}) \to (X, \mathbf{X})$. *Book:* (7.25)–(7.29), pp. 354–355, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (restated):* the sc-diffeomorphisms of the sheets enter through their topological content (V7.96); (7.28) is read through $u$, an index-preserving homeomorphism onto the open set $u(X)$; off composable pairs $m$ takes a junk value (Findings, item 140).

**V7.98 (Theorem 7.6.9, its topological part;** `Polyfold.EpGroupoid.metrizableSpace_bObj`, `Polyfold.EpGroupoid.metrizableSpace_bMor`, `Polyfold.EpGroupoid.continuous_bSrc`, `Polyfold.EpGroupoid.continuous_bTgt`, `Polyfold.EpGroupoid.continuous_bInv`, `Polyfold.EpGroupoid.con` `Polyfold.EpGroupoid.isLocalHomeomorph_bSrc`, `Polyfold.EpGroupoid.isLocalHomeomorph_bTgt`, `Polyfold.EpGroupoid.continuousOn_bComp`, `Polyfold.EpGroupoid.exists_isProperMap_bTgt`, `Polyfold.EpGroupoid.theorem_7_6_9_2021`, `Polyfold.EpGroupoid.bSrc_trivial`, `Polyfold.EpGroupoid.tri` For a tame ep-groupoid: (1) $bX$ and $b\mathbf{X}$ are metrizable, every $U(x, V, \theta)$ is open and every $\Phi_{(x,V,\theta)}$ an open embedding, and these properties determine the topologies; (3) both maps $B$ are continuous; (4) a point of index $d$ has exactly $d$ preimages under $B$, on objects and on morphisms; (5) $s, t$ are local homeomorphisms, $u, \iota$ are continuous, $m$ is continuous on the composable pairs,

$(bX, b\mathbf{X})$ has the properness property of Definition 7.1.3 (3), and $B$ is a functor. The trivial ep-groupoid of the tame quadrant $[0,\infty)^n \oplus W$, $n \geq 1$, is tame, with $bX$, $b\mathbf{X}$ nonempty and $n$ classes over the corner. *Book:* Theorem 7.6.9 and the characterization after it, p. 355, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (narrower):* the topological part only; (2), the sc-smoothness in (3), the index $d_{bX}$ in (4) beyond the fibre count, and the sc-smoothness and tameness in (5) depend on 2021 Theorem 2.7.13 (2) and are not covered here (they are V7.99–V7.101, `Chapter7/TameBoundaryEpStructure.lean`); "$\Phi$ is a homeomorphism onto $\theta$" holds as we read it only for $V$ centred at $x$, as for (2.44), and is used as an open embedding (Findings, items 140, 141).

**V7.99 (sc-smooth maps of tame boundaries; `Polyfold.MPolyfold.isScSmoothOn_of_bMap`, `Polyfold.MPolyfold.isScSmoothMap_bLift`, `Polyfold.MPolyfold.isLiftOf_bLift_sheet`, `Polyfold.MPolyfold.isLocalScDiffeo_bLift`).** Let $A$, $B$ be tame M-polyfolds with the tame M-polyfold structures of Part I's Theorem 2.7.13 (2) on $bA$, $bB$ (`Polyfold.MPolyfold.tameBoundaryPolyfold`), and $f : A \to B$ sc-smooth on an open $U$. A map $g : bA \to bB$, continuous on an open $O \subseteq B^{-1}(U)$ with $B \circ g = f \circ B$ on $O$, is sc-smooth on $O$; in the charts $p \mapsto \pi(\varphi(Bp))$ of $bA$ and $bB$ it reads $\pi' \circ (\varphi' \circ f \circ \varphi^{-1}) \circ \iota$. Hence the lift $bf$, $[y, V, \theta] \mapsto [f(y), f(V), f(\theta)]$ (V7.96), of an sc-smooth index-preserving local homeomorphism is sc-smooth, and the lift of a local sc-diffeomorphism is a local sc-diffeomorphism: over a sheet $f : W \to V$ with inverse $\sigma$, $bf$ maps $B^{-1}(W)$ onto $B^{-1}(V)$, with an inverse that covers $\sigma$ and undoes $bf$. *Book:* §7.6.5.2, p. 355, new in the 2021 edition: the proof sketch of Theorem 7.6.9, which reads the structure maps in the charts $\Phi$ as sc-diffeomorphisms $s : \theta \to s(\theta)$ of faces. *Hidden hypotheses:* none. *Deviation (extra):* a tool of the formalization; it replaces the sketch's sc-diffeomorphism of faces by the computation in the charts of $bA$, which needs only that $f$ is sc-smooth (Findings, item 146).

**V7.100 (Theorem 7.6.9 (2), (3), (5): the boundary ep-groupoid, and $B$ an ep-functor; `Polyfold.EpGroupoid.paracompactSpace_bObj`, `Polyfold.EpGroupoid.paracompactSpace_bMor`, `Polyfold.EpGroupoid.bPObj`, `Polyfold.EpGroupoid.bPMor`, `Polyfold.EpGroupoid.isLocalScDiffeo_bSrc`, `Polyfold.EpGroupoid.isLocalScDiffeo_bTgt`, `Polyfold.EpGroupoid.isScSmoothMap_bUnit`, `Polyfold.EpGroupoid.isScSmoothMap_bInv`, `Polyfold.EpGroupoid.isScSmoothCompSheet_bComp`, `Polyfold.EpGroupoid.isScSmoothComp_bComp`, `Polyfold.EpGroupoid.boundaryEpGroupoid`, `Polyfold.EpGroupoid.isTame_boundaryEpGroupoid`, `Polyfold.EpGroupoid.boundaryEpFunctor`, `Polyfold.EpGroupoid.phiTot_bSrc`).** For a tame ep-groupoid $(X, \mathbf{X})$, give $bX$ and $b\mathbf{X}$ the tame M-polyfold structures of Theorem 2.7.13 (2) for the object and the morphism M-polyfold ($\mathbf{X}$ is tame, V7.97); both are paracompact. Then $s, t : b\mathbf{X} \to bX$ of (7.25) are local sc-diffeomorphisms, $u$ of (7.28) and $\iota$ of (7.27) are sc-smooth, and $m$ of (7.29) is sc-smooth on $b\mathbf{X}\,{}_s\!\times_t b\mathbf{X}$; with the properness of V7.98, $(bX, b\mathbf{X})$ is an ep-groupoid (Definition 7.1.3 (2017: 7.3)), the boundary ep-groupoid, and it is tame. Both maps $B$ form an ep-functor $(bX, b\mathbf{X}) \to (X, \mathbf{X})$, a morphism of the category $\mathcal{EP}$ (Definition 7.1.7 (2017: 7.7)). For a representative $(\varphi, \mathbf{V}, \theta)$ with $\mathbf{V}$ in a sheet of $s$, $s$ maps $U(\varphi, \mathbf{V}, \theta)$ into $U(x, s(\mathbf{V}), s(\theta))$ and $\Phi_{(x,s(\mathbf{V}),s(\theta))} \circ s = s \circ \Phi_{(\varphi,\mathbf{V},\theta)}$ there. *Book:* Theorem 7.6.9 (2), (3), (5), p. 355, new in the 2021 edition, whose verification is left to the reader, and the diagram of its proof sketch. *Hidden hypotheses:* none. *Deviation (stronger):* the text asks $B$ to be sc-smooth; it is also a functor (V7.97), so an ep-functor. Condition (4) of Definition 7.1.3, the sc-smoothness of $m$, is proved in the sheet form of V7.3: near $b$, $b' \mapsto m(\sigma(tb'), b')$ covers $\gamma \mapsto m(\sigma_1(t\gamma), \gamma)$ for a sheet $\sigma_1$ of $s$ on $\mathbf{X}$, and V7.99 applies (Findings, items 145, 146). *Proof:* $s$, $t$, $u$, $\iota$ are lifts of local sc-diffeomorphisms or of sc-smooth index-preserving local homeomorphisms of $(X, \mathbf{X})$ (V7.97), so V7.99 applies; tameness is that of Theorem 2.7.13 (2).

**V7.101 (Theorem 7.6.9 in full; `Polyfold.EpGroupoid.theorem_7_6_9_full_2021`, `Polyfold.EpGroupoid.trivia`
`Polyfold.MPolyfold.theorem_2_7_13_structure_2021`).** For a tame ep-groupoid $(X, \mathbf{X})$: (1)

$bX$ and $b\mathbf{X}$ are metrizable; (2) they carry the tame M-polyfold structures of V7.100, characterized as in Theorem 2.7.13: for every admissible triple with $V$ centred at $x$, $U(x,V,\theta)$ is open and $\Phi_{(x,V,\theta)}$ is an sc-diffeomorphism onto the face $\theta$ with its induced structure, inverse to the homeomorphism of V7.98, and every M-polyfold structure with this property has an atlas equivalent to it, for $bX$ and for $b\mathbf{X}$; (3) both maps $B$ are sc-smooth; (4) $d_X(B(p)) - 1 = d_{bX}(p)$ on $bX$ and $d_{\mathbf{X}}(B(a)) - 1 = d_{b\mathbf{X}}(a)$ on $b\mathbf{X}$; (5) the boundary ep-groupoid is tame, its structure maps are those of (7.25)–(7.29), $s$, $t$ are local sc-diffeomorphisms, $u$, $\iota$ are sc-smooth and $m$ is sc-smooth on the fibred product; and $B$ is an ep-functor. The trivial ep-groupoid of the tame quadrant $[0,\infty)^n \oplus W$, $n \geq 1$, satisfies the hypothesis, its boundary ep-groupoid is tame with nonempty $bX$, $b\mathbf{X}$, and the classes over the corner have $d_{bX} = n - 1$. *Book:* Theorem 7.6.9 and the characterization after it, p. 355, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (narrower):* the characterization and its uniqueness are for representatives whose $V$ is centred at $x$ (V7.98); in Part I they are verified since for every representative, with image $\theta \cap V^x$ (V2.104; gap G.78, closed), here not; "natural" in (2) is read through that characterization and its uniqueness. (4) is stated as printed, with the subtraction of natural numbers ($B$ takes values in $\partial X$, where $d_X \geq 1$); the additive form is `Polyfold.MPolyfold.degIndex_tameBoundaryMap`. The hypotheses of the two uniqueness statements are satisfied by the structures of (2) themselves (Findings, items 145, 147). *Proof:* (1) and the topology are V7.98; (2)–(4) are Theorem 2.7.13 (2)–(4) for the object and the morphism M-polyfold; (5) and the ep-functor are V7.100.

**V7.102 (Proposition 7.4.9, and Corollary 7.4.11, Theorems 7.4.13 and 7.4.24 with (ZHOU) for $X$ only;** `Polyfold.MPolyfold.continuous_levelDown`, `Polyfold.MPolyfold.shiftSmoothMap`, `Polyfold.MPolyfold.shiftSmoothInv`, `Polyfold.MPolyfold.shiftSmoothHomeomorph`, `Polyfold.EpGroupoid.isotropyLevelVal`, `Polyfold.EpGroupoid.levelVal_Φ_liftOne`, `Polyfold.EpGroupoid.tangentLower_tangentMor_liftOne`, `Polyfold.EpGroupoid.isZhou'_liftOne`, `Polyfold.EP.isZhou'_lift`, `Polyfold.EP.proposition_7_4_9_2021`, `Polyfold.EP.corollary_7_4_11_all_20` `Polyfold.EP.theorem_7_4_13_all_2021`, `Polyfold.EP.theorem_7_4_24_all_2021`, `Polyfold.EpGroupoid.Exa` `Polyfold.EpGroupoid.Examples.theorem_7_4_13_all_2021_pointEP`, `Polyfold.EpGroupoid.Examples.theor` The inclusion $X_{m+1} \to X_m$ is continuous, and $(X^1)_\infty \to X_\infty$ is a homeomorphism. A natural representation of $X^1$ at $x$ inside $U(x) \cap X_1$ acts by the natural action of $X$, and the tangent action of $\mathbf{X}^1$ is that of $\mathbf{X}$ under the inclusion $T(X^1) \to TX$. (ZHOU') for $X$ gives (ZHOU') for $X^1$, hence for every $X^i$. So if $X$ satisfies (ZHOU), $(T_xX)_\infty$ is dense in $T_xX$ at every smooth $x$, and every $X^i$ satisfies the Part I facts `SmoothRegular` of V7.86: every $X^i$ satisfies (ZHOU) (Proposition 7.4.9) and has LUC (Corollary 7.4.11), and $X$ has (SLUC) (Theorem 7.4.13) and (SLR) (Theorem 7.4.24). The point ep-groupoid satisfies the hypotheses. *Book:* Proposition 7.4.9, p. 333, Corollary 7.4.11 and Theorem 7.4.13, p. 334, Theorem 7.4.24, p. 340, new in the 2021 edition. *Hidden hypotheses:* the Part I facts of V7.86 for every $X^i$, and $(T_xX)_\infty$ dense in $T_xX$ for $X$ (gap G.85). *Deviation (explicit):* "(ZHOU') is the same for $X$ and $X^i$" is proved in the direction the proof uses, from $X$ to $X^i$, through $(X^1)_\infty \cong X_\infty$, Part I's Proposition 2.3.17 for $i = 1$ and the comparison of natural representations; (ZHOU) gives (ZHOU') for $X$ by Lemma 7.4.6, and (ZHOU') gives (ZHOU) for $X^i$ by Proposition 7.4.8 (V7.89). This closes gap G.84 for these statements (Findings, item 148). *Proof:* induction on $i$; the rest as in V7.90, V7.91, V7.94.

**V7.103 (Propositions 7.4.16 and 7.4.17 with (VSLUC) for $X$ only;** `Polyfold.MPolyfold.DenseSmooth.shift`, `Polyfold.EP.denseSmooth_lift`, `Polyfold.EpGroupoid.isVSLUC_liftOne`, `Polyfold.EP.isVSLUC_lift`, `Polyfold.EP.proposition_7_4_16_all_2021`, `Polyfold.EP.proposition_7_4_17_all_2021`, `Polyfold.EpGroupoid.Examples.isVSLUC_pointEP`, `Polyfold.EpGroupoid.Examples.proposition_7_4_17_al` (VSLUC) for $X$ gives (VSLUC) for $X^1$, with $V'(x) = V(x) \cap U'(x)$ under $(X^1)_\infty = X_\infty$, hence for every $X^i$; the density of $X_\infty$ in every $X_m$ passes from $X$ to every $X^i$. So if $X$ has (VSLUC),

$X_\infty$ is dense in every $X_m$ and every $X^i$ satisfies `SmoothConnected`, every $X^i$ has (VSLUC) and LUC (Proposition 7.4.16) and $X$ has (SLUC) (Proposition 7.4.17). The point ep-groupoid has (VSLUC) and satisfies the hypotheses. *Book:* Propositions 7.4.16 and 7.4.17, p. 336, new in the 2021 edition: (VSLUC) is phrased in terms of $X_\infty = X^i_\infty$. *Hidden hypotheses:* the density of $X_\infty$ for $X$ (gap G.73) and `SmoothConnected` for every $X^i$ (gap G.85), which Lemma 7.4.15 uses. *Deviation (explicit):* "$X_\infty = X^i_\infty$" is the homeomorphism of V7.102; "in particular (LUC)" is Lemma 7.4.15 (V7.92) for every $X^i$ (Findings, item 149).

**V7.104 (Proposition 7.4.18, (VSLUC) □ (SLUC);** `Polyfold.MPolyfold.levelDownTo`, `Polyfold.MPolyfold.cont` `Polyfold.MPolyfold.exists_level_nhds_smooth`, `Polyfold.EpGroupoid.isotropyLevelMk`, `Polyfold.EpGroupoid.isotropyLevelMk_levelVal`, `Polyfold.EpGroupoid.isotropyLevelVal_mul`, `Polyfold.EpGroupoid.isotropyLevelVal_inv`, `Polyfold.EpGroupoid.NaturalRep.liftOne`Φ, `Polyfold.EpGroupoid.NaturalRep.levelVal_liftOne`Φ, `Polyfold.EpGroupoid.NaturalRep.mapsTo_liftOne`Φ `Polyfold.EpGroupoid.NaturalRep.isScDiffeoOn_liftOne`Φ, `Polyfold.EpGroupoid.NaturalRep.liftOneRep`, `Polyfold.EpGroupoid.NaturalRep.eqOn_of_liftOneRep`, `Polyfold.EpGroupoid.NaturalRep.LevelConnected` `Polyfold.EpGroupoid.NaturalRep.levelConnected_liftOneRep`, `Polyfold.EP.eqOn_of_level_smooth`, `Polyfold.EP.isVSLUC_of_isSLUC`, `Polyfold.EP.proposition_7_4_18_2021`, `Polyfold.EpGroupoid.Examples.` `Polyfold.EpGroupoid.Examples.proposition_7_4_18_2021_pointEP`**).** The sets $O \cap X_\infty$, $O$ open in some $X_{N+1}$, form a neighbourhood basis of $X_\infty$. A natural representation $U(x)$ of $x \in X_1$ with $U(x) \cap X_1$ connected in $X^1$ restricts to a natural representation of $x$ in $X^1$, the isotropy groups of $x$ in $X$ and in $X^1$ being identified; if every $U(x) \cap X_m$ is connected in $X_m$, so it is for the restriction. Under LUC for $X^{N+1}$, if $\hat{g} = \mathrm{id}$ on the smooth points of a nonempty open subset of $U(x) \cap X_{N+1}$, then $\hat{g} = \mathrm{id}$ on $U(x)$. Hence, if $X_\infty$ is dense in every $X_m$ and every smooth $x$ has a natural representation with every $U(x) \cap X_m$ connected, (SLUC) implies (VSLUC); with `SmoothConnected` for every $X^i$ as well, (VSLUC) and (SLUC) are equivalent. Both hold for the point ep-groupoid, which satisfies the hypotheses. *Book:* Proposition 7.4.18, p. 336, new in the 2021 edition. *Hidden hypotheses:* the density of $X_\infty$ (gap G.73); the natural representations with every $U(x) \cap X_m$ connected, which the proof takes "by construction" and which are a hypothesis here (gap G.85). *Deviation (explicit):* LUC for $X^N$ is applied to the restriction of $U(x)$ to $X^N$, built one level at a time; the direction (VSLUC) □ (SLUC) is V7.103 (Findings, item 150). *Proof:* the text's: an open subset of $V(x) = U(x) \cap X_\infty$ contains $W' \cap X_\infty$ for a nonempty open $W'$ of some $X_N$, $\hat{g} = \mathrm{id}$ on $W'$ by density and continuity, and LUC for $X^N$ on the restricted $U(x)$ gives $\hat{g} = \mathrm{id}$ on $U(x)$.

**What is not verified in this chapter**

- p. 289 (2017 p. 227), Definition 7.1.4 (2017: 7.4), the tangent $T\varphi$ as an sc-isomorphism: that it is well defined, equal to $T\hat{\varphi}(s(\varphi))$, a bounded linear isomorphism on level 0 for $\varphi$ on level 1, and an sc-operator for smooth $\varphi$; $T\varphi$ is only defined (`Polyfold.EpGroupoid.tangentMor`). Its linearity needs the linear structure of $T_xX$ (gap G.7); deferred to Chapter 8 (V7.5) (gap G.70; Findings, item 110).
- 2017 p. 227, the comparison with the classical properness of $(s,t) : \mathbf{X} \to X \times X$; no formal content.
- 2017 p. 228, the filtration $|X| = |X_0| \supset |X_1| \supset \cdots \supset |X_\infty| = \bigcap_k |X_k|$: $|X_k|$ is defined (`Polyfold.EpGroupoid.orbitLevel`), $|X_\infty|$ and the identity are not stated (V7.6) (gap G.72; Findings, item 112).
- 2017 p. 227, "the set $X_\infty$ is dense in every space $X_k$": stated without proof; Part I proves it only for sc-Banach spaces, not for M-polyfolds; not formalized (V7.6) (gap G.73; Findings,

item 116).

- pp. 290–292 (2017 pp. 228–229), the remark after Definition 7.1.7 (2017: 7.7): it does not hold as we read it; only the form for functors continuous on morphisms is verified (V7.10) (Findings, item 9). The 2021 text (p. 291) assumes continuity on morphisms, and its form is verified (V7.69) (Findings, item 117).
- 2017 pp. 237–238, the tangent facts (3) and (4) recalled before Proposition 7.1.22 (2017: 7.7) ($T_aA$ has an sc-complement in $T_aX$; $T_aA = Tr(a)T_aX$): not restated, since $T_aA$ and $T_aX$ carry no linear structure here; in a chart they are Part I's `Polyfold.proposition_2_21_3`, `Polyfold.proposition_2_21_4` (V7.30) (gap G.7; Findings, item 40).
- pp. 313–324 (2017 pp. 242–244), Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017) for a general ep-groupoid: that $\mathbf{X}^R$ is Hausdorff and paracompact is a hypothesis, discharged for effective $X$ and the trivial ep-groupoid. Hausdorffness fails in general (`Polyfold.EpGroupoid.Examples.exists_not_t2Space_reducedMor`, formalized after the round-trip audit); not formalized: paracompactness when $\mathbf{X}^R$ is Hausdorff, and whether Hausdorffness with $\mathbf{X}$ second countable suffices (V7.39, V7.40) (gap G.71; Findings, items 56, 60, 107, 111). *2021 edition:* Proposition 7.2.12 (p. 317) proves $T_R$ paracompact for every ep-groupoid, verified (V7.77), so the question of paracompactness is settled; Theorem 7.2.14 (p. 318) assumes $T_R$ Hausdorff, exactly the hypothesis of V7.39 and V7.40 (V7.78), and Theorem 7.2.22 (p. 323) characterizes it (V7.81); the counterexample is of the kind of Remark 7.2.8 and Appendix 7.6.3 (V7.82) (gap G.71, closed; Findings, items 129, 143).
- p. 340 (2017 p. 249), "It is presently not known whether the property (5) is a consequence of the properties (1)–(4)": an open question of the text, not formalized and not given a gap, since nothing depends on it; Remark 7.5.2 (2017: 7.1) has no formal content (V7.53) (Findings, items 82, 113).
- p. 342 (2017 p. 251), Remark 7.5.6 (2017: 7.2): no formal content (V7.57).
- 2017 p. 253, the contractibility of $N^t_g$, $N^s_g$ in the proof of Theorem 7.6.1 (2017: 7.5): not needed (V7.62) (Findings, item 29).
- p. 347 (2017 p. 256), Corollary 7.6.6 (2017: 7.3) with Part I's Theorem 5.5.16 (2017: 5.14) as an explicit hypothesis (V7.68) (gap G.53; Findings, item 95).
- Gaps of Part I taken up: G.15 (Part I's Lemma 2.7.3 (2017: 2.7) and Proposition 2.7.4 (2017: 2.17)) is now formalized (V7.50); of G.56, the local path connectedness of M-polyfolds is now formalized (V7.18), the sc$^+$-smooth paths in a path component are not; G.53 holds for finite-dimensional spaces (`Polyfold.godefroyTroyanskiWhitfieldZizlerStatement_of_finiteDimensional`) (Findings, item 114).
- 2017 p. 234, Proposition 7.5 (2) (2017) as printed, $s, t : \mathbf{U}(\psi) \to U(y)$ with one $y$: it does not hold as we read it; the 2021 form, $U(s(\psi))$ and $U(t(\psi))$, is verified (V7.71) (Findings, item 119).
- p. 315, Lemma 7.2.11 for an ep-groupoid whose $T_R$ is not Hausdorff: it does not hold as we read it; verified with $T_R$ Hausdorff (V7.76) (Findings, item 124).
- pp. 329–340, the Part I facts used in §7.4 ($X_\infty$ dense in every $X_m$, gap G.73; $U \cap X_\infty$ connected for connected open $U$; $p : TX \to X^1$ open, gap G.8; tangent maps determined by $X_\infty$; $(T_xX)_\infty$ dense in $T_xX$) and the connectedness of $U(x) \cap X_1$ in $X^1$ that the proof of Lemma 7.4.4 uses: explicit, satisfiable hypotheses (V7.83, V7.85, V7.86), for the results on every $X^i$ taken on every $X^i$ (`SmoothRegular`, `SmoothConnected`; `DenseSmooth` passes from $X$ to every $X^i$ by proof), and, for Proposition 7.4.18, the choice of $U(x)$ with every $U(x) \cap X_m$ connected (`NaturalRep.LevelConnected`, V7.104) (gap G.85; Findings, items 133, 134, 148–150).
- pp. 333–336, §7.4 for the index lifts: in Proposition 7.4.9, "(ZHOU') is the same for $X$ and $X^i$" is verified from $X$ to $X^i$, the direction the proof uses (V7.102). Proposition 7.4.9, Corol-

lary 7.4.11, Theorems 7.4.13, 7.4.24 and Propositions 7.4.16–7.4.18 are verified for every $X^i$ (V7.102–V7.104; gap G.84, closed; Findings, items 137, 148–150).

- p. 348, Appendix 7.6.3: Zhou's retract itself and the exact non-effective part of its ep-groupoid; verified for an analogue (V7.82), one point of $\mathrm{cl}(\mathbf{X}^T) \setminus \mathbf{X}^T$; that it fails (ZHOU) (p. 331) is verified given the Part I facts for its object M-polyfold (V7.95) (gap G.83; Findings, items 130, 139).
- pp. 348–351, Appendix 7.6.4, Theorem 7.6.7 and Corollary 7.6.8: the identification $(X^1)_R = (X_R)^1$ as an sc-diffeomorphism, and the case $i \geq 2$; verified for $i = 1$ as sets, with a continuous map (V7.83) (gap G.82; Findings, item 131).
- p. 355, after Theorem 7.6.9: the characterization of the structures of $bX$, $b\mathbf{X}$ for representatives whose $V$ is not centred at $x$, as formerly in Part I (gap G.78, closed there since by V2.104, with image $\theta \cap V^x$); "$\Phi$ is a homeomorphism onto $\theta$" holds as we read it only for $V$ centred at $x$ (V7.98). Theorem 7.6.9 (2)–(5) themselves, which depend on 2021 Theorem 2.7.13 (2), are verified (V7.99–V7.101) (Findings, items 140, 141, 145–147).

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**9. Chapter 8 (HWZ, Part II, §§8.1–8.6 (2017 §§8.1–8.5)): verified except the sc-differential forms of Definitions 8.2.1 and 8.2.2 (2017 Definitions 8.2 and 8.3), the linearity of $T\varphi$ in Remark 8.1.10 (2017 Remark 8.1), and Lemma 8.1.2 (2017 Lemma 8.1) in the form $m_{T\mathbf{X}} = Tm$, and, of the 2021 additions, the step $T(g,h) = \mathrm{id} \Rightarrow Tg = \mathrm{id}$ of Proposition 8.1.16, Proposition 8.4.1 (1) □ (2), (3), Proposition 8.4.3 and the strong bundle structure of $F^*V$ in Lemma 8.5.10**

Chapter 8 (pp. 357–400 (2017 pp. 257–289); Part II has no 2014 preprint, so numbers and pages are those of the book (Springer 2021), with the 2017 text's in parentheses) treats bundles over ep-groupoids and covering functors. §8.1 equips the tangent $T(X, \mathbf{X}) = (TX, T\mathbf{X})$ with the structure maps (8.1)–(8.9) and proves that it is an ep-groupoid, tame if $X$ is, with an sc-smooth projection functor to $X^1$ (Theorem 8.1.1 (2017: 8.1), Lemmas 8.1.2–8.1.9 (2017: 8.1–8.8)); it describes the morphisms of $T\mathbf{X}$ by the pairs $(T\varphi, a)$ (Remark 8.1.10 (2017: 8.1)), proves that the tangent of an sc-smooth functor is one and that $T : \mathcal{EP} \to \mathcal{EP}$ is a functor (Theorems 8.1.11, 8.1.12 (2017: 8.2, 8.3)), defines natural transformations and natural equivalence (Definition 8.1.13 (2017: 8.1)), proves that tangents of natural transformations are natural transformations (Proposition 8.1.14 (2017: 8.1)), and that paracompactness of $|X|$ passes to $|TX|$ (Theorem 8.1.15 (2017: 8.4)). §8.2 defines sc-differential forms on ep-groupoids and the sc-de Rham complex (Definitions 8.2.1, 8.2.2 (2017: 8.2, 8.3)). §8.3 defines strong bundles over ep-groupoids (Definition 8.3.1 (2017: 8.4)) with their functorial picture (pp. 375–377 (2017 pp. 271–272)), strong bundle maps and strong bundle functors (Definitions 8.3.2, 8.3.3 (2017: 8.5, 8.6)) and section functors (Definition 8.3.4 (2017: 8.7)). §8.5 (2017 §8.4) defines proper covering functors (Definition 8.5.1 (2017: 8.8)) with the degeneracy remark after it, proves that they compose (Proposition 8.5.2 (2017: 8.2)), describes their local structure (Theorem 8.5.3 (2017: 8.5), Remark 8.5.4 (2017: 8.2)), defines covering functors of finite groupoids and geometric lifts (Definition 8.5.5 (2017: 8.9)), constructs the proper covering functor of a geometric lift (Proposition 8.5.7 (2017: 8.3)), shows that every proper covering functor is locally of this form (Theorem 8.5.8 (2017: 8.6)), and defines proper strong bundle covering functors (Definition 8.5.9 (2017: 8.10)). Appendix 8.6 (2017 8.5) counts the fibres of a proper covering functor (§8.6.1 (2017 §8.5.1), Lemmas 8.6.1–8.6.3 (2017: 8.9–8.11), Proposition 8.6.4 (2017: 8.4)) and describes strong bundle coverings fibrewise (§8.6.2 (2017 §8.5.2)).

What is verified. **Theorems 8.1.1–8.5.8 (2017: 8.1–8.6) hold**, with the readings recorded below, and

so do Propositions 8.1.14, 8.5.2, 8.6.4 (2017: 8.1, 8.2, 8.4), Proposition 8.5.7 (2017: 8.3) for nonempty $D$, Lemmas 8.1.2–8.6.3 (2017: 8.1–8.11) (Lemma 8.1.2 (2017: 8.1) in the sheet form of its proof), Remark 8.1.10 (2017: 8.1) without its linearity claims, and Remark 8.5.4 (2017: 8.2). Every definition of the chapter except Definitions 8.2.1 and 8.2.2 (2017: 8.2 and 8.3) is formalized: Definitions 8.1.13 (2017: 8.1; V8.15), 8.3.1 (2017: 8.4; V8.20), 8.3.2 (2017: 8.5; V8.23), 8.3.3 (2017: 8.6; V8.24), 8.3.4 (2017: 8.7; V8.26), 8.5.1 (2017: 8.8; V8.27), 8.5.5 (2017: 8.9; V8.32), 8.5.9 (2017: 8.10; V8.37). Several places of the text needed more than a reading. (a) *Theorem 8.5.3 (5) (2017: 8.5).* As printed, the right vertical arrow of the square is $t$ and its lower right corner $U(t(F(\psi_0)))$; with $t$ the square says $F(s(\psi)) = t(F(\psi))$ on $\mathbf{U}(\psi_0)$, which does not hold as we read it, since $F(s(\psi)) = s(F(\psi))$ for a functor (checked in Lean on $\mathbb{Z}/2\ltimes\mathbb{R}$, `Polyfold.CoveringExamples.theorem_8_5_five_as_printed`). The square is verified with $s$ on both vertical arrows, together with the companion square with $t$ on both (V8.30). (b) *$D$ nonempty.* Definition 8.5.5 (2017: 8.9) allows $D = \emptyset$, and then the functor of Proposition 8.5.7 (2017: 8.3) is surjective on objects only if $U(x) = \emptyset$; Proposition 8.5.7 (2017: 8.3) is verified for nonempty $D$, which holds in Theorem 8.5.8 (2017: 8.6) (V8.35). (c) *The degeneracy remark after Definition 8.5.1 (2017: 8.8).* The equality $d_{\mathbf{X}\,{}_s\times_F Y} = \frac{1}{2}d_{\mathbf{X}\times Y}$ uses the product formula $d_{\mathbf{X}\times Y} = d_{\mathbf{X}} + d_Y$, which Part I has for tame M-polyfolds; it is verified for tame $X$, $Y$, the inequality in general (V8.28). (d) *Omitted steps.* The tangents of maps defined on open sets, which the proofs of §8.1 differentiate without a definition (V8.2), the tameness of $Tr$ in Lemma 8.1.9 (2017: 8.8), asserted without argument (V8.11), the claim that $\simeq$ is an equivalence relation (V8.16), the ep-groupoid of the functorial picture, asserted without proof (V8.22), the equivalence of Definitions 8.3.2 and 8.3.3 (2017: 8.5 and 8.6) (V8.25), the open embedding $G_x\backslash TU(x) \to |TX|$ in the proof of Theorem 8.1.15 (2017: 8.4) (V8.18), the "one verifies" of Proposition 8.5.7 (2017: 8.3) (V8.35) and of Theorem 8.5.8 (2017: 8.6) (V8.36) are proved. (e) *Other routes.* Theorem 8.1.15 (2017: 8.4) is proved by metrizability, not by the text's closed paracompact cover (V8.18); Theorem 8.5.3 (2017: 8.5) through Definition 8.5.1 (3) (2017: 8.8), not through sheets and natural representations in $Y$ (V8.30); Lemma 8.1.2 (2017: 8.1) and condition (4) of Lemma 8.1.7 (2017: 8.6) in the sheet form of Chapter 7 (V8.4, V8.9). (f) *Readings.* Part I has strong bundle maps only between local models, and the global notion is a reading (V8.19); in Theorem 8.5.8 (2017: 8.6) the text's "geometric lift $A : K \to L$" is read as a geometric lift with its associated functor $A$ (V8.36); in Definition 8.5.9 (3) (2017: 8.10) the strong bundle structure of $\mathbf{V}\,{}_s\times_A W$, which the text does not give, is read as a pull-back (V8.37). Not verified: Definitions 8.2.1 and 8.2.2 (2017: 8.2 and 8.3), the linearity of $T_xX$ and of $T\varphi$, the identification $T(\mathbf{X}\,{}_s\times_t \mathbf{X}) = T\mathbf{X}\,{}_{Ts}\times_{Tt} T\mathbf{X}$, and the M-polyfold structure transported along $G$ in Remark 8.1.10 (2017: 8.1) (see the list at the end of this section).

The setting. Ep-groupoids, sc-smooth functors, natural representations, $X^1$ and paracompact orbit spaces are those of Chapter 7 (`Polyfold.EpGroupoid`, `Polyfold.EpFunctor`, `Polyfold.EpGroupoid.NaturalRep`, `Polyfold.EpGroupoid.liftOne`, `Polyfold.EpGroupoid.HasParacompactOr`). $TX$ and $T\mathbf{X}$ are Part I's tangent M-polyfolds (`Polyfold.MPolyfold.TangentSpace`, `Polyfold.MPolyfold.tangen`) which for $X$ in `Type u` live in `Type (max u 1)`; their fibres $T_xX$ carry no linear structure (gap G.7), so no statement of this chapter about the linearity of a tangent map is formalized. Strong bundles over M-polyfolds and their pull-backs are Part I's (`Polyfold.StrongBundle`, `Polyfold.StrongBundle.pullback`). "$f : U \to V$ is an sc-diffeomorphism" between open sets is Part I's `Polyfold.MPolyfold.IsScDiffeoOn`, with an inverse. Part I results are cited by their 2017 numbers. Findings referred to below are those of `Exposition/Findings.md`, "Chapter 8".

**V8.1 (the tangent data (8.1)–(8.9); `Polyfold.EpGroupoid.tSrc`, `Polyfold.EpGroupoid.tTgt`, `Polyfold.EpGroupoid.tUnit`, `Polyfold.EpGroupoid.tInv`, `Polyfold.EpGroupoid.tBase`, `Polyfold.EpGroupoid.tComp`, `Polyfold.EpGroupoid.tangent_ext`, `Polyfold.EpGroupoid.exists_tSrc_eq`, `Polyfold.EpGroupoid.src_tBase_eq`, `Polyfold.EpGroupoid.tComp_spec`, `Polyfold.EpGroupoid.tBase_tComp`**

`Polyfold.EpGroupoid.tSrc_tComp`, `Polyfold.EpGroupoid.tComp_eq`, `Polyfold.EpGroupoid.tUnit_eq`, `Polyfold.MPolyfold.tangentMap`, `Polyfold.MPolyfold.tangentProj`). The objects of $T(X,\mathbf{X})$ are $TX$, the morphisms $T\mathbf{X}$; $s_T = Ts$, $t_T = Tt$ (8.3); for $Ts(k) = Tt(h)$ the composite $k \circ h$ is the vector over $p(k) \circ p(h)$ with $Ts(k \circ h) = Ts(h)$ (8.5), (8.6); $u_T(a) = (1_x, Ts(1_x)^{-1}a)$ (8.7), (8.8); $\iota_T = T\iota$ (8.9). *Book:* §8.1, pp. 357–359 (2017 pp. 257–258). *Hidden hypotheses:* none. *Deviation (restated):* the composition is defined by choice and is total, with a junk value off composable pairs, as Chapter 7's composition; on composable pairs it is unique, since $Ts$ is injective on every fibre $T_\varphi\mathbf{X}$ (`Polyfold.EpGroupoid.tangent_ext`); $u_T$ is $Tu$, which is (8.8). $TX$ and $T\mathbf{X}$ live in `Type (max u 1)`. (8.5) writes $T_{\varphi\circ\psi}\mathbf{X}$ for $T_{\psi\circ\varphi}\mathbf{X}$ (possible misprint) (Findings, items 1, 2).

**V8.2 (tangents of maps sc-smooth on open sets and of local sc-diffeomorphisms;** `Polyfold.MPolyfold.tangentI` `Polyfold.MPolyfold.isOpen_tangentPreimage`, `Polyfold.MPolyfold.tangentMapOn`, `Polyfold.MPolyfold.le` `Polyfold.MPolyfold.tangentMapOn_eq_tangentMap`, `Polyfold.MPolyfold.tangentMapOn_comp`, `Polyfold.MPolyfold.tangentMap_tangentMapOn`, `Polyfold.MPolyfold.tangentMapOn_isScSmoothOn`, `Polyfold.MPolyfold.IsScDiffeoOn.tangent`, `Polyfold.MPolyfold.IsScDiffeoOn.tangentMap`, `Polyfold.MPolyfold.IsScDiffeoOn.bijOn_tangentFiber`, `Polyfold.MPolyfold.IsLocalScDiffeo.tangentMa` `Polyfold.MPolyfold.IsLocalScDiffeo.surjective_tangentMap`, `Polyfold.MPolyfold.IsLocalScDiffeo.bij` `Polyfold.MPolyfold.isScDiffeoOn_tangentMap_subtypeVal`, `Polyfold.MPolyfold.isOpenEmbedding_tanger` `Polyfold.MPolyfold.nonempty_tangentSpace`). For open $U$, $TU = p^{-1}(U)$ is open in $TX$ and is the tangent of the open sub-M-polyfold $U$; a map sc-smooth on $U$ has a tangent on $p^{-1}(U)$, sc-smooth, over the map, with the chain rule, equal to Part I's tangent for a globally sc-smooth map; the tangent of an sc-diffeomorphism $U \to V$ is an sc-diffeomorphism $p^{-1}(U) \to p^{-1}(V)$; the tangent of a local sc-diffeomorphism is one, bijective on every fibre, and surjective if the map is. *Book:* used without statement in (8.5), (8.8) and in the proofs of Lemmas 8.1.2, 8.1.3, 8.1.5, 8.1.7 and 8.1.8 (2017: 8.1, 8.2, 8.4, 8.6 and 8.7) (pp. 358–364 (2017 pp. 258–263)). *Hidden hypotheses:* none. *Deviation (not in the book):* Part I defines the tangent only of globally sc-smooth maps; the tangent of a map defined on an open set, such as $(s|\mathbf{U}(\psi))^{-1}$, is built from the charts here (Findings, item 3).

**V8.3 (Theorem 8.1.1 (2017: 8.1), the tangent ep-groupoid;** `Polyfold.EpGroupoid.tangentGroupoidStr`, `Polyfold.EpGroupoid.tangentEpGroupoid`, `Polyfold.EpGroupoid.tangentProjFunctor`, `Polyfold.EpGroupoid.theorem_8_1`, `Polyfold.EpGroupoid.theorem_8_1_tame`). $T(X,\mathbf{X}) = (TX, T\mathbf{X})$ with the structure maps (8.3)–(8.9) is an ep-groupoid, tame if $(X,\mathbf{X})$ is, and $(p,\mathbf{p}) : T(X,\mathbf{X}) \to (X,\mathbf{X})^1$ is an sc-smooth functor. *Book:* Theorem 8.1.1 (2017: 8.1), p. 359 (2017 p. 258), proved on pp. 363–365 (2017 pp. 263–264) from Lemmas 8.1.2–8.1.9 (2017: 8.1–8.8). *Hidden hypotheses:* none. *Deviation (restated):* `Polyfold.EpGroupoid.theorem_8_1` records that `Polyfold.EpGroupoid.tangentEpGroupoid` has the M-polyfolds $TX$, $T\mathbf{X}$ and the text's structure maps, with the base point and source of a composite; the conditions of Definition 7.1.3 (2017: 7.3) are Lemmas 8.1.3–8.1.8 (2017: 8.2–8.7) (V8.5–V8.10), the tameness sentence is `Polyfold.EpGroupoid.theorem_8_1_tame` (Lemma 8.1.9 (2017: 8.8), V8.11); $(X,\mathbf{X})^1$ is Chapter 7's `Polyfold.EpGroupoid.liftOne` (Findings, items 2, 9).

**V8.4 (Lemma 8.1.2 (2017: 8.1), the composition as a tangent map;** `Polyfold.EpGroupoid.lemma_8_1`, `Polyfold.EpGroupoid.tComp_eq_tangentMapOn`, `Polyfold.EpGroupoid.tangentMapOn_srcInv_tSrc`, `Polyfold.EpGroupoid.mem_tangentPreimage_tgt`, `Polyfold.EpGroupoid.isScSmoothCompSheet`). For a sheet $s : W \to V$ with inverse $\sigma$ and composable $k, h$ with $k$ over $W$: $k = T\sigma(Tt(h))$, $k \circ h = Tg(h)$ for $g(\varphi') = m(\sigma(t\varphi'), \varphi')$, and $k \circ h$ lies over $p(k) \circ p(h)$ with $Ts(k \circ h) = Ts(h)$. *Book:* Lemma 8.1.2 (2017: 8.1) and its proof, 2017 p. 259 ($(\psi,k) \circ (\varphi,h) = Tm(\psi,\varphi)(k,h)$ on $T(\mathbf{X}\,{}_s\!\times_t \mathbf{X}) = T\mathbf{X}\,{}_{Ts}\!\times_{Tt} T\mathbf{X}$). *Hidden hypotheses:* none. *Deviation (restated):* in the form the proof

uses, $\sigma(t(\varphi'))$ being the text's $F(\varphi') = (s|\mathbf{U}(\psi))^{-1}(t(\varphi'))$; the identification of $T(\mathbf{X}\,{}_s\!\times_t \mathbf{X})$ with the fibred product of the tangents, and $m_{T\mathbf{X}} = Tm$ on it, are not formalized (gap G.75). The proof writes $s|\mathbf{U}(\varphi \circ \psi)$ for $s|\mathbf{U}(\psi \circ \varphi)$ (possible misprint) (Findings, items 5, 111).

**V8.5 (Lemma 8.1.3 (2017: 8.2), source and target of a composite; `Polyfold.EpGroupoid.lemma_8_2`, `Polyfold.EpGroupoid.tTgt_tangentMapOn`).** $s_T(\Psi \circ \Phi) = s_T(\Phi)$ and $t_T(\Psi \circ \Phi) = t_T(\Psi)$. *Book:* Lemma 8.1.3 (2017: 8.2) and its proof, pp. 359–361 (2017 pp. 259–260). *Hidden hypotheses:* none. *Deviation:* none; the target is computed as in the text, with the tangent of $t \circ \sigma \circ t$ on $t^{-1}(V)$. The proof has "==", ": $\mathbf{U}(\psi \circ \varphi) \to U(x)$" without $s$, and "$Tt)\varphi)(h)$" (possible misprints) (Findings, items 4, 6).

**V8.6 (Lemma 8.1.4 (2017: 8.3), associativity; `Polyfold.EpGroupoid.lemma_8_3`).** The composition of $T\mathbf{X}$ is associative. *Book:* Lemma 8.1.4 (2017: 8.3), pp. 361 ff. (2017 pp. 260–261). *Hidden hypotheses:* none. *Deviation:* none; by fibre injectivity of $Ts$ (Findings, item 4).

**V8.7 (Lemma 8.1.5 (2017: 8.4), units; `Polyfold.EpGroupoid.tSrc_tUnit`, `Polyfold.EpGroupoid.tTgt_tUnit`, `Polyfold.EpGroupoid.tComp_tUnit`, `Polyfold.EpGroupoid.tUnit_tComp`, `Polyfold.EpGroupoid.lemma_8_4`).** $1_q = (1_x, Ts(1_x)^{-1}q)$ is a morphism $q \to q$ and a two-sided unit. *Book:* Lemma 8.1.5 (2017: 8.4), p. 361 (2017 p. 261). *Hidden hypotheses:* none. *Deviation:* none; $Tt(1_q) = q$ by the chain rule for $t \circ u = \mathrm{id}$ instead of $t \circ (s|\mathbf{U}(1_x))^{-1} = \mathrm{id}$ near $x$ (Findings, items 4, 6).

**V8.8 (Lemma 8.1.6 (2017: 8.5), inverses; `Polyfold.EpGroupoid.tSrc_tInv`, `Polyfold.EpGroupoid.tTgt_tInv`, `Polyfold.EpGroupoid.tComp_tInv`, `Polyfold.EpGroupoid.tInv_tComp`, `Polyfold.EpGroupoid.lemma_8_5`).** $\iota_T(\varphi, k) = (\varphi^{-1}, T\iota(\varphi)k)$ is a two-sided inverse. *Book:* Lemma 8.1.6 (2017: 8.5), pp. 362 ff. (2017 pp. 261–262). *Hidden hypotheses:* none. *Deviation:* none (Findings, item 4).

**V8.9 (Lemma 8.1.7 (2017: 8.6), the structure maps; `Polyfold.EpGroupoid.isLocalScDiffeo_tSrc`, `Polyfold.EpGroupoid.isLocalScDiffeo_tTgt`, `Polyfold.EpGroupoid.surjective_tSrc`, `Polyfold.EpGroupoid.surjective_tTgt`, `Polyfold.EpGroupoid.isScSmoothMap_tUnit`, `Polyfold.EpGroupoid.isScSmoothMap_tInv`, `Polyfold.EpGroupoid.isScSmoothCompSheet_tangent`, `Polyfold.EpGroupoid.isScSmoothComp_tangent`, `Polyfold.EpGroupoid.lemma_8_6`, `Polyfold.isScSmoothCom`).** $s_T$, $t_T$ are surjective local sc-diffeomorphisms, $u_T$ and $\iota_T$ are sc-smooth, and the multiplication is sc-smooth on $T\mathbf{X}\,{}_{s_T}\!\times_{t_T} T\mathbf{X}$. *Book:* Lemma 8.1.7 (2017: 8.6), p. 362 (2017 p. 262). *Hidden hypotheses:* none. *Deviation (restated):* "by construction" is V8.2; the sc-smoothness of the multiplication is proved in Chapter 7's sheet form and from it on the fibred product, not through "$m_{T\mathbf{X}}$ is precisely $Tm$" (gap G.75) (Findings, item 7).

**V8.10 (Lemma 8.1.8 (2017: 8.7), properness; `Polyfold.EpGroupoid.lemma_8_7`).** Every $a \in TX$ has an open $V' \ni a$ with $t_T : s_T^{-1}(\mathrm{cl}\, V') \to TX$ proper, $V' = p^{-1}(V)$ for $V$ as in Definition 7.1.3 (3) (2017: 7.3) for $X$. *Book:* Lemma 8.1.8 (2017: 8.7), pp. 363–364 (2017 pp. 262–263). *Hidden hypotheses:* none. *Deviation:* none; the step "$\varphi_k \to \varphi$ in $\mathbf{X}^1$" is not needed. The proof writes $t(\varphi_k, h_k) = Ts(\varphi_k)(h_k)$ for $Tt(\varphi_k)(h_k)$ (possible misprint) (Findings, item 7).

**V8.11 (Lemma 8.1.9 (2017: 8.8) and the tameness of $Tr$; `Polyfold.ScIso.tangentPres`, `Polyfold.tangentSet_eq_tangentPres`, `Polyfold.degIndex_tangentSet`, `Polyfold.mem_minimalSubspaceWit`, `Polyfold.ScRetraction.scDeriv_tangentRetraction_fst`, `Polyfold.ScRetraction.scDeriv_tangentRetrac`, `Polyfold.ScRetraction.IsTameWith.tangent`, `Polyfold.ScRetraction.IsTame.tangentRetraction`, `Polyfold.IsTameRetract.tangent`, `Polyfold.MPolyfold.IsTame.tangent`, `Polyfold.EpGroupoid.lemma_8_8`, `Polyfold.EpGroupoid.isTame_tangentEpGroupoid_trivial_rn`).** If $(X, \mathbf{X})$ is tame, so is $(TX, T\mathbf{X})$; the tangent $Tr$ of a tame retraction is tame, and the tangent of a tame M-polyfold is tame. *Book:* Lemma 8.1.9 (2017: 8.8), p. 364 (2017 p. 263). *Hidden hypotheses:* none. *Deviation:* none.

"$Tr$ is a tame retraction", asserted without argument and not in Part I, is proved from Part I's Definition 2.5.2 (2017: 2.17) for one presentation of $TC$ with the complement $A' = (A \cap E_1) \oplus A$, without the second derivative of $r$, and passed to all presentations by Part I's Lemma 1.3.4 (2017: 1.1) (`Polyfold.ScRetraction.IsTameWith.isTame`) (Findings, items 8, 104).

**V8.12 (Remark 8.1.10 (2017: 8.1), the morphisms of $T\mathbf{X}$ as pairs $(T\varphi, a)$; `Polyfold.EpGroupoid.tangentMor`, `Polyfold.EpGroupoid.tangentMor_tSrc`, `Polyfold.EpGroupoid.tangentMor_comp`, `Polyfold.EpGroupoid.rem` `Polyfold.EpGroupoid.remark_8_1`).** $t_T \circ (s_T|T_\varphi\mathbf{X})^{-1} = T\varphi$; $T(\psi \circ \varphi) = T\psi \circ T\varphi$ on $T_{s(\varphi)}X$; $G(\varphi, h) = (T\varphi, Ts(h))$ is a bijection onto the pairs with $\varphi \in \mathbf{X}_1$, $a \in T_{s(\varphi)}X$, compatible with the compositions. *Book:* Remark 8.1.10 (2017: 8.1), pp. 364 ff. (2017 pp. 264–265). *Hidden hypotheses:* none. *Deviation (narrower):* $T\varphi$ is Chapter 7's `Polyfold.EpGroupoid.tangentMor`; the first entry of $G$ is read as $T\varphi$ labelled by $\varphi$, since as a bare map $G$ need not be injective. Not formalized: that $T\varphi$ and $t_T \circ (s_T|T_xX)^{-1}$ are linear isomorphisms, and that $T\varphi$ is $T\widehat{\varphi}(x)$ (gaps G.7, G.70); the M-polyfold structure transported along $G$, and the triples $(a, T\varphi, b)$. The remark writes "$T\varphi : T_xX \to T_yY$" for $T_yX$, and calls $T\varphi$ the linearization "at $x$" and later "at $\varphi$" (possible misprints) (Findings, items 10, 112, 113).

**V8.13 (Theorem 8.1.11 (2017: 8.2), the tangent of an sc-smooth functor; `Polyfold.EpFunctor.tangent`, `Polyfold.EpFunctor.theorem_8_2`).** $(T\Phi, T\Phi)$ is an sc-smooth functor $T(X, \mathbf{X}) \to T(Y, \mathbf{Y})$. *Book:* Theorem 8.1.11 (2017: 8.2) and its proof, pp. 365–367 (2017 pp. 265–266). *Hidden hypotheses:* none. *Deviation:* none; compatibility with sources, targets, units and the composition is stated; the units, not checked in the text, are proved (Findings, items 4, 11).

**V8.14 (Theorem 8.1.12 (2017: 8.3), the functor $T : \mathcal{EP} \to \mathcal{EP}$; `Polyfold.EP.tangent`, `Polyfold.tangentFunctor`, `Polyfold.theorem_8_3`, `Polyfold.theorem_8_3_tame`).** $T$ is a functor $\mathcal{EP} \to \mathcal{EP}$, $X \mapsto TX$, $\Phi \mapsto T\Phi$, and $TX$ is tame if $X$ is. *Book:* Theorem 8.1.12 (2017: 8.3), p. 366 (2017 p. 266). *Hidden hypotheses:* none. *Deviation (restated):* a functor `EP.{max u 1}  EP.{max u 1}`, since $TX$ lives in `Type (max u 1)`; the tameness sentence is `Polyfold.theorem_8_3_tame` (Findings, items 2, 11).

**V8.15 (Definition 8.1.13 (2017: 8.1), natural transformations and natural equivalence; `Polyfold.EpNatTrans`, `Polyfold.EpFunctor.NaturallyEquivalent`).** A natural transformation $\tau : F \to G$ is an sc-smooth $\tau : X \to \mathbf{Y}$ with $\tau(x) : F(x) \to G(x)$ and $\tau(x') \circ F(h) = G(h) \circ \tau(x)$; $F \simeq G$ if one exists. *Book:* Definition 8.1.13 (2017: 8.1), p. 367 (2017 p. 266). *Hidden hypotheses:* none. *Deviation:* none (Findings, item 12).

**V8.16 (the equivalence relation $\simeq$; `Polyfold.EpNatTrans.refl`, `Polyfold.EpNatTrans.symm`, `Polyfold.EpNatTrans.trans`, `Polyfold.EpNatTrans.isScSmoothMap_comp_app`, `Polyfold.EpFunctor.Natural` `Polyfold.EpFunctor.NaturallyEquivalent.symm`, `Polyfold.EpFunctor.NaturallyEquivalent.trans`, `Polyfold.EpFunctor.naturallyEquivalent_equivalence`, `Polyfold.EpNatTrans.whiskerLeft`, `Polyfold.EpNatTrans.whiskerRight`, `Polyfold.EpFunctor.NaturallyEquivalent.comp_left`, `Polyfold.EpFunctor.NaturallyEquivalent.comp_right`).** $\simeq$ is an equivalence relation on the sc-smooth functors $X \to Y$. *Book:* 2017 p. 267, without proof. *Hidden hypotheses:* none. *Deviation:* none; the sc-smoothness of $x \mapsto \tau'(x) \circ \tau(x)$ comes from Chapter 7's sheet form of Definition 7.1.3 (4) (2017: 7.3); whiskering, not in the text, is added (Findings, item 12).

**V8.17 (Proposition 8.1.14 (2017: 8.1), the tangent of a natural transformation; `Polyfold.EpNatTrans.tangent`, `Polyfold.proposition_8_1`).** If $\tau : \Phi \to \Psi$ is a natural transformation, $T\tau : TX \to T\mathbf{Y}$ is a natural transformation $T\Phi \to T\Psi$; hence $\Phi \simeq \Psi$ implies $T\Phi \simeq T\Psi$. *Book:* Proposition 8.1.14 (2017: 8.1) and its proof, pp. 367–372 (2017 pp. 267–268). *Hidden hypotheses:* none. *Deviation:* none;

naturality by fibre injectivity of $Ts$ instead of differentiating the naturality square. The proof has "$T\tau(x_0(h))$" (possible misprint), and labels the lower arrow of the last diagram $T\tau(x_0)h$ for $T\tau(x_0')h'$ (possible misprint or extraction artefact) (Findings, items 4, 13).

**V8.18 (Theorem 8.1.15 (2017: 8.4), the orbit space of $TX$; `Polyfold.EpGroupoid.tangentOrbitProj`, `Polyfold.EpGroupoid.continuous_tangentOrbitProj`, `Polyfold.EpGroupoid.preimage_tangentOrbitProj_i` `Polyfold.EpGroupoid.NaturalRep.tangentU`, `Polyfold.EpGroupoid.NaturalRep.tΦ`, `Polyfold.EpGroupoid.N` `Polyfold.EpGroupoid.NaturalRep.tangentMulAction`, `Polyfold.EpGroupoid.NaturalRep.tΓ`, `Polyfold.EpGroupoid.NaturalRep.tangent_ _eq_iff`, `Polyfold.EpGroupoid.NaturalRep.isOpenEmbedding_` `Polyfold.EpGroupoid.NaturalRep.range_tangentOrbitQuotientMap`, `Polyfold.EpGroupoid.NaturalRep.ima` `Polyfold.EpGroupoid.NaturalRep.metrizableSpace_image_tangentU`, `Polyfold.EpGroupoid.NaturalRep.me` `Polyfold.EpGroupoid.metrizableSpace_tangent_orbitSpace`, `Polyfold.EpGroupoid.theorem_8_4`, `Polyfold.EpGroupoid.hasParacompactOrbitSpace_tangent_trivial`).** If $|X|$ is paracompact, so is $|TX|$. *Book:* Theorem 8.1.15 (2017: 8.4) and its proof, 2017 p. 268. *Hidden hypotheses:* none. *Deviation (stronger):* $|TX|$ is metrizable. The proof differs from the text's: Smirnov's theorem pulled back along $|p| : |TX| \to |X^1|$ (Part I's `Polyfold.metrizableSpace_of_pullback`), as for Proposition 7.5.3 (2017: 7.12), with the metrizable $|TU(x)| = |p|^{-1}(|U(x)_1|)$; the closed paracompact $\tilde{A}_z$, the regularity of $|TX|$ and Part I's Proposition 2.7.4 (2017: 2.17) are not used. The open embedding $G_x\backslash TU(x) \to |TX|$, left implicit, is proved; the local bundle model $TO \to O^1$ is not used, and the linearity of the action of $G_x$ on $TU(x)$ is not formalized (gap G.7). "parcompact" (possible misprint) (Findings, items 20, 21, 104).

**V8.19 (strong bundle maps between strong bundles over M-polyfolds, and the pull-back along $s$; `Polyfold.StrongBundle.pullback`, `Polyfold.pullbackSpace`, `Polyfold.StrongBundle.paracompactSpace_` `Polyfold.StrongBundle.pullback_atlas`, `Polyfold.PullbackChartData.exists_mem_atlas`, `Polyfold.StrongBundle.isScSmoothMap_pullbackProj`, `Polyfold.StrongBundle.isScSmoothMap_pullbackPr` `Polyfold.StrongBundle.isScSmoothOn_to_pullback_iff`, `Polyfold.StrongBundle.isScSmoothMap_pullback` `Polyfold.pullbackPair`, `Polyfold.StrongBundle.isScDiffeoOn_pullbackProj'`, `Polyfold.StrongBundle.is` `Polyfold.StrongBundle.IsStrongBundleMap`, `Polyfold.fiberRestrict`, `Polyfold.StrongBundle.IsStrongBu` `Polyfold.StrongBundle.IsStrongBundleMap.isScSmoothMap_base`, `Polyfold.StrongBundle.IsStrongBundle` `Polyfold.StrongBundle.IsStrongBundleMap.mapsTo_doubleLevel`, `Polyfold.StrongBundle.IsStrongBundle` `Polyfold.StrongBundle.isStrongBundleMap_pullbackProj'`, `Polyfold.MPolyfold.isScSmoothMap_of_foral`** A strong bundle map $\Phi : Y \to Y'$ covering $f$ covers $f$, is linear on the fibres, sc-smooth on $Y[0]$, maps $Y_{0,1}$ into $Y'_{0,1}$ and is sc-smooth on $Y[1]$; it then preserves the double filtration, has bounded fibre maps and an sc-smooth base map. The pull-back $\mathbf{X}_s\times_P W$ is Part I's, with its universal property on $[0]$ and $[1]$, and over a sheet of $s$ the projection to $W$ is an sc-diffeomorphism. *Book:* used in Definitions 8.3.1–8.3.3, 8.5.9 (2017: 8.4–8.6, 8.10) (pp. 373–396 (2017 pp. 270–285)); Part I's Definition 2.6.1 (2017: 2.22) and Proposition 2.6.8 (2017: 2.16). *Hidden hypotheses:* none. *Deviation (not in the book):* Part I defines strong bundle maps only between local models; the global notion is a reading (Findings, items 30, 31).

**V8.20 (Definition 8.3.1 (2017: 8.4), strong bundles over ep-groupoids; `Polyfold.EpStrongBundle`, `Polyfold.EpStrongBundle.P_`, `Polyfold.EpGroupoid.unitPair`, `Polyfold.EpStrongBundle.continuous_`, `Polyfold.EpStrongBundle.trivial`).** A strong bundle $(P, \mu)$ over $X$: a strong bundle $P : W \to X$ and a strong bundle map $\mu : \mathbf{X}_s\times_P W \to W$ covering $t$, with (1) $\mu$ a surjective local sc-diffeomorphism, linear on the fibres, (2) $\mu(1_x, w) = w$, (3) $\mu(\psi\circ\gamma, w) = \mu(\psi, \mu(\gamma, w))$. *Book:* Definition 8.3.1 (2017: 8.4), p. 375 (2017 p. 271). *Hidden hypotheses:* none. *Deviation (restated):* on Part I's pull-back strong bundle, with the strong bundle maps of V8.19; the surjectivity in (1) is kept although it follows from (2) (Findings, items 32, 102).

**V8.21 (the fibre maps $\mu(\psi,\cdot)$; `Polyfold.EpStrongBundle.fiberMor`, `Polyfold.EpStrongBundle.fiberMor_appl` `Polyfold.EpStrongBundle._add`, `Polyfold.EpStrongBundle._smul`, `Polyfold.EpStrongBundle._inv_`, `Polyfold.EpStrongBundle._ _inv`, `Polyfold.EpStrongBundle.fiberMorEquiv`).** For $\psi : x \to y$, $\mu(\psi,\cdot) : W_x \to W_y$ is a linear isomorphism. *Book:* 2017 p. 271, after Definition 8.3.1 (2017: 8.4). *Hidden hypotheses:* none. *Deviation (stronger):* a topological linear isomorphism, with inverse $\mu(\psi^{-1},\cdot)$ (Findings, item 32).

**V8.22 (the functorial picture, the ep-groupoid $(W,\mathbf{W})$; `Polyfold.EpStrongBundle.morInv`, `Polyfold.EpStrongBundle.morComp`, `Polyfold.EpStrongBundle.morComp_val`, `Polyfold.EpStrongBundle._m` `Polyfold.EpStrongBundle.toGroupoidStr`, `Polyfold.EpStrongBundle.exists_isProperMap_tgt`, `Polyfold.EpStrongBundle.isScSmoothCompSheet_toGroupoidStr`, `Polyfold.EpStrongBundle.toEpGroupoid`, `Polyfold.EpStrongBundle.projFunctor`, `Polyfold.EpStrongBundle.isStrongBundleMap_src`, `Polyfold.EpStrongBundle.isStrongBundleMap_tgt`, `Polyfold.EpStrongBundle.exists_isomorphic_of_ison` `Polyfold.EpStrongBundle.surjective_orbitMap_projFunctor`, `Polyfold.EpStrongBundle.continuous_orbi`).** $(W,\mathbf{W})$ with $\mathbf{W} = \mathbf{X}_s\times_P W$, $s_{\mathbf{W}}(\psi,w) = w$, $t_{\mathbf{W}} = \mu$, the units $(1_x,w)$, the inverses $(\psi^{-1},\mu(\psi,w))$ and the composition $(\psi,w')\circ(\gamma,w) = (\psi\circ\gamma,w)$ is an ep-groupoid; $s_{\mathbf{W}}$, $t_{\mathbf{W}}$ are strong bundle maps covering $s$, $t$; $(P,\pi_1)$ is an sc-smooth functor; $|P| : |W| \to |X|$ is continuous and surjective. *Book:* pp. 375–377 (2017 pp. 271–272). *Hidden hypotheses:* none. *Deviation (restated):* "we can define the ep-groupoid", stated without proof, is proved (properness by sequences, condition (4) in sheet form); from here on $W$ lives in the universe of the objects and morphisms of $X$ (Findings, item 33).

**V8.23 (Definition 8.3.2 (2017: 8.5), strong bundle maps over ep-groupoids; `Polyfold.EpStrongBundle.IsMap`, `Polyfold.EpStrongBundle.isMap_id`).** A strong bundle map $\Phi : W \to W'$ covering the object map of an sc-smooth functor $\phi$, with $\Phi(\mu(\gamma,w)) = \mu'(\phi(\gamma),\Phi(w))$. *Book:* Definition 8.3.2 (2017: 8.5) and the paragraph after it, 2017 p. 273. *Hidden hypotheses:* none. *Deviation:* none (Findings, item 34).

**V8.24 (Definition 8.3.3 (2017: 8.6), strong bundle functors; `Polyfold.EpStrongBundle.IsStrongBundleFunctor`, `Polyfold.EpStrongBundle.IsStrongBundleFunctor.projFunctor_comp`, `Polyfold.EpStrongBundle.IsStrong` `Polyfold.EpStrongBundle.isStrongBundleFunctor_id`).** A functor $(W,\mathbf{W}) \to (W',\mathbf{W}')$, linear on the fibres, covering an sc-smooth functor $\phi$ and inducing strong bundle maps on objects and morphisms, with $P'\circ\Phi = \phi\circ P$. *Book:* Definition 8.3.3 (2017: 8.6), p. 378 (2017 p. 274). *Hidden hypotheses:* none. *Deviation (restated):* "linear on the fibres", "covers $\phi$" and "induces strong bundle maps" are read together as two conditions `IsStrongBundleMap`, on objects (for $P$, $P'$) and on morphisms (for the pull-backs $\mathbf{W}\to\mathbf{X}$, $\mathbf{W}'\to\mathbf{X}'$) (Findings, item 34).

**V8.25 (the equivalence of Definitions 8.3.2 and 8.3.3 (2017: 8.5 and 8.6); `Polyfold.EpStrongBundle.IsStrongBun` `Polyfold.EpStrongBundle.liftMor`, `Polyfold.EpStrongBundle.IsMap.liftFunctor`, `Polyfold.EpStrongBund` `Polyfold.EpStrongBundle.IsStrongBundleFunctor.eq_liftFunctor`, `Polyfold.EpStrongBundle.isMap_iff_` `Polyfold.StrongBundle.pullbackProj'One`, `Polyfold.StrongBundle.isScSmoothMap_to_pullback_one`, `Polyfold.isLinearMap_fiberRestrict_pullbackMap`).** $\Phi : W \to W'$ is a strong bundle map of Definition 8.3.2 (2017: 8.5) iff it is the object map of a strong bundle functor of Definition 8.3.3 (2017: 8.6), and the functor, $\Phi(\gamma,w) = (\phi(\gamma),\Phi(w))$, is unique. *Book:* pp. 377–379 (2017 pp. 273–274) ("these are identical notions"). *Hidden hypotheses:* none. *Deviation:* none; the $[1]$-smoothness of the lifted morphism map needs the universal property of the pull-back on level $[1]$, proved here (Findings, item 34).

**V8.26 (Definition 8.3.4 (2017: 8.7), section functors; `Polyfold.EpStrongBundle.IsFunctorialSection`, `Polyfold.EpStrongBundle.IsScSmoothSectionFunctor`, `Polyfold.EpStrongBundle.IsScPlusSectionFunctor`**

`Polyfold.EpStrongBundle.IsScFredholmSectionFunctor`, `Polyfold.EpStrongBundle.sectionFunctors`, `Polyfold.EpStrongBundle.plusSectionFunctors`, `Polyfold.EpStrongBundle.fredholmSectionFunctors`, `Polyfold.EpStrongBundle.fiberZero_mem_sectionFunctors`, `Polyfold.EpStrongBundle.sectionAdd_mem_se` `Polyfold.EpStrongBundle.sectionSmul_mem_sectionFunctors`, `Polyfold.EpStrongBundle.plusSectionFunc` `Polyfold.EpStrongBundle.fiberZero_mem_plusSectionFunctors`, `Polyfold.EpStrongBundle.sectionAdd_me` `Polyfold.EpStrongBundle.sectionSmul_mem_plusSectionFunctors`, `Polyfold.EpStrongBundle.fredholmSec` `Polyfold.StrongBundle.IsScSmoothSection.sectionAdd_of_isScSmoothSection`, `Polyfold.StrongBundle.I` `Polyfold.StrongBundle.isScSmoothSection_fiberZero`). (1) An sc-smooth section functor is an sc-smooth section $f$ with $f(y) = \mu(\varphi, f(x))$ for $\varphi : x \to y$; (2) an sc$^+$-section functor takes values in $W_{0,1}$ and is sc-smooth into $W[1]$; (3) an sc-Fredholm section functor is an sc-Fredholm section; $\Gamma(P,\mu)$ is a vector space and $\Gamma^+(P,\mu) \subseteq \Gamma(P,\mu)$ a subspace. *Book:* Definition 8.3.4 (2017: 8.7), pp. 379 ff. (2017 pp. 274–275). *Hidden hypotheses:* none. *Deviation (restated):* "vector space" is closure under the zero section, sums and constant multiples, with Part I's fibrewise operations; no `Module` structure is built; (3) uses Part I's Definition 3.1.16 (2017: 3.8) in its germ reading. The sum of two sc-smooth sections and constant multiples, which Part I does not state, are proved here in the charts (Findings, items 35, 103).

**V8.27 (Definition 8.5.1 (2017: 8.8), proper covering functors;** `Polyfold.EpFunctor.IsProperCovering`, `Polyfold.EpFunctor.fibredMap`, `Polyfold.MPolyfold.isSubMPolyfold_fibredProduct`, `Polyfold.EpFunctor.isProperCovering_id`). (1) $F$ is a surjective local sc-diffeomorphism on objects; (2) the fibres $F^{-1}(x)$ are finite and evenly covered; (3) $\varphi \mapsto (F(\varphi), s(\varphi))$ is an sc-diffeomorphism $\mathbf{Y} \to \mathbf{X}_{s\times_F} Y$. *Book:* Definition 8.5.1 (2017: 8.8), p. 383 (2017 p. 275). *Hidden hypotheses:* none. *Deviation (restated):* in (2) $U(x)$ is open and the $U(y)$ carry local inverses $\sigma(y)$; (3) on Part I's fibred-product sub-M-polyfold of $\mathbf{X} \times Y$, as in Definition 7.1.3 (4) (2017: 7.3) (Findings, items 50, 102).

**V8.28 (the degeneracy remark after Definition 8.5.1 (2017: 8.8);** `Polyfold.EpFunctor.IsProperCovering.degInd` `Polyfold.EpFunctor.IsProperCovering.degIndex_mor`, `Polyfold.EpFunctor.IsProperCovering.degIndex_p` `Polyfold.EpFunctor.IsProperCovering.degIndex_prod_fibredMap`, `Polyfold.MPolyfold.degIndex_prod_le` `Polyfold.MPolyfold.isTame_prod`, `Polyfold.proposition_2_24`). $d_{\mathbf{Y}}(\varphi) = d_{\mathbf{X}_{s\times_F Y}}(F(\varphi), s(\varphi))$, $d_{\mathbf{X}\times Y}(F(\varphi), s(\varphi)) = 2\, d_{\mathbf{Y}}(\varphi)$ and $d_{\mathbf{X}_{s\times_F Y}} = \frac{1}{2} d_{\mathbf{X}\times Y}$ at $(F(\varphi), s(\varphi))$. *Book:* pp. 379–385 (2017 pp. 275–276). *Hidden hypotheses:* $X$ and $Y$ tame, for the product formula $d_{\mathbf{X}\times Y} = d_{\mathbf{X}} + d_Y$, which Part I proves for tame M-polyfolds (Part I's Proposition 2.5.20 (2017: 2.15) states its formula $d = \frac{1}{2} d_{X\times Y}$ for tame $X$, $Y$). *Deviation (narrower):* the first equality and $d_{\mathbf{X}}(F(\varphi)) = d_{\mathbf{Y}}(\varphi)$ hold in general; the other two for tame $X$, $Y$; in general $d_{\mathbf{X}\times Y}(F(\varphi), s(\varphi)) \le 2\, d_{\mathbf{Y}}(\varphi)$ (Findings, item 51).

**V8.29 (Proposition 8.5.2 (2017: 8.2), composites of proper covering functors;** `Polyfold.EpFunctor.IsProperCove` `Polyfold.EpFunctor.IsProperCovering.isScDiffeo_fibredMap_comp`, `Polyfold.EpFunctor.IsProperCoveri` If $F : Y \to X$ and $H : Z \to Y$ are proper covering functors, so is $F \circ H$. *Book:* Proposition 8.5.2 (2017: 8.2) and its proof, pp. 384–386 (2017 pp. 276–277). *Hidden hypotheses:* none. *Deviation:* none; (2), "follows easily", is proved by shrinking $U(x)$; (3) through $\tilde{K}$, $\tilde{L}$ as in the text. "fo" for "for" (possible misprint) (Findings, item 52).

**V8.30 (Theorem 8.5.3 (2017: 8.5), Proper Covering I;** `Polyfold.EpFunctor.ProperCoveringChart`, `Polyfold.EpFunctor.ProperCoveringLocal.Data`, `Polyfold.EpFunctor.ProperCoveringLocal.exists_data`, `Polyfold.EpFunctor.ProperCoveringLocal.Data.toChart`, `Polyfold.EpFunctor.ProperCoveringLocal.isSc` `Polyfold.EpFunctor.theorem_8_5`, `Polyfold.EpFunctor.theorem_8_5_of_naturalRep`). For a proper covering functor $F : Y \to X$ and $x \in X$, a connected, sufficiently small $U(x)$

with the natural $G_x$-action carries sets $U(y)$, $y \in F^{-1}(x)$, and $\mathbf{U}(\psi_0)$, $\psi_0$ a morphism between points of $F^{-1}(x)$, with (0)–(5). *Book:* Theorem 8.5.3 (2017: 8.5), p. 385 (2017 p. 277), proved on pp. 386–388 (2017 pp. 278–279). *Hidden hypotheses:* none. *Deviation (stronger):* "sufficiently small" is proved in two forms, a chart with $U(x)$ inside every open $V \ni x$, and one open $W \ni x$ that works for every natural representation on a connected $U(x) \subseteq W$; $\mathbf{U}(\varphi)$ for $\varphi \in \mathbf{U}(x)$ is the connected component of $\mathbf{U}(x)$ at $\varphi$. (5) as printed, with $t$ on the right and $U(t(F(\psi_0)))$, does not hold as we read it (possible misprint; counterexample `Polyfold.CoveringExamples.theorem_8_5_five_as_printed`, the identity functor of $\mathbb{Z}/2 \ltimes \mathbb{R}$ at $\psi_0 = (-1, 0)$, for every chart, which exists by `Polyfold.CoveringExamples.theorem_8_5_five_as_printed_no`r it is formalized with $s$ on both vertical arrows (`comm_src`), with the companion square with $t$ (`comm_tgt`). The proof differs from the text's: Definition 8.5.1 (3) (2017: 8.8) gives the sheets $\mathbf{U}(\psi_0)$ as images of continuous sections of $s$; the finiteness of $F^{-1}(x)$ and Definition 8.5.1 (1) (2017: 8.8) are not needed. In the proof "$g \in G_y$", "$G_y$-invariant" and "$F(z') = s(F(\varphi))$" for $G_x$, $G_x$-invariant and $t(F(\varphi))$, "admitting admitting", and "morphismsm" in (2) (possible misprints) (Findings, items 53, 54, 55).

**V8.31 (Remark 8.5.4 (2017: 8.2);** `Polyfold.EpFunctor.remark_8_2_a`, `Polyfold.EpFunctor.isotropyMor`, `Polyfold.EpFunctor.remark_8_2_b`, `Polyfold.EpFunctor.ProperCoveringChart.connectedComponentIn_U`, `Polyfold.EpFunctor.ProperCoveringChart.connectedComponentIn_sheet`**).** (a) A morphism between points of $F^{-1}(x)$ is mapped into $G_x$; (b) everything in Theorem 8.5.3 (2017: 8.5) is determined by $U(x)$. *Book:* Remark 8.5.4 (2017: 8.2), pp. 386–387 (2017 pp. 277–278). *Hidden hypotheses:* none. *Deviation (restated):* (b) as: two charts with the same $U(x)$ have the same $\Phi$ and $\Gamma$ on $U(x)$, the same $U(y)$ and the same $\mathbf{U}(\psi_0)$, and these are connected components; its remarks on the naturality of Theorem 7.1.13's (2017: 7.1) construction have no further formal content (Findings, item 56).

**V8.32 (the category** $\mathrm{Diff}_{sc}$ **and Definition 8.5.5 (2017: 8.9), covering functors and geometric lifts;** `Polyfold.MPolyfoldCat`, `Polyfold.ScDiffeo`, `Polyfold.ScDiffeo.id`, `Polyfold.ScDiffeo.comp`, `Polyfold.ScDiffeo.ext`, `Polyfold.coveringMap`, `Polyfold.IsCoveringFunctor`, `Polyfold.GeometricLift`, `Polyfold.IsCoveringFunctor.existsUnique_lift`, `Polyfold.IsCoveringFunctor.lift`, `Polyfold.isCoveringFunctor_of_isEmpty`, `Polyfold.CoveringExamples.isCoveringFunctor_id`**).** $\mathrm{Diff}_{sc}$ has the M-polyfolds as objects and the sc-diffeomorphisms as morphisms; $f : D \to C$, $C$ with one object, is a covering functor if $\psi \mapsto (f(\psi), s(\psi))$ is a bijection $\mathbf{D} \to C \,{}_s\!\times_f D$; a geometric lift is a pair of functors $L_D$, $L_C$ into $\mathrm{Diff}_{sc}$ with a natural transformation $\tau : L_D \to L_C \circ f$. *Book:* 2017 p. 279 and Definition 8.5.5 (2017: 8.9), pp. 388–389 (2017 pp. 279–280). *Hidden hypotheses:* none. *Deviation (restated):* $D$ a Mathlib groupoid in `Type`, \$C = \$ `SingleObj G` for a group $G$ in `Type`, so $C \,{}_s\!\times_f D = G \times D$; the finiteness of $D$, its hom-sets and $G$ is carried by the results that use it; $\mathrm{Diff}_{sc}$ has bundled M-polyfolds of one universe; $D = \emptyset$ is allowed (Findings, items 57, 100).

**V8.33 (the discussion after Definition 8.5.5 (2017: 8.9):** $\varphi_{yy'}$**, the actions and equivariance;** `Polyfold.GeometricLift.Ux`, `Polyfold.GeometricLift. y`, `Polyfold.GeometricLift.isScDiffeoOn_ y`, `Polyfold.GeometricLift. `, `Polyfold.GeometricLift.isScDiffeoOn_ `, `Polyfold.GeometricLift. _comp`, `Polyfold.GeometricLift. _id`, `Polyfold.GeometricLift. _inv_ `, `Polyfold.GeometricLift.smul_def`, `Polyfold.GeometricLift.smul_def'`, `Polyfold.GeometricLift.isScSmoothMap_smul`, `Polyfold.GeometricLift. y_ `, `Polyfold.GeometricLift. _equivariant`**).** $\tau(y) : U(y) \to U(x)$ and $\varphi_{yy'}(\psi) = L_D(\psi)$ are sc-diffeomorphisms, $\varphi_{yy''}(\psi \circ \gamma) = \varphi_{y'y''}(\psi) \circ \varphi_{yy'}(\gamma)$, $G_y$ and $G_x$ act on $U(y)$, $U(x)$, and $\tau(y)(h * z) = f(h) * \tau(y)(z)$. *Book:* pp. 388–390 (2017 pp. 280–281). *Hidden hypotheses:* none. *Deviation:* none; the action of $G_x$ is a left action through `SingleObj G`'s composition (Findings, item 57).

**V8.34 (disjoint unions of M-polyfolds;** `Polyfold.paracompactSpace_sigma`, `Polyfold.MChart.pushforward`, `Polyfold.MChart.compatible_of_disjoint`, `Polyfold.MPolyfold.sigmaChart`, `Polyfold.MPolyfold.sigma`, `Polyfold.MPolyfold.isChart_sigmaChart_iff`, `Polyfold.MPolyfold.isScDiffeoOn_sigmaMk`, `Polyfold.MPolyfold.isScSmoothMap_sigma_iff`, `Polyfold.MPolyfold.isLocalScDiffeo_sigma_iff`, `Polyfold.MPolyfold.degIndex_sigma`, `Polyfold.MPolyfold.isTame_sigma_iff`, `Polyfold.MPolyfold.isScDi` `Polyfold.isScDiffeoOn_prod_discrete_sigma`**).** A disjoint union $\bigsqcup_i X_i$ of M-polyfolds, for any index type, is an M-polyfold whose charts are those of the summands; the summands are open and sc-diffeomorphic to the $X_i$; sc-smoothness, local sc-diffeomorphisms, the degeneracy index and tameness are summand-wise; products distribute over the union, and $G \times X \cong \bigsqcup_G X$ for discrete $G$. *Book:* p. 390 (2017 p. 281), "possesses a natural M-polyfold structure", in the proof of Proposition 8.5.7 (2017: 8.3). *Hidden hypotheses:* none. *Deviation (not in the book):* the structure is not described in the text (Findings, item 58).

**V8.35 (Proposition 8.5.7 (2017: 8.3), the proper covering functor of a geometric lift;** `Polyfold.GeometricLift.YObj`, `Polyfold.GeometricLift.YMor`, `Polyfold.GeometricLift.sourceStr`, `Polyfold.GeometricLift.yObjPolyfold`, `Polyfold.GeometricLift.yMorPolyfold`, `Polyfold.GeometricLift.` `Polyfold.GeometricLift.isLocalScDiffeo_yTgt`, `Polyfold.GeometricLift.isProperMap_yTgt`, `Polyfold.GeometricLift.isScSmoothCompSheet_y`, `Polyfold.geometricLiftSource`, `Polyfold.geometricLif` `Polyfold.geometricLiftFunctor`, `Polyfold.geometricLiftFunctor_surjective`, `Polyfold.geometricLiftFu` `Polyfold.geometricLiftFunctor_isLocalScDiffeo`, `Polyfold.geometricLiftFunctor_finite_fiber`, `Polyfold.geometricLiftFunctor_evenlyCovered`, `Polyfold.GeometricLift.fibredInv`, `Polyfold.GeometricLift.isScSmoothMap_fibredInv`, `Polyfold.geometricLiftFunctor_isScDiffeo_fibredM` `Polyfold.proposition_8_3`, `Polyfold.EpGroupoid.Examples.actionEpGroupoid`, `Polyfold.CoveringExample` A covering functor $f : D \to C$ with a geometric lift gives a proper covering functor $F : Y \to X$, $X = G_x \ltimes U(x)$, $Y = \bigsqcup U(y)$, $\mathbf{Y} = \bigsqcup \mathrm{mor}(y, y') \times U(y)$, $F|U(y) = \tau(y)$, $F(\psi, z) = (f(\psi), \tau(y)(z))$. *Book:* Proposition 8.5.7 (2017: 8.3) and its proof, pp. 389–391 (2017 pp. 281–282). *Hidden hypotheses:* $D$ nonempty; for $D = \emptyset$, which Definition 8.5.5 (2017: 8.9) allows, $F$ is surjective on objects only if $U(x) = \emptyset$ (`Polyfold.geometricLiftFunctor_surjective_iff`). *Deviation (narrower):* for nonempty $D$; the finiteness of Definition 8.5.5 (2017: 8.9) enters as instance arguments; $X$ is Chapter 7's action ep-groupoid, $G$ with the discrete topology and lifted by `ULift`; the "one verifies" (composition sc-smooth, properness, here with $V = Y$) is carried out; the inverse of (8.16) is shown sc-smooth directly. The proof's "$F : X \to Y$" is read as $F : Y \to X$ (possible misprint) (Findings, items 59, 101).

**V8.36 (Theorem 8.5.8 (2017: 8.6), Proper Covering II;** `Polyfold.GroupoidStr.FullOb`, `Polyfold.EpFunctor.FiberCat`, `Polyfold.EpFunctor.fiberIsotropy`, `Polyfold.EpFunctor.restrictFiber`, `Polyfold.EpFunctor.covering_restrict`, `Polyfold.EpFunctor.fintypeFiberCat`, `Polyfold.EpFunctor.none` `Polyfold.MPolyfoldCat.ofOpen`, `Polyfold.ScDiffeo.ofOpen`, `Polyfold.EpFunctor.IsFullyFaithfulOpenEmb` `Polyfold.EpFunctor.ProperCoveringChart.geometricLift`, `Polyfold.EpFunctor.ProperCoveringChart.Φ`, `Polyfold.EpFunctor.ProperCoveringChart.Ψ`, `Polyfold.EpFunctor.ProperCoveringChart.isFullyFaithful` `Polyfold.EpFunctor.ProperCoveringChart.isFullyFaithfulOpenEmbedding_Ψ`, `Polyfold.EpFunctor.Proper` `Polyfold.EpFunctor.ProperCoveringChart.range_Φ_mor`, `Polyfold.EpFunctor.ProperCoveringChart.range` `Polyfold.EpFunctor.ProperCoveringChart.range_Ψ_mor`, `Polyfold.EpFunctor.ProperCoveringChart.comp_` `Polyfold.EpFunctor.theorem_8_6`**).** For a proper covering functor $F : Y \to X$ and $x \in X$: $f = F|D : D \to C$ is a covering functor ($C$ on $x$, $D$ on $F^{-1}(x)$), and for $U(x)$ as in Theorem 8.5.3 (2017: 8.5) there are a geometric lift with associated proper covering functor $A : K \to L$ and fully faithful open embeddings $\Phi : L \to X$, $\Psi : K \to Y$ with images the full subcategories on $U(x)$ and $F^{-1}(U(x))$ and $F \circ \Psi = \Phi \circ A$. *Book:* Theorem 8.5.8 (2017: 8.6) and its proof, 2017 pp. 283–284. *Hidden hypotheses:* none. *Deviation (narrower):* for ep-groupoids whose objects and morphisms are

in `Type`, since Definition 8.5.5 (2017: 8.9) is stated there; "a sufficiently small $U(x)$ as in Theorem 8.5" is any chart of Theorem 8.5.3 (2017: 8.5) (which exist inside every open $V \ni x$, part (b)); "a geometric lift $A : K \to L$" is a geometric lift with its functor $A$ of Proposition 8.5.7 (2017: 8.3); $L_D$ on the morphism $\psi_0$ of $D$ is $t \circ (s|\mathbf{U}(\psi_0))^{-1}$ (the text's "$\psi \in \mathbf{U}(\psi_0)$" read as $\psi = \psi_0$); both "one verifies" are carried out. "there exists exists", "$\tau(t)$" for $\tau(y)$, and "$A(\psi, z) = (f(z), F(z))$" for $(f(\psi), F(z))$ (possible misprints) (Findings, items 2, 60).

**V8.37 (Definition 8.5.9 (2017: 8.10), proper strong bundle covering functors;** `Polyfold.StrongBundle.IsStrongB` `Polyfold.StrongBundle.IsStrongBundleIsoOn.fiberEquiv`, `Polyfold.EpFunctor.fibredSnd`, `Polyfold.EpFunctor.isScSmoothMap_fibredSnd`, `Polyfold.EpStrongBundle.fibredLift`, `Polyfold.EpStrongBundle.fibredBundleEquiv`, `Polyfold.EpStrongBundle.pullbackProj_fibredBundleEqui` `Polyfold.EpStrongBundle.fibredBundleEquiv_symm_fibredLift`, `Polyfold.EpStrongBundle.IsProperStron` `Polyfold.EpStrongBundle.isProperStrongBundleCovering_id`, `Polyfold.TrivialBundle.epBundle_isPrope` A strong bundle map $A$, $Q \circ A = F \circ P$, with (1) $A$ surjective and a local strong bundle isomorphism covering a proper covering functor $F$; (2) $A : W|U(y) \to V|U(x)$ strong bundle isomorphisms over an even covering; (3) $(\psi, w) \mapsto ((F(\psi), A(w)), w)$ a strong bundle isomorphism $\mathbf{W} \to \mathbf{V}_s\times_A W$ covering $\mathbf{Y} \to \mathbf{X}_s\times_F Y$. *Book:* Definition 8.5.9 (2017: 8.10), p. 394 (2017 p. 285). *Hidden hypotheses:* none. *Deviation (restated):* $A$ is a strong bundle functor (Definition 8.3.3 (2017: 8.6), V8.24); the finiteness in (2) is that of Definition 8.5.1 (2) (2017: 8.8); the strong bundle structure of $\mathbf{V}_s\times_A W$, not given in the text, is read as the pull-back of $P$ along $\mathrm{pr}_2 : \mathbf{X}_s\times_F Y \to Y$, with the text's projection $((\varphi, v), w) \mapsto (\varphi, P(w))$ (`Polyfold.EpStrongBundle.fibredBundleEquiv`) (Findings, items 70, 102, 103).

**V8.38 (§8.6.1 (2017 §8.5.1), the morphisms over a fibre and the isomorphism classes;** `Polyfold.EpFunctor.fiberClass`, `Polyfold.EpFunctor.fiberIndex`, `Polyfold.EpFunctor.fiberClasses`, `Polyfold.EpFunctor.iUnion_fiberClass`, `Polyfold.EpFunctor.pairwiseDisjoint_fiberClass`, `Polyfold.EpFunctor.nat_card_isotropy_eq_of_mem_fiberClass`, `Polyfold.EpFunctor.IsProperCovering.b` `Polyfold.EpFunctor.IsProperCovering.existsUnique_lift`, `Polyfold.EpFunctor.IsProperCovering.morOn` `Polyfold.EpFunctor.IsProperCovering.ncard_morOn_fiber`, `Polyfold.EpFunctor.IsProperCovering.finit` $\varphi \mapsto (F(\varphi), s(\varphi))$ is a bijection $\mathrm{mor}(F^{-1}(x)) \to G_x \times F^{-1}(x)$, so $\#\mathrm{mor}(F^{-1}(x)) = \#G_x \cdot \#F^{-1}(x)$; $F^{-1}(x) = \Theta_1 \cup \cdots \cup \Theta_q$, with isomorphic isotropy groups within a class. *Book:* §8.6.1 (2017 §8.5.1), 2017 p. 286. *Hidden hypotheses:* none. *Deviation (restated):* the classes are indexed by their images in $|Y|$, $\Theta_c = F^{-1}(x) \cap \pi^{-1}(c)$; $\#$ is the cardinality of finite sets (Findings, item 71).

**V8.39 (Lemma 8.6.1 (2017: 8.9);** `Polyfold.EpFunctor.morOn_fiberClass_eq`, `Polyfold.EpFunctor.IsProperCov` `Polyfold.EpFunctor.IsProperCovering.lemma_8_9`**).** $\#G_x \cdot \#\Theta_j = \#\{\psi \mid s(\psi), t(\psi) \in \Theta_j\} = \#G_{y_0} \cdot (\#\Theta_j)^2$ and $\#G_x = \#G_{y_0} \cdot \#\Theta_j$ for $y_0 \in \Theta_j$. *Book:* Lemma 8.6.1 (2017: 8.9) and its proof, pp. 396–398 (2017 pp. 286–287). *Hidden hypotheses:* none. *Deviation:* none; (8.17) writes "$\psi \to (F\psi), s(\psi))$" (possible misprint) (Findings, items 71, 72).

**V8.40 (Lemma 8.6.2 (2017: 8.10);** `Polyfold.EpFunctor.IsProperCovering.exists_isomorphic_of_isomorphic`, `Polyfold.EpFunctor.IsProperCovering.fiberIndex_eq`, `Polyfold.EpFunctor.IsProperCovering.lemma_8_1` `Polyfold.EpFunctor.IsProperCovering.lemma_8_10_equiv`**).** For isomorphic $x, x'$ the isomorphism classes of $F^{-1}(x)$ and $F^{-1}(x')$ correspond one to one, and isomorphic $y \in F^{-1}(x)$, $y' \in F^{-1}(x')$ lie in corresponding classes for exactly one index. *Book:* Lemma 8.6.2 (2017: 8.10) and the paragraph before it, 2017 p. 287. *Hidden hypotheses:* none. *Deviation (restated):* the correspondence is "the same index $c \in |Y|$"; as a bijection of the sets of classes it is `Polyfold.EpFunctor.IsProperCovering.lemma_8_10_equiv` (Findings, items 71, 73).

**V8.41 (Lemma 8.6.3 (2017: 8.11);** `Polyfold.EpFunctor.IsProperCovering.ncard_fiber_eq_sum, Polyfold.EpFunctor.IsProperCovering.lemma_8_11, Polyfold.EpFunctor.IsProperCovering.ncard_fiber_`
For isomorphic $x, x'$, $\#\Theta_j = \#\Theta'_j$ and $\#F^{-1}(x) = \#F^{-1}(x')$. *Book:* Lemma 8.6.3 (2017: 8.11) and the paragraph before it, p. 398 (2017 p. 287). *Hidden hypotheses:* none. *Deviation:* none; "the third claim of Lemma 8.10" for that of Lemma 8.6.1 (2017: 8.9), and the summation index $i$ with summands $\#\Theta_j$ (possible misprints) (Findings, item 73).

**V8.42 (Proposition 8.6.4 (2017: 8.4), the number of sheets;** `Polyfold.EpFunctor.IsProperCovering.one_le_nca Polyfold.EpFunctor.IsProperCovering.eventually_ncard_fiber_eq, Polyfold.EpFunctor.IsProperCoveri Polyfold.EpFunctor.IsProperCovering.fiberCardOrbit, Polyfold.EpFunctor.IsProperCovering.isLocall Polyfold.EpFunctor.IsProperCovering.proposition_8_4, Polyfold.EpFunctor.IsProperCovering.fiberCa Polyfold.EpGroupoid.trivial, Polyfold.discretePolyfold`**).** $\#F^{-1}(x)$ is constant on the preimage of every connected component of $|X|$, and $x \mapsto \#F^{-1}(x)$ is an sc-smooth functor $X \to \mathbb{N}$. *Book:* Proposition 8.6.4 (2017: 8.4) and the paragraph before it, 2017 p. 288. *Hidden hypotheses:* none. *Deviation (restated):* $\mathbb{N}$ is Lean's, with 0, as the trivial ep-groupoid of the discrete M-polyfold; the values are $\geq 1$; only the local constancy of $\#F^{-1}(x)$ is used, not "the structure of isomorphism classes stays the same" (Findings, item 74).

**V8.43 (§8.6.2 (2017 §8.5.2), the structure of strong bundle coverings;** `Polyfold.EpStrongBundle.IsProperStrong Polyfold.EpStrongBundle.IsProperStrongBundleCovering.fiberEquiv, Polyfold.EpStrongBundle.IsStro Polyfold.EpStrongBundle.section_8_5_2`**).** $A(\varphi, w) = (F(\varphi), A_{s(\varphi)}(w))$ with $A_y$ topological linear isomorphisms, and $\mu_X(F(\varphi)) \circ A_{s(\varphi)} = A_{t(\varphi)} \circ \mu_Y(\varphi)$: $y \mapsto A_y$ is a natural transformation $\mu_Y \to \mu_X \circ F$ of functors into Ban. *Book:* §8.6.2 (2017 §8.5.2), 2017 pp. 288–289. *Hidden hypotheses:* none. *Deviation (restated):* the natural transformation is stated as its naturality identity; of Definition 8.5.9 (2017: 8.10) only the strong bundle functor part enters the identity (Findings, item 75).

### The additions of the 2021 edition (V8.44–V8.52)

The Springer edition (2021) adds Propositions 8.1.16 and 8.1.17 (the Zhou condition and unique continuation for the tangent $TX$), the auxiliary norms of Definition 8.3.5, §8.4 (Proposition 8.4.1 on paracompact orbit spaces of strong bundles, the local regularity property of Definition 8.4.2 and Proposition 8.4.3), Example 8.5.6, condition (0) of Definition 8.5.9 and Lemma 8.5.10; Proposition 8.2.3, a Poincaré lemma, belongs to the sc-differential forms of §8.2 (gap G.74). Numbers and pages below are the 2021 text's. Lean files: `Chapter8/Book2021.lean`, `Chapter8/TangentZhou.lean`, and, for Proposition 8.1.17 (2) on every $(TX)^i$, `Chapter8/TangentZhouLevels.lean` with Part I's `Chapter2/TangentLiftAll.lean`.

**V8.44 (Proposition 8.1.16, (ZHOU) for $TX$;** `Polyfold.MPolyfold.levelVal_tangentProj_mem_smoothSet, Polyfold.EpGroupoid.mem_nonEffectivePart_tangent, Polyfold.EpGroupoid.TangentZhouReflect, Polyfold.EpGroupoid.proposition_8_1_16_2021, Polyfold.EpGroupoid.tangentMor_trivial, Polyfold.EpGroupoid.tangentZhouReflect_trivial, Polyfold.EpGroupoid.Examples.proposition_8_1_16_`
A smooth point of $TX$ lies over a smooth point of $X$; if $s = t$ near the base $p(v)$ of $v \in T\mathbf{X}$, then $v \in (T\mathbf{X})^T$. If $X$ satisfies (ZHOU) (Definition 7.4.5, V7.88) and `TangentZhouReflect` holds — for a smooth $b \in TX$ and $(g, h) \in G_b$, if the tangent of $(g, h)$ in the ep-groupoid $TX$ is the identity on $T_b(TX)$, then $Tg$ is the identity on $T_{p(b)}X$ — then $TX$ satisfies (ZHOU). The hypotheses hold for the trivial ep-groupoid of a point (`TangentZhouReflect` for every trivial ep-groupoid). *Book:* Proposition 8.1.16, p. 369, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (narrower):* the step "$T(g, h) = \mathrm{id}$ implies $Tg = \mathrm{id}$", which the text obtains by differentiating (8.16) and using

that $T\tau$ is onto, is the hypothesis `TangentZhouReflect`: the second tangent bundle $T(TX)$ and $T\tau$ are not formalized (gap G.88, open for this step; its Proposition 8.1.17 (2) part is closed, V8.52); as in Chapter 7, tangents of isotropy elements are set maps, no linear structure is used (Findings, item 115).

**V8.45 (Proposition 8.1.17, LUC for $TX$; `Polyfold.EpGroupoid.nonEffectivePart_tangent_eq`, `Polyfold.EpGroupoid.proposition_8_1_17_2021_one`, `Polyfold.EpGroupoid.proposition_8_1_17_2021_two` `Polyfold.EpGroupoid.isOpenMap_tangentProj_of_subsingleton`, `Polyfold.EpGroupoid.Examples.proposit`** Assume that every $x \in X_1$ has a natural representation with $U(x) \cap X_1$ connected in $X^1$ and dense in $U(x)$, and that $p : T\mathbf{X} \to \mathbf{X}^1$ is open. (1) If $X^1$ has LUC, then $(T\mathbf{X})^T = p^{-1}((\mathbf{X}^1)^T)$ and $TX$ has LUC. (2) If $X$ has (SLUC), $TX$ has LUC. The hypotheses hold for the point ep-groupoid. *Book:* Proposition 8.1.17, p. 371, new in the 2021 edition: (1) as stated; (2) (SLUC) for $X$ gives (SLUC) for $TX$. *Hidden hypotheses:* the connectedness and density of $U(x) \cap X_1$ (as in Appendix 7.6.4, V7.83; the density is gap G.73), and the openness of $p$ (Part I, gap G.8). *Deviation (narrower):* (1) is proved through Proposition 7.2.21 for $TX$; (2) here only for $(TX)^0 = TX$; for every $(TX)^i$, through $T(X^i) \cong (TX)^i$ as ep-groupoids, it is V8.52 (gap G.88, closed for Proposition 8.1.17 (2)) (Findings, items 116, 124).

**V8.46 (Definition 8.3.5, auxiliary norms; `Polyfold.EpStrongBundle.IsAuxNorm`, `Polyfold.EpStrongBundle.ex` `Polyfold.EpStrongBundle.exists_isAuxNorm_trivial`, `Polyfold.TrivialBundle.exists_isAuxNorm_epBund`** For a strong bundle $(P : W \to X, \mu)$ over an ep-groupoid, an auxiliary norm is a continuous $N : W_{0,1} \to \mathbb{R}$ which (1) is a complete norm on every fibre, (2) has $h_k \to 0_x$ in $W_{0,1}$ whenever $P(h_k) \to x$ and $N(h_k) \to 0$ — (1), (2) and continuity are Part I's auxiliary norm — and (3) satisfies $N(h) = N(k)$ whenever $h = \mu(\varphi, k)$, $s(\varphi) = P(k)$. Condition (3) is not vacuous: $\mu(\varphi, k) \in W_{0,1}$ for $k \in W_{0,1}$. Over the trivial ep-groupoid every auxiliary norm of Part I is one, and $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ has one. *Book:* Definition 8.3.5, p. 379, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (restated):* values in $[0, \infty)$ follow from (1); in (3), $h$ is given in $W_{0,1}$ and $h = \mu(\varphi, k)$ is an equation in $W$ (Findings, item 117).

**V8.47 (Proposition 8.4.1, (2) implies (1); `Polyfold.EpStrongBundle._fiberZero`, `Polyfold.EpStrongBundle.ze` `Polyfold.EpStrongBundle.continuous_zeroOrbitMap`, `Polyfold.EpStrongBundle.leftInverse_zeroOrbitMa` `Polyfold.EpStrongBundle.isEmbedding_zeroOrbitMap`, `Polyfold.EpStrongBundle.proposition_8_4_1_2021`** For a strong bundle over an ep-groupoid, $\mu(\varphi, 0_{s(\varphi)}) = 0_{t(\varphi)}$, the zero section induces a continuous $|0| : |X| \to |W[0]|$ with left inverse $|P|$, a topological embedding, and if $|W[0]|$ is paracompact so is $|X|$. *Book:* Proposition 8.4.1, p. 380, new in the 2021 edition: $|X|$, $|W[0]|$ and $|W[1]|$ are paracompact together. *Hidden hypotheses:* none. *Deviation (narrower):* only (2) implies (1); $W[1]$ is not built as an ep-groupoid, and (1) implies (2), (3), which needs $G_x$-equivariant bundle charts with invariant metrics, is not formalized (gap G.86). *Proof:* $|W[0]|$ is metrizable (Theorem 7.3.1 and Smirnov's theorem), so is its subspace $|X|$, hence paracompact, instead of the closed-retract argument (Findings, item 118).

**V8.48 (Definition 8.4.2, the local regularity property of a strong bundle; `Polyfold.EpStrongBundle.natLift`, `Polyfold.EpStrongBundle.IsLR`, `Polyfold.EpStrongBundle.isLR_trivial`).** The natural lift of the $G_x$-action to $W|U(x)$ is $g * w = \mu(\Gamma(g, P(w)), w)$. A strong bundle has the local regularity property if inside every open $O \ni x$ there is a natural representation $U(x)$ such that every strong bundle map $\Phi : W \to W$, covering some $\beta$, which restricts to a strong bundle isomorphism of $W|U(x)$ and satisfies $|\Phi(w)| = |w|$ for $P(w) \in U(x)$, is $w \mapsto g * w$ on $W|U(x)$ for some $g \in G_x$. Every strong bundle over a trivial ep-groupoid has it. *Book:* Definition 8.4.2, p. 382, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (restated):* "strong bundle isomorphism

$B : W|U(x) \to W|U(x)$" is read as Part I's `IsStrongBundleIsoOn` for a strong bundle map $W \to W$, as for Definition 8.5.9 (V8.37), and $|B(w)| = |w|$ as equality of orbits in $(W, \mathbf{W})$. Proposition 8.4.3 (effective $X$ with (LR) gives the local regularity property) is not formalized (gap G.86) (Findings, item 119).

**V8.49 (the remark after Definition 8.5.5 and Example 8.5.6; `Polyfold.IsCoveringFunctor.injective_map_end`, `Polyfold.Example856D`, `Polyfold.example856Functor`, `Polyfold.example_8_5_6_2021`).** For a covering functor $f : D \to C$ and $d \in D$, $f : G_d \to G_x$ is injective. For $D$ with two objects $a, b$ and one morphism between any two objects, $C = \{1_x, \varepsilon\}$ and $f(\varphi) = f(\varphi^{-1}) = \varepsilon$: $f$ is a covering functor, $G_a$, $G_b$ are trivial, $G_x \cong \mathbb{Z}/2$ is not, and $f : G_d \to G_x$ is injective and not surjective. *Book:* the remark after Definition 8.5.5 and Example 8.5.6, p. 388 (Definition 8.5.5: 2017 Definition 8.9; Example 8.5.6 new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (same):* none; $C$ is $\mathbb{Z}^\times$ as a one-object category, $\varepsilon = -1$ (Findings, item 120).

**V8.50 (Definition 8.5.9 in the 2021 form; `Polyfold.EpStrongBundle.IsProperStrongBundleCovering2021`, `Polyfold.EpStrongBundle.IsProperStrongBundleCovering.isProperStrongBundleCovering2021`, `Polyfold.EpStrongBundle.isProperStrongBundleCovering2021_iff`, `Polyfold.EpStrongBundle.isProperSt` `Polyfold.TrivialBundle.epBundle_isProperStrongBundleCovering2021`).** For strong bundles $(W, \mu)$ over $Y$ and $(V, \tau)$ over $X$ and an ep-functor $F : Y \to X$, a map $A : W \to V$ is a proper strong bundle covering functor if it is a strong bundle map with $Q \circ A = F \circ P$ satisfying (0) $\tau(F(\psi), A(w)) = A(\mu(\psi, w))$, and (1)–(3) of V8.37. This holds iff $A$ is the object map of a unique strong bundle functor which is a proper strong bundle covering functor in the sense of V8.37. The identity of every strong bundle over an ep-groupoid is one. *Book:* Definition 8.5.9, p. 394 (2017: 8.10, p. 285), with (0) new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (restated):* stated for the object map, (0) being the strong bundle map property of Definition 8.3.2 (2017: 8.5); V8.37 had read 2017's "strong bundle map $A$" as a strong bundle functor, which contains (0), and the two readings are equivalent (Findings, items 70, 121).

**V8.51 (Lemma 8.5.10; `Polyfold.EpStrongBundle.pullback` , `Polyfold.EpStrongBundle.IsProperStrongBundl` `Polyfold.EpStrongBundle.IsProperStrongBundleCovering2021.bundleB_val`, `Polyfold.EpStrongBundle.Is` `Polyfold.EpStrongBundle.IsProperStrongBundleCovering2021.eq_of_isStrongBundleIsoOn`, `Polyfold.EpStrongBundle.IsProperStrongBundleCovering2021.bijective_bundleB`, `Polyfold.EpStrongBur` `Polyfold.EpStrongBundle.IsProperStrongBundleCovering2021.bundleBInv_eq_of`, `Polyfold.EpStrongBunc` `Polyfold.EpStrongBundle.IsProperStrongBundleCovering2021.mapsTo_oneSet_bundleBInv`, `Polyfold.EpStrongBundle.IsProperStrongBundleCovering2021.isStrongBundleMap_bundleBInv`, `Polyfold.EpStrongBundle.IsProperStrongBundleCovering2021.lemma_8_5_10_2021`).** For a proper strong bundle covering functor $A : W \to V$ covering $F : Y \to X$ (V8.50), on the pull-back $F^*V = \{(y, v) \mid F(y) = Q(v)\}$ with $\mu_{F^*V}(\varphi, (y, v)) = (t(\varphi), \mu_X(F(\varphi), v))$, the map $B(w) = (P(w), A(w))$ is a strong bundle isomorphism $W \to F^*V$ covering the identity of $Y$, with inverse $B^{-1}$, and $B(\mu_Y(\varphi, w)) = \mu_{F^*V}(\varphi, B(w))$ (8.23); $A = \bar{A} \circ B$ with $\bar{A}(y, v) = v$. *Book:* Lemma 8.5.10, p. 395, new in the 2021 edition. *Hidden hypotheses:* none (the paracompactness of $F^*V$, a hypothesis of Part I's pull-back, always holds). *Deviation (narrower):* "strong bundle isomorphism over the ep-groupoid $Y$" is read as a strong bundle isomorphism $W \to F^*V$ together with (8.23); that $\mu_{F^*V}$ makes $F^*V$ a strong bundle over the ep-groupoid $Y$ is not proved, only its formula is used (gap G.89). *Proof:* $B^{-1}$ is near every point $(y, v) \mapsto \Psi(v)$ for the inverse $\Psi$ of a local strong bundle isomorphism of Definition 8.5.9 (1) (Findings, item 122).

**V8.52 (Proposition 8.1.17 (2) for every $(TX)^i$; `Polyfold.MPolyfold.ScEquiv.toHomeomorph`, `Polyfold.EP.tangentEP`, `Polyfold.EpGroupoid.EpEquiv`, `Polyfold.EpGroupoid.EpEquiv.refl`,**

`Polyfold.EpGroupoid.EpEquiv.trans`, `Polyfold.EpGroupoid.EpEquiv.shift`, `Polyfold.EpGroupoid.EpEquiv`
`Polyfold.EpGroupoid.EpEquiv.isLUC`, `Polyfold.EpGroupoid.tangentShiftEpEquiv`, `Polyfold.EP.tangentLi`
`Polyfold.EP.isLUC_tangent_lift`, `Polyfold.EP.proposition_8_1_17_two_all_2021`,
`Polyfold.EpGroupoid.hasConnectedDenseLevelOneReps_of_subsingleton`, `Polyfold.EpGroupoid.Examples.`
`Polyfold.BundledMPolyfold.tangentLiftEquiv`, `Polyfold.proposition_2_3_17_all_2021`**).** An equivalence of ep-groupoids, here, is a pair of sc-diffeomorphisms of the object and the morphism M-polyfolds commuting with source and target; it maps $\mathbf{X}^T$ onto $\mathbf{X}'^T$ and so transports LUC, by Proposition 7.2.21, and it passes to the index raises. Part I's Proposition 2.3.17 for $i = 1$, on objects and morphisms, gives an equivalence $T(X^1) \cong (TX)^1$, and by recursion $T(X^i) \cong (TX)^i$ for every $i$. So if $X$ has (SLUC), every $X^i$ satisfies the hypothesis of V8.45 (natural representations with $U(x) \cap (X^i)_1$ connected in $(X^i)^1$ and dense in $U(x)$), and every $p : T(\mathbf{X}^i) \to (\mathbf{X}^i)^1$ is open, then Proposition 8.1.17 (1) for $X^i$ gives LUC for $T(X^i)$, hence for $(TX)^i$: $TX$ has (SLUC). The point ep-groupoid satisfies the hypotheses. *Book:* Proposition 8.1.17 (2), p. 371, new in the 2021 edition. *Hidden hypotheses:* those of (1) on every level: the connectedness and density of $U(x) \cap (X^i)_1$ for every $X^i$ (gap G.73) and the openness of every $p$ (Part I, gap G.8). *Deviation (explicit):* "naturally $(TX)^i = T(X^i)$" is used through source and target only; the compatibility of the equivalence with unit, inverse and composition is not formalized, since LUC does not need it. This closes gap G.88 for Proposition 8.1.17 (2); the step of Proposition 8.1.16 stays open (V8.44) (Findings, item 124). *Proof:* the text's.

**What is not verified in this chapter**

- pp. 372–373 (2017 pp. 268–270), Definitions 8.2.1 and 8.2.2 (2017: 8.2 and 8.3), sc-differential forms on ep-groupoids, the direct limit $\Omega^*_{ep,\infty}(X)$ and the sc-de Rham complex: not formalized. They need the Whitney sums $\oplus^k TX$, the linear structure of $T_xX$ (gap G.7), linear $T\varphi$ (gap G.70), the levels $X^i$ for $i \geq 2$ (gap G.6, closed since: V2.93, V2.97) and the sc-differential forms, the Lie bracket and the exterior derivative of Part I's §4.4, which are not formalized either (gaps G.37–G.39; the Lie bracket on an M-polyfold is formalized since, V4.76, gap G.38 closed) (gap G.74; Findings, item 110).
- p. 359 (2017 p. 259), Lemma 8.1.2 (2017: 8.1) as printed, $T(\mathbf{X}_s\times_t \mathbf{X}) = T\mathbf{X}_{Ts}\times_{Tt} T\mathbf{X}$ and $m_{T\mathbf{X}} = Tm$; also used in the proof of Lemma 8.1.7 (2017: 8.6) (p. 362 (2017 p. 262)). The sheet form is verified (V8.4, V8.9) (gap G.75; Findings, item 111).
- pp. 364 ff. (2017 pp. 264–265), Remark 8.1.10 (2017: 8.1): that $T\varphi$ and $t_T \circ (s_T|T_xX)^{-1}$ are linear isomorphisms, and that $T\varphi$ is the tangent of $\hat{\varphi}$ at $x$ (V8.12) (gaps G.7, G.70; Findings, item 112); the M-polyfold structure transported along $G$ and the description by triples $(a, T\varphi, b)$, not formalized and not given a gap, since nothing depends on them (Findings, item 113).
- 2017 p. 268, the proof of Theorem 8.1.15 (2017: 8.4): the sc-isomorphism of $TU(x_z)$ with a local bundle model $TO \to O^1$ and the linearity of the bundle action of $G_{x_z}$; not used, since the proof here does not need them (V8.18) (gap G.7; Findings, items 21, 113).
- p. 385 (2017 p. 277), Theorem 8.5.3 (5) (2017: 8.5) as printed, with $t$ on the right arrow: it does not hold as we read it; the square with $s$ and the companion square with $t$ are verified (V8.30) (Findings, item 54).
- p. 390 (2017 p. 281), Proposition 8.5.7 (2017: 8.3) for $D = \emptyset$: the functor is then not surjective on objects unless $U(x) = \emptyset$; the proposition is verified for nonempty $D$ (V8.35) (Findings, item 59).
- pp. 379–385 (2017 pp. 275–276), the degeneracy remark for non-tame $X, Y$: only $d_{\mathbf{X}\times Y} \leq 2\, d_{\mathbf{Y}}$ (V8.28) (Findings, item 51).
- Witnesses: every hypothesis of the chapter is satisfiable (Findings, item 105), but the proper

covering functors constructed have fibres $F^{-1}(x)$ with one point: no proper covering functor with $\#F^{-1}(x) > 1$, and no proper strong bundle covering functor other than identities, is constructed (Findings, items 101, 102).

- Gaps taken up: G.70 in part, the well-definedness and functoriality of $T\varphi$ being now formalized (V8.12); G.7 remains open (Findings, items 112, 114).
- p. 369, Proposition 8.1.16: the step "$T(g,h) = \mathrm{id}$ implies $Tg = \mathrm{id}$" (differentiating (8.16), $T\tau$ onto) is the hypothesis `TangentZhouReflect` (V8.44) (gap G.88, open for this step; Findings, item 115). Proposition 8.1.17 (2) for every $(TX)^i$ is verified (V8.52; G.88 closed for it); of "naturally $(TX)^i = T(X^i)$" only the sc-diffeomorphisms of objects and morphisms commuting with source and target are formalized, not the compatibility with unit, inverse and composition, which (LUC) does not use, nor the linearity on the fibres (gap G.7) (Findings, items 116, 124).
- p. 373, Proposition 8.2.3 (Poincaré lemma for sc-differential forms on ep-groupoids): not formalized, with the forms of §8.2 (gaps G.87, G.74; Findings, item 123).
- p. 380, Proposition 8.4.1, (1) implies (2) and (3), and the ep-groupoid $W[1]$; p. 382, Proposition 8.4.3: they need $G_x$-equivariant bundle charts with invariant metrics and the openness of $P : W \to X$ (V8.47, V8.48) (gap G.86; Findings, items 118, 119).
- p. 395, Lemma 8.5.10: that $F^*V$ with $\mu_{F^*V}$ is a strong bundle over the ep-groupoid $Y$; only the formula of $\mu_{F^*V}$ is used (V8.51) (gap G.89; Findings, item 122).

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## 10. Chapter 9 (HWZ, Part II, §§9.1–9.5 and §9.6.1 (2017 §§9.1–9.5)): verified for §§9.1–9.4 and §9.6.1 with the readings and hypotheses recorded below, unconditionally for ep-groupoids whose object M-polyfold is $\mathbb{R}^m$, except the global part of Proposition 9.1.11, Proposition 9.1.9, Lemma 9.4.3 and the boundary-orientation geometry of Definitions 9.3.8 and 9.3.10; for §9.5 the sets of measure zero, the σ-algebras and the isotropy factors are verified, branched integration, the boundary measures and Stokes' theorem are not

Chapter 9 (pp. 401–458 (2017 pp. 291–332); numbers and pages are those of the book (Springer 2021), with the 2017 text's in parentheses) treats branched ep$^+$-subgroupoids, the objects that carry the solution sets of perturbed sc-Fredholm problems with symmetries. §9.1 defines them as functors $\Theta : X \to \mathbb{Q}^+$ that are locally weighted sums of indicator functions of properly embedded finite-dimensional submanifolds (Definition 9.1.2 (2017: 9.1)), proves that $|\mathrm{supp}(\Theta)|$ is locally closed (Lemma 9.1.5 (2017: 9.1)), constructs symmetric local branching structures (Lemma 9.1.7 (2017: 9.2)), defines closed, compact, manifold- and orbifold-type $\Theta$ (Definitions 9.1.6, 9.1.8 (2017: 9.2, 9.3)), and decomposes $\Theta$ by dimension (Propositions 9.1.11 (2017: 9.2) and 9.1.12, the latter new in the 2021 edition). §9.2 introduces the category $\mathrm{Gr}(X)$ of formal sums of tangent spaces and the tangent section $T_\Theta$ (Theorem 9.2.1 (2017: 9.1)), good and bad points (Definitions 9.2.2, 9.2.6, Lemmas 9.2.3–9.2.7 (2017: 9.4, 9.5, 9.3–9.5)), the graph representation near a bad point (Proposition 9.2.8, Lemmas 9.2.9–9.2.11 (2017: 9.3, 9.6–9.8)), the tangent $T\Theta$ (Definition 9.2.12 (2017: 9.6)), the position of a submanifold relative to $\partial X$ (Lemma 9.2.13, Remark 9.2.14 (2017: 9.9, 9.5)) and the boundary $\partial\Theta$ (Proposition 9.2.15, Definition 9.2.16, Remark 9.2.17 (2017: 9.4, 9.7, 9.6)). §9.3 defines orientations (the category $\widehat{\mathrm{Gr}}(X)$, Definition 9.3.2 (2017: 9.8), Remark 9.3.3, Lemma 9.3.4, Example 9.3.5 (2017: 9.8, 9.10, 9.1)), tame $\Theta$ (Definition 9.3.6 (2017: 9.10), Lemma 9.3.7, new) and boundary orientations ((9.11), Definition 9.3.8, Proposition 9.3.9, Definition 9.3.10 (2017: 9.9, —, 9.11)). §9.4 studies the geometry of local branching structures: essential points, sets of measure zero and the decomposition of an oriented structure (Definitions 9.4.1, 9.4.5, 9.4.7, Lemmas 9.4.2–9.4.8, Proposi-

tions 9.4.9, 9.4.10 (2017: Definitions 9.12–9.14, Lemmas 9.11–9.15, Propositions 9.5, 9.6)). §9.5 integrates sc-differential forms over branched ep⁺-subgroupoids and proves Stokes' theorem; verified here are its measure-theoretic part (Definitions 9.5.1, 9.5.3, 9.5.14, Lemma 9.5.2 (2017: Definitions 9.15–9.17, Lemma 9.16)) and the isotropy and set parts of Lemmas 9.5.9–9.5.11 and Proposition 9.5.13 (2017: Lemmas 9.18, 9.19; Lemma 9.5.10 and Proposition 9.5.13 new in the 2021 edition), not the integrals. Appendix §9.6.1 (new in the 2021 edition) proves Proposition 9.1.12 (Lemmas 9.6.1, 9.6.2, 9.6.4, Definition 9.6.3, Proposition 9.6.5); §§9.6.2–9.6.3 discuss open questions and have no formal content.

What is verified. Every definition of §§9.1–9.4, of the measure-theoretic part of §9.5 and of §9.6.1 is formalized: Definitions 9.1.2 (V9.1), 9.1.6 (V9.6), 9.1.8 (V9.8), 9.2.2 (V9.15), 9.2.6 (V9.19), 9.2.12 (V9.23), 9.2.16 (V9.27), 9.3.2 (V9.30), 9.3.6 (V9.34), 9.3.8 (V9.36), 9.3.10 (V9.38), 9.4.1 (V9.39), 9.4.5 (V9.43), 9.4.7 (V9.45), 9.5.1 (V9.49), 9.5.3 (V9.51), 9.5.14 (V9.52) and 9.6.3 (V9.60), with Property (T) (V9.41). **Theorem 9.2.1 (2017: 9.1) holds** for every branched ep⁺-subgroupoid of an ep-groupoid whose object M-polyfold is $\mathbb{R}^m$, and in general under the graph-chart hypotheses below (V9.13, V9.14); so do Propositions 9.1.12, 9.2.15, 9.3.9, 9.4.9, 9.4.10 and 9.6.5 in the forms stated below, Lemmas 9.1.5, 9.1.7, 9.2.3, 9.2.5, 9.2.7, 9.2.13 (read at points of $\partial X$), 9.3.4 (for $n$-dimensional $\Theta$), 9.3.7, 9.4.2, 9.4.4 (for sheets of one dimension), 9.4.6, 9.4.8, 9.6.1, 9.6.2 and 9.6.4, Remarks 9.1.3 (b), 9.2.4, 9.2.14, 9.2.17 and 9.3.3, and Example 9.3.5 with the corrections recorded in V9.33. In §9.5, Lemma 9.5.2 (2017: 9.16) is verified for structures with sheets of one class (V9.50), the remark after Definition 9.5.3 as the completion by null sets (V9.51), and the parts of Lemmas 9.5.9–9.5.11 and Proposition 9.5.13 that do not involve integrals: the orders of the isotropy groups, (9.23) and the decompositions of the preimages (V9.54–V9.57). Several places of the text needed more than a reading. (a) *Proposition 9.1.11 (2017: 9.2).* The uniqueness of the dimensional decomposition, as stated, does not hold as we read it: on the trivial ep-groupoid of $\mathbb{R}^2$ the indicator of the $x$-axis has two decompositions (`Polyfold.EpGroupoid.not_proposition_9_1_11_rn`); the local construction is verified, the global gluing is not formalized (V9.9). (b) *Lemma 9.2.13 (2017: 9.9).* As printed, (2) □ (1) does not hold as we read it at interior points (`Polyfold.MPolyfold.not_lemma_9_2_13_two_one`); the lemma is verified for $x \in M \cap \partial X$, the case the text uses, together with $d_M(x) = 0 \Leftrightarrow T_xM \subseteq T^R_xX$ at every $x \in M$ (V9.24). (c) *Definition 9.2.16 (2017: 9.7).* (1) and (2) do not characterize $\partial\Theta$ uniquely as we read them (the zero functor satisfies them; `Polyfold.not_unique_isBoundaryCharacterized_halfLine`); uniqueness holds among functors vanishing off $\mathrm{supp}(\Theta)$ (V9.27). (d) *Lemma 9.3.4 (2017: 9.10)* does not hold as we read it without equidimensionality (`Polyfold.lemma_9_3_4_mixed_dim_rn`); it is verified for $n$-dimensional $\Theta$ (V9.32). (e) *Example 9.3.5 (2017: 9.1).* The two orientations also coincide on $\{x < 0\}$, and there are at least five orientations with weights 1, not four (V9.33). (f) *Lemma 9.4.4 (2017: 9.13)* does not hold as we read it for sheets of different dimensions (`Polyfold.lemma_9_4_4_mixed_dim`) (V9.42), and neither does *Lemma 9.5.2 (2017: 9.16)*: two sheets of different dimensions do not induce the same null sets on their overlap (`Polyfold.isNull_transfer_mixed_dim`) (V9.50). (g) *Omitted steps.* The step of Lemma 9.6.4 "similarly as good points" and the independence of the formula on p. 457 from the structure (V9.61, V9.62), the step "the image is $M$ near 0" in the converse of Lemma 9.2.13 (V9.24), and the facts on the $TM_i$ needed for Definition 9.2.12 (V9.23) are proved or, where Part I lacks the tools, made explicit hypotheses. (h) *Other routes.* The "measure zero" arguments of Lemmas 9.6.1 and 9.6.2 are replaced by a topological covering lemma (V9.58, V9.59); cones in Proposition 9.2.8 are replaced by rays and slices (V9.22); Proposition 9.4.10 is proved without Property (T) (V9.48). Not verified: the global part of Proposition 9.1.11, Remark 9.1.4, Proposition 9.1.9 and Remark 9.1.10, Lemma 9.4.3, the identification $T_y\partial M = T^R_yM$ with the outward direction and the continuity of $o_{\partial M}$, the classification of all orientations in Example

9.3.5, and in §9.5 the forms, integrals and measures: Theorems 9.5.4, 9.5.5 and 9.5.15–9.5.18 with the measure identities of Lemmas 9.5.7, 9.5.9–9.5.11 and Proposition 9.5.13.

The hypotheses. Part I has no level-0 sc-calculus inside partial quadrants and no linear structure on $T_xX$ (gap G.7), so four facts of the text's §9.2 are taken as hypotheses on a general M-polyfold, each proved for $\mathbb{R}^m$: the graph representation of the $M_i$ near a point (`Polyfold.MPolyfold.HasGraphChartAt`, `Polyfold.MPolyfold.HasGraphChartsAt`, `Polyfold.MPolyfold.HasGraphBallAt`; Lemmas 9.2.9–9.2.11), the local constancy of the dimension (`Polyfold.MPolyfold.HasLocallyConstantDimAt`; Remark 9.1.1), invariance of domain (`Polyfold.MPolyfold.HasDomainInvarianceAt`; Lemma 9.2.7 (2)) and the level-1 continuity of frame fields (`Polyfold.MPolyfold.HasContinuousLevelOneOn`; Lemma 9.3.4). Their witnesses are `Polyfold.MPolyfold.hasGraphChartAt_rn`, `Polyfold.MPolyfold.hasGraphChartsAt_rn`, `Polyfold.hasGraphBallAt_rn`, `Polyfold.MPolyfold.hasLocallyConstantDimAt_rn`, `Polyfold.MPolyfold.has` and `Polyfold.hasContinuousLevelOneOn_rn`. On $\mathbb{R}^m$ the results that use them are proved without hypotheses: `Polyfold.EpGroupoid.proposition_9_1_12_rn`, `Polyfold.EpGroupoid.hasIndependentTange` `Polyfold.EpGroupoid.lemma_9_2_7_2_rn`, `Polyfold.EpGroupoid.lemma_9_3_4_rn`. Definition 9.2.12 keeps, also on $\mathbb{R}^m$, the hypothesis that the $TM_i$ are smooth submanifolds of $TX$ (gap G.99).

The setting. Ep-groupoids, natural representations of the isotropy groups, the orbit space and paracompact orbit spaces are those of Chapter 7 (`Polyfold.EpGroupoid`, `Polyfold.EpGroupoid.NaturalRep`, `Polyfold.EpGroupoid.HasParacompactOrbitSpace`); the tangent ep-groupoid is Chapter 8's (`Polyfold.EpGroupoid.tangentEpGroupoid`). Smooth finite-dimensional submanifolds (images of sc⁺-retractions), the smooth set, the boundary and the degeneracy index are Part I's (`Polyfold.MPolyfold.IsSmoothSubmanifold`, `Polyfold.MPolyfold.smoothSet`, `Polyfold.MPolyfold.boundary`, `Polyfold.MPolyfold.degIndex`). A functor $\Theta : X \to \mathbb{Q}^+$ is a function on objects with $\Theta(s(\varphi)) = \Theta(t(\varphi))$ (`Polyfold.EpGroupoid.IsFunctorial`), with values in $\mathbb{Q}$ and nonnegativity a separate condition. The tangent space $T_yM$ of a submanifold is a set of tangent vectors (`Polyfold.MPolyfold.submanifoldTangent`); linear algebra is done in charts (`Polyfold.MPolyfold.chartTangent`). Witnesses on $\mathbb{R}^m$ are ep-groupoids with object M-polyfold Part I's $\mathbb{R}^m$ (`Polyfold.rnPolyfold`), most often the trivial one (`Polyfold.EpGroupoid.trivial`). Findings referred to below are those of `Exposition/Findings.md`, "Chapter 9".

**V9.1 (Definition 9.1.2 (2017: 9.1), branched ep⁺-subgroupoids; `Polyfold.EpGroupoid.IsFunctorial`, `Polyfold.EpGroupoid.supp`, `Polyfold.EpGroupoid.orbitSupp`, `Polyfold.EpGroupoid.LocalBranchingStruct` `Polyfold.EpGroupoid.IsBranchedEpSubgroupoid`, `Polyfold.EpGroupoid.isBranchedEpSubgroupoid_zero`, `Polyfold.EpGroupoid.localBranchingStructure_indicator`, `Polyfold.EpGroupoid.isBranchedEpSubgroupo` `Polyfold.rnPolyfold_isSmoothSubmanifold_singleton`).** A branched ep⁺-subgroupoid is a functor $\Theta \geq 0$ whose support lies in the smooth set, whose orbit support $|\mathrm{supp}(\Theta)|$ is paracompact, and which at every $x \in \mathrm{supp}(\Theta)$ has a local branching structure: an open $U(x)$, finitely many smooth submanifolds $M_i \subseteq U(x)$ through $x$ with proper inclusions, weights $\sigma_i > 0$, and $\Theta(y) = \sum_{y \in M_i} \sigma_i$ on $U(x)$. The zero functor and the indicator of a smooth point of $\mathbb{R}^n$ are examples. *Book:* Definition 9.1.2, p. 402 (2017: Definition 9.1, p. 292). *Hidden hypotheses:* none. *Deviation (restated):* $\Theta$ takes values in $\mathbb{Q}$ with $\Theta \geq 0$ a field; "the topology induced on $|\mathrm{supp}(\Theta)|$ is paracompact" is a `ParacompactSpace` instance on the subtype; "the inclusion $M_i \to U(x)$ is proper" is `IsProperMap` of the inclusion. The 2017 text's "(ii) and (iii)" in (4) is "(2) and (3)" in 2021 (possible misprint, corrected) (Findings, item 1).

**V9.2 (Remark 9.1.1 (2017: 9.1) and the last paragraph of Definition 9.1.2, the dimension of a submanifold and $n$-dimensional $\Theta$; `Polyfold.MPolyfold.HasDimAt`, `Polyfold.MPolyfold.finrank_tangentSpace`**

`Polyfold.MPolyfold.HasDimAt.finrank_eq`, `Polyfold.MPolyfold.HasDimAt.unique`, `Polyfold.MPolyfold.Is`
`Polyfold.MPolyfold.hasDimAt_inter_isOpen_iff`, `Polyfold.MPolyfold.hasDimAt_image_iff`,
`Polyfold.MPolyfold.IsSubmanifoldOfDim`, `Polyfold.EpGroupoid.LocalBranchingStructure.IsOfDim`,
`Polyfold.EpGroupoid.LocalBranchingStructure.dim_unique`, `Polyfold.EpGroupoid.IsNDimensional`,
`Polyfold.EpGroupoid.isNDimensional_zero`, `Polyfold.MPolyfold.HasLocallyConstantDimAt`,
`Polyfold.MPolyfold.hasLocallyConstantDimAt_rn`). A submanifold has dimension $k$ at $a$ if the tangent space of its sub-retract at the chart image of $a$ has dimension $k$; the value does not depend on the data and the chart, is unique, exists at every point, and is invariant under restriction to open sets and under sc-diffeomorphisms. $\Theta$ is $n$-dimensional if at every point of its support some local branching structure has only $n$-dimensional $M_i$; the zero functor is $n$-dimensional for every $n$. The dimensions of the $M_i$ of one structure through $x$ are determined. *Book:* Remark 9.1.1, p. 402 (2017: Remark 9.1, p. 291); Definition 9.1.2, last paragraph and footnote, p. 402. *Hidden hypotheses:* none. *Deviation (narrower):* the dimension is measured in a chart, since $T_aA$ carries no linear structure (gap G.7). The local constancy of $a \mapsto \dim T_aA$ is not proved in general; it is the hypothesis `HasLocallyConstantDimAt` where it is needed, proved on $\mathbb{R}^m$ from the trace of $Ds(y)$, and elsewhere the dimensions of the $M_i$ enter as labellings $\dim : I \to \mathbb{N}$. The footnote ($n$-dimensional $\Theta \neq 0$ is not $m$-dimensional, $m \neq n$) and "locally equidimensional" are not formalized (gap G.93) (Findings, items 2, 3).

**V9.3 (branched ep⁺-subgroupoids of ep-groupoids over $\mathbb{R}^n$;** `Polyfold.EpGroupoid.paracompactSpace_orbitSu`
`Polyfold.EpGroupoid.localBranchingStructure_sum_rn`, `Polyfold.EpGroupoid.isBranchedEpSubgroupoid_`
`Polyfold.EpGroupoid.isNDimensional_sum_rn`, `Polyfold.EpGroupoid.localBranchingStructure_indicator`
`Polyfold.EpGroupoid.isBranchedEpSubgroupoid_indicator_rn`, `Polyfold.rnPolyfold_isSmoothSubmanifol`
`Polyfold.rnPolyfold_hasDimAt_range`, `Polyfold.graph2`, `Polyfold.rnPolyfold_isSubmanifoldOfDim_graph`
`Polyfold.EpGroupoid.rnPolyfold_isSubmanifoldOfDim_singleton`). On the trivial ep-groupoid of $\mathbb{R}^n$, finite weighted sums of indicators of closed submanifolds through which the sum is taken, and indicators of locally closed submanifolds, are branched ep⁺-subgroupoids, $n$-dimensional when all submanifolds are. The image of a smooth idempotent $s$ of $\mathbb{R}^n$ is a closed smooth submanifold of dimension rank $Ds(a)$; graphs of smooth $\beta : \mathbb{R} \to \mathbb{R}$ in $\mathbb{R}^2$ are closed 1-dimensional submanifolds; points are 0-dimensional. *Book:* not in the book. *Hidden hypotheses:* none. *Deviation (extra):* witnesses for the hypotheses of the chapter (Findings, item 100).

**V9.4 (Remark 9.1.3 (b) (2017: 9.2 (b)), branching structures are not unique;** `Polyfold.EpGroupoid.Remark913Hyp`
`Polyfold.EpGroupoid.remark913Hyp_expNegInvGlue`, `Polyfold.EpGroupoid.thetaGraphs`,
`Polyfold.EpGroupoid.thetaGraphs_eq`, `Polyfold.EpGroupoid.Remark913Hyp.structure`,
`Polyfold.EpGroupoid.Remark913Hyp.structure`, `Polyfold.EpGroupoid.Remark913Hyp._pos`,
`Polyfold.EpGroupoid.Remark913Hyp.eq_add_of_thetaGraphs_eq`, `Polyfold.EpGroupoid.remark_9_1_3_b`).
For smooth $\beta_1, \beta_2 : \mathbb{R} \to \mathbb{R}$ with the sign conditions of the text (satisfiable, from `expNegInvGlue`), $\Theta(p) = \#\{i \mid p \in \operatorname{graph} \beta_i\}$ is a 1-dimensional branched ep⁺-subgroupoid of the trivial ep-groupoid of $\mathbb{R}^2$, equal to the one given by the graphs of 0 and $\beta_1 + \beta_2$, and no submanifold of the one structure agrees with a submanifold of the other on any neighbourhood of the origin. *Book:* Remark 9.1.3 (b), p. 403 (2017: Remark 9.2 (b), p. 292). *Hidden hypotheses:* none. *Deviation (restated):* the text gives $\gamma_2$ only by $\gamma_2(s) > 0$ for $s \neq 0$; describing the same $\Theta$ forces $\gamma_2 = \beta_1 + \beta_2$; "not unique, even close to $x$" is read as the last conjunct (Findings, item 4).

**V9.5 (Lemma 9.1.5 (2017: 9.1), $|\operatorname{supp}(\Theta)|$ is locally closed;** `Polyfold.EpGroupoid.preimage_ _orbitSupp`,
`Polyfold.EpGroupoid.LocalBranchingStructure.mem_supp_iff`, `Polyfold.EpGroupoid.LocalBranchingStru`
`Polyfold.EpGroupoid.isLocallyClosed_supp`, `Polyfold.EpGroupoid.lemma_9_1_5`). For a branched ep⁺-subgroupoid $\Theta$, $\pi^{-1}(|\operatorname{supp}(\Theta)|) = \operatorname{supp}(\Theta)$, $\operatorname{supp}(\Theta) \cap U(x) = \bigcup_i M_i$ is closed

in $U(x)$, $\mathrm{supp}(\Theta)$ is locally closed in $X$ and $|\mathrm{supp}(\Theta)|$ is locally closed in $|X|$. *Book:* Lemma 9.1.5, p. 403 (2017: Lemma 9.1, p. 293). *Hidden hypotheses:* none. *Deviation (same):* the openness of $\pi$ (Chapter 7) replaces the text's neighbourhood with the properness property (Findings, item 5).

**V9.6 (Definition 9.1.6 (2017: 9.2), closed and compact $\Theta$; `Polyfold.EpGroupoid.IsClosedBranched`, `Polyfold.EpGroupoid.IsCompactBranched`, `Polyfold.EpGroupoid.isCompactBranched_indicator`).** $\Theta$ is closed if $|\mathrm{supp}(\Theta)|$ is closed in $|X|$, compact if it is compact; the indicator of a point of $\mathbb{R}^n$ is compact. *Book:* Definition 9.1.6, p. 404 (2017: Definition 9.2, p. 293). *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 7).

**V9.7 (the restriction of a local branching structure, and Lemma 9.1.7 (2017: 9.2), symmetric structures; `Polyfold.EpGroupoid.LocalBranchingStructure.restrict`, `Polyfold.EpGroupoid.LocalBranchingSt`
`Polyfold.isProperMap_inclusion_iff`, `Polyfold.EpGroupoid.SymmetricBranchingStructure`,
`Polyfold.EpGroupoid.IsFunctorial.apply_Φ`, `Polyfold.EpGroupoid.LocalBranchingStructure.symmetrize`
`Polyfold.EpGroupoid.lemma_9_1_7`, `Polyfold.EpGroupoid.lemma_9_1_7'`).** A local branching structure at $x$ restricts to every open $V \subseteq U(x)$ containing $x$. For a branched ep⁺-subgroupoid and $x \in \mathrm{supp}(\Theta)$ there is, inside every open $V \ni x$, a symmetric local branching structure: the domain of a natural representation $(\Phi, \Gamma)$ of $G_x$, a $G_x$-action on the index set with $\Phi(g)(M_i) = M_{g\cdot i}$ and $\sigma_{g\cdot i} = \sigma_i$, and $\Theta(y) = \sum_{y\in M_i} \sigma_i$ there; it is obtained with $J = I \times G_x$, $M_{(i,g)} = \Phi(g)(M_i \cap U)$ and $\sigma_{(i,g)} = \sigma_i/|G_x|$. *Book:* the remark before Lemma 9.1.7, p. 404, and Lemma 9.1.7, p. 405 (2017: Lemma 9.2, p. 294). *Hidden hypotheses:* none. *Deviation (stronger):* the symmetric structure lies inside any given $V$. The restriction remark is formalized at $x$ itself, not at a point $x_0$ of $V(x_0) \subseteq U(x)$, where the $N_i$ may miss $x_0$ (reading). That $\Phi(g)(M_i \cap U)$ is a submanifold properly embedded in $U$ through $x$, used without statement in the text, is proved (Findings, items 5, 6).

**V9.8 (Definition 9.1.8 (2017: 9.3), manifold- and orbifold-type; `Polyfold.EpGroupoid.IsOrbifoldType`, `Polyfold.EpGroupoid.IsManifoldType`, `Polyfold.EpGroupoid.isManifoldType_indicator`).** $\Theta$ is of orbifold type if it takes only the values 0 and 1, of manifold type if moreover between two objects of $\mathrm{supp}(\Theta)$ there is at most one morphism; the indicator of a point of $\mathbb{R}^n$ is of manifold type. *Book:* Definition 9.1.8, p. 405 (2017: Definition 9.3, p. 294). *Hidden hypotheses:* none. *Deviation (same):* none; Proposition 9.1.9 and Remark 9.1.10, which build on it, are not formalized (gap G.95) (Findings, item 7).

**V9.9 (Proposition 9.1.11 (2017: 9.2), the dimensional decomposition; `Polyfold.EpGroupoid.LocalBranchingStr`
`Polyfold.EpGroupoid.LocalBranchingStructure.dimPartStructure`, `Polyfold.EpGroupoid.LocalBranching`
`Polyfold.EpGroupoid.LocalBranchingStructure.eq_sum_dimPart`, `Polyfold.EpGroupoid.LocalBranchingSt`
`Polyfold.EpGroupoid.LocalBranchingStructure.dimPart_eq_of_isOfDim`, `Polyfold.EpGroupoid.Symmetric`
`Polyfold.EpGroupoid.SymmetricBranchingStructure.dimPart_Φ`, `Polyfold.EpGroupoid.IsDimDecompositi`
`Polyfold.EpGroupoid.IsLocallyFiniteFamily`, `Polyfold.EpGroupoid.Proposition_9_1_11`,
`Polyfold.EpGroupoid.axis2`, `Polyfold.EpGroupoid.decompA`, `Polyfold.EpGroupoid.decompB`,
`Polyfold.EpGroupoid.decompA_isDimDecomposition`, `Polyfold.EpGroupoid.decompB_isDimDecomposition`,
`Polyfold.EpGroupoid.not_proposition_9_1_11_rn`).** For a local branching structure with dimension labelling, $\Theta_k(y) = \sum_{\dim M_i = k,\, y\in M_i} \sigma_i$ is represented on $U(x)$ by the $k$-dimensional structure $(M_i)_{\dim M_i = k}$, $\Theta = \sum_k \Theta_k$ on $U(x)$, and for a symmetric structure $G_x$ preserves dimension and $\Theta_k$ is $G_x$-invariant. The statement of the proposition, "a unique family $(\Theta_k)$ of branched ep⁺-subgroupoids, $\Theta_k$ $k$-dimensional, with $\Theta = \sum_k \Theta_k$ pointwise, and every such family locally finite", is `Proposition_9_1_11`; on the trivial ep-groupoid of $\mathbb{R}^2$ it does not hold as we read it: for the indicator $\Theta$ of the $x$-axis $L$, both $\Theta_0 = 1_{\{0\}}$, $\Theta_1 = 1_{L\setminus\{0\}}$ and $\Theta_0 = 0$, $\Theta_1 = 1_L$ are such

families. *Book:* Proposition 9.1.11, pp. 407–408 (2017: Proposition 9.2, p. 297). *Hidden hypotheses:* the dimension labellings of V9.2. *Deviation (narrower):* only the local construction is verified; the gluing of the local $\Theta_k$ into functors on $X$ (compatibility across morphisms, paracompactness of $|\mathrm{supp}(\Theta_k)|$, local finiteness) and a uniqueness statement under an extra condition (for instance that each $\Theta_k$ be represented near every point by a subfamily of a local branching structure of $\Theta$) are not formalized (gap G.96) (Findings, item 8).

**V9.10 (Proposition 9.1.12, the decomposition does not depend on the structure;** `Polyfold.EpGroupoid.LocalBra` `Polyfold.EpGroupoid.LocalBranchingStructure.DimPartsAgreeNear.exists_naturalRep`, `Polyfold.EpGroupoid.Proposition_9_1_12`, `Polyfold.EpGroupoid.LocalBranchingStructure.dimPartsAgre` `Polyfold.EpGroupoid.LocalBranchingStructure.dimPartsAgreeNear_of_topMultiplicity`, `Polyfold.EpGroupoid.proposition_9_1_12_of_topMultiplicity`, `Polyfold.EpGroupoid.LocalBranchingStr` `Polyfold.EpGroupoid.proposition_9_1_12_of_hasGraphChartAt`, `Polyfold.EpGroupoid.proposition_9_1_` `Polyfold.proposition_9_1_12_rn_zero`, `Polyfold.proposition_9_1_12_rn_one`, `Polyfold.exists_mixed_rn` `Polyfold.EpGroupoid.Remark913Hyp.exists_mixed_rn_two`**).** For two local branching structures of $\Theta$ at $x$ with dimension labellings there is an open $V \ni x$ in both domains on which their dimensional decompositions agree, and $V$ can be taken to be the domain of a natural representation of $G_x$. This holds for equidimensional structures without hypothesis, for every M-polyfold with `TopMultiplicity` in every dimension or with `HasGraphChartAt` at every point, and, without hypothesis, for every ep-groupoid whose object M-polyfold is $\mathbb{R}^m$. Structures with sheets of different dimensions exist on $\mathbb{R}^1$ and $\mathbb{R}^2$. *Book:* Proposition 9.1.12, p. 408, proof in §9.6.1, pp. 452–457, new in the 2021 edition (the 2017 proof of Proposition 9.2 calls the uniqueness obvious). *Hidden hypotheses:* the graph representation `HasGraphChartAt` (Lemmas 9.2.9–9.2.11) on a general M-polyfold. *Deviation (explicit):* "open, $G_x$-invariant, with the usual properties" is any open $V$ in both domains, shrunk to a natural representation by `exists_naturalRep`; the hypothesis is discharged on $\mathbb{R}^m$ (gap G.97 in general) (Findings, items 9, 90–93).

**V9.11 (the category** $\mathrm{Gr}(X)$**, p. 409;** `Polyfold.EpGroupoid.tangentMor_unit_of_eq`, `Polyfold.EpGroupoid.tang` `Polyfold.EpGroupoid.tangentMor_eq_tangentMapOn_hat`, `Polyfold.EpGroupoid.Gr`, `Polyfold.EpGroupoid.g` `Polyfold.EpGroupoid.grPush_mem`, `Polyfold.EpGroupoid.grPush_comp`, `Polyfold.EpGroupoid.grPush_unit`, `Polyfold.EpGroupoid.GrObj`, `Polyfold.EpGroupoid.SmoothCat`, `Polyfold.EpGroupoid.grProj`**).** $\mathrm{Gr}(x)$ is the set of finite formal sums $\sum_L \sigma_L L$, $\sigma_L \in \mathbb{Q}$, of subsets $L$ of $T_xX$; a morphism $\varphi : x \to y$ of $X^\infty$ pushes them forward by the tangent $T\varphi$ of Chapter 7, functorially; the objects $(x, a)$, $a \in \mathrm{Gr}(x)$, and the pairs $(a, \varphi)$ form a category with the projection functor $\pi$ to the full subcategory $X^\infty$ of smooth objects. *Book:* §9.2, p. 409 (2017 pp. 297–298). *Hidden hypotheses:* none. *Deviation (restated):* the generators are subsets of $T_xX$ rather than linear subspaces, since $T_xX$ carries no linear structure (gap G.7); in a chart the tangent spaces of submanifolds are linear subspaces (V9.12) (Findings, item 20).

**V9.12 (the tangent space** $T_yM$ **of a submanifold, set form and chart form;** `Polyfold.MPolyfold.submanifoldTan` `Polyfold.MPolyfold.mem_submanifoldTangent_iff`, `Polyfold.MPolyfold.submanifoldTangent_subset_tang` `Polyfold.MPolyfold.submanifoldTangent_inter_isOpen`, `Polyfold.MPolyfold.submanifoldTangent_image`, `Polyfold.MPolyfold.chartTangent`, `Polyfold.MPolyfold.chartTangent_eq`, `Polyfold.MPolyfold.exists_su` `Polyfold.MPolyfold.finrank_chartTangent`, `Polyfold.MPolyfold.chartTangent_chart_change`, `Polyfold.MPolyfold.exists_linearEquiv_chartTangent`, `Polyfold.EpGroupoid.LocalBranchingStructure.` $T_yM$ is the set of tangent vectors at $y$ fixed by the tangent of the sc$^+$-retraction that presents $M$; it does not depend on the retraction, is local, and is carried by sc-diffeomorphisms. In a chart its image is a linear subspace of the dimension of $M$ at $y$, carried linearly by chart changes; the generators $T_xM_i$ of $T_\Theta(x)$ are linear subspaces in charts. *Book:* §9.2, the tangent spaces $T_xM_i$

before Theorem 9.2.1, p. 409 (2017 p. 298), with Part I's Remark 4.2.2 (preprint: 4.14). *Hidden hypotheses:* none. *Deviation (restated):* sets instead of subspaces (gap G.7) (Findings, item 20).

**V9.13 (Theorem 9.2.1 (2017: 9.1), the tangent section $T_\Theta$; `Polyfold.EpGroupoid.LocalBranchingStructure.tan` `Polyfold.EpGroupoid.LocalBranchingStructure.transport`, `Polyfold.EpGroupoid.LocalBranchingStructu` `Polyfold.EpGroupoid.HasIndependentTangentSums`, `Polyfold.EpGroupoid.tangentValue`, `Polyfold.EpGroupoid.tangentValue_eq`, `Polyfold.EpGroupoid.grPush_tangentValue`, `Polyfold.EpGroupoid.tangentSection`, `Polyfold.EpGroupoid.theorem_9_2_1`).** If the sums $\sum_i \sigma_i\, T_x M_i$ of any two local branching structures of $\Theta$ at $x$ agree (`HasIndependentTangentSums`), there is a functor $T_\Theta : X^\infty \to \mathrm{Gr}(X)$ with $\pi \circ T_\Theta = \mathrm{Id}$, $T_\Theta(x) = \sum_i \sigma_i\, T_x M_i$ for every local branching structure at $x \in \mathrm{supp}(\Theta)$, $T_\Theta(x) = 0$ off the support, and $T_\Theta(\varphi) = (T_\Theta(s(\varphi)), \varphi)$. *Book:* Theorem 9.2.1, p. 409 (2017: Theorem 9.1, p. 298), proved on pp. 412–418 (2017 pp. 301–306). *Hidden hypotheses:* the independence, which the text proves through Proposition 9.2.8 (V9.14). *Deviation (explicit):* the functoriality on morphisms uses the transport of structures along $\hat\varphi$ and $\Theta \circ \hat\varphi = \Theta$ (Findings, items 21, 22).

**V9.14 (Theorem 9.2.1 (2017: 9.1), the independence of the local branching structure; `Polyfold.EpGroupoid.LocalBranchingStructure.tangentSum_eq_of_isGoodPoint`, `Polyfold.EpGroupoid.ha` `Polyfold.EpGroupoid.hasIndependentTangentSums_of_isOfDim`, `Polyfold.EpGroupoid.hasIndependentTang` `Polyfold.EpGroupoid.hasIndependentTangentSums_of_isOfDim_of_hasGraphChartsAt`, `Polyfold.EpGroupoid.hasIndependentTangentSums_of_hasGraphChartAt_of_hasLocallyConstantDimAt`, `Polyfold.EpGroupoid.hasIndependentTangentSums_rn`, `Polyfold.EpGroupoid.Remark913Hyp.tangentSum_st` `Polyfold.hasIndependentTangentSums_indicator`).** The sums are independent at good points without hypothesis; for equidimensional structures under the slice hypothesis `HasTangentSlicesAt`, which follows from `HasGraphChartsAt`; in general under `HasGraphChartAt` and `HasLocallyConstantDimAt` on the support; and for every $\Theta$ on an ep-groupoid whose object M-polyfold is $\mathbb{R}^m$, without hypothesis. In the example of Remark 9.1.3 (b) the two structures have the same tangent sum at the bad point $0$. *Book:* Theorem 9.2.1, p. 409 (2017: Theorem 9.1, p. 298), the proof on pp. 412–418 (2017 pp. 301–306). *Hidden hypotheses:* the graph representation and the local constancy of the dimension on a general M-polyfold. *Deviation (explicit):* the text reduces to the equidimensional case by Proposition 9.1.11; the formalization does the same through Proposition 9.1.12 (V9.10) with dimension labellings (gaps G.93, G.97) (Findings, items 21, 25, 100).

**V9.15 (Definition 9.2.2 (2017: 9.4), good points; `Polyfold.EpGroupoid.LocalBranchingStructure.MU`, `Polyfold.EpGroupoid.LocalBranchingStructure.IsGoodPoint`, `Polyfold.EpGroupoid.LocalBranchingStruc` `Polyfold.EpGroupoid.LocalBranchingStructure.nbhdAt`, `Polyfold.EpGroupoid.LocalBranchingStructure.`** $y \in M_U = \bigcup_i M_i$ is good for a local branching structure if near $y$ all $M_i$ through $y$ coincide; equivalently, $y \in \mathrm{supp}(\Theta)$ and $\Theta$ is constant on $\mathrm{supp}(\Theta)$ near $y$. *Book:* Definition 9.2.2, p. 410 (2017: Definition 9.4, p. 299). *Hidden hypotheses:* none. *Deviation (restated):* stated for every $\Theta$ with a local branching structure, not only equidimensional ones; the intrinsic form is the argument of the proof of Lemma 9.2.3 (Findings, item 23).

**V9.16 (Lemma 9.2.3 (2017: 9.3), being good does not depend on the structure; `Polyfold.EpGroupoid.LocalBranc`** If $y$ lies in the domains of two local branching structures of $\Theta$, at possibly different points, and is good for one, it is good for the other. *Book:* Lemma 9.2.3, p. 410 (2017: Lemma 9.3, p. 299). *Hidden hypotheses:* none. *Deviation (stronger):* any two structures whose domains contain $y$, any $\Theta$; in the proof $Q(z)$ is presumably $Q(y)$ (possible misprint) (Findings, item 23).

**V9.17 (Remark 9.2.4 (2017: 9.4); `Polyfold.EpGroupoid.remark_9_2_4_a`, `Polyfold.EpGroupoid.mem_goodSet_i` `Polyfold.EpGroupoid.remark_9_2_4_b`).** (a) A point good for some structure is good for every

structure whose domain contains it. (b) Near a good point $y$, $\Theta = \Theta(y) \cdot 1_M$ for one submanifold $M$ properly embedded in an open $V \ni y$. *Book:* Remark 9.2.4, p. 411 (2017: Remark 9.4, p. 300). *Hidden hypotheses:* none. *Deviation (restated):* (b) with $M = M_i \cap V(y)$ for any $i$ with $y \in M_i$ (Findings, item 23).

**V9.18 (Lemma 9.2.5 (2017: 9.4), good points are preserved by morphisms; `Polyfold.EpGroupoid.Sheet.apply_h` `Polyfold.EpGroupoid.src_mem_goodSet_of_tgt`, `Polyfold.EpGroupoid.lemma_9_2_5`).** For every morphism $\varphi$, $s(\varphi)$ is good iff $t(\varphi)$ is. *Book:* Lemma 9.2.5, p. 411 (2017: Lemma 9.4, p. 300). *Hidden hypotheses:* none. *Deviation (stronger):* for every branched ep⁺-subgroupoid, not only equidimensional ones; the transport of structures is replaced by $\Theta \circ \hat{\varphi} = \Theta$ on the sheet of $\varphi$ (Findings, item 23).

**V9.19 (Definition 9.2.6 (2017: 9.5), the good and the bad set; `Polyfold.EpGroupoid.goodSet`, `Polyfold.EpGroupoid.badSet`, `Polyfold.EpGroupoid.preimage_ _image_goodSet`, `Polyfold.EpGroupoid.pre`** $S^g_\Theta$ is the set of points of $\operatorname{supp}(\Theta)$ good for some structure whose domain contains them, $S^b_\Theta = \operatorname{supp}(\Theta) \setminus S^g_\Theta$; both are saturated, so they descend to $|X|$. *Book:* Definition 9.2.6, p. 411 (2017: Definition 9.5, p. 300). *Hidden hypotheses:* none. *Deviation (restated):* the text cites "Remark 9.2.4 (c)" for the descent; Remark 9.2.4 has (a) and (b) only, and Lemma 9.2.5 is what is used (possible misprint) (Findings, item 23).

**V9.20 (Lemma 9.2.7 (1) (2017: 9.5 (1)), the good set is open and dense; `Polyfold.EpGroupoid.lemma_9_2_7_1`, `Polyfold.EpGroupoid.dense_goodSet`, `Polyfold.EpGroupoid.isNowhereDense_badSet`, `Polyfold.EpGroupoid.lemma_9_2_7_1_dense`).** In $\operatorname{supp}(\Theta)$ the good set is open and dense and the bad set is closed and nowhere dense. *Book:* Lemma 9.2.7 (1), p. 412 (2017: Lemma 9.5 (1), p. 300). *Hidden hypotheses:* none. *Deviation (stronger):* for every branched ep⁺-subgroupoid, without equidimensionality and without (2), from a topological covering argument (Findings, item 24).

**V9.21 (Lemma 9.2.7 (2) (2017: 9.5 (2)), good points accumulate on every sheet; `Polyfold.MPolyfold.HasDomainIn`r `Polyfold.MPolyfold.hasDomainInvarianceAt_rn`, `Polyfold.EpGroupoid.lemma_9_2_7_2`, `Polyfold.EpGroupoid.lemma_9_2_7_2_rn`, `Polyfold.EpGroupoid.Remark913Hyp.zero_mem_badSet`, `Polyfold.EpGroupoid.Remark913Hyp.exists_seq_goodSet_structure` ).** For an $n$-dimensional local branching structure at $x$ and every $i$ there is a sequence of good points in $M_i$ converging to $x$, under invariance of domain for $n$-dimensional submanifolds on $U(x)$, which holds on $\mathbb{R}^m$; on $\mathbb{R}^m$ the statement has no hypothesis. In Remark 9.1.3 (b) the origin is bad and is approached by good points of the graph of $\beta_1$. *Book:* Lemma 9.2.7 (2), p. 412 (2017: Lemma 9.5 (2), p. 301). *Hidden hypotheses:* invariance of domain (`HasDomainInvarianceAt`) on a general M-polyfold. *Deviation (explicit):* the statement holds at every $x$, not only at bad points (gap G.98) (Findings, item 24).

**V9.22 (Proposition 9.2.8 and Lemmas 9.2.9–9.2.11 (2017: Proposition 9.3, Lemmas 9.6–9.8), the graph representation and tangent slices; `Polyfold.MPolyfold.HasTangentSlicesAt`, `Polyfold.MPolyfold.IsGraphChartAt`, `Polyfold.MPolyfold.HasGraphChartsAt`, `Polyfold.MPolyfold.hasTan`**
**`Polyfold.exists_mem_graph_of_hasStrictFDerivAt`, `Polyfold.MPolyfold.isGraphChartAt_rn`, `Polyfold.MPolyfold.hasGraphChartsAt_rn`, `Polyfold.MPolyfold.hasTangentSlicesAt_rn`, `Polyfold.MPolyfold.HasGraphChartAt`, `Polyfold.MPolyfold.hasGraphChartAt_rn`, `Polyfold.MPolyfold.exi`**
**`Polyfold.MPolyfold.exists_generic_slice`, `Polyfold.MPolyfold.IsGraphBallAt`, `Polyfold.MPolyfold.Has`**
**`Polyfold.hasGraphBallAt_rn`, `Polyfold.MPolyfold.hasOrientedGoodPointsAt_of_hasGraphBallAt`).**
If near $x$ every $n$-dimensional submanifold through $x$ is, in one chart, a graph over a ray (or ball) in its tangent space (`IsGraphChartAt`, `HasGraphChartAt` for all dimensions at once, `HasGraphBallAt`), then for every class $\Xi$ of sheets with one tangent space $L$ at $x$ and every neighbourhood $W$ of $x$

there is a finite slice of points in $W$ meeting each $M_i$, $i \in \Xi$, once and missing the other $M_{i'}$, so that $\sum_{i\in\Xi} \sigma_i = \sum_{y\in F} \Theta(y)$ (`HasTangentSlicesAt`), and the slice can be chosen generic. On $\mathbb{R}^m$ the graph representation holds at every point and in every dimension, by the inverse function theorem. *Book:* Proposition 9.2.8, p. 413 (2017: Proposition 9.3, p. 302), Lemmas 9.2.9–9.2.11, pp. 414–417 (2017: Lemmas 9.6–9.8, pp. 302–305). *Hidden hypotheses:* the graph representation on a general M-polyfold. *Deviation (narrower):* Proposition 9.2.8 is formalized in slice form, not with the text's subsets $I^e$ and sequences of good points; rays and balls replace the cones around $h_0$; Lemmas 9.2.9–9.2.11 are the hypotheses, proved on $\mathbb{R}^m$ (gap G.97) (Findings, item 25).

**V9.23 (Definition 9.2.12 (2017: 9.6), the tangent $T\Theta$; `Polyfold.EpGroupoid.tangentSubmanifold`, `Polyfold.EpGroupoid.tangentTheta`, `Polyfold.EpGroupoid.tangentTheta_eq`, `Polyfold.EpGroupoid.sum_ta` `Polyfold.EpGroupoid.tangentTheta_nonneg`, `Polyfold.EpGroupoid.isFunctorial_tangentTheta`, `Polyfold.EpGroupoid.TangentBranchingHyp`, `Polyfold.EpGroupoid.isBranchedEpSubgroupoid_tangentThet` `Polyfold.EpGroupoid.isClosed_tangentSubmanifold`, `Polyfold.EpGroupoid.paracompactSpace_orbitSupp_` `Polyfold.EpGroupoid.tangentBranchingHyp_of_isSmoothSubmanifold`, `Polyfold.EpGroupoid.isBranchedEp` `Polyfold.EpGroupoid.LocalBranchingStructure.isSubMPolyfold_tangentSubmanifold`, `Polyfold.EpGroupoid.isBranchedEpSubgroupoid_tangentTheta_rn`, `Polyfold.EpGroupoid.tangentBranchin` `Polyfold.EpGroupoid.isBranchedEpSubgroupoid_tangentTheta_zero`).** Under the independence of V9.13, $T\Theta(h) = \sum_{h\in TM_i} \sigma_i$ for $h$ over $U(x)$, and 0 over the complement of the support, is well defined, nonnegative and a functor on the tangent ep-groupoid; it is a branched ep$^+$-subgroupoid of $TX$ if the $TM_i$ are smooth submanifolds of $TX$ (closedness and the paracompactness of $|\mathrm{supp}(T\Theta)|$ for paracompact $|X|$ are proved), and on $\mathbb{R}^m$ the $TM_i$ are sub-M-polyfolds of $TX$, retracts of $Ts$. *Book:* Definition 9.2.12, p. 418 (2017: Definition 9.6, p. 306). *Hidden hypotheses:* that the $TM_i$ are smooth submanifolds of $TX$ (the tangent of the sc$^+$-retraction is sc$^+$ on $TX$), and the paracompactness of $|X|$, neither discussed in the text. *Deviation (explicit):* $h \in TM_i$ is read as $h \in T_{p(h)}M_i$; only the zero functor is a witness of the full hypothesis (gap G.99) (Findings, item 26).

**V9.24 (Lemma 9.2.13 (2017: 9.9), $d_M(x) = 0$ and reduced tangent position; `Polyfold.IsReducedTangentAt`, `Polyfold.isReducedTangentAt_iff_subset_reduced`, `Polyfold.isReducedTangentAt_iff_subset_partialCo` `Polyfold.MPolyfold.subDegIndex`, `Polyfold.MPolyfold.subDegIndex_le`, `Polyfold.MPolyfold.IsTangentRe` `Polyfold.MPolyfold.isReducedTangentAt_chart_change`, `Polyfold.MPolyfold.isTangentReducedAt_iff_fo` `Polyfold.MPolyfold.lemma_9_2_13_chart`, `Polyfold.MPolyfold.lemma_9_2_13`, `Polyfold.MPolyfold.lemma_` `Polyfold.MPolyfold.lemma_9_2_13_iff`, `Polyfold.MPolyfold.subDegIndex_eq_zero_iff`, `Polyfold.ScRetraction.IsScPlus.exists_injOn_proj`, `Polyfold.ScRetraction.IsScPlus.exists_flat`, `Polyfold.MPolyfold.not_lemma_9_2_13_two_one`).** For a smooth submanifold $M$ and $x \in M \cap \partial X$: $d_M(x) = 0$ iff $T_xM \subseteq T^R_xX$ (in every chart, the chart image of $T_xM$ lies in the reduced tangent space). For every $x \in M$: $d_M(x) = 0$ iff $T_xM \subseteq T^R_xX$. As printed, with "$x \in \partial X$" in (1), (2) □ (1) does not hold as we read it at a point $p$ of $\mathbb{R}^n$, which satisfies (2) and lies off $\partial\mathbb{R}^n$. *Book:* Lemma 9.2.13, p. 419 (2017: Lemma 9.9, p. 307). *Hidden hypotheses:* none. *Deviation (stronger):* read with $x \in M \cap \partial X$ as the standing assumption, which is where the text uses it, and proved at every $x \in M$ in the form `subDegIndex_eq_zero_iff`. The converse needs a chart of $M$ without boundary near $x$: by the inverse function theorem on $T_xM$ and the step that the projection is injective on $M$ near $x$ ("the image is $M$ near 0"), which the text does not spell out; the text's complement $Y$ and map $F$ are not needed (Findings, items 40, 41).

**V9.25 (Remark 9.2.14 (2017: 9.5), $\partial M$; `Polyfold.MPolyfold.remark_9_2_14`, `Polyfold.MPolyfold.remark_9_2_` `Polyfold.MPolyfold.mem_boundary_subPolyfold_iff`, `Polyfold.MPolyfold.remark_9_2_14_not_isTangentR` `Polyfold.MPolyfold.remark_9_2_14_iff`, `Polyfold.MPolyfold.remark_9_2_14_not_mem_boundary_iff`).**

$x \in \partial M$ iff $x \in \partial X$ and $T_xM \not\subseteq T^R_xX$; in particular $\partial M \subseteq \partial X$, and $d_M(x) \geq 1$ at such points. *Book:* Remark 9.2.14, p. 419 (2017: Remark 9.5, p. 307). *Hidden hypotheses:* none. *Deviation (restated):* $\partial M$ is the boundary of the induced M-polyfold structure; the text's "not contained in $C_xX$" is equivalent to the reduced-tangent form (`isReducedTangentAt_iff_subset_partialQuadrant`) (Findings, item 42).

**V9.26 (Proposition 9.2.15 (2017: 9.4), the boundary weights;** `Polyfold.MPolyfold.submanifoldBoundary`, `Polyfold.MPolyfold.mem_submanifoldBoundary_iff`, `Polyfold.EpGroupoid.boundaryPart`, `Polyfold.EpGroupoid.boundaryTheta`, `Polyfold.EpGroupoid.boundaryTheta_eq`, `Polyfold.EpGroupoid.prop` `Polyfold.EpGroupoid.proposition_9_2_15_2`, `Polyfold.EpGroupoid.isFunctorial_boundaryTheta`, `Polyfold.EpGroupoid.proposition_9_2_15_3`, `Polyfold.EpGroupoid.filter_tangentValue_eq`**).** (1) Under the independence of V9.13, $\sum_{y\in\partial M_i}\sigma_i$ is the same for any two local branching structures whose domains contain $y$. (2) At $y$ with $d_X(y)=0$ this sum and the corresponding formal sum of tangent spaces vanish, for every structure. (3) $\partial\Theta$, the part of $T_\Theta(y)$ on tangent spaces not contained in $T^R_yX$, is a functor, nonnegative, supported in $\partial X$, and equal to $\sum_{y\in\partial M_i}\sigma_i$ on every structure. *Book:* Proposition 9.2.15, p. 420 (2017: Proposition 9.4, p. 308). *Hidden hypotheses:* the independence of Theorem 9.2.1 for (1) and (3). *Deviation (explicit):* (1) and (2) need neither $x \in \operatorname{supp}(\Theta) \cap \partial X$ nor that $\Theta$ be branched, and (2) needs no independence (Findings, item 43).

**V9.27 (Definition 9.2.16 (2017: 9.7), the boundary $\partial\Theta$;** `Polyfold.EpGroupoid.IsBoundaryCharacterized`, `Polyfold.EpGroupoid.isBoundaryCharacterized_boundaryTheta`, `Polyfold.EpGroupoid.isBoundaryCharact` `Polyfold.EpGroupoid.eq_boundaryTheta`, `Polyfold.not_unique_isBoundaryCharacterized_halfLine`**).** $\partial\Theta$ satisfies the text's (1) $\operatorname{supp}(\partial\Theta) \subseteq \partial X$ and (2) the local formula at points of $\operatorname{supp}(\partial\Theta)$; a functor that vanishes off $\operatorname{supp}(\Theta)$ and satisfies (2) at the points of $\operatorname{supp}(\Theta)$ equals $\partial\Theta$. As stated, (1) and (2) do not characterize $\partial\Theta$ uniquely as we read them: the zero functor satisfies them for every $\Theta$, and for $\Theta = 1$ on $[0,\infty)$ both 0 and $\partial\Theta \neq 0$ do. *Book:* Definition 9.2.16, p. 421 (2017: Definition 9.7, p. 309). *Hidden hypotheses:* the independence of Theorem 9.2.1. *Deviation (narrower):* "uniquely characterized" is verified with (2) asked on $\operatorname{supp}(\Theta)$. "In view of Proposition 9.2.13" before the definition (2017: "Proposition 9.9") is presumably Proposition 9.2.15, and "branched ep$^+$-groupoid" is "ep$^+$-subgroupoid" (possible misprints) (Findings, items 44, 45).

**V9.28 (Remark 9.2.17 (2017: 9.6);** `Polyfold.EpGroupoid.boundaryTheta_le`, `Polyfold.boundaryTheta_halfLin` `Polyfold.boundaryTheta_halfLine_one`, `Polyfold.EpGroupoid.boundaryTheta_indicator_of_mem`, `Polyfold.EpGroupoid.isBranchedEpSubgroupoid_one`, `Polyfold.EpGroupoid.boundaryTheta_one`**).** $\partial\Theta \leq \Theta$; $\partial\Theta$ is in general not the restriction of $\Theta$ to $\partial X$: for the indicator of the corner 0 of $[0,\infty)$, $\Theta(0)=1$ and $\partial\Theta(0)=0$, while for $\Theta = 1$ on $[0,\infty)$, $\partial\Theta(0)=1$. *Book:* Remark 9.2.17, p. 421 (2017: Remark 9.6, p. 310). *Hidden hypotheses:* the independence of Theorem 9.2.1, which the witnesses satisfy. *Deviation (restated):* the forward reference "In Section 9.5, Definition 9.3.6" is to §9.3 (possible misprint, also in 2017) (Findings, items 45, 46).

**V9.29 (the category $\widehat{\mathrm{Gr}}(X)$ and oriented tangent spaces, p. 422;** `Polyfold.MPolyfold.chartVec`, `Polyfold.MPolyfold.OrTangent`, `Polyfold.MPolyfold.chartFrames`, `Polyfold.MPolyfold.IsChartOrientati` `Polyfold.MPolyfold.IsOrientedAt`, `Polyfold.MPolyfold.IsOrientedAt.isChartOrientation`, `Polyfold.MPolyfold.isOrientedAt_iff_forall`, `Polyfold.MPolyfold.exists_two_orientations`, `Polyfold.MPolyfold.OrTangent.push`, `Polyfold.MPolyfold.IsOrientedAt.push_tangentMapOn`, `Polyfold.EpGroupoid.isOrientedAt_push_tangentMor`, `Polyfold.EpGroupoid.GrHat`, `Polyfold.EpGroupoid.grHatPush`, `Polyfold.EpGroupoid.GrHatObj`, `Polyfold.EpGroupoid.grHatProj`, `Polyfold.EpGroupoid.grForget`, `Polyfold.EpGroupoid.grForgetFunctor`, `Polyfold.EpGroupoid.grForgetFu` An oriented tangent space at $y$ is a subset $L \subseteq T_yX$ with a set of positive frames, oriented in a

chart and then in every chart; every finite-dimensional tangent space has exactly two orientations; morphisms push oriented tangent spaces forward. $\widehat{\mathrm{Gr}}(X)$ is the category of formal sums of oriented tangent spaces over $X^\infty$, with the forgetful functor to $\mathrm{Gr}(X)$ over $X^\infty$. *Book:* §9.3, p. 422 (2017 p. 310), with Remark 9.3.1 (2017: 9.7), which is historical. *Hidden hypotheses:* none. *Deviation (restated):* frames instead of orientations of vector spaces (gap G.7) (Findings, item 50).

**V9.30 (Definition 9.3.2 (2017: 9.8), orientations of $\Theta$;** `Polyfold.MPolyfold.SubmanifoldOrientation`, `Polyfold.EpGroupoid.OrientedBranchingStructure`, `Polyfold.EpGroupoid.OrientedBranchingStructure.o` `Polyfold.EpGroupoid.OrientedBranchingStructure.grForget_orientedSum`, `Polyfold.EpGroupoid.IsOrien` `Polyfold.EpGroupoid.IsOrientable`, `Polyfold.EpGroupoid.IsOrientation.mk'`, `Polyfold.EpGroupoid.orie` `Polyfold.EpGroupoid.orientedSection_comp_grHatProj`, `Polyfold.EpGroupoid.orientedSection_comp_grH` `Polyfold.EpGroupoid.isOrientation_zero`, `Polyfold.EpGroupoid.isOrientation_sum_rn`, `Polyfold.EpGroupoid.isOrientation_sum_trivial_rn`**).** An orientation of $\Theta$ is a natural assignment $\hat{T}_\Theta$ of formal sums of oriented tangent spaces at smooth points, lifting $T_\Theta$, which near every $x \in \mathrm{supp}(\Theta)$ is $\sum_i \sigma_i T_y(M_i, o_i)$ for an oriented local branching structure (continuous orientations $o_i$ of the $M_i$); it defines a section functor of $\widehat{\mathrm{Gr}}(X)$ over $X^\infty$. The zero functor and finite sums of oriented closed submanifolds of $\mathbb{R}^n$ are oriented. *Book:* Definition 9.3.2, p. 423 (2017: Definition 9.8, p. 311). *Hidden hypotheses:* the lift condition uses $T_\Theta$, hence the independence of Theorem 9.2.1, discharged on $\mathbb{R}^m$. *Deviation (restated):* the continuity of $o_i$ is the local constancy of the chart frames (Findings, item 51).

**V9.31 (Remark 9.3.3 (2017: 9.8), orientations are not unique;** `Polyfold.EpGroupoid.remark_9_3_3`**).** The indicator of a point of $\mathbb{R}^n$ has infinitely many orientations, pairwise distinct, $T_q$ for $q \in (0,1) \cap \mathbb{Q}$. *Book:* Remark 9.3.3, p. 423 (2017: Remark 9.8, p. 311). *Hidden hypotheses:* none. *Deviation (stronger):* infinitely many, not only more than one (Findings, item 52).

**V9.32 (Lemma 9.3.4 (2017: 9.10), orientations are determined by good points;** `Polyfold.EpGroupoid.IsOrientat` `Polyfold.MPolyfold.HasOrientedGoodPointsAt`, `Polyfold.EpGroupoid.lemma_9_3_4`, `Polyfold.hasOrientedGoodPointsAt_rn_zero`, `Polyfold.hasOrientedGoodPointsAt_rn`, `Polyfold.MPolyfold.HasContinuousLevelOneOn`, `Polyfold.hasContinuousLevelOneOn_rn`, `Polyfold.frameDet`, `Polyfold.orientation_eq_iff_frameDet`, `Polyfold.MPolyfold.SubmanifoldOrientatio` `Polyfold.EpGroupoid.lemma_9_3_4_of_hasGraphBallAt`, `Polyfold.EpGroupoid.lemma_9_3_4_rn`, `Polyfold.mixedTheta`, `Polyfold.lemma_9_3_4_mixed_dim`, `Polyfold.lemma_9_3_4_mixed_dim_rn`**).** Two orientations of an $n$-dimensional $\Theta$ that agree at the good points agree at every smooth point, under `HasOrientedGoodPointsAt` on the support, which follows from the ball graph representation `HasGraphBallAt` and the level-1 continuity `HasContinuousLevelOneOn`; on $\mathbb{R}^m$ without hypothesis. Without equidimensionality the lemma does not hold as we read it: for $\Theta = 1_{\mathbb{R}\times 0} + 1_{(0,0)}$ on $\mathbb{R}^2$ two orientations agree at all good points and differ at the origin. *Book:* Lemma 9.3.4, p. 423 (2017: Lemma 9.10, p. 311). *Hidden hypotheses:* $n$-dimensional $\Theta$; on a general M-polyfold the graph representation and the continuity of frame fields on level 1. *Deviation (narrower):* the proof compares the two structures on one family of slices, applying Proposition 9.2.8 to both families together, and compares frames of different sheets by level-1 continuity; the text takes the good-point sequences from the first structure only, a step that needs an argument it does not give (gaps G.97, G.100) (Findings, item 53).

**V9.33 (Example 9.3.5 (2017: 9.1);** `Polyfold.betaPP`, `Polyfold.betaMM`, `Polyfold.betaPM`, `Polyfold.betaMP`, `Polyfold.example935Theta`, `Polyfold.example_9_3_5_count`, `Polyfold.example935Theta_` `Polyfold.exampleT1`, `Polyfold.exampleT2`, `Polyfold.example_9_3_5`, `Polyfold.example_9_3_5_rn`, `Polyfold.example_9_3_5_eq_origin`, `Polyfold.example_9_3_5_ne_of_pos`, `Polyfold.example_9_3_5_cannot`

`Polyfold.example_9_3_5_eq_of_neg`, `Polyfold.orientedLift1_injective`, `Polyfold.exampleT2_ne_oriente` `Polyfold.exists_goodSet_exampleT1_ne_exampleT2`). For $\beta^{++} = e(s) + e(-s)$ ($e =$ `expNegInvGlue`), $\beta^{--} = -\beta^{++}$ and the two mixed functions, the graphs define a branched ep$^+$-subgroupoid $\Theta$ of the trivial ep-groupoid of $\mathbb{R}^2$ by two structures; the orientations $\hat{T}^1$, $\hat{T}^2$ given by $\pm(1, \beta')$ coincide at the origin, differ at $(s, \beta^{++}(s))$ for $s > 0$, and no sign choice for the second structure reproduces $\hat{T}^1$ near the origin. They also coincide wherever $x < 0$, and the four sign choices of the first structure give four different orientations, none equal to $\hat{T}^2$. *Book:* Example 9.3.5, p. 425 (2017: Example 9.1, p. 313). *Hidden hypotheses:* none on $\mathbb{R}^2$. *Deviation (narrower):* "otherwise different" and "four different possibilities" do not hold as we read them (at least five orientations with weights 1); the classification of all orientations of $\Theta$ is not formalized (gap G.101) (Findings, item 54).

**V9.34 (Definition 9.3.6 (2017: 9.10), tame $\Theta$;** `Polyfold.EpGroupoid.TameBranchingStructure`, `Polyfold.EpGroupoid.IsTameBranched`, `Polyfold.EpGroupoid.isTameBranched_zero`, `Polyfold.EpGroupoid.isTameBranched_indicator`, `Polyfold.MPolyfold.hasTameInducedStructure_of_degl` $\Theta$ is tame if at every point of its support there is a local branching structure on a domain with the natural $G_x$-action and the properness property all of whose $M_i$ have a tame induced structure; the zero functor and the indicator of a point of $\mathbb{R}^n$ are tame. *Book:* Definition 9.3.6, p. 425 (2017: Definition 9.10, p. 314). *Hidden hypotheses:* none. *Deviation (restated):* "an equivalent tame structure" is Part I's tame induced structure (Definition 2.5.18); the consequence "in particular every $M_i$ is a manifold with boundary with corners" is not formalized (gap G.102) (Findings, item 55).

**V9.35 (Lemma 9.3.7, equal tangent spaces have equal degeneracy;** `Polyfold.MPolyfold.subDegIndex_eq_quotD` `Polyfold.MPolyfold.subDegIndex_eq_of_tangentSpace_eq`, `Polyfold.MPolyfold.lemma_9_3_7`, `Polyfold.MPolyfold.subDegIndex_eq_of_submanifoldTangent_eq`). For submanifolds $M_1$, $M_2$ with tame induced structures and $y \in M_1 \cap M_2$, $T_yM_1 = T_yM_2$ implies $d_{M_1}(y) = d_{M_2}(y)$. *Book:* Lemma 9.3.7, p. 426, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (stronger):* neither "$Y$ tame" nor "of the same dimension" is needed; in the proof "$T_0O \cap C = T_0^R O$" is read through the linearity of the chart image of $T_0M$ (Findings, item 56).

**V9.36 (Definition 9.3.8 (2017: 9.9), the boundary orientation of an oriented $M$;** `Polyfold.exists_basis_orienta` `Polyfold.exists_boundary_orientation`, `Polyfold.det_cons_smul_add`, `Polyfold.MPolyfold.boundaryFram` `Polyfold.MPolyfold.isOrientedAt_boundaryFrames`, `Polyfold.MPolyfold.boundaryFrames_eq`, `Polyfold.exists_boundaryFrames_graph`). For an oriented $(M, o)$, a hyperplane $K$ of $T_yM$ and $e \notin K$, the frames $(a_1, \dots, a_{n-1})$ of $K$ with $(e, a_1, \dots, a_{n-1})$ positive orient $K$, and the orientation is unchanged under $e \mapsto ce + k$, $c > 0$, $k \in K$; both orientations occur. *Book:* Definition 9.3.8, p. 427 (2017: Definition 9.9, p. 313). *Hidden hypotheses:* none. *Deviation (narrower):* $T_y\partial M$ and the outward vector enter as data $K$, $e$; that $T_y\partial M = T_y^R M$ and the outward half-space are determined by $M$ (through the cones $C_yM$) is not formalized (gap G.103) (Findings, item 57).

**V9.37 ((9.11) and Proposition 9.3.9, the boundary orientation does not depend on the structure;** `Polyfold.MPolyfold.HasSubDegree`, `Polyfold.MPolyfold.SubmanifoldOrientation.hasSubDegree_tangent_` `Polyfold.EpGroupoid.degreePart`, `Polyfold.EpGroupoid.degreePart_orientedSum`, `Polyfold.EpGroupoid.e` `Polyfold.EpGroupoid.proposition_9_3_9`). For two oriented local branching structures with tame $M_i$ and the same oriented sum at $y$, the parts of degree $\ell = d_{M_i}(y)$ agree for every $\ell$ (9.11); hence (9.12) gives the same value for every structure, and vanishes at smooth points off $\mathrm{supp}(\Theta)$. *Book:* (9.11), p. 427, and Proposition 9.3.9, p. 428, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (narrower):* the boundary operation $T_y(M, o) \mapsto T_y(\partial M, o_{\partial M})$ is a function `bd`

of the oriented tangent space, which is what "the discussion preceding it" provides, given G.103 (Findings, item 58).

**V9.38 (Definition 9.3.10 (2017: 9.11), the oriented boundary $\hat{T}_{\partial\Theta}$; `Polyfold.EpGroupoid.boundaryLift`).** $\hat{T}_{\partial\Theta}(y) = 0$ off $\operatorname{supp}(\Theta)$ and $\sum_{d_{M_i}(y)=1} \sigma_i \, T_y(\partial M_i, o_{\partial M_i})$ on a structure (9.12). *Book:* Definition 9.3.10, p. 428 (2017: Definition 9.11, p. 314, with (9.12) numbered (9.11) there). *Hidden hypotheses:* none. *Deviation (narrower):* with `bd` as in V9.37; the continuity of $o_{\partial M}$ along $\partial M$ and its extension to the faces (Figure 9.1) are not formalized (gap G.103); Remark 9.3.11 has no formal content (Findings, item 58).

**V9.39 (Definition 9.4.1 (2017: 9.12), the relative tangent space and essential points; `Polyfold.chartImage`, `Polyfold.relTangent`, `Polyfold.span_tangentConeAt_chartImage_eq_map`, `Polyfold.relTangent_eq_map`, `Polyfold.relTangent_eq_top_iff`, `Polyfold.essentialSet`, `Polyfold.essentialSet_eq`, `Polyfold.crossLines`, `Polyfold.essentialSet_crossLines`, `Polyfold.nonEssential_crossLines`).** For a subset $K$ of a finite-dimensional manifold with corners, $T^K_z M$ is the span of the limits of unit directions $\varphi(z_k)/|\varphi(z_k)|$, $z_k \to z$ in $K \setminus \{z\}$, in a chart centred at $z$, independent of the chart; $z \in K$ is essential if it is a limit of $z_k \in K$ with $T^K_{z_k} M = T_{z_k} M$. For two crossing lines in $\mathbb{R}^2$ the essential set is $\{0\}$. *Book:* Definition 9.4.1, p. 429 (2017: Definition 9.12, p. 315). *Hidden hypotheses:* none. *Deviation (restated):* on Mathlib manifolds with corners, the span of Mathlib's tangent cone in a chart (Findings, item 70).

**V9.40 (Lemma 9.4.2 (2017: 9.11), the essential set is closed; `Polyfold.lemma_9_4_2`, `Polyfold.lemma_9_4_2_open`).** $K^e$ is closed in $K$ and $K \setminus K^e$ is open in $K$. *Book:* Lemma 9.4.2, p. 430 (2017: Lemma 9.11, p. 315). *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 70).

**V9.41 (Property (T) (2017 p. 316); `Polyfold.EpGroupoid.OrientedBranchingStructure.PropertyT`, `Polyfold.MPolyfold.OrTangent.eq_or_eq_or_eq_of_isOrientedAt`, `Polyfold.MPolyfold.exists_ne_isOrie`, `Polyfold.propertyT_example935B1`).** An oriented local branching structure has Property (T) on $V$ if, whenever two sheets have the same tangent space at a point of $V$ and their orientations there are related by $\varepsilon$, they are related by $\varepsilon$ at every point of $V$ where their tangent spaces agree; the first structure of Example 9.3.5 has it. *Book:* Property (T), p. 430 (2017 p. 316). *Hidden hypotheses:* none. *Deviation (restated):* Lemma 9.4.3, that (T) holds after shrinking $U(x)$, is not formalized (gap G.104) (Findings, item 71).

**V9.42 (Lemma 9.4.4 (2017: 9.13), essential in $M_i$ iff essential in $M_{i'}$; `Polyfold.MPolyfold.SheetGeometry`, `Polyfold.graphSheets`, `Polyfold.graphSheetGeometry`, `Polyfold.lemma_9_4_4_mixed_dim`).** For sheets of one dimension the essential points of $K(i,i') = M_i \cap M_{i'}$ do not depend on the reference sheet; this is the field `ess_comm` of a sheet geometry, which exists for smooth graphs in $\mathbb{R}^2$. For sheets of different dimensions the lemma does not hold as we read it: $\mathbb{R} \times 0$ is essential in itself as a subset of $\mathbb{R}$ and has no essential point as a subset of $\mathbb{R}^2$. *Book:* Lemma 9.4.4, p. 431 (2017: Lemma 9.13, p. 317). *Hidden hypotheses:* sheets of one dimension. *Deviation (narrower):* for M-polyfold sheets, which carry no Mathlib manifold structure, Lemmas 9.4.4 and 9.4.6 are the fields of `SheetGeometry` (gap G.105); the note in Definition 9.4.7 ($i \sim_y i'$ implies $T_y M_i = T_y M_{i'}$) is its field `submanifoldTangent_eq` (Findings, item 72).

**V9.43 (Definition 9.4.5 (2017: 9.13), sets of measure zero; `Polyfold.IsMeasureZero`, `Polyfold.isMeasureZero_iUnion`, `Polyfold.isMeasureZero_of_forall_mem`, `Polyfold.not_isMeasureZero_u`).** $K \subseteq M$ has measure zero if its image in every chart has Lebesgue measure zero; measure zero is monotone and closed under countable unions, can be checked on a cover by charts, and $\mathbb{R}^2$ itself does not have it. *Book:* Definition 9.4.5, p. 431 (2017: Definition 9.13, p. 317). *Hidden hypotheses:*

none. *Deviation (restated):* for Mathlib manifolds with corners (Findings, item 70).

**V9.44 (Lemma 9.4.6 (2017: 9.14), non-essential points form a null set; `Polyfold.unitDirections`, `Polyfold.span_unitDirections`, `Polyfold.exists_pos_measure_cone_lt`, `Polyfold.addHaar_setOf_span_ta` `Polyfold.isMeasureZero_setOf_relTangent_ne_top`, `Polyfold.lemma_9_4_6`).** For every subset $K$ of a finite-dimensional manifold, $K \setminus K^e$ has measure zero, by a Lebesgue density argument at points where the relative tangent space is not everything. *Book:* Lemma 9.4.6, p. 431 (2017: Lemma 9.14, p. 317). *Hidden hypotheses:* none. *Deviation (stronger):* any $K$, not only $K(i, i')$; for M-polyfold sheets it is the field `isNull_diff_ess` of `SheetGeometry` (gap G.105) (Findings, items 70, 72).

**V9.45 (Definition 9.4.7 (2017: 9.14), $y$-equivalence; `Polyfold.MPolyfold.SheetGeometry.IsLink`, `Polyfold.MPolyfold.SheetGeometry.YEquiv`, `Polyfold.MPolyfold.SheetGeometry.equivalence_yEquiv`, `Polyfold.MPolyfold.SheetGeometry.ySetoid`, `Polyfold.MPolyfold.SheetGeometry.submanifoldTangent_ec` `Polyfold.MPolyfold.SheetGeometry.numClasses`).** $i \sim_y i'$ if a chain of sheets through $y$ links them with $y$ essential in each consecutive intersection; it is an equivalence relation on $I_y$, $y$-equivalent sheets have the same tangent space at $y$, and $\#P_y$ is the number of classes. *Book:* Definition 9.4.7, p. 432 (2017: Definition 9.14, p. 318). *Hidden hypotheses:* none. *Deviation (restated):* none (Findings, item 72).

**V9.46 (Lemma 9.4.8 (2017: 9.15), points with two classes form a null set; `Polyfold.MPolyfold.SheetGeometry.Is` `Polyfold.MPolyfold.SheetGeometry.lemma_9_4_8`, `Polyfold.MPolyfold.SheetGeometry.MStar`, `Polyfold.MPolyfold.SheetGeometry.isNullIn_diff_MStar`).** $\{y \in M_U \mid \#P_y \geq 2\}$ is null in the sheets, and $M_U \setminus M_U^*$ is null. *Book:* Lemma 9.4.8, p. 432 (2017: Lemma 9.15, p. 318). *Hidden hypotheses:* none. *Deviation (restated):* "measure zero" in $M_U$ is null in every sheet through the point; the last step of the proof uses $K(i,j)^{ne}$ inside $M_i$, while $K(j,j')^{ne}$ is null only in $M_j$ and $M_{j'}$ (reading) (Findings, item 73).

**V9.47 (Proposition 9.4.9 (2017: 9.5), the decomposition of $M_U^*$; `Polyfold.EpGroupoid.LocalBranchingStructur` `Polyfold.MPolyfold.SheetGeometry.MStarPart`, `Polyfold.EpGroupoid.proposition_9_4_9`).** For an oriented local branching structure with Property (T), the independence of Theorem 9.2.1 and sheets in a sheet geometry: $M_U \setminus M_U^*$ is null, $M_U^*$ is the disjoint union of the $M_U^J = \{y \in M_U^* \mid I_y = J\}$, and on $M_U^J$: $\Theta = \sigma_J$, $T_\Theta = \sigma_J T^J$, $\hat{T}_\Theta = a_J(T^J, o_J^a) + b_J(T^J, o_J^b)$ with $a_J + b_J = \sigma_J$ and $o_J^a = -o_J^b$. *Book:* Proposition 9.4.9, p. 434 (2017: Proposition 9.5, p. 319). *Hidden hypotheses:* the independence of Theorem 9.2.1; the sheet geometry. *Deviation (explicit):* the footnote's "Lemma 9.5.2" is a forward reference (Findings, items 72, 74).

**V9.48 (Proposition 9.4.10 (2017: 9.6), comparing two oriented structures; `Polyfold.EpGroupoid.proposition_9_` `Polyfold.EpGroupoid.tangent_eq_iff_of_mem_ess`, `Polyfold.example935B1`, `Polyfold.example935B2`, `Polyfold.example935_two_le_numClasses`, `Polyfold.example935_hypotheses_9_4_10`, `Polyfold.example935_MU_eq`).** For two oriented local branching structures on the same $U(x)$ with (a) the same weights and (b) the same oriented sums at every point, $M_U = M_U'$ and there are $M_U^\dagger \subseteq M_U$ with null complement in both families and a finite partition $A_1 \sqcup \cdots \sqcup A_\ell$ of $M_U^\dagger$ with (2)–(5) of the text. The two structures of Example 9.3.5 satisfy (a) and (b), and the origin has two classes. *Book:* Proposition 9.4.10, p. 435 (2017: Proposition 9.6, p. 321). *Hidden hypotheses:* the sheet geometry. *Deviation (stronger):* Property (T) is not needed (Findings, item 74).

**V9.49 (Definition 9.5.1 (2017: 9.15), sets of measure zero in $S = |\mathrm{supp}(\Theta)|$; `Polyfold.SheetNullSets`, `Polyfold.SheetNullSets.isNull_union`, `Polyfold.SheetNullSets.isNull_iUnion_countable`, `Polyfold.SheetNullSets.isNull_of_cover`, `Polyfold.MPolyfold.SheetMeasureGeometry`,**

`Polyfold.MPolyfold.SheetMeasureGeometry.toSheetNullSets`, `Polyfold.EpGroupoid.LocalBranchingStruc`
`Polyfold.EpGroupoid.LocalBranchingStructure.hasNullPreimages_iff_isNullIn`, `Polyfold.EpGroupoid.I`
`Polyfold.EpGroupoid.IsMeasureZeroSupp`, `Polyfold.EpGroupoid.IsMeasureZeroSuppAny`, `Polyfold.EpGroupoid.IsMeasureZeroSuppAll`). For a system of null sets in the sheets of a class $\mathcal{S}$ of submanifolds (a σ-ideal in every sheet, local, the same on the overlap of two sheets of $\mathcal{S}$, a nonempty sheet not null in itself), $A \subseteq |X|$ has measure zero if at every $x \in \operatorname{supp}(\Theta)$ some tame local branching structure (Definition 9.3.6) with sheets in $\mathcal{S}$ has preimages $M_i \cap \pi^{-1}(A)$ null in $M_i$; the variants ask this of some local branching structure, or of every structure with sheets in $\mathcal{S}$. A `SheetMeasureGeometry` is a §9.4 sheet geometry whose null sets are such a system. *Book:* Definition 9.5.1, p. 437 (2017: Definition 9.15, p. 322). *Hidden hypotheses:* none. *Deviation (restated):* "measure zero in $M_i$" is the abstract system `SheetNullSets`, satisfiable by Lebesgue null sets of graphs in $\mathbb{R}^2$ and of open subsets of $\mathbb{R}^2$ (V9.53); "a suitable $U(x)$" is a tame structure; compactness and orientation of Θ are not used (gap G.106) (Findings, item 80).

**V9.50 (Lemma 9.5.2 (2017: 9.16), measure zero does not depend on the choice;** `Polyfold.EpGroupoid.isMeasureZ`
`Polyfold.EpGroupoid.isMeasureZeroSuppAny_of_isMeasureZeroSupp`, `Polyfold.EpGroupoid.lemma_9_5_2`,
`Polyfold.EpGroupoid.isSigmaIdeal_isMeasureZeroSupp`, `Polyfold.EpGroupoid.isSigmaIdeal_isMeasureZe`
`Polyfold.EpGroupoid.not_isMeasureZeroSuppAny_orbitSupp`, `Polyfold.EpGroupoid.not_isMeasureZeroSup`
`Polyfold.isNull_transfer_mixed_dim`). If every point of $\operatorname{supp}(\Theta)$ has a tame local branching structure with sheets in $\mathcal{S}$, then "null preimages for a choice of tame structures", "for a choice of structures" and "for every structure with sheets in $\mathcal{S}$" agree; the sets of measure zero form a σ-ideal, and $S \neq \emptyset$ does not have measure zero. The axis $\mathbb{R} \times 0$ is null in the sheet $\mathbb{R}^2$ and not null in itself. *Book:* Lemma 9.5.2, p. 437 (2017: Lemma 9.16, p. 322), cited from the literature without proof. *Hidden hypotheses:* the sheets of all structures compared lie in one class $\mathcal{S}$ on which two sheets induce the same null sets on their overlap. *Deviation (narrower):* this transfer property holds for sheets of one dimension and does not hold across dimensions (`Polyfold.isNull_transfer_mixed_dim`), so the lemma is proved for structures with sheets in one class, as for the $n$-dimensional structures of an $n$-dimensional Θ; for every structure of an $n$-dimensional Θ it would need Proposition 9.1.12 in general (gaps G.97, G.106) (Findings, item 81).

**V9.51 (Definition 9.5.3 (2017: 9.16) and the remark after it, the σ-algebra $L(S)$;** `Polyfold.borelNullSigma`,
`Polyfold.IsSigmaIdeal`, `Polyfold.borel_le_borelNullSigma`, `Polyfold.measurableSet_borelNullSigma_of`
`Polyfold.borelNullCompletion`, `Polyfold.measurableSet_borelNullSigma_iff`, `Polyfold.EpGroupoid.meas`
`Polyfold.EpGroupoid.lebesgueSigma`, `Polyfold.EpGroupoid.isSigmaIdeal_measureZeroSets`,
`Polyfold.EpGroupoid.remark_9_5_3`, `Polyfold.EpGroupoid.measurableSet_lebesgueSigma_of_isMeasureZe`
$L(S)$ is the smallest σ-algebra on the subspace $S$ containing the Borel sets and the subsets of measure zero; for a σ-ideal $\mathcal{N}$ the σ-algebra generated by the Borel sets and $\mathcal{N}$ is $\{A \mid A \,\Delta\, B \in \mathcal{N}$ for a Borel $B\}$, so, under the standing assumption of V9.50, $K \in L(S)$ iff $K \,\Delta\, B$ has measure zero for some Borel $B$. *Book:* Definition 9.5.3 and the remark after it, p. 438 (2017: Definition 9.16, p. 323). *Hidden hypotheses:* none. *Deviation (restated):* the remark that $L(S)$ is obtained by adding the sets of measure zero is read as the completion $K \,\Delta\, B$ null; the form $B \cup N$ would need Borel null hulls (Findings, item 82).

**V9.52 (Definition 9.5.14 (2017: 9.17) and measure zero in $\partial S$, p. 448;** `Polyfold.EpGroupoid.IsMeasureZeroBound`
`Polyfold.EpGroupoid.IsMeasureZeroBoundaryAll`, `Polyfold.EpGroupoid.isMeasureZeroBoundaryAll_of`,
`Polyfold.EpGroupoid.boundaryMeasureZeroSets`, `Polyfold.EpGroupoid.lebesgueSigmaBoundary`,
`Polyfold.EpGroupoid.orbitSupp_boundaryTheta_subset`). For a second null-set system on a class of boundaries $\partial M_i$, $K \subseteq |X|$ has measure zero in $\partial S = |\operatorname{supp}(\partial\Theta)|$ if at every point of $\operatorname{supp}(\partial\Theta)$ a tame structure has boundary preimages $\partial M_i \cap \pi^{-1}(K)$ null in $\partial M_i$; then every

structure with boundaries in the class has; $\partial S \subseteq S$; $L(\partial S)$ is the σ-algebra generated by the Borel sets of $\partial S$ and these sets. *Book:* p. 448 and Definition 9.5.14, p. 448 (2017: Definition 9.17, p. 327). *Hidden hypotheses:* the independence of Theorem 9.2.1 (through which $\partial\Theta$ is defined). *Deviation (explicit):* $(n-1)$-dimensional measure zero in $\partial M_i$ is an abstract system; its description face by face is not formalized, and no witness with $\partial S \neq \emptyset$ is given (gap G.108) (Findings, item 83).

**V9.53 (witnesses for §9.5 (not in the book);** `Polyfold.isMeasureZero_image_of_local`, `Polyfold.isMeasureZero_of_subsingleton`, `Polyfold.isMeasureZero_rn2_iff`, `Polyfold.exists_tameBranc` `Polyfold.hasTameSheetsIn_trivial_rn`, `Polyfold.injective_ _trivial_rn`, `Polyfold.graphSheetMeasureG` `Polyfold.graphNull`, `Polyfold.theta935`, `Polyfold.isBranchedEpSubgroupoid_theta935`, `Polyfold.hasTameSheetsIn_theta935`, `Polyfold.zero_mem_supp_theta935`, `Polyfold.isMeasureZeroSupp_th` `Polyfold.not_isMeasureZeroSupp_theta935`, `Polyfold.lebesgueSigma_theta935`, `Polyfold.planeSheets`, `Polyfold.planeNull`, `Polyfold.planeSheet`, `Polyfold.thetaPlane`, `Polyfold.isSmoothSubmanifold_univ_rn` `Polyfold.isBranchedEpSubgroupoid_thetaPlane`, `Polyfold.mem_supp_thetaPlane`, `Polyfold.hasTameSheets` `Polyfold.exists_mem_lebesgueSigma_not_borel_thetaPlane`**).** On the trivial ep-groupoid of $\mathbb{R}^n$ every local branching structure restricts to a tame one. The graph sheets of §9.4 carry a `SheetMeasureGeometry`; for the $\Theta$ of the two graphs $G^{++}$, $G^{--}$ of Example 9.3.5 the touching point $|0|$ is a nonempty set of measure zero, $S$ is not null and $S \in L(S)$; for $\Theta = 1$ on $\mathbb{R}^2$ with the Lebesgue null sets there is a member of $L(S)$ that is not Borel. *Book:* not in the book. *Hidden hypotheses:* none. *Deviation (extra):* the hypotheses of V9.49–V9.51 are satisfiable, and $L(S)$ is strictly larger than the Borel σ-algebra (Findings, item 82).

**V9.54 (Lemma 9.5.9 (2017: 9.18), restrictions in the $G_x$-case: the isotropy and set part;** `Polyfold.EpGroupoid.NaturalRep. _Φ`, `Polyfold.EpGroupoid.NaturalRep.mem_stabilizer_iff_Φ`, `Polyfold.EpGroupoid.NaturalRep.card_quotient_stabilizer_mul_card_isotropy`, `Polyfold.EpGroupoid.N` `Polyfold.EpGroupoid.NaturalRep.mapsTo_Φ_MU`, `Polyfold.EpGroupoid.NaturalRep.MU_inter_preimage_eq_` `Polyfold.EpGroupoid.NaturalRep.MU_inter_preimage_eq_iUnion_quotient`, `Polyfold.EpGroupoid.Natural` For a natural representation of $G_x$ on $U(x)$ and $y \in U(x)$ with stabilizer $H$: $\#(G_x/H) \cdot \#G_y = \#G_x$; there is an open $V \ni y$ inside any given neighbourhood with $\varphi_g(V) = V$ for $g \in H$ and $\varphi_g(V) \cap V = \emptyset$ otherwise; for $K \subseteq |M_U \cap V|$ the preimage $M_U \cap \pi^{-1}(K)$ is the union over coset representatives $r$ of $\varphi_r(M_U \cap V \cap \pi^{-1}(K))$, and the union is disjoint. *Book:* Lemma 9.5.9, p. 442 (2017: Lemma 9.18, p. 326). *Hidden hypotheses:* none. *Deviation (narrower):* the equality of the measures $\mu^U_\omega$, $\mu^V_\omega$ is not stated (forms and integrals, gap G.107); the representatives $R$ are `Quotient.out` on $G_x/H$, and the restriction $\tilde{M}_j = V \cap M_j$ keeps the sheets through $y$ (`restrictAt`) (Findings, item 84).

**V9.55 (Lemma 9.5.10, the effective orders and (9.23);** `Polyfold.EpGroupoid.card_effIsotropy_mul_card_trivi` `Polyfold.EpGroupoid.NaturalRep.kerΦ_eq_trivialSubgroup`, `Polyfold.EpGroupoid.NaturalRep.kerΦ_le_s` `Polyfold.EpGroupoid.NaturalRep.stabilizerEquivIsotropy_mem_trivialSubgroup_iff`, `Polyfold.EpGroupoid.NaturalRep.card_trivialSubgroup_eq`, `Polyfold.EpGroupoid.NaturalRep.card_quot` $\#T_x \cdot \#G^{\mathrm{eff}}_x = \#G_x$; under LUC for $U(x)$, $T_x$ is the kernel of the representation, it lies in the stabilizer of every $y \in U(x)$, it corresponds to $T_y$ under $H \cong G_y$, so $\#T_y = \#T_x$, and $\#(G_x/H) \cdot \#G^{\mathrm{eff}}_y = \#G^{\mathrm{eff}}_x$, which is (9.23). *Book:* Lemma 9.5.10, p. 444, and (9.23), p. 445, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (narrower):* the equality of the effective measures is not stated (gap G.107); LUC is `NaturalRep.HasLUC` of the given $U(x)$ (Definition 7.2.18) (Findings, item 84).

**V9.56 (Lemma 9.5.11 (2017: 9.19), morphism invariance: the isotropy and set part;** `Polyfold.EpGroupoid.card_i` `Polyfold.EpGroupoid.card_effIsotropy_eq_of_mor`, `Polyfold.EpGroupoid.IsFunctorial.eq_of_ _eq`,

`Polyfold.EpGroupoid.image_ _MU_eq`, `Polyfold.EpGroupoid.NaturalRep.Φ_hyps`**).** For a morphism $\varphi : x \to x'$, $\#G_x = \#G_{x'}$ and $\#G_x^{\mathrm{eff}} = \#G_{x'}^{\mathrm{eff}}$; for local branching structures on $U, U'$ and a map $\widehat{\varphi}$ of $U$ onto $U'$ with $|\widehat{\varphi}(z)| = |z|$, $|M_U| = |M_{U'}|$; the maps $\varphi_g$ of a natural representation satisfy these hypotheses. *Book:* Lemma 9.5.11, p. 445 (2017: Lemma 9.19, p. 326). *Hidden hypotheses:* none. *Deviation (narrower):* the equality $\mu_\omega^U(K) = \mu_\omega^{U'}(K)$, "a change of variables", is not stated (gap G.107) (Findings, item 84).

**V9.57 (Proposition 9.5.13, the normalizing factors;** `Polyfold.EpGroupoid.card_effIsotropy_eq_of_isEffecti` `Polyfold.EpGroupoid.isFunctorial_reduced_iff`, `Polyfold.EpGroupoid.LocalBranchingStructure.toRedu` `Polyfold.EpGroupoid.card_effIsotropy_eq_reduced`**).** For effective $X$, $\#G_x^{\mathrm{eff}} = \#G_x$; for the reduced ep-groupoid $X_R$ (with Hausdorff paracompact morphism space), functors and local branching structures of $X$ and $X_R$ coincide and $\#G_x^{\mathrm{eff}} = \#G_x^R = \#(G_x^R)^{\mathrm{eff}}$. *Book:* Proposition 9.5.13, p. 446, new in the 2021 edition. *Hidden hypotheses:* the morphism space of $X_R$ Hausdorff and paracompact (instances; Hausdorff under LUC). *Deviation (narrower):* the equality of the integrals in (1) and (2) is not stated (gap G.107); what is proved is the equality of the normalizing factors on which the proof rests (Findings, item 84).

**V9.58 (Lemma 9.6.1 and $\dim_\Theta$, the top dimension;** `Polyfold.MPolyfold.HasDimAt.le_of_inter_subset`, `Polyfold.MPolyfold.exists_mem_forall_notMem`, `Polyfold.EpGroupoid.LocalBranchingStructure.topDim`, `Polyfold.EpGroupoid.LocalBranchingStructure.lemma_9_6_1`, `Polyfold.EpGroupoid.localTopDim`, `Polyfold.EpGroupoid.localTopDim_eq`, `Polyfold.EpGroupoid.localTopDim_eq_bot`**).** Two local branching structures at $x$ with dimension labellings have the same top dimension, which defines $\dim_\Theta(x)$. *Book:* Lemma 9.6.1, p. 453, and $\dim_\Theta$, p. 454, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (stronger):* the "measure zero" argument is replaced by a topological covering lemma: a nonempty open piece of a $k$-dimensional submanifold is not covered by finitely many submanifolds of smaller dimension; no Lebesgue measure and no Baire argument (Findings, item 90).

**V9.59 (Lemma 9.6.2, the union of the top-dimensional sheets;** `Polyfold.EpGroupoid.LocalBranchingStructure` `Polyfold.EpGroupoid.LocalBranchingStructure.lemma_9_6_2`, `Polyfold.EpGroupoid.LocalBranchingStruc` On $U \cap U'$ the unions $M_U^k$ of the top-dimensional sheets of two structures agree. *Book:* Lemma 9.6.2, p. 454, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (same):* by the covering lemma of V9.58 (Findings, item 90).

**V9.60 (Definition 9.6.3, top-good and top-bad points;** `Polyfold.EpGroupoid.LocalBranchingStructure.IsTopG` `Polyfold.EpGroupoid.LocalBranchingStructure.IsTopBadPoint`, `Polyfold.EpGroupoid.LocalBranchingStr` $y \in M_U^k$ is top-good if near $y$ all top-dimensional sheets through $y$ coincide; at top-good points the top-dimensional parts $\Theta_k$ of two structures agree, without hypothesis. *Book:* Definition 9.6.3, p. 455, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (restated):* none (Findings, item 91).

**V9.61 (Lemma 9.6.4, top-goodness does not depend on the structure;** `Polyfold.MPolyfold.TopMultiplicity`, `Polyfold.MPolyfold.topMultiplicity_zero`, `Polyfold.topMultiplicity_rn_of_le`, `Polyfold.EpGroupoid.L` `Polyfold.EpGroupoid.LocalBranchingStructure.isTopBadPoint_iff`, `Polyfold.MPolyfold.TopMultiplicit` `Polyfold.MPolyfold.topMultiplicityAt_of_hasGraphChartAt`, `Polyfold.MPolyfold.topMultiplicity_rn`, `Polyfold.EpGroupoid.LocalBranchingStructure.lemma_9_6_4_of_hasGraphChartAt`**).** A point in both domains is top-good for one structure iff for the other, under `TopMultiplicity` (the top weight of a second structure at a point equals the sum of the weights of its sheets there), which holds for $k = 0$, for $k \geq n$ on $\mathbb{R}^n$, and wherever `HasGraphChartAt` holds, so on $\mathbb{R}^m$ in all dimensions. *Book:* Lemma 9.6.4, p. 455, new in the 2021 edition. *Hidden hypotheses:* the graph

representation on a general M-polyfold. *Deviation (explicit):* the step "similarly as good points" needs that the top weight of the second structure at $y$ is $\sigma$, an argument the text does not give; it is `TopMultiplicityAt`, proved from `HasGraphChartAt` by grouping the sheets of both families by tangent space (gap G.97) (Findings, item 91).

**V9.62 (Proposition 9.6.5 and the formula on p. 457; `Polyfold.EpGroupoid.LocalBranchingStructure.IsTopSli` `Polyfold.EpGroupoid.LocalBranchingStructure.proposition_9_6_5`, `Polyfold.EpGroupoid.LocalBranchin` `Polyfold.EpGroupoid.LocalBranchingStructure.dimPart_topDim_eq_sum_slices`, `Polyfold.EpGroupoid.Lo` `Polyfold.EpGroupoid.LocalBranchingStructure.dimPart_topDim_eq_of_hasGraphChartAt`).** Under `HasGraphChartAt` at $x$, for every class of top-dimensional sheets with one tangent space at $x$ and every neighbourhood $W$ of $x$ there is a finite top slice in $W$; the top-dimensional part is $\Theta_k(x) = \sum_{\text{classes}} \sum_{y \in \text{slice}} \Theta(y)$, and the class sums agree for any two structures. *Book:* Proposition 9.6.5, p. 456, and the formula on p. 457, new in the 2021 edition. *Hidden hypotheses:* the graph representation on a general M-polyfold. *Deviation (narrower):* slices replace the subsets $I^e$ and the sequences of top-good points; the independence of the right-hand side from the structure, which the text does not show, is proved with one slice for the sheets of both structures. In (1) "$I_k = I^1 \sqcup \cdots \sqcup I^p$" reads as $\Xi = I^1 \sqcup \cdots \sqcup I^p$, the points $x_e^k$ are not introduced, and "This completes the proof of Proposition 9.6.5" refers to Proposition 9.1.12 (possible misprints) (gap G.97) (Findings, items 91, 92).

**What is not verified in this chapter**

- p. 402, Remark 9.1.1 on a general M-polyfold: the local constancy of $\dim T_m M$ (the hypothesis `HasLocallyConstantDimAt`, proved on $\mathbb{R}^m$); the footnote to Definition 9.1.2 and "locally equidimensional" (V9.2) (gap G.93; Findings, items 2, 3).
- p. 403, Remark 9.1.4: the metrizability of a paracompact $L \subseteq |X|$ and the compactness criterion for the paracompactness of $|\mathrm{supp}(\Theta)|$ (gap G.94; Findings, item 7).
- p. 405, Proposition 9.1.9 (2017: 9.1) and Remark 9.1.10 (2017: 9.3): the natural manifold and orbifold structures on $|\mathrm{supp}(\Theta)|$ (gap G.95; Findings, item 7).
- pp. 407–408, Proposition 9.1.11 (2017: 9.2): the gluing of the local $\Theta_k$ into branched ep$^+$-subgroupoids and the local finiteness; the uniqueness, which does not hold as we read it, under a repaired condition (V9.9) (gap G.96; Findings, item 8).
- pp. 413–417, Proposition 9.2.8 as printed and Lemmas 9.2.9–9.2.11 (2017: Proposition 9.3, Lemmas 9.6–9.8) on a general M-polyfold: the graph representation is the hypothesis `HasGraphChartAt` (`HasGraphChartsAt`, `HasGraphBallAt`), proved on $\mathbb{R}^m$; on a general M-polyfold Proposition 9.1.12, the independence in Theorem 9.2.1, Lemma 9.3.4, Lemma 9.6.4 and Proposition 9.6.5 rest on it (V9.10, V9.14, V9.22, V9.32, V9.61, V9.62) (gap G.97; Findings, items 25, 100).
- p. 412, Lemma 9.2.7 (2) (2017: 9.5 (2)) on a general M-polyfold: invariance of domain (`HasDomainInvarianceAt`, proved on $\mathbb{R}^m$) (V9.21) (gap G.98; Findings, item 24).
- p. 418, Definition 9.2.12 (2017: 9.6): that the $TM_i$ are smooth submanifolds of $TX$; also on $\mathbb{R}^m$ (V9.23) (gap G.99; Findings, item 26).
- p. 423, Lemma 9.3.4 (2017: 9.10) on a general M-polyfold: the level-1 continuity of frame fields (`HasContinuousLevelOneOn`, proved on $\mathbb{R}^m$) (V9.32) (gap G.100; Findings, item 53).
- p. 425, Example 9.3.5 (2017: 9.1): a classification of all orientations of $\Theta$ (V9.33) (gap G.101; Findings, item 54).
- p. 426, Definition 9.3.6 (2017: 9.10): that a tame $M_i$ is a manifold with boundary with corners (V9.34) (gap G.102; Findings, item 55).

- pp. 427–429, Definitions 9.3.8 and 9.3.10 (2017: 9.9 and 9.11): $T_y\partial M = T_y^R M$ and the outward direction from the cones $C_y M$, the continuity of $o_{\partial M}$ and Figure 9.1; Remark 9.3.11 has no formal content (V9.36–V9.38) (gap G.103; Findings, items 57, 58).
- p. 430, Lemma 9.4.3 (2017: 9.12): Property (T) after shrinking $U(x)$ (V9.41) (gap G.104; Findings, item 71).
- pp. 431–432, Lemmas 9.4.4 and 9.4.6 (2017: 9.13, 9.14) for sheets in an M-polyfold: the fields of `SheetGeometry`; proved for Mathlib manifolds (V9.42, V9.44) (gap G.105; Findings, item 72).
- p. 437, Definition 9.5.1 and Lemma 9.5.2 (2017: Definition 9.15, Lemma 9.16) with Lebesgue null sets in the sheets of a general M-polyfold: "measure zero in $M_i$" is the abstract system `SheetNullSets` (or the fields of `SheetMeasureGeometry`), satisfiable on $\mathbb{R}^2$; Lemma 9.5.2 for structures with sheets of several dimensions (V9.49, V9.50) (gap G.106; Findings, items 80, 81).
- pp. 438–447, branched integration: Theorems 9.5.4 and 9.5.5 (2017: Theorem 9.2 and new), Definition 9.5.6, (9.21)–(9.22), the measure identities of Lemma 9.5.7, Lemmas 9.5.9–9.5.11 and Proposition 9.5.13, and Remark 9.5.8; they need the sc-differential forms of §8.2 and Part I (gaps G.74, G.80) (V9.54–V9.57) (gap G.107; Findings, items 84, 85).
- p. 448, $(n-1)$-dimensional measure zero in the boundary faces of the $M_i$ and a witness for $L(\partial S)$ with $\partial S \neq \emptyset$ (V9.52) (gap G.108; Findings, item 83).
- p. 449, Theorems 9.5.15 and 9.5.16 (2017: Theorem 9.3 and new), the canonical boundary measures (gap G.109; Findings, item 85).
- pp. 449–452, Theorems 9.5.17 and 9.5.18 (2017: Theorem 9.4 and new), Stokes' theorem, and Remark 9.5.20 (gap G.110; Findings, item 85). Remarks 9.5.12 (history and open questions) and 9.5.19 (on the partition-of-unity hypothesis) have no formal content.
- pp. 457–458, §§9.6.2–9.6.3: open questions, no formal content.
- Witnesses: every hypothesis of the chapter is satisfiable, and those of §9.2 and §9.6.1 hold on every ep-groupoid over $\mathbb{R}^m$; the hypothesis of Definition 9.2.12 has only the zero functor as a witness, and no witness of Lemma 9.2.13 at a boundary point is constructed (Findings, items 100–102).

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**11. Chapter 10 (HWZ, Part II, §§10.1–10.4 and Appendix §§10.6.1–10.6.2 (2017 §§10.1–10.4)): verified for equivalences, natural equivalences, the weak fibred product, the localization $\mathcal{EP}(E^{-1})$ and strong bundle equivalences, with the readings recorded below, except Remark 10.1.8, Propositions 10.1.9 and 10.1.10, the local regularity part of Proposition 10.1.19, the universal property of the localization, Proposition 10.3.16, Theorem 10.3.19, the (LR) and (SLR) parts of Theorem 10.3.21, the Fred part of Theorem 10.4.5 and Propositions 10.4.6 and 10.4.7; the strong bundle localization (§10.5) is not yet covered by this section**

Chapter 10 (pp. 461–520 (2017 pp. 333–374); numbers and pages are those of the book (Springer 2021), with the 2017 text's in parentheses) treats equivalences of ep-groupoids, the sc-smooth form of equivalences of categories, and prepares their inversion. §10.1 defines equivalences (Definition 10.1.1 (2017: 10.1)), characterizes them (Lemma 10.1.2 (2017: 10.1)), refines an ep-groupoid along an open covering (Proposition 10.1.3 (2017: 10.1)), recalls natural equivalences (Definition 10.1.4 (2017: 10.2)) and shows that they and compositions respect equivalences (Propositions 10.1.5, 10.1.6 (2017: 10.2, 10.3)), that the tangent of an equivalence is one (Theorem 10.1.7 (2017: 10.1)),

and, mostly new in the 2021 edition, which properties of Chapter 7 are invariant under equivalences: the local regularity property (Propositions 10.1.9, 10.1.10 (2017: 10.4), Remark 10.1.11 (2017: 10.1)), the non-effective part and the reduced ep-groupoid (Remark 10.1.12, Propositions 10.1.13, 10.1.14 (2017: 10.5), Example 10.1.15 (2017: 10.1)), (LUC), (ZHOU), (SLUC) and (SLR) (Propositions 10.1.16–10.1.19). §10.2 builds the weak fibred product $X \times_Y Z$ (Definition 10.2.1 (2017: 10.3)) and shows that it is an ep-groupoid with an equivalence to $Z$ when $F$ is an equivalence (Theorem 10.2.2 (2017: 10.2), Remark 10.2.3 (2017: 10.2)). §10.3 inverts the equivalences: diagrams $X \overset{F}{\longleftarrow} A \overset{\Phi}{\longrightarrow} Y$ with $F$ an equivalence, their refinements and equivalence (Definitions 10.3.1, 10.3.3, 10.3.5, Lemmas 10.3.2, 10.3.4 (2017: Definitions 10.4–10.6, Lemmas 10.3, 10.4)), the induced maps of orbit spaces (Lemma 10.3.6, Definition 10.3.7 (2017: 10.5, 10.7)), the composition of classes (Theorem 10.3.8 (2017: 10.3), proved in Appendix §10.6.1), generalized maps and isomorphisms (Definition 10.3.9, Theorem 10.3.10 (2017: 10.8, 10.4), the latter proved in Appendix §10.6.2), the category $\mathcal{EP}(E^{-1})$ with the functors $i : \mathcal{EP} \to \mathcal{EP}(E^{-1})$ and $|\cdot| : \mathcal{EP}(E^{-1}) \to \mathrm{TOP}$ (Proposition 10.3.11, Definition 10.3.12, Theorems 10.3.14, 10.3.15 (2017: Proposition 10.6, Definition 10.9, Theorems 10.5, 10.6)), and, mostly new in 2021, the invariance of the properties of Chapter 7 under generalized isomorphisms (Propositions 10.3.16, 10.3.17, Remark 10.3.18 (2017: 10.3), Theorem 10.3.19 (2017: 10.7), Definition 10.3.20, Theorem 10.3.21). §10.4 defines strong bundle equivalences (Definition 10.4.1 (2017: 10.10)), the pull-back and push-forward of section functors and their properties (Propositions 10.4.2–10.4.4 (2017: 10.7–10.9), Theorem 10.4.5 (2017: 10.8)), natural equivalences of strong bundle functors (Definitions 10.4.8, 10.4.10, Remark 10.4.9 (2017: Definitions 10.11, 10.12, Remark 10.4)) and the category $\mathcal{SEP}$ (Definition 10.4.11 (2017: 10.13)). §10.5, the localization of strong bundles, is not covered by this section yet.

What is verified. Every definition of §§10.1, 10.2 and 10.4 is formalized: Definitions 10.1.1 (V10.1), 10.1.4 (V10.4), 10.2.1 (V10.17), 10.4.1 (V10.22), 10.4.8 (V10.28), 10.4.10 (V10.30) and 10.4.11 (V10.31), with the class $E$ of equivalences, the refinement $X_{\mathcal{U}}$ (V10.3) and the pull-back and push-forward of section functors (V10.23). Lemma 10.1.2, Propositions 10.1.3, 10.1.5, 10.1.6, 10.1.13, 10.1.14, 10.1.16, 10.1.17 and 10.1.18, Theorem 10.1.7, Remark 10.1.12 and Remark 10.2.3 hold as stated; Remark 10.1.11 and Example 10.1.15 hold in stronger forms (V10.8, V10.12); Theorem 10.2.2 holds in all its claims, its first claim under the weaker hypothesis that $F$ or $G$ is a local sc-diffeomorphism on objects (V10.18–V10.20); Propositions 10.4.2–10.4.4 hold for all section functors, without sc-smoothness (V10.24–V10.26); Theorem 10.4.5 holds without its Fred part (V10.27); Remark 10.4.9 and the equivalence of Definitions 10.4.8 and 10.4.10 hold for strong bundle functors (V10.29, V10.30). Proposition 10.1.19 (SLR) is reduced to Proposition 10.1.9 for the lifts (V10.16). In §10.3 every definition is formalized (V10.33, V10.35, V10.37, V10.39, V10.41, V10.44, V10.49); the properties (E1)–(E3), Lemmas 10.3.2, 10.3.4, 10.3.6, Theorems 10.3.8, 10.3.10, 10.3.14, 10.3.15, Propositions 10.3.11 and 10.3.17 hold as stated, with the proofs of Appendix §§10.6.1–10.6.2 (V10.32–V10.47); Remark 10.3.18 is verified without its non-uniqueness claim (V10.48), and Theorem 10.3.21 for (LUC), (SLUC) and (ZHOU) (V10.50). Several places of the text needed more than a reading. (a) *Proposition 10.1.6 (2017: 10.3).* The argument on objects covers only the points $F(x)$; the transport to every $y$ along a morphism $F(x) \to y$ is supplied (V10.6). (b) *Theorem 10.1.7 (2017: 10.1).* Fullness and faithfulness of $TF$, left to the reader, are proved (V10.7); Proposition 10.1.5, quoted from the literature, is proved (V10.5). (c) *Propositions 10.1.13 and 10.1.16.* The closedness of $\mathbf{X}^T$ passes along equivalences by a closure argument with a conjugation by a local inverse of $t$ in place of sequences (V10.10, V10.13). (d) *Theorem 10.2.2 (2017: 10.2).* The ep-groupoid structure of $X \times_Y Z$ needs of $F$ only that it is a local sc-diffeomorphism on objects, its properness holds for any $F$ and $G$, and its M-polyfold structures are read as iterated sub-M-polyfolds of Part I's fibred products (V10.18); the composite in the fullness step is read with the ends it must have (V10.19). (e) *Defini-*

*tion 10.4.1 (2017: 10.10).* $W[1]$ is not an ep-groupoid in the project; clause (1) at level $[1]$ is read as "$|\Phi| : |W[1]| \to |W'[1]|$ is a homeomorphism", the rest of it being a consequence of (2) (V10.22). (f) *Theorem 10.4.5 (2017: 10.8).* The sc-smoothness of $\Phi_* f$, "easily verified" in the text, and the preservation of $\mathrm{sc}^+$-sections are proved (V10.27). (g) *Theorem 10.3.8 (2017: 10.3), Appendix §10.6.1.* The functor $L$ between weak fibred products is shown to be an equivalence by two out of three, and the text's identification of the two iterated weak fibred products ("the same space") is a reassociation functor, an equivalence (V10.40). (h) *Theorem 10.3.10 (2017: 10.4), Appendix §10.6.2.* In claim (C2) the text uses that a map $\Gamma'$ is a local sc-diffeomorphism, which, as we read it, already needs the other half of (C2); the local sc-diffeomorphism property of $\Phi$ is proved instead from a local sc-smooth right inverse and local injectivity (V10.42). (i) *Theorem 10.3.14 (2017: 10.5).* With Definition 10.3.1 as stated, it is the composite diagram (10.13) that refines $d_{\Psi\circ\Phi}$, not the reverse; the equality of classes is unchanged (V10.45). Possible misprints are recorded in V10.3, V10.6, V10.9, V10.14, V10.15, V10.17, V10.19, V10.24, V10.27, V10.40 and V10.42. Not verified: the subcategory $\mathcal{EP}_{lreg}$, Remark 10.1.8 with its example in Appendix §10.6.3, Propositions 10.1.9 and 10.1.10, Proposition 10.1.19 itself, the agreement of the M-polyfold structure of $X \times_Y Z$ with one induced from the triple products, the universal property of the localization, Proposition 10.3.16 with the (LR) and (SLR) parts of Theorem 10.3.21, the non-uniqueness claim of Remark 10.3.18, Theorem 10.3.19, the Fred part of Theorem 10.4.5, and Propositions 10.4.6 and 10.4.7 (gaps G.111–G.122).

The hypotheses. No hypothesis beyond the text's is taken on a general ep-groupoid, apart from three technical ones, each with a witness: the object set is nonempty in Proposition 10.1.3 (the open-subset M-polyfold structure of Part I needs a nonempty set; the trivial covering of any nonempty ep-groupoid and a two-set covering of $\mathbb{Z}/2 \ltimes \mathbb{R}$ are witnesses); the weak fibred product is given its ep-groupoid structure under `Polyfold.WeakFibredProduct.Admissible` ($F$ or $G$ a local sc-diffeomorphism on objects), which every equivalence supplies (`Polyfold.WeakFibredProduct.Admissible.of_i` `Polyfold.WeakFibredProduct.Admissible.of_isEquivalence_right`); and the reduction of Proposition 10.1.19 assumes Proposition 10.1.9 for the lifts, a hypothesis satisfied when the two ep-groupoids agree (`Polyfold.EP.isLR_iff_lift_self`). Proposition 10.1.14 keeps the text's hypothesis that both reduced morphism spaces are Hausdorff. In §10.3 no hypothesis is added; a diagram is a structure one universe above the object and morphism types of the ep-groupoids, so that the classes of diagrams are sets and $\mathcal{EP}(E^{-1})$ is a large category like $\mathcal{EP}$.

The setting. Ep-groupoids, sc-smooth functors, the orbit spaces with their quotient topologies, the category $\mathcal{EP}$ (with Mathlib's composition order: $f \gg g$ is the text's $g \circ f$, and so is $d.\mathrm{comp}\, e$ for diagrams) and the lifts $X^i$ are those of Chapter 7 (`Polyfold.EpGroupoid`, `Polyfold.EpFunctor`, `Polyfold.EP`, `Polyfold.EP.lift`); (LUC), (ZHOU), (LR), (SLUC) and (SLR) are Chapter 7's `Polyfold.EpGroupoid.IsLUC`, `Polyfold.EpGroupoid.IsZhou`, `Polyfold.EpGroupoid.IsLR`, `Polyfold.EP.IsSLUC`, `Polyfold.EP.IsSLR`, and the reduced ep-groupoid is `Polyfold.EpGroupoid.reducedEpGroupoid`. Natural transformations, natural equivalence and the tangent functor are Chapter 8's (`Polyfold.EpNatTrans`, `Polyfold.EpFunctor.NaturallyEquivalen` `Polyfold.EpFunctor.tangent`), and so are strong bundles over ep-groupoids, strong bundle functors and local strong bundle isomorphisms (`Polyfold.EpStrongBundle`, `Polyfold.EpStrongBundle.IsStrongBu` `Polyfold.StrongBundle.IsStrongBundleIsoOn`). "Full", "faithful", "essentially surjective" and "equivalence of categories" are Mathlib's notions for the underlying functor of the groupoids. Findings referred to below are those of `Exposition/Findings.md`, "Chapter 10".

**V10.1 (Definition 10.1.1 (2017: 10.1), equivalences and the class $E$;** `Polyfold.EpFunctor.IsEquivalence`, `Polyfold.EP.equivalences`, `Polyfold.GroupoidFunctor.full_iff`, `Polyfold.GroupoidFunctor.faithful_if` `Polyfold.GroupoidFunctor.essSurj_iff`, `Polyfold.EpFunctor.IsEquivalence.essSurj`,

`Polyfold.EpFunctor.IsEquivalence.exists_iso`, `Polyfold.EpFunctor.IsEquivalence.exists_mor_eq`, `Polyfold.EpFunctor.IsEquivalence.mor_injective`, `Polyfold.EpFunctor.IsEquivalence.id`, `Polyfold.EpGroupoid.Examples.pairToPoint`, `Polyfold.EpGroupoid.Examples.isEquivalence_pairToPoint`
An equivalence $F : X \to Y$ of ep-groupoids is an sc-smooth functor which is full and faithful (every $\psi : F(x) \to F(x')$ is $F(\varphi)$ for some $\varphi : x \to x'$, and $F$ is injective on each set of morphisms $x \to x'$), induces a homeomorphism $|F| : |X| \to |Y|$ of the orbit spaces, and is a local sc-diffeomorphism on objects. The equivalences form a class $E$ of morphisms of $\mathcal{EP}$ containing the identities and closed under composition. An equivalence is essentially surjective: every $y$ is the target of a morphism from some $F(x)$. The identity and the functor from the pair groupoid of $\mathbb{Z}$ to the point are equivalences. *Book:* Definition 10.1.1, p. 461 (2017: Definition 10.1, p. 333), with the category notions recalled on pp. 461–462. *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 1).

**V10.2 (Lemma 10.1.2 (2017: 10.1) and the remarks around it;** `Polyfold.EpFunctor.lemma_10_1_2`, `Polyfold.EpFunctor.isEquivalence_iff`, `Polyfold.EpFunctor.IsEquivalence.isEquivalence_toFunctor`, `Polyfold.EpFunctor.IsEquivalence.comp`, `Polyfold.GroupoidFunctor.injective_orbitMap_of_full`, `Polyfold.GroupoidFunctor.surjective_orbitMap_iff`, `Polyfold.GroupoidFunctor.bijective_orbitMap_of` `Polyfold.EpFunctor.continuous_orbitMap`, `Polyfold.EpFunctor.isOpenMap_orbitMap`, `Polyfold.EpGroupoid.Examples.not_exists_leftInverse_pairToPoint`**).** An sc-smooth functor which is an equivalence of categories and a local sc-diffeomorphism on objects is an equivalence, and conversely every equivalence is both. For a functor of groupoids, $|F|$ is injective if $F$ is full, surjective if and only if $F$ is essentially surjective, and bijective for an equivalence of categories; $|F|$ is continuous, and open if $F$ is open on objects. The composite of two equivalences is an equivalence, and the equivalence from the pair groupoid of $\mathbb{Z}$ to the point has no left inverse on objects. *Book:* Lemma 10.1.2, p. 462 (2017: Lemma 10.1, p. 334), and the remarks on pp. 461–462. *Hidden hypotheses:* none. *Deviation (stronger):* the lemma is stated with its converse; "not invertible as a functor" gets a witness.

**V10.3 (Proposition 10.1.3 (2017: 10.1), the refinement $X_{\mathcal{U}}$;** `Polyfold.EpGroupoid.OpenCover`, `Polyfold.EpGroupoid.OpenCover.univ`, `Polyfold.EpGroupoid.OpenCover.objSet`, `Polyfold.EpGroupoid.OpenCover.groupoidStr`, `Polyfold.EpGroupoid.OpenCover.objPolyfold`, `Polyfold.EpGroupoid.OpenCover.morPolyfold`, `Polyfold.EpGroupoid.OpenCover.isLocalScDiffeo_src`, `Polyfold.EpGroupoid.OpenCover.isLocalScDiffeo_tgt`, `Polyfold.EpGroupoid.OpenCover.exists_isProper` `Polyfold.EpGroupoid.OpenCover.isScSmoothCompSheet`, `Polyfold.EpGroupoid.OpenCover.epGroupoid`, `Polyfold.EpGroupoid.OpenCover.functor`, `Polyfold.EpGroupoid.OpenCover.proposition_10_1_3`, `Polyfold.EpGroupoid.Examples.reflectionCover`, `Polyfold.EpGroupoid.Examples.isEquivalence_reflect` `Polyfold.EpGroupoid.Examples.not_injective_reflectionCover_obj`**).** For a covering of the object M-polyfold of $X$ by open sets $U_i$, $i \in \iota$, the refinement $X_{\mathcal{U}}$ has as objects the pairs $(i, x)$ with $x \in U_i$, an open subset of $\iota \times X$ with $\iota$ discrete, and as morphisms the triples $(i, \varphi, j)$ with $s(\varphi) \in U_i$, $t(\varphi) \in U_j$, an open subset of $(\iota \times \iota) \times \mathbf{X}$, with $s(i, \varphi, j) = (i, s(\varphi))$, $t(i, \varphi, j) = (j, t(\varphi))$, $(j, \psi, k) \circ (i, \varphi, j) = (i, \psi \circ \varphi, k)$, units $(i, 1_x, i)$ and inverses $(j, \varphi^{-1}, i)$. If $X$ has an object, $X_{\mathcal{U}}$ is an ep-groupoid and $F : X_{\mathcal{U}} \to X$, $(i, x) \mapsto x$, $(i, \varphi, j) \mapsto \varphi$, is an equivalence. For the covering $\{x > -1\}$, $\{x < 1\}$ of $\mathbb{Z}/2 \ltimes \mathbb{R}$ this $F$ is an equivalence which is not injective on objects. *Book:* Proposition 10.1.3, p. 462 (2017: Proposition 10.1, p. 334), with the construction of $X_{\mathcal{U}}$ on p. 462. *Hidden hypotheses:* the object set is nonempty (the open-subset M-polyfold structure needs a nonempty set; for an empty $X$ the statement is trivial). *Deviation (explicit):* the covering is indexed by a type, the text's index set being the covering itself; "induced from $X$ and $\mathbf{X}$" is read as open subsets of products with a discrete factor; the printed $(U, \varphi, v)$ in

the formula for $F$ is read $(U, \varphi, V)$ (possible misprint) (Findings, item 2). *Proof:* the properness at $(i, x)$ takes a closed neighbourhood of $x$ inside $U_i$ and a neighbourhood with the properness property of $X$, and is proved with ultrafilters instead of sequences.

**V10.4 (Definition 10.1.4 (2017: 10.2), natural equivalence;** `Polyfold.definition_10_1_4`, `Polyfold.EpNatTrans`, `Polyfold.EpFunctor.NaturallyEquivalent`, `Polyfold.EpFunctor.naturallyEquivale` `Polyfold.EpFunctor.NaturallyEquivalent.comp_left`, `Polyfold.EpFunctor.NaturallyEquivalent.comp_ri` `Polyfold.EpNatTrans.toNatIso`**).** A natural transformation $\tau : F \to G$ of sc-smooth functors $X \to Y$ is an sc-smooth map $\tau : X \to \mathbf{Y}$ with $\tau(x) : F(x) \to G(x)$ and $\tau(x') \circ F(\varphi) = G(\varphi) \circ \tau(x)$ for $\varphi : x \to x'$; $F \simeq G$ if one exists. $\simeq$ is an equivalence relation, and $F \simeq G$ implies $F \circ \Psi \simeq G \circ \Psi$ and $\Phi \circ F \simeq \Phi \circ G$. A natural transformation is a natural isomorphism of the underlying functors. *Book:* Definition 10.1.4, p. 463 (2017: Definition 10.2, p. 335), and the claims after it, p. 464. *Hidden hypotheses:* none. *Deviation (same):* none; the definition is Definition 8.1.13 (Findings, item 3).

**V10.5 (Proposition 10.1.5 (2017: 10.2), natural equivalence preserves equivalences;** `Polyfold.EpFunctor.propos` `Polyfold.EpFunctor.IsEquivalence.of_natTrans`, `Polyfold.EpNatTrans.continuous_app`, `Polyfold.EpNatTrans.isLocalScDiffeo_app`, `Polyfold.EpNatTrans.isLocalScDiffeo_obj`**).** If $F \simeq G$, then $F$ is an equivalence if and only if $G$ is. *Book:* Proposition 10.1.5, p. 464 (2017: Proposition 10.2, p. 336), quoted from the literature without proof. *Hidden hypotheses:* none. *Deviation (same):* none. *Proof:* $s \circ \tau = F$ makes $\tau$ a local sc-diffeomorphism, so $G = t \circ \tau$ is one on objects; the underlying functors are naturally isomorphic, so $G$ is an equivalence of categories, and Lemma 10.1.2 applies (Findings, item 4).

**V10.6 (Proposition 10.1.6 (2017: 10.3), the two-out-of-three property;** `Polyfold.EpFunctor.proposition_10_1_6` `Polyfold.MPolyfold.IsLocalScDiffeoAt`, `Polyfold.MPolyfold.IsLocalScDiffeoAt.comp`, `Polyfold.MPolyfold.IsLocalScDiffeoAt.congr`, `Polyfold.MPolyfold.IsLocalScDiffeoAt.of_comp_left`, `Polyfold.MPolyfold.IsLocalScDiffeoAt.of_comp_right`, `Polyfold.EpFunctor.isLocalScDiffeoAt_mor`, `Polyfold.EpFunctor.isLocalScDiffeo_mor`, `Polyfold.EpFunctor.isLocalScDiffeoAt_obj_tgt`**).** If $F : X \to Y$ and $\Phi \circ F : X \to Z$ are equivalences and $\Phi : Y \to Z$ is sc-smooth, then $\Phi$ is an equivalence. A functor which is a local sc-diffeomorphism on objects at $s(\varphi)$ is one at $t(\varphi)$, and one on morphisms at $\varphi$. *Book:* Proposition 10.1.6, p. 464 (2017: Proposition 10.3, p. 336). *Hidden hypotheses:* none. *Deviation (same):* the text's argument on objects gives the local sc-diffeomorphism property of $\Phi$ only at the points $F(x)$; the transport along a morphism $F(x) \to y$ is added. The proof writes $|\Phi| : |X| \to |Y|$ for $|\Phi| : |Y| \to |Z|$ (possible misprint) (Findings, item 5). *Proof:* $\Phi$ is an equivalence of categories since $F$ and $\Phi \circ F$ are, and Lemma 10.1.2 applies.

**V10.7 (Theorem 10.1.7 (2017: 10.1), the tangent of an equivalence;** `Polyfold.EpFunctor.theorem_10_1_7`, `Polyfold.EpFunctor.tangent`**).** If $F$ is an equivalence, its tangent $TF : TX \to TY$ is an equivalence of the tangent ep-groupoids. *Book:* Theorem 10.1.7, p. 465 (2017: Theorem 10.1, p. 337). *Hidden hypotheses:* none. *Deviation (same):* none; fullness and faithfulness of $TF$, left to the reader, are proved (Findings, item 6). *Proof:* by Lemma 10.1.2; a tangent morphism is determined by its base morphism and its source vector, so faithfulness comes from that of $F$; fullness lifts the base morphism given by fullness of $F$ and uses the injectivity of $TF$ on tangent fibres; essential surjectivity lifts $\psi^{-1}$ for $\psi : F(x) \to y$ and uses that $TF$ maps $T_x X$ onto $T_{F(x)} Y$.

**V10.8 (Remark 10.1.11 (2017: 10.1), $\alpha_x$ cannot be prescribed;** `Polyfold.EpGroupoid.Examples.pointReflection` `Polyfold.EpGroupoid.Examples.negFunctor`, `Polyfold.EpGroupoid.Examples.isEquivalence_negFunctor`, `Polyfold.EpGroupoid.Examples.orbitMap_id_eq_negFunctor`, `Polyfold.EpGroupoid.Examples.remark_10_1` `Polyfold.EpGroupoid.Examples.remark_10_1_11`, `Polyfold.EpGroupoid.Examples.remark_10_1_11_scSmoot` On the translation groupoid of $\mathbb{Z}/2$ acting on $\mathbb{R}^2$ by $z \mapsto -z$, with $F = \mathrm{id}$ and $G = -\mathrm{id}$ (both

equivalences, $|F| = |G|$), $\alpha_z = (-1, z) : F(z) \to G(z)$ is an sc-smooth family, but no family $\alpha_z : F(z) \to G(z)$ defined near 0 with $\alpha_0 = 1_0$ is continuous at 0, and none is sc-smooth on an open neighbourhood of 0. *Book:* Remark 10.1.11, p. 470 (2017: Remark 10.1, p. 340). *Hidden hypotheses:* none. *Deviation (stronger):* "not in an sc-smooth local family" is proved in the form "not in a family continuous at 0"; that the conclusion of Proposition 10.1.10 (1) holds in the example is added (Findings, item 8).

**V10.9 (Remark 10.1.12, isotropy groups under equivalences;** `Polyfold.EpFunctor.isotropyHom`, `Polyfold.EpFunctor.isotropyHom_mem_trivialSet_iff`, `Polyfold.EpFunctor.IsEquivalence.bijective_is` `Polyfold.EpFunctor.IsEquivalence.remark_10_1_12`**).** For a functor which is a local sc-diffeomorphism on objects, $g \mapsto F(g)$, $G_x \to G_{F(x)}$, maps $g$ into $T_{F(x)}$ if and only if $g \in T_x$; for an equivalence it is bijective, and for $\psi : F(x) \to y$ there is a group isomorphism $G_x \to G_y$, $g \mapsto \psi \circ F(g) \circ \psi^{-1}$, mapping $T_x$ onto $T_y$. *Book:* Remark 10.1.12, p. 472 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (same):* the printed $G_x \to Y_y$ is read $G_x \to G_y$ (possible misprint) (Findings, item 9).

**V10.10 (Proposition 10.1.13, the reduced morphism spaces;** `Polyfold.EpFunctor.mor_mem_nonEffectivePart`, `Polyfold.EpFunctor.mem_nonEffectivePart_of_mor`, `Polyfold.EpFunctor.nonEffectivePart_eq`, `Polyfold.EpGroupoid.continuousAt_comp`, `Polyfold.EpFunctor.IsEquivalence.isClosed_nonEffectivePar` `Polyfold.EpFunctor.reducedMap`, `Polyfold.EpFunctor.continuous_reducedMap`, `Polyfold.EpFunctor.isOpe` `Polyfold.EpFunctor.proposition_10_1_13`, `Polyfold.EpGroupoid.theorem_7_2_22_2021`**).** For a functor which is a local sc-diffeomorphism on objects, $\mathbf{X}^T = F^{-1}(\mathbf{Y}^T) \cap \{s = t\}$ and $F_R : \mathbf{X}^R \to \mathbf{Y}^R$, $[\varphi] \mapsto [F(\varphi)]$, is well defined, continuous and open. For an equivalence $F$, $\mathbf{X}^T$ is closed if and only if $\mathbf{Y}^T$ is, and $\mathbf{X}^R$ is Hausdorff if and only if $\mathbf{Y}^R$ is. *Book:* Proposition 10.1.13, p. 472 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (same):* (1) is proved for any functor which is a local sc-diffeomorphism on objects (Findings, item 10). *Proof:* (2) through Theorem 7.2.22 ($T_R$ Hausdorff if and only if $\mathbf{X}^T$ closed), as in the text; the closedness passes from $\mathbf{X}^T$ to $\mathbf{Y}^T$ by conjugating with a local inverse of $t$ at a morphism $\sigma : F(x) \to y$, with closures in place of sequences.

**V10.11 (Proposition 10.1.14 (2017: 10.5), the reduced functor;** `Polyfold.EpFunctor.reducedGroupoidFunctor`, `Polyfold.EpFunctor.reduced`, `Polyfold.EpFunctor.reduced _eq_of_mor`, `Polyfold.EpFunctor.proposition` If $F$ is an equivalence and the reduced morphism spaces $\mathbf{X}^R$, $\mathbf{Y}^R$ are Hausdorff, then $F_R = F$ on objects, $F_R([\varphi]) = [F(\varphi)]$ is a well-defined sc-smooth functor $X_R \to Y_R$ of the reduced ep-groupoids and an equivalence. *Book:* Proposition 10.1.14, p. 473 (2017: Proposition 10.5, p. 341). *Hidden hypotheses:* none. *Deviation (same):* none; by Proposition 10.1.13 (2) one Hausdorff hypothesis would suffice (Findings, item 11).

**V10.12 (Example 10.1.15 (2017: 10.1), a functor that does not pass to the reduced groupoids;** `Polyfold.EpGroupoid.Examples.example1015X`, `Polyfold.EpGroupoid.Examples.example1015Y`, `Polyfold.EpGroupoid.Examples.example1015Functor`, `Polyfold.EpGroupoid.Examples.example1015X_two_n` `Polyfold.EpGroupoid.Examples.example1015Y_two_not_mem_nonEffectivePart`, `Polyfold.EpGroupoid.Exam` `Polyfold.EpGroupoid.Examples.example1015Y_reduced _one_ne_three`, `Polyfold.EpGroupoid.Examples.ex` For $X = \mathbb{Z}/4 \ltimes \mathbb{R}$ (1 and 3 acting by $x \mapsto -x$, 2 trivially), $Y = \mathbb{Z}/4 \ltimes \mathbb{R}^2$ (rotations by multiples of 90°) and the sc-smooth functor $\Phi(x) = 0$, $\Phi(i, x) = (i, 0)$: $(2, x) \in \mathbf{X}^T$, $(2, 0) \notin \mathbf{Y}^T$, $[(1, x)] = [(3, x)]$ in $\mathbf{X}^R$ and $[(1, 0)] \neq [(3, 0)]$ in $\mathbf{Y}^R$, so no map $f : \mathbf{X}^R \to \mathbf{Y}^R$ satisfies $f([\varphi]) = [\Phi(\varphi)]$. *Book:* Example 10.1.15, p. 474 (2017: Example 10.1, p. 341). *Hidden hypotheses:* none. *Deviation (stronger):* no Hausdorff hypothesis and no topology on the reduced spaces is involved; no map at all is compatible with $\Phi$ (Findings, item 12).

**V10.13 (Proposition 10.1.16, (LUC) under equivalences;** `Polyfold.EpFunctor.proposition_10_1_16`, `Polyfold.EpFunctor.IsEquivalence.isClosed_nonEffectivePart_iff`, `Polyfold.EpGroupoid.proposition_`
If $F : X \to Y$ is an equivalence, $X$ has the local unique continuation property if and only if $Y$ has it. *Book:* Proposition 10.1.16, p. 474 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 10). *Proof:* (LUC) holds if and only if $\mathbf{X}^T$ is closed (Proposition 7.2.21, a part of Theorem 7.2.22), and the closedness passes along equivalences (V10.10).

**V10.14 (Proposition 10.1.17, the Zhou condition under equivalences;** `Polyfold.EpGroupoid.mem_smoothSet_sr` `Polyfold.EpGroupoid.mem_smoothSet_tgt_iff`, `Polyfold.EpFunctor.tangentMor_mor`, `Polyfold.EpFunctor.IsEquivalence.isZhou_of_isZhou`, `Polyfold.EpFunctor.IsEquivalence.isZhou_of_is` `Polyfold.EpFunctor.proposition_10_1_17`**).** If $F : X \to Y$ is an equivalence, $X$ satisfies the Zhou condition (for smooth $x$ and $g \in G_x$, $Tg = \mathrm{id}$ on $T_xX$ implies $g \in T_x$) if and only if $Y$ does. *Book:* Proposition 10.1.17, p. 475 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (same):* the proof uses the naturality $T(F(\varphi)) \circ TF = TF \circ T\varphi$, the composition rule for $T$ and the bijectivity of $TF$ on tangent fibres instead of the natural representations; the text's $U(F(y))$ and "$x \in U(x)$" in (b) are read $U(F(x))$ and $a \in U(x)$ (possible misprints) (Findings, item 13).

**V10.15 (Proposition 10.1.18, (SLUC) under equivalences;** `Polyfold.EpFunctor.liftOne`, `Polyfold.EpFunctor.IsEquivalence.liftOne`, `Polyfold.EP.exists_isEquivalence_lift`, `Polyfold.EP.proposition_10_1_18`**).** The restriction $F^1 : X^1 \to Y^1$ of an sc-smooth functor to level 1 is an sc-smooth functor, and an equivalence if $F$ is; hence for every $i$ there is an equivalence $X^i \to Y^i$, and $X$ satisfies (SLUC) if and only if $Y$ does. *Book:* Proposition 10.1.18, p. 476 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (same):* the proof cites Proposition 10.1.9 for the step (LUC) along $F^i$, which is Proposition 10.1.16 (possible misprint) (Findings, item 14).

**V10.16 (Proposition 10.1.19, (SLR) under equivalences, reduced to Proposition 10.1.9;** `Polyfold.EP.isSLR_iff_of_isLR_iff`, `Polyfold.EP.isLR_iff_lift_self`**).** If $F$ is an equivalence and, for every $i$, every equivalence $X^i \to Y^i$ carries (LR) in both directions, then $X$ satisfies (SLR) if and only if $Y$ does; the hypothesis holds for $Y = X$. *Book:* Proposition 10.1.19, p. 477 (new in the 2021 edition). *Hidden hypotheses:* Proposition 10.1.9 for the lifts, as a hypothesis. *Deviation (narrower):* Proposition 10.1.9 is not formalized (gap G.113), so only the reduction, which is the text's proof, is verified (gap G.115; Findings, item 15).

**V10.17 (Definition 10.2.1 (2017: 10.3), the weak fibred product as a groupoid;** `Polyfold.WeakFibredProduct.obj` `Polyfold.WeakFibredProduct.objSet`, `Polyfold.WeakFibredProduct.Obj`, `Polyfold.WeakFibredProduct.mor` `Polyfold.WeakFibredProduct.morSet`, `Polyfold.WeakFibredProduct.Mor`, `Polyfold.WeakFibredProduct.con` `Polyfold.WeakFibredProduct.conj_comp_mor`, `Polyfold.WeakFibredProduct.conj_recover`, `Polyfold.WeakFibredProduct.src`, `Polyfold.WeakFibredProduct.tgt`, `Polyfold.WeakFibredProduct.unit`, `Polyfold.WeakFibredProduct.inv`, `Polyfold.WeakFibredProduct.comp`, `Polyfold.WeakFibredProduct.src_e` `Polyfold.WeakFibredProduct.groupoid`**).** For sc-smooth functors $F : X \to Y$ and $G : Z \to Y$, the objects of $X \times_Y Z$ are the triples $(x, \phi, z)$ with $\phi : F(x) \to G(z)$, the morphisms the triples $(h, \phi, k)$ with $\phi : F(s(h)) \to G(s(k))$; $s(h, \phi, k) = (s(h), \phi, s(k))$, $t(h, \phi, k) = (t(h), G(k) \circ \phi \circ F(h)^{-1}, t(k))$, the unit at $(x, \phi, z)$ is $(1_x, \phi, 1_z)$, the inverse is $(h^{-1}, G(k) \circ \phi \circ F(h)^{-1}, k^{-1})$ and the product of a composable pair is $(h \circ h', \phi', k \circ k')$. This is a groupoid. *Book:* Definition 10.2.1, p. 477 (2017: Definition 10.3, p. 342). *Hidden hypotheses:* none. *Deviation (restated):* objects and morphisms are elements of iterated fibred products, $X \,{}_F\!\times_s (\mathbf{Y}\,{}_t\!\times_G Z)$ and $\mathbf{X}\,{}_{F\circ s}\!\times_s (\mathbf{Y}\,{}_t\!\times_{G\circ s} \mathbf{Z})$; the multiplication is a total map whose value off composable pairs is not used; the printed inverse $\iota(h, \psi, k) = (h^{-1}, G(k) \circ \phi \circ F(h)^{-1}, k^{-1})$ is read with $\psi = \phi$ (possible misprint) (Findings, item 20).

**V10.18 (Theorem 10.2.2 (2017: 10.2), first claim: $X \times_Y Z$ is an ep-groupoid;** `Polyfold.WeakFibredProduct.Admis`
`Polyfold.WeakFibredProduct.Admissible.of_isEquivalence_left`, `Polyfold.WeakFibredProduct.Admissib`
`Polyfold.WeakFibredProduct.PA`, `Polyfold.WeakFibredProduct.PB`, `Polyfold.WeakFibredProduct.PObj`,
`Polyfold.WeakFibredProduct.PMor`, `Polyfold.WeakFibredProduct.isSubMPolyfold_objSet`,
`Polyfold.WeakFibredProduct.isSubMPolyfold_morSet`, `Polyfold.WeakFibredProduct.isLocalScDiffeo_z`,
`Polyfold.WeakFibredProduct.isLocalScDiffeo_k`, `Polyfold.WeakFibredProduct.isLocalScDiffeo_x`,
`Polyfold.WeakFibredProduct.isLocalScDiffeo_h`, `Polyfold.WeakFibredProduct.isLocalScDiffeo_src`,
`Polyfold.WeakFibredProduct.isLocalScDiffeo_tgt`, `Polyfold.WeakFibredProduct.isScSmoothMap_unit`,
`Polyfold.WeakFibredProduct.isScSmoothMap_inv`, `Polyfold.WeakFibredProduct.isScSmoothComp`,
`Polyfold.WeakFibredProduct.exists_isProperMap`, `Polyfold.WeakFibredProduct.epGroupoid`,
`Polyfold.MPolyfold.IsLocalScDiffeo.of_comm`, `Polyfold.MPolyfold.isLocalScDiffeo_snd_fibredProduct`
`Polyfold.MPolyfold.isLocalScDiffeo_fst_fibredProduct`, `Polyfold.EpGroupoid.isScSmoothOn_comp_of`,
`Polyfold.EpGroupoid.tendsto_comp`**).** If $F$ or $G$ is a local sc-diffeomorphism on objects, in particular if one of them is an equivalence, the groupoid of Definition 10.2.1 with the object M-polyfold $X \,{}_F\!\times_s (\mathbf{Y}\,{}_t\!\times_G Z)$ and the morphism M-polyfold $\mathbf{X} \,{}_{F\circ s}\!\times_s (\mathbf{Y}\,{}_t\!\times_{G\circ s} \mathbf{Z})$, iterated sub-M-polyfolds of products, is an ep-groupoid. For $F$ a local sc-diffeomorphism on objects, $(x,\phi,z) \mapsto z$ and $(h,\phi,k) \mapsto k$ are local sc-diffeomorphisms ((10.9), (10.10)); for $G$ one, $(x,\phi,z) \mapsto x$ and $(h,\phi,k) \mapsto h$ are. The properness of Definition 7.1.3 (3) holds for any $F$ and $G$. *Book:* Theorem 10.2.2, p. 477 (2017: Theorem 10.2, p. 343), first claim, and its proof, pp. 478–481 (2017 pp. 343–346). *Hidden hypotheses:* none. *Deviation (stronger):* of $F$ the construction uses only that it is a local sc-diffeomorphism on objects. The text's "sub-M-polyfolds of $X \times \mathbf{Y} \times Z$ and $\mathbf{X} \times \mathbf{Y} \times \mathbf{Z}$" is read as the iterated sub-M-polyfold structures of Part I's fibred products (Proposition 2.5.20); their agreement with a structure induced directly from the triple products is not formalized (gap G.116); the text's "$t : s^{-1}(U) \to X$ proper" is read with $\mathrm{cl}(U)$, as in Definition 7.1.3 (3) (Findings, items 21, 22). *Proof:* $s$ and $t$ are local sc-diffeomorphisms by the diagrams $\pi_Z \circ s = s_Z \circ \pi_{\mathbf{Z}}$ and $\pi_Z \circ t = t_Z \circ \pi_{\mathbf{Z}}$; the properness is the text's sequential argument, with $\phi_n = G(k_n)^{-1} \circ (G(k_n) \circ \phi_n \circ F(h_n)^{-1}) \circ F(h_n)$.

**V10.19 (Theorem 10.2.2 (2017: 10.2), the case $F$ an equivalence;** `Polyfold.WeakFibredProduct.projLeft`,
`Polyfold.WeakFibredProduct.projRight`, `Polyfold.WeakFibredProduct.natTrans`, `Polyfold.WeakFibredPro`
`Polyfold.WeakFibredProduct.naturallyEquivalent`, `Polyfold.WeakFibredProduct.isEquivalence_projRig`
`Polyfold.WeakFibredProduct.degIndex_eq_z`, `Polyfold.WeakFibredProduct.theorem_10_2_2`,
`Polyfold.EpFunctor.IsEquivalence.degIndex_obj`, `Polyfold.EpFunctor.IsEquivalence.isTame_of_isTame`
`Polyfold.EpFunctor.IsEquivalence.hasParacompactOrbitSpace`, `Polyfold.WeakFibredProduct.Examples.t`
The projections $\pi_X$ and $\pi_Z$ are sc-smooth functors and $\sigma(x,\phi,z) = \phi$ is a natural transformation $F \circ \pi_X \to G \circ \pi_Z$ whenever $X \times_Y Z$ is defined. If $F$ is an equivalence and $G$ is sc-smooth, then $\pi_Z$ is an equivalence, $F \circ \pi_X \simeq G \circ \pi_Z$, $d_{X\times_Y Z}(x,\phi,z) = d_Z(z)$, and if $Z$ is tame, or has a paracompact orbit space, so has $X \times_Y Z$. An equivalence preserves the degeneracy index and pulls back tameness and the paracompactness of the orbit space. The weak fibred product of the equivalence from the pair groupoid of $\mathbb{Z}$ to the point with the identity of the point is a nonempty ep-groupoid whose projection to the point is an equivalence. *Book:* Theorem 10.2.2, pp. 477–478 (2017: Theorem 10.2, p. 343), with the paragraph after it, p. 478, and its proof, pp. 481–482. *Hidden hypotheses:* none. *Deviation (same):* in the fullness step the printed $\phi' \circ G(k) \circ \phi^{-1}$, which does not compose, is read $\phi'^{-1} \circ G(k) \circ \phi : F(x) \to F(x')$ (possible misprint); essential surjectivity is taken from that of $F$ (Findings, item 23). *Proof:* $\pi_Z$ is full and faithful because the unique morphism over $k : z \to z'$ is $(h,\phi,k)$ with $F(h) = \phi'^{-1} \circ G(k) \circ \phi$, essentially surjective because every $G(z)$ receives a morphism from some $F(x)$, and Lemma 10.1.2 applies.

**V10.20 (Theorem 10.2.2 (2017: 10.2), the cases $G$ an equivalence and both;** `Polyfold.WeakFibredProduct.isEqui`

`Polyfold.WeakFibredProduct.degIndex_eq_x`, `Polyfold.WeakFibredProduct.theorem_10_2_2_right`, `Polyfold.WeakFibredProduct.theorem_10_2_2_both`**).** If $G$ is an equivalence, $\pi_X$ is an equivalence, $F \circ \pi_X \simeq G \circ \pi_Z$, $d_{X\times_Y Z}(x,\phi,z) = d_X(x)$, and tameness and a paracompact orbit space pass from $X$ to $X \times_Y Z$; if both are, $d_X(x) = d_{X\times_Y Z}(x,\phi,z) = d_Z(z)$. Since the admissibility is a proposition, the two cases give the same ep-groupoid. *Book:* Theorem 10.2.2, p. 478 (2017: Theorem 10.2, p. 343). *Hidden hypotheses:* none. *Deviation (stronger):* for $G$ an equivalence the text states only that $\pi_X$ is an equivalence and the degeneracy index; the natural equivalence and the transfer of tameness and paracompactness are added (Findings, item 24).

**V10.21 (Remark 10.2.3 (2017: 10.2), two equivalences;** `Polyfold.WeakFibredProduct.remark_10_2_3`, `Polyfold.WeakFibredProduct.Examples.remark_10_2_3_negFunctor`**).** For equivalences $F : X \to Y$ and $G : Z \to Y$, both projections of $X \times_Y Z$ are equivalences and $\tau(x,\phi,z) = \phi$ is a natural transformation $F \circ \pi_X \to G \circ \pi_Z$. For the identity and $-\mathrm{id}$ of $\mathbb{Z}/2 \ltimes \mathbb{R}^2$ the weak fibred product is nonempty and both projections are equivalences. *Book:* Remark 10.2.3, pp. 482–483 (2017: Remark 10.2, p. 347). *Hidden hypotheses:* none. *Deviation (same):* none; the closing sentences on "smaller models" have no formal content (Findings, item 25).

**V10.22 (Definition 10.4.1 (2017: 10.10), strong bundle equivalences;** `Polyfold.EpStrongBundle.IsStrongBundle`
`Polyfold.EpStrongBundle.OneOrbitSpace`, `Polyfold.EpStrongBundle.oneOrbitMap`, `Polyfold.EpStrongBund`
`Polyfold.EpStrongBundle.IsStrongBundleEquivalence.covers`, `Polyfold.EpStrongBundle.IsStrongBundle`
`Polyfold.EpStrongBundle.IsStrongBundleEquivalence.eq_of_obj_eq`, `Polyfold.EpStrongBundle.IsStrong`
`Polyfold.EpStrongBundle.IsStrongBundleEquivalence.isLocalScDiffeo_one`, `Polyfold.EpStrongBundle.i`
`Polyfold.EpStrongBundle.IsStrongBundleEquivalence.comp`, `Polyfold.EpStrongBundle.isStrongBundleEc`
`Polyfold.StrongBundleEquivalenceExamples.isStrongBundleEquivalence_pullbackProj'`,
`Polyfold.StrongBundleEquivalenceExamples.trivialNeg_obj_ne_id`**).** A strong bundle equivalence $\Phi : (P : W \to X, \mu) \to (P' : W' \to X', \mu')$ is a strong bundle functor $\Phi : (W, \mathbf{W}) \to (W', \mathbf{W}')$ (linear on fibres, satisfying (10.14)) covering an equivalence $\phi : X \to X'$, with (1) $\Phi : W[0] \to W'[0]$ an equivalence of ep-groupoids and $|\Phi| : |W[1]| \to |W'[1]|$ a homeomorphism, and (2) every point of $X$, and every morphism of $\mathbf{X}$, having an open neighbourhood over which $\Phi$ on objects, respectively on morphisms, is a strong bundle isomorphism covering $\phi$. The fibre maps $W_x \to W'_{\phi(x)}$ are then topological linear isomorphisms and $\Phi : W[1] \to W'[1]$ is a local sc-diffeomorphism. The identity and composites of strong bundle equivalences are strong bundle equivalences, and so is every isomorphism of strong bundles over ep-groupoids; the projection $(-\mathrm{id})^*(\mathbb{R}^n \times \mathbb{R}^k) \to \mathbb{R}^n \times \mathbb{R}^k$ over $-\mathrm{id}$ is one, not over the identity for $n \geq 1$. *Book:* Definition 10.4.1, p. 496 (2017: Definition 10.10, p. 357). *Hidden hypotheses:* none. *Deviation (restated):* $W[1]$ is not built as an ep-groupoid in the project; (1) at level $[1]$ is read as the homeomorphism of $|W[1]|$, the quotient of $W_{0,1}$ by the orbit relation of $W$; the local sc-diffeomorphism property at level $[1]$ follows from (2), and fullness and faithfulness at level $[1]$ are those at level $[0]$ restricted (Findings, item 40).

**V10.23 (the pull-back and the push-forward of section functors, (10.15)–(10.17) and (10.22);**
`Polyfold.EpStrongBundle.IsStrongBundleEquivalence.pullback`, `Polyfold.EpStrongBundle.IsStrongBunc`
`Polyfold.EpStrongBundle.IsStrongBundleEquivalence.pullback_eq`, `Polyfold.EpStrongBundle.IsStrongB`
`Polyfold.EpStrongBundle.IsStrongBundleEquivalence.preObj`, `Polyfold.EpStrongBundle.IsStrongBundle`
`Polyfold.EpStrongBundle.IsStrongBundleEquivalence.pushforward`, `Polyfold.EpStrongBundle.IsStrongB`
`Polyfold.EpStrongBundle.IsStrongBundleEquivalence.isFunctorialSection_pushforward`,
`Polyfold.EpStrongBundle.IsStrongBundleEquivalence.pullback_pushforward`, `Polyfold.EpStrongBundle.`
`Polyfold.EpStrongBundle._injective_snd`, `Polyfold.EpStrongBundle.exists_eq_of_orbitMap_eq`,
`Polyfold.EpStrongBundle.orbitMap_eq_of_forall_exists`**).** For a strong bundle equivalence

$\Phi$ covering $\phi$, $\Phi^*(f')(x) = \Phi_x^{-1}(f'(\phi(x)))$ is the unique $w \in W_x$ with $\Phi(w) = f'(\phi(x))$, and $(\Phi_* f)(x') = \mu'(\sigma, \Phi(f(x)))$ for a chosen $x$ and $\sigma : \phi(x) \to x'$. For a section functor $f$ the value does not depend on the choice: $(\Phi_* f)(x') = \mu'(\sigma', \Phi(f(y)))$ for every $y$ and $\sigma' : \phi(y) \to x'$ ((10.22)). Pull-backs and push-forwards of section functors are section functors, and $\Phi^*\Phi_* f = f$, $\Phi_*\Phi^* f' = f'$. $|\Phi| = |\Psi|$ if and only if every $\Phi(w)$ is connected to $\Psi(w)$ by a morphism $(\sigma, \Phi(w))$ with $\mu'(\sigma, \Phi(w)) = \Psi(w)$ ((10.15)). *Book:* §10.4, pp. 496–498 (2017 pp. 357–360). *Hidden hypotheses:* none. *Deviation (restated):* $\Phi^*$ and $\Phi_*$ are defined on all maps, by the zero of the fibre where the value is not in the right fibre (never for sections), and the choice of $x$ and $\sigma$ is made once for all (Findings, item 41).

**V10.24 (Proposition 10.4.2 (2017: 10.7), pull-backs along $|\Phi| = |\Psi|$; `Polyfold.EpStrongBundle.proposition_10` `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.pullback_eq_of_orbitMap_eq`).** If $\Phi$ and $\Psi$ are strong bundle equivalences $(P, \mu) \to (P', \mu')$ covering $\phi$ and $\psi$ with $|\Phi| = |\Psi|$, then $\Phi^*(f') = \Psi^*(f')$ for every section functor $f'$ of $(P', \mu')$, sc-smooth or not. *Book:* Proposition 10.4.2, p. 497 (2017: Proposition 10.7, p. 358). *Hidden hypotheses:* none. *Deviation (stronger):* sc-smoothness of $f'$ is not used; the printed "$f' : W' \to X'$" is read $f' : X' \to W'$ (possible misprint) (Findings, item 42). *Proof:* with $g = \Psi^* f'$ and a morphism $(\sigma, \Phi(g(x)))$ with $\mu'(\sigma, \Phi(g(x))) = \Psi(g(x))$, functoriality of $f'$ gives $\mu'(\sigma, \Phi(g(x))) = \mu'(\sigma, \Phi(\Phi^* f'(x)))$; $\mu'(\sigma, \cdot)$ and $\Phi_x$ are injective.

**V10.25 (Proposition 10.4.3 (2017: 10.8), push-forwards along $|\Phi| = |\Psi|$; `Polyfold.EpStrongBundle.proposition` `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.pushforward_eq_of_orbitMap_eq`).** Under the hypotheses of Proposition 10.4.2, $\Phi_* f = \Psi_* f$ for every section functor $f$ of $(P, \mu)$, sc-smooth or not. *Book:* Proposition 10.4.3, p. 499 (2017: Proposition 10.8, p. 360). *Hidden hypotheses:* none. *Deviation (stronger):* sc-smoothness of $f$ is not used (Findings, item 43).

**V10.26 (Proposition 10.4.4 (2017: 10.9), functoriality; `Polyfold.EpStrongBundle.proposition_10_4_4`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.comp`).** For strong bundle equivalences $\Phi : (P, \mu) \to (P', \mu')$ and $\Psi : (P', \mu') \to (P'', \mu'')$, $(\Psi \circ \Phi)^* = \Phi^* \circ \Psi^*$ on all sections of $P''$ and $(\Psi \circ \Phi)_* = \Psi_* \circ \Phi_*$ on all section functors of $(P, \mu)$. *Book:* Proposition 10.4.4, p. 499 (2017: Proposition 10.9, p. 360). *Hidden hypotheses:* none. *Deviation (stronger):* the pull-back identity holds for all sections and the push-forward identity for all section functors, without sc-smoothness (Findings, item 43).

**V10.27 (Theorem 10.4.5 (2017: 10.8), without the Fred part; `Polyfold.EpStrongBundle.theorem_10_4_5`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.eqOn_pullback`, `Polyfold.EpStrongBundle.IsStron` `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.isScPlusSection_pullback`, `Polyfold.EpStrongBun` `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.isScSmoothSection_pushforward`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.isScPlusSection_pushforward`, `Polyfold.StrongBundle.isScSmoothOn_to_pullback_one`, `Polyfold.MPolyfold.isScSmoothOn_of_forall_ch` `Polyfold.StrongBundle.IsStrongBundleIsoOn.restrict`, `Polyfold.StrongBundle.IsStrongBundleIsoOn.co` `Polyfold.StrongBundle.IsStrongBundleIsoOn.exists_comp`, `Polyfold.StrongBundleEquivalenceExamples.`** For a strong bundle equivalence $\Phi$ covering $\phi$, $\Phi_* : \Gamma(P, \mu) \to \Gamma(P', \mu')$ and $\Phi^* : \Gamma(P', \mu') \to \Gamma(P, \mu)$ are mutually inverse bijections of the sc-smooth section functors; if $\Psi$ is another strong bundle equivalence with $|\Phi| = |\Psi|$, then $\Phi_* = \Psi_*$ and $\Phi^* = \Psi^*$ on them; and $\Phi_*$, $\Phi^*$ map $\mathrm{sc}^+$-section functors to $\mathrm{sc}^+$-section functors. For the projection $(-\mathrm{id})^*(\mathbb{R}^n \times \mathbb{R}^k) \to \mathbb{R}^n \times \mathbb{R}^k$ the push-forward is a bijection of the section functors. *Book:* Theorem 10.4.5, pp. 500–501 (2017: Theorem 10.8, p. 362), with the paragraph before it, p. 500. *Hidden hypotheses:* none. *Deviation (narrower):* the preservation of sc-Fredholm section functors is not formalized (gap G.121); the second display's $\Phi^* : \Gamma(P', \mu') \to \Gamma(P, \mu)$ is read with $\Gamma^+$ (possible misprint) (Findings, item

44). *Proof:* the sc-smoothness of $\Phi_* f$, called easily verified in the text, comes from the local form $\Phi_* f = \mu' \circ (\tau, \Phi \circ f \circ g_1 \circ s' \circ \tau)$ near $x'_0$, with $\tau$ a local inverse of $t'$ and $g_1$ one of $\phi$; at level $[1]$ through the universal property of the pull-back on an open set.

**V10.28 (Definition 10.4.8 (2017: 10.11), natural equivalence of strong bundle functors;** `Polyfold.EpStrongBundle.StrongBundleNatTrans`**,** `Polyfold.EpStrongBundle.StrongBundleNatEquiv`**,** `Polyfold.EpStrongBundle.StrongBundleNatTrans.refl`**,** `Polyfold.StrongBundleEquivalenceExamples.stro`
A natural transformation between strong bundle functors $\Phi, \Psi : (W, \mathbf{W}) \to (W', \mathbf{W}')$ covering $\phi, \psi$ is a map $T$ assigning to each $w$ a morphism $T(w) : \Phi(w) \to \Psi(w)$ of $(W', \mathbf{W}')$ with $T(w') \circ \Phi(h) = \Psi(h) \circ T(w)$ for $h : w \to w'$, together with a natural transformation $\tau : \phi \to \psi$ with $\tau(P(w)) = P'(T(w))$; $\Phi \simeq \Psi$ if one exists. Every strong bundle functor is naturally equivalent to itself. *Book:* Definition 10.4.8, p. 502 (2017: Definition 10.11, p. 362). *Hidden hypotheses:* none. *Deviation (restated):* $T$ is data without smoothness, as in the text, and $\tau$ is sc-smooth (Definition 10.1.4) (Findings, item 46).

**V10.29 (Remark 10.4.9 (2017: 10.4) and (10.23);** `Polyfold.EpStrongBundle.StrongBundleNatTrans.T_val`**,** `Polyfold.EpStrongBundle.StrongBundleNatTrans.T_eq`**,** `Polyfold.EpStrongBundle.StrongBundleNatTrans.unique`**,** `Polyfold.EpStrongBundle.StrongBundleNatTrans.isStrongBundleMap_T`**).** For a natural transformation $(T, \tau)$ from $\Phi$ to $\Psi$: $T(w) = (\tau(P(w)), \Phi(w))$ and $\Psi(w) = \mu'(\tau(P(w)), \Phi(w))$; $\Psi$ on objects and $T$ are determined by $\Phi$ and $\tau$, and so is $\Psi$ itself among strong bundle functors covering $\psi$; for a strong bundle functor $\Phi$, $T : W \to \mathbf{X}'_{\,s} \times_{P'} W'$ is a strong bundle map covering $\tau$. *Book:* Remark 10.4.9, pp. 502–503 (2017: Remark 10.4, p. 363). *Hidden hypotheses:* that $\Psi$ is a strong bundle functor, for the determination of $\Psi$ as a functor. *Deviation (explicit):* the second line of (10.23), printed with $\mu$, is read with $\mu'$ (Findings, item 46).

**V10.30 (Definition 10.4.10 (2017: 10.12) and its equivalence with Definition 10.4.8;** `Polyfold.EpStrongBundle.Is`
`Polyfold.EpStrongBundle.StrongBundleNatTrans.isStrongBundleNatTransMap`**,** `Polyfold.EpStrongBundle.`
`Polyfold.EpStrongBundle.StrongBundleNatTrans.orbitMap_eq`**,** `Polyfold.EpStrongBundle.StrongBundleNa`
A natural transformation in the sense of Definition 10.4.10 is a strong bundle map $T : W \to \mathbf{X}'_{\,s} \times_{P'} W'$ covering a natural transformation $\tau : \phi \to \psi$, with $T(w) = (\tau(P(w)), \Phi(w))$ and $\Psi(w) = \mu'(\tau(P(w)), \Phi(w))$. For strong bundle functors $\Phi$, $\Psi$ the two definitions agree; $\Phi \simeq \Psi$ implies $|\Phi| = |\Psi|$. *Book:* Definition 10.4.10 and the sentence after it, p. 503 (2017: Definition 10.12, p. 363). *Hidden hypotheses:* that $\Phi$ and $\Psi$ are strong bundle functors, for Definition 10.4.10 □ Definition 10.4.8. *Deviation (explicit):* none beyond the hypothesis (Findings, item 46).

**V10.31 (Definition 10.4.11 (2017: 10.13), the category $\mathcal{SEP}$;** `Polyfold.SEP`**,** `Polyfold.SEP.Hom`**,** `Polyfold.SEP.equivalences`**,** `Polyfold.SEP.toEP`**,** `Polyfold.SEP.toEP_map_mem_equivalences`**,** `Polyfold.StrongBundleEquivalenceExamples.negSEPHom`**,** `Polyfold.StrongBundleEquivalenceExamples.neg`
$\mathcal{SEP}$ is a category whose objects are the strong bundles over ep-groupoids and whose morphisms are the strong bundle functors $(\phi, \Phi)$; the strong bundle equivalences form a class $F$ containing the identities and closed under composition; the forgetful functor $\mathcal{SEP} \to \mathcal{EP}$ maps $F$ into $E$. The projection $(-\mathrm{id})^*(\mathbb{R}^n \times \mathbb{R}^k) \to \mathbb{R}^n \times \mathbb{R}^k$ is a morphism in $F$. *Book:* Definition 10.4.11, p. 503 (2017: Definition 10.13, p. 364). *Hidden hypotheses:* none. *Deviation (restated):* a morphism carries both $\phi$ and $\Phi$, since $\Phi$ determines $\phi$ only on the image of $P$ (Findings, item 47).

**V10.32 (the properties (E1)–(E3) of the class $E$;** `Polyfold.EPLocalization.isEquivalence_of_isIso`**,** `Polyfold.EpFunctor.IsEquivalence.comp`**,** `Polyfold.EpFunctor.proposition_10_1_5`**).** Every isomorphism of the category $\mathcal{EP}$ is an equivalence; the composite of two equivalences is an equivalence; a functor naturally equivalent to an equivalence is an equivalence. *Book:* §10.3,

(E1)–(E3), p. 483 (2017 p. 347). *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 30).

**V10.33 (Definition 10.3.1 (2017: 10.4), diagrams and refinements;** `Polyfold.EPLocalization.Diagram`, `Polyfold.EPLocalization.Diagram.IsRefinement`, `Polyfold.EPLocalization.Diagram.IsRefinement.refl`, `Polyfold.EPLocalization.Diagram.precomp`, `Polyfold.EPLocalization.Diagram.isRefinement_precomp`, `Polyfold.EPLocalization.Diagram.isRefinement_of_naturallyEquivalent`**).** A diagram $d : X \xleftarrow{F} A \xrightarrow{\Phi} Y$ consists of an ep-groupoid $A$, an equivalence $F : A \to X$ and an sc-smooth functor $\Phi : A \to Y$. A diagram $d' : X \xleftarrow{F'} A' \xrightarrow{\Phi'} Y$ is a refinement of $d$ if there is an equivalence $H : A' \to A$ with $F \circ H \simeq F'$ and $\Phi \circ H \simeq \Phi'$. Every diagram refines itself; for an equivalence $H : A' \to A$, $X \xleftarrow{F\circ H} A' \xrightarrow{\Phi\circ H} Y$ refines $d$; for $G \simeq F$ (an equivalence) and $\Psi \simeq \Phi$, $X \xleftarrow{G} A \xrightarrow{\Psi} Y$ refines $d$. *Book:* Definition 10.3.1, p. 484 (2017: Definition 10.4, p. 348), with the diagrams of p. 483 and the two examples after the definition. *Hidden hypotheses:* none. *Deviation (restated):* the middle ep-groupoid is an object of $\mathcal{EP}$, so a diagram is a structure one universe above the types of objects and morphisms (Findings, item 30).

**V10.34 (Lemma 10.3.2 (2017: 10.3), refinements compose;** `Polyfold.EPLocalization.Diagram.lemma_10_3_2`, `Polyfold.EPLocalization.Diagram.IsRefinement.trans`**).** A refinement of a refinement of $d$ is a refinement of $d$. *Book:* Lemma 10.3.2, p. 484 (2017: Lemma 10.3, p. 348). *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 31).

**V10.35 (Definition 10.3.3 (2017: 10.5), common refinements;** `Polyfold.EPLocalization.Diagram.IsCommonRefi` $d''$ is a common refinement of $(d, d')$ if it refines both. *Book:* Definition 10.3.3, p. 485 (2017: Definition 10.5, p. 349). *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 31).

**V10.36 (Lemma 10.3.4 (2017: 10.4), transitivity;** `Polyfold.EPLocalization.Diagram.lemma_10_3_4`, `Polyfold.EPLocalization.Diagram.Equiv.trans`, `Polyfold.EPLocalization.Diagram.IsRefinement.equiv`**)** If $(d, d')$ and $(d', d'')$ have common refinements, so has $(d, d'')$: for $W$ refining $d$ by $H$ and $d'$ by $H'$, and $W'$ refining $d'$ by $K$ and $d''$ by $K'$, the diagram (10.11) $X \leftarrow W \times_{A'} W' \to Y$, formed with the weak fibred product of $H'$ and $K$, refines $d$ by $H \circ \pi_W$ and $d''$ by $K' \circ \pi_{W'}$. A refinement is equivalent to the refined diagram. *Book:* Lemma 10.3.4, p. 485 (2017: Lemma 10.4, p. 349), with (10.11), p. 486. *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 31).

**V10.37 (Definition 10.3.5 (2017: 10.6), equivalence of diagrams;** `Polyfold.EPLocalization.Diagram.Equiv`, `Polyfold.EPLocalization.Diagram.Equiv.refl`, `Polyfold.EPLocalization.Diagram.Equiv.symm`, `Polyfold.EPLocalization.Diagram.setoid`, `Polyfold.EPLocalization.Diagram.equiv_iff`, `Polyfold.EPLocalization.Examples.equiv_pairToPoint_id`**).** Two diagrams are equivalent if they have a common refinement; this is an equivalence relation on the diagrams $X \to Y$. The diagram pt $\leftarrow$ $\mathbb{Z}$-pair $\to$ pt with both legs the equivalence from the pair groupoid of $\mathbb{Z}$ is equivalent to $1_{\mathrm{pt}}$. *Book:* Definition 10.3.5, p. 486 (2017: Definition 10.6, p. 350). *Hidden hypotheses:* none. *Deviation (restated):* the equivalence relation is a setoid on the type of diagrams; its quotient lives in the universe of the objects of $\mathcal{EP}$ (Findings, item 31).

**V10.38 (Lemma 10.3.6 (2017: 10.5), the map of orbit spaces of a diagram;** `Polyfold.EPLocalization.Diagram.orb` `Polyfold.EPLocalization.Diagram.continuous_orbitMap`, `Polyfold.EPLocalization.Diagram.orbitMap_or` `Polyfold.EPLocalization.Diagram.lemma_10_3_6`, `Polyfold.EPLocalization.Diagram.Equiv.orbitMap_eq`, `Polyfold.EpFunctor.NaturallyEquivalent.orbitMap_eq`, `Polyfold.EpFunctor.orbitMap_comp`**).** For a diagram $d$, $|d| = |\Phi| \circ |F|^{-1} : |X| \to |Y|$ is continuous; naturally equivalent functors induce the same map of orbit spaces; if $d'$ refines $d$, then $|d| = |d'|$, and equivalent diagrams have the same map. *Book:* Lemma 10.3.6, p. 488 (2017: Lemma 10.5, p. 351), with $|d|$ on p. 487. *Hidden*

*hypotheses:* none. *Deviation (same):* none (Findings, item 32).

**V10.39 (Definition 10.3.7 (2017: 10.7), the map $||[d]||$; `Polyfold.EPLocalization.Diagram.orbitMapQ`, `Polyfold.EPLocalization.Diagram.orbitMapQ_mk`, `Polyfold.EPLocalization.Diagram.isHomeomorph_orbit`** $||[d]|| : |X| \to |Y|$ is $|\Phi| \circ |F|^{-1}$ for any representative; if $\Phi$ is an equivalence too, it is a homeomorphism. *Book:* Definition 10.3.7 and the remark after it, p. 488 (2017: Definition 10.7, p. 351). *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 32).

**V10.40 (Theorem 10.3.8 (2017: 10.3), the composition of classes, with its proof in Appendix §10.6.1; `Polyfold.EPLocalization.Diagram.theorem_10_3_8`, `Polyfold.EPLocalization.Diagram.id`, `Polyfold.EPLocalization.Diagram.comp`, `Polyfold.EPLocalization.Diagram.admissible`, `Polyfold.EPLocalization.wfp`, `Polyfold.EPLocalization.wfpMap`, `Polyfold.EPLocalization.projLeft_com`, `Polyfold.EPLocalization.projRight_comp_wfpMap`, `Polyfold.EPLocalization.isEquivalence_wfpMap`, `Polyfold.EPLocalization.assocFunctor`, `Polyfold.EPLocalization.isEquivalence_assocFunctor`, `Polyfold.EPLocalization.Diagram.IsRefinement.comp`, `Polyfold.EPLocalization.Diagram.Equiv.comp`, `Polyfold.EPLocalization.Diagram.isRefinement_assoc`, `Polyfold.EPLocalization.Diagram.equiv_assoc`, `Polyfold.EPLocalization.Diagram.isRefinement_id_comp`, `Polyfold.EPLocalization.Diagram.isRefineme`, `Polyfold.EpFunctor.IsEquivalence.of_comp_left`).** For $d : X \xleftarrow{F} A \xrightarrow{\Phi} Y$ and $e : Y \xleftarrow{G} B \xrightarrow{\Psi} Z$, the composite $X \xleftarrow{F\circ\pi_A} A \times_Y B \xrightarrow{\Psi\circ\pi_B} Z$, with the weak fibred product of $\Phi$ and $G$, is a diagram whose class depends only on the classes of $d$ and $e$; the composition of classes is associative, and the classes of $1_X = X \xleftarrow{1} X \xrightarrow{1} X$ are units. If $d'$ refines $d$ by $H$ and $e'$ refines $e$ by $K$, the composite of $d'$ and $e'$ refines that of $d$ and $e$ by the functor $L(a', \phi, b') = (H(a'), \tau_{B'}(b') \circ \phi \circ \tau_{A'}(a')^{-1}, K(b'))$, an equivalence. If $G$ and $G \circ L$ are equivalences, so is $L$. *Book:* Theorem 10.3.8, p. 488 (2017: Theorem 10.3, p. 352), proved in Appendix §10.6.1, pp. 509–515. *Hidden hypotheses:* none. *Deviation (restated):* $d$.comp $e$ is the text's $[e] \circ [d]$; the two iterated weak fibred products of the associativity are related by a reassociation functor, an equivalence, where the text identifies them; $L$ is shown to be an equivalence by two out of three ($\pi_A \circ L = H \circ \pi_{A'}$); in the proof, $\tau_A$ in the naturality of $\tau_{B'}$, "$(\Phi \circ \pi_B) \circ L$", $[1_X] \circ [d]$ and $\tau(a, \phi, a)$ are read $\tau_{B'}$, $(\Psi \circ \pi_B) \circ L$, $[d] \circ [1_X]$ and $\tau(x, \phi, a)$ (possible misprints) (Findings, items 33, 60).

**V10.41 (Definition 10.3.9 (2017: 10.8), generalized maps and isomorphisms; `Polyfold.EPLocalization.EPLoc.h`, `Polyfold.EPLocalization.EPLoc.IsGeneralizedIso`, `Polyfold.EPLocalization.EPLoc.isGeneralizedIso_i`** A generalized map $X \to Y$ is a class $[d]$ of diagrams; it is a generalized isomorphism if some generalized map $g : Y \to X$ composes with it to the units on both sides. The generalized isomorphisms are the isomorphisms of $\mathcal{EP}(E^{-1})$. *Book:* Definition 10.3.9, p. 489 (2017: Definition 10.8, p. 352). *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 34).

**V10.42 (Theorem 10.3.10 (2017: 10.4), generalized isomorphisms, with its proof in Appendix §10.6.2; `Polyfold.EPLocalization.EPLoc.theorem_10_3_10`, `Polyfold.EPLocalization.Diagram.symm`, `Polyfold.EPLocalization.diagFunctor`, `Polyfold.EPLocalization.isEquivalence_diagFunctor`, `Polyfold.EPLocalization.Diagram.isRefinement_comp_symm`, `Polyfold.EPLocalization.Diagram.isRefine`, `Polyfold.EPLocalization.Diagram.equiv_comp_symm`, `Polyfold.EPLocalization.Diagram.equiv_symm_comp`, `Polyfold.EPLocalization.Diagram.isEquivalence_Φ_of_equiv_id`, `Polyfold.EPLocalization.isEquivaler`, `Polyfold.EPLocalization.isLocalScDiffeo_obj_of_comp`, `Polyfold.EPLocalization.Diagram.isEquivaler`, `Polyfold.EPLocalization.EPLoc.isIso_homMk_iff`, `Polyfold.EPLocalization.EPLoc.isIso_homMk`, `Polyfold.EPLocalization.EPLoc.inv_homMk`, `Polyfold.MPolyfold.isLocalScDiffeoAt_of_injOn`, `Polyfold.MPolyfold.IsLocalScDiffeoAt.exists_injOn`).** (1) $[X \xleftarrow{F} A \xrightarrow{\Phi} Y]$ is a generalized

isomorphism if and only if $\Phi$ is an equivalence; (2) every inverse is then $[Y \xleftarrow{\Phi} A \xrightarrow{F} X]$. An sc-smooth map injective near $x$ with an sc-smooth local right inverse through $x$ is a local sc-diffeomorphism at $x$. *Book:* Theorem 10.3.10, p. 489 (2017: Theorem 10.4, p. 353), proved in Appendix §10.6.2, pp. 515–519. *Hidden hypotheses:* none. *Deviation (same):* in (C1) the text's $A \times_X B$, $\mathrm{mor}_Y(\Psi(a), \Psi(a'))$ and "$x$" are read $A \times_Y B$, $\mathrm{mor}_X(\Psi(b), \Psi(b'))$ and "$y$", and in (C2) $(s|U)^{-1}(b')$ is read $(s|U)^{-1}(\Psi(b'))$ (possible misprints); the step of (C2) that $\Gamma'$ is a local sc-diffeomorphism does not hold as we read it without the other half of (C2), and the local sc-diffeomorphism property of $\Phi$ is proved from a local right inverse obtained from the equivalence $\Phi \circ \pi_{A'}$ and from local injectivity (Findings, items 34, 61).

**V10.43 (Proposition 10.3.11 (2017: 10.6), orbit maps compose;** `Polyfold.EPLocalization.EPLoc.orbitMap`, `Polyfold.EPLocalization.EPLoc.orbitMap_homMk`, `Polyfold.EPLocalization.EPLoc.continuous_orbitMap`, `Polyfold.EPLocalization.EPLoc.orbitMap_id`, `Polyfold.EPLocalization.Diagram.orbitMap_comp`, `Polyfold.EPLocalization.EPLoc.proposition_10_3_11`**).** For generalized maps $f : X \to Y$ and $g : Y \to Z$, $|g \circ f| = |g| \circ |f|$; $|f|$ is continuous and $|1_X| = \mathrm{id}$. *Book:* Proposition 10.3.11, p. 490 (2017: Proposition 10.6, p. 353), with the computation before it. *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 35).

**V10.44 (Definition 10.3.12 (2017: 10.9), the category $\mathcal{EP}(E^{-1})$;** `Polyfold.EPLocalization.EPLoc`, `Polyfold.EPLocalization.EPLoc.toEP_of`, `Polyfold.EPLocalization.EPLoc.of_toEP`, `Polyfold.EPLocalization.EPLoc.homMk_comp`, `Polyfold.EPLocalization.EPLoc.id_eq`, `Polyfold.EPLocalization.EPLoc.homMk_eq_homMk_iff`, `Polyfold.EPLocalization.EPLoc.homMk_surjective` $\mathcal{EP}(E^{-1})$ is a category with the ep-groupoids as objects and the generalized maps as morphisms, composed by Theorem 10.3.8, with units $[1_X]$; $[d] = [d']$ if and only if $d$ and $d'$ are equivalent, and every morphism is some $[d]$. *Book:* Definition 10.3.12, p. 490 (2017: Definition 10.9, p. 354). *Hidden hypotheses:* none. *Deviation (restated):* the objects are a copy of those of $\mathcal{EP}$ (a one-field structure), and $[d] \gg [e] = [d.\mathrm{comp}\, e]$ is the text's $[e] \circ [d]$ (Findings, item 35).

**V10.45 (Theorem 10.3.14 (2017: 10.5), the localization functor;** `Polyfold.EPLocalization.Diagram.ofFunctor`, `Polyfold.EPLocalization.Diagram.isRefinement_ofFunctor_comp`, `Polyfold.EPLocalization.EPLoc.local` `Polyfold.EPLocalization.EPLoc.localizationFunctor_obj`, `Polyfold.EPLocalization.EPLoc.localizatio` `Polyfold.EPLocalization.EPLoc.theorem_10_3_14`, `Polyfold.EPLocalization.EPLoc.isInvertedBy_locali` `Polyfold.EPLocalization.EPLoc.inv_localizationFunctor_map`, `Polyfold.EPLocalization.EPLoc.homMk_e` `Polyfold.EPLocalization.Examples.pairToPointEP`, `Polyfold.EPLocalization.Examples.not_isIso_pairT` `Polyfold.EPLocalization.Examples.isGeneralizedIso_pairToPoint`, `Polyfold.EPLocalization.Examples.`

There is a functor $i : \mathcal{EP} \to \mathcal{EP}(E^{-1})$, the identity on objects, with $i(\Phi) = [X \xleftarrow{1} X \xrightarrow{\Phi} Y]$; for an equivalence $F$, $i(F)$ is a generalized isomorphism and every inverse is $[Y \xleftarrow{F} X \xrightarrow{1} X]$. $i$ inverts the class $E$ in the sense of Mathlib, and every generalized map is $i(\Phi) \circ i(F)^{-1}$. The equivalence from the pair groupoid of $\mathbb{Z}$ to the point is not an isomorphism of $\mathcal{EP}$, but its image under $i$ is invertible. *Book:* Theorem 10.3.14, p. 491 (2017: Theorem 10.5, p. 354), with p. 487 for $[d] = i(\Phi) \circ i(F)^{-1}$. *Hidden hypotheses:* none. *Deviation (same):* in the proof, the text calls $d_{\Psi \circ \Phi}$ a refinement of (10.13) by $\pi_X$; with Definition 10.3.1 it is (10.13) that refines $d_{\Psi \circ \Phi}$, which is what is proved; the universal property of the localization, mentioned on p. 483, is not formalized (gap G.117) (Findings, item 36).

**V10.46 (Theorem 10.3.15 (2017: 10.6), the orbit space functor;** `Polyfold.EPLocalization.EPLoc.orbitFunctor`, `Polyfold.EPLocalization.EPLoc.theorem_10_3_15`**).** There is a functor $|\cdot| : \mathcal{EP}(E^{-1}) \to \mathrm{TOP}$ taking $X$ to $|X|$ and $[X \xleftarrow{F} A \xrightarrow{\Phi} Y]$ to $|\Phi| \circ |F|^{-1}$. *Book:* Theorem 10.3.15, p. 492 (2017: Theorem

10.6, p. 355). *Hidden hypotheses:* none. *Deviation (same):* TOP is Mathlib's `TopCat` (Findings, item 35).

**V10.47 (Proposition 10.3.17, (LUC) under generalized isomorphisms;** `Polyfold.EPLocalization.EPLoc.exists` `Polyfold.EPLocalization.EPLoc.iff_of_isGeneralizedIso`, `Polyfold.EPLocalization.EPLoc.proposition` A generalized isomorphism is represented by a diagram whose two legs are equivalences; hence a property of ep-groupoids invariant under equivalences is invariant under generalized isomorphisms, and for a generalized isomorphism $X \to Y$, $X$ has (LUC) if and only if $Y$ has it. *Book:* Proposition 10.3.17, p. 492 (new in the 2021 edition), with the argument before Proposition 10.3.16. *Hidden hypotheses:* none. *Deviation (same):* "if one has it, so has the other" is stated as an equivalence (Findings, item 38).

**V10.48 (Remark 10.3.18 (2017: 10.3), equivalence of pairs $(X, \beta)$;** `Polyfold.EPLocalization.OrbitChart`, `Polyfold.EPLocalization.OrbitChart.Rel`, `Polyfold.EPLocalization.OrbitChart.Rel.refl`, `Polyfold.EPLocalization.OrbitChart.Rel.symm`, `Polyfold.EPLocalization.OrbitChart.Rel.trans`, `Polyfold.EPLocalization.OrbitChart.setoid`, `Polyfold.EPLocalization.OrbitChart.rel_pairEP_pointEP` For a space $Z$, pairs $(X, \beta)$ with $\beta : |X| \to Z$ a homeomorphism, with $(X', \beta')$ equivalent to $(X, \beta)$ if some generalized isomorphism $f : X \to X'$ has $\beta' \circ |f| = \beta$, form a set with an equivalence relation. The pair groupoid of $\mathbb{Z}$ with the orbit map of its equivalence to the point, and the point with the identity, are equivalent pairs. *Book:* Remark 10.3.18, pp. 492–493 (2017: Remark 10.3, p. 355). *Hidden hypotheses:* none. *Deviation (narrower):* the claim that $f$ is in general not unique is not formalized (gap G.119); the metrizability of $Z$ plays no role in the relation and is not assumed (Findings, item 39).

**V10.49 (Definition 10.3.20, the subcategories of ep-groupoids with (LR);** `Polyfold.EPLocalization.EPreg`, `Polyfold.EPLocalization.EPLocReg`, `Polyfold.EPLocalization.Examples.pointReg`, `Polyfold.EPLocalization.Examples.pointLocReg`**).** $\mathcal{EP}_{reg}$ and $\mathcal{EP}_{reg}(E^{-1})$ are the full subcategories of $\mathcal{EP}$ and $\mathcal{EP}(E^{-1})$ on the ep-groupoids with the local regularity property; the point ep-groupoid is an object of both. *Book:* Definition 10.3.20, p. 494 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 39).

**V10.50 (Theorem 10.3.21, invariance under generalized isomorphisms, for (LUC), (SLUC) and (ZHOU);** `Polyfold.EPLocalization.EPLoc.theorem_10_3_21`**).** For a generalized isomorphism $X \to Y$, $X$ has (LUC), (SLUC), respectively (ZHOU), if and only if $Y$ has it. *Book:* Theorem 10.3.21, p. 495 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (narrower):* the parts (LR) and (SLR) rest on Proposition 10.1.9 and are not formalized (gap G.118); the proof, left to the reader in the text, applies Propositions 10.1.16, 10.1.17 and 10.1.18 to the two legs of a representing diagram (Findings, item 38).

**What is not verified in this chapter**

- p. 461, the full subcategory $\mathcal{EP}_{lreg}$ of ep-groupoids with the local regularity property: not formalized, not a numbered item (gap G.111; Findings, item 7).
- p. 467, Remark 10.1.8 and Appendix §10.6.3, p. 519 (Zhou's example of a $\beta$ with $|\beta(y)| = |y|$ not given by the group): it needs the weighted Sobolev sc-structure and the sc-retraction of the appendix (gap G.112; Findings, item 7).
- p. 468, Proposition 10.1.9 (new in the 2021 edition): (LR) under equivalences; it needs a natural representation transported along $F$ and a morphism $F(x) \to y$, which the Chapter 7 API of natural representations does not provide (gap G.113; Findings, item 7).

- p. 470, Proposition 10.1.10 (2017: 10.4): sc-smooth families $\alpha_z : F(z) \to G(z)$ for $|F| = |G|$, and $|TF| = |TG|$; it assumes (LR) and uses Proposition 10.1.9; the example of Remark 10.1.11 satisfies (1) (V10.8) (gap G.114; Findings, items 7, 8).
- p. 477, Proposition 10.1.19 (new in the 2021 edition) itself: only its reduction to Proposition 10.1.9 is verified (V10.16) (gap G.115; Findings, item 15).
- pp. 478–481, Theorem 10.2.2: the agreement of the iterated sub-M-polyfold structures of $X \times_Y Z$ with structures induced directly from $X \times \mathbf{Y} \times Z$ and $\mathbf{X} \times \mathbf{Y} \times \mathbf{Z}$; nothing depends on it (V10.18) (gap G.116; Findings, item 21).
- p. 483, the universal property of the localization $i : \mathcal{EP} \to \mathcal{EP}(E^{-1})$ (not a numbered statement): only the inversion of $E$ is proved (V10.45); the universal property would need that every functor inverting $E$ identifies naturally equivalent sc-smooth functors, through a cylinder ep-groupoid the project does not have (gap G.117; Findings, item 36).
- p. 491, Remark 10.3.13 (new in the 2021 edition), $X$ as an atlas on $|X|$: no formal content (Findings, item 37).
- p. 492, Proposition 10.3.16 (new in the 2021 edition), and p. 495, the parts (LR) and (SLR) of Theorem 10.3.21: they rest on Proposition 10.1.9 (G.113); `Polyfold.EPLocalization.EPLoc.iff_of_isGene` reduces them to it, as the text does (gap G.118; Findings, item 38).
- pp. 492–493, Remark 10.3.18 (2017: 10.3), the claim that a generalized isomorphism $f$ with $\beta' \circ |f| = \beta$ is in general not unique: no example is formalized (the one behind Remark 10.1.8 is G.112) (gap G.119; Findings, item 39).
- p. 493, Theorem 10.3.19 (2017: 10.7): generalized isomorphisms with the same orbit map agree when one side has (LR); it rests on Propositions 10.1.10 and 10.3.16 (gap G.120; Findings, item 39).
- pp. 500–501, Theorem 10.4.5 (2017: 10.8), the Fred part: it needs the invariance of sc-Fredholm sections of strong bundles over M-polyfolds under local strong bundle isomorphisms, of which Part I has only chart-level germ statements (V10.27) (gap G.121; Findings, item 44).
- pp. 501–502, Propositions 10.4.6 and 10.4.7 (new in the 2021 edition): the strong bundle versions of Propositions 10.1.9 and 10.1.10, the proof of 10.4.6 left to the reader in the text (gap G.122; Findings, item 45).
- p. 483, the closing sentences of Remark 10.2.3 ("smaller models"): no formal content.
- Witnesses: every hypothesis taken is satisfiable, and every definition has a non-identity witness (Findings, item 70).

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## 12. Sign-off

The statements below are the ones the reader is asked to confirm say what HWZ mean. Each is a summary of the V-item it names; the standing hypotheses of that item are not repeated.

1. (V1.1) An sc-Banach space is a sequence of Banach spaces $E_0 \supset E_1 \supset \cdots$ with compact inclusions and $E_\infty$ dense in every $E_m$, taken as an indexed family with injective compact inclusions; finite-dimensional spaces carry the constant structure, sums have levels $E_m \oplus F_m$, and $E^k$ has levels $E_{k+m}$.
2. (V1.2) An sc-operator is a linear map continuous $E_m \to F_m$ on every level, given by its level maps; an sc-isomorphism is one that is a continuous linear equivalence on every level.
3. (V1.3) An sc$^+$-operator is an sc-operator $E \to F^1$; it is compact $E_m \to F_m$ on every level.
4. (V1.4) An sc-subspace is a closed subspace whose smooth points are dense in each $F \cap E_m$; an sc-splitting $E = F \oplus G$ consists of two sc-subspaces with $F_m$ and $G_m$ complementary

on every level.

5. (V1.5) A finite-dimensional subspace is an sc-subspace iff it lies in $E_\infty$, and then has an sc-complement.
6. (V1.6) For a closed subspace $A$ of an sc-Banach space $E$, the quotients $E_m/A_m$ form an sc-Banach space $E/A$ with the quotient map an sc-operator whose level kernels are the $A_m$.
7. (V1.7) An sc-Fredholm operator is an sc-operator with sc-splittings $E = \ker T \oplus X$, $F = C \oplus \operatorname{range} T$, $\ker T$ and $C$ finite-dimensional, and $T$ bijective $X_m \to \operatorname{range}(T)_m$ on every level; its index $\dim\ker T - \dim C$ is $\dim\ker T_0 - \dim\operatorname{coker} T_0$, and each $T_m$ is Fredholm.
8. (V1.8) An sc-Fredholm operator is regularizing: $Te \in F_m$ implies $e \in E_m$; conversely, an sc-operator that is Fredholm on level 0 and regularizing is sc-Fredholm.
9. (V1.9) The sum of an sc-Fredholm operator and an sc$^+$-operator is sc-Fredholm.
10. (V1.10) A partial quadrant is a set $C = T^{-1}([0,\infty)^n \oplus W)$ for an sc-isomorphism $T : E \to \mathbb{R}^n \oplus W$; a tuple $(U, C, E)$ has $U = O \cap C$ with $O$ open.
11. (V1.11) A map is sc$^0$ on $U$ if it maps $U_m$ into $F_m$ continuously for every $m$.
12. (V1.12) The tangent $TE$ is $E^1 \oplus E$ and $TU = U^1 \oplus E$, with levels $U_{m+1} \times E_m$.
13. (V1.13) A map is sc$^1$ on $U$ if it is sc$^0$, differentiable from $U_1$ into $F_0$ at every point of $U_1$ with a derivative $Df(x)$ bounded on $E_0$, and $(x,h) \mapsto (f(x), Df(x)h)$ is sc$^0$ on $TU$; $Df(x)$ is unique on relatively open subsets of partial quadrants; $x \mapsto Df(x)$ need not be norm continuous (an example on a weighted $c_0$ scale).
14. (V1.14) $f$ is sc^{k+1} if it is sc$^1$ with sc^k tangent map, and sc-smooth if sc^k for all $k$.
15. (V1.15) An sc$^0$ map on an sc-domain is sc$^1$ iff on every level $f : U_{m+1} \to F_m$ is differentiable with a derivative extending to a bounded $Df(x) : E_m \to F_m$ that depends jointly continuously on $(x,h) \in U_{m+1} \times E_m$; then $f : U_{m+1} \to F_m$ is $C^1$, and $Df(x)$ is an sc-operator at smooth points.
16. (V1.16) If $f$ is sc^k on $U$ then $f : U^1 \to F^1$ is sc^k.
17. (V1.17) If $f$ is sc^k then $f : U_{m+k} \to F_m$ is $C^k$, and $f : U_{m+l} \to F_m$ is $C^l$ for $l \le k$.
18. (V1.18) If $f : U_{m+l} \to F_m$ is $C^{l+1}$ for all $m \ge 0$ and $0 \le l \le k$, then $f$ is sc^{k+1}, as in the text; if this holds for every $l$, $f$ is sc-smooth.
19. (V1.19) A map into $\mathbb{R}^N$ that is $C^{l+1}$ on $U_l$ for $0 \le l \le k$ is sc^{k+1}, as in the text; one that is $C^{l+1}$ on every $U_l$, or $C^\infty$ on $U$, is sc-smooth.
20. (V1.20) The composition of sc$^1$ maps between relatively open subsets of partial quadrants is sc$^1$ with $T(g \circ f) = Tg \circ Tf$; sc^k and sc-smooth maps compose likewise.
21. (V1.21) The degeneracy index $d_C(x)$ is the number of vanishing half-line coordinates of $T(x)$.
22. (V1.22) $d_C$ does not depend on the presentation $T$, and neither does the number $n$ of half-line factors.
23. (V1.23) An sc$^1$ map with sc$^1$ inverse between relatively open subsets of partial quadrants preserves the degeneracy index at every point.
24. (V1.24) For a compact operator $c$ on a Banach space, $1 + c$ is Fredholm (Riesz–Schauder); a continuous linear map with finite-dimensional kernel and closed range of finite codimension is Fredholm.
25. (V1.25) A compact perturbation of a Fredholm operator between Banach spaces is Fredholm.
26. (V1.26) If $(x,y) \mapsto A(x)y$ is $C^k$ on $s \times Y$ and $\iota : Y' \to Y$ is compact, then $x \mapsto A(x) \circ \iota$ is $C^k$ into $\mathcal{L}(Y', Z)$; for an sc$^1$ map the level operators $Df(x) : E_m \to F_m$ exist for $x \in U_{m+1}$ and are the derivatives of $f : U_{m+1} \to F_m$.
27. (V1.27) An sc-domain has level sets that are open-in-convex with unique differentiability; relatively open subsets of partial quadrants are sc-domains, and $T$ and the level shift preserve the property.

28. (V1.28) The index $\dim\ker - \dim\operatorname{coker}$ is additive under composition, and the Fredholm index between Banach spaces is invariant under small and under compact perturbations.
29. (V1.29) Every level map $T_m$ of an sc-Fredholm operator is Fredholm of index $\operatorname{ind}(T) = \dim K - \dim C$.
30. (V1.30) For $T$ sc-Fredholm and $S$ sc$^+$, $\operatorname{ind}(T+S) = \operatorname{ind}(T)$, on every level.
31. (V1.31) An sc-structure on a finite-dimensional space is sc-isomorphic, through its inclusions, to the constant one; an onto inclusion $E_{m+1} \to E_m$ forces $E_m$ finite-dimensional.
32. (V1.32) $E = A \oplus B$ is an sc-splitting iff some idempotent sc-operator has range $A$ and kernel $B$.
33. (V1.33) For an sc-subspace $A$, a class $x + A$ meeting every $E_j$ contains a smooth point: the smooth points of $E/A$ are the classes of smooth points.
34. (V1.34) $(x,h) \mapsto A(x)h$ is jointly continuous iff $x \mapsto A(x)$ is continuous for uniform convergence on compact sets; so $Df$ is compact-open continuous on $U_1$, and not in general norm continuous (V1.13).
35. (V1.35) Between finite-dimensional spaces, on relatively open subsets of quadrants, sc$^1$ is $C^1$ within $U$ (one-sided derivatives at the boundary) and sc-smooth is $C^\infty$.
36. (V1.36) $C_1$ is a partial quadrant in $E^1$ and $TC$ one in $TE$; $U_1$ and $TU$ are relatively open in them.
37. (V1.37) If $f : U_{m+l} \to F_m$ is $C^{l+1}$ for all $m$ and $l \le k$, the derivatives $d^i f(x)$ on $E_{m+l}$ into $F_m$ ($i \le l+1$, $i \le k+1$) are restrictions of one base-level map, continuous in $x$, and differentiable in $x$ with derivative $d^{i+1}f(x)$ for $i \le l$, $i \le k$.
38. (V1.38) An elementary map $z \mapsto d^i f(\pi_0 z)(\pi_1 z, \dots, \pi_i z)$ with $\pi_0 : X \to E^p$, $\pi_a : X \to E^{n_a}$, into $F^m$, with $m + i \le p$, $m + i \le n_a + 1$, $i \le k$ and an sc-domain as domain, is sc^s for $i + s \le k + 1$; its tangent map is a pair of elementary maps and sums of them.
39. (V1.39) Sums of sc^k maps on an sc-domain are sc^k; the evaluation of continuous multilinear maps has the product-rule derivative.
40. (V1.40) Two norms on a vector space that induce the same topology are equivalent; two complete norms one of which dominates the other induce the same topology (open mapping theorem).
41. (V1.41) For $T$ sc-Fredholm and $S$ sc$^+$, $T+S$ is sc-Fredholm of index $\operatorname{ind}(T)$, and every level map $(T+S)_m$ is Fredholm of that index.
42. (V2.1) An sc-smooth retraction is an sc-smooth $r : U \to U$, $U$ relatively open in a partial quadrant, with $r \circ r = r$; its retract $O = r(U)$ is its fixed-point set, and $Dr(x)$ is a projection for $x \in O_1$.
43. (V2.2) $(O, C, E)$ is a retract if $O = r(U)$ for such an $r$.
44. (V2.3) Two retractions onto the same $O$ have the same tangent image $Tr(TU) = Tr'(TU')$.
45. (V2.4) $TO = Tr(TU) = \{(r(x), Dr(x)h)\}$, and $T_xO$ is the range of the projection $Dr(x)$.
46. (V2.5) Sc-smoothness of $f \circ r$ and $T(f \circ r)$ on $TO$ do not depend on the retraction $r$ onto $O$, and $T(f \circ r)$ maps $TO$ into $TO'$.
47. (V2.6) $f : O \to O'$ is sc-smooth if $f(O) \subseteq O'$ and $f \circ r$ is sc-smooth for a (every) retraction onto $O$; $Tf = T(f \circ r)|_{TO}$.
48. (V2.7) Sc-smooth maps between retracts compose, with $T(g \circ f) = Tg \circ Tf$.
49. (V2.8) $(TO, TC, TE)$ is a retract with retraction $Tr$, and $Tf$ is sc-smooth between tangents.
50. (V2.9) Open subsets of retracts, and images of idempotent sc-smooth maps on retracts, are retracts.
51. (V2.10) A chart is a homeomorphism of an open set onto a retract; charts are compatible when both transition maps are sc-smooth between the overlap retracts.
52. (V2.11) An atlas is a set of pairwise compatible charts covering $X$; atlases are equivalent when

their union is an atlas, and this is an equivalence relation.

53. (V2.12) An M-polyfold is a Hausdorff paracompact space with an atlas, its charts being all charts compatible with it.
54. (V2.13) $f$ is sc-smooth if it is continuous and all its coordinate representations are sc-smooth between retracts; one pair of charts per point suffices.
55. (V2.14) $X_m$ consists of the points that lie on level $m$ in one (every) chart, with the topology generated by the chart preimages of open subsets of $O_m$.
56. (V2.15) The inclusions $X_{m+1} \to X_m$ are continuous.
57. (V2.16) The level charts $(V_1, \varphi, (O_1, C_1, E^1))$ make $X_1$ an M-polyfold.
58. (V2.17) For every M-polyfold, $X$ and every $X_m$ are metrizable, $X_m$ is paracompact, and $X_\infty$ (initial topology of the inclusions) is metrizable; no hypothesis beyond the text's.
59. (V2.18) Tangent tuples $(x, \varphi, h)$ are equivalent when one is the transport of the other by $T$ of the transition map; this is an equivalence relation.
60. (V2.19) $TX$ is the set of classes, with $p : TX \to X_1$ and the bijections $T\varphi : TV \to TO$.
61. (V2.20) $TX$ is Hausdorff, metrizable and paracompact, and $p$ is continuous (openness of $p$ not verified).
62. (V2.21) The charts $(TV, T\varphi, (TO, TC, TE))$ make $TX$ an M-polyfold, with transition maps $T(\varphi' \circ \varphi^{-1})$, and $p : TX \to X^1$ is sc-smooth.
63. (V2.22) An sc-smooth $f : X \to Y$ has a well-defined sc-smooth tangent map $Tf : TX \to TY$ covering $f$, with $T(g \circ f) = Tg \circ Tf$.
64. (V2.23) A sub-M-polyfold is locally the image of an sc-smooth retraction of an open set.
65. (V2.24) A sub-M-polyfold, with the subspace topology, is an M-polyfold for which the inclusion and the maps $i^{-1} \circ r$ are sc-smooth.
66. (V2.25) In a chart, at a smooth $a$, $T_aO = Dt(a)E \oplus (Dr(a) - Dt(a))E$ is an sc-splitting with $Dt(a)E = T_aA$, and $Ts$ maps $T_aO$ onto $T_aA$.
67. (V2.26) $d_X(x)$ is the least $d_C(\varphi(x))$ over the charts $\varphi$ around $x$.
68. (V2.27) $d_X$ is upper semicontinuous.
69. (V2.28) Sc-diffeomorphisms between open sets preserve $d_X$.
70. (V2.29) $\partial X = \{d_X \geq 1\}$ is closed; $X$ is without boundary iff $d_X \equiv 0$.
71. (V2.30) On a sub-M-polyfold $A$, $d_A \leq d_X$.
72. (V2.31) $E_x$ is the closed subspace cut out by the coordinates vanishing at $x$, of codimension $d_C(x)$; $C_x$ the cone on which they are $\geq 0$.
73. (V2.32) $T^R_xO = T_xO \cap E_x$ and $C_xO = T_xO \cap C_x$, defined at every $x \in O_1$.
74. (V2.33) $T^R_xO \subseteq C_xO \subseteq T_xO$, and $C_xO$ is a partial quadrant in $T_xO$ at every smooth point of every sc-smooth retract: the remark's claim that in general it is not is false.
75. (V2.34) For an sc¹ map between quadrants, $Df(x)(E_x) \subseteq E'_{f(x)}$ and $Df(x)(C_x) \subseteq C'_{f(x)}$ at every level-1 point.
76. (V2.35) A germ of sc¹-diffeomorphisms between retracts maps $T^R_xO$ onto $T^R_yO'$ and $C_xO$ onto $C_yO'$, at every level-1 point.
77. (V2.36) A nonempty retract, and a nonempty relatively open subset of a partial quadrant, is an M-polyfold with the chart id.
78. (V2.37) $\dim(T_xO/T^R_xO) \leq d_X(x)$ in every chart of every M-polyfold at points on level 1; the second statement holds without its hypothesis (V2.33).
79. (V2.38) On a relatively open subset of a partial quadrant, the degeneracy index of the M-polyfold equals $d_C$.
80. (V2.39) $C_e$ and $E_e$ are the cone and the subspace cut out by the coordinates vanishing at $e$, in a presentation of $C$.

81. (V2.40) A retraction is tame if it preserves $d_C$ and, at smooth points of $O$, $T_xO$ has an sc-complement inside $E_x$ (for every presentation of $C$).
82. (V2.41) For a tame $r$ at a smooth $x \in O$, $(1 - Dr(x))E \subseteq E_x$ is an sc-complement of $T_xO$; only the second tameness condition is used.
83. (V2.42) For a tame retract, $\dim(T_xO/T_x^RO) = d_O(x)$ at smooth points, where $T_x^RO$ is an sc-subspace and $C_xO$ a partial quadrant in $T_xO$, and $d_O = d_C$ everywhere; a tame chart of any M-polyfold realizes $d_X$.
84. (V2.43) A tame M-polyfold has an equivalent atlas of charts with tame models.
85. (V2.44) A face is the closure of a component of $\{d_X = 1\}$; face-structured means every $x$ lies in exactly $d_X(x)$ faces.
86. (V2.45) Near a point of a tame retract, the retraction maps points of a coordinate hyperplane $f_i$ into the closure of a single component of $\{d_C = 1\} \cap O \cap f_i$.
87. (V2.46) The faces of a tame retract are the components of the sets $f_i \cap O$, distinct for distinct $i$ through a common point.
88. (V2.47) A tame retract is face-structured.
89. (V2.48) In a tame M-polyfold each point has a tame chart realizing $d_X$ whose model is face-structured, and lies in at most $d_X(x)$ global faces.
90. (V2.49) A face of a tame retract is the retract of $r \circ P_i$ on a neighbourhood of the face, hence a sub-M-polyfold, and with its induced structure it is tame with $d_F = d_O - 1$.
91. (V2.50) Every point of a tame M-polyfold lies in exactly $d_X(x)$ local faces, each tame with $d_F(x) = d_X(x) - 1$; the intersection of $\#\sigma$ of them is, near $x$, a tame sub-M-polyfold $A$ with $d_A = d_X - \#\sigma$.
92. (V2.51) In a tame M-polyfold the interior $F \cap \{d_X = 1\}$ of a face is a component $G$ and a sub-M-polyfold; if $X$ is face-structured, every face is a tame sub-M-polyfold with $d_X = d_F + 1$.
93. (V2.52) $X \times Y$ is an M-polyfold with the product charts, $d_{X\times Y} \leq d_X + d_Y$, with equality and $X \times Y$ tame when $X$ and $Y$ are tame.
94. (V2.53) For a local sc-diffeomorphism $f$ and an sc-smooth $g$ the fibred product is a sub-M-polyfold of $X \times Y$ with $d = d_Y$, tame if $Y$ is; if both are local sc-diffeomorphisms, $d = d_X = d_Y$, and $2d = d_{X\times Y}$ for tame $X, Y$.
95. (V2.54) A strong bundle map preserves the double filtration, is linear in the fibre and sc-smooth on $[0]$ and $[1]$; a strong bundle retraction is an idempotent one, covering an sc-smooth retraction.
96. (V2.55) A local strong bundle retract $K = R(U \lhd F)$ and its part $K[1]$ are sc-smooth retracts, with sc-smooth projections to $O$.
97. (V2.56) Sections of $K$ are sc-smooth into $K[0]$, sc$^+$ into $K[1]$; sc$^+$ implies sc-smooth.
98. (V2.57) A strong bundle chart is a fibrewise linear homeomorphism $P^{-1}(V) \to K$ over a chart of $X$; compatibility means sc-smooth transitions on $[0]$ and $[1]$; equivalence of strong bundle atlases is an equivalence relation, and charts restrict to open subsets of the base.
99. (V2.58) A strong bundle is $P : Y \to X$ with a strong bundle atlas; $Y[0]$ and $Y[1]$ are M-polyfolds, $P[0]$ and $P[1]$ are sc-smooth, and one chart decides whether $y \in Y_{m,k}$; the charts compatible with the atlas depend only on its class.
100. (V2.59) Sections of a strong bundle are sc-smooth into $Y[0]$, sc$^+$ in the charts $\Phi[1]$, which is the same as sc-smooth into $Y[1]$.
101. (V2.60) The pull-back $f^*Y$ is the fibre product of $P$ and $f$.
102. (V2.61) The pull-back of a strong bundle along an sc-smooth map is a strong bundle, with the pulled-back charts and retractions $R_g$.
103. (V2.62) A paracompact Hausdorff locally metrizable space is metrizable (Smirnov), proved in a pull-back form.

104. (V2.63) Sc^k-smoothness restricts to smaller domains and is a local property.
105. (V2.64) $Y[1]$ is Hausdorff and paracompact (metrizable), and the inclusion $Y[1] \to Y[0]$ is sc-smooth.
106. (V2.65) For retractions $r$, $s$ onto $O$ and $f(O) \subseteq O'$: $f \circ r$ is sc-smooth iff $f \circ s$ is, and then $T(f \circ r) = T(f \circ s)$ on $TO$ and $T(f \circ r)$ maps $TO$ into $TO'$.
107. (V2.66) Every point of a retract has open $V \subseteq W$ with $\mathrm{cl}_O(V) \subseteq W$ and $\mathrm{cl}_O(V)$ complete for $|\cdot|_0$, $W$ arbitrarily small.
108. (V2.67) A locally complete space has, around every point, $V$, $W$ with $\mathrm{cl}(V) \subseteq W$ and a metric on $W$ for which $\mathrm{cl}(V)$ is complete; it is automatically Hausdorff.
109. (V2.68) If $\mathrm{cl}_X(V) \subseteq W$ then $\mathrm{cl}_W(V) = \mathrm{cl}_X(V)$, and local completeness implies its modified form with $\mathrm{cl}_W(V)$.
110. (V2.69) Sc-smooth retracts are locally complete.
111. (V2.70) Local completeness passes to locally closed subspaces, homeomorphic spaces, open subspaces, and from an open cover with condition (b) to the whole space.
112. (V2.71) Locally complete spaces are first countable and completely regular.
113. (V2.72) Every level $U_k$, $k \in \mathbb{N} \cup \{\infty\}$, of a connected relatively open $U$ is path-connected, $U_\infty$ with the topology of all the norms.
114. (V2.73) $W = V \cap r^{-1}(V)$ is a relatively open neighbourhood of $x$ with $r(W) = O \cap V \subseteq W$, and retracts are locally path-connected.
115. (V2.74) Retracts are almost locally contractible, by $\Phi(t, y) = r((1-t)y + tx)$.
116. (V2.75) The footnote's paracompactness is Mathlib's, so it changes nothing for M-polyfolds; with "subcovering" instead of "refinement", $\mathbb{R}$ would not be paracompact.
117. (V2.76) In a regular space every open $U \ni x$ contains an open $V \ni x$ with $\mathrm{cl}(V) \subseteq U$, arbitrarily small in a Hausdorff paracompact space.
118. (V2.77) M-polyfolds are locally complete, with arbitrarily small $W(x)$, and locally path-connected.
119. (V2.78) A tame sub-M-polyfold is given locally by a retraction preserving $d_X$ whose image has a tangent complement in the reduced tangent space, in a chart; it is a sub-M-polyfold.
120. (V2.79) Faces of open sets are defined from $d_X$; the degeneracy index of a tame M-polyfold satisfies the corner axioms from which all of §2.7.4 is proved.
121. (V2.80) Every point of a tame M-polyfold has an open $V$ with $d_X \le d_X(x)$, every $y$ in exactly $d_X(y)$ faces of $V$, and a neighbourhood $U(x)$ whose faces in $V$ all contain $x$.
122. (V2.81) $N_{X,x}$ is nonempty, and $N_X$ and $N_{X,x}$ are closed under nonempty open subsets and intersections as stated; every open subset of a quadrant is face-structured.
123. (V2.82) $L_X$ is a thin category and $F_X$ a functor to sets with injective maps, bijective over a fixed point.
124. (V2.83) Equivalence of admissible triples is an equivalence relation; $B : bX \to X$ has image $\partial X$ and fibres of $d_X(x)$ points.
125. (V2.84) $\Phi_{(x,V,\theta)}$ is injective with image $\theta \cap V^x$, and a bijection onto $\theta$ when every face of $V$ contains $x$; such $V$ are cofinal in $N_{X,x}$.
126. (V2.85) In $[0,\infty)^3$, for $x = (0,0,1)$ and a thin slab $V \in N_{X,x}$ along the segment to $(0,1,0)$, $\Phi_{(x,V,\theta)}$ misses $(0,1,0) \in \theta$.
127. (V2.86) $T_{bX}$, read with $O$ containing the class, is a topology.
128. (V2.87) The $U(x,V,\theta)$ form a basis of open sets, each $\Phi$ is an open embedding into $\theta$, and transition maps are the identity on points of $X$ between open subsets of $\theta$ and $\theta'$.
129. (V2.88) $bX$ is Hausdorff and metrizable.
130. (V2.89) $T_{bX}$ is the unique topology making every $U(x,V,\theta)$ open and every $\Phi$ open onto its

image; $B$ is continuous onto $\partial X$ with fibres of $d_X(x)$ points.

131. (V2.90) The quadrant $[0,\infty)^n \oplus W$ is tame, its corner has $n$ preimages in $bX$, and it is a tame sub-M-polyfold of itself.
132. (V2.91) $\mathbb{R}$ with the K-topology is Hausdorff and satisfies the modified notion of Remark 2.1.6 but is not locally complete; so the modified notion is strictly weaker.
133. (V2.92) Bundle retractions and bundle maps are defined by their shape $(r(u), \rho(u)v)$, $(a(u), \phi(u)v)$; $Tr$ is a bundle retraction covering $r^1$, and $Tf$ a bundle map covering $f^1$.
134. (V2.93) For every $m$, $(X^1)_m$ is $X_{m+1}$ with its topology and charts, through the identity of points.
135. (V2.94) $p : TX \to X^1$ is sc-smooth, locally the bundle retraction $Tr$ over $r^1$, with transition maps bundle maps covering the level-1 transition maps of $X$.
136. (V2.95) $T(X^1)$ and $(TX)^1$ are sc-diffeomorphic through a map commuting with the projections to $(X^1)^1$, independent of the chart and the identity in charts.
137. (V2.96) An sc-diffeomorphism raises to an sc-diffeomorphism of the index lifts; $T(X^2) \cong (TX)^2$ over $(X^1)^2 = (X^2)^1$.
138. (V2.97) For every $i$, $\Phi_i : T(X^i) \to (TX)^i$ and $J_i : (X^1)^i \to (X^i)^1$ are sc-diffeomorphisms with $J_i \circ p^i \circ \Phi_i = p$, compatible with the inclusions, unique, and natural in $X$.
139. (V2.98) The diagonal of $[0,\infty)^3 \oplus W$ is a retract sc-diffeomorphic to the half-line, with $d_C = 3$ and $d_\Delta = 1$ at the corner, and one chart onto $[0,\infty) \oplus W$ realizes $d_\Delta$.
140. (V2.99) Condition (3) of Definition 2.5.19 holds in every chart, every presentation and for every sub-M-polyfold retraction once it holds in one chart.
141. (V2.100) A tame sub-M-polyfold of a tame M-polyfold has a tame induced structure.
142. (V2.101) $A = \{(x, y, 0)\}$ is a tame sub-M-polyfold, $B = \{(x, x, 0)\}$ is a sub-M-polyfold that is not tame as a sub-M-polyfold but has a tame induced structure, for any $W$ in place of $\mathbb{R}$.
143. (V2.102) For tame $X$, the charts $\pi \circ \varphi \circ B$ make $bX$ a tame M-polyfold, $B$ is sc-smooth and $d_X(B(p)) = d_{bX}(p) + 1$.
144. (V2.103) For centred representatives, $\Phi$ is an sc-diffeomorphism of $U(x, V, \theta)$ onto $\theta$, the transition maps are sc-diffeomorphisms, and this characterizes the structure of $bX$.
145. (V2.104) For every representative, $\Phi$ is an sc-diffeomorphism onto the open $\theta \cap V^x$, Proposition 2.7.12 holds as printed, and the characterization holds with this image.
146. (V2.105) For the slab triple of V2.85, $\Phi$ is not onto $\theta$.
147. (V2.106) The iterated boundaries $b^k X$ are tame M-polyfolds, $B^k$ is sc-smooth with $d_X \circ B^k = d_{b^k X} + k$, and its image is $\{d_X \geq k\}$.
148. (V2.107) For the quadrant, $n \geq 1$, chart data of $bX$ exist and $d_{bX} = n - 1$ over the corner.
149. (V3.1) A germ of neighbourhoods is a nested sequence of relatively open neighbourhoods $U_m$ of 0 in $C_m$, with tangent germ $(U_{m+1} \times E_m)_m$; a germ is a fixed sequence of sets.
150. (V3.2) sc⁰-, sc¹-, sc^k- and sc-smooth germs are the notions of Chapter 1 with $U \cap E_m$ replaced by $U_m$.
151. (V3.3) Germs restrict, compose (when the inner germ maps $U_m$ into $V_m$ level by level), pair, and sc-operators are sc-smooth germs.
152. (V3.4) Propositions 1.2.1–1.2.3 (preprint 1.16–1.18) hold for germs.
153. (V3.5) An sc-smooth section germ is a section whose principal part composed with $r$ is an sc-smooth germ, on some germ of neighbourhoods inside $U$, in some centred strong bundle chart compatible with the atlas.
154. (V3.6) Sums of sections have sums of principal parts; sc⁺-sections have sc-smooth representatives $O \to K[1]$ in every compatible chart.
155. (V3.7) A filling is an sc-smooth germ satisfying (1)–(3) of Definition 3.1.4 (preprint 3.4) near 0, the map in (3) an sc¹-germ (the text's germ reading).

156. (V3.8) For a filler with $g(0) = 0$, $Tf(0) = Dg(0)|_{Dr(0)E}$ is onto iff $Dg(0)$ is, Fredholm iff $Dg(0)$ is, with the same index; if $Dg(0)$ is sc-Fredholm, so is $Tf(0)$, and condition (3) holds on every level.
157. (V3.9) A filled version is a filling of the principal part in the chart of Definition 3.1.11 (preprint 3.11), an sc-smooth germ whose levels may shrink with $m$.
158. (V3.10) A basic germ is an sc-smooth germ with $f(0) = 0$ whose $W$-component is $w - B(a, w)$ with $B$ an sc$^0$-contraction germ.
159. (V3.11) For sc$^1$ sc$^0$-contraction germs, Lemma 3.1.9 (preprint 3.9) as stated: $|D_2B(x)\zeta|_m \leq \varepsilon|\zeta|_m$ for $x \in U_{m+1}$ small on level $m$, for every $\varepsilon$ and $m$; and $D_2B(0, 0) = 0$.
160. (V3.12) $D(h + s)(0)$ is sc-Fredholm of index $n - N$ for a basic germ $h$ and an sc$^+$-germ $s$, and on every level $m$, $D(h + s)(x)$ is Fredholm of index $n - N$ for $x \in U_{m+1}$ small on level $m$.
161. (V3.13) An sc-Fredholm germ has a filling germ $g$ and an sc$^+$-germ $s$, $s(0) = g(0)$, with $g - s$ conjugated to a basic germ near 0 (the text's germ reading).
162. (V3.14) Near a zero of an sc-Fredholm germ, the closures of the zeros in $O(i)$ are compact subsets of $X_i$.
163. (V3.15) Zeros converging to the base point (or, for sc-Fredholm sections, to any point) converge on every level.
164. (V3.16) An sc-Fredholm section is sc-smooth, regularizing and an sc-Fredholm germ at every smooth point.
165. (V3.17) An sc-Fredholm section plus an sc$^+$-section is sc-Fredholm, for any strong bundle.
166. (V3.18) $s \circ r$ agrees with $s$ on $O$ and $R$ fixes $(r(u), s(r(u)))$.
167. (V3.19) $g + s \circ r$ is a filling germ of $f + s$ when $g$ is one of $f$, for $s$ an sc-smooth section of $K$ (a germ suffices).
168. (V3.20) Good position: $N \cap C$ has interior in $N$ and an algebraic complement satisfies the $\varepsilon$-condition; the text's sc-complement form is also formalized and holds for smooth $N$ on a partial quadrant.
169. (V3.21) If $N$ is in good position to $C$, then $N \cap C$ is a partial quadrant in $N$.
170. (V3.22) The tangent germ property, with $Y$ closed and "sc^∞-germ" read with $k$-dependent neighbourhoods.
171. (V3.23) The tangent germ property is invariant under sc-diffeomorphisms.
172. (V3.24) Under the regularity conditions (a) $\delta$ sc-smooth on $V$, (b) $\pi(O) \subseteq C$, a set with the tangent germ property is a sub-M-polyfold, tame, with corner charts $\pi\varphi$; without (a), (b) the text's proof fails in examples.
173. (V3.25) An sc$^0$-contraction germ has the contraction estimate on small balls on every level, and a working form with explicit regions.
174. (V3.26) The solution germ exists, is sc$^0$, and is unique in the contraction region.
175. (V3.27) If $B$ is sc$^1$, the solution germ is sc$^1$ on the germ shifted by one level.
176. (V3.28) $f^{(1)}$ is a contraction germ of class sc^{min(k, j)} with solution germ $T\delta$; it satisfies Definition 3.3.1 (preprint 3.28).
177. (V3.29) If $B$ is sc^k, the solution germ is sc^j on some shifted germ for every $j \leq k$.
178. (V3.30) If $B$ is sc^k, the solution germ is sc^k on the germ shifted by $2^k - 1$ levels; the text's hypotheses suffice.
179. (V3.31) If $B$ is sc-smooth, $\delta$ is $C^k$ into $W_m$ on a neighbourhood $V_{m,k}$ of 0; $\delta$ is an sc-smooth germ on one germ.
180. (V3.32) For a quadrant-like $\widetilde{C}$, any representative of the solution germ of an sc^k (sc-smooth) contraction germ is sc^k (an sc-smooth germ) on a smaller germ.
181. (V3.33) The basic class is Definition 3.1.7 (preprint 3.7).

182. (V3.34) A linear coordinate change makes the index-raised push-forward of a basic germ plus an sc$^+$-germ a basic germ, with $n - N$ unchanged.
183. (V3.35) The solutions near 0 of a contraction germ plus an sc$^+$-germ have compact closures on every level, for closed $\widetilde{C}$.
184. (V3.36) Zeros of $h + s$ near 0 lie on level 1, where the linearization is surjective Fredholm of index $n - N$.
185. (V3.37) Zeros of the push-forward near 0 have their $X$-component small on level 1.
186. (V3.38) Under Lemma 3.5.5's (preprint 3.42) regularity, the zeros of $g$ near 0 form the graph of a $C^1$ map $\tau$ over $\ker Dg(0) \cap C'$ into a good complement, $\tau(0) = 0$, $D\tau(0) = 0$, and $\tau$ is $C^k$ into level $k$ near 0 for every $k$.
187. (V3.39) $\ker Dg(0)$ is the graph of $D\delta(0)$ over $\ker DH(0)$.
188. (V3.40) Linear isomorphisms carrying $C$ onto $C'$ carry good complements and good position.
189. (V3.41) $\ker DH(0)$ has a good complement $Z$ with $Z \times W$ a good complement of $K'$.
190. (V3.42) The zeros of $f + s$ near 0 form the graph of a $C^1$ map $\sigma$ over $\ker D(f + s)(0) \cap C$, $\sigma(0) = 0$, $D\sigma(0) = 0$, $C^k$ into level $k$ near 0 for every $k$.
191. (V3.43) Definition 3.6.1 (preprint 3.48) is Definition 3.1.4 (preprint 3.4).
192. (V3.44) After shrinking, (1) and (2) of the filler hold on all of $O \cap W$, $U \cap W$.
193. (V3.45) Definitions 3.6.3, 3.6.4 (preprint 3.50, 3.51) are Definitions 3.1.11, 3.1.16 (preprint 3.11, 3.14) for a tame strong bundle; the theorems of §3.6 use them with the chart of the germ data tame (V3.103).
194. (V3.46) In a centred chart, $Dg(0)$ and $Tf(x_0)$ are sc-Fredholm with index $n - N$ on every level, the dimensions of any basic germ to which $g - s$ is conjugated; at a zero of an sc-Fredholm section the linearization is sc-Fredholm in every chart.
195. (V3.47) Good position of the germ: $Tf(0)$ onto and $\ker Dg(0)$ in good position to $T_0O \cap C$ in $T_0O$, in a centred chart; also with the text's sc-complement, the same at zeros of sc-Fredholm sections with tame germ data.
196. (V3.48) Near a zero in good position in a tame chart, the zero set is carried injectively by $\psi\varphi$ onto the graph of a $C^1$ map $\sigma$ over $\ker D(h + t)(0) \cap C'$, with $Y$ closed, the kernel smooth, and $C^k$-lifts of $\sigma$ into level $k$.
197. (V3.49) Along the graph of $\tau$ near 0, $Y$ (closed) stays a good complement of $N_v$ with respect to $C_{v+\tau(v)}$; also for $\tau$ on a relatively open $V$ and any presentation of $C$.
198. (V3.50) Without boundary, a zero set with the regular tangent germ property is a manifold without boundary.
199. (V3.51) A zero set with the regular tangent germ property is a tame sub-M-polyfold and a manifold with boundary with corners.
200. (V3.52) Injectivity of $L(0)$ and the compactness property make $L(b)$ injective for $b$ near 0; in Theorem 3.7.1 (preprint 3.58) both hypotheses follow from its condition (2).
201. (V3.53) On a compact parameter set with injective $L(b)_m$ there is a uniform lower bound $c_m$.
202. (V3.54) $(v, k) \mapsto L(v)^{-1}k$ is an sc-smooth germ under uniform lower bounds $c_m$.
203. (V3.55) A good complement has vanishing coordinates in $\Sigma$.
204. (V3.56) $d_{N\cap C} = d_C$ off $N \cap W$, and $d_{N\cap C} + \dim(N \cap W) = \dim N$ on it.
205. (V3.57) $d_{N\cap C}(v) = d_C(v + \tau(v))$ for $v \notin N \cap W$ near 0.
206. (V3.58) The quadrant inverse function theorem holds for $C^j$ maps.
207. (V3.59) A $C^j$ map with $\|Df - 1\| \le \frac{1}{2}$ preserving interior points is a $C^j$-diffeomorphism onto a relatively open set.
208. (V3.60) Such a map is bi-Lipschitz, injective, and has relatively open image.
209. (V3.61) Such a map is a homeomorphism onto its image.

210. (V3.62) Its inverse has derivative $[Df(g(y))]^{-1}$.
211. (V3.63) The implicit function theorem in partial quadrants, with $f(k+\tau(k)) = 0$ and $k+y \in U$.
212. (V3.64) There is a region on which $B(k,\cdot)$ is a $\frac{1}{2}$-contraction and fixed points are unique.
213. (V3.65) $B(k,\cdot)$ maps $\{|y| \le \varepsilon|k|\}$ into itself.
214. (V3.66) $\tau$ is continuous with $|\tau(k)| \le \varepsilon|k|/2$, Lipschitz with constant $\varepsilon/(1+\varepsilon) \le \varepsilon$ on all of $V_0$.
215. (V3.67) $\tau$ is $C^j$ on the whole $V_0$, with the derivative formula.
216. (V3.68) Under the filler condition at the nearby zeros (a consequence of good position, V3.102), near a zero in good position in a tame chart of an sc-Fredholm section with tame germ data over an M-polyfold without boundary the zeros are smooth points in good position in tame charts and form a manifold without boundary; the forms with the text's hypotheses, the index and the kernel are V3.95, V3.97, V3.99, V3.101.
217. (V3.69) Near a zero in good position in a tame chart whose nearby zeros are in good position in tame charts (or, for tame germ data, under the filler condition of V3.82), the local solution set consists of smooth points and is a tame sub-M-polyfold equivalent to a $d$-dimensional manifold with boundary with corners.
218. (V3.70) With good position in a tame chart at every zero ($S \neq \emptyset$), the solution set is a sub-M-polyfold with tame induced structure, a manifold with boundary with corners of locally constant dimension; no hypothesis is added.
219. (V3.71) The image of a tame retract under an sc-diffeomorphism, and an open subset of a tame retract, are tame retracts.
220. (V3.72) A nonempty set with regular data of Definition 3.2.1 (preprint 3.25) of some dimension at every point is a sub-M-polyfold, tame, with an atlas of finite-dimensional corner charts $\pi\varphi$.
221. (V3.73) Two graph descriptions of one set, the first $C^1$ and the second centred at a point of it with $C^k$-lifts at 0, have the same dimension, and the first is $C^k$ into every level near that point.
222. (V3.74) Data of Definition 3.2.1 (preprint 3.25) with $\pi(O) \subseteq C$, at a point whose neighbours all have data, can be shrunk to regular data, and the dimension is locally constant.
223. (V3.75) A finite-dimensional subspace in good position has a closed good complement along which the projection maps the partial quadrant into itself.
224. (V3.76) If $g$ is conjugated to a basic germ and $s$ is an sc$^+$-map with $s(0) = 0$, then $(g+s)^1$ is conjugated to a basic germ, by $\Lambda \circ \Phi^1$.
225. (V3.77) Theorems 3.5.6, 3.5.11 (preprint 3.43, 3.47) and the graph description of Theorem 3.6.8 (preprint 3.54) hold with a closed good complement whose projection maps the partial quadrant into itself.
226. (V3.78) At a zero, the linearizations in two strong bundle charts are conjugate by isomorphisms respecting the partial quadrants: surjectivity, Fredholm index and good position of the kernel do not depend on the chart.
227. (V3.79) Surjectivity onto a dense part of $K_z$ and good position of $\ker Dg(z) \cap T_zO$, in a chart not centred at the zero $y$ ($z$ in the germ of $g$), give good position of $(f,y)$ in a tame chart, for tame germ data.
228. (V3.80) In the chart at $x_0$, at the zeros near $x_0$, $D(h+t)$ is surjective Fredholm of index $n-N$ and the tangent space $N_v$ of the graph is in good position with complement $Y$.
229. (V3.81) In the chart at $x_0$, at the zeros near $x_0$, $\ker Dg(z_0) = D\psi^{-1}(N_v)$ lies in $T_{z_0}O$ and is in good position there; no hypothesis is added.

230. (V3.82) For tame germ data, if the filler in the tame chart at $x_0$ also satisfies condition (3) of Definition 3.1.4 (preprint 3.4) at its zeros near 0, every zero near $x_0$ is in good position in a tame chart; the text's argument needs that condition, which holds whenever $(f, x_0)$ is in good position (V3.102).
231. (V3.83) A good complement of $\mathbb{R} \times N$ projects into the lineality space; the tilted kernel of the parameter extension is in good position.
232. (V3.84) At a zero $x_0$ in good position in a tame chart, the solution set near $x_0$ has data of Definition 3.2.1 (preprint 3.25) at $x_0$ in the chart $\psi\varphi$, with $\pi(O) \subseteq C$.
233. (V3.85) If $x_0$ and the zeros near it are in good position in tame charts, the solution set has regular data at $x_0$ of a dimension $d$ that every nearby datum shares; Theorem 3.6.8 (1)–(2) (preprint 3.54) under the filler condition of V3.82.
234. (V3.86) Over an M-polyfold without boundary, with good position in a tame chart at every zero, the solution set is a manifold without boundary of locally constant dimension.
235. (V3.87) With good position in a tame chart at every zero (the text's hypotheses), the solution set is a sub-M-polyfold with tame induced structure, a manifold with boundary with corners of locally constant dimension, the disjoint union of its open and closed parts $S_n$ of dimension $n$.
236. (V3.88) Given sc-isomorphisms $L(b) = P_Y Df(b,0)|_X$ with uniform lower bounds and condition (3), $\Phi_*(f - f(0))$ is a basic germ with contraction $x - L(b)^{-1}P_Y(f - f(0))$.
237. (V3.89) Under conditions (1)–(3) of Theorem 3.7.1 (preprint 3.58), in the coordinates of its proof, $f - f(0)$ is conjugated near 0 to a basic germ.
238. (V3.90) A $C^j$ map with $\|Df - 1\| \leq \frac{1}{2}$ on a convex set, whose image has unique differentiability, has a $C^j$ inverse with $Dg = [Df \circ g]^{-1}$.
239. (V3.91) For a strong bundle (tameness not needed), an sc-Fredholm section $f$ and sc$^+$-sections $s_i$, $(r, x) \mapsto f(x) + \sum r_i s_i(x)$ is an sc-Fredholm section of the pull-back bundle over $\mathbb{R}^n \times X$.
240. (V3.92) In the product chart at $(0, x)$ the linearization of the perturbed section is $(\delta r, \xi) \mapsto f'(x)\xi + \sum \delta r_i s_i(x)$ (chart form); at $(r_0, x)$ the first term is that of $f + \sum r_{0,i} s_i$.
241. (V3.93) In the chart at a zero $x_0$ in good position, at every zero $z_0$ of the filler near 0 and for every smooth $k$, $\rho(z_0)k = Dg(z_0)x$ for some $x \in T_{z_0}O$, by a curve of zeros of $g + \mu\rho(r(\cdot))k$ through $z_0$ from an implicit function theorem at $z_0$.
242. (V3.94) With the text's hypotheses (tame germ data, good position in a tame chart), every zero near a zero in good position is in good position in a tame chart, and the solution set has there regular data of a dimension that every nearby datum shares; also with good position in the text's sense (sc-complement) in the conclusion.
243. (V3.95) With the text's hypotheses (tame germ data, good position in a tame chart), the local solution set near a zero in good position consists of smooth points in good position in tame charts and is a tame sub-M-polyfold, a $d$-dimensional manifold with boundary with corners, without boundary if $d_X \equiv 0$.
244. (V3.96) At a zero, the linearization in every chart is sc-Fredholm of the index $n - N$ of any conjugating basic germ on every level, and onto on every level with $\dim\ker = n - N$ if the germ is in good position.
245. (V3.97) The dimension of the solution set near a zero in good position in a tame chart is the Fredholm index, $\dim\ker f'(y) = \operatorname{ind} f'(y)$, at every zero near it and in every chart; also in Theorems 3.1.26, 3.1.22, 3.6.11 (preprint 3.23, 3.19, 3.57).
246. (V3.98) For tame germ data, near a zero in good position in a tame chart, every zero $y$ has $f'(y)$ onto with $\operatorname{ind} f'(y) = \operatorname{ind} f'(x_0)$: the index is locally constant along the solution set; for tame germ data, without bump functions, the index is locally constant along the zero

set without good position, the index of Definition 6.2.1 (preprint 6.1) with a locally defined sc$^+$-section is well defined and exists at every smooth point, and it is constant along every sc-smooth path $\mathbb{R} \to X$ (Proposition 6.2.2 (preprint 6.2)); under bump functions the local and the global reading of Definition 6.2.1 (preprint 6.1) agree.

247. (V3.99) At the zeros near $x_0$, $D\psi$ maps $\ker f'(y)$, read in the chart at $x_0$, onto the tangent space of the solution graph.
248. (V3.100) For an sc-Fredholm section with tame germ data and a zero $y$ with $d_X(y) = 0$, $(f, y)$ is in good position in a tame chart if and only if $f'(y)$ is onto.
249. (V3.101) Over an M-polyfold without boundary, Theorems 3.1.22 and 3.6.10 (preprint 3.19 and 3.56) hold for tame germ data with the hypothesis "$f'(x)$ surjective", with tame induced structure and the dimension the index.
250. (V3.102) At a point of the retract with $\ker Dg \subseteq T_{z_0}O$, $\rho(z_0)F \subseteq Dg(T_{z_0}O)$ implies condition (3) of the filler there (and conversely when $Dg$ is onto); so the filler condition at the nearby zeros holds whenever $f$ has tame germ data and $(f, x_0)$ is in good position in a tame chart.
251. (V3.103) The theorems of §3.6 assume, as Definitions 3.6.3, 3.6.4 and 3.6.6 (preprint 3.50, 3.51 and 3.53) do, germ data and good position in a strong bundle chart whose base retraction is tame; compatible charts whose base retraction is the identity of $U$ are tame, so the hypotheses can be met.
252. (V3.104) The product of the constant retraction onto $(1, 0)$ of $[0, \infty) \oplus W$ with any sc-smooth retraction of a nonempty set is not tame: the degeneracy index drops by one under it.
253. (V3.105) If $f$ is an sc-Fredholm section with tame germ data and $s$ an sc$^+$-section, $f + s$ has tame germ data, in the same charts.
254. (V3.106) If $f$ has tame germ data, $(r, x) \mapsto f(x) + \sum r_i s_i(x)$ is an sc-Fredholm section of $\pi^*P$ with tame germ data: the product chart of a tame chart is tame.
255. (V3.107) The good position of $(f, y)$ obtained in one chart holds in every chart carrying germ data at $y$, in particular in the tame one.
256. (V3.108) The 2021 filling is the filling without tameness of $r$, with $r(0) = 0$; it is the project's filling, and a 2017 filling is a 2021 filling.
257. (V3.109) Over $r(a, b) = (0, b)$ on $[0, \infty) \times \mathbb{R}$, which is not tame, $g(a, b) = a$ fills the zero section: a 2021 filling need not be a 2017 one.
258. (V3.110) The 2021 sc-Fredholm germs and sections are the project's, and the 2017 germs those with tame germ data; the definitions are satisfiable.
259. (V3.111) For a 2021 filling with $\widehat{f}(0) = 0$, $T\widehat{f}(0)$ and $Dg(0)$ are onto, and Fredholm, together, with the same index.
260. (V3.112) Local compactness and convergence on every level hold for 2021 sc-Fredholm germs.
261. (V3.113) 2021 sc-Fredholm sections are stable under sc$^+$-perturbations and parametrized perturbations.
262. (V3.114) At a zero of a 2021 sc-Fredholm germ, the linearization is Fredholm of index $n - N$.
263. (V3.115) Without a tame chart, good position of $\ker T\widehat{f}(0)$ in $T_0O$ need not give good position of $\ker Dg(0)$ in $E$, the step of the proof of Theorem 3.6.8 on p. 138.
264. (V3.116) Zeros of a basic germ plus an sc$^+$-germ converging to 0 on level 0 lie, from some index on, on level $m$ and converge there.
265. (V3.117) A pointed $N \cap C$, $N$ in good position, is a quadrant of $N$, its extreme rays generated by the preimages of the unit vectors.
266. (V3.118) Vanishing coordinates of points of $N \cap C$ are vanishing coordinates of extreme-ray generators, and good complements vanish there.

267. (V3.119) Either $N \cap (\mathbb{R}^{\Gamma^c} \oplus W) = 0$, or $N$ is a line in $\mathbb{R}^{\Gamma^c} \oplus W$ and $\Gamma = \emptyset$; the case $N = 0$ belongs to the first alternative.
268. (V3.120) If $N \cap (\mathbb{R}^{\Gamma^c} \oplus W) = 0$, the projection to $\mathbb{R}^{\Gamma}$ carries $(N, N \cap C)$ onto a quadrant; if $0 \neq N \subseteq \mathbb{R}^{\Gamma^c} \oplus W$, $(N, N \cap C) \cong (\mathbb{R}, [0, \infty))$.
269. (V3.121) If $N$ is in good position and $N \cap C$ is a closed convex cone, $N \cap C$ is a quadrant of $N$.
270. (V4.1) A corner chart is a chart onto a relatively open subset of a partial quadrant in a finite-dimensional space with the constant structure; a corner structure is an equivalent atlas of corner charts, of possibly varying dimension, and makes $X$ tame.
271. (V4.2) The transition maps between compatible corner charts are $C^\infty$ on their relatively open overlaps.
272. (V4.3) An sc-Banach space with finite-dimensional $E_0$ has all levels equal and is sc-isomorphic to $E_0$ with the constant structure; all its points are smooth.
273. (V4.4) Corner structures are the atlases of charts onto relatively open subsets of partial quadrants in sc-Banach spaces with finite-dimensional $E_0$; those of fixed dimension $n$ are the manifolds with corners of Theorem 3.2.3 (preprint 3.27).
274. (V4.5) An M-polyfold, tame or not, has a corner structure iff all its points are smooth, $1^1_0$ is sc-smooth, and every point is the centre of a tame chart whose $T_0O$ is finite-dimensional and in good position.
275. (V4.6) $1^0_1 : X^1 \to X$ is sc-smooth; if $X = X_1$ and $1^1_0$ is sc-smooth it is an sc-diffeomorphism; some M-polyfold with $X_0 = X_\infty$ has $1^1_0$ not sc-smooth; a finite-dimensional space with the constant structure has $V^1 = V$, and all its retracts are sc$^+$-retracts.
276. (V4.7) An sc$^+$-map is a map with sc-smooth lift into $F^1$; it raises levels, is sc-smooth, and is preserved by composition with sc-smooth maps on either side.
277. (V4.8) At a smooth point the derivative of an sc$^+$-map is an sc$^+$-operator, compact on every level; a compact projection has finite-dimensional range.
278. (V4.9) An sc$^+$-retraction is an sc-smooth retraction that is an sc$^+$-map (on an open subset of an M-polyfold: with values in $X_1$ and $1^1_0 \circ s$ sc-smooth).
279. (V4.10) A retraction with the same retract as an sc$^+$-retraction is an sc$^+$-retraction.
280. (V4.11) An sc$^+$-retract is the retract of some, equivalently of every, retraction, which is sc$^+$.
281. (V4.12) The retract of an sc$^+$-retraction consists of smooth points and its tangent spaces are finite-dimensional.
282. (V4.13) A retract is an sc$^+$-retract iff it consists of smooth points and the lift $O \to O^1$ is sc-smooth between retracts.
283. (V4.14) A chart's model is an sc$^+$-retract iff its domain is smooth and $1^1_0$ is sc-smooth on it; "$X_0 = X_\infty$ and $1^1_0$ sc-smooth" iff every chart's model is an sc$^+$-retract.
284. (V4.15) $C_e$ is a partial quadrant with $C_e - e = C_e$; near $e$ it agrees with $C$, and re-centring preserves degeneracy indices, minimal subspaces and the cones $C_v$.
285. (V4.16) Translation by a smooth $e$ is an sc-diffeomorphism from relatively open subsets of $C$ near $e$ onto relatively open subsets of $C_e$, and carries retracts (tame retracts) to retracts (tame retracts).
286. (V4.17) Every smooth point of a chart's domain is the centre of a re-centred chart of $X$, tame if the original is.
287. (V4.18) Re-centring a strong bundle chart in the base gives a strong bundle chart; every smooth point of the base is the centre of one.
288. (V4.19) An sc$^+$-retraction is $C^1$ on every level $k+1$ with operator-norm-continuous derivative, and satisfies the mean value estimate with any constant $\theta$ on small balls of $C_{k+1}$.
289. (V4.20) For an sc$^+$-retraction onto $O \ni 0$ with $T_0O$ in good position, a closed good comple-

ment $Y$ and the projection $p$ along $Y$, $p$ is an sc-diffeomorphism from $O\cap W$ onto a relatively open $V\ni 0$ of $T_0O\cap C$, with inverse $\Gamma$, $\Gamma(a)-a\in Y$.

290. (V4.21) Near 0, $(1-p)$ contracts differences of points of $O$ on level $k+1$, so two such points with the same projection coincide.
291. (V4.22) For small $a\in N\cap C$ there is a point $u\in O$ with $p(u)=a$, $u-a\in Y$ small on level 1.
292. (V4.23) Projections of points of $O$ near 0 lie in $C$, and the neighbourhoods $W$, $V$ and the inverse $\Gamma$ exist with the required level-1 control.
293. (V4.24) $\Gamma$ is Lipschitz on level 1, continuous into every level, and differentiable into every level with the frame formula $D\Gamma(a)=Dt(\Gamma(a))|_N(p\,Dt(\Gamma(a))|_N)^{-1}$.
294. (V4.25) $\Gamma$ is $C^\infty$ into every level, hence sc-smooth on the fixed $V$; $p$ and $\Gamma$ are sc-smooth between the retracts.
295. (V4.26) A chart composed with an sc-diffeomorphism between retracts is a chart of the M-polyfold.
296. (V4.27) Corner charts re-centre to tame centred charts whose $T_0O$ is the whole finite-dimensional space, in good position; so (i) implies (ii).
297. (V4.28) Under (ii), $p\circ\psi$ (with a closed good complement) is a corner chart of $X$ at every point; so (ii) implies (i).
298. (V4.29) A smooth finite-dimensional submanifold is locally the image of an sc$^+$-retraction on an open set; it is a sub-M-polyfold.
299. (V4.30) In every chart, $\varphi\circ s\circ\varphi^{-1}\circ r$ is an sc$^+$-retraction onto the chart image of $A$, whose points are smooth.
300. (V4.31) In a chart, $T_aA$ is the image of $T_aX$ under the tangent map of $s$, lies in $T_aX$, and is finite-dimensional.
301. (V4.32) At points with $d_X=0$ a submanifold is in good position; if $d_X=0$ on all of $A$, $A$ has a corner structure, without tameness of $X$.
302. (V4.33) $A\subseteq X_\infty$, and all points of $A$ are smooth for its induced structure.
303. (V4.34) $A$ is locally closed in $X$.
304. (V4.35) $\partial A\subseteq\partial X$.
305. (V4.36) $1^1_0:A\to A^1$ is sc-smooth for the induced structure.
306. (V4.37) Tangent spaces of retracts are germs, are monotone under inclusion of retracts, and are mapped into each other by derivatives of maps between retracts.
307. (V4.38) Good position of $A$ at $a$ means, in a chart, that $T_zA$ is in good position to $C_z$ inside $T_zO$.
308. (V4.39) Good position of $A$ at $a$ does not depend on the chart, the data of Definition 4.2.1 (preprint 4.13) or the presentation.
309. (V4.40) A tame submanifold is one whose induced M-polyfold structure is tame.
310. (V4.41) A smooth finite-dimensional submanifold of a tame M-polyfold, in good position everywhere, has a corner structure and is tame.
311. (V4.42) In a tame centred chart, good position inside $T_0O$ gives good position in $E$, with good complement $Y\oplus(1-\pi)E$ and constant $\varepsilon/(\|\pi\|+1)$.
312. (V4.43) For tame $X$, good position at a point holds at all nearby points of $A$.
313. (V4.44) General position of $A$ at $a$ means, in a chart, an sc-complement of $T_zA$ in $T_zO$ inside $T^R_zO$, level by level.
314. (V4.45) General position does not depend on the chart, the data or the presentation.
315. (V4.46) At the corner of a centred chart, $N+E^{\min}_0=E_0$ gives onto coordinates, good position with a closed complement in $E^{\min}_0$, and $d_{N\cap C}=d_C$ on $N\cap C$.
316. (V4.47) A finite-dimensional smooth $N$ has in $N+R$ an sc-complement inside the

sc-subspace $R$; $T_yQ + E_0^{\min} = E_0$ gives general position at $y$.

317. (V4.48) In a tame retract, general position at a smooth point implies good position.
318. (V4.49) In a tame centred chart, general position at the centre gives the graph description with $Y \subseteq E_0^{\min}$, equal degeneracy indices along the graph, and general position along it.
319. (V4.50) Theorem 4.2.11 (1)–(4) (preprint 4.22) holds in a tame chart centred at a point of general position, with $F_\sigma$ a germ inside $U$; (1) and (4) for the faces of the chart model, not of $U$ itself.
320. (V4.51) $\Gamma \circ \kappa_\sigma \circ p \circ t$ is an sc$^+$-retraction onto $A \cap F_\sigma$ in the chart; its corner model is sc$^+$ and in general position in the face model.
321. (V4.52) An sc$^+$-retraction in a chart is an sc$^+$-retraction on an open subset of $X$; $A \cap F_\sigma$ is a tame submanifold of the tame $F_\sigma$, in general position in it.
322. (V4.53) A stratified regular point is one where the projection maps the tangent space of the stratum through it onto $\mathbb{R}^m$, in some product chart.
323. (V4.54) Theorem 4.3.1 (preprint 4.23): almost every $\lambda$ with $|\lambda| < \varepsilon$ is a stratified regular value whose fibre is a compact submanifold in general position with a tame corner structure.
324. (V4.55) In a tame chart $\operatorname{codim} T_x^R X = d_X(x)$; general position at $a$ gives $d_X(a) \leq \dim T_a A$, and a zero-dimensional submanifold in general position avoids $\partial X$.
325. (V4.56) Sard: the critical values of a $C^\infty$ map between finite-dimensional spaces form a Haar-null set.
326. (V4.57) The strata of $\{\ell \geq 0\}$ have dimension $\dim V - \#\sigma$, and the stratified critical values of a $C^\infty$ map form a null set.
327. (V4.58) Theorem 3.8.22 (preprint 3.78) for $C^\infty$ data has a $C^\infty$ solution.
328. (V4.59) At a stratified regular point the kernel of $Df$ is in good position to the local partial quadrant, with a closed complement in the stratum.
329. (V4.60) Near a stratified regular point the fibre of $f$ is the image of a $C^\infty$ retraction whose derivative at the point is the identity on $\ker Df$.
330. (V4.61) $e \mapsto (\Gamma(\rho(p(t(c,e)))))_2$ is an sc$^+$-retraction onto the chart image of $A_c$, whose tangent space contains the vertical tangent vectors of $A$.
331. (V4.62) General position of $A$ and stratified regularity at $z$ give general position of the fibre at $z_2$.
332. (V4.63) Every point of $A$ has a neighbourhood and a null set of parameters off which the conclusions of Theorem 4.3.1 (preprint 4.23) hold locally.
333. (V4.64) Countably many such neighbourhoods cover $A$, so for any parameter set the good parameters have null complement.
334. (V4.65) In the 2021 form of Theorem 4.2.11 (1) (preprint 4.22), each of the $d_X(x)$ coordinate faces of the centred chart has a germ that contains $x$ and lies in $U$.
335. (V4.66) Theorem 4.3.1 (preprint 4.23) in the 2021 form: with compact, possibly empty, fibres over $|\lambda| \leq \varepsilon$, the good parameters contain a measurable set of full measure in $\{|\lambda| < \varepsilon\}$.
336. (V4.67) In a local model, a vector field is a map $A : E_1 \to E_0$ with $x \mapsto (x, A(x))$ sc-smooth from $O^1$ to $TO$; then $Dr(x)A(x) = A(x)$ on $O_1$, and 0 is one.
337. (V4.68) On an M-polyfold, an sc-smooth vector field is an sc-smooth section $X^1 \to TX$ of the tangent projection; the zero section is one.
338. (V4.69) $T(O^1) = (TO)^1$, and the level map $A_1$ of a vector field $A$ on $O$ is a vector field on $O^1$.
339. (V4.70) For an sc$^2$ map on a relatively open subset of a partial quadrant, $D^2 f(z)(a,b)$, $z \in U_2$, $a, b \in E_1$, is symmetric in $a$ and $b$.
340. (V4.71) In a local model, $[A,B](x) = D\tilde{A}(x)B(x) - D\tilde{B}(x)A(x)$ satisfies $Dr(x)[A,B](x) =$

$[A,B](x)$ at every $x \in O_2$.

341. (V4.72) In a local model, $x \mapsto (x, [A,B](x))$ is an sc-smooth map from $O^2$ to $TO$.
342. (V4.73) For $x \in O_2$ the bracket $[A,B](x)$ need not lie in $E_1$: two sc-operators on a weighted sequence space.
343. (V4.74) In a local model, an sc-smooth map that relates two pairs of vector fields on level 1 relates their brackets on level 2, also on an open part $O \cap W$ of the retract.
344. (V4.75) The principal parts of an sc-smooth vector field are vector fields on the models, and their brackets in two charts are related by the transition map.
345. (V4.76) On an M-polyfold, $[A,B] : X^2 \to TX$, given in any chart by the bracket of the principal parts, is an sc-smooth section over the inclusion $X^2 \to X^1$, independent of the chart.
346. (V4.77) For an sc-smooth $f : X \to Y$ with $Tf \circ A = A' \circ f_1$ and $Tf \circ B = B' \circ f_1$ on $X^1$, $Tf \circ [A,B] = [A',B'] \circ f_2$ on $X^2$.
347. (V4.78) $A^{(1)} = \Phi^{-1} \circ A_1$ is an sc-smooth vector field on $X^1$, and $\Phi(A^{(1)}(x)) = A(x)$ for $x \in X^2$.
348. (V4.79) For every $i$, $A^{(i)}$, the index of $A$ raised $i$ times, is an sc-smooth vector field on $X^i$, and it is $A$ through the identifications of Proposition 2.3.17.
349. (V5.1) Every strong bundle chart compatible with the atlas reads $Y_{0,1}$ by the fibre component in $F_1$, and its $\Phi[1]$ is a chart of $Y[1]$.
350. (V5.2) The points of $P^{-1}(x)$ in $Y_{0,1}$ form a linear subspace, complete for the chart norm $|\cdot|_1$; the norms of two charts are locally uniformly comparable.
351. (V5.3) An auxiliary norm is a continuous function on $Y[1]$, a complete norm on every fibre, with $P(w_k) \to x$ and $N(w_k) \to 0$ forcing $w_k \to 0_x$.
352. (V5.4) A continuous fibrewise complete norm is auxiliary iff near every point it is two-sided comparable with $|h|_1$ in some strong bundle chart, also literally on the whole domain of the chart; auxiliary norms are so in every chart.
353. (V5.5) Every strong bundle has an auxiliary norm, glued from the chart norms by a partition of unity.
354. (V5.6) Two auxiliary norms are comparable by a continuous positive function $f$: $fN_0 \le N_1 \le f^{-1}N_0$.
355. (V5.7) Compact solution set, weakly proper and proper are as in the text, with $N = \infty$ off $Y_{0,1}$; for continuous $f$ proper implies weakly proper implies compact solution set.
356. (V5.8) Near 0, the points where a basic germ (on a closed $\tilde{C}$) has level-1 norm $\le c$ lie in a compact set, on a neighbourhood independent of $c$.
357. (V5.9) Strong bundle maps have locally uniform $(0,1)$ fibre bounds; at an sc-Fredholm germ the points with $|\Phi(f(x))_2|_1 \le C$ near $x_0$ have compact closure, for every $C$.
358. (V5.10) An sc-Fredholm section with compact solution set is proper, without tameness; proper, weakly proper and compact solution set are equivalent for sc-Fredholm sections.
359. (V5.11) At a zero of an sc-Fredholm section there are nested neighbourhoods $U(i)$ with $\mathrm{cl}\{y \in U(i) \mid f(y) = 0\}$ compact in $X_i$.
360. (V5.12) With the initial topology of the inclusions, $X_\infty$ is metrizable, and a compact solution set of an sc-Fredholm section is compact in $X_\infty$, without tameness.
361. (V5.13) For an sc-Fredholm section with compact solution set, $f^1$ is an sc-Fredholm section of $P^1$ with compact solution set, and so is $f^2$ of $P^2$; the solution set is compact in every $X_i$.
362. (V5.14) $P^1 : Y_{1,1} \to X^1$ with the index raises of the strong bundle charts is a strong bundle, and for every sc-Fredholm section $f$, $f^1$ is an sc-Fredholm section of it, the filler condition on level 1 included.

363. (V5.15) $\mathbb{R}$ with the constant structure is an M-polyfold; sc-smoothness of maps is checked in local charts; $C^\infty$ functions of sc-smooth functions are sc-smooth.
364. (V5.16) Admitting sc-smooth bump functions means a bump with $f(x) \neq 0$ inside every neighbourhood of $x$; Definition 5.5.2 (preprint 5.29) as written is weaker, follows from the density of such points, and gives them at the smooth points of an sc-Banach space.
365. (V5.17) With bump functions, every $x$ has an sc-smooth $f : X \to [0,1]$ supported in a given neighbourhood, equal to $1$ near $x$.
366. (V5.18) The zero section, products of sc-smooth functions and sc$^+$-sections, and finite sums of sc$^+$-sections are sc$^+$-sections; sc$^+$-sections take values in $Y_{0,1}$.
367. (V5.19) A local sc$^+$-section of a chart of the atlas, cut off by an sc-smooth function supported in the chart domain and extended by $0$, is an sc$^+$-section.
368. (V5.20) With bump functions and an auxiliary norm, every smooth $e$ over a smooth $x$ is the value at $x$ of an sc$^+$-section supported in $U$ with $N(s(y)) < N(e) + \varepsilon$ everywhere.
369. (V5.21) $\Gamma^{+,1}_U(P)$ is the set of sc$^+$-sections supported in $U$ with $N \leq 1$; $\rho = \sup_x N(s(x) - s'(x))$ is a metric on it, not complete for the trivial bundle over $\mathbb{R}$, complete for $U = \emptyset$.
370. (V5.22) A section vanishing off an adapted $U$ with $N(s) \leq 1$ has the zeros of $f + s$ in $\{x \in U \mid N(f(x)) \leq 1\}$, with compact closure; for $s$ in the completion (uniform $N$-limits of allowable sections) the solution set of $f + s$ is closed, hence compact.
371. (V5.23) Transversality to the zero section is surjectivity of the linearization at every zero, in some and then every chart.
372. (V5.24) At a zero of an sc-Fredholm section, finitely many allowable sc$^+$-sections supported in a given $U$ span a complement of the range of the linearization.
373. (V5.25) The slices $g + \sum \lambda_i t_i$ carry the zeros of $F$, are sc-Fredholm, and stay allowable for $m|\lambda| \leq \varepsilon$ when $N(s_0) \leq 1 - \varepsilon$.
374. (V5.26) A slice over a parameter where $\ker \mathrm{Lin}_F$ projects onto $\mathbb{R}^m$ is transversal; spanning sections make $F$ transversal at $(0, y)$, also after adding sections.
375. (V5.27) $\mathbb{R}^m \times X$ has no boundary if $X$ has none; transversality of $F$ spreads to nearby zeros and gives good position on an open $V \supseteq \{0\} \times S_{g'}$ which contains the zeros over small $\lambda$.
376. (V5.28) Near a zero of $F$ in good position, the slices are transversal at their zeros over all $\lambda$ off a null set.
377. (V5.29) Theorem 5.3.5 (preprint 5.13): for $\partial X = \emptyset$, $f$ with tame germ data, there is $U$ such that allowable perturbations have compact solution sets and $s_0 + \sum \lambda_i t_i$ is transversal with a manifold solution set for small $\lambda$ off a null set.
378. (V5.30) Theorem 5.3.5 (preprint 5.13), dense form: such good allowable perturbations exist within $\rho$-distance $\varepsilon$ of every allowable $s'$.
379. (V5.31) General position of $(f, y)$: in a tame chart, $\mathrm{Lin}(z)$ onto and its kernel with an sc-complement inside $T^R_z O$, for every presentation.
380. (V5.32) Regular tangent germ data give sc$^+$-retractions, so solution sets in good position are smooth finite-dimensional submanifolds (Definition 4.2.1 (preprint 4.13)).
381. (V5.33) Near a zero in good position, the tangent space of the solution set equals $\ker \mathrm{Lin}(z)$ in every chart.
382. (V5.34) Near a zero in good position, general position of $(f, y)$ is general position of the solution set at $y$, and is chart independent.
383. (V5.35) For $X$ tame, general position of $(f, x_0)$ persists at the nearby zeros, with the solution set a submanifold in general position there.
384. (V5.36) Properties (i)–(ii) are general position of $F$ at $(0, y)$, survive added sections, and are achieved by finitely many allowable sections.

385. (V5.37) For $X$ tame, a family with $F$ in general position along $\{0\} \times S_{f+s_0}$ gives, for small $\lambda$ off a null set, $f + s_\lambda$ with all the conclusions of Theorem 5.3.10 (preprint 5.18).
386. (V5.38) Theorem 5.3.10 (preprint 5.18), generic form, for $X$ tame with bump functions and $f$ with tame germ data.
387. (V5.39) Theorem 5.3.10 (preprint 5.18), dense form: perturbations in general position exist within $\rho$-distance $\varepsilon$ of every allowable $s'$.
388. (V5.40) Theorem 5.3.12 (preprint 5.19): with good position at the boundary zeros, small $\sum \lambda_i t_i$ vanishing near $\partial X$ give transversality and good position everywhere, a compact manifold with corners.
389. (V5.41) A generalized compact homotopy is a tame $X$, $f$ with tame germ data and compact solution set, and $t$ onto $[0,1]$ with tame fibres on which $f$ restricts to sc-Fredholm sections of the pull-backs.
390. (V5.42) Lemma 5.3.16 (preprint 5.23) is linear algebra, and the one-dimensional complement $Z$ of $\ker dt(x)$ exists iff $dt(x) \neq 0$.
391. (V5.43) The product rule and the affine chain rule hold at level-1 points; $\beta \cdot \ell(\varphi(\cdot) - \varphi(x))$ is sc-smooth.
392. (V5.44) Lemma 5.3.17 (preprint 5.24): for $\mathrm{Lin}_f(z) \circ b$ with values in the line of a smooth $e$, an sc$^+$-section vanishing at $x$ with small support cancels it on $K$.
393. (V5.45) Theorem 5.25 (preprint): for $X$ tame and $f$ proper with tame germ data, some allowable $s$ puts $f + s$ in good position at every zero.
394. (V5.46) Lemma 5.26 (preprint): general position implies good position, with the interior condition, without tameness of $X$.
395. (V5.47) Admitting sc-smooth partitions of unity: every open cover has a subordinate partition of unity of sc-smooth functions.
396. (V5.48) Sums, products, locally finite sums and chart-supported extensions by 0 of sc-smooth functions are sc-smooth.
397. (V5.49) Sc-smooth partitions of unity give sc-smooth bump functions.
398. (V5.50) $C^k$-bump functions and $C^k$ partitions of unity of a Banach space are as in the text.
399. (V5.51) The sc-smooth bump function property of a local model, with $f(x) \neq 0$; a retract is closed near each of its points.
400. (V5.52) Theorem 5.5.7 (preprint 5.34): bump functions on $X$ iff a covering family of charts with the local bump property.
401. (V5.53) A local model on a Hilbert $E_0$ (or a norm dominated by a continuous quadratic form) has the bump property; M-polyfolds on sc-Hilbert spaces admit bump functions.
402. (V5.54) The sc-smooth approximation property of a local model, with $V = O \cap W$ and closures in $E_0$.
403. (V5.55) With sc-smooth partitions of unity, charts restricted to sets with closure in the chart domain have the approximation property.
404. (V5.56) A covering family of charts with the approximation property gives sc-smooth Urysohn functions and sc-smooth partitions of unity.
405. (V5.57) Theorem 5.5.10 (preprint 5.37): sc-smooth partitions of unity iff a covering family of charts with the approximation property.
406. (V5.58) If $E_0$ admits $C^\infty$ partitions of unity, every local model on $E$ has the approximation property.
407. (V5.59) $c_0(\Gamma)$ and coordinate-wise $C^k$ embeddings are as in the text; Theorem 5.5.13 (preprint 5.40) is only stated.
408. (V5.60) WCG: a weakly compact set with dense span; reflexive: the bidual map is onto.

409. (V5.61) Reflexive and separable Banach spaces are WCG.
410. (V5.62) Theorem 5.5.16 (preprint 5.43) is only stated; Hilbert spaces have $C^\infty$-bump functions and are reflexive.
411. (V5.63) Corollary 5.5.17 (preprint 5.44) holds with Theorem 5.5.16 (preprint 5.43) for $E_0$ as an explicit hypothesis.
412. (V5.64) In the 2021 Definition 5.2.1, weakly proper with the bound $N(s) \leq 1$ on $U$ only is the earlier notion; weakly proper$_{<\infty}$ and proper$_{<\infty}$ are as in the text, with $N = \infty$ off $Y_{0,1}$.
413. (V5.65) Proper$_{<\infty}$ implies proper, weakly proper$_{<\infty}$ implies weakly proper, and for continuous $f$ proper$_{<\infty}$ implies weakly proper$_{<\infty}$.
414. (V5.66) An sc-Fredholm section with compact solution set is proper$_{<\infty}$; for sc-Fredholm sections the five compactness properties are equivalent.
415. (V5.67) In Theorem 5.3.5 (preprint 5.13) the transversal perturbations have solution sets that are compact sub-M-polyfolds.
416. (V5.68) Definition 5.3.13 (preprint 5.20) in the 2021 form, with $dt(x)$ onto read in a chart; the earlier data together with (3) give it.
417. (V5.69) Under the 2021 condition (3), $dt(x) \neq 0$ in a chart, and Lemma 5.3.16 (preprint 5.23) holds with a one-dimensional complement $Z$ of its kernel.
418. (V5.70) On $\mathbb{R}$, the graph section of the identity with $t(x) = 1/2 + \arctan(x)/\pi$ is a generalized compact homotopy in the 2021 sense but not in the earlier one.
419. (V6.1) At a chart point, Lin$(f, x)$ is $(f + s)'(x)$ plus the sc$^+$-operators; its members are Fredholm on level 0 of one index, and it is convex and, with sections through $f(x)$, nonempty.
420. (V6.2) $\lambda(E)$ and $\lambda^*(E)$ are lines; wedges change by determinants; $\lambda(\Phi)$ is functorial and nonzero iff $\Phi$ is bijective.
421. (V6.3) $\iota : \lambda(E^*) \to \lambda^*(E)$ is given by $\det(e_i^*(a_j))$ and maps the dual-basis wedge to $(e_1 \wedge \cdots \wedge e_n)^*$.
422. (V6.4) $\lambda^*(\Phi) \circ \iota = \iota \circ \lambda(\Phi^*)$ for an isomorphism $\Phi$.
423. (V6.5) $\Phi_E$ of a four-term exact sequence is the isomorphism of lines given by (91) for one choice.
424. (V6.6) Formula (91) holds for every choice.
425. (V6.7) $\Psi_E$ maps the wedge of the basis of $C$, in the order of (91), and $a \otimes d^*$ to the wedge of the basis of $B$.
426. (V6.8) $\det(T) = \lambda(\ker T) \otimes \lambda^*(\operatorname{coker} T)$ is a line for $T$ Fredholm.
427. (V6.9) An orientation of $T$ is a ray of $\det(T)$; there are exactly two.
428. (V6.10) Good left-projections are as in Definition 6.4.3 (preprint 6.9), $\leq$ is a partial order, the remarks after the definition hold, and $\Pi_T \neq \emptyset$ for $T$ Fredholm.
429. (V6.11) Two good left-projections have a common lower bound in $\Pi_T$, for any bounded $T$; the text's candidate need not be one.
430. (V6.12) The sequence $E(T, P)$ is exact.
431. (V6.13) $\gamma_T^P = \Phi_{E(T,P)}$ is an isomorphism $\det(T) \to \det(PT)$ given on wedges by (91).
432. (V6.14) For $P \leq Q$, $\gamma_{QT}^P \circ \gamma_T^Q = \gamma_T^P$.
433. (V6.15) $(\gamma_{RT}^P)^{-1} \circ \gamma_{ST}^P = \gamma_T^R \circ (\gamma_T^S)^{-1}$ for $P \leq S, R$.
434. (V6.16) The stabilization sequence of $T_\varphi$ is exact, and stabilizations exist.
435. (V6.17) $\psi((c_1 \wedge \cdots \wedge c_n)^*) = \det[c]^{-1} 1^*$.
436. (V6.18) $\iota_\varphi$ is an isomorphism given by the formula of Proposition 6.4.15 (preprint 6.15), and $m + l = n$ always holds.
437. (V6.19) $T \oplus S$ is Fredholm and $\det(T) \otimes \det(S) \to \det(T \oplus S)$ is the isomorphism given by the formula for every choice.

438. (V6.20) Lemma 6.5.1 (preprint 6.17) holds, without the hypothesis that $T$ is Fredholm.
439. (V6.21) The kernels $\ker(PS)$ near $T$ are trivialized continuously by $k \mapsto k + y(k, S)$, with continuous inverse.
440. (V6.22) The frames $\omega(S)$ are nowhere zero, and the frames of two charts with the same $P$ differ by a continuous nowhere-zero factor.
441. (V6.23) $\gamma_{PS}^{Q}$ maps the $P$-frame to a continuous nowhere-zero multiple of the $Q$-frame, for any two charts.
442. (V6.24) The charts $\gamma_{(T,P,\varepsilon)}$, read in frames, form a vector prebundle with fibres $\det(S)$.
443. (V6.25) $\mathcal{B}$ is the collection of Definition 6.5.6 (preprint 6.22), read in the frames.
444. (V6.26) $\mathcal{B}$ is a basis of the topology of $\mathrm{DET}(E, F)$.
445. (V6.27) $\mathrm{DET}(E, F)$ is a topological line bundle, trivialized by the $\gamma_{(T,P,\varepsilon)}$.
446. (V6.28) The transition maps are multiplications by continuous nowhere-zero functions.
447. (V6.29) A line bundle over a nonempty simply connected, locally path-connected base has exactly two orientations.
448. (V6.30) $\mathrm{DET}(E, F)$ over a nonempty convex set of Fredholm operators is a line bundle with exactly two orientations.
449. (V6.31) $\mathrm{Or}_x^f$, the orientations of $\mathrm{DET}(f, x)$, has exactly two elements, each determined by its value at one $L$.
450. (V6.32) Proposition 6.6.3 (preprint 6.28) holds with (1′) in place of (1), with global sections over $\mathbb{R}^n \times X$.
451. (V6.33) Finitely many sc$^+$-sections put $F$ in general position at all points $(0, (q, \Phi(q)))$, $q$ near $[0, 1]^n$.
452. (V6.34) The kernels of the linearizations along a continuous family of zeros in good position form a continuous family of subspaces.
453. (V6.35) Continuous families of subspaces have determinant line bundles, and orientations are continued along $[0, 1]$.
454. (V6.36) Kernels, images and cuts of continuous families are continuous families, with their orientations.
455. (V6.37) In the product chart the slice kernel is $\ker(T \oplus \varphi)$ with $T \in \mathrm{Lin}(f, x)$, and $T \oplus \varphi$ is onto.
456. (V6.38) The sequence (96) is exact.
457. (V6.39) Continuation and the stabilization isomorphisms compose to a bijection of the rays at the two ends.
458. (V6.40) With the sign $(-1)^{\dim \operatorname{coker} T}$ the orientations carried by $\Phi_E$ along a family are continuous; without it they need not be.
459. (V6.41) The push-forward along stabilized kernels is defined with that sign and is compatible with pull-back and concatenation.
460. (V6.42) With the sections of Lemma 6.6.4 (preprint 6.29) the stabilized kernels along a family form a continuous family.
461. (V6.43) Push-forwards along sc-smooth paths exist in the sets of Proposition 6.6.3 (preprint 6.28) and split along halves.
462. (V6.44) Two push-forwards along the same path agree.
463. (V6.45) Appending sections does not change the push-forward.
464. (V6.46) Affine reparametrizations keep sc$^+$-sections, and adding sections keeps general position.
465. (V6.47) Push-forwards along the two edges of a square of stabilized kernels with fixed end points agree.

466. (V6.48) Every smooth point has an open neighbourhood on which push-forwards exist and depend only on the end points.
467. (V6.49) Nearby smooth points are joined by sc-smooth paths in a chart, and concatenations are sc-smooth.
468. (V6.50) An orientable neighbourhood (a pair $V \subseteq U$) has exactly two continuous orientations, and every smooth point has one.
469. (V6.51) Reparametrizations of a path are homotopic to it with the end points fixed.
470. (V6.52) The sc-smooth homotopy classes of paths form a groupoid $PX$ on the smooth points.
471. (V6.53) Two chart points at $x$ identify $\mathrm{Lin}(f,x)$ and $\mathrm{Or}^f_x$, with the cocycle rules.
472. (V6.54) Under this identification the local propagations of two charts agree.
473. (V6.55) $\mathrm{Or}^f_x$ read in a chosen chart has two elements and is identified with that of any chart at $x$.
474. (V6.56) Local push-forwards give a transport along paths, homotopy invariant, functorial, and unique.
475. (V6.57) There is exactly one functor $\Gamma_f$ on $PX$ with property (□ ).
476. (V6.58) Orientability in the sense of Definition 6.6.14 (preprint 6.38) is the existence of a family $o_y$ preserved by $\Gamma_f$.
477. (V6.59) Definitions 6.6.14 and 6.6.15 (preprint 6.38 and 6.39) are equivalent.
478. (V6.60) Orientations of $f$ are the families preserved by $\Gamma_f$; they exist iff $f$ is orientable, two on each class.
479. (V6.61) For a proper $f$ in general position at its zeros, $S$ is a compact manifold with boundary with corners, and the tangent orientations $o_x$ of $T_xS = \ker f'(x)$ vary continuously along the sc-smooth paths of zeros in a chart.
480. (V6.62) An orientation of $\mathrm{DET}(f,x)$ gives exactly one orientation of $\mathrm{DET}(\hat f,(t,x))$, by $h \mapsto \tilde h$ with the sign $(-1)^{\dim\operatorname{coker} L}$, independent of $L$.
481. (V6.63) The proof of Lemma 6.3.3 (preprint 6.5) in §6.8.1 (preprint §6.7.1) goes through, with homogeneity in the bases of $A$ and $D$.
482. (V6.64) The proof of Proposition 6.4.11 (preprint 6.12) in §6.8.2 (preprint §6.7.2) goes through, with the spanning proved directly in $F$.
483. (V6.65) Near $(T_0,P_0)$, $T$ is Fredholm, $P \in \Pi_T$, and $\mathrm{ind}(PT) = \mathrm{ind}(T) = \mathrm{ind}(T_0) = \mathrm{ind}(P_0T_0)$, the indices taken as maps $E \to F$.
484. (V6.66) Two finite-dimensional subspaces of the same dimension have a common algebraic complement.
485. (V6.67) For $P,Q \in \Pi_T$ with the same range, the text's $A_T$ and $B_T$ are isomorphisms making the diagram of $E(T,P)$ and $E(T,Q)$ commute; their continuity in $T$ is V6.70.
486. (V6.68) Appending zero sections (Lemma 6.6.8) or arbitrary sections (Lemma 6.6.9) does not change the push-forward of orientations.
487. (V6.69) Two finite-dimensional subspaces of the same dimension of a normed space have a common closed, topological complement.
488. (V6.70) With such a closed $X$, $A_S$ (on $E$) and $B_S$ (lifted to $F$) depend continuously on $S$ near $T$, also jointly with the vector, and for every such $S$ they are the text's isomorphisms filling the diagram of $E(S,P)$ and $E(S,Q)$.
489. (V6.71) $O_f$ is the set of $(x,o)$, $x \in X_\infty$, $o \in \mathrm{Or}^f_x$, over $X_\infty$ with the topology of Theorem 2.3.10, and properties (1)–(3) of Theorem 6.6.17 are stated as the text does, (3) with the push-forward along segments of the path.
490. (V6.72) Sc-smooth maps $\mathbb{R}^n \to X$ are continuous into $X_\infty$, and points of $X_\infty$ near $x$ are joined to $x$ by sc-smooth paths inside any given neighbourhood of $x$ in $X_\infty$.

491. (V6.73) The topology defined by the continuous orientations is Hausdorff and makes $\pi_f$ a local homeomorphism.
492. (V6.74) Over M-polyfolds whose smooth points have small open neighbourhoods with connected smooth part ($\mathbb{R}^n$, for instance), that topology has exactly two continuous sections over such neighbourhoods.
493. (V6.75) Under the hypotheses of Theorem 6.6.13, it has property (3), it is the only topology with (1) and (3), and so the only Hausdorff one with (1)–(3); (2) is proved under the hypothesis of V6.74.
494. (V7.1) A groupoid is a structure of maps with the laws on composable pairs, and $|X|$ is its orbit space with $\pi$.
495. (V7.2) The isotropy group $G_x$ has the multiplication $gh = g \circ h$.
496. (V7.3) An ep-groupoid satisfies (1)–(4) of Definition 7.1.3 (2017: 7.3), with (3) existential and (4) on the fibred product of Part I's Proposition 2.5.20 (2017: 2.15), equivalently in sheet form.
497. (V7.4) Properness of $t : s^{-1}(Q) \to X$ is equivalent to the sequential criterion and passes to smaller closures.
498. (V7.5) $\widehat{\varphi} = t \circ (s|\mathbf{U}(\varphi))^{-1}$ is an sc-diffeomorphism with $\widehat{\varphi}(x) = y$; $T\varphi$ is only defined.
499. (V7.6) $s, t$ preserve levels and the degeneracy index, so $ml$ and $d_X$ are constant on classes.
500. (V7.7) $d_{|X|}(|x|) = d_X(x)$ is well defined.
501. (V7.8) If $X$ is tame, so is $\mathbf{X}$.
502. (V7.9) The ep-groupoids and the functors sc-smooth on objects and morphisms form the category $\mathcal{EP}$.
503. (V7.10) A functor sc-smooth on objects and continuous on morphisms is sc-smooth on morphisms; without continuity it need not be.
504. (V7.11) $\pi : X \to |X|$ is open.
505. (V7.12) Every point of $|X|$ has a monotone countable basis of open neighbourhoods.
506. (V7.13) $\pi(\mathrm{cl}_X A) = \mathrm{cl}_{|X|}\pi(A)$ for every $A \subseteq \mathrm{cl}_X U$, $U$ with the properness property.
507. (V7.14) Only finitely many $\psi$ have $s(\psi) = x$ and $t(\psi) \in U(x_0)$, and likewise with $s$ and $t$ exchanged.
508. (V7.15) $G_x$ is finite.
509. (V7.16) $g \mapsto \varphi g \varphi^{-1}$ is an isomorphism $G_x \to G_y$, unique up to an inner automorphism of $G_y$.
510. (V7.17) Inside every open $V \ni x$ there is a natural representation on a connected open $U$ with (1)–(3) of Theorem 7.1.13 (2017: 7.1).
511. (V7.18) M-polyfolds are locally path connected.
512. (V7.19) $\mathbf{U}(x)$ is the disjoint union of the connected open $\mathbf{U}(g)$, on which $s, t$ are sc-diffeomorphisms onto $U(x)$.
513. (V7.20) $G \times U$ is an M-polyfold on which sc-smoothness is sc-smoothness on each slice.
514. (V7.21) $G_x \ltimes U(x)$ is an ep-groupoid.
515. (V7.22) $\Gamma : G_x \ltimes U(x) \to X$ is a fully faithful functor onto the full subcategory on $U(x)$.
516. (V7.23) $\widehat{\Phi}$ is a homomorphism into the sc-diffeomorphisms of $\mathbf{U}(x)$ with the two commutative diagrams.
517. (V7.24) Two natural representations at $x$ agree on the component of $x$ in $U \cap U'$, and they restrict to invariant connected open neighbourhoods.
518. (V7.25) Every finite set of mutually isomorphic objects has a natural representation on neighbourhoods inside a given neighbourhood.
519. (V7.26) Definition 7.1.18 (2017: 7.8) is the datum of Proposition 7.1.15 (2017: 7.5).
520. (V7.27) $\tau : G_x \backslash U(x) \to \pi(U(x))$ is a homeomorphism, and the functor $G_x \ltimes U(x) \to X$ is

fully faithful.

521. (V7.28) Every natural representation is a local uniformizer, local uniformizers exist inside every neighbourhood, and they are injective on objects and morphisms.
522. (V7.29) An ep-subgroupoid is the full subcategory of a saturated sub-M-polyfold.
523. (V7.30) The inclusion $A \to X$ and the maps $i^{-1} \circ r$ are sc-smooth.
524. (V7.31) For saturated $A$, $\mathbf{A} = s^{-1}(A)$ is a sub-M-polyfold of $\mathbf{X}$.
525. (V7.32) The full subcategory of a saturated sub-M-polyfold is an ep-groupoid with the restricted structure maps.
526. (V7.33) Effectiveness and regularity are as in Definition 7.2.1 (2017: 7.11), and the targets of morphisms from regular objects are regular.
527. (V7.34) $X_{reg}$ is open.
528. (V7.35) $X$ is effective iff natural representations with injective $\Phi$ exist inside every neighbourhood.
529. (V7.36) For effective $X$, $X_{reg}$ is open and dense; conversely, density implies effectiveness.
530. (V7.37) $T_x$ is a normal subgroup, the kernel of $\Phi$ for small $U(x)$, characterized by $s = t$ near $g$, and conjugation carries $T_x$ to $T_y$.
531. (V7.38) $\sim$ is an equivalence relation, and $\mathbf{X}^R$ is the quotient.
532. (V7.39) The charts $\psi \circ \sigma \circ s$ of $\mathbf{X}^R$ have sc-smooth transitions, and $\mathbf{X}^R$ is an M-polyfold if Hausdorff and paracompact.
533. (V7.40) Under the same hypotheses $X_R$ is an effective ep-groupoid.
534. (V7.41) The isotropy group of $x$ in $X_R$ is $G_x/T_x = G_x^{\text{eff}}$.
535. (V7.42) $|X|$ is locally metrizable, regular and Hausdorff, and metrizable if paracompact.
536. (V7.43) $|U(x)|$ is metrizable.
537. (V7.44) Every point of $|X|$ has a monotone countable basis of open neighbourhoods.
538. (V7.45) $|X|$ is Hausdorff.
539. (V7.46) $|X|$ is regular.
540. (V7.47) A second countable $|X|$ is metrizable.
541. (V7.48) Subsets of $|X|$ are locally metrizable, and paracompact ones are metrizable.
542. (V7.49) Component compact means compact in every connected component.
543. (V7.50) Michael's characterization of paracompactness and the paracompactness of locally finite unions of closed paracompact sets hold for regular spaces.
544. (V7.51) A component compact $K$ has an open neighbourhood with paracompact metrizable closure.
545. (V7.52) $X^1$ is an ep-groupoid, tame if $X$ is.
546. (V7.53) Definition 7.5.1 (2017: 7.15) is paracompactness of $|X|$, implied by second countability of $X$.
547. (V7.54) If $|X|$ is paracompact, $|X^1|$ is metrizable, hence paracompact.
548. (V7.55) For paracompact $|X|$, saturated open covers have invariant continuous partitions of unity.
549. (V7.56) Definition 7.5.5 (2017: 7.16) is the existence of invariant sc-smooth partitions of unity with saturated local finiteness.
550. (V7.57) Remark 7.5.6 (2017: 7.2) has no formal content.
551. (V7.58) Paracompact $|X|$ and sc-smooth partitions of unity on $X$ give sc-smooth partitions of unity of $X$.
552. (V7.59) Proper subsets and the properness property are as in Definition 7.5.8 (2017: 7.17).
553. (V7.60) Proper subsets are closed.
554. (V7.61) $\pi$ maps proper subsets to closed sets, and for paracompact $|X|$ every closed set is so

obtained.

555. (V7.62) Theorem 7.6.1 (2017: 7.5) is Theorem 7.1.13 (2017: 7.1), proved along Appendix 7.6.1 (2017 7.5.1).
556. (V7.63) Morphisms between points of a small $U_1$ lie in the given open $N_g$.
557. (V7.64) $U = \bigcup_g t_g s_g^{-1}(U_2)$ is a connected invariant neighbourhood inside $U_1$.
558. (V7.65) Theorem 7.6.4 (2017: 7.6) is Theorem 7.5.7 (2017: 7.4).
559. (V7.66) Invariant continuous $\beta$ supported in $U(x)$ have invariant sc-smooth majorants supported in $\mathrm{cl}_X U(x)$.
560. (V7.67) The proof of Theorem 7.5.7 (2017: 7.4) goes through, with the support bound of Lemma 7.6.5 (2017: 7.10) and Mathlib's bump coverings.
561. (V7.68) A Hilbert atlas and Part I's Theorem 5.5.16 (2017: 5.14) give sc-smooth partitions of unity of $X$.
562. (V7.69) A functor sc-smooth on objects and continuous on morphisms is sc-smooth on morphisms, with the local formula of the 2021 remark.
563. (V7.70) The data (7.3) exist, and on a small invariant $V(x)$ the natural representation is $\Gamma(g, v) = s_g^{-1}(v)$, $\Phi(g)(v) = t(s_g^{-1}(v))$.
564. (V7.71) Proposition 7.1.15 holds in its 2021 form; the printed 2017 form of (2) would force $s(\psi) = t(\psi)$.
565. (V7.72) Limits of sequences in $\mathbf{U}(\psi)$ whose sources or targets converge in $U(s(\psi)) \cup U(t(\psi))$ stay in $\mathbf{U}(\psi)$.
566. (V7.73) Corollary 7.1.17 (4)–(7): the natural action, equivariance along $\gamma_\varphi$, the multiplication as an sc-diffeomorphism onto $\mathbf{U}(\psi \circ \varphi)$, and the properness property.
567. (V7.74) $T_x$ is the maximal subgroup acting trivially on a small $W(x)$, and conjugation by $\varphi$ maps $T_x$ onto $T_y$.
568. (V7.75) $T_R$ is the finest topology making $\pi$ continuous, and $\pi$ is an open quotient map.
569. (V7.76) The conclusion of Lemma 7.2.11 holds at every morphism iff $T_R$ is Hausdorff.
570. (V7.77) $T_R$ is paracompact for every ep-groupoid.
571. (V7.78) The natural charts give an M-polyfold structure iff $T_R$ is Hausdorff, and then $X_R$ is an effective ep-groupoid.
572. (V7.79) $T_R$ is Hausdorff iff $\mathbf{X}^T$ is closed in $\mathbf{X}$.
573. (V7.80) LUC is as in Definition 7.2.18, independent of the natural representation, and satisfiable, also for non-effective $X$.
574. (V7.81) $T_R$ Hausdorff, LUC and $\mathbf{X}^T$ closed are equivalent, and under LUC $X_R$ is well defined.
575. (V7.82) In the analogue of Zhou's example $\mathbf{X}^T$ is not closed, $T_R$ is not Hausdorff and LUC fails.
576. (V7.83) Under LUC for $X^1$ and connected dense $U(x) \cap X_1$, $(\mathbf{X}^1)^T = (\mathbf{X}^T)^1$ and $(\mathbf{X}^1)^R \to \mathbf{X}^R$ is a continuous injection onto $\pi(\mathbf{X}_1)$.
577. (V7.84) $X^{i+1} = (X^i)^1$, and under LUC every natural representation has the LUC property.
578. (V7.85) If $X^i$ has LUC and the level-one sets $U(x) \cap X_1$ are connected and dense, every $X^k$, $k \le i$, has LUC.
579. (V7.86) The Part I facts used in §7.4 are explicit hypotheses, satisfiable on a point.
580. (V7.87) $Tg$, defined without charts, is $T\Phi(g)$ on $T_x X$ for every natural representation.
581. (V7.88) (ZHOU), $T_x^\infty$ and (ZHOU') are as in Definitions 7.4.5 and 7.4.7, and hold when $T_x = G_x$.
582. (V7.89) With the Part I facts, (ZHOU) and (ZHOU') are equivalent.
583. (V7.90) With the Part I facts, (ZHOU) gives LUC for $X$ and $X^1$, and for every $X^i$ if (ZHOU) holds for every $X^i$.

584. (V7.91) (SLUC) is LUC for every $X^i$, and (ZHOU) with the Part I facts for every $X^i$ gives (SLUC).
585. (V7.92) (VSLUC) implies LUC for $X$ and $X^1$, (VSLUC) for every $X^i$ gives (SLUC), and (ZHOU) implies (VSLUC).
586. (V7.93) (LR) and (LR′) are as in Definitions 7.4.19, 7.4.21 and are equivalent by Lemma 7.4.20.
587. (V7.94) With the Part I facts, (ZHOU) implies (LR), and (ZHOU) for every $X^i$ gives (SLR).
588. (V7.95) Given the Part I facts for its object M-polyfold, the analogue of Zhou's example fails (ZHOU).
589. (V7.96) Index-preserving local homeomorphisms induce local homeomorphisms of the tame boundaries.
590. (V7.97) (7.25)–(7.29) define the groupoid $(bX, b\mathbf{X})$, and $B$ is a functor.
591. (V7.98) Theorem 7.6.9 holds in its topological part: metrizable $bX$, $b\mathbf{X}$, continuous $B$ with $d$ preimages, local homeomorphisms $s, t$, continuous $u, \iota, m$, properness.
592. (V7.99) A continuous map of tame boundaries that covers an sc-smooth map is sc-smooth; lifts of local sc-diffeomorphisms are local sc-diffeomorphisms.
593. (V7.100) For a tame ep-groupoid, $(bX, b\mathbf{X})$ with (7.25)–(7.29) is a tame ep-groupoid, and $B$ is an ep-functor.
594. (V7.101) Theorem 7.6.9 holds in full: (1)–(5), with the characterization and its uniqueness for centred $V$.
595. (V7.102) Under (ZHOU) for $X$ and the Part I facts on every level, every $X^i$ satisfies (ZHOU) and has LUC, and $X$ has (SLUC) and (SLR).
596. (V7.103) (VSLUC) for $X$ passes to every $X^i$; with the Part I facts, every $X^i$ has LUC and $X$ has (SLUC).
597. (V7.104) With the density of $X_\infty$ and level-connected natural representations, (SLUC) implies (VSLUC); so (VSLUC) and (SLUC) are equivalent.
598. (V8.1) The tangent data are Part I's tangent maps of $s, t, u, \iota$ and the composition (8.5), unique on composable pairs.
599. (V8.2) Maps sc-smooth on open sets and local sc-diffeomorphisms have tangents with the chain rule, and local sc-diffeomorphisms have local sc-diffeomorphisms as tangents, bijective on the fibres.
600. (V8.3) $T(X, \mathbf{X})$ is an ep-groupoid with the text's structure maps, tame if $X$ is, and $(p, \mathbf{p})$ is an sc-smooth functor into $X^1$.
601. (V8.4) On a sheet of $s$, the composition of $T\mathbf{X}$ is the tangent of $\varphi' \mapsto m(\sigma(t\varphi'), \varphi')$.
602. (V8.5) $s_T(\Psi \circ \Phi) = s_T(\Phi)$ and $t_T(\Psi \circ \Phi) = t_T(\Psi)$.
603. (V8.6) The composition of $T\mathbf{X}$ is associative.
604. (V8.7) $1_q = (1_x, Ts(1_x)^{-1}q)$ is a two-sided unit.
605. (V8.8) $(\varphi^{-1}, T\iota(\varphi)k)$ is a two-sided inverse of $(\varphi, k)$.
606. (V8.9) $s_T, t_T$ are surjective local sc-diffeomorphisms and all structure maps of $T(X, \mathbf{X})$ are sc-smooth.
607. (V8.10) $T(X, \mathbf{X})$ has the properness property, with $V' = p^{-1}(V)$.
608. (V8.11) If $X$ is tame, $Tr$ is a tame retraction and $T(X, \mathbf{X})$ is tame.
609. (V8.12) $T\varphi = t_T \circ (s_T|T_\varphi\mathbf{X})^{-1}$ is functorial and $G$ is a bijection onto the labelled pairs, compatible with the compositions; linearity is not verified.
610. (V8.13) The tangent of an sc-smooth functor is an sc-smooth functor.
611. (V8.14) $T : \mathcal{EP} \to \mathcal{EP}$ is a functor preserving tameness.
612. (V8.15) Natural transformations and natural equivalence are as in Definition 8.1.13 (2017: 8.1).

613. (V8.16) Natural equivalence is an equivalence relation.
614. (V8.17) The tangent of a natural transformation is a natural transformation between the tangent functors.
615. (V8.18) If $|X|$ is paracompact, $|TX|$ is metrizable, hence paracompact.
616. (V8.19) Strong bundle maps between strong bundles over M-polyfolds, read globally, preserve the double filtration, and the pull-back along $s$ has its universal property on $[0]$ and $[1]$.
617. (V8.20) A strong bundle over an ep-groupoid is a structure map $\mu$ on the pull-back with (1)–(3) of Definition 8.3.1 (2017: 8.4).
618. (V8.21) $\mu(\psi,\cdot) : W_x \to W_y$ is a topological linear isomorphism.
619. (V8.22) $(W, \mathbf{X}\,{}_s\!\times_P W)$ is an ep-groupoid and $(P, \pi_1)$ an sc-smooth functor.
620. (V8.23) A strong bundle map over ep-groupoids is compatible with $\mu$ as in Definition 8.3.2 (2017: 8.5).
621. (V8.24) A strong bundle functor is a functor of the functorial pictures inducing strong bundle maps on objects and morphisms.
622. (V8.25) Definitions 8.3.2 and 8.3.3 (2017: 8.5 and 8.6) are equivalent, and the strong bundle functor is determined by its object map.
623. (V8.26) Section functors are as in Definition 8.3.4 (2017: 8.7), and $\Gamma$, $\Gamma^+$ are closed under the linear operations.
624. (V8.27) A proper covering functor satisfies (1)–(3) of Definition 8.5.1 (2017: 8.8), with (3) on Part I's fibred product.
625. (V8.28) $L_F$ preserves the degeneracy index, and for tame $X$, $Y$, $d_{\mathbf{X}\,{}_s\!\times_F Y} = \frac{1}{2} d_{\mathbf{X}\times Y}$ along its image.
626. (V8.29) Composites of proper covering functors are proper covering functors.
627. (V8.30) Theorem 8.5.3 (2017: 8.5) holds for every natural representation on a small enough connected $U(x)$, with (5) read with $s$.
628. (V8.31) Morphisms over $F^{-1}(x)$ go to $G_x$, and the data of Theorem 8.5.3 (2017: 8.5) are determined by $U(x)$.
629. (V8.32) $\mathrm{Diff}_{sc}$, covering functors and geometric lifts are as in Definition 8.5.5 (2017: 8.9), with $C = \texttt{SingleObj G}$.
630. (V8.33) The $\varphi_{yy'}$ are functorial and the $\tau(y)$ are equivariant along $f : G_y \to G_x$.
631. (V8.34) Disjoint unions of M-polyfolds are M-polyfolds, with summand-wise sc-smoothness, degeneracy and tameness.
632. (V8.35) For nonempty $D$, a geometric lift of a covering functor gives a proper covering functor onto $G_x \ltimes U(x)$.
633. (V8.36) For ep-groupoids in `Type`, a proper covering functor is near $x$ and $F^{-1}(x)$ the functor of a geometric lift, up to fully faithful open embeddings.
634. (V8.37) Proper strong bundle covering functors are as in Definition 8.5.9 (2017: 8.10), with (3) on a pull-back.
635. (V8.38) $\#\mathrm{mor}(F^{-1}(x)) = \#G_x \cdot \#F^{-1}(x)$, and the classes of $F^{-1}(x)$ are indexed by their images in $|Y|$.
636. (V8.39) $\#G_x \cdot \#\Theta_j = \#G_{y_0} \cdot (\#\Theta_j)^2$ and $\#G_x = \#G_{y_0} \cdot \#\Theta_j$.
637. (V8.40) For isomorphic $x, x'$ the classes of $F^{-1}(x)$ and $F^{-1}(x')$ correspond, with a unique common index.
638. (V8.41) For isomorphic $x, x'$, corresponding classes have equal cardinality and $\#F^{-1}(x) = \#F^{-1}(x')$.
639. (V8.42) $\#F^{-1}(x)$ is constant on connected components of $|X|$ and an sc-smooth functor $X \to$

$\mathbb{N}$.

640. (V8.43) $\mu_X(F(\varphi)) \circ A_{s(\varphi)} = A_{t(\varphi)} \circ \mu_Y(\varphi)$ for every morphism $\varphi$ of $Y$.
641. (V8.44) If $X$ satisfies (ZHOU) and $T(g,h) = \mathrm{id}$ implies $Tg = \mathrm{id}$, then $TX$ satisfies (ZHOU).
642. (V8.45) With $U(x) \cap X_1$ connected and dense and $p$ open, LUC for $X^1$, or (SLUC) for $X$, gives LUC for $TX$.
643. (V8.46) Auxiliary norms over ep-groupoids are Part I's auxiliary norms invariant under $\mu$.
644. (V8.47) If $|W[0]|$ is paracompact, so is $|X|$, through the embedding $|0| : |X| \to |W[0]|$.
645. (V8.48) The local regularity property of strong bundles is as in Definition 8.4.2, and holds over trivial ep-groupoids.
646. (V8.49) A covering functor is injective on isotropy groups, and Example 8.5.6 shows it need not be surjective.
647. (V8.50) Definition 8.5.9 in the 2021 form is equivalent to the reading of Definition 8.10 (2017) in V8.37.
648. (V8.51) $B(w) = (P(w), A(w))$ is a strong bundle isomorphism $W \to F^*V$ covering the identity, compatible with the structure maps by (8.23).
649. (V8.52) If $X$ has (SLUC), with the hypotheses of Proposition 8.1.17 (1) on every $X^i$, then $TX$ has (SLUC), through $T(X^i) \cong (TX)^i$.
650. (V9.1) A branched ep$^+$-subgroupoid is a nonnegative functor, supported in the smooth set with paracompact orbit support, locally a weighted sum of indicators of properly embedded smooth submanifolds through the point.
651. (V9.2) The dimension of a submanifold at a point is the dimension of its tangent space in a chart, independent of the chart; $\Theta$ is $n$-dimensional if it has $n$-dimensional local branching structures.
652. (V9.3) On $\mathbb{R}^n$, finite weighted sums of indicators of closed submanifolds and indicators of locally closed submanifolds are branched ep$^+$-subgroupoids.
653. (V9.4) The graphs of $\beta_1$, $\beta_2$ and those of 0, $\beta_1 + \beta_2$ are two local branching structures of one $\Theta$ on $\mathbb{R}^2$ with no common sheet near the origin.
654. (V9.5) $|\mathrm{supp}(\Theta)|$ is locally closed in $|X|$.
655. (V9.6) $\Theta$ is closed or compact if $|\mathrm{supp}(\Theta)|$ is closed or compact in $|X|$.
656. (V9.7) At every point of the support there are symmetric local branching structures inside every neighbourhood.
657. (V9.8) $\Theta$ is of orbifold type if its values are 0 and 1, of manifold type if moreover there is at most one morphism between two points of its support.
658. (V9.9) The local dimensional decomposition $\Theta = \sum_k \Theta_k$ is constructed; the uniqueness of the global decomposition, as stated, does not hold as we read it.
659. (V9.10) The dimensional decompositions of two local branching structures agree near the point, unconditionally over $\mathbb{R}^m$ and under the graph representation in general.
660. (V9.11) $\mathrm{Gr}(X)$ is the category of formal rational sums of tangent sets at smooth points, with the push-forward by morphisms.
661. (V9.12) $T_yM$ is the set of tangent vectors fixed by the tangent of the retraction; in a chart it is a linear subspace of the dimension of $M$.
662. (V9.13) If the tangent sums do not depend on the structure, $T_\Theta$ is a section functor of $\mathrm{Gr}(X)$ with the text's (1) and (2).
663. (V9.14) The tangent sums do not depend on the structure, unconditionally over $\mathbb{R}^m$ and under the graph representation and local constancy of dimension in general.
664. (V9.15) A point is good if near it all sheets through it coincide, equivalently if $\Theta$ is constant on the support near it.

665. (V9.16) Being a good point does not depend on the local branching structure.
666. (V9.17) Good points are good for every structure, and near a good point $\Theta$ is a multiple of the indicator of one submanifold.
667. (V9.18) $s(\varphi)$ is good iff $t(\varphi)$ is good.
668. (V9.19) The good and bad sets are saturated subsets of the support.
669. (V9.20) The good set is open and dense in the support, the bad set closed and nowhere dense.
670. (V9.21) Every sheet of an $n$-dimensional structure at $x$ contains good points converging to $x$, unconditionally over $\mathbb{R}^m$ and under invariance of domain in general.
671. (V9.22) Under the graph representation, each class of sheets with one tangent space at $x$ has finite slices near $x$ whose $\Theta$-values sum to the weights of the class.
672. (V9.23) $T\Theta$ is a well-defined nonnegative functor on $TX$, and a branched ep$^+$-subgroupoid if the $TM_i$ are smooth submanifolds of $TX$ and $|X|$ is paracompact.
673. (V9.24) For $x \in M \cap \partial X$, $d_M(x) = 0$ iff $T_xM \subseteq T^R_xX$; this also holds at every $x \in M$, while the printed (2) $\square$ (1) does not hold as we read it at interior points.
674. (V9.25) $x \in \partial M$ iff $x \in \partial X$ and $T_xM \not\subseteq T^R_xX$.
675. (V9.26) The boundary weights do not depend on the structure, vanish where $d_X = 0$, and define a nonnegative functor $\partial\Theta$ supported in $\partial X$.
676. (V9.27) $\partial\Theta$ satisfies (1) and (2) and is the only functor vanishing off $\mathrm{supp}(\Theta)$ with (2) on $\mathrm{supp}(\Theta)$; (1) and (2) alone do not determine it.
677. (V9.28) $\partial\Theta \le \Theta$, and $\partial\Theta$ need not be $\Theta$ restricted to $\partial X$.
678. (V9.29) Oriented tangent spaces are tangent sets with a chart-independent class of positive frames, and $\widehat{\mathrm{Gr}}(X)$ is the category of their formal sums.
679. (V9.30) An orientation of $\Theta$ is a natural lift of $T_\Theta$ that is locally the oriented sum of an oriented local branching structure.
680. (V9.31) The indicator of a point has infinitely many orientations.
681. (V9.32) Two orientations of an $n$-dimensional $\Theta$ that agree at good points agree at all smooth points; without equidimensionality the lemma does not hold as we read it.
682. (V9.33) In Example 9.3.5 the two orientations agree at the origin, differ for $x > 0$, agree for $x < 0$, and there are at least five orientations with weights $1$.
683. (V9.34) $\Theta$ is tame if it has local branching structures with tame induced structures on domains with the natural $G_x$-action and the properness property.
684. (V9.35) Tame submanifolds with the same tangent space at $y$ have the same degeneracy index at $y$.
685. (V9.36) The outward-first convention orients a hyperplane of $T_yM$, independently of the outward vector chosen.
686. (V9.37) The degree-$\ell$ parts of the oriented sums of two structures agree, so the boundary orientation (9.12) does not depend on the structure.
687. (V9.38) $\widehat{T}_{\partial\Theta}$ is $0$ off the support and the weighted sum of the boundary orientations of the sheets with $d_{M_i}(y) = 1$.
688. (V9.39) The relative tangent space $T^K_zM$ is the span of limiting unit directions in a chart, independent of the chart, and essential points are limits of points where it is everything.
689. (V9.40) The essential set of $K$ is closed in $K$.
690. (V9.41) Property (T) says that the orientation relation of two sheets with the same tangent space is the same wherever their tangent spaces agree.
691. (V9.42) For sheets of one dimension essentiality in $M_i \cap M_{i'}$ does not depend on the reference sheet; for different dimensions it does.
692. (V9.43) A set has measure zero if its chart images have Lebesgue measure zero.

693. (V9.44) The non-essential points of any subset of a finite-dimensional manifold form a set of measure zero.
694. (V9.45) $y$-equivalence is an equivalence relation on the sheets through $y$, and equivalent sheets have the same tangent space.
695. (V9.46) The points of $M_U$ with at least two $y$-classes form a null set.
696. (V9.47) Off a null set, $M_U$ splits into pieces $M_U^J$ on which $\Theta$, $T_\Theta$ and $\hat{T}_\Theta$ have the text's constant form.
697. (V9.48) Two oriented structures with the same weights and oriented sums agree, off a null set, piece by piece as in (1)–(5) of the text, without Property (T).
698. (V9.49) A subset of $S = |\mathrm{supp}(\Theta)|$ has measure zero if, for a system of null sets in the sheets of a class, a tame local branching structure at every point of the support has null preimages.
699. (V9.50) For structures with sheets of one class, measure zero does not depend on the structures chosen; across dimensions the transfer of null sets between sheets does not hold.
700. (V9.51) $L(S)$ is the σ-algebra generated by the Borel sets and the sets of measure zero, and its members are the sets that differ from a Borel set by a set of measure zero.
701. (V9.52) Measure zero in $\partial S$ and $L(\partial S)$ are defined from null sets in the boundaries of the sheets and do not depend on the structures chosen.
702. (V9.53) The hypotheses of §9.5 are satisfiable on $\mathbb{R}^2$, with nonempty sets of measure zero and members of $L(S)$ that are not Borel.
703. (V9.54) For a natural representation of $G_x$ and $y \in U(x)$, $\#(G_x/H) \cdot \#G_y = \#G_x$, and preimages decompose disjointly over coset representatives on a slice around $y$.
704. (V9.55) $\#T_x \cdot \#G_x^{\mathrm{eff}} = \#G_x$, and under LUC (9.23) holds.
705. (V9.56) The orders of the isotropy groups and of the effective isotropy groups are invariant along morphisms, and $|M_U| = |M_{U'}|$.
706. (V9.57) For effective $X$ the effective and full isotropy orders agree, and the effective order on $X$ is the isotropy order on $X_R$.
707. (V9.58) Two structures at $x$ have the same top dimension.
708. (V9.59) The unions of the top-dimensional sheets of two structures agree on the common domain.
709. (V9.60) At top-good points the top-dimensional parts of two structures agree.
710. (V9.61) Top-goodness does not depend on the structure, unconditionally over $\mathbb{R}^m$ and under the graph representation in general.
711. (V9.62) The top-dimensional part at $x$ is the sum of $\Theta$ over finite top slices, and the class sums agree for two structures.
712. (V10.1) An equivalence of ep-groupoids is a full and faithful sc-smooth functor, a local sc-diffeomorphism on objects, inducing a homeomorphism of orbit spaces.
713. (V10.2) An sc-smooth functor is an equivalence iff it is an equivalence of categories and a local sc-diffeomorphism on objects.
714. (V10.3) The refinement of a nonempty ep-groupoid along an open covering is an ep-groupoid, and its projection to the ep-groupoid is an equivalence.
715. (V10.4) Natural equivalence of sc-smooth functors is an equivalence relation compatible with composition on both sides.
716. (V10.5) A functor naturally equivalent to an equivalence is an equivalence.
717. (V10.6) If $F$ and $\Phi \circ F$ are equivalences, so is $\Phi$.
718. (V10.7) The tangent of an equivalence is an equivalence.
719. (V10.8) For id and $-$id on $\mathbb{Z}/2 \ltimes \mathbb{R}^2$, no family through $1_0$ is even continuous at $0$, while $\alpha_z = (-1, z)$ is sc-smooth.

720. (V10.9) For an equivalence and $\psi : F(x) \to y$, $g \mapsto \psi \circ F(g) \circ \psi^{-1}$ is an isomorphism $G_x \to G_y$ mapping $T_x$ onto $T_y$.
721. (V10.10) For an equivalence, $F_R$ is continuous and open, and one reduced morphism space is Hausdorff iff the other is.
722. (V10.11) An equivalence with Hausdorff reduced morphism spaces descends to an equivalence of the reduced ep-groupoids.
723. (V10.12) The functor of Example 10.1.15 is compatible with no map of the reduced morphism spaces.
724. (V10.13) (LUC) holds for one side of an equivalence iff it holds for the other.
725. (V10.14) The Zhou condition holds for one side of an equivalence iff it holds for the other.
726. (V10.15) (SLUC) holds for one side of an equivalence iff it holds for the other.
727. (V10.16) (SLR) passes along an equivalence once (LR) passes along the equivalences of the lifts.
728. (V10.17) The weak fibred product of two sc-smooth functors is a groupoid with the structure maps of Definition 10.2.1.
729. (V10.18) If $F$ or $G$ is a local sc-diffeomorphism on objects, the weak fibred product is an ep-groupoid with the iterated sub-M-polyfold structures.
730. (V10.19) For $F$ an equivalence, $\pi_Z$ is an equivalence, $F \circ \pi_X \simeq G \circ \pi_Z$, $d = d_Z \circ \pi_Z$, and tameness and paracompactness of the orbit space pass from $Z$.
731. (V10.20) For $G$ an equivalence, $\pi_X$ is an equivalence and $d = d_X \circ \pi_X$; for two equivalences, $d_X(x) = d(x, \phi, z) = d_Z(z)$.
732. (V10.21) For two equivalences both projections of the weak fibred product are equivalences, and $\tau(x, \phi, z) = \phi$ is natural.
733. (V10.22) A strong bundle equivalence is a strong bundle functor over an equivalence, an equivalence at level $[0]$, a homeomorphism of $|W[1]|$, and locally a strong bundle isomorphism on objects and morphisms.
734. (V10.23) Push-forward and pull-back of section functors along a strong bundle equivalence are well defined and mutually inverse.
735. (V10.24) $|\Phi| = |\Psi|$ implies $\Phi^* f' = \Psi^* f'$ for every section functor $f'$.
736. (V10.25) $|\Phi| = |\Psi|$ implies $\Phi_* f = \Psi_* f$ for every section functor $f$.
737. (V10.26) $(\Psi \circ \Phi)^* = \Phi^* \circ \Psi^*$ and $(\Psi \circ \Phi)_* = \Psi_* \circ \Phi_*$.
738. (V10.27) $\Phi_*$ and $\Phi^*$ are mutually inverse bijections of sc-smooth section functors, depend only on $|\Phi|$, and preserve $\mathrm{sc}^+$-section functors.
739. (V10.28) A natural transformation of strong bundle functors is a natural family $T(w) : \Phi(w) \to \Psi(w)$ over a natural transformation $\tau$ of the base functors.
740. (V10.29) $T(w) = (\tau(P(w)), \Phi(w))$ and $\Psi(w) = \mu'(\tau(P(w)), \Phi(w))$, so $T$ and $\Psi$ are determined by $\Phi$ and $\tau$, and $T$ is a strong bundle map.
741. (V10.30) Definitions 10.4.8 and 10.4.10 agree for strong bundle functors, and naturally equivalent strong bundle functors induce the same map of orbit spaces.
742. (V10.31) Strong bundles over ep-groupoids and strong bundle functors form a category with the strong bundle equivalences as a distinguished class.
743. (V10.32) Isomorphisms of $\mathcal{EP}$ and composites of equivalences are equivalences, and so is a functor naturally equivalent to one.
744. (V10.33) A refinement of a diagram $X \leftarrow A \to Y$ is given by an equivalence of the middle ep-groupoids commuting with both legs up to natural equivalence.
745. (V10.34) A refinement of a refinement is a refinement.
746. (V10.35) A common refinement of two diagrams refines both.
747. (V10.36) Having a common refinement is transitive, through the weak fibred product (10.11).

748. (V10.37) Equivalence of diagrams is an equivalence relation.
749. (V10.38) The map $|\Phi| \circ |F|^{-1}$ of orbit spaces is continuous and unchanged under refinement.
750. (V10.39) $||[d]||$ is well defined, and a homeomorphism when both legs are equivalences.
751. (V10.40) The composition of classes of diagrams through weak fibred products is well defined, associative and unital.
752. (V10.41) Generalized isomorphisms are the isomorphisms of $\mathcal{EP}(E^{-1})$.
753. (V10.42) $[X \leftarrow A \xrightarrow{\Phi} Y]$ is a generalized isomorphism iff $\Phi$ is an equivalence, with the reversed diagram as inverse.
754. (V10.43) $|g \circ f| = |g| \circ |f|$ for generalized maps.
755. (V10.44) Ep-groupoids and generalized maps form the category $\mathcal{EP}(E^{-1})$.
756. (V10.45) $i : \mathcal{EP} \to \mathcal{EP}(E^{-1})$ is a functor inverting the equivalences, with $i(F)^{-1} = [Y \xleftarrow{F} X \xrightarrow{1} X]$.
757. (V10.46) The orbit spaces and the maps $||[d]||$ form a functor $\mathcal{EP}(E^{-1}) \to$ TOP.
758. (V10.47) (LUC) is invariant under generalized isomorphisms.
759. (V10.48) Equivalence of pairs $(X, \beta)$ by generalized isomorphisms is an equivalence relation.
760. (V10.49) The ep-groupoids with (LR) span full subcategories of $\mathcal{EP}$ and $\mathcal{EP}(E^{-1})$.
761. (V10.50) (LUC), (SLUC) and (ZHOU) are invariant under generalized isomorphisms.